

Gasoline Profiles and Impact of Ethanol Blending in Europe

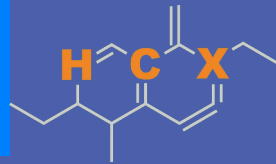
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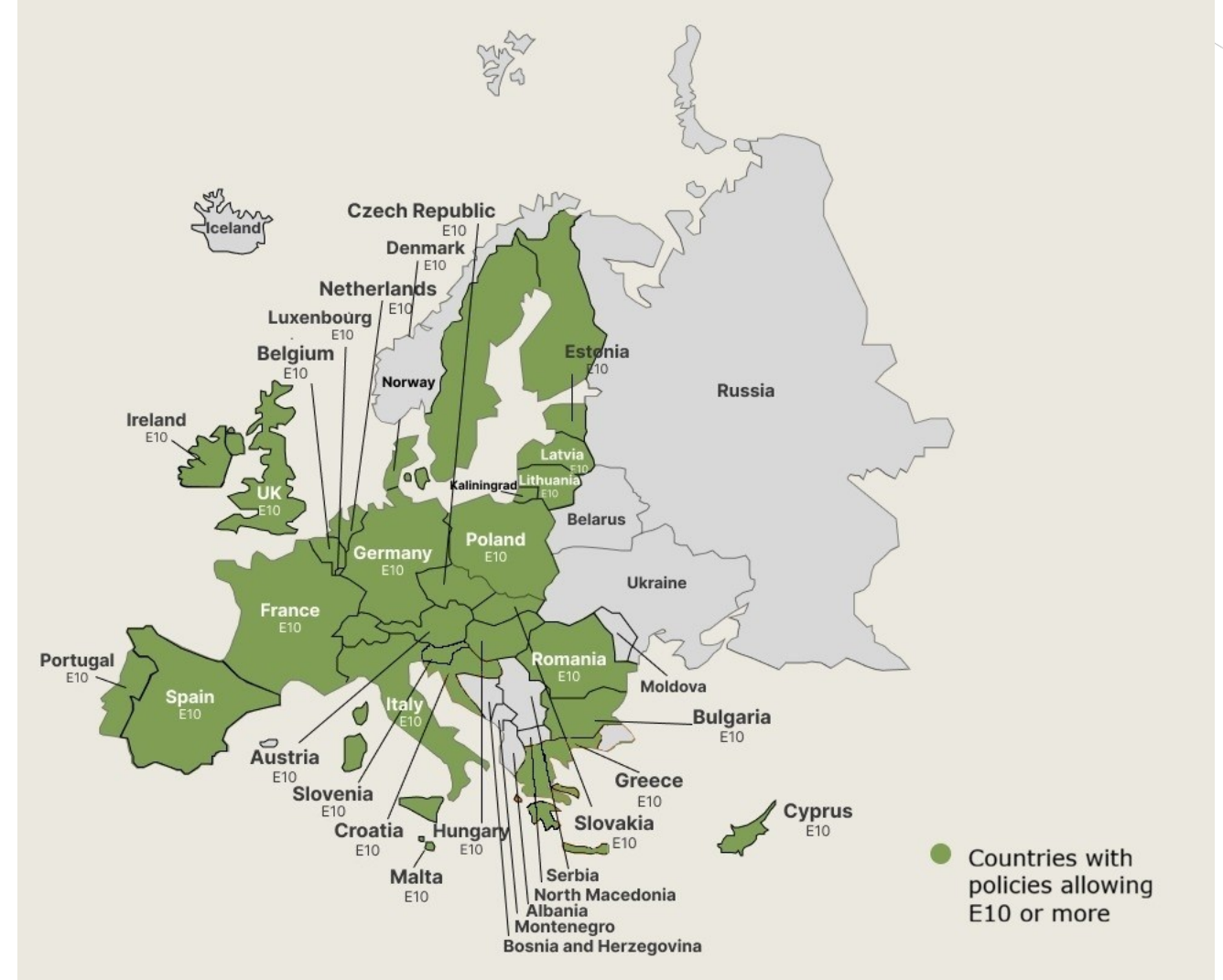


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Ethanol Blending in Europe

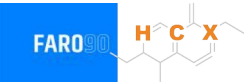
There are important fuel quality and environmental impact of vehicle emission challenges in the Region.

- The use of ethanol improves gasoline quality and creates flexibility in gasoline production.
- Ethanol use is a cost-effective way to increase gasoline octane and to replace more expensive gasoline components.
- Ethanol contributes to transport decarbonization and air quality improvement.
- There are opportunities across Europe to increase the ethanol blend level and implement new policies on the use of gasoline-ethanol blends.



Twenty eight countries with potential and additional use of ethanol were studied: 1) Gasoline market profiles; 2) Optimization of gasoline blends with ethanol and, 3) Environmental impact of gasolines blended with ethanol.

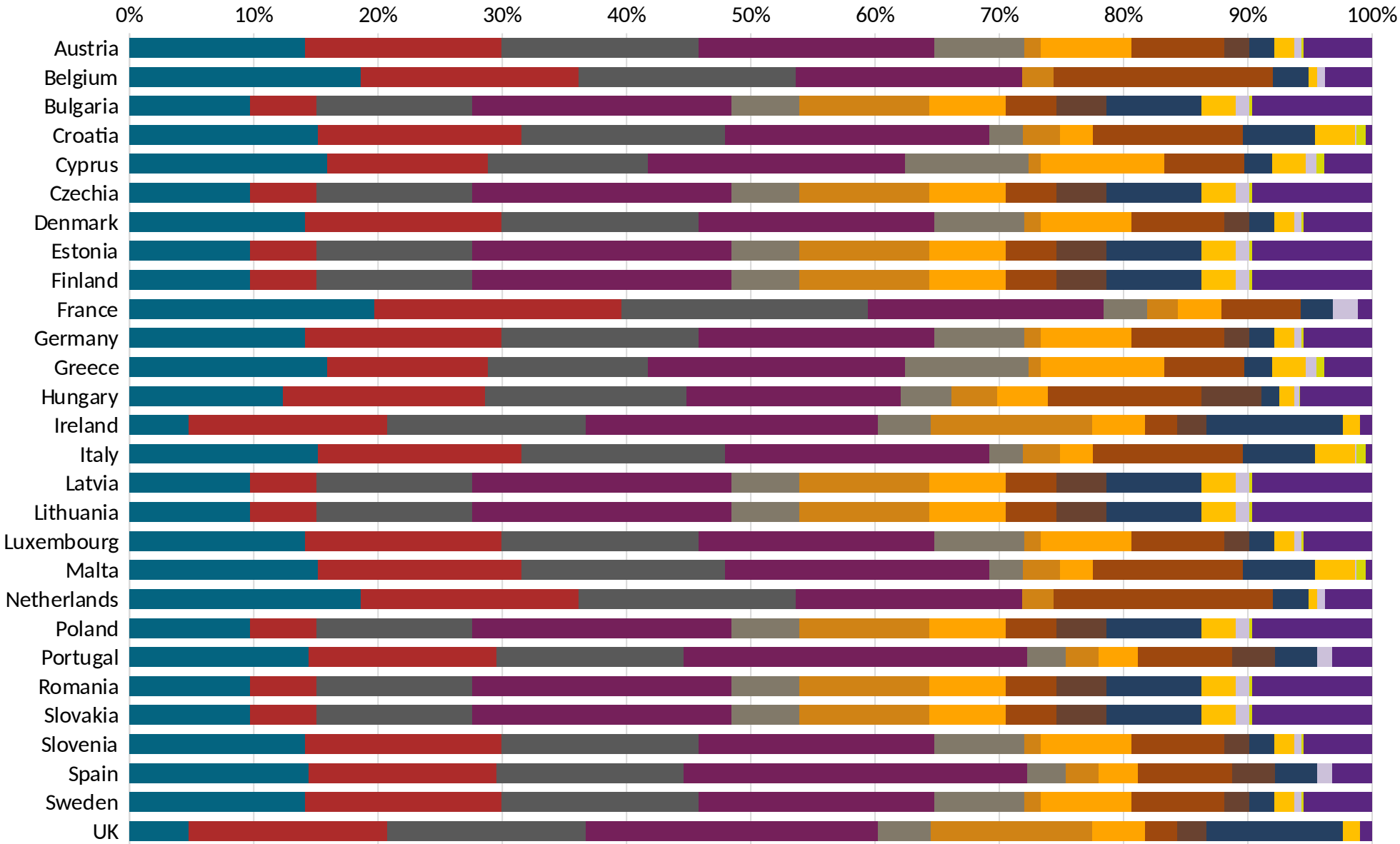
Gasoline Component Blending in Europe



Gasoline is a blend of a base gasoline and other components. Each component provides different properties to the final blend, for example, isomerates, alkylates and butanes increase the octane. The components commonly used in Europe are:

- Catalytic Gasoline
- Light Primary Naphta
- Heavy Primary Naphta
- Reformate
- Isopentane
- Alkylate
- Isomerate
- Hydrocracked Gasoline
- Coker Naphta
- n-Butane
- MTBE
- ETBE
- TAME
- Aromatics

Source:
Faro90



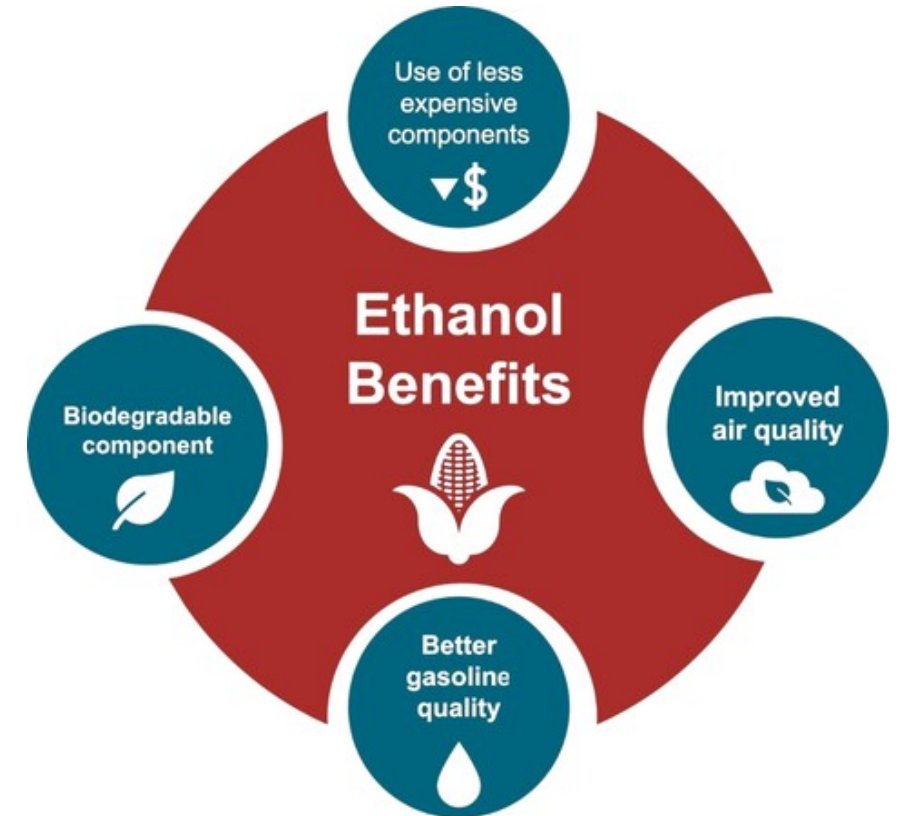
Gasoline Blending Optimization

In some parts of the world, ethanol is added to gasoline as a blending component. The advantages of ethanol include that it is a renewable fuel made of biomass; that it is an octane booster that helps to dilute sulfur; and that it allows the fulfillment of environmental objectives. To determine the optimal components to be blended with ethanol, a **blending model** was used. This model selects the components to add in the gasoline/ethanol blend based on:

- Components prices,
- Properties each component affects,
- Quality parameters by country, and
- Component availability by country.

Through iterations, the model obtains the %v/v of the components to be blended with 10%, 15%, 20%, 25% and 30% of ethanol, in such a way that the final blend complies with the required properties of a finished gasoline by country.

The blending model uses gasoline component spot average prices 2024 and provides fuel prices that do not include country distribution costs, local taxes and subsidies and import or gas station margins.



Vehicle Emission Impact for Ethanol Gasoline Blending

The model used in this analysis takes as a reference the [International Vehicle Emissions Model \(IVE\)](#).

The model uses the Base Emission Rates from IVE model, as well as its Adjustment Factors based on:

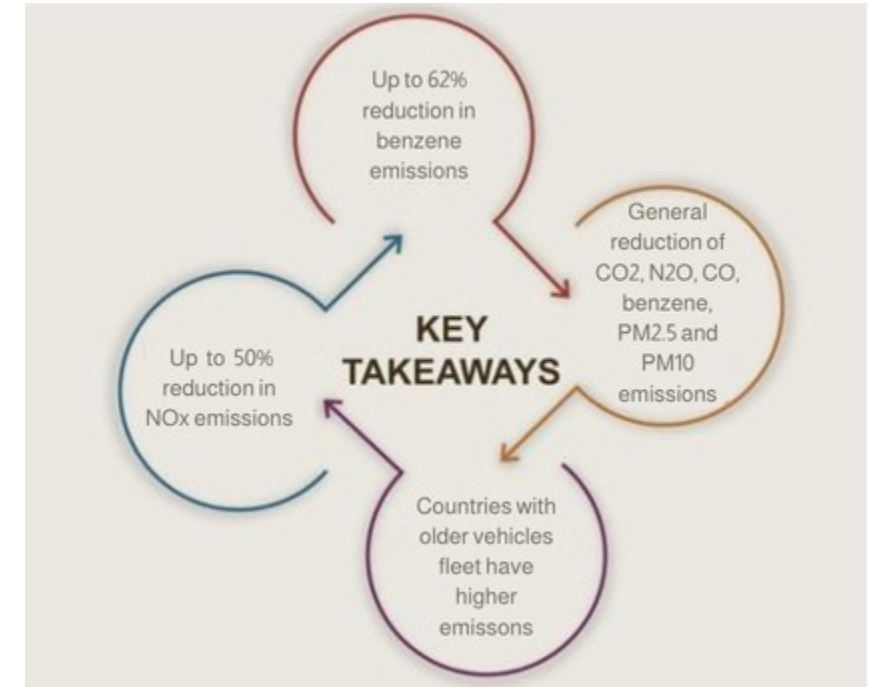
- Vehicle technology (cars, trucks, buses, motorcycles),
- Vehicle fleet average age,
- Average traveled distance per vehicle by country, as well as
- Geographical and climatic conditions (altitude, humidity, temperature).

Emissions of criteria pollutants, toxic pollutants, and greenhouse gases (GHG) were calculated and calibrated with emission inventories, using real gasoline quality data. The reduction rates for gasoline/ethanol blends were obtained from various sources (IPCC, US Grains, among others).

Emission estimations for different pollutants for gasoline and gasoline/ethanol blends (10%, 15%, 20%, 25% and 30% ethanol) were determined using the IVE Model. A comparison between the results and the European (Euro 6) requirements is made. Results are also compared with real emissions of the United States vehicle fleet*.

**Source: Bureau of transportation statistics.*

Main Results



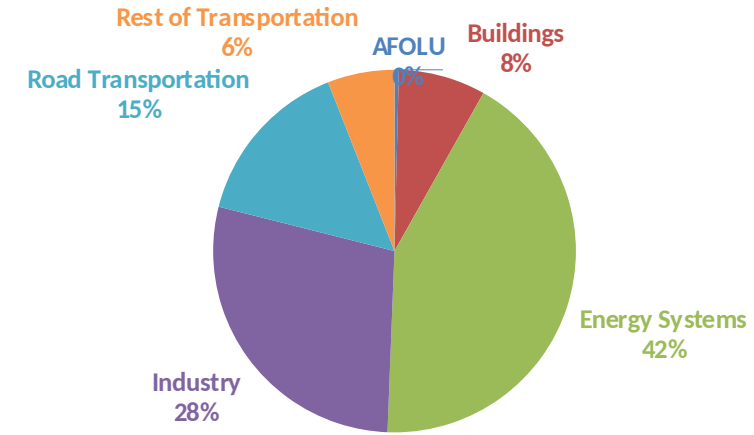
Life Cycle GHG Emissions

The Global Agenda for Climate Action calls for countries, cities and regions, businesses, and members of civil society around the world to achieve a low-carbon and climate-resilient economies in support of the Paris Agreement. Road Transportation accounts for 15% of total GHG emissions (IPCC,2023).

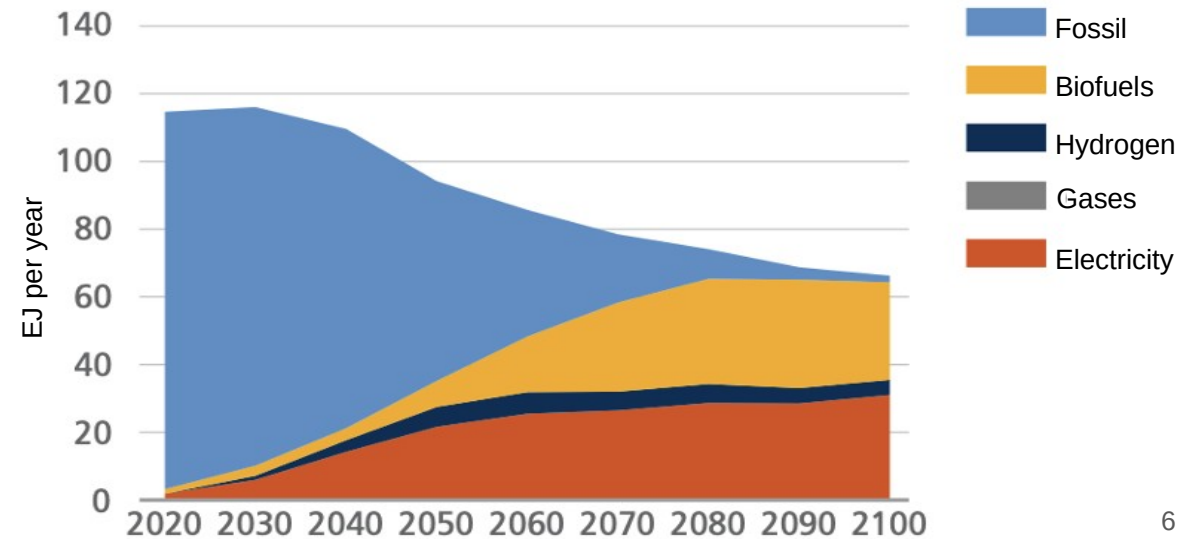
Ethanol gasoline blends are an internationally adopted practice that offer significant short-term benefits. Biofuels are an important solution to achieve reduction targets in the transportation sector.

Country GHG emission reductions are calculated using **RED II/III** and **GREET** life cycle emission methodologies and the **International Vehicular Emissions Model (IVE)**. Results are compared to the current country targets for 2035. Current 2024 country emission are those published by **IPCC**.

Current GHG Global Emissions Distribution



IPCC Transport Sector Sustainable Development Goals and climate policies Scenario, 2023

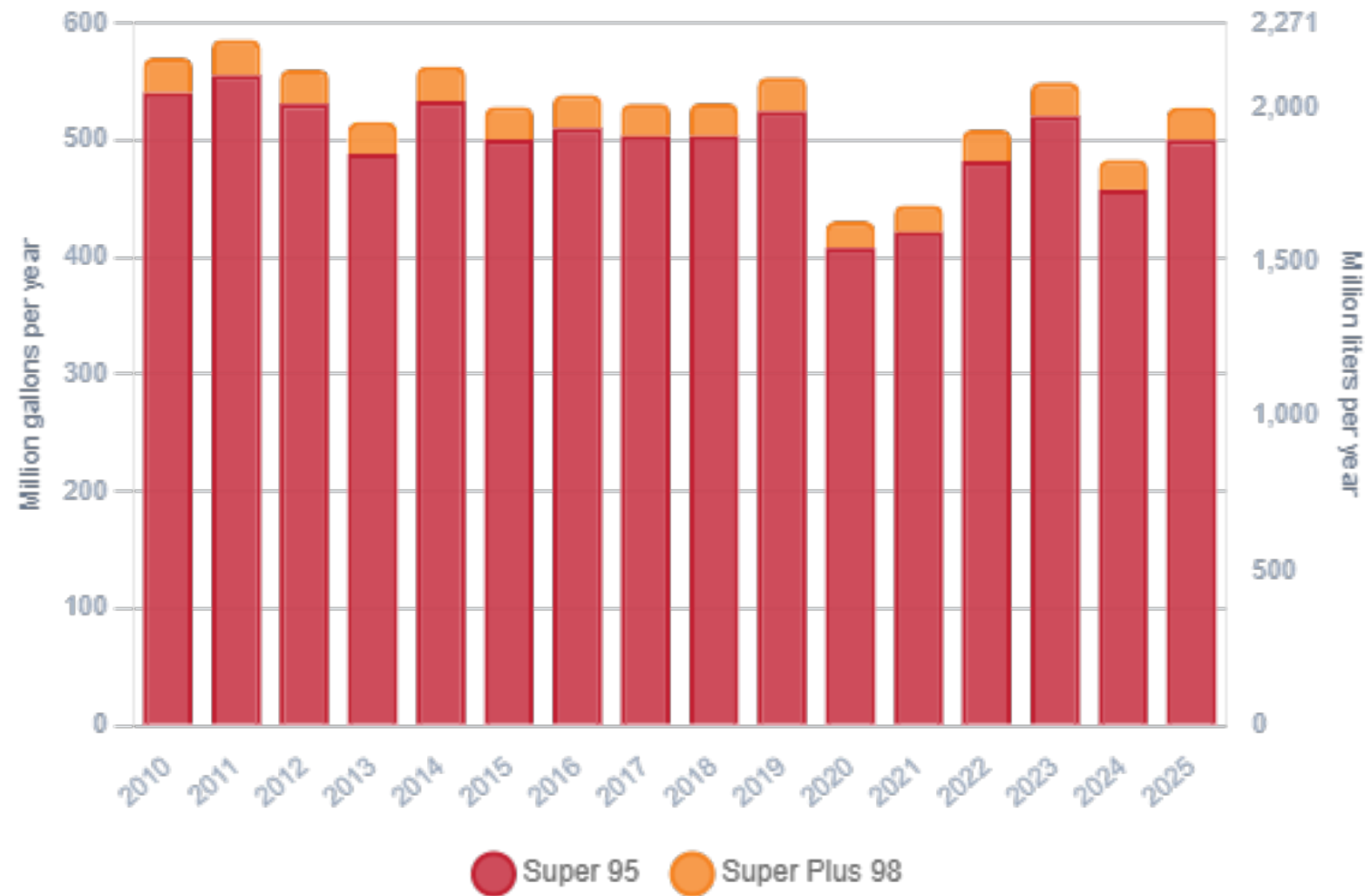


Case Studies



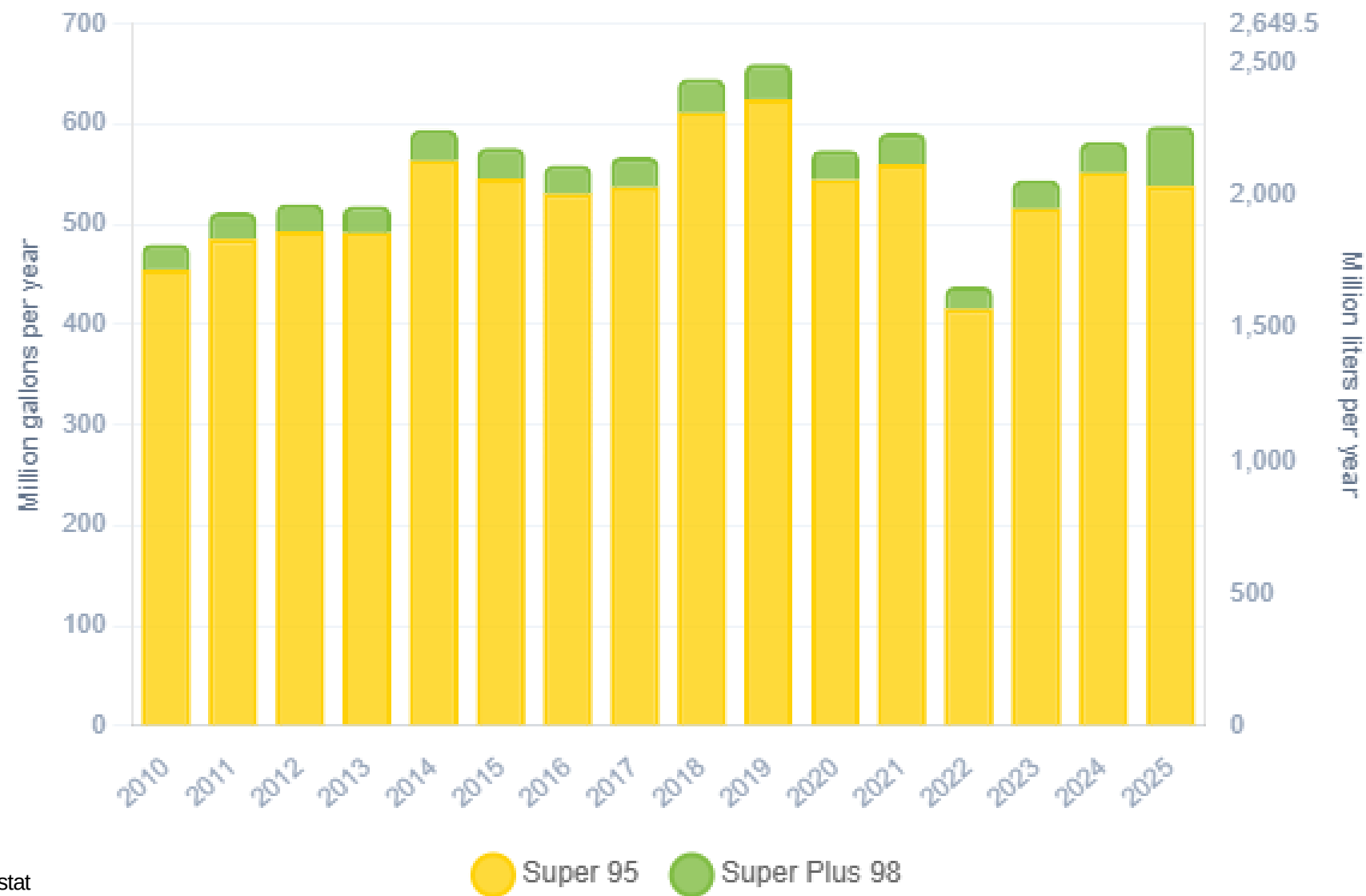
- | | | |
|------------------------------------|--|---|
| 1. <u>Austria</u> | 11. <u>Germany</u> | 21. <u>Poland</u> |
| 2. <u>Belgium</u> | 12. <u>Greece</u> | 22. <u>Portugal</u> |
| 3. <u>Bulgaria</u> | 13. <u>Hungary</u> | 23. <u>Romania</u> |
| 4. <u>Croatia</u> | 14. <u>Ireland</u> | 24. <u>Slovakia</u> |
| 5. <u>Cyprus</u> | 15. <u>Italy</u> | 25. <u>Slovenia</u> |
| 6. <u>Czechia</u> | 16. <u>Latvia</u> | 26. <u>Spain</u> |
| 7. <u>Denmark</u> | 17. <u>Lithuania</u> | 27. <u>Sweden</u> |
| 8. <u>Estonia</u> | 18. <u>Luxembourg</u> | 28. <u>United Kingdom</u> |
| 9. <u>Finland</u> | 19. <u>Malta</u> | |
| 10. <u>France</u> | 20. <u>Netherlands</u> | |

Gasoline Demand in Austria



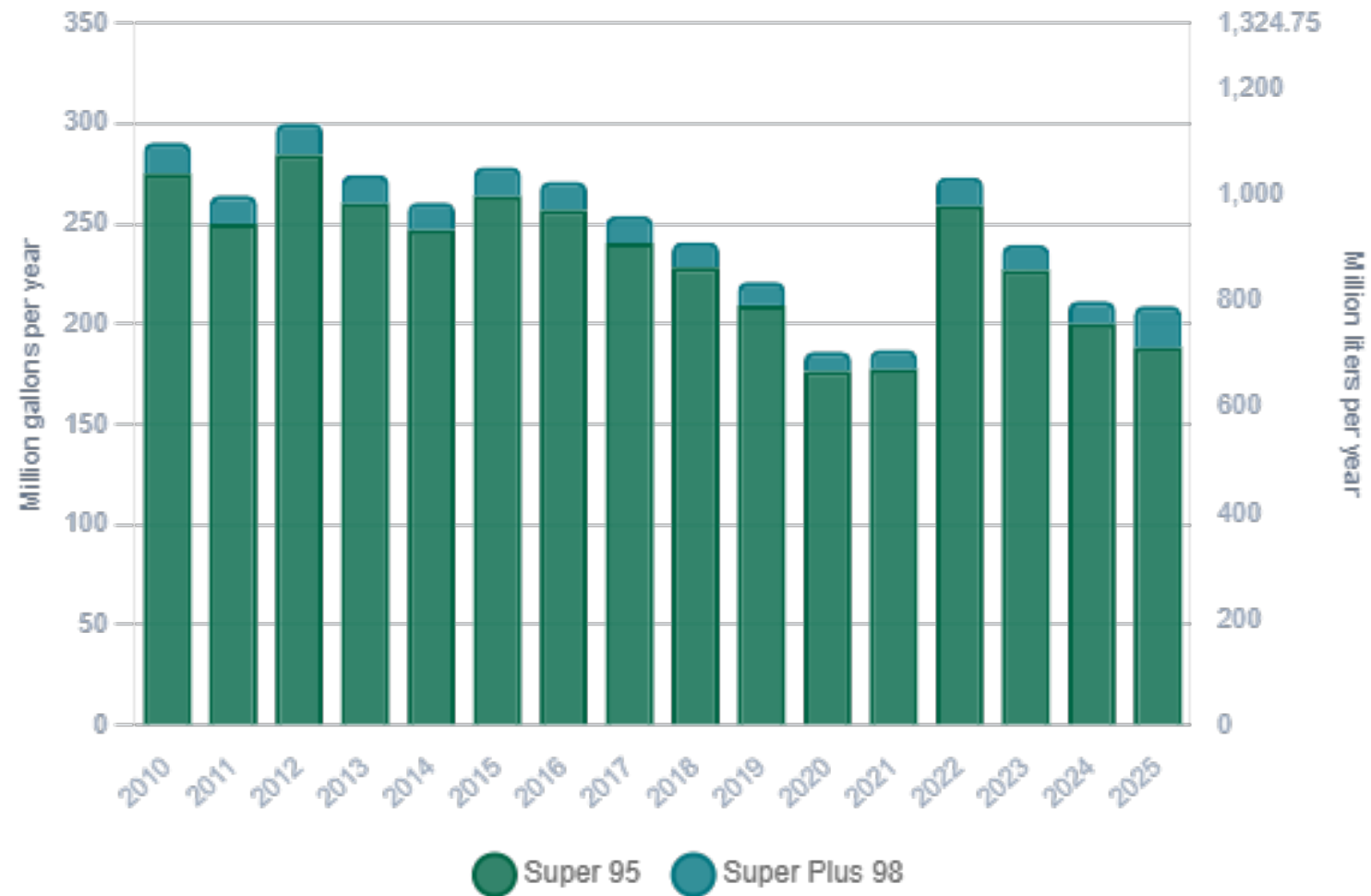
Source: Eurostat

Gasoline Production in Austria



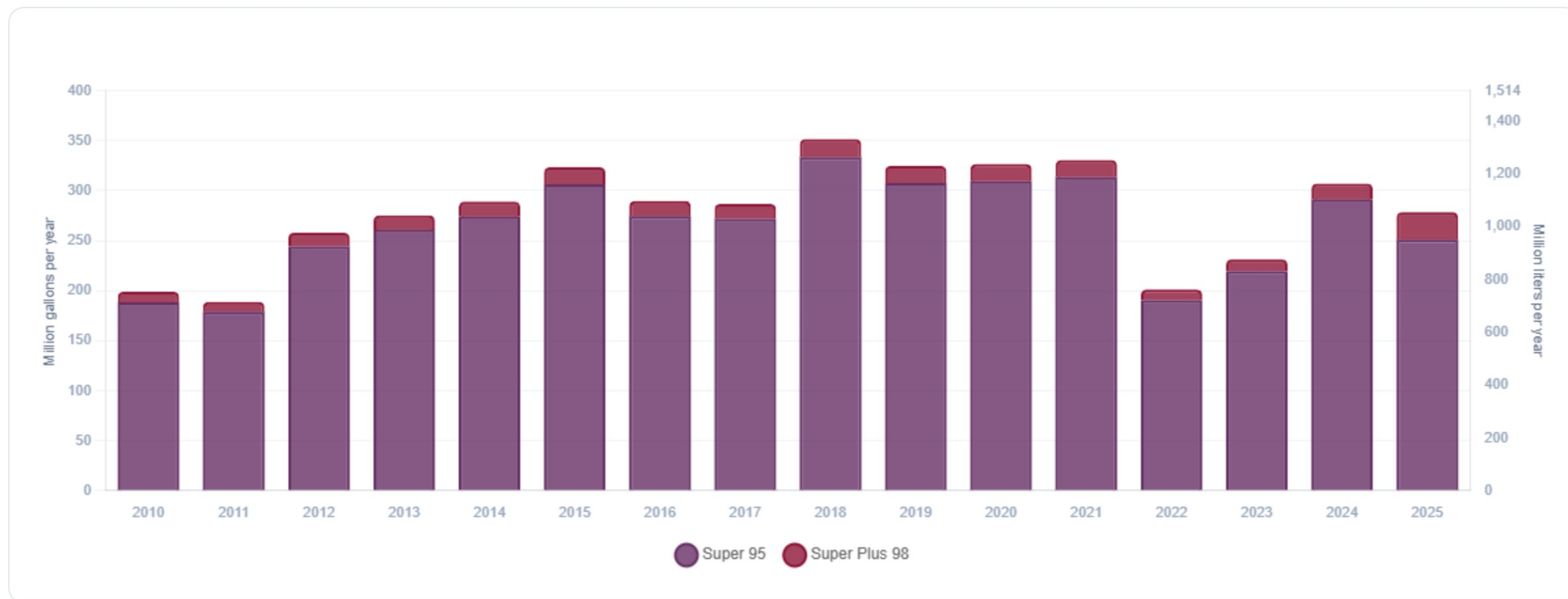
Source: Eurostat

Gasoline Imports in Austria



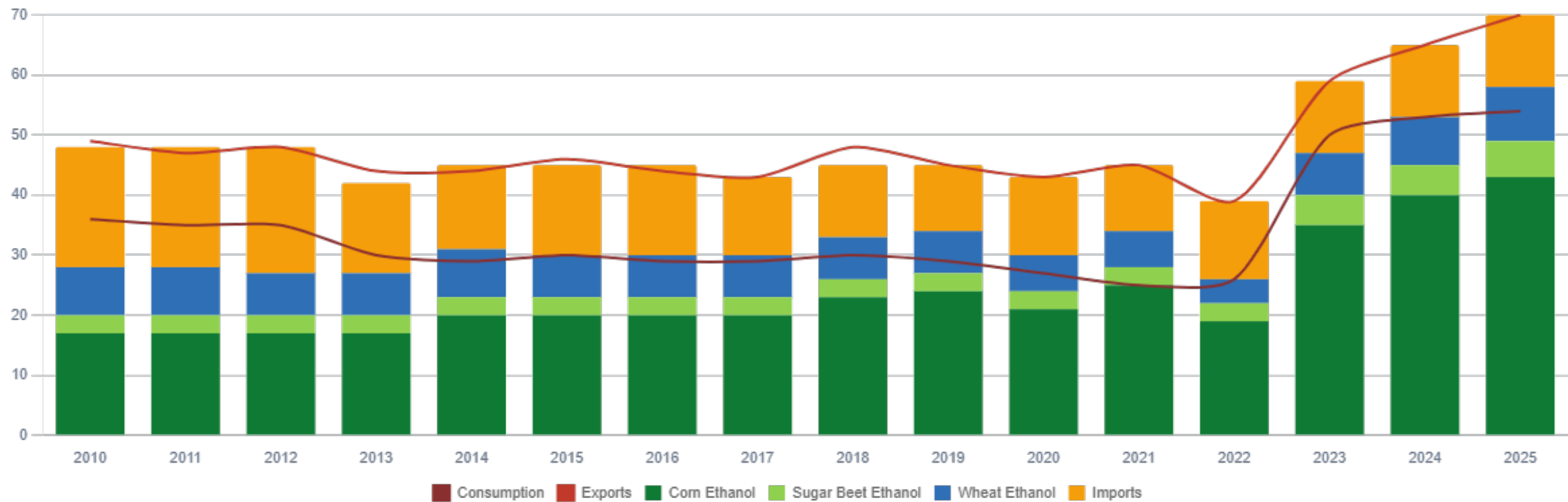
Source: Eurostat

Gasoline Exports in Austria



Source: Eurostat

Ethanol Balance in Austria



Source: Eurostat

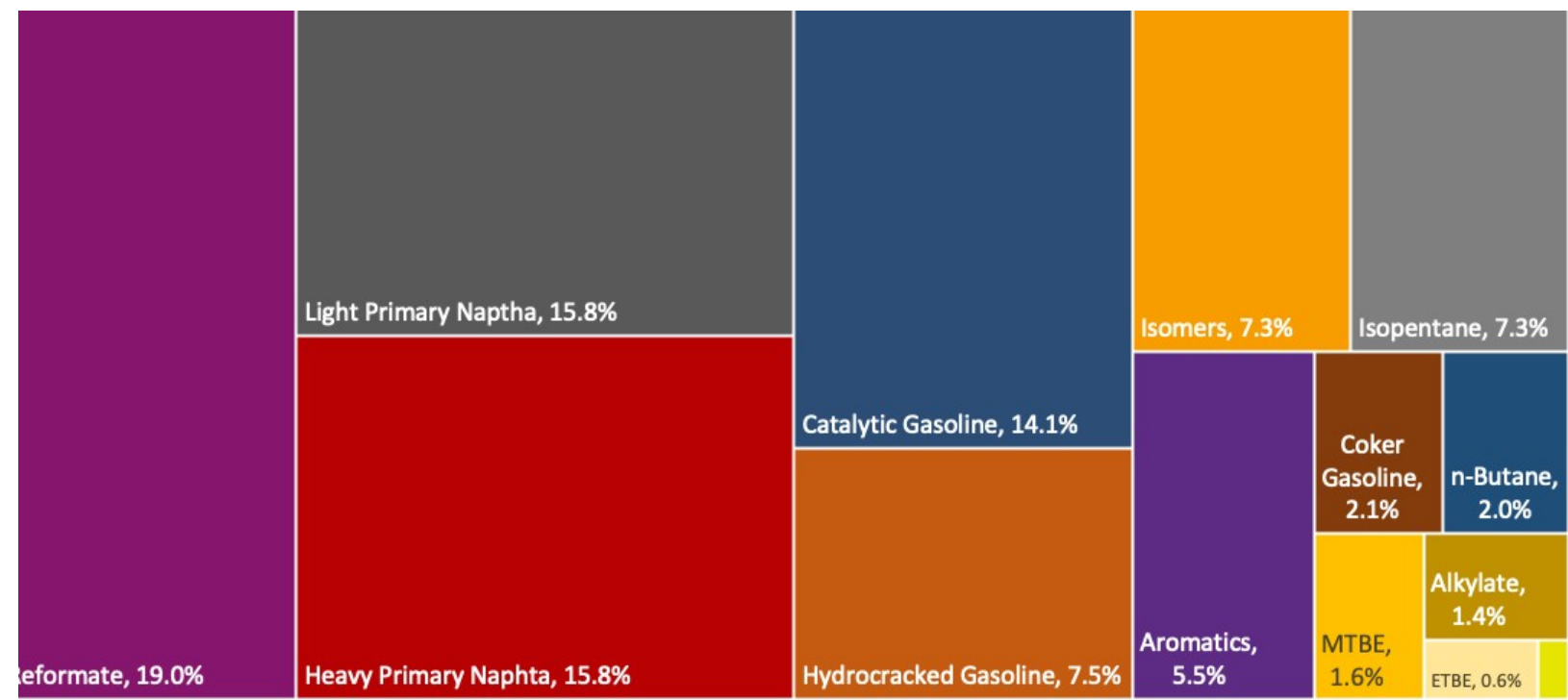
Gasoline Quality in Austria

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	3.4							
GHG Emission Reduction (%)	7.5							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

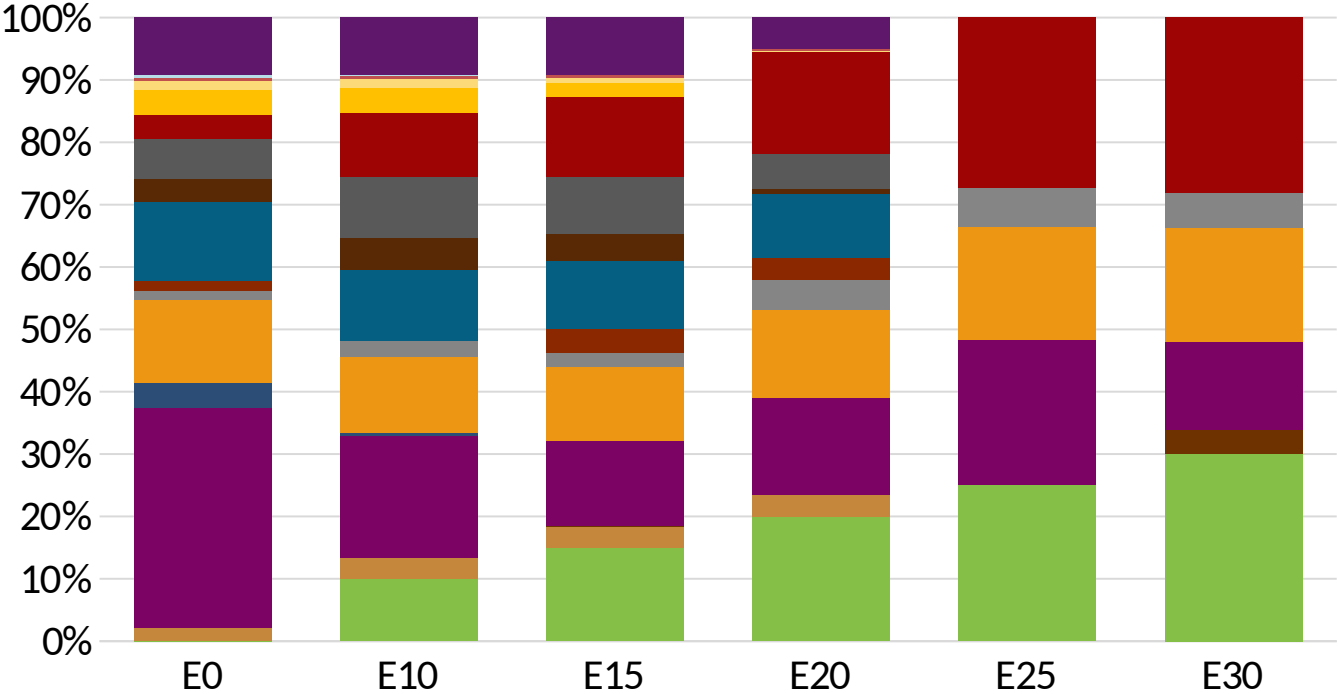
Source: European Commision

Austria Available Blending Components



TAME,
0.2%

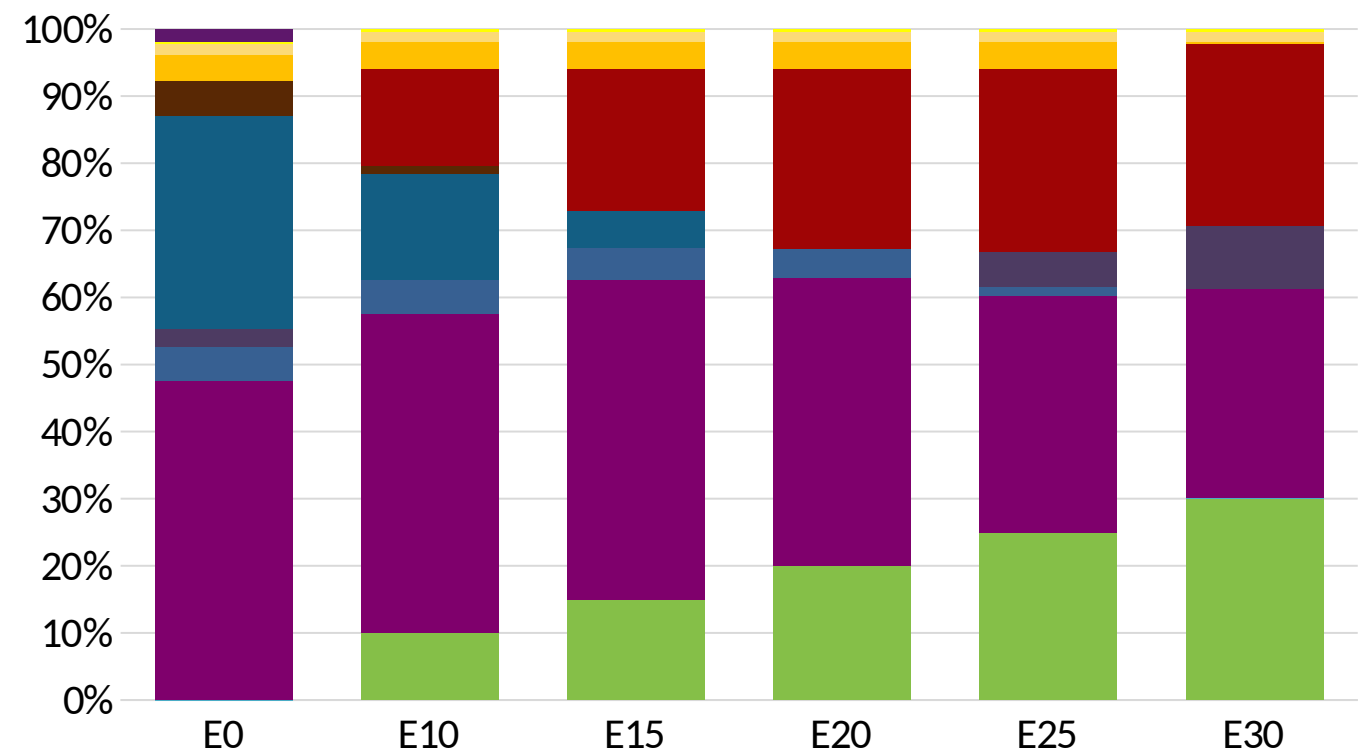
Austria – Gasoline 95 – Constant Octane



Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.069	\$ 1.941	\$ 1.880	\$ 1.799	\$ 1.677	\$ 1.612

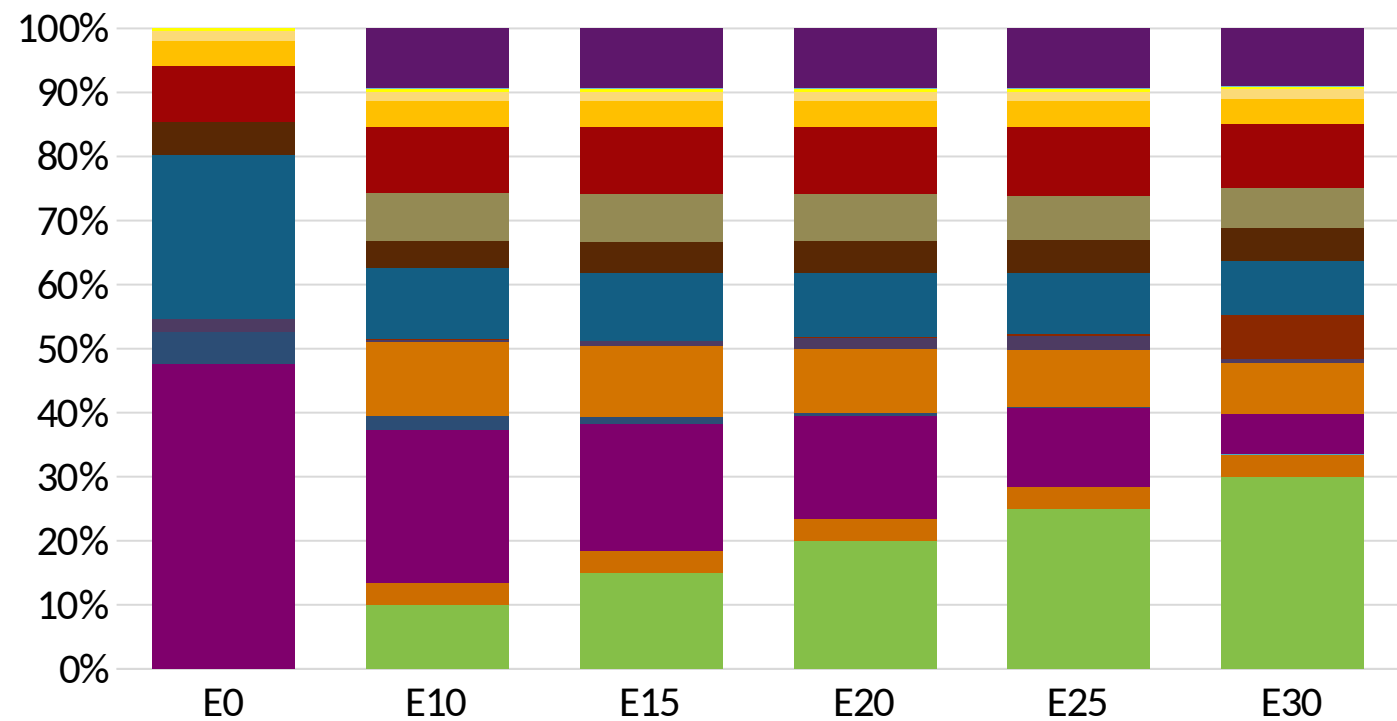
Austria - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.144	\$ 2.015	\$ 1.943	\$ 1.866	\$ 1.789	\$ 1.722

Average CIF 2025
Prices

Austria – Gasoline 95 – Increased Octane

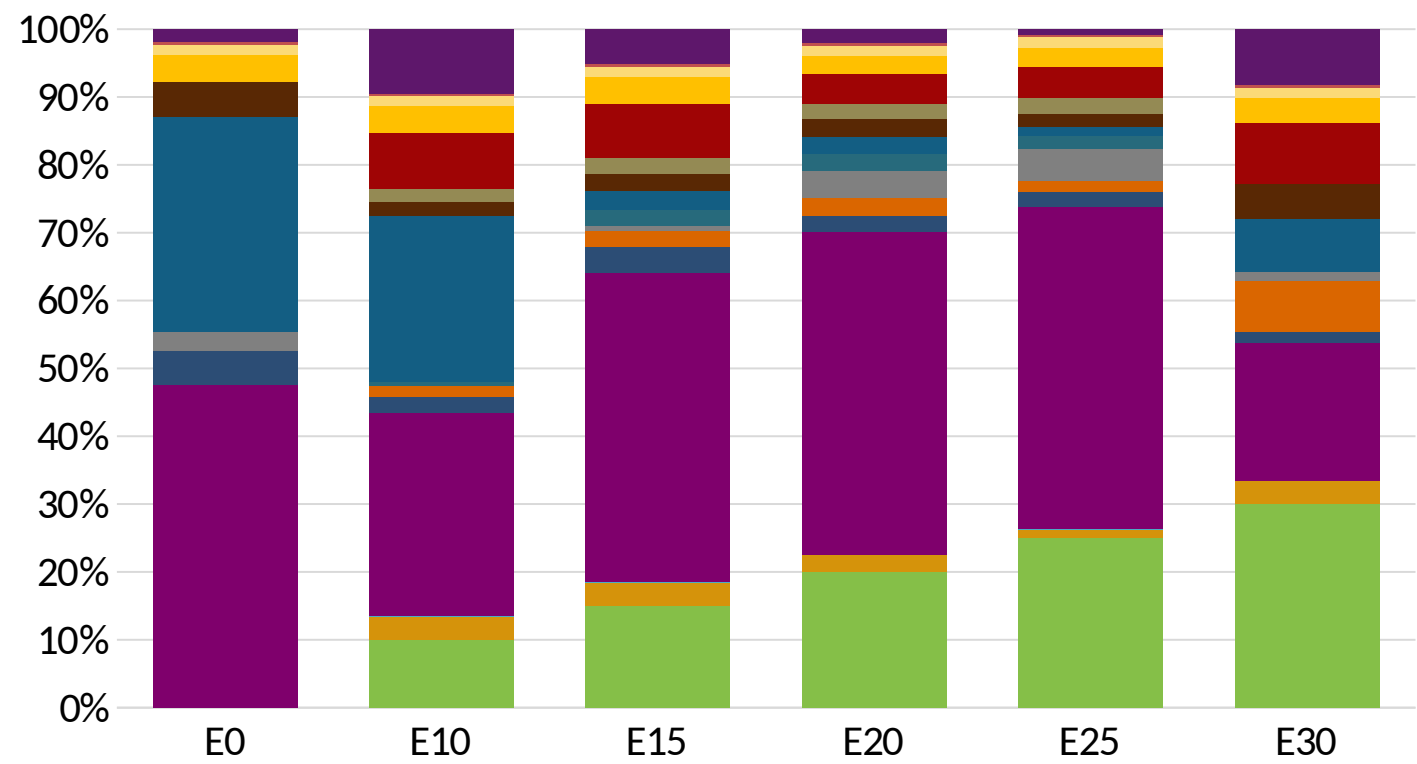


Average CIF 2025
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.006	\$ 2.006	\$ 2.006	\$ 2.006	\$ 2.006	\$ 2.006

Source:
Faro90

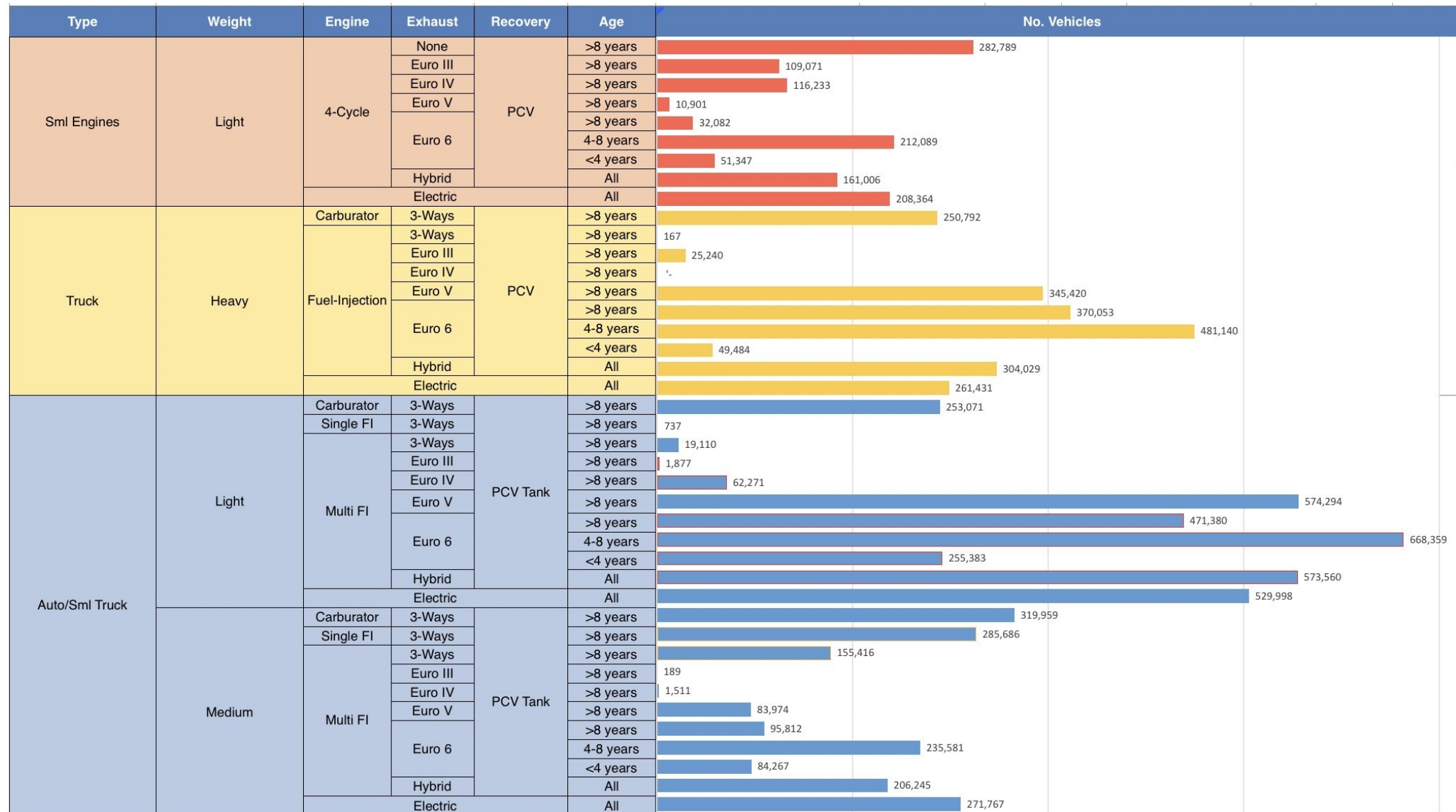
Austria – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Source:
Faro90

Gasoline Vehicle Fleet - Austria

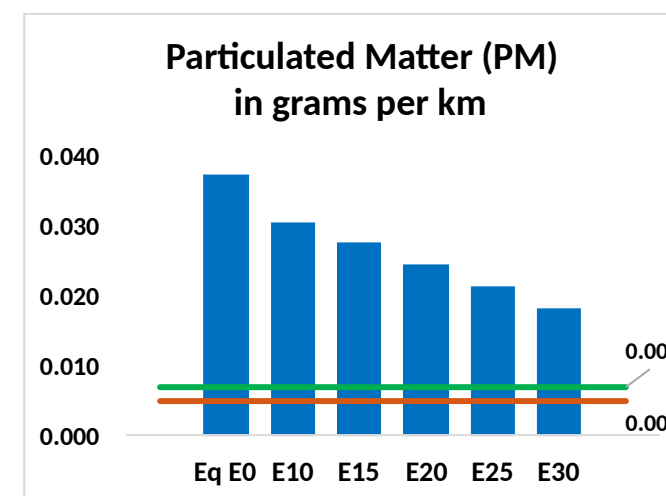
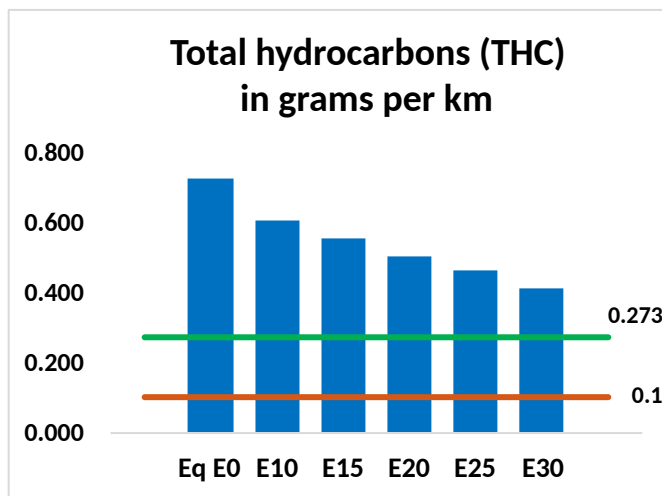
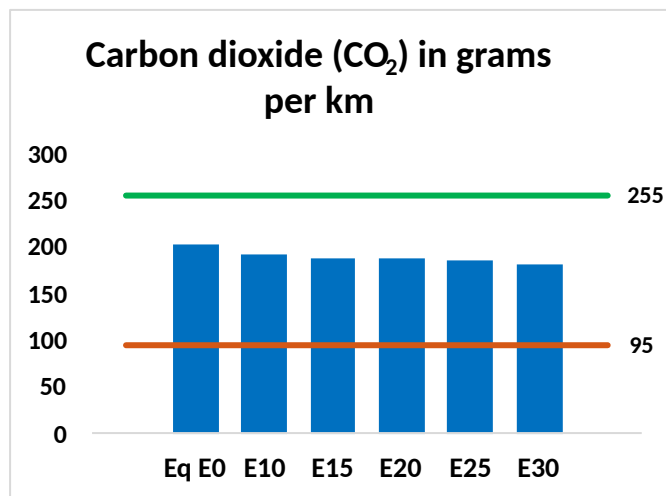
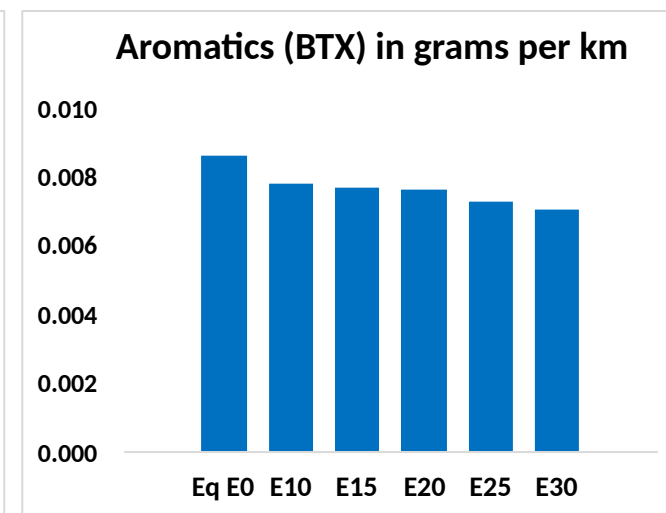
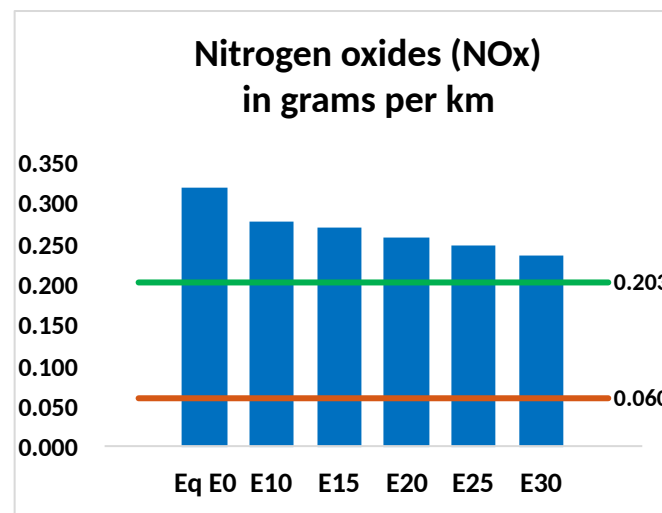
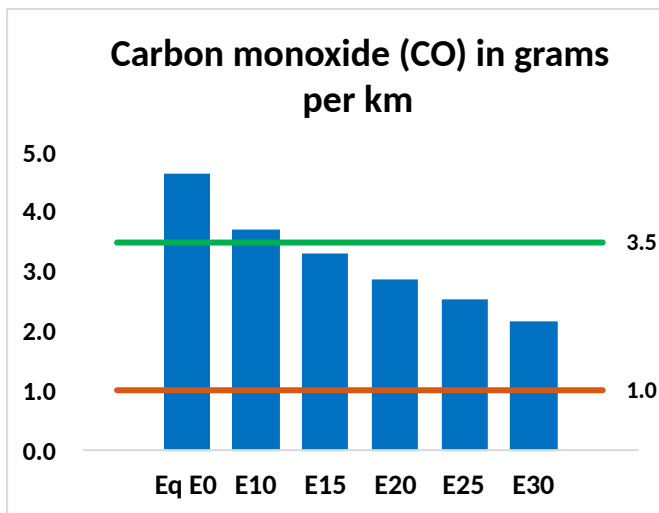


Vehicle Fleet: 8,066,135
Gasoline Vehicle Fleet: 2,845,832

Source: Eurostat, ACEA

Austria – Vehicular Emissions

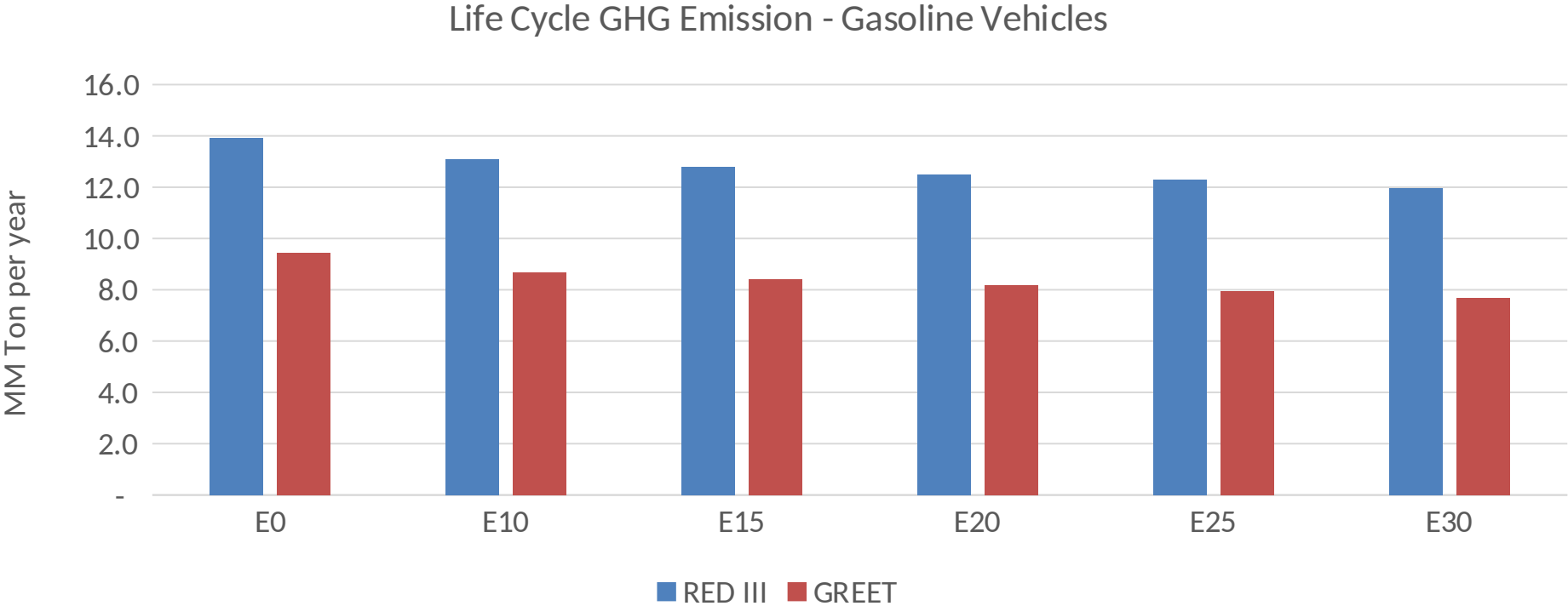
— TIER III BIN 125 United States
— Euro 6 Standard



Austria – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	347,248	311,593	275,938	244,972	213,005	188,182	162,523	-21%	-39%
CO2	15,102,742	14,714,386	14,347,605	14,059,142	13,916,400	13,833,377	13,578,388	-5%	-8%
VOC	53,987	49,541	45,096	41,395	37,644	34,752	30,921	-16%	-30%
NOx	23,878	22,047	20,826	20,221	19,228	18,458	17,546	-13%	-19%
SOx	172	159	146	135	124	113	101	-15%	-28%
NH3	5,532	5,478	5,423	5,449	5,510	5,553	5,589	-2%	0%
N2O	265	263	262	263	269	272	275	-1%	2%
CH4	11,299	11,299	11,299	11,468	11,717	11,932	12,135	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	328	311	293	280	268	258	245	-11%	-18%
Acetaldehyde	946	1,086	1,593	2,426	3,305	3,777	4,463	68%	249%
Formaldehyde	3,479	3,383	3,943	4,630	4,851	5,366	5,842	13%	39%
Benzene	642	617	584	575	570	546	528	-9%	-11%
PM2.5	890	809	728	657	588	513	435	-18%	-34%
PM10	1,893	1,721	1,549	1,397	1,249	1,090	924	-18%	-34%

Austria – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	67.5
Target @ 2035 (MMT/year)	38.0
Delta	29.5

Source: Greet, RED II/III, climateactiontracker.org, F90

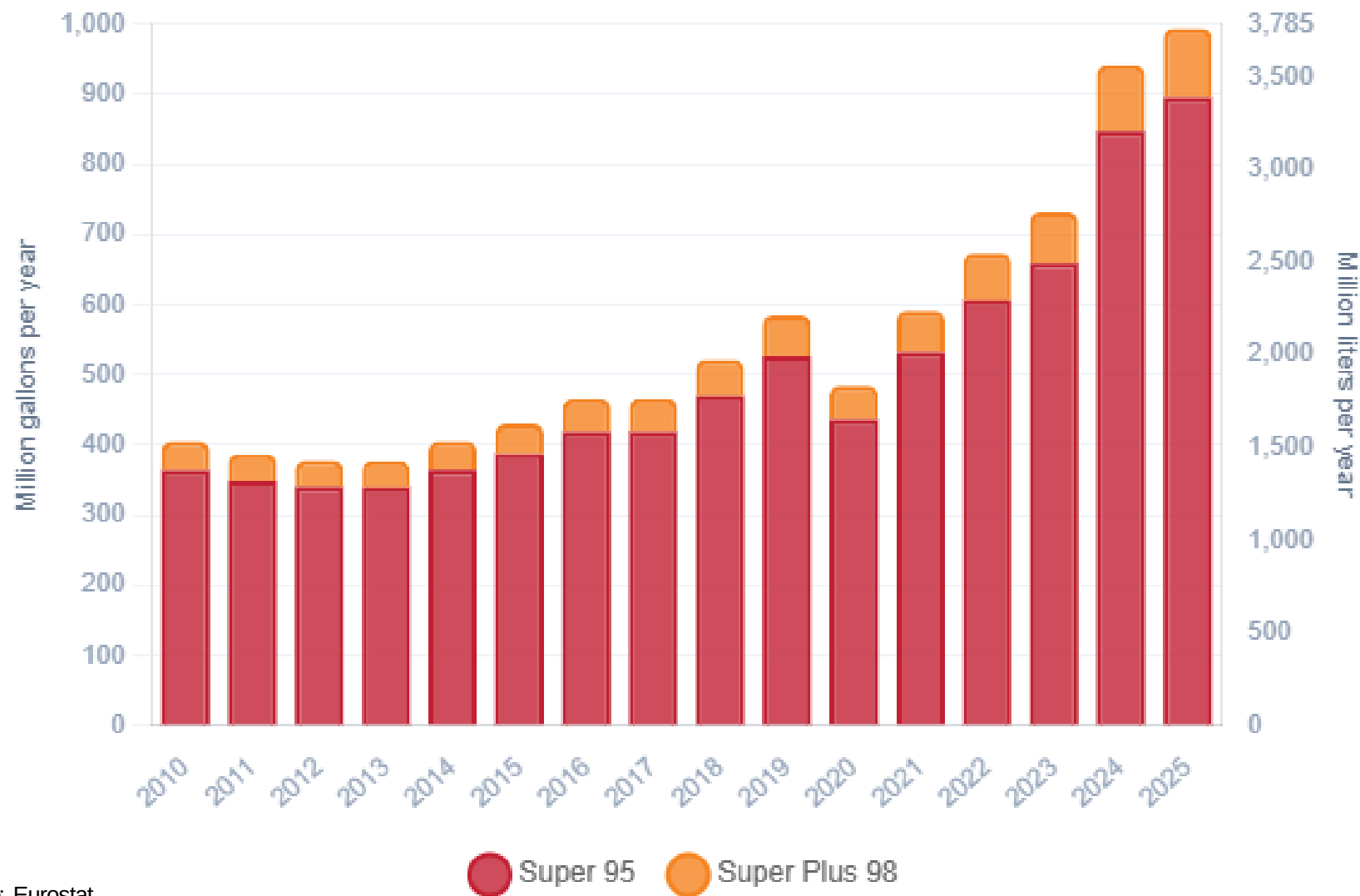
Ethanol Gasoline Blending in Belgium



In 2025, gasoline consumption in Belgium was 3,761 million liters (993 million gallons) and growth rate was 6.2%. Gasoline production and net imports were 4,397 and -4,172 million liters (1,162 and -1,102 million gallons) correspondingly. Belgium complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.3 RON.

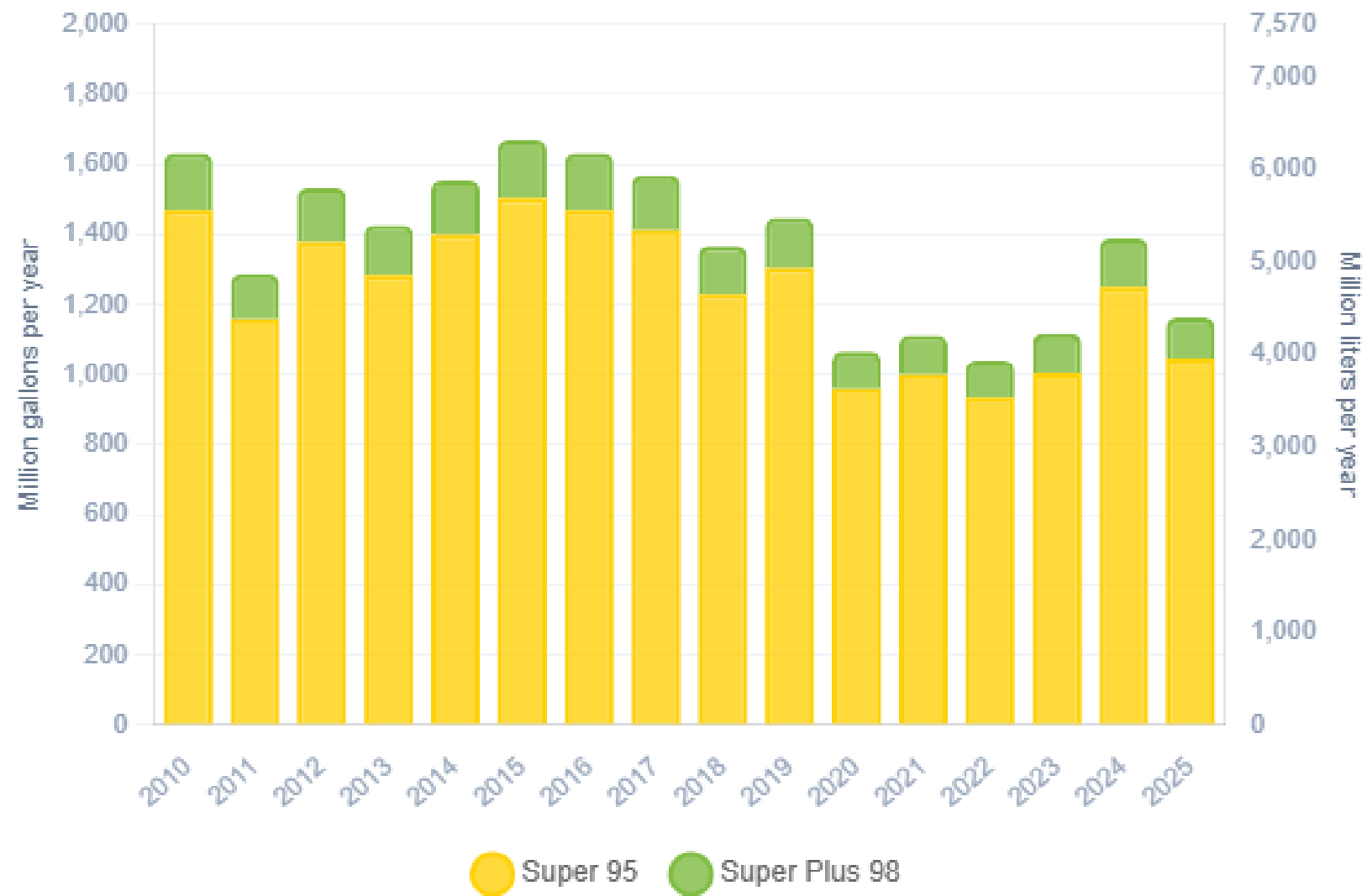
In 2025, ethanol consumption in Belgium was 165 million liters (43 million gallons) representing 4% of gasoline demand. Ethanol production and Net Imports were 620 and -105 million liters (164 and -28 million gallons) correspondingly. Ethanol sources are Sugar Cane 20% and Cereals 80%.

Gasoline Demand in Belgium



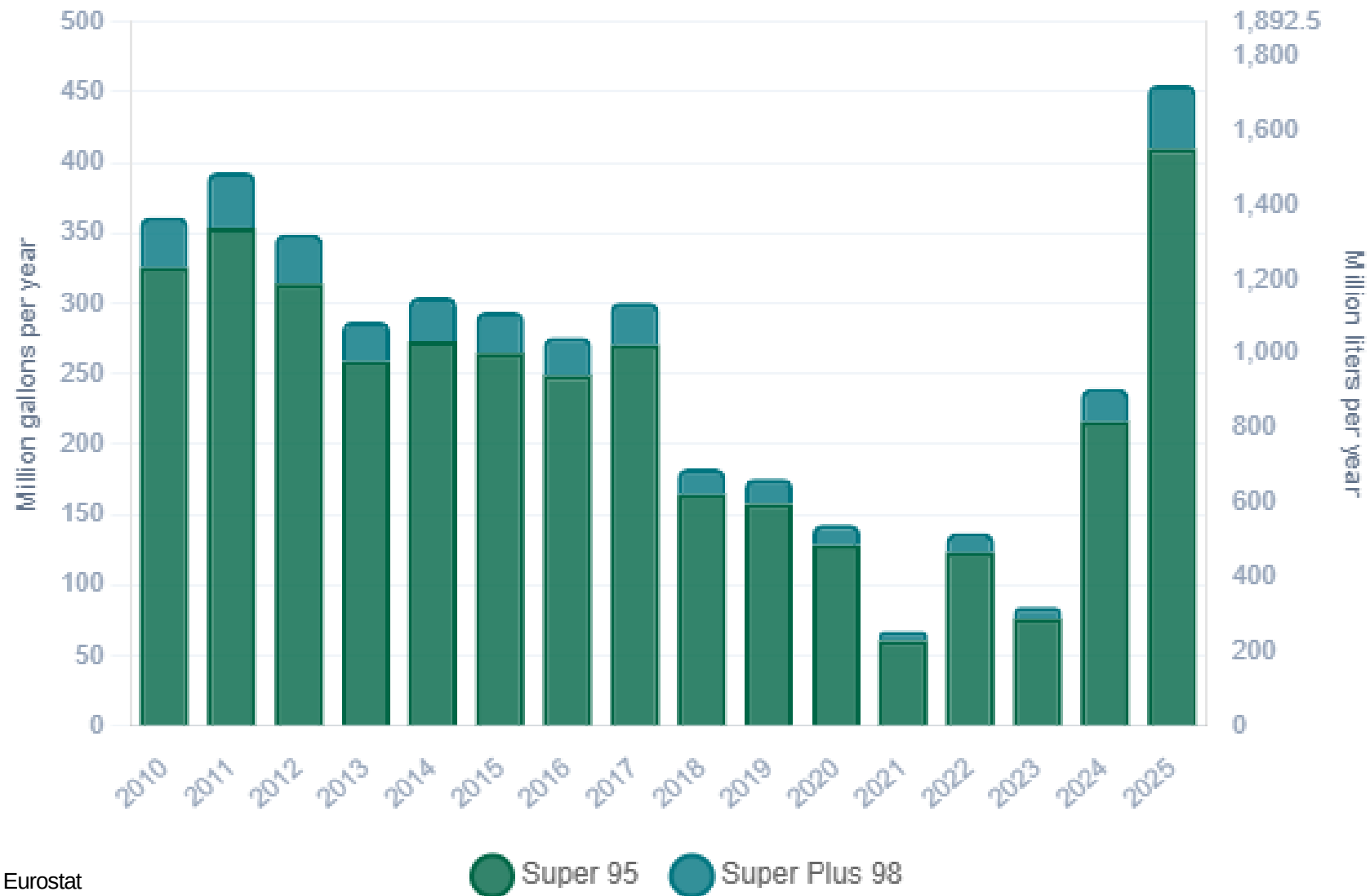
Source: Eurostat

Gasoline Production in Belgium



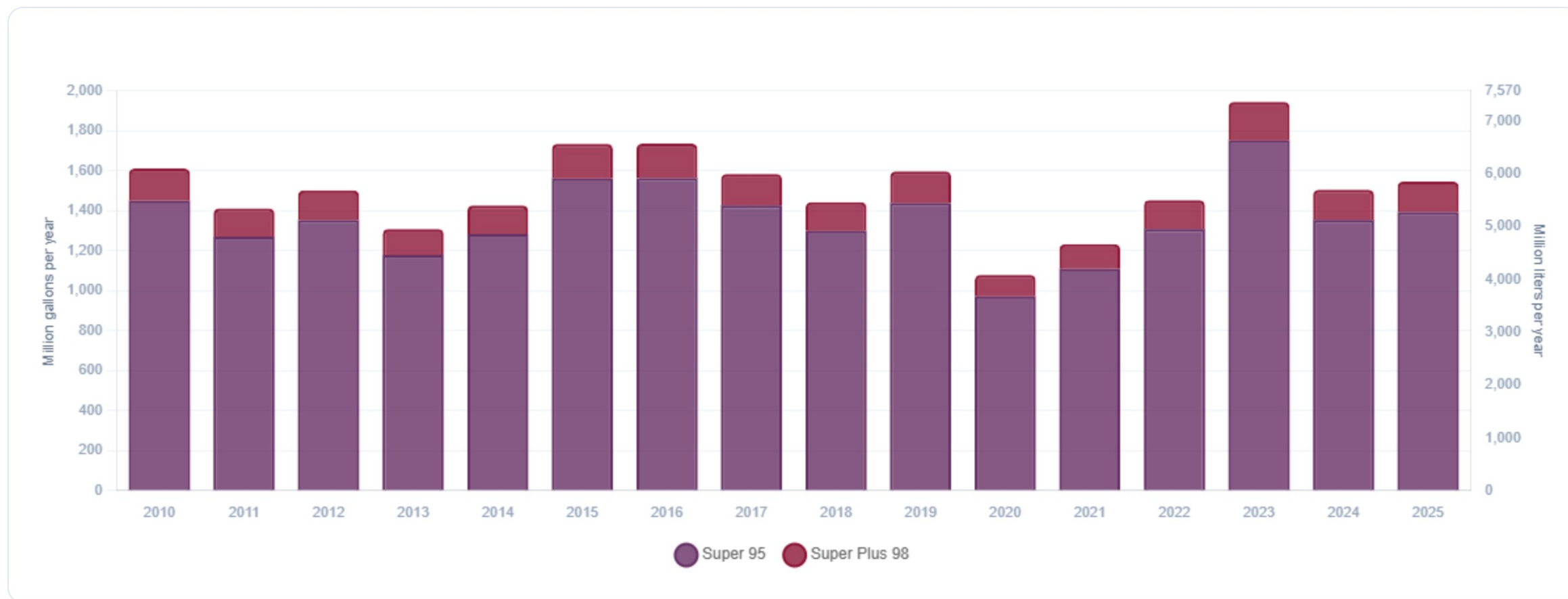
Source: Eurostat

Gasoline Imports in Belgium



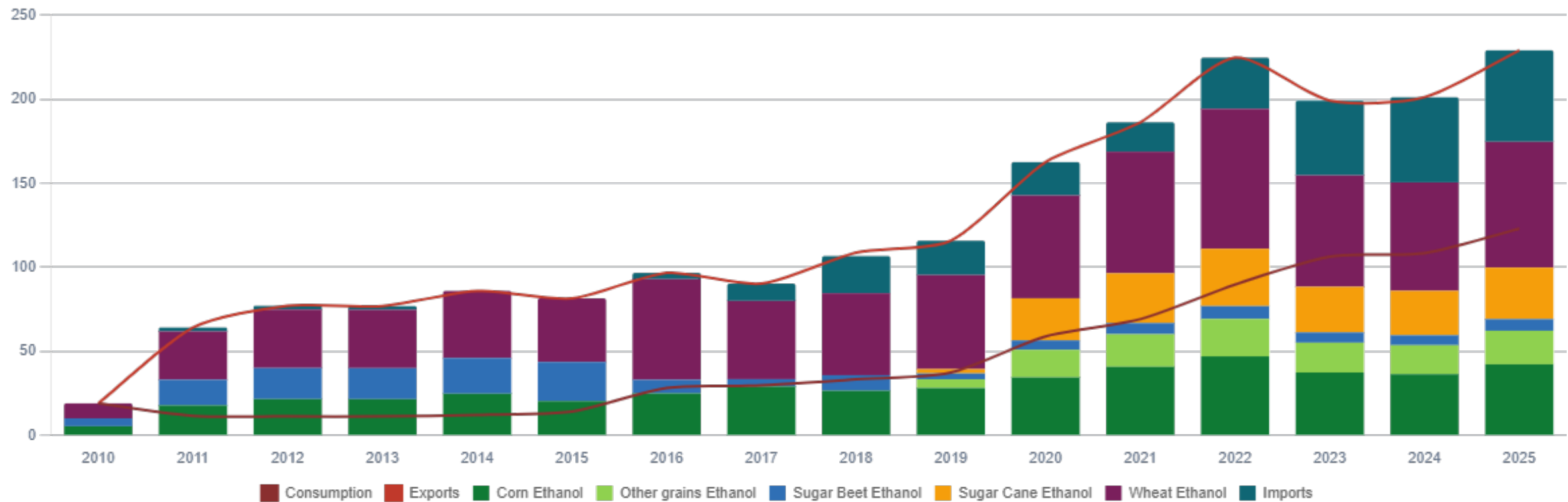
Source: Eurostat

Gasoline Exports in Belgium



Source: Eurostat

Ethanol Balance in Belgium



Source: Eurostat

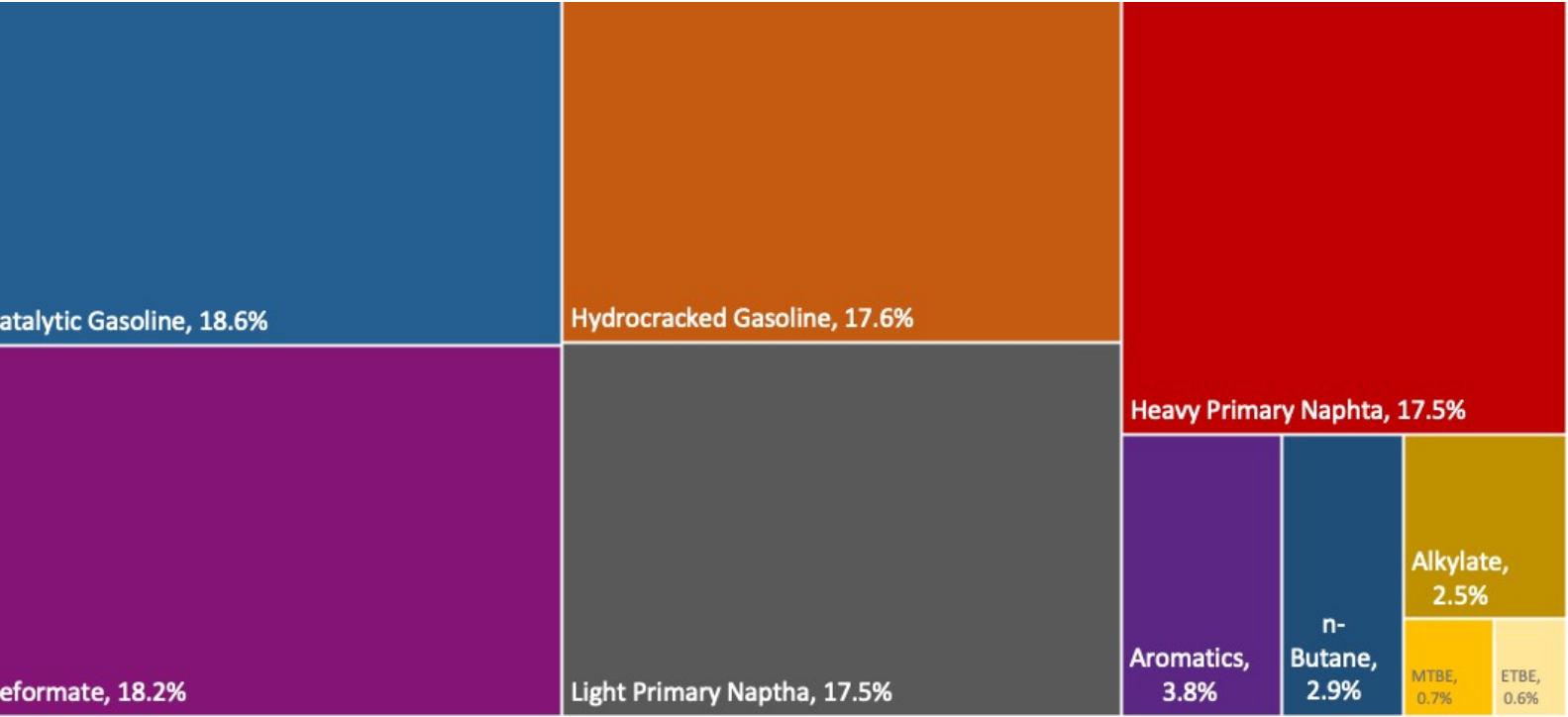
Gasoline Quality in Belgium

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.5							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	5.7							
GHG Emission Reduction (%)	-							

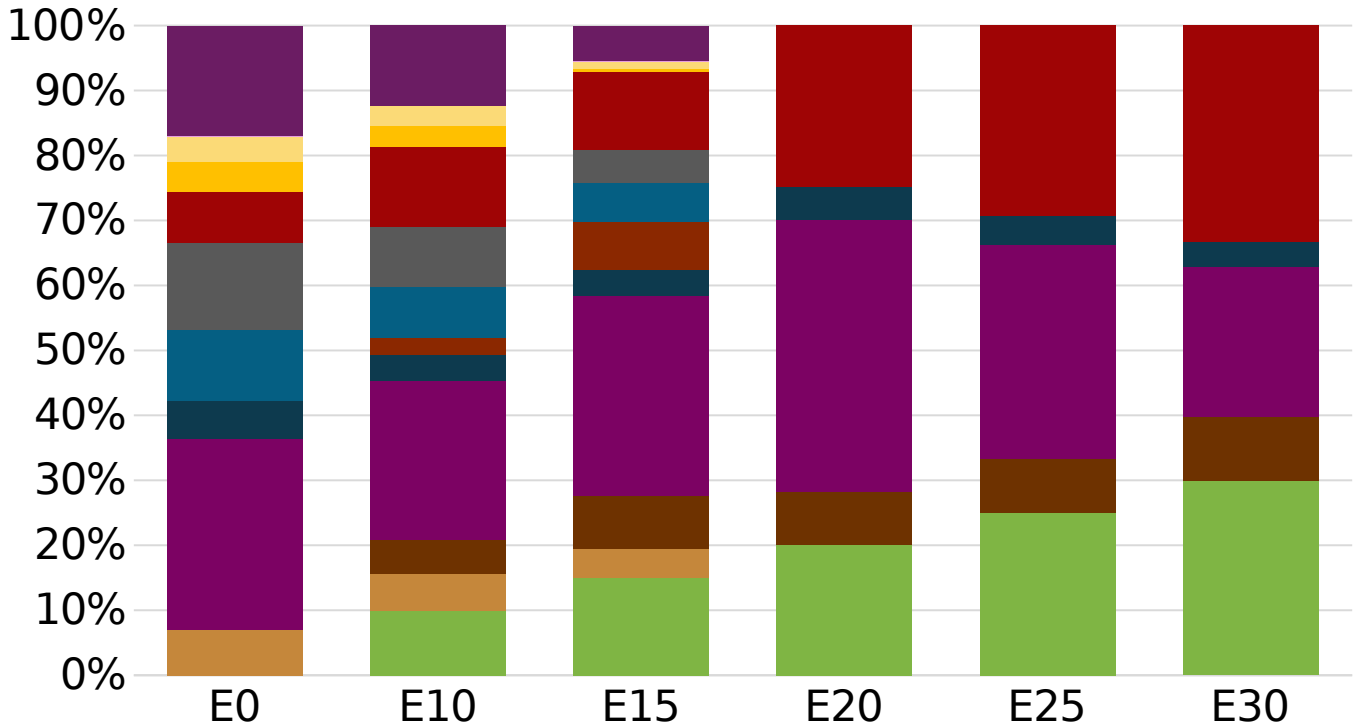
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Belgium Available Blending Components



Belgium – Gasoline 95 – Constant Octane

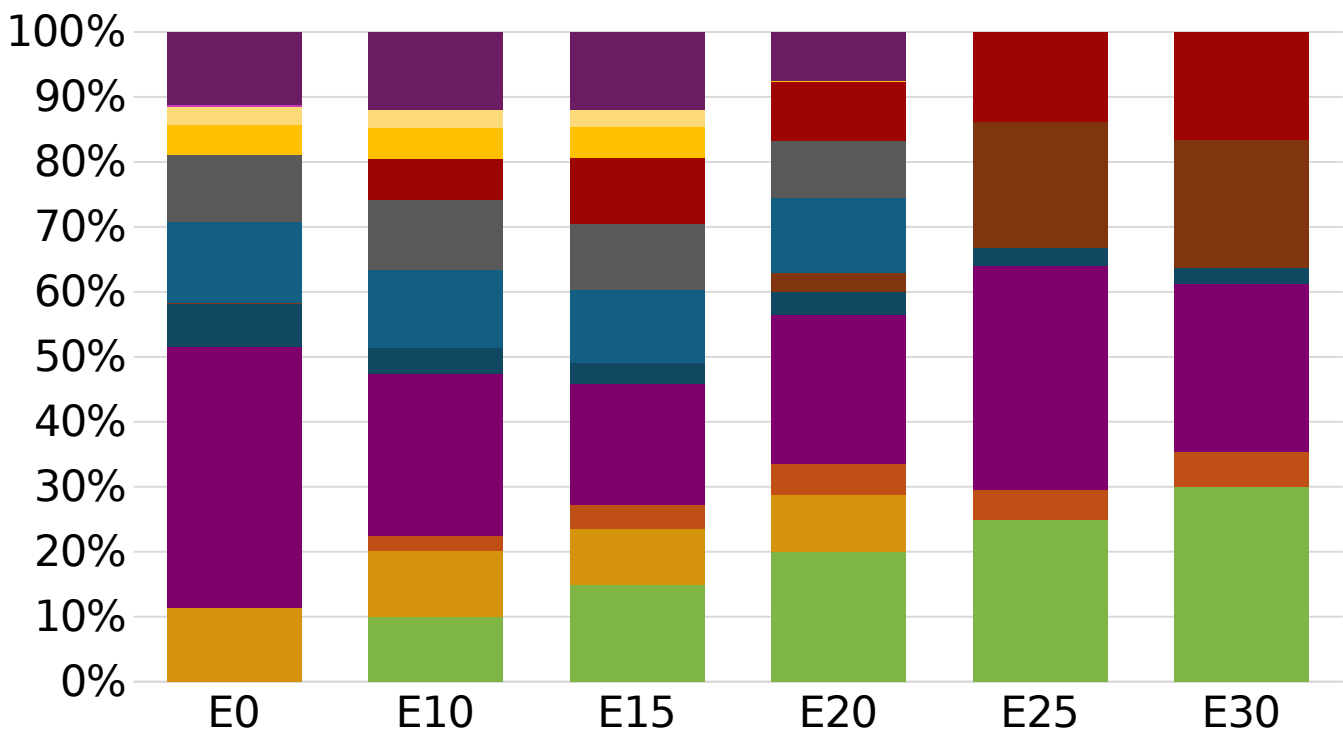


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.1	95.3	95.3
Price (US\$/gal)	\$ 2.078	\$ 1.944	\$ 1.869	\$ 1.748	\$ 1.672	\$ 1.577

Source:
Faro90

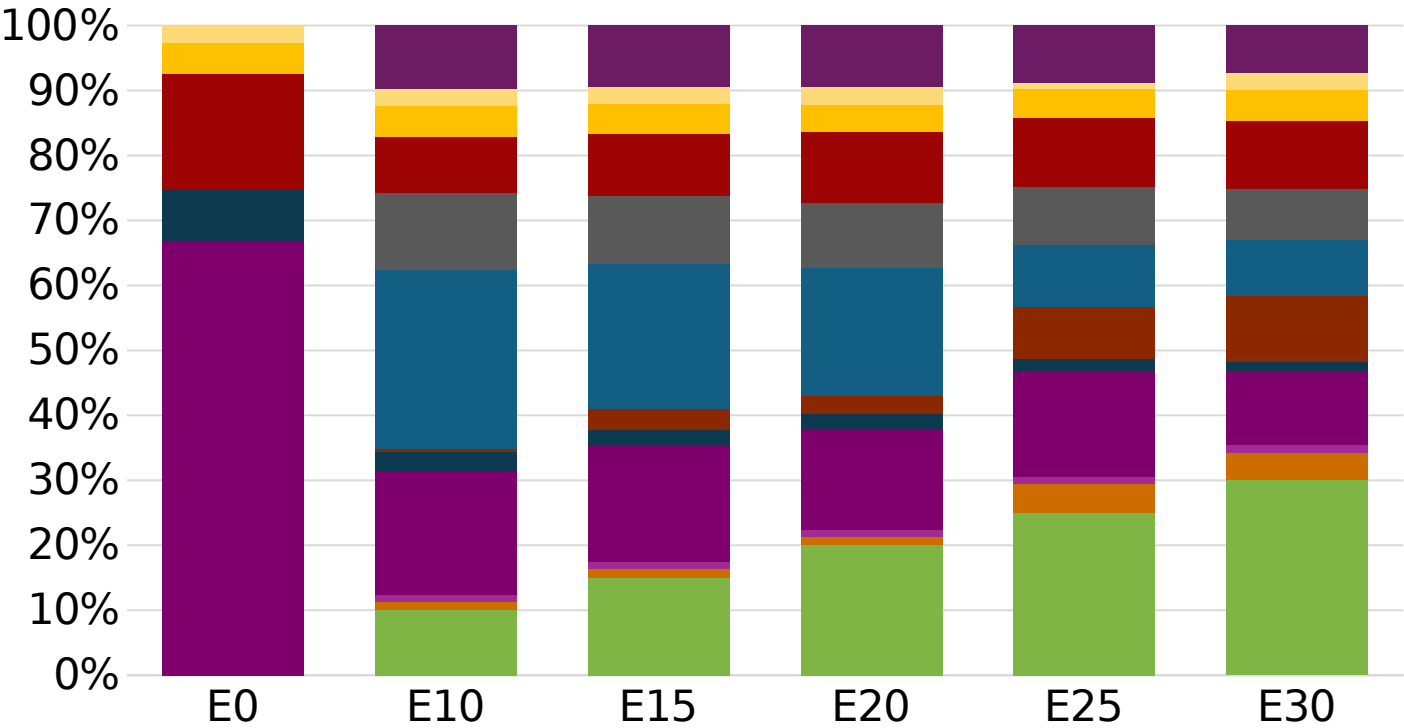
Belgium - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.1	98.5
Price (US\$/gal)	\$ 2.206	\$ 2.087	\$ 2.013	\$ 1.961	\$ 1.854	\$ 1.791

Average CIF 2025
Prices

Belgium – Gasoline 95 – Increased Octane

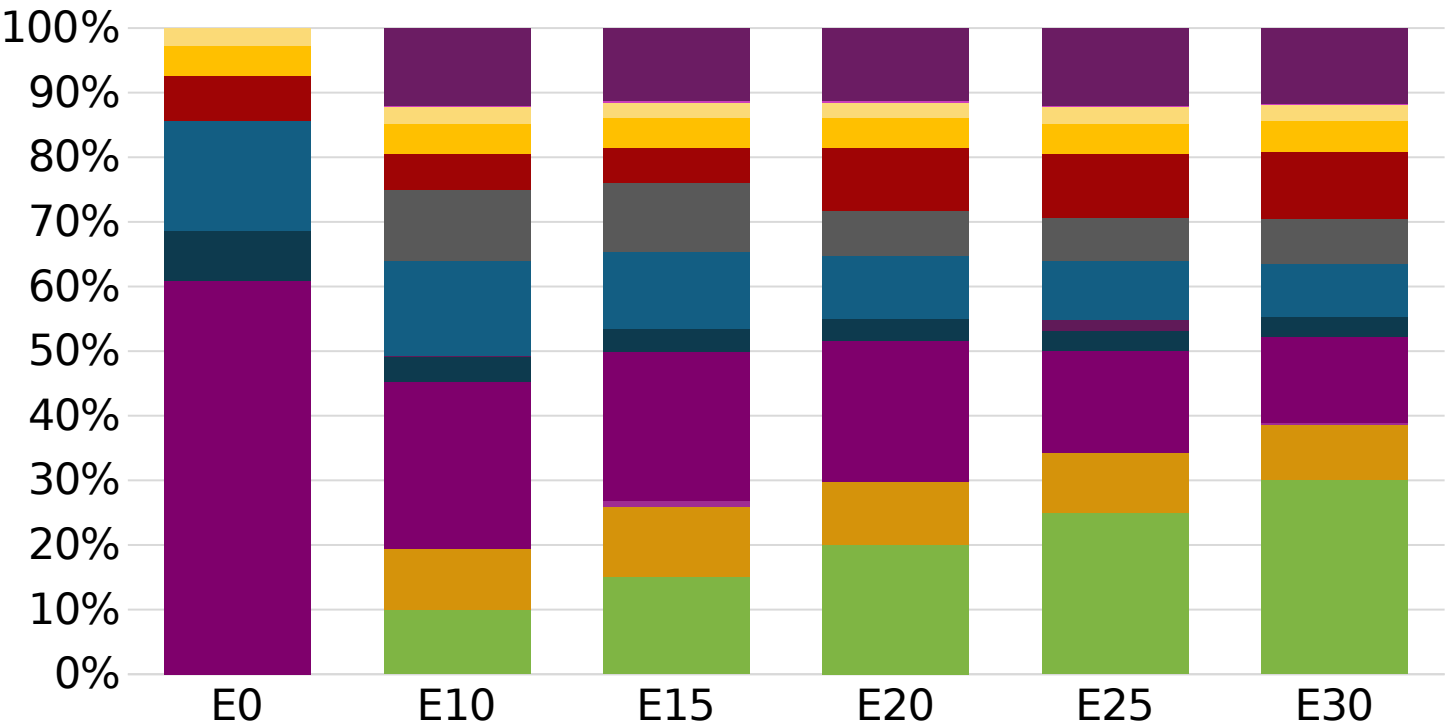


Average CIF 2025
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.5	102.1
Price (US\$/gal)	\$ 1.990	\$ 1.990	\$ 1.990	\$ 1.990	\$ 1.990	\$ 1.990

Source:
Faro90

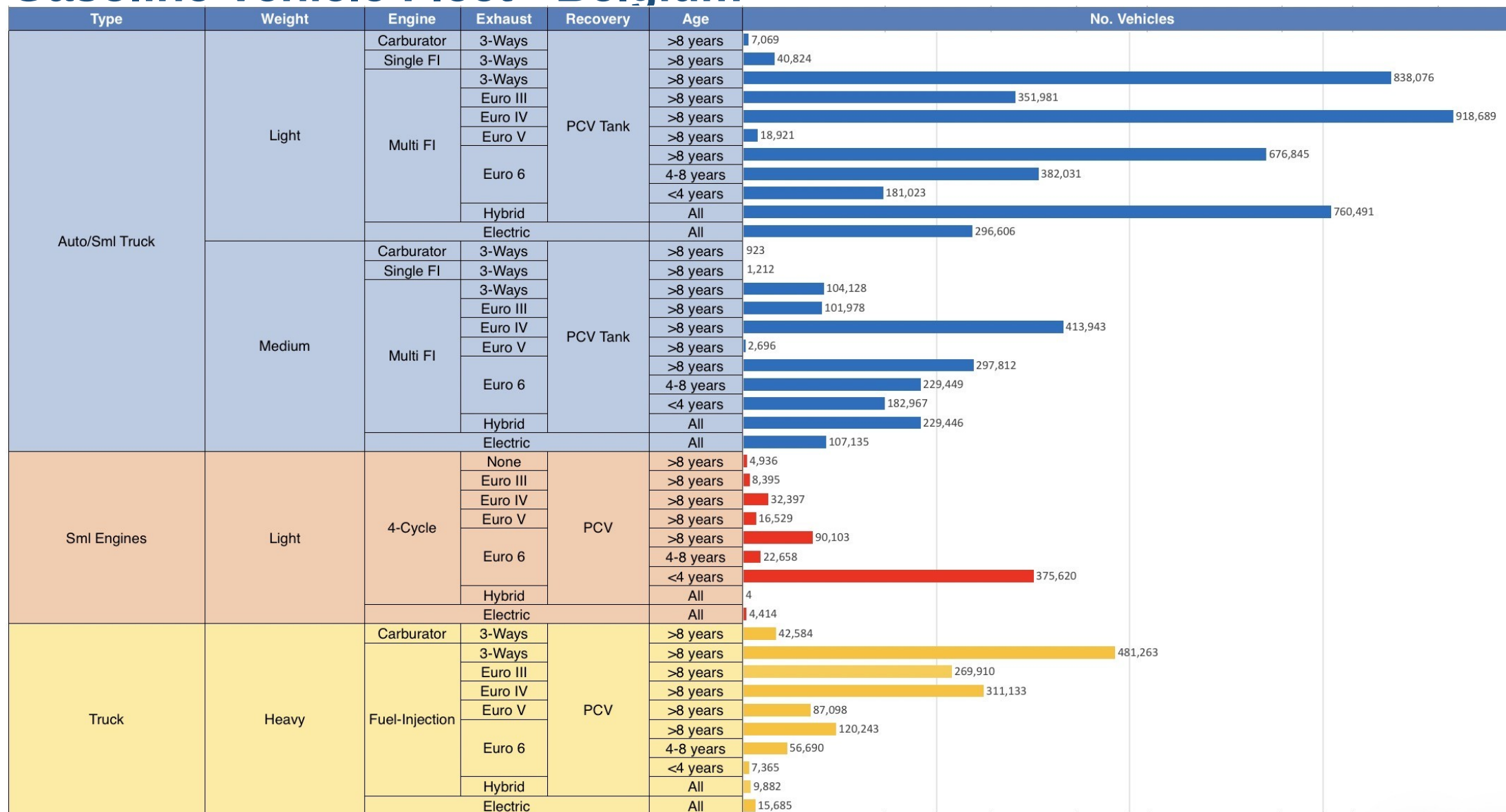
Belgium – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Source:
Faro90

Gasoline Vehicle Fleet - Belgium

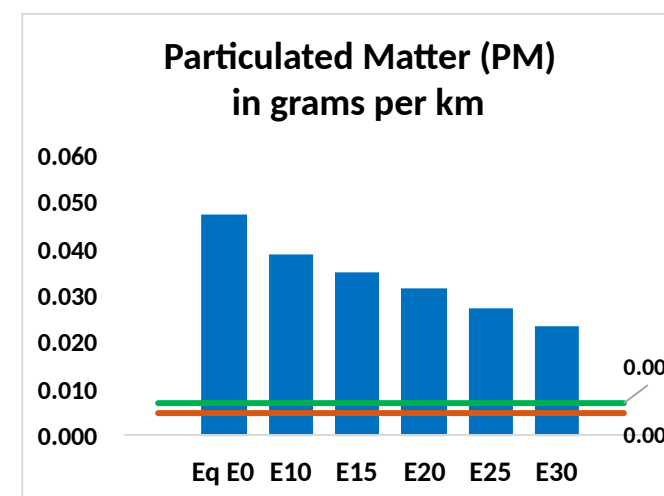
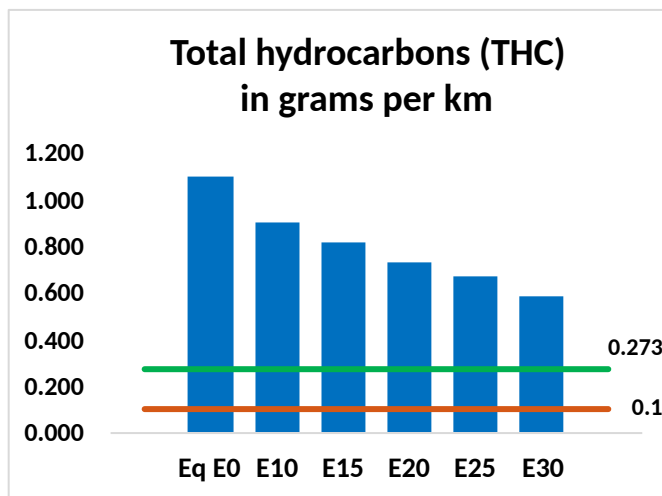
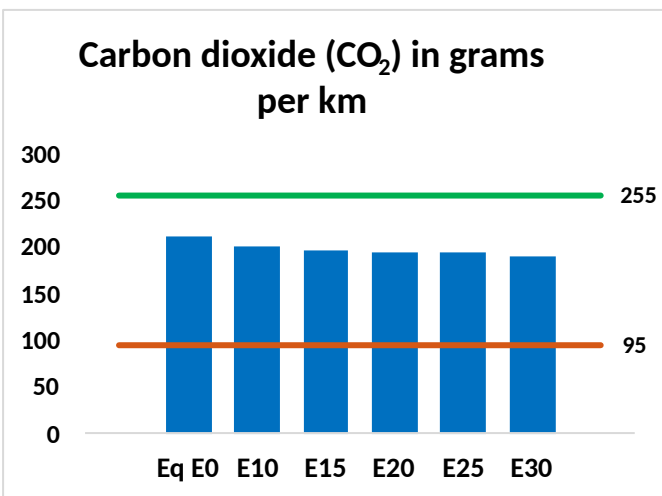
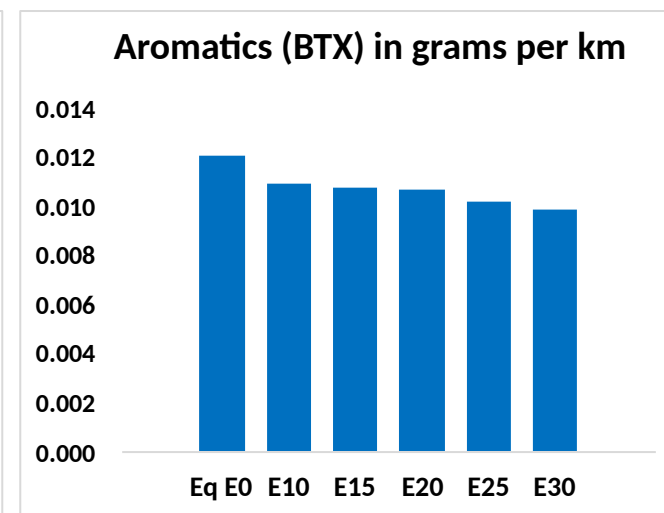
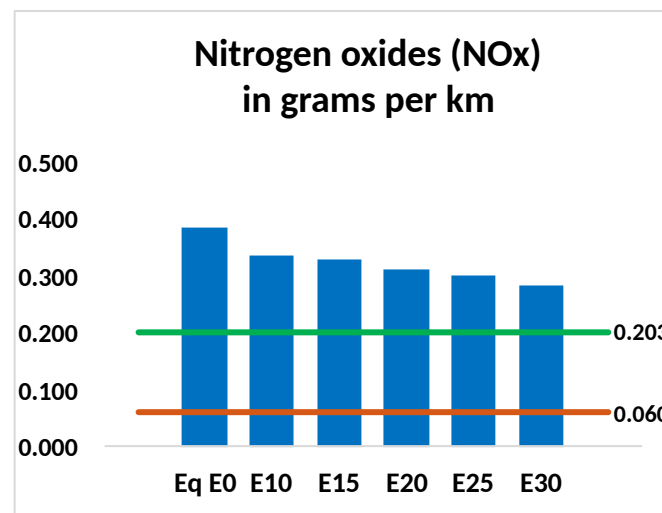
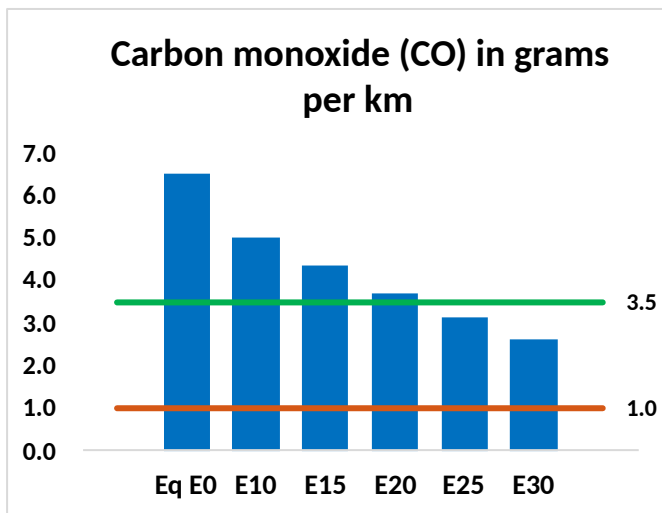
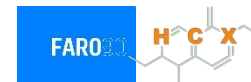


Vehicle Fleet: 8,101,154
Gasoline Vehicle Fleet: 3,299,143

Source: Eurostat, ACEA

Belgium – Vehicular Emissions

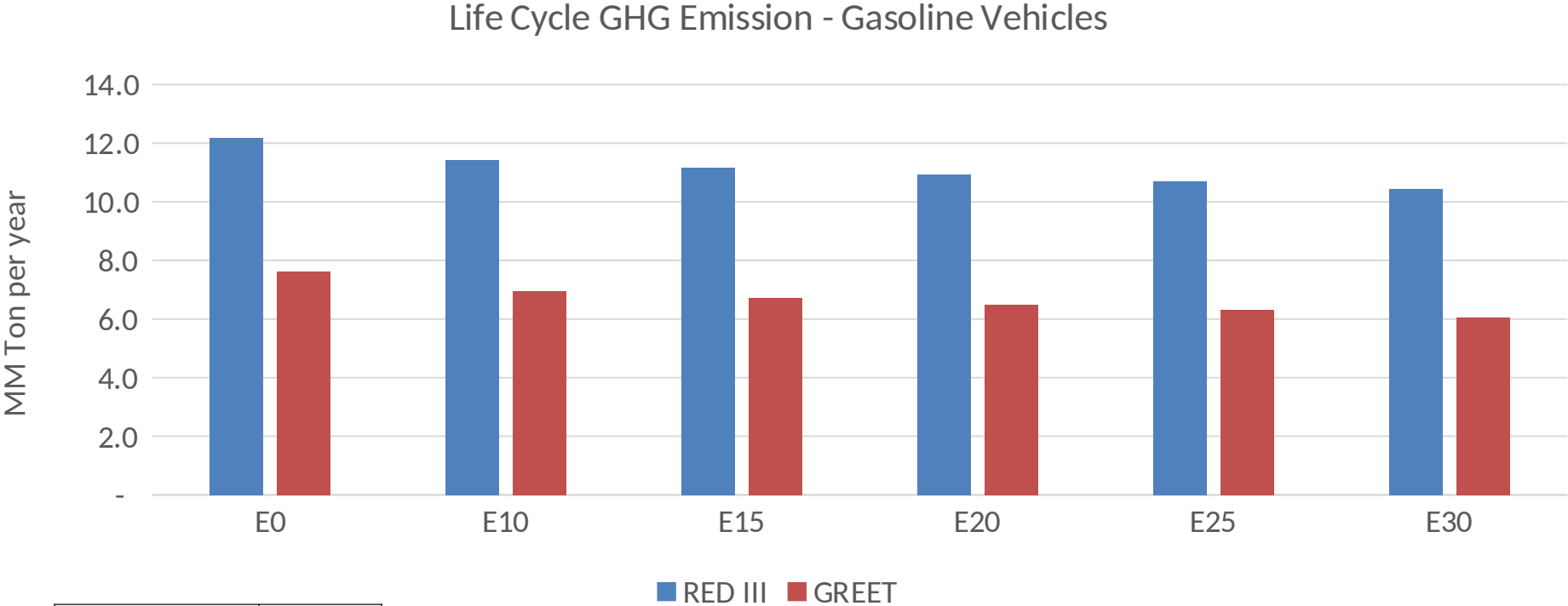
— TIER III BIN 125 United States
— Euro 6 Standard



Belgium – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	554,479	490,937	427,395	371,243	312,858	267,397	221,422	-23%	-44%
CO2	17,996,550	17,533,782	17,096,723	16,752,989	16,582,895	16,482,982	16,179,153	-5%	-8%
VOC	93,593	85,189	76,784	69,661	62,384	56,755	49,360	-18%	-33%
NOx	33,022	30,490	28,823	28,009	26,658	25,618	24,381	-13%	-19%
SOx	205	189	174	161	148	135	120	-15%	-28%
NH3	6,828	6,760	6,693	6,725	6,800	6,853	6,898	-2%	0%
N2O	353	351	349	351	358	362	366	-1%	2%
CH4	18,449	18,449	18,449	18,726	19,132	19,482	19,814	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	480	451	422	401	379	363	341	-12%	-21%
Acetaldehyde	1,511	1,735	2,545	3,876	5,281	6,034	7,130	68%	249%
Formaldehyde	5,654	5,497	6,408	7,524	7,883	8,720	9,493	13%	39%
Benzene	1,026	986	934	919	912	872	844	-9%	-11%
PM2.5	1,036	942	848	765	684	597	506	-18%	-34%
PM10	3,022	2,748	2,473	2,231	1,994	1,741	1,475	-18%	-34%

Belgium – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	105.6
Target @ 2035 (MMT/year)	59.4
Delta	46.1

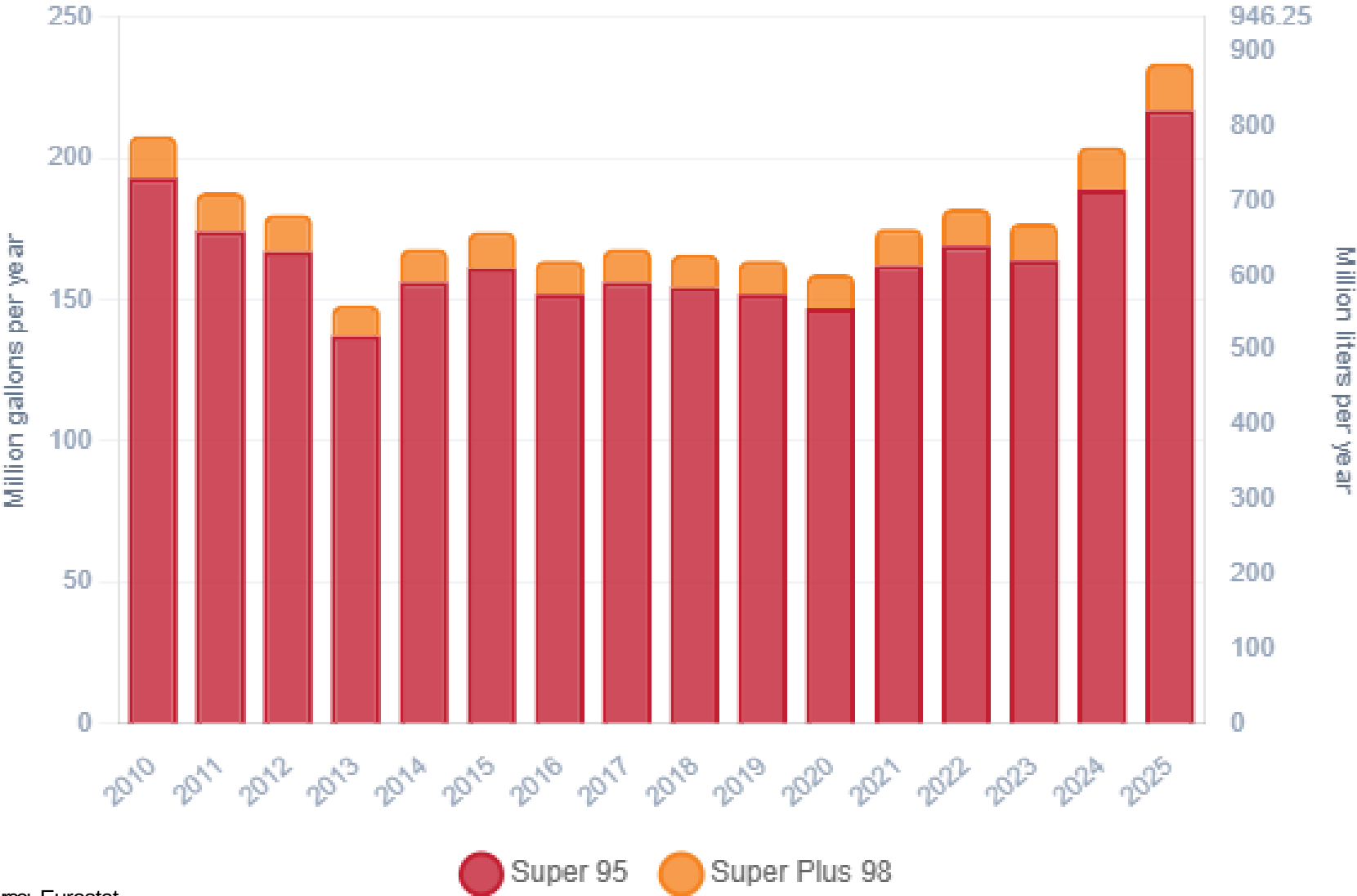
Ethanol Gasoline Blending in Bulgaria



In 2025, gasoline consumption in Bulgaria was 885 million liters (234 million gallons) and growth rate was 0.79%. Gasoline production and net imports were 1,664 and -735 million liters (440 and -194 million gallons) correspondingly. Bulgaria complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

In 2025, ethanol consumption in Bulgaria was 54 million liters (14 million gallons) representing 6% of gasoline demand. Ethanol Net Imports were 15 million liters each (4 million gallons). Ethanol sources are Corn 45% and Cereals 55%.

Gasoline Demand in Bulgaria



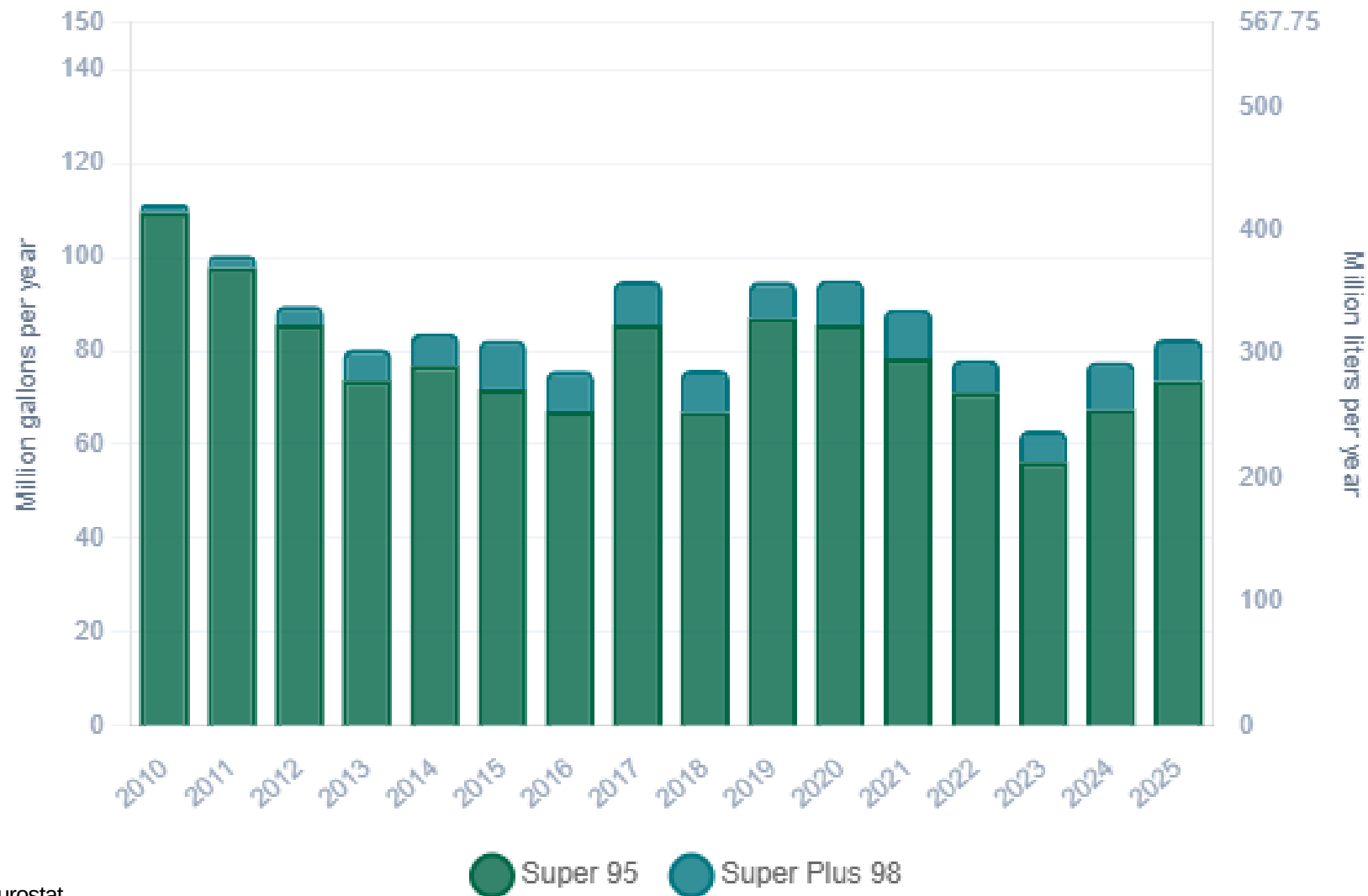
Source: Eurostat

Gasoline Production in Bulgaria



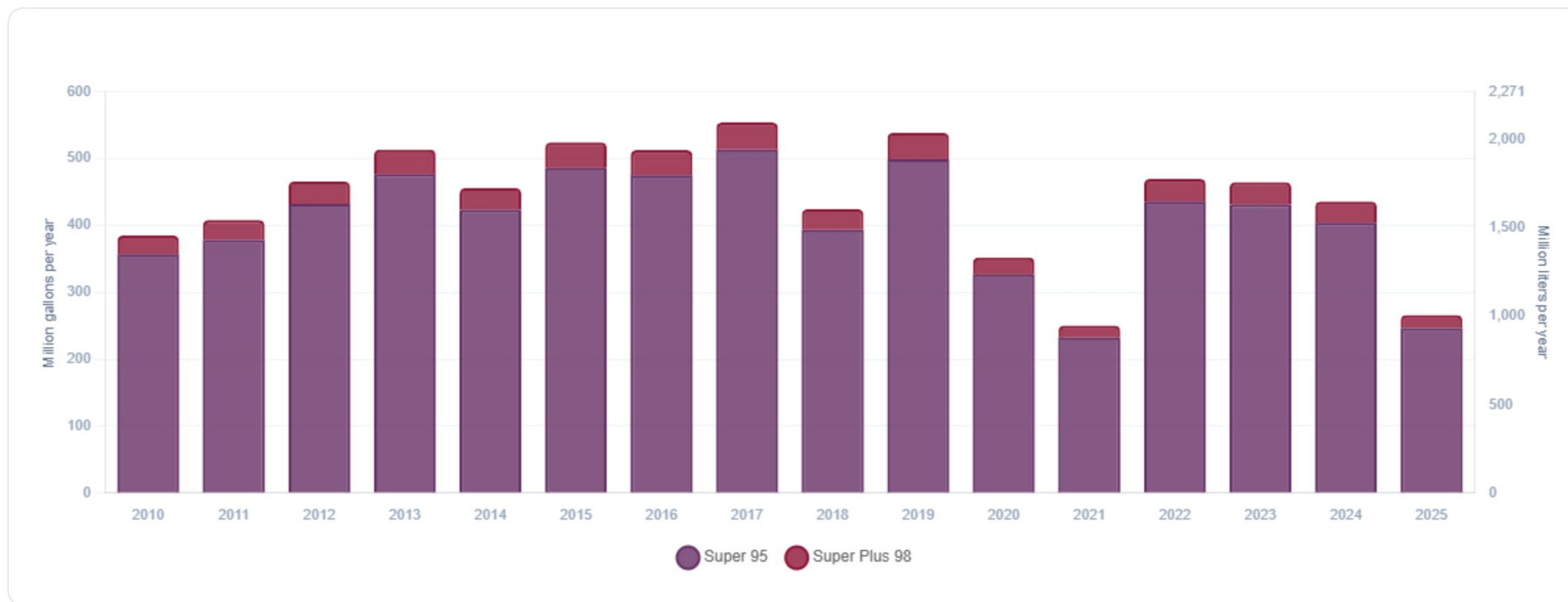
Source: Eurostat

Gasoline Imports in Bulgaria



Source: Eurostat

Gasoline Exports in Bulgaria



Source: Eurostat

The FARO90 logo is shown in a blue box, and a chemical structure is displayed to its right. The chemical structure features a central carbon atom (C) bonded to a hydrogen atom (H), a methyl group (CH₃), and a side chain. The side chain includes a double bond and a terminal group labeled 'X'.



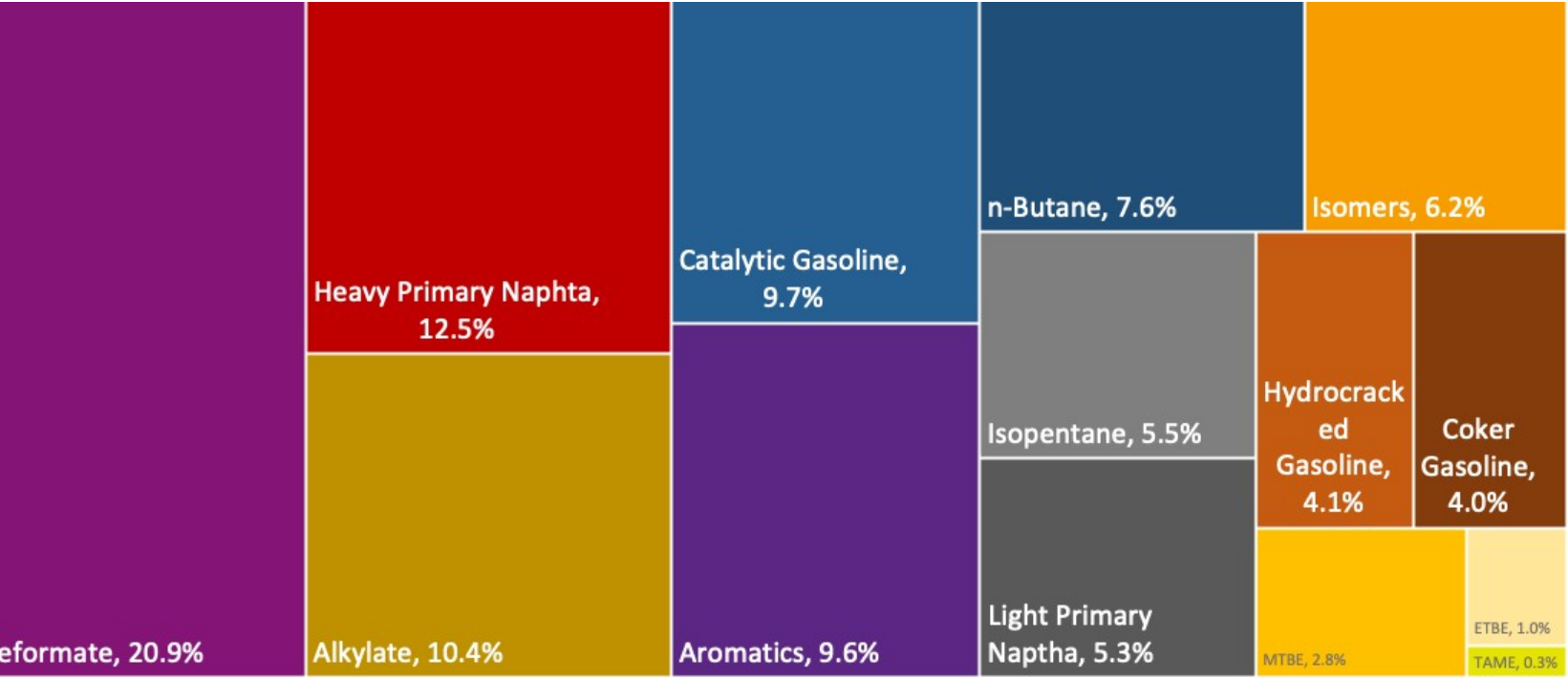
Gasoline Quality in Bulgaria

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Countries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

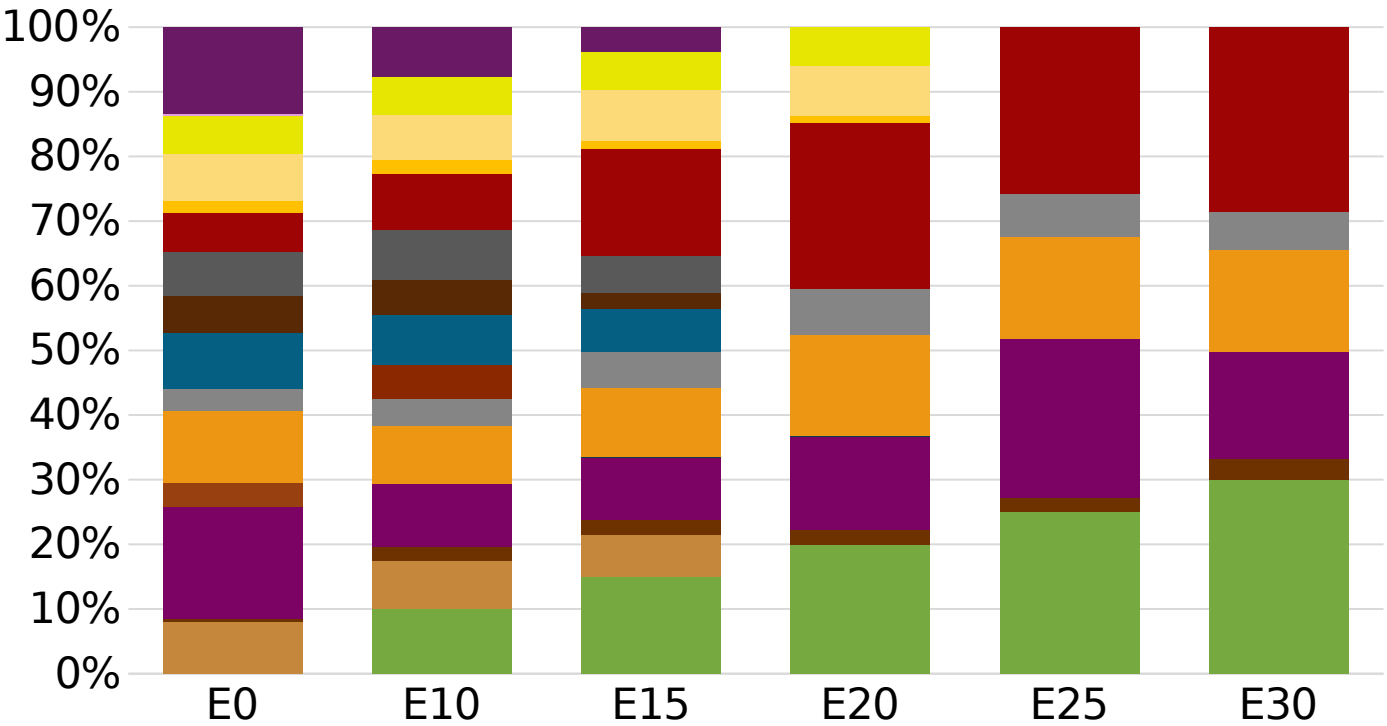
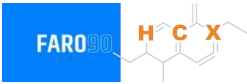
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commission

Bulgaria Available Blending Components



Bulgaria – Gasoline 95 – Constant Octane

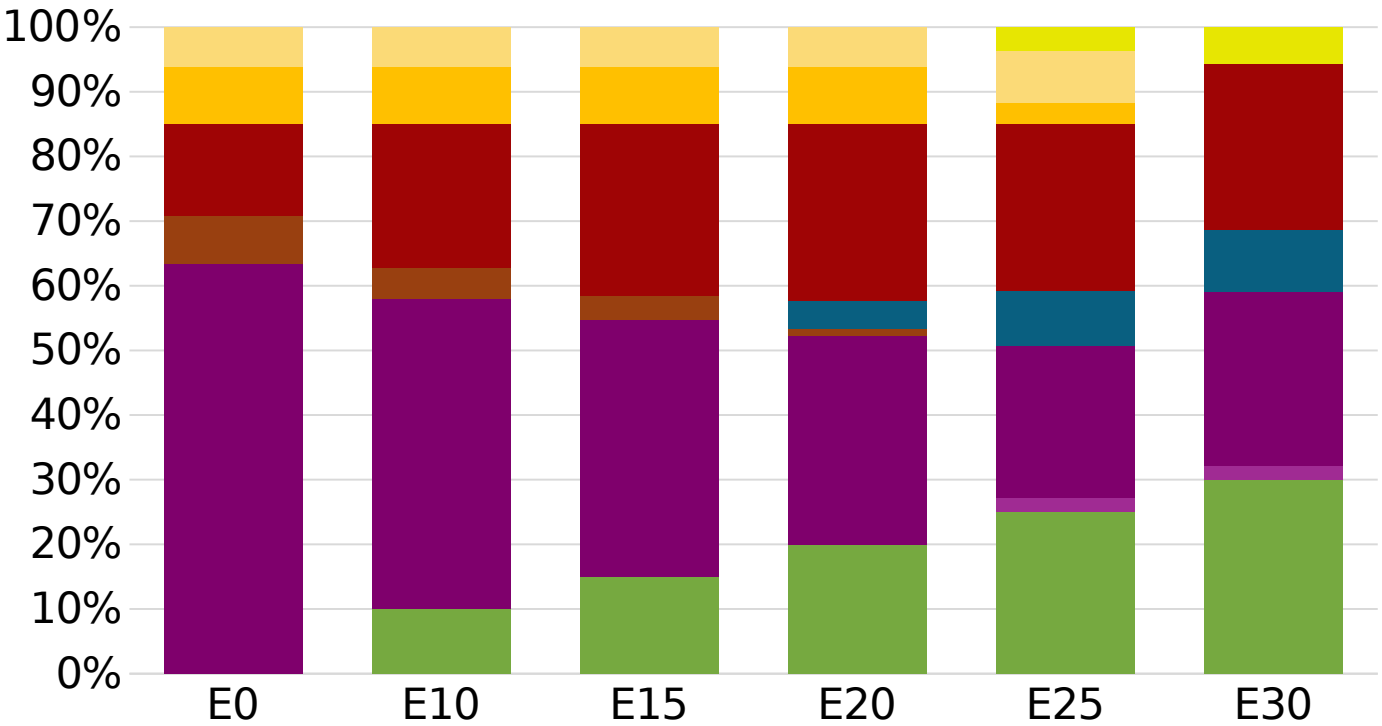


Average CIF 2025 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.091	\$ 1.953	\$ 1.853	\$ 1.738	\$ 1.692	\$ 1.639

Source:
Faro90

Bulgaria - Gasoline 98 - Constant Octane

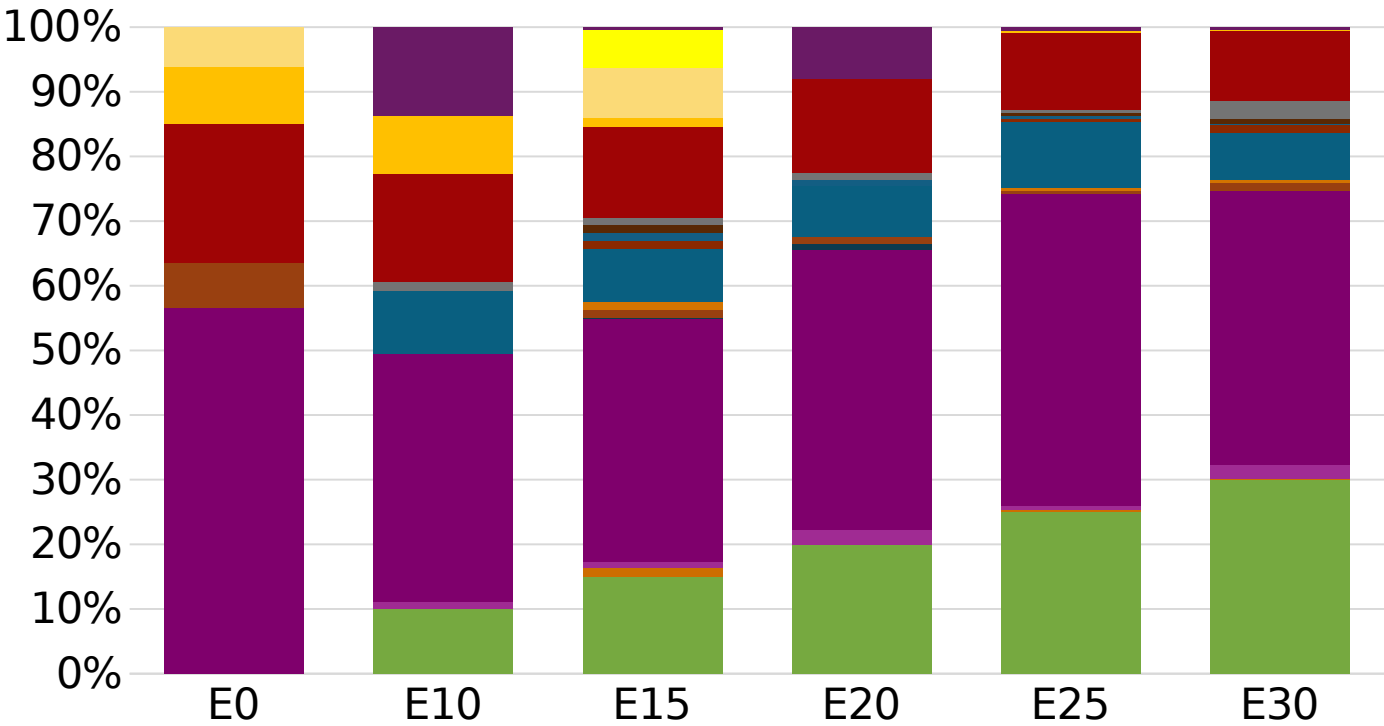


Average CIF 2025
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.111	\$ 1.994	\$ 1.925	\$ 1.853	\$ 1.788	\$ 1.738

Source:
Faro90

Bulgaria – Gasoline 95 – Increased Octane

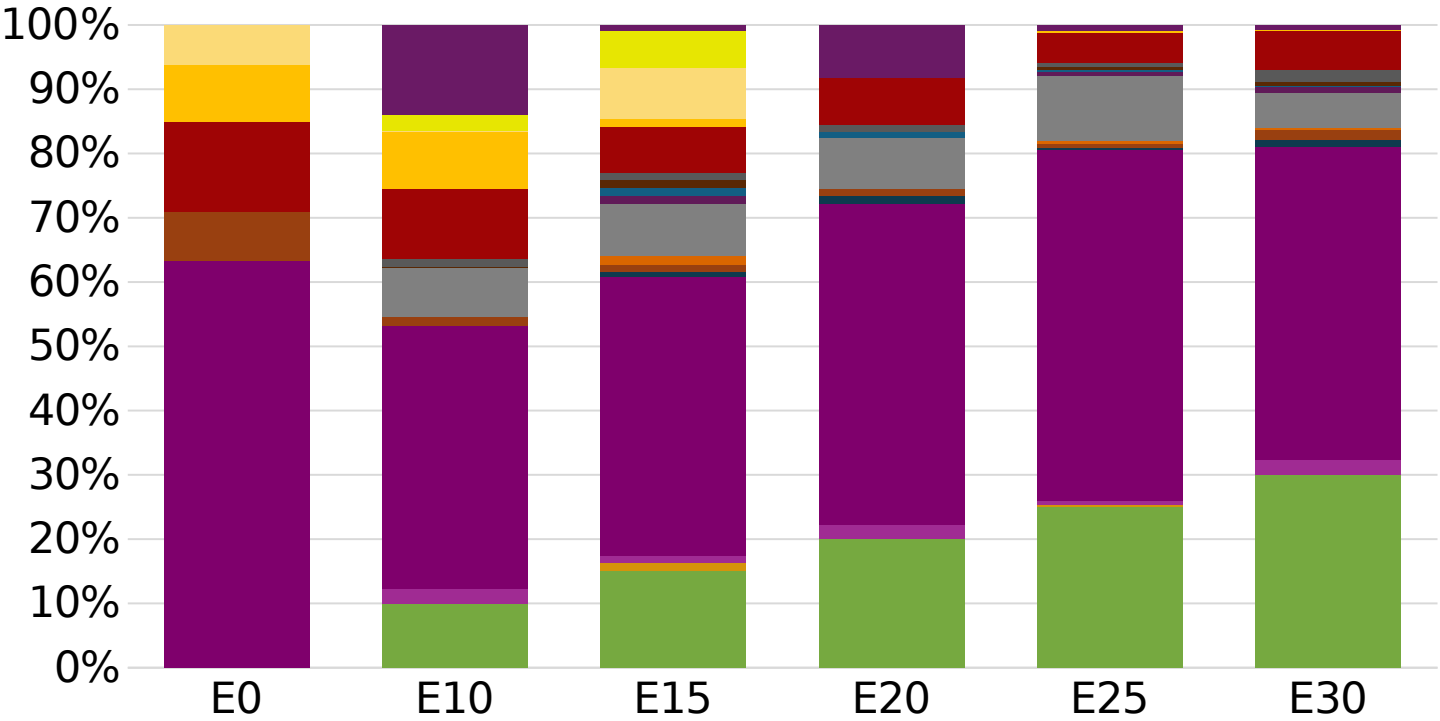


Average CIF 2025
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 1.979	\$ 1.979	\$ 1.979	\$ 1.979	\$ 1.979	\$ 1.979

Source:
Faro90

Bulgaria – Gasoline 98 – Increased Octane

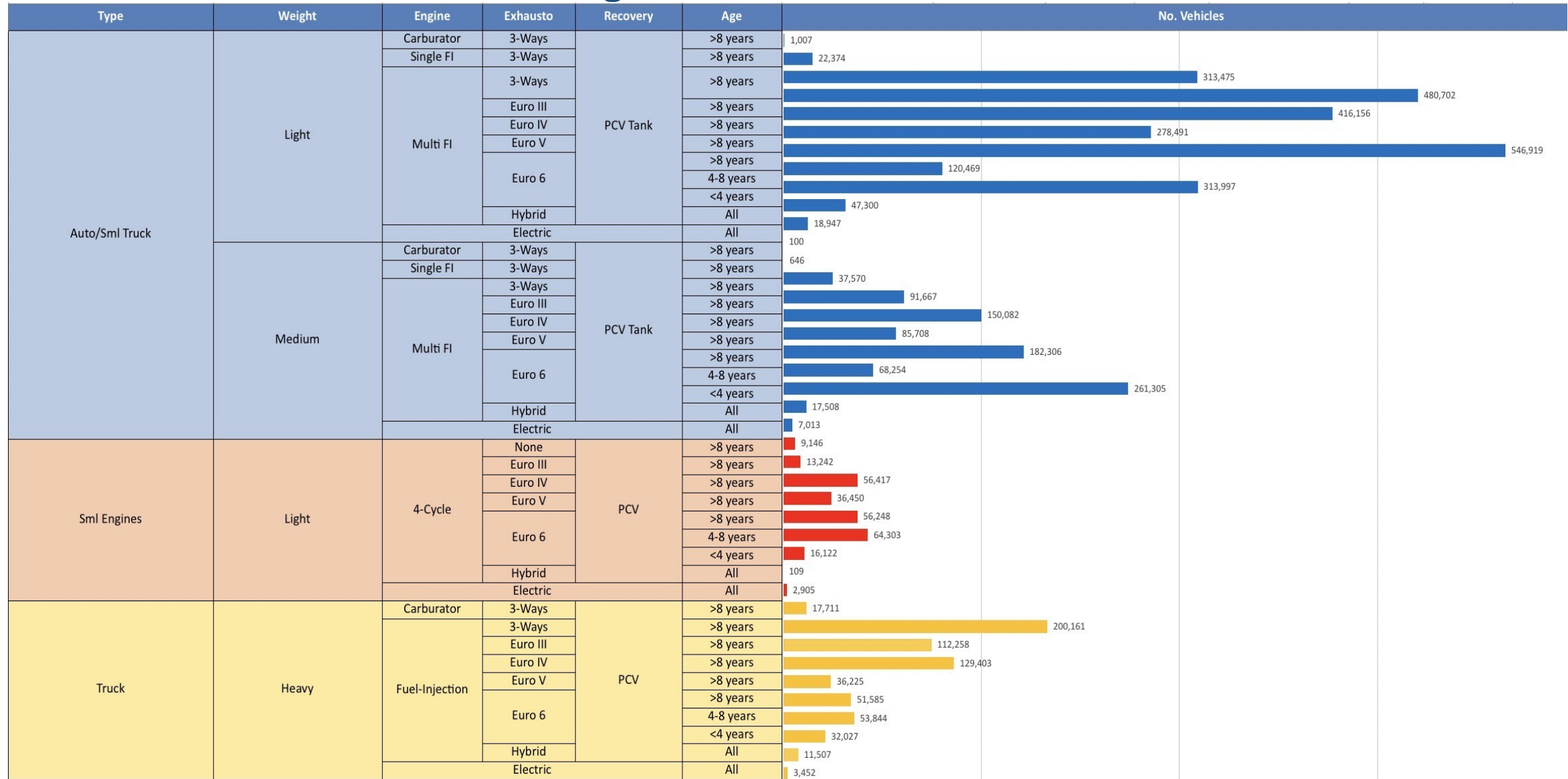


Average CIF 2025
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.111	\$ 2.111	\$ 2.111	\$ 2.111	\$ 2.111	\$ 2.111

Source:
Faro90

Gasoline Vehicle Fleet - Bulgaria

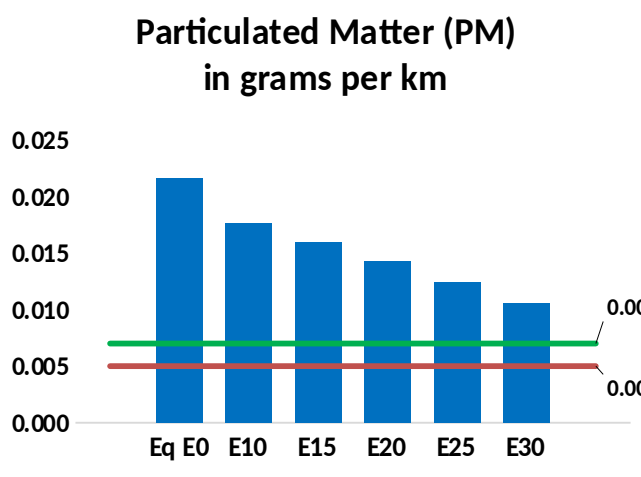
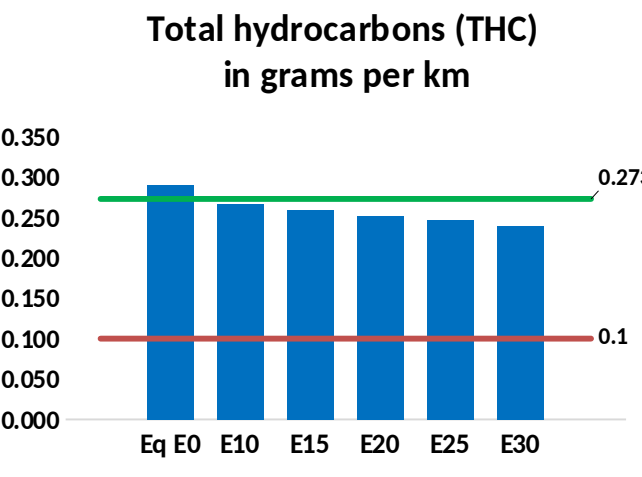
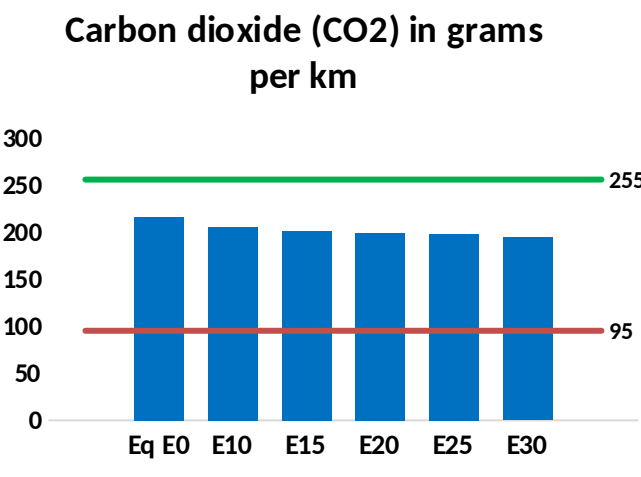
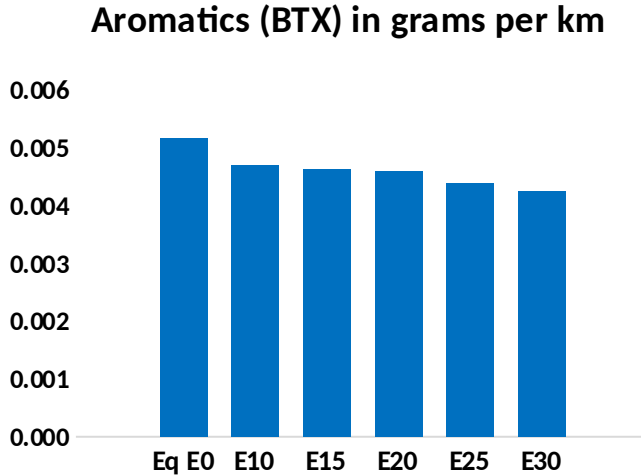
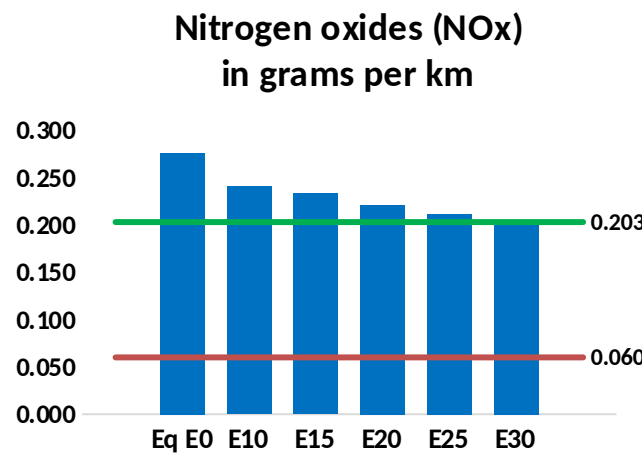
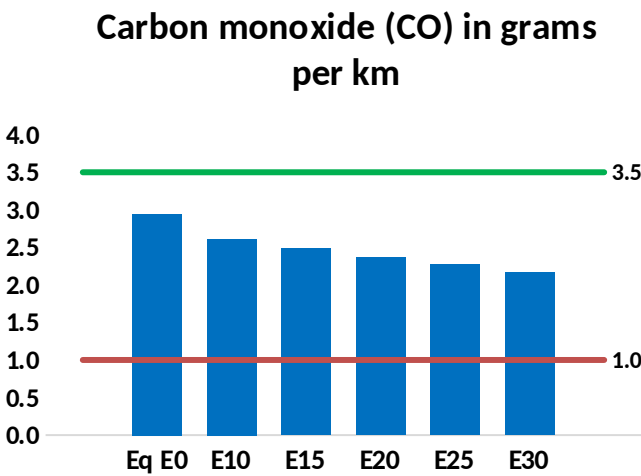
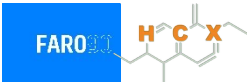


Vehicle Fleet: 4,365,110
Gasoline Vehicle Fleet: 2,909,802

Source: Eurostat, ACEA

Bulgaria – Vehicular Emissions

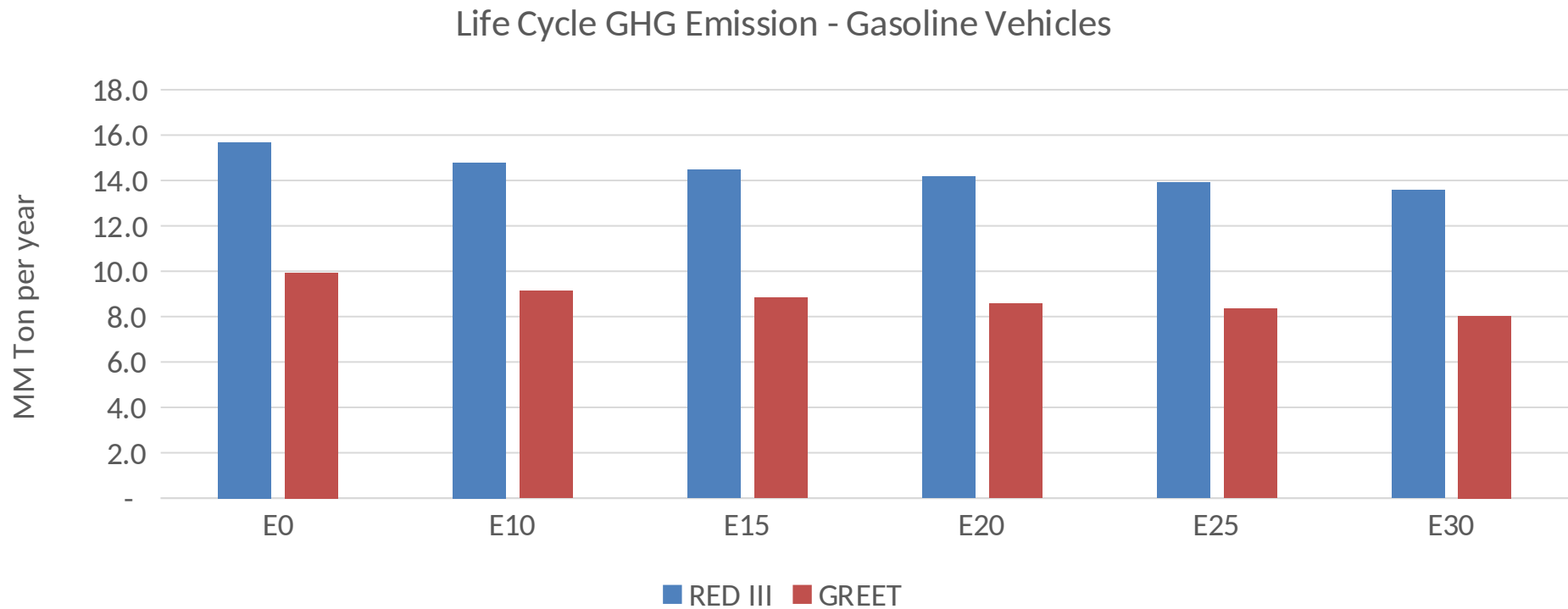
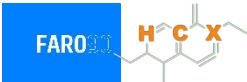
TIER III BIN 125 United States
Euro 6 Standard



Bulgaria – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	218,139	205,812	193,485	184,251	175,357	168,639	160,529	-11%	-20%
CO2	15,979,135	15,568,243	15,180,179	14,874,977	14,723,951	14,636,070	14,366,285	-5%	-8%
VOC	21,538	20,654	19,771	19,197	18,689	18,319	17,752	-8%	-13%
NOx	20,472	18,902	17,825	17,277	16,393	15,700	14,884	-13%	-20%
SOx	182	169	155	143	132	120	107	-15%	-28%
NH3	5,390	5,337	5,283	5,309	5,368	5,410	5,445	-2%	0%
N2O	283	281	280	281	287	291	294	-1%	2%
CH4	4,892	4,892	4,892	4,966	5,073	5,166	5,254	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	223	218	213	211	210	210	209	-5%	-6%
Acetaldehyde	464	533	782	1,190	1,622	1,853	2,189	68%	249%
Formaldehyde	1,435	1,395	1,627	1,910	2,001	2,214	2,410	13%	39%
Benzene	383	368	349	343	341	326	315	-9%	-11%
PM2.5	926	842	758	684	611	534	452	-18%	-34%
PM10	672	611	550	496	444	387	328	-18%	-34%

Bulgaria – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	15.3
Target @ 2035 (MMT/year)	3.1
Delta	12.1

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

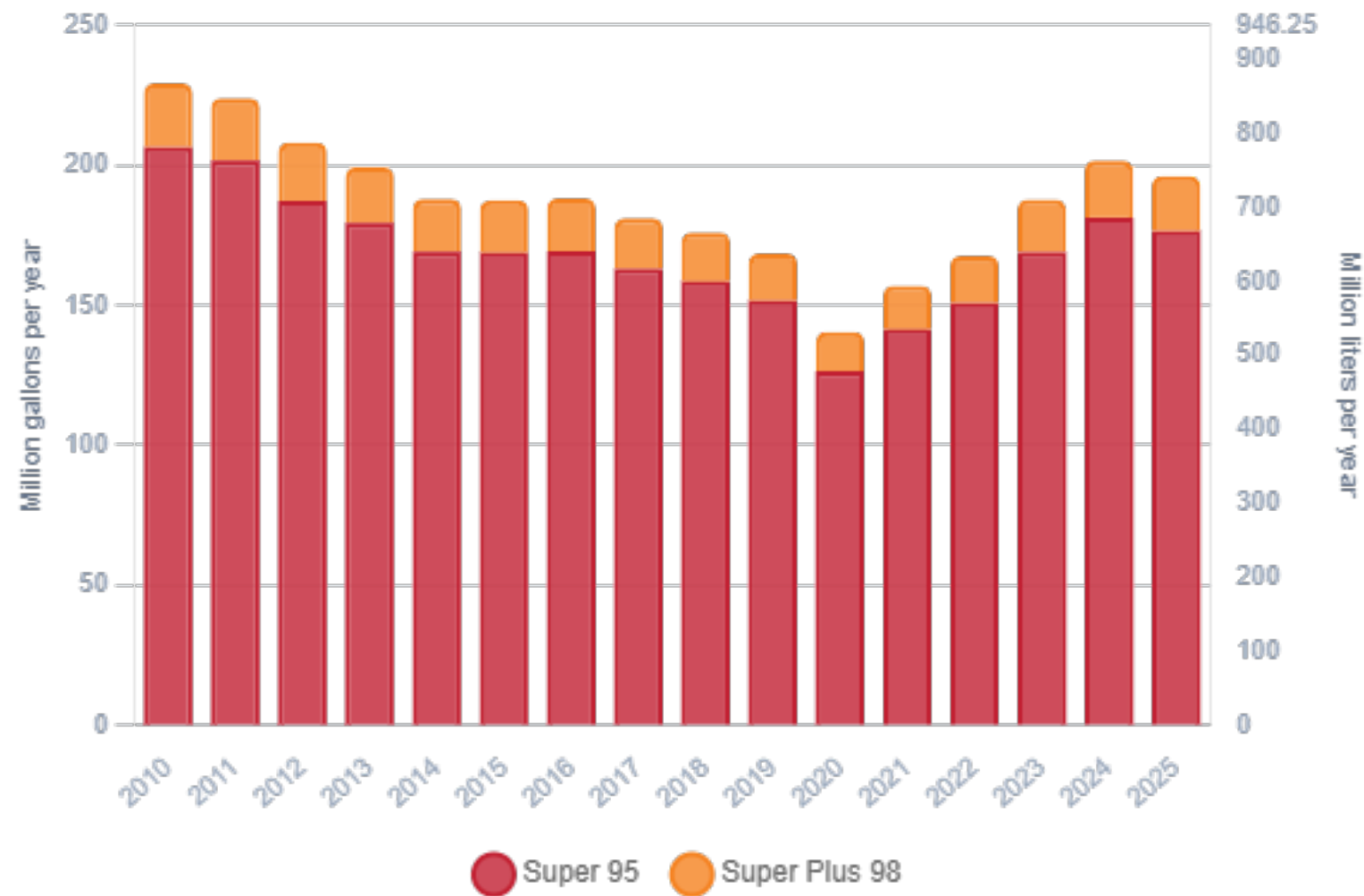
Ethanol Gasoline Blending in Croatia



In 2025, gasoline consumption in Croatia was 772 million liters (204 million gallons) and growth rate was -0.8%. Gasoline production and net imports were 1,310 and -510 million liters (346 and -135 million gallons) correspondingly. Croatia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.3 RON.

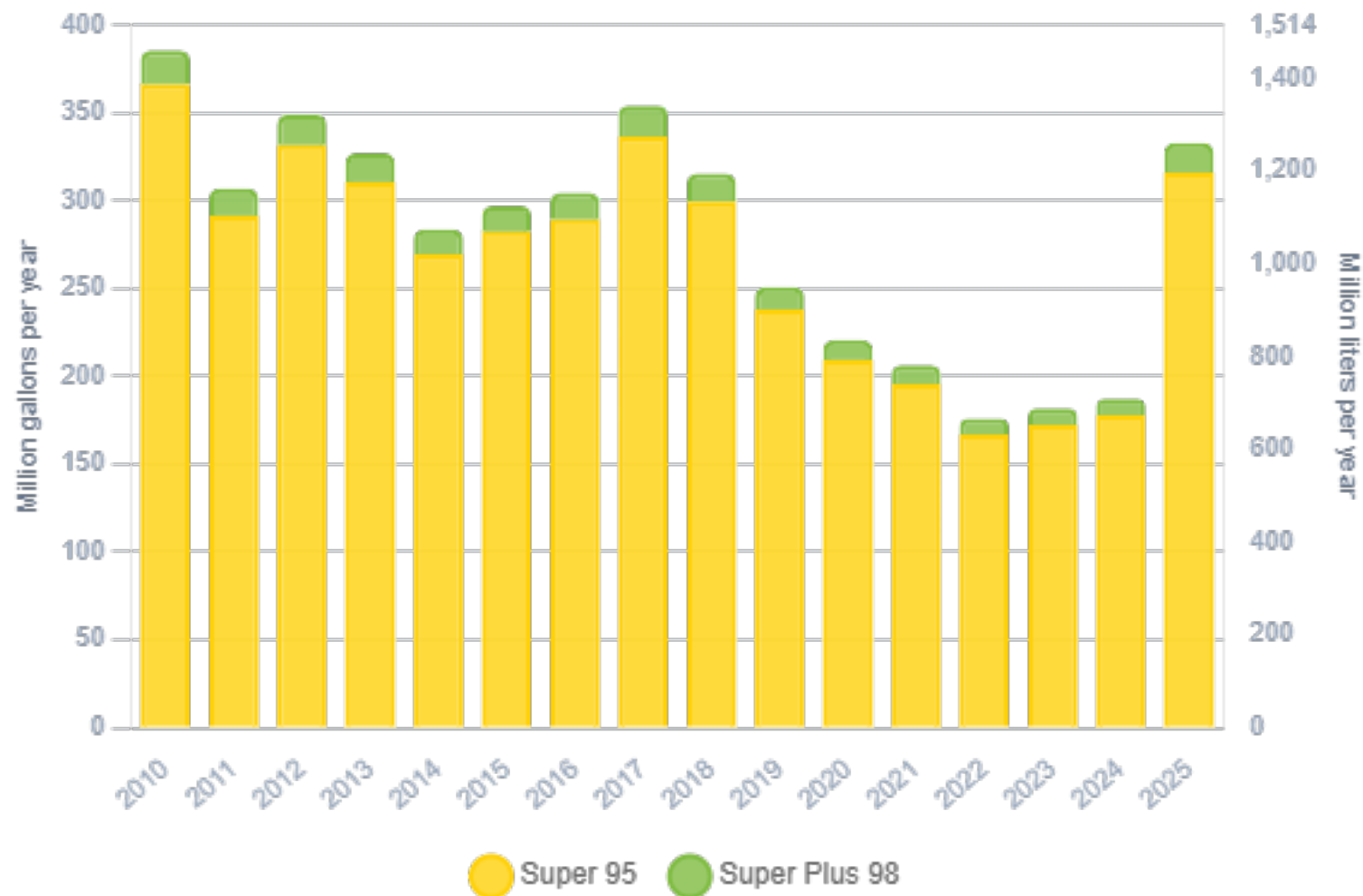
Although E10 is allowed in Croatia, it does not produce or consumes bioethanol in a commercial level. There is a small production of generic ethanol that is exported in minimal quantities to Italy, Slovenia, Austria and Bosnia.

Gasoline Demand in Croatia



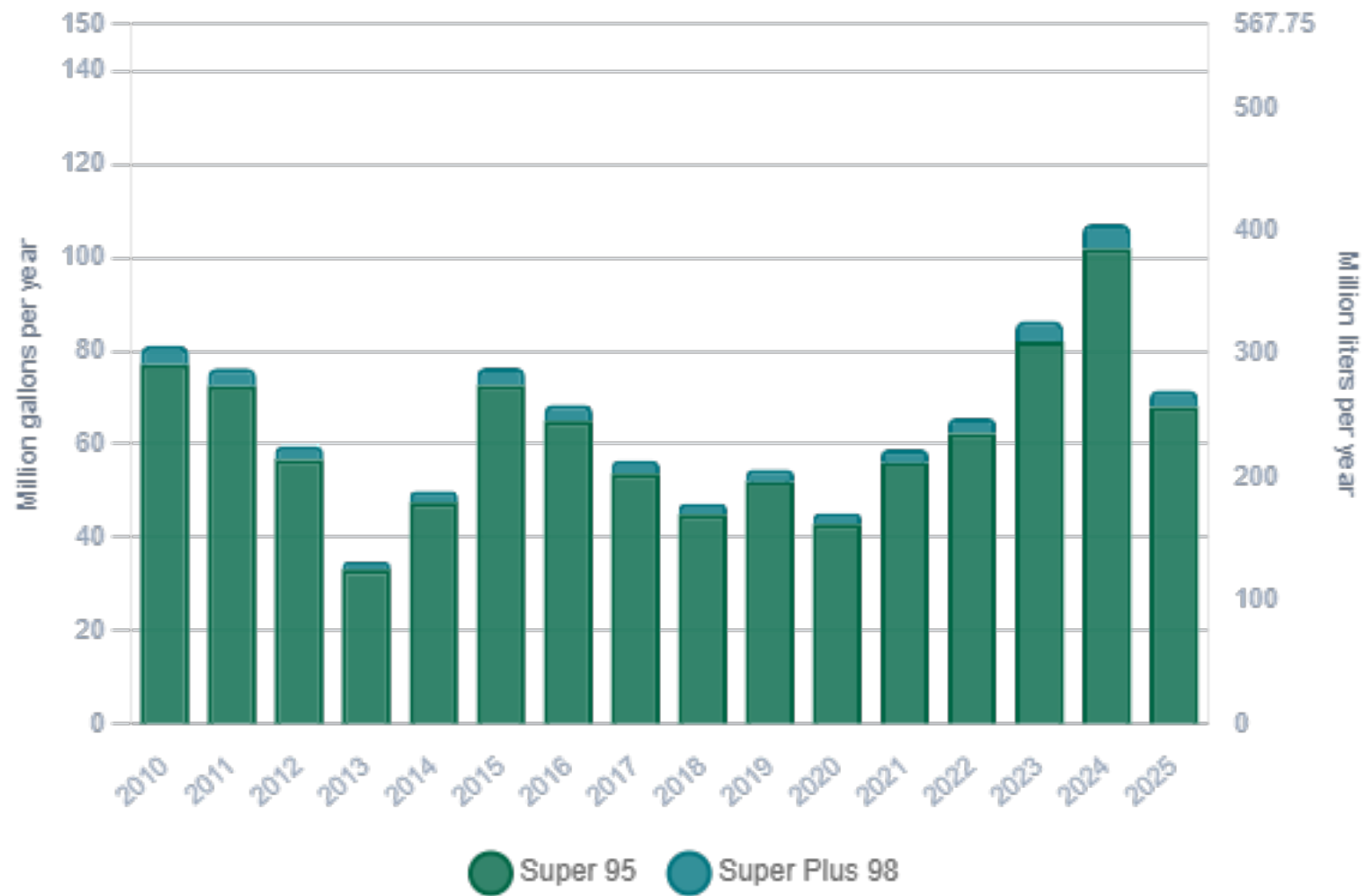
Source: Eurostat

Gasoline Production in Croatia



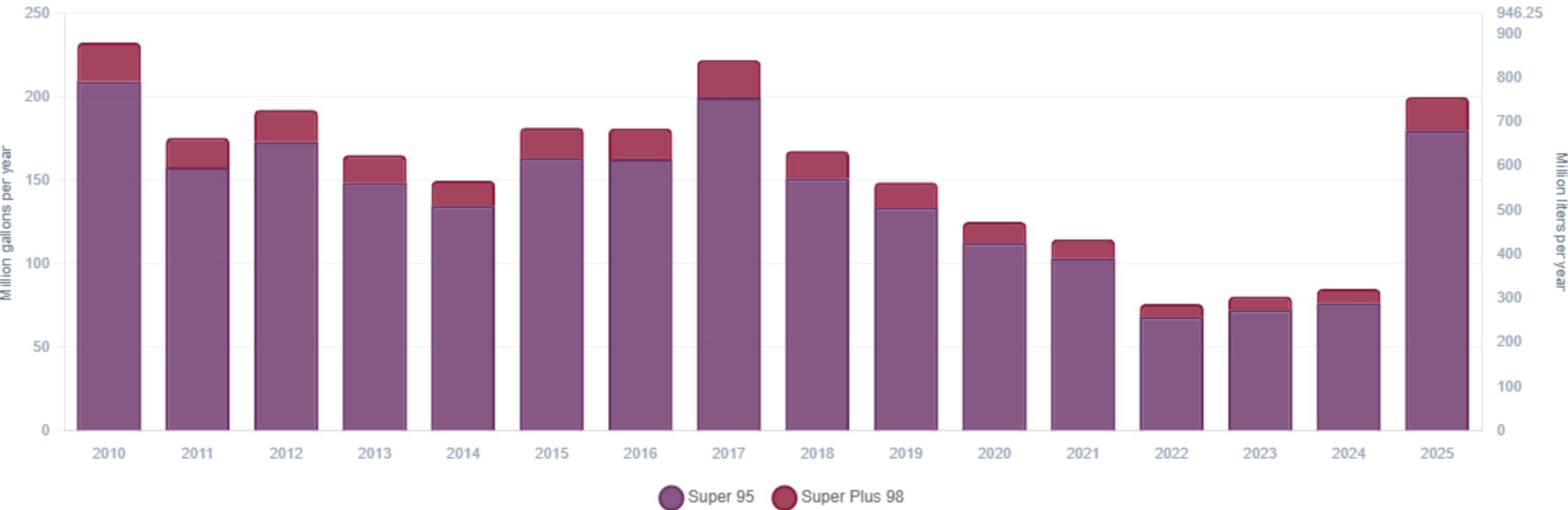
Source: Eurostat

Gasoline Imports in Croatia



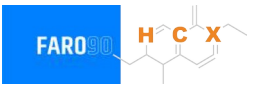
Source: Eurostat

Gasoline Exports in Croatia



Source: Eurostat

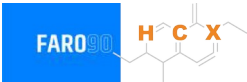
Ethanol Balance in Croatia



Croatia does not produce or consumes bioethanol in a commercial level. There is a small production of generic ethanol that is exported in minimal quantities to Italy, Slovenia, Austria and Bosnia.

Source: The Global Economy, WITS/Comtrade, Total Croatia News, Axens

Gasoline Quality in Croatia

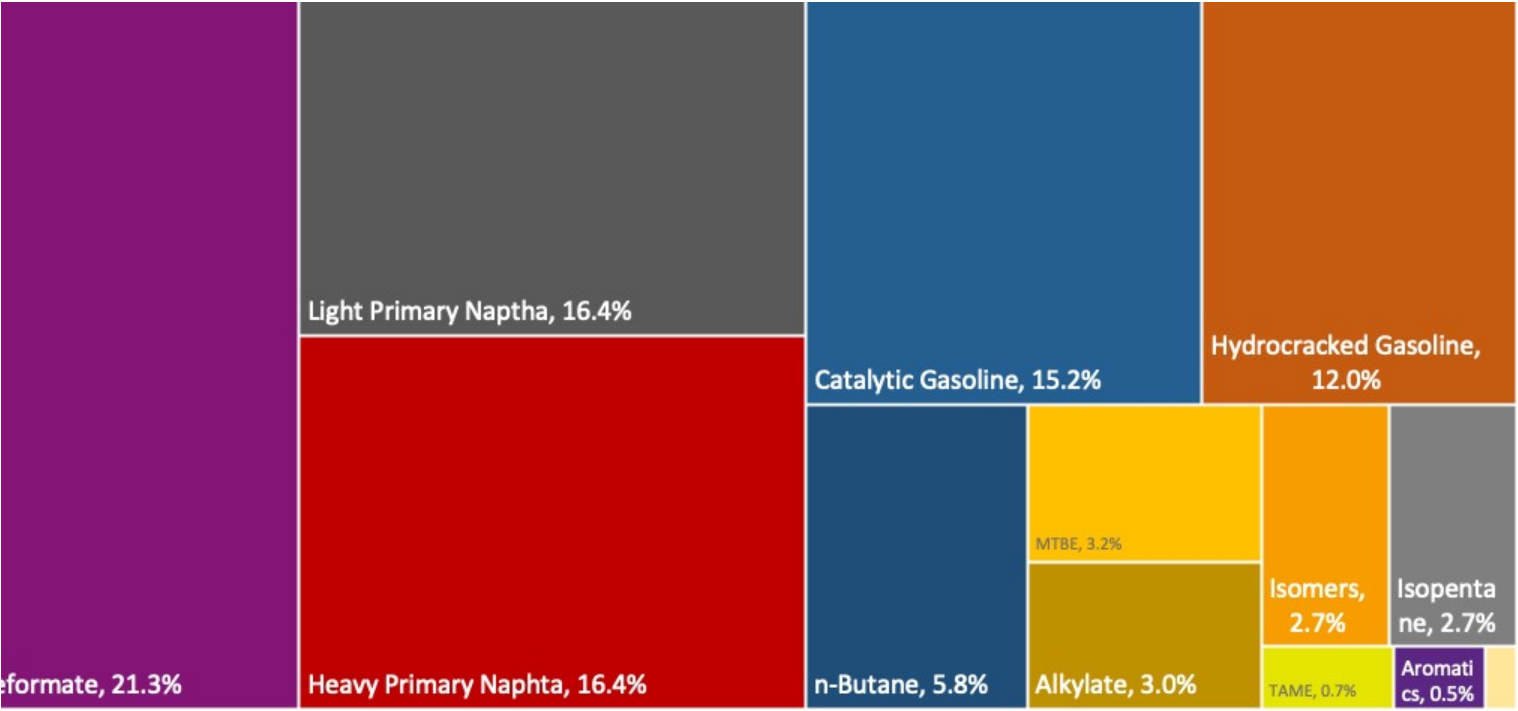


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	8.8							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	1.0							
GHG Emission Reduction (%)	-							

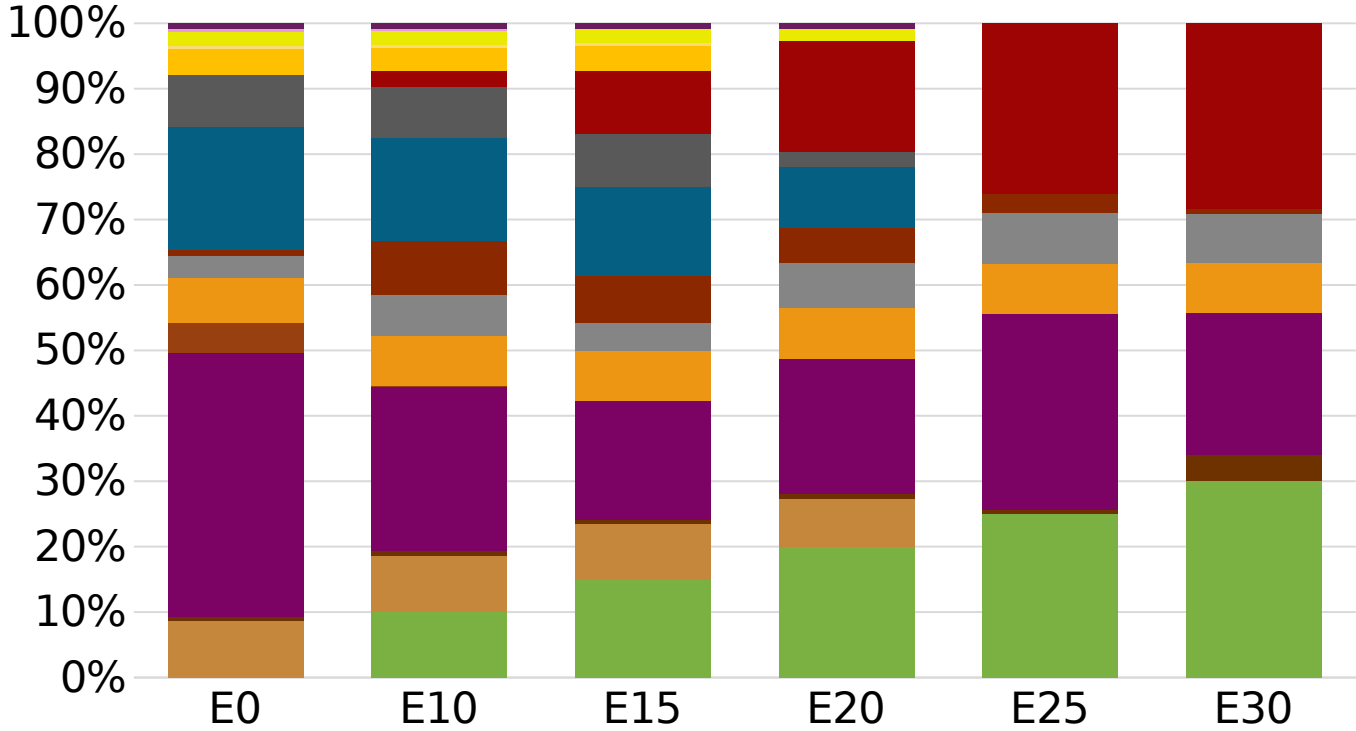
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Croatia Available Blending Components



Croatia – Gasoline 95 – Constant Octane

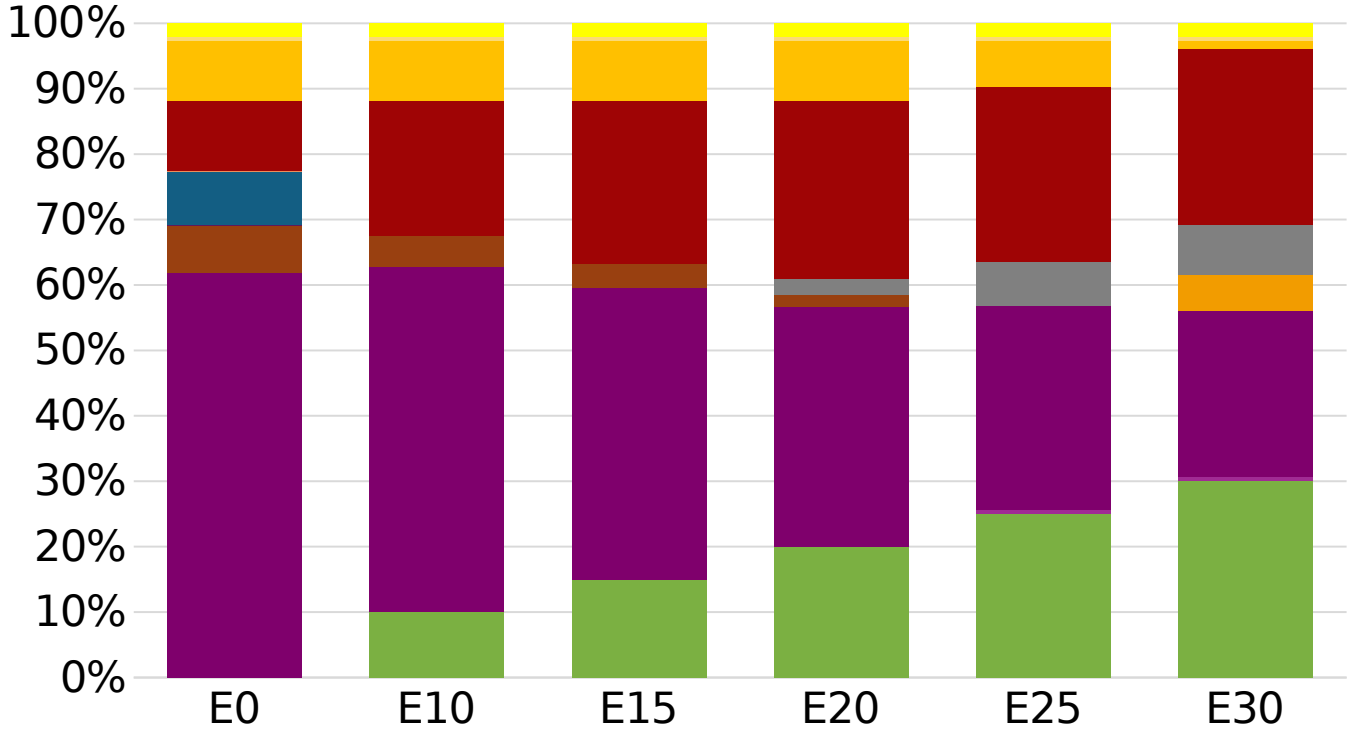


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US\$/gal)	\$ 2.055	\$ 1.942	\$ 1.859	\$ 1.783	\$ 1.688	\$ 1.614

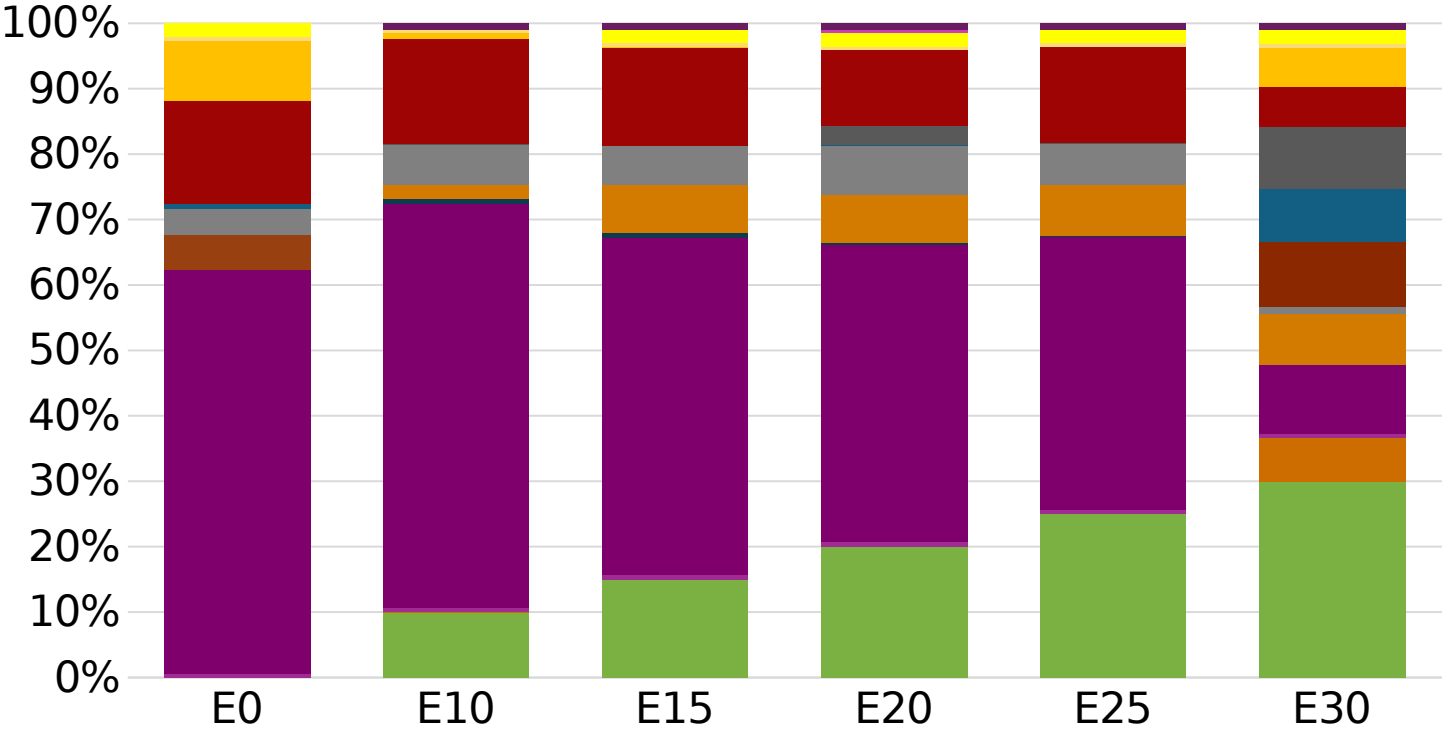
Source:
Faro90

Croatia - Gasoline 98 - Constant Octane



Average CIF 2025
Prices

Croatia – Gasoline 95 – Increased Octane

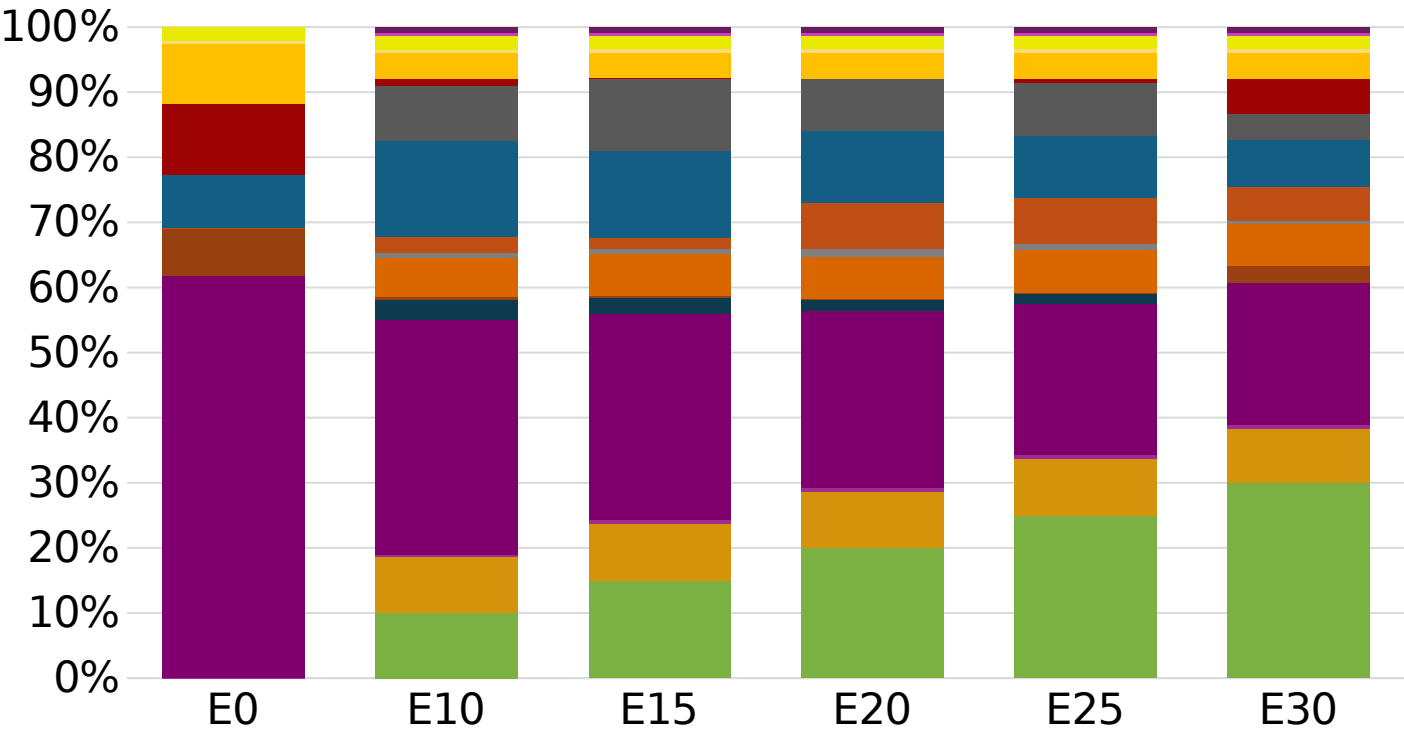


Average CIF 2025
Prices

Octane (RON)	95.0	95.9	97.5	99.0	100.5	101.8
Price (US\$/gal)	\$ 1.974	\$ 1.974	\$ 1.974	\$ 1.974	\$ 1.974	\$ 1.974

Source:
Faro90

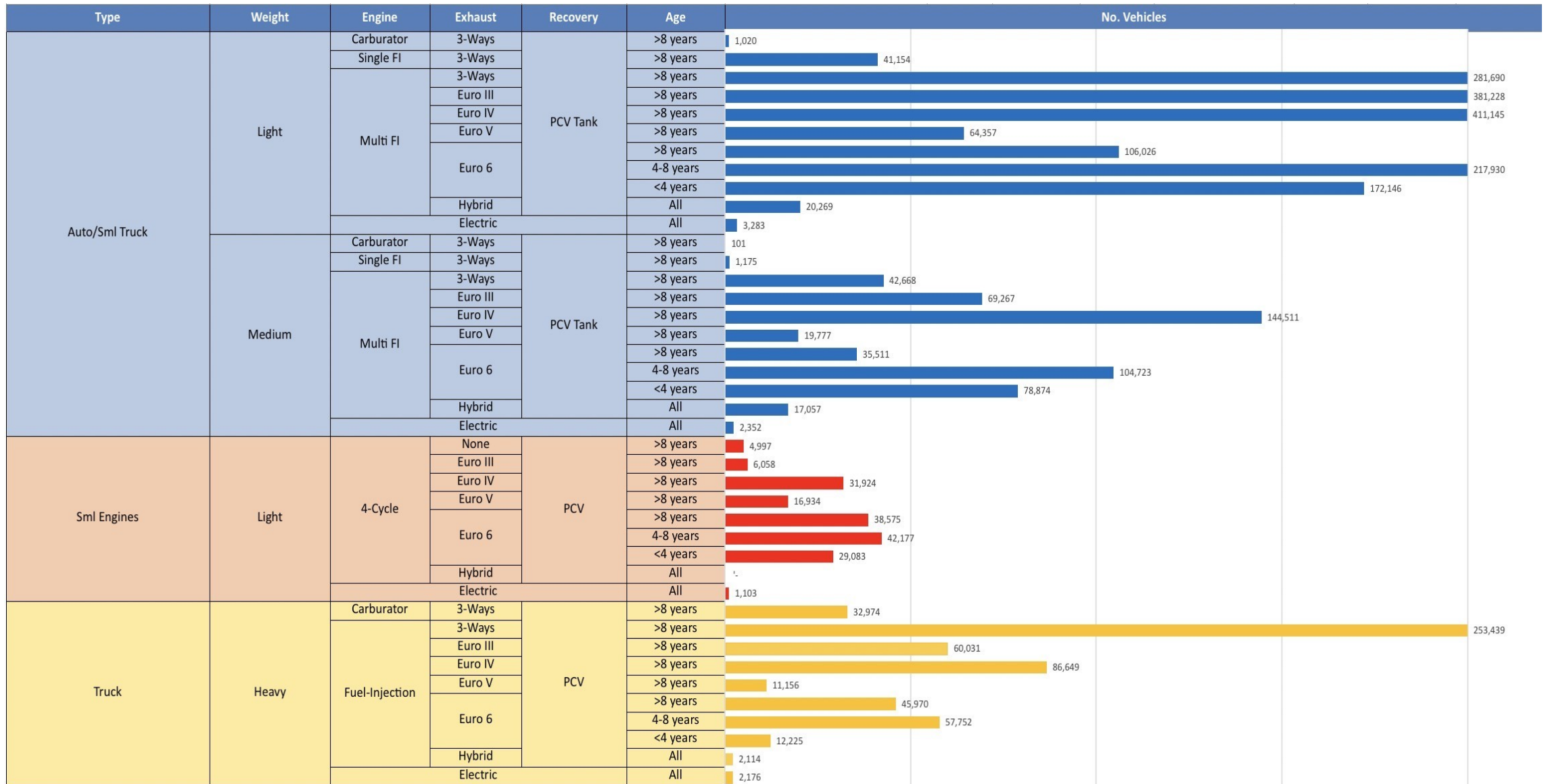
Croatia – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Source:
Faro90

Gasoline Vehicle Fleet - Croatia

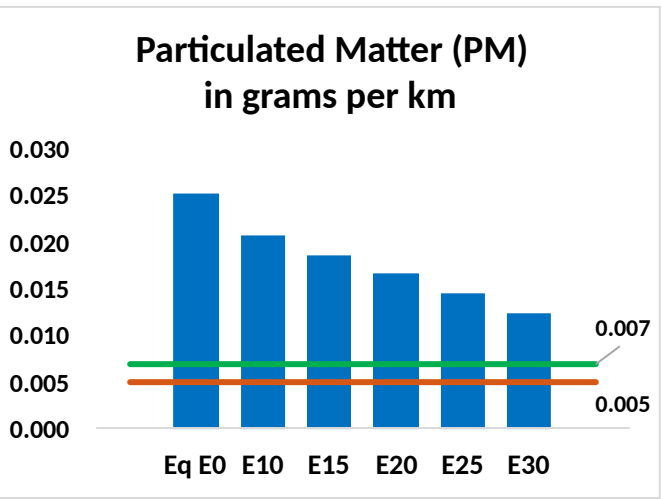
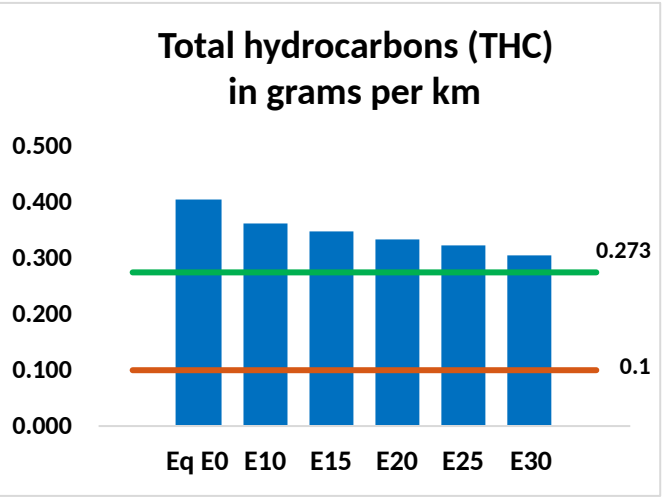
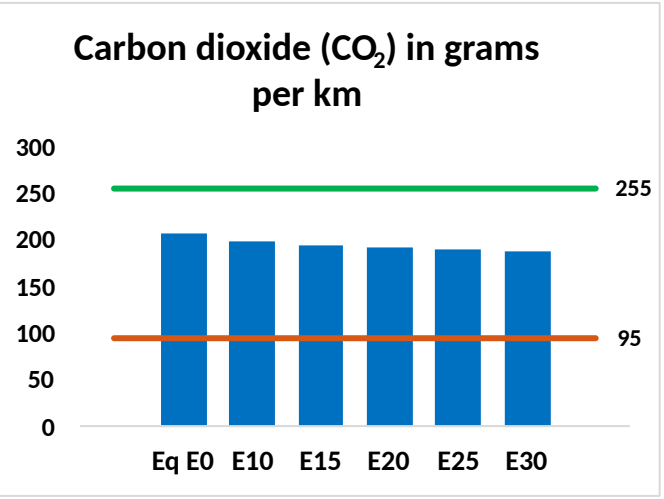
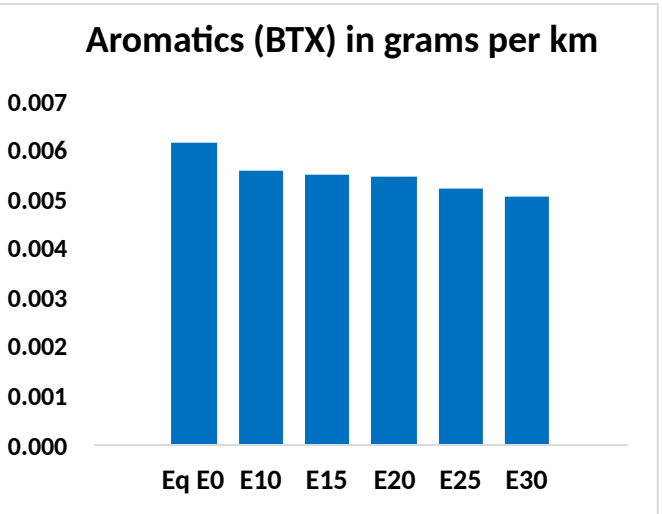
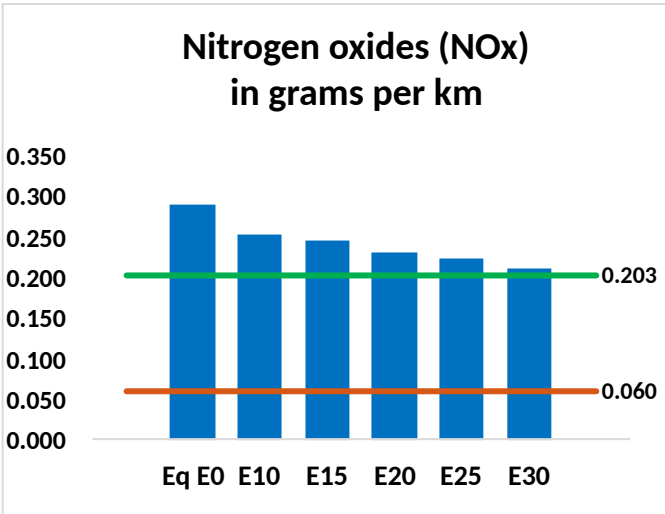
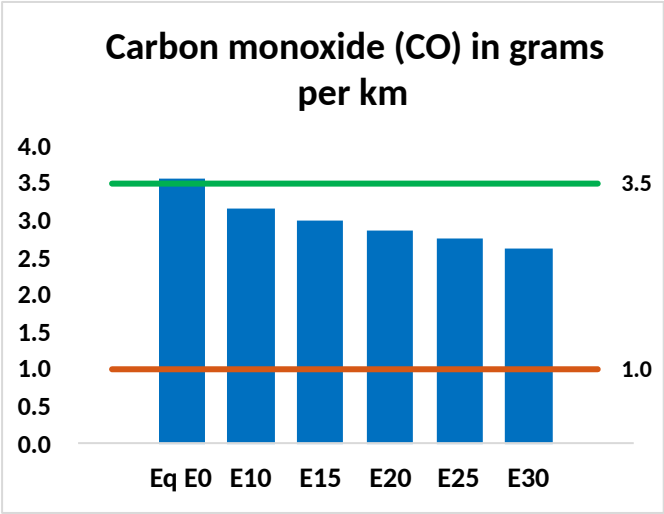
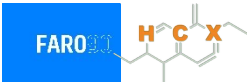


Vehicle Fleet: 2,951,603
Gasoline Vehicle Fleet: 909,203

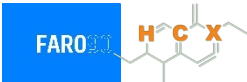
Source: UNECE, Eurostat, ACEA

Croatia – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard

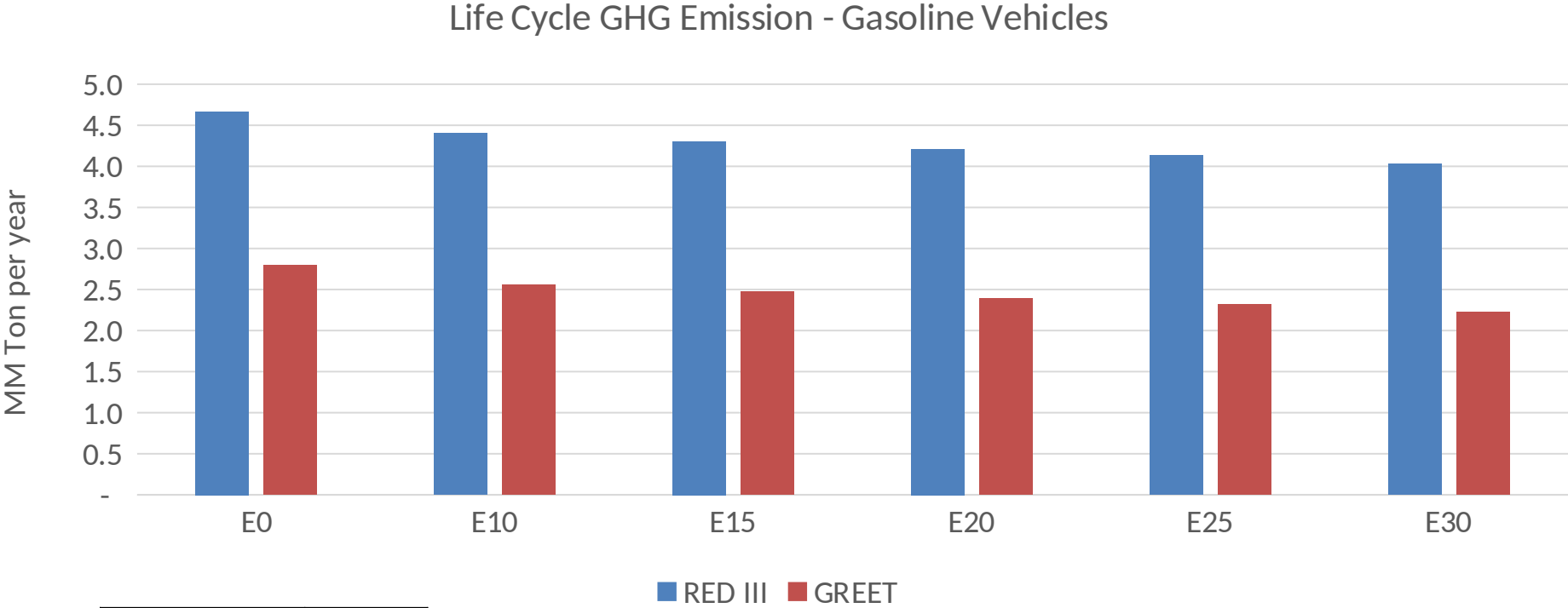


Croatia – Vehicular Emissions



Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	86,529	81,621	76,713	73,032	69,483	66,802	63,456	-11%	-20%
CO2	5,060,220	4,930,100	4,807,209	4,710,559	4,662,733	4,634,205	4,548,784	-5%	-8%
VOC	9,859	9,341	8,824	8,446	8,088	7,820	7,438	-10%	-18%
NOx	7,061	6,520	6,150	5,963	5,660	5,423	5,144	-13%	-20%
SOx	57	53	49	45	42	38	34	-15%	-28%
NH3	1,754	1,737	1,719	1,727	1,747	1,760	1,772	-2%	0%
N2O	71	71	71	71	72	73	74	-1%	2%
CH4	2,170	2,170	2,170	2,203	2,250	2,292	2,331	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	85	82	80	79	79	78	77	-5%	-7%
Acetaldehyde	200	230	338	514	700	800	946	68%	249%
Formaldehyde	657	639	745	875	917	1,014	1,104	13%	39%
Benzene	150	144	136	134	133	127	123	-9%	-11%
PM2.5	300	273	246	222	198	173	147	-18%	-34%
PM10	315	287	258	233	208	182	154	-18%	-34%

Croatia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	17.9
Target @ 2035 (MMT/year)	10.1
Delta	7.8

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

Ethanol Gasoline Blending in Cyprus



In 2025, gasoline consumption in Cyprus was 465 million liters (123 million gallons) and growth rate was -0.74%. Gasoline production and net imports were 0 and 481 million liters (0 and 127 million gallons) correspondingly. Cyprus complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.3 RON.

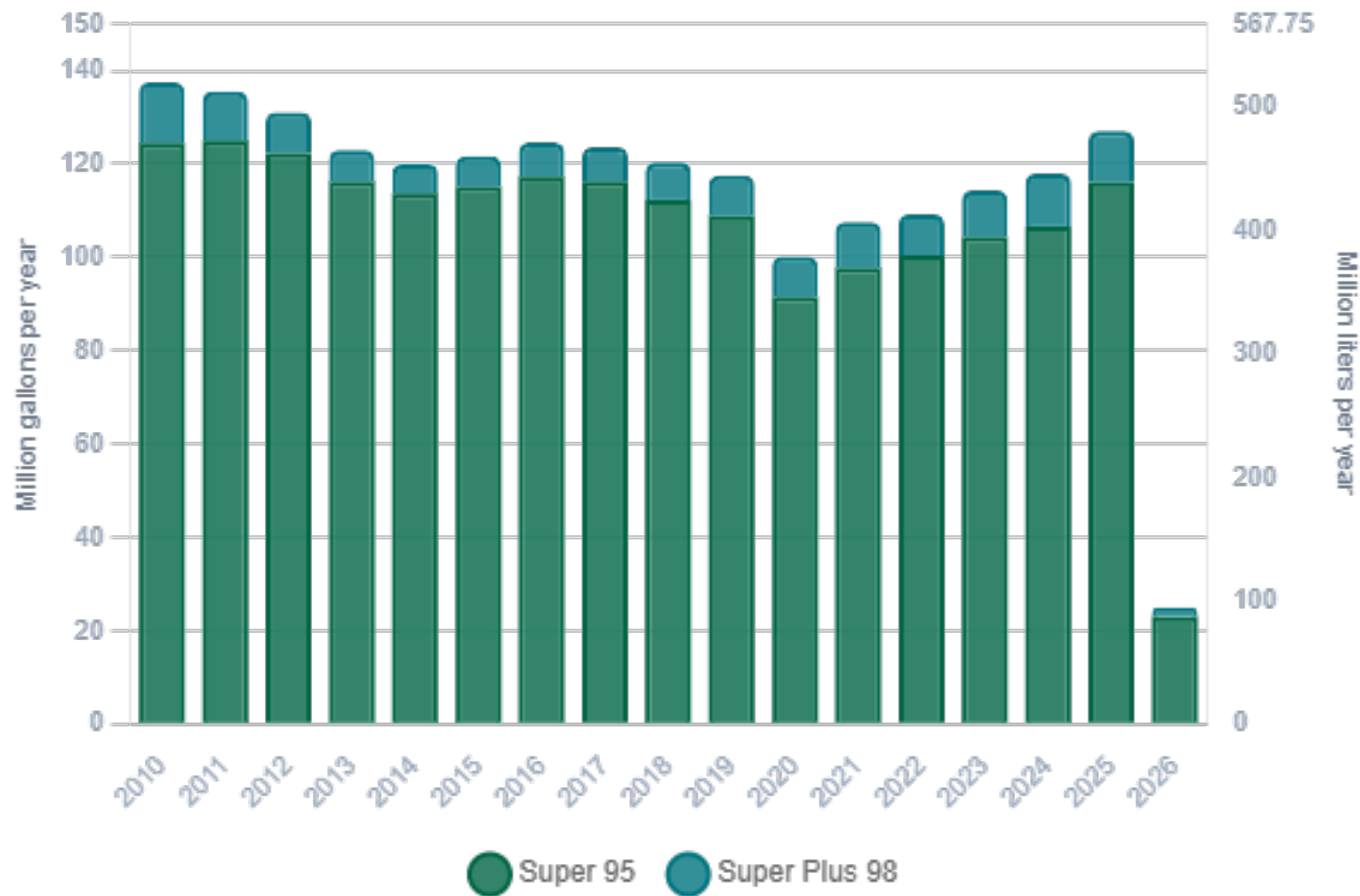
Cyprus has no specific ethanol mandate and a target of 14% biofuels in the transport sector by 2030. Ethanol demand was 19.2 million liters in 2025 (5.1 million gallons) equivalent to 4.1% of gasoline demand. Ethanol is supplied by imports.

Gasoline Demand in Cyprus



Source: Eurostat

Gasoline Imports in Cyprus



Source: Eurostat

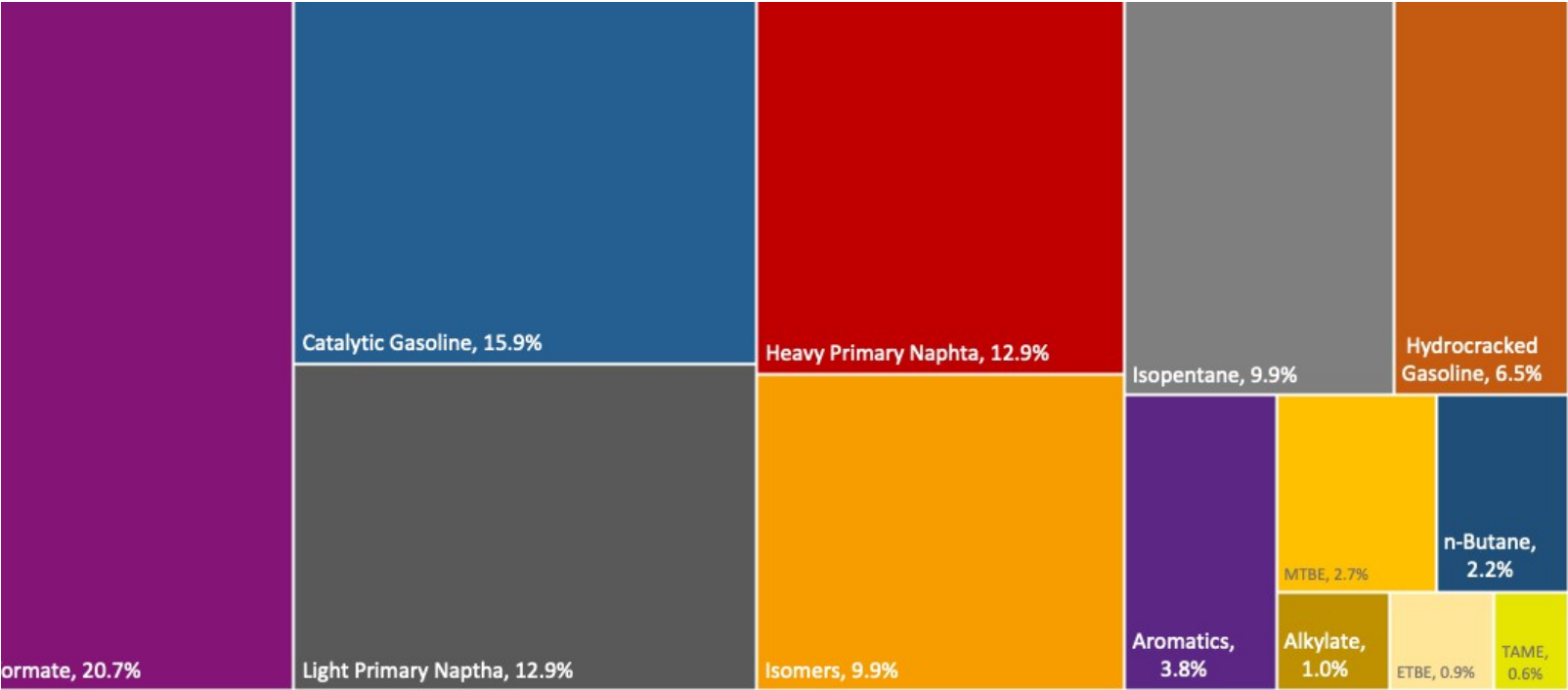
Gasoline Quality in Cyprus

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

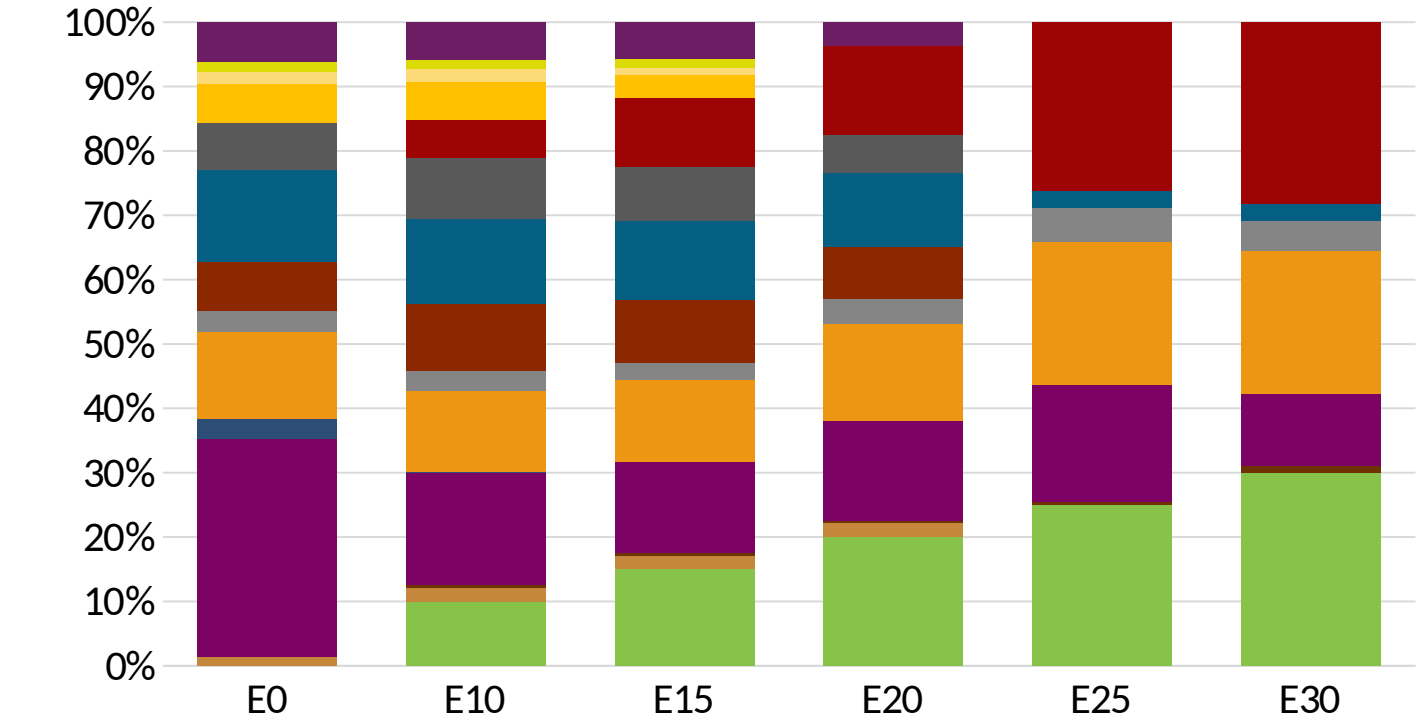
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Cyprus Available Blending Components



Cyprus – Gasoline 95 – Constant Octane

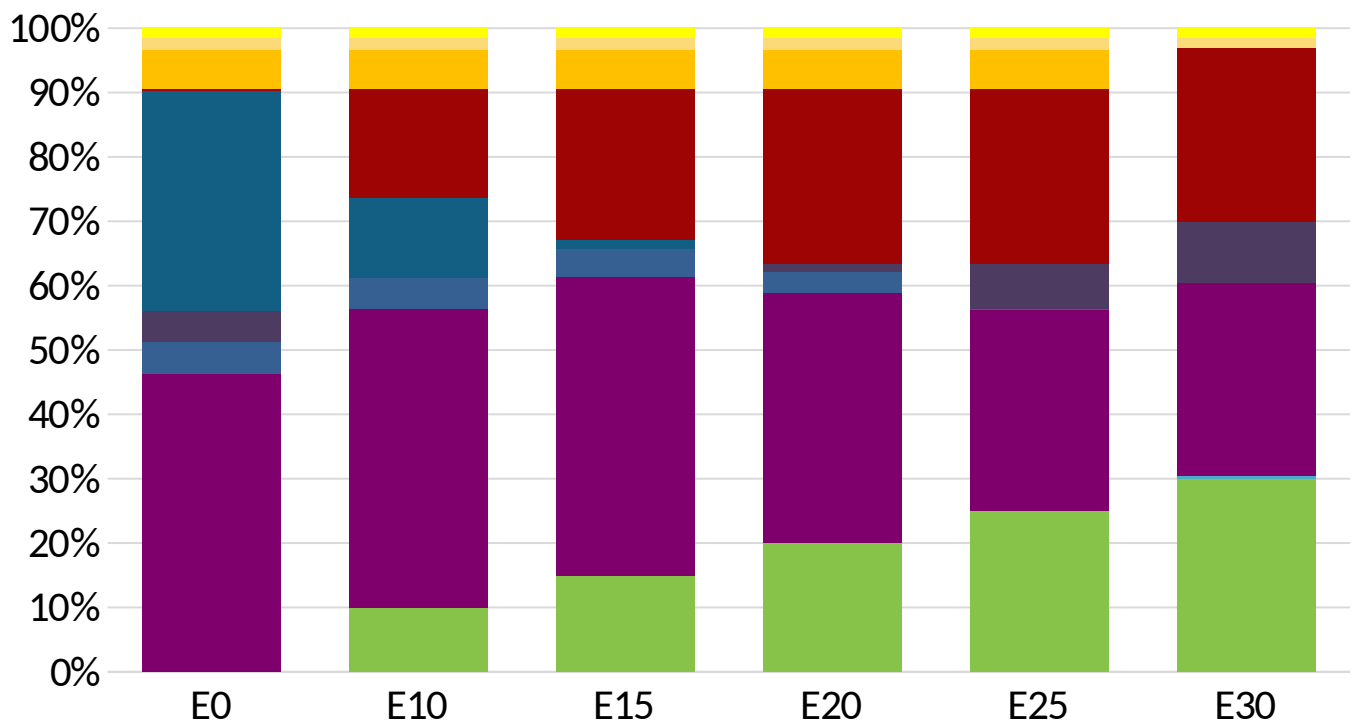


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	96.0
Price (US\$/gal)	\$ 2.055	\$ 1.932	\$ 1.866	\$ 1.799	\$ 1.678	\$ 1.642

Source:
Faro90

Cyprus - Gasoline 98 - Constant Octane

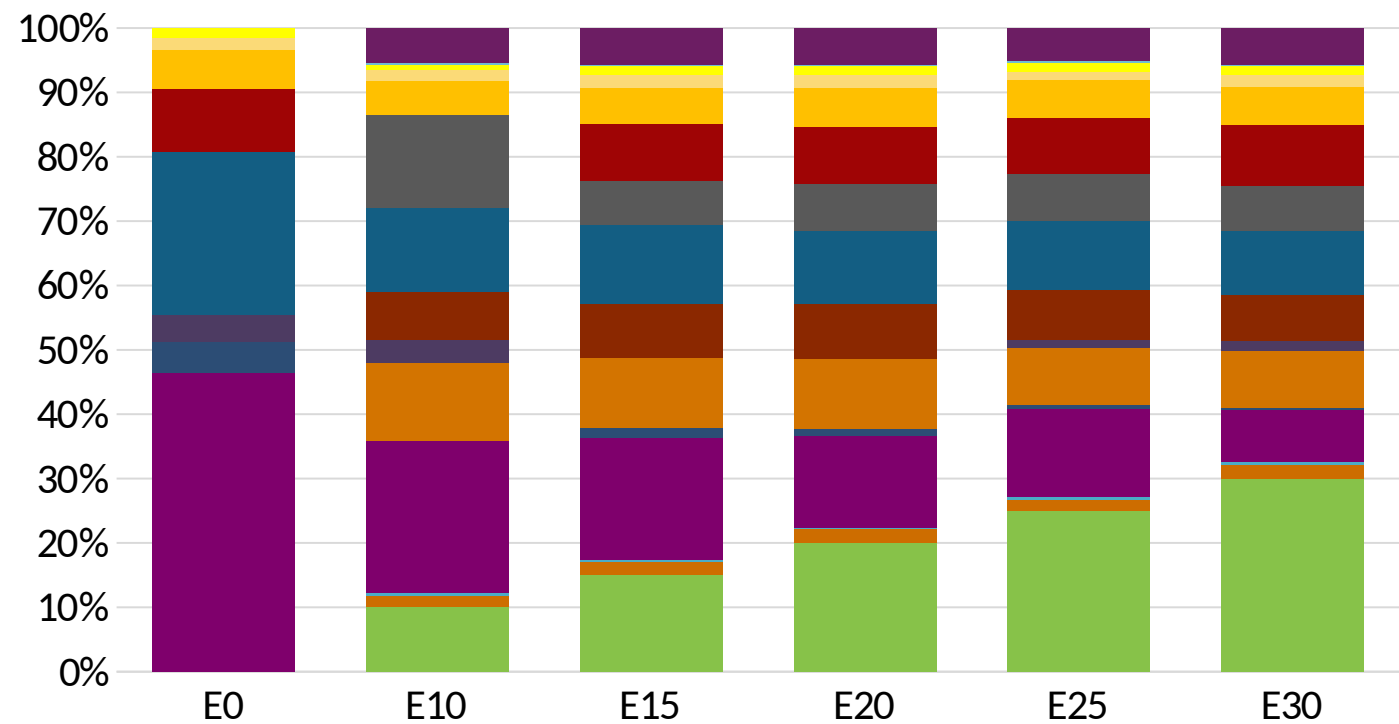


Average CIF 2025
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.127	\$ 2.001	\$ 1.929	\$ 1.854	\$ 1.777	\$ 1.721

Source:
Faro90

Cyprus – Gasoline 95 – Increased Octane

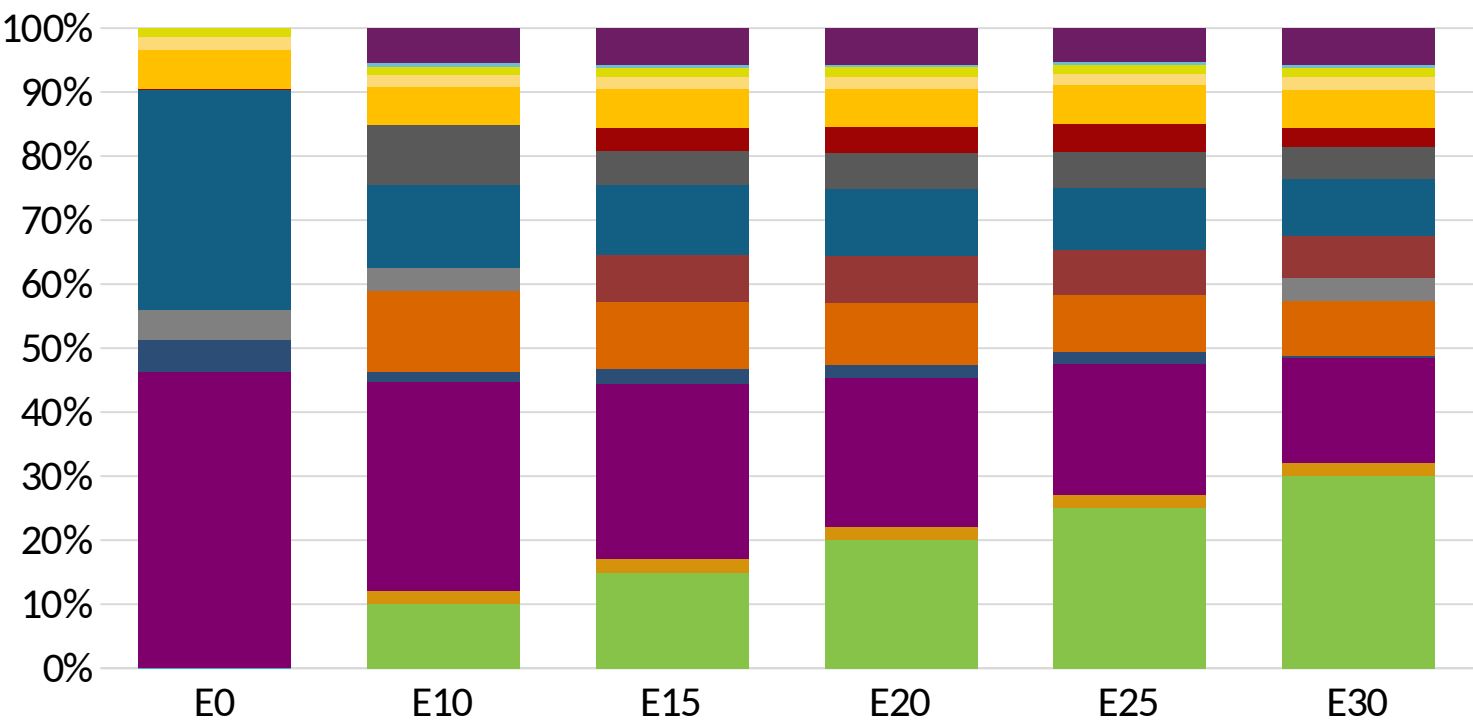


Average CIF 2025
Prices

Octane (RON)	95.0	96.2	97.9	99.4	100.8	102.2
Price (US\$/gal)	\$ 1.993	\$ 1.993	\$ 1.993	\$ 1.993	\$ 1.993	\$ 1.993

Source:
Faro90

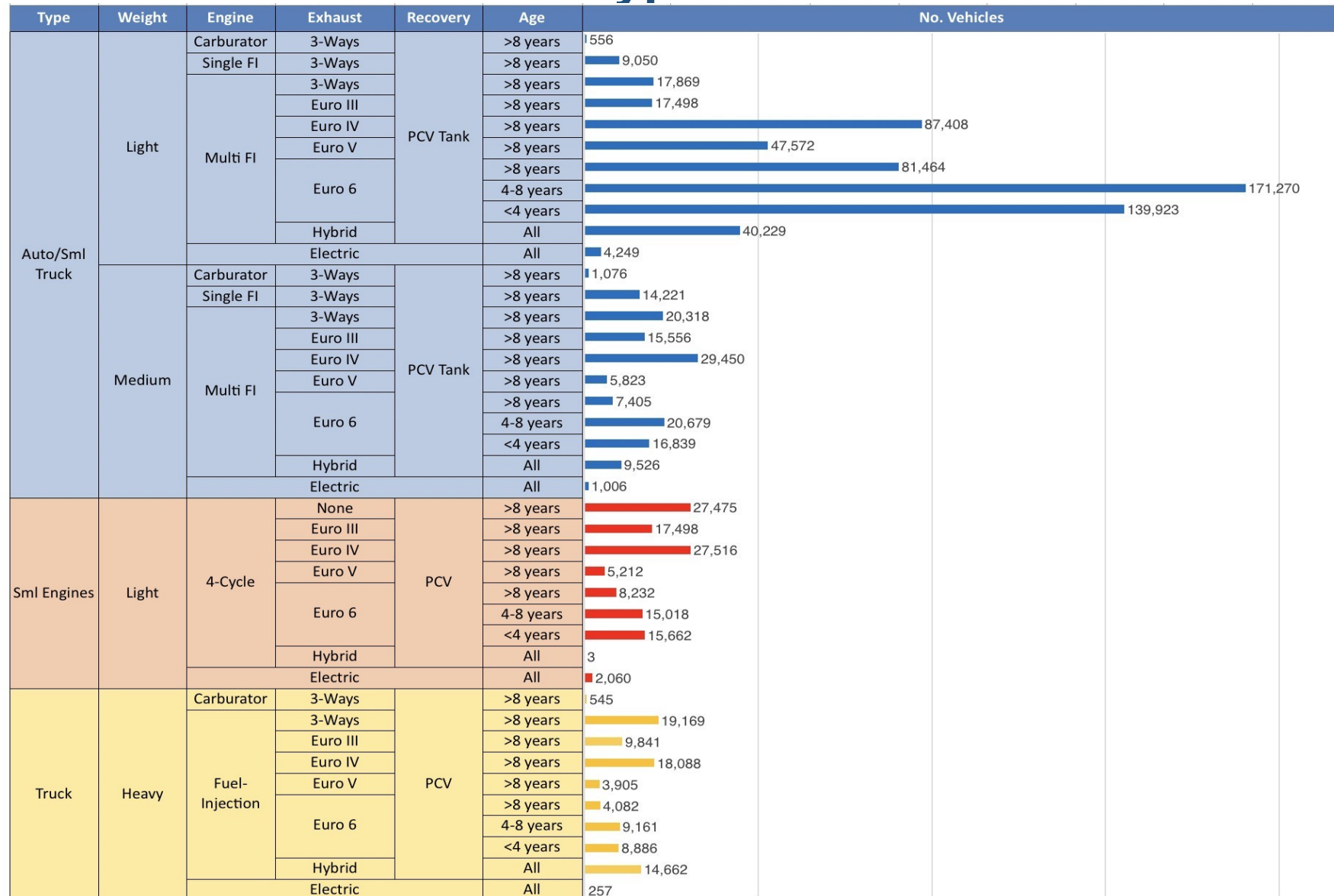
Cyprus – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Source:
Faro90

Gasoline Vehicle Fleet - Cyprus

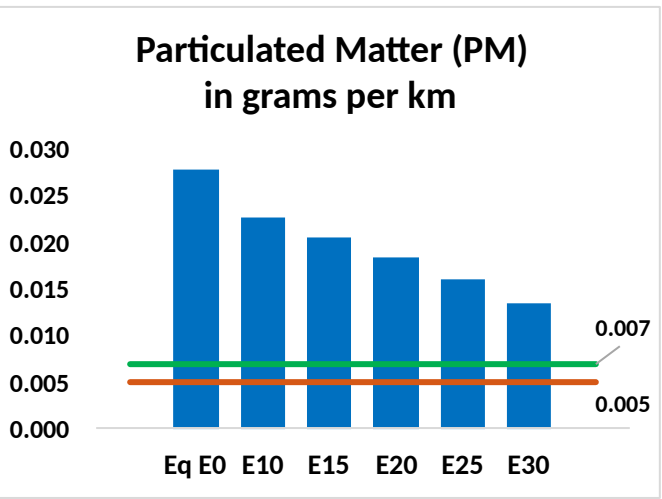
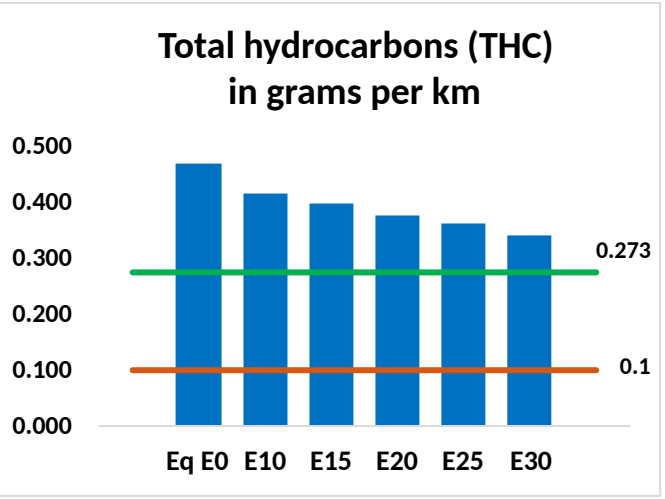
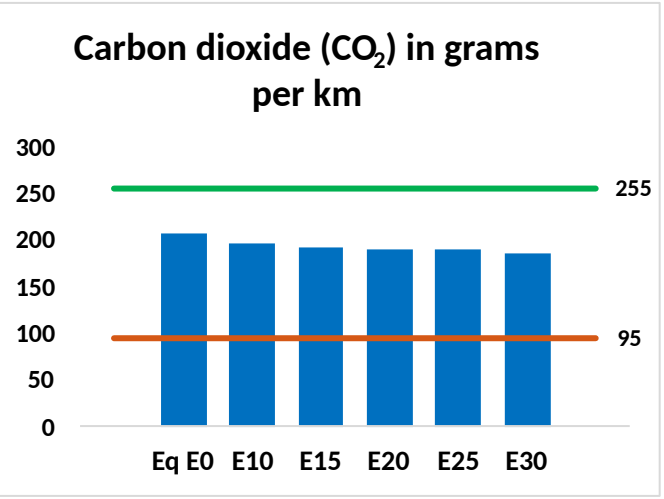
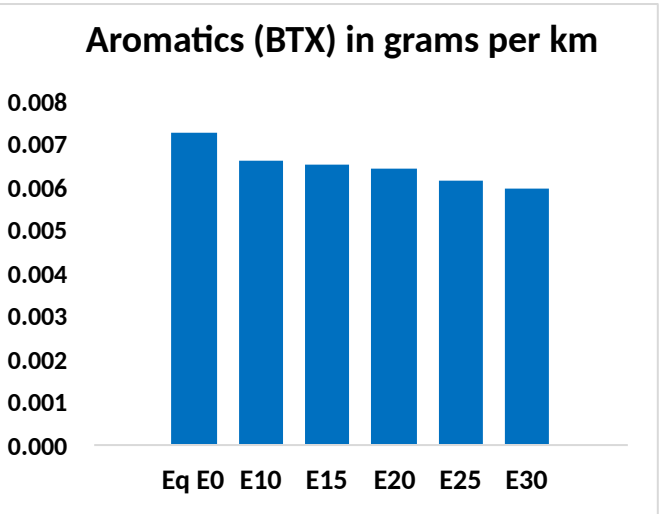
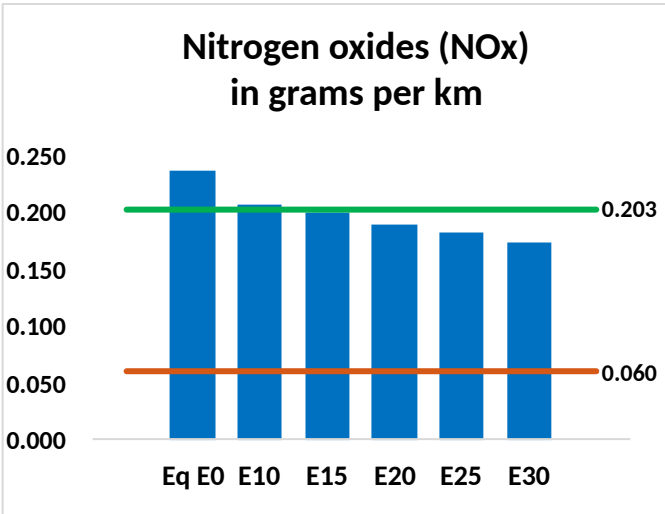
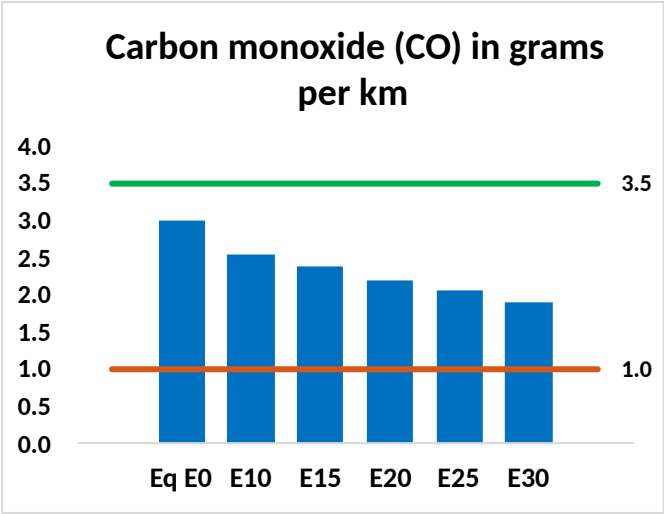
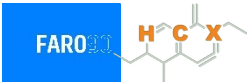


Vehicle Fleet: 966,257
Gasoline Vehicle Fleet: 739,718

Source: Eurostat, ACEA

Cyprus – Vehicular Emissions

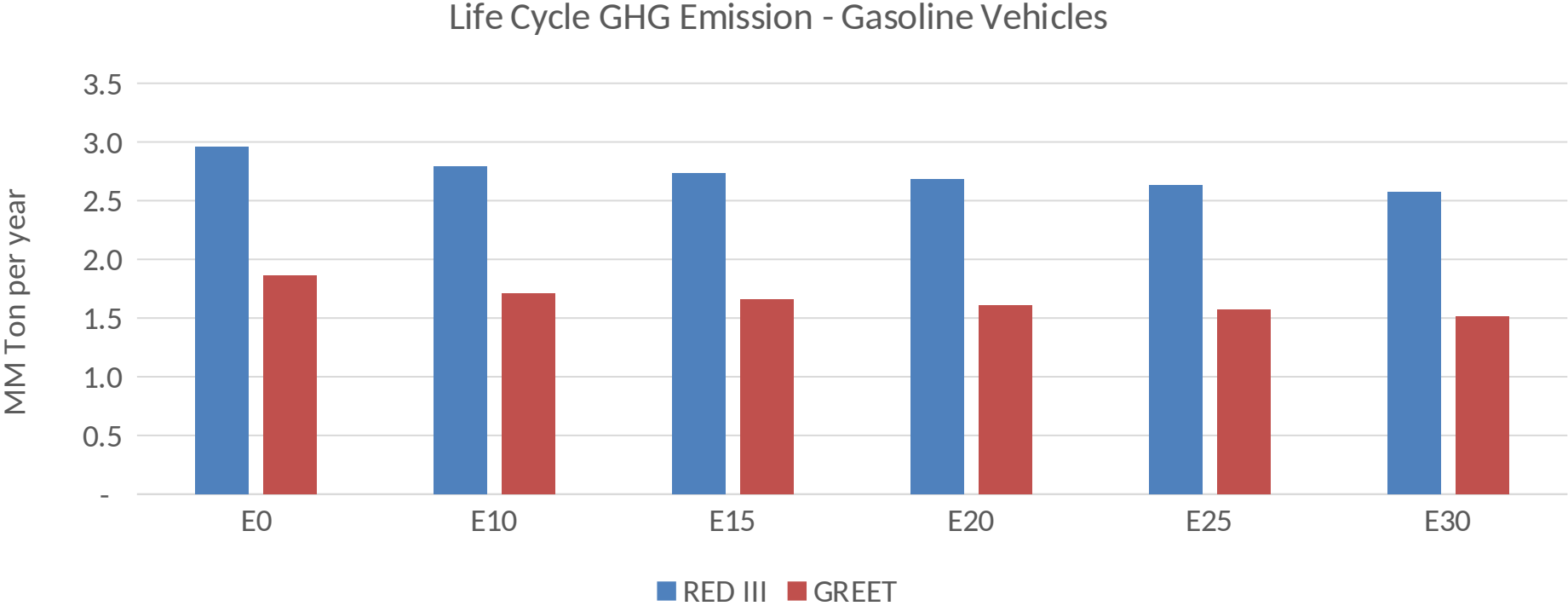
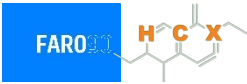
TIER III BIN 125 United States
Euro 6 Standard



Cyprus – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	50,907	47,165	43,424	40,391	37,354	35,024	32,475	-15%	-27%
CO2	3,517,206	3,426,763	3,341,345	3,274,167	3,240,924	3,224,611	3,165,173	-5%	-8%
VOC	8,016	7,556	7,095	6,749	6,414	6,160	5,808	-11%	-20%
NOx	4,059	3,748	3,537	3,430	3,257	3,123	2,964	-13%	-20%
SOx	40	37	34	31	29	26	23	-15%	-28%
NH3	1,259	1,247	1,234	1,240	1,254	1,264	1,272	-2%	0%
N2O	48	48	48	48	49	50	50	-1%	2%
CH4	1,790	1,790	1,790	1,817	1,856	1,890	1,922	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	43	40	38	37	35	34	33	-10%	-17%
Acetaldehyde	114	131	192	293	399	456	539	68%	249%
Formaldehyde	418	407	474	557	583	645	702	13%	39%
Benzene	124	119	113	111	110	105	102	-9%	-11%
PM2.5	210	191	172	155	138	121	102	-18%	-34%
PM10	264	240	216	195	174	152	129	-18%	-34%

Cyprus – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	8.3
Target @ 2035 (MMT/year)	4.7
Delta	3.6

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

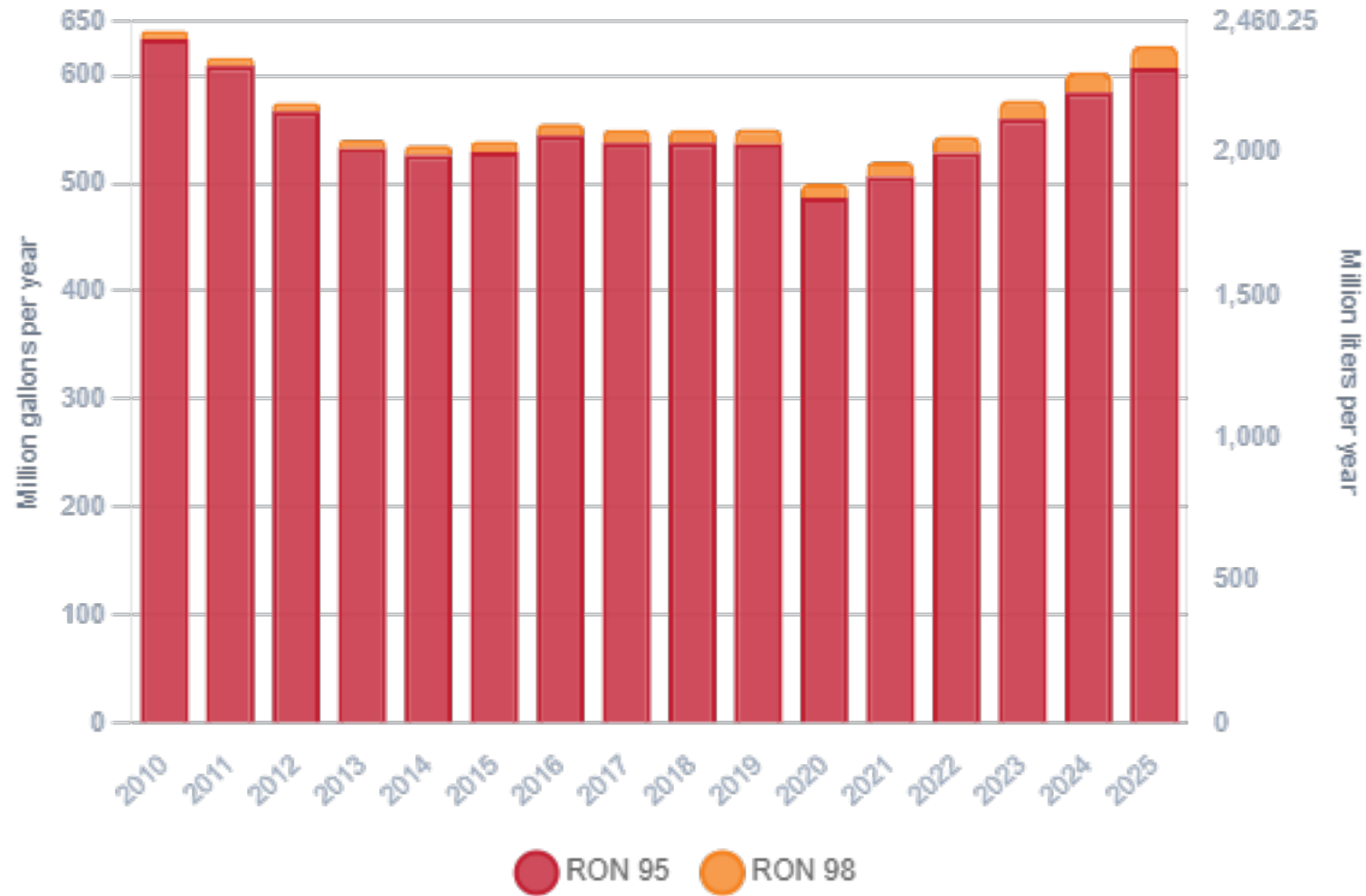
Ethanol Gasoline Blending in Czechia



In 2025, gasoline consumption in Czechia was 2,445 million liters (646 million gallons) and growth rate was -0.19%. Gasoline production and net imports were 2,033 and 214 million liters (537 and 56 million gallons) correspondingly. Czechia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.1 RON.

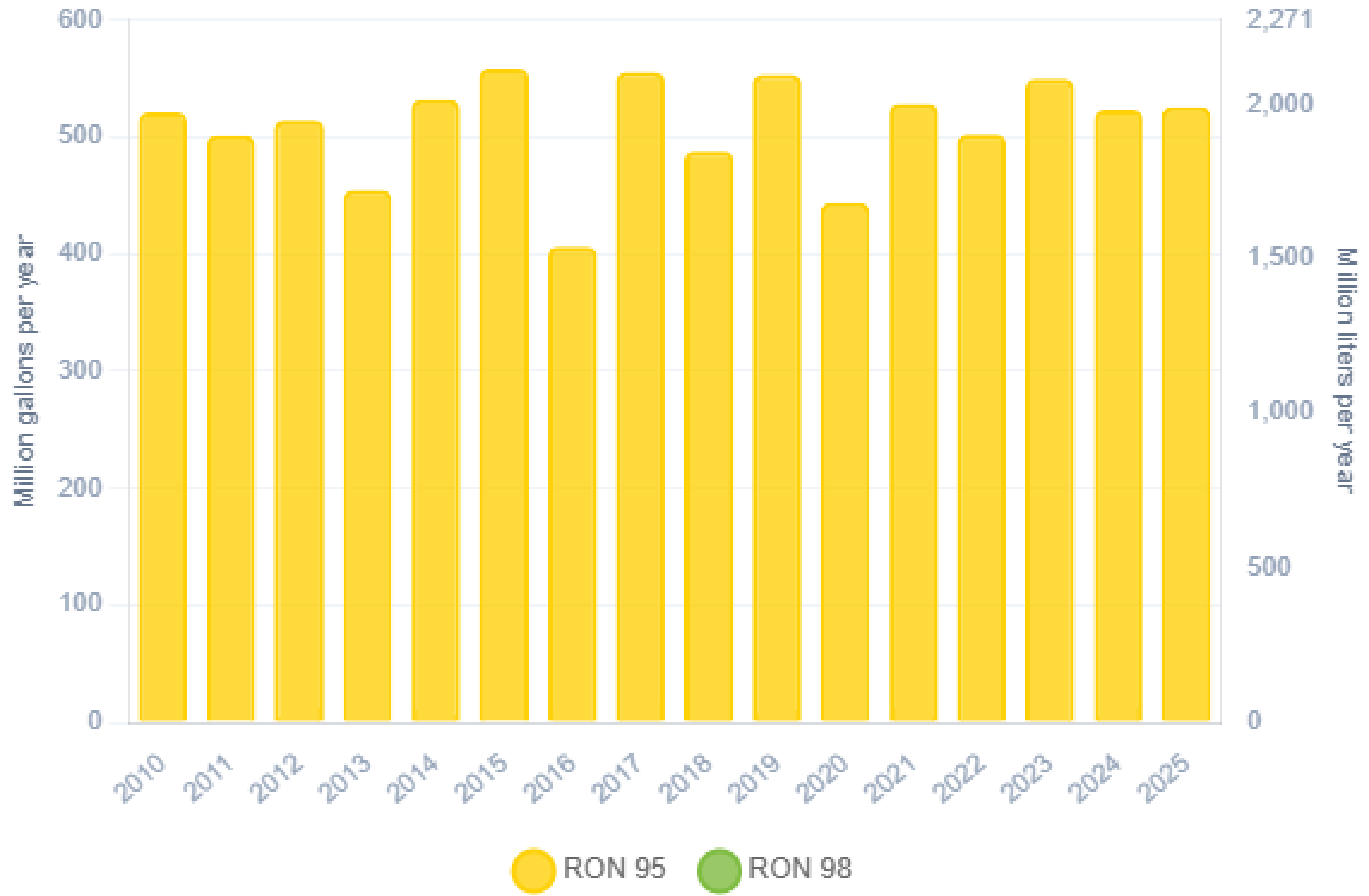
In 2025, ethanol consumption in Czechia was 227 million liters (60 million gallons) representing 10% of gasoline demand. Ethanol production and Net Imports were 47 and 123 million liters (12 and 32 million gallons) correspondingly. Ethanol sources are Corn 50% and Sugar Beet 50%.

Gasoline Demand in Czechia



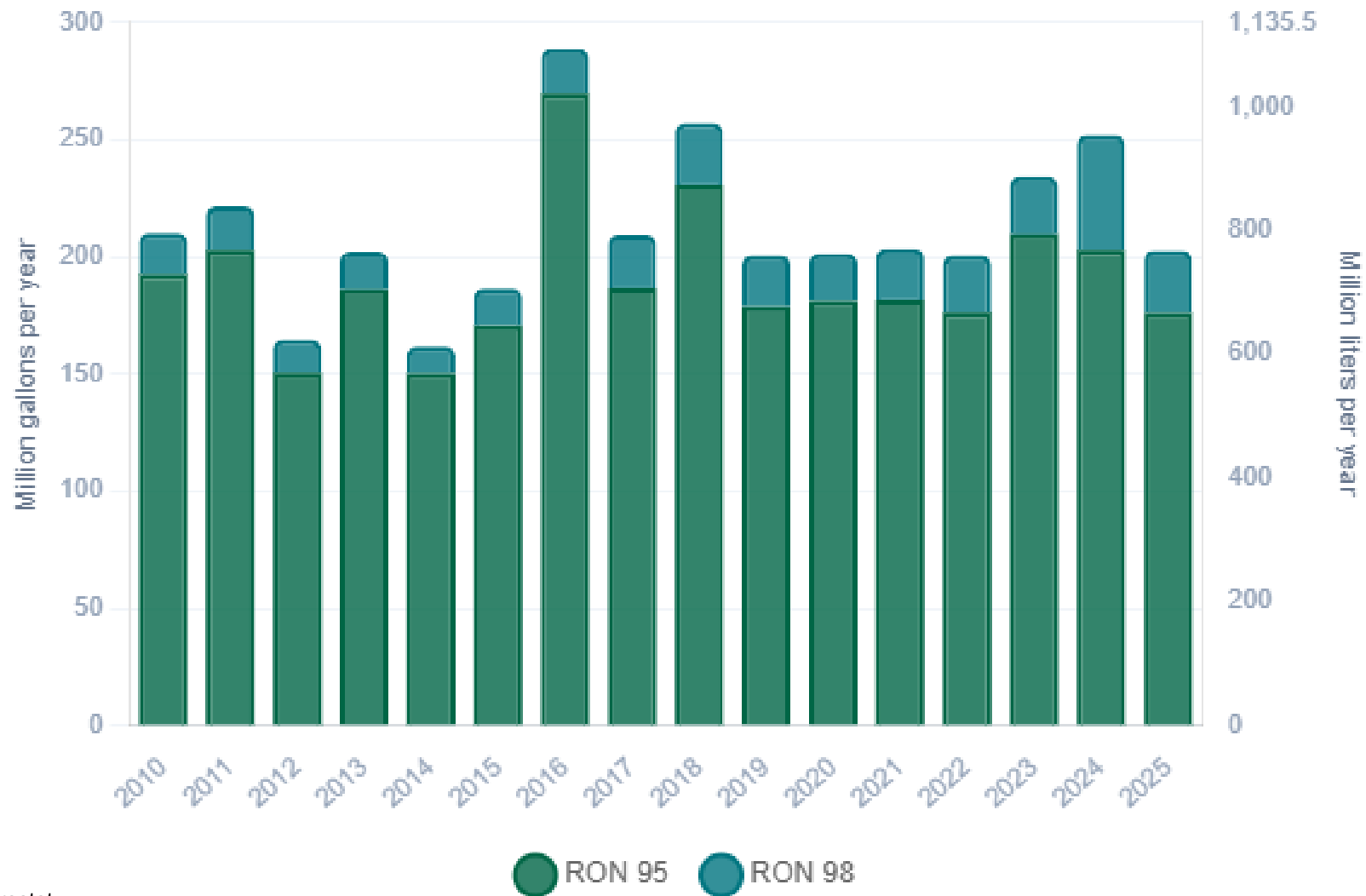
Source: Eurostat

Gasoline Production in Czechia



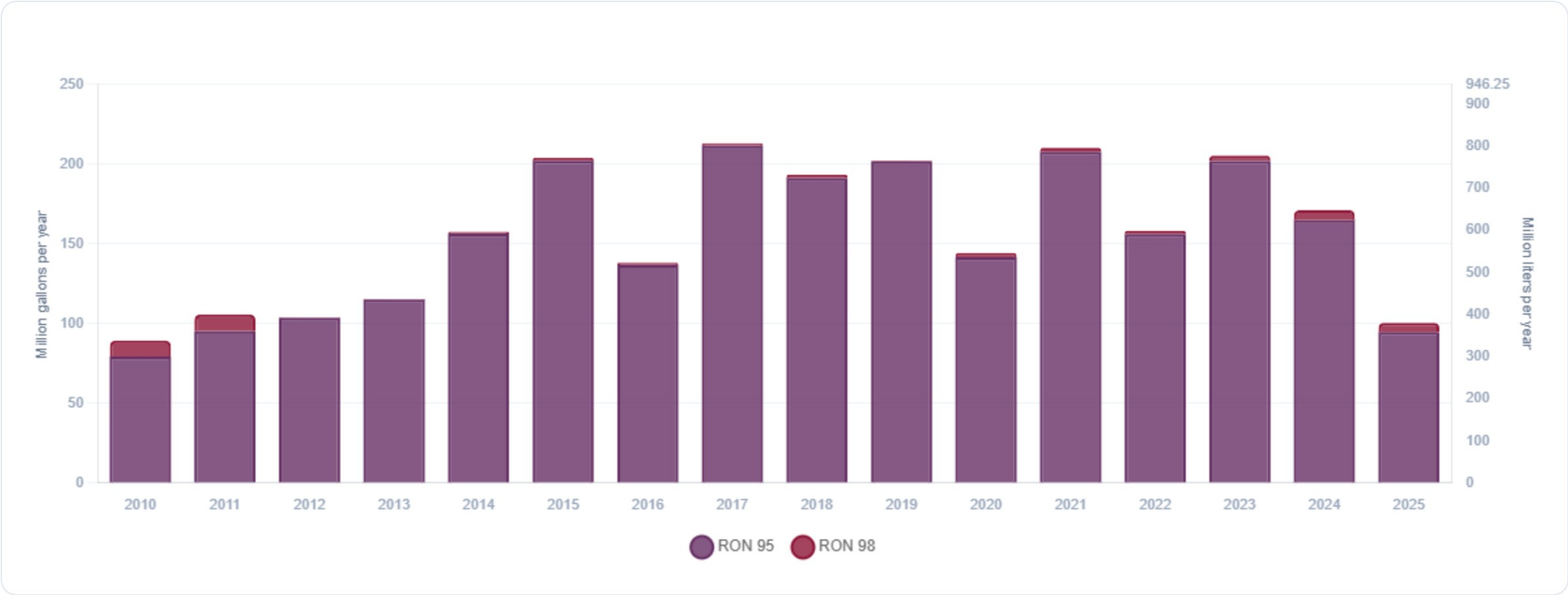
Source: Eurostat

Gasoline Imports in Czechia



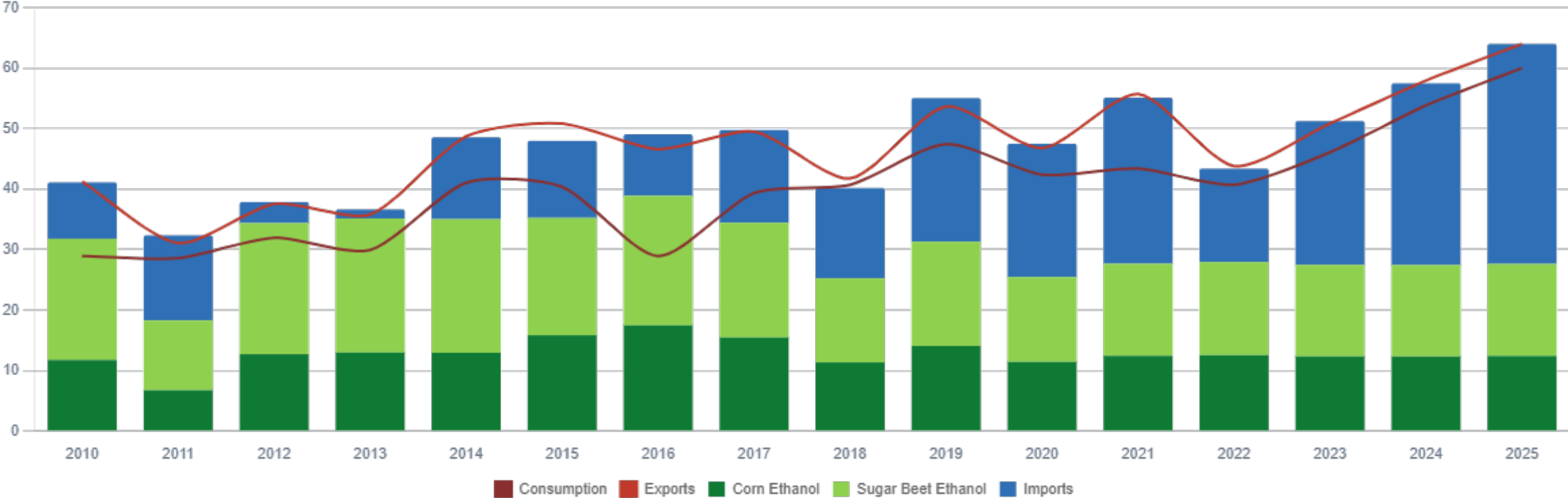
Source: Eurostat

Gasoline Exports in Czechia



Source: Eurostat

Ethanol Balance in Czechia



Source: Eurostat

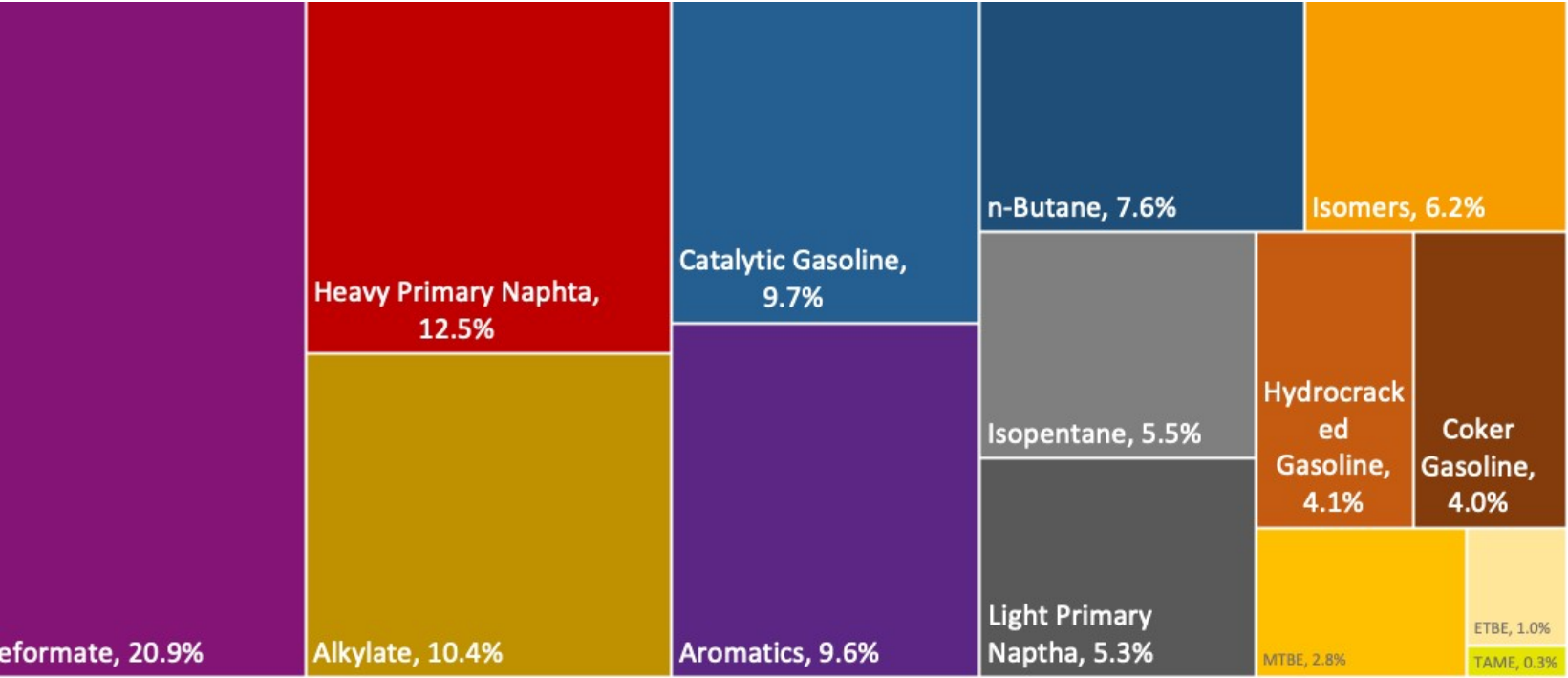
Gasoline Quality in Czechia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.1							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	6.0							

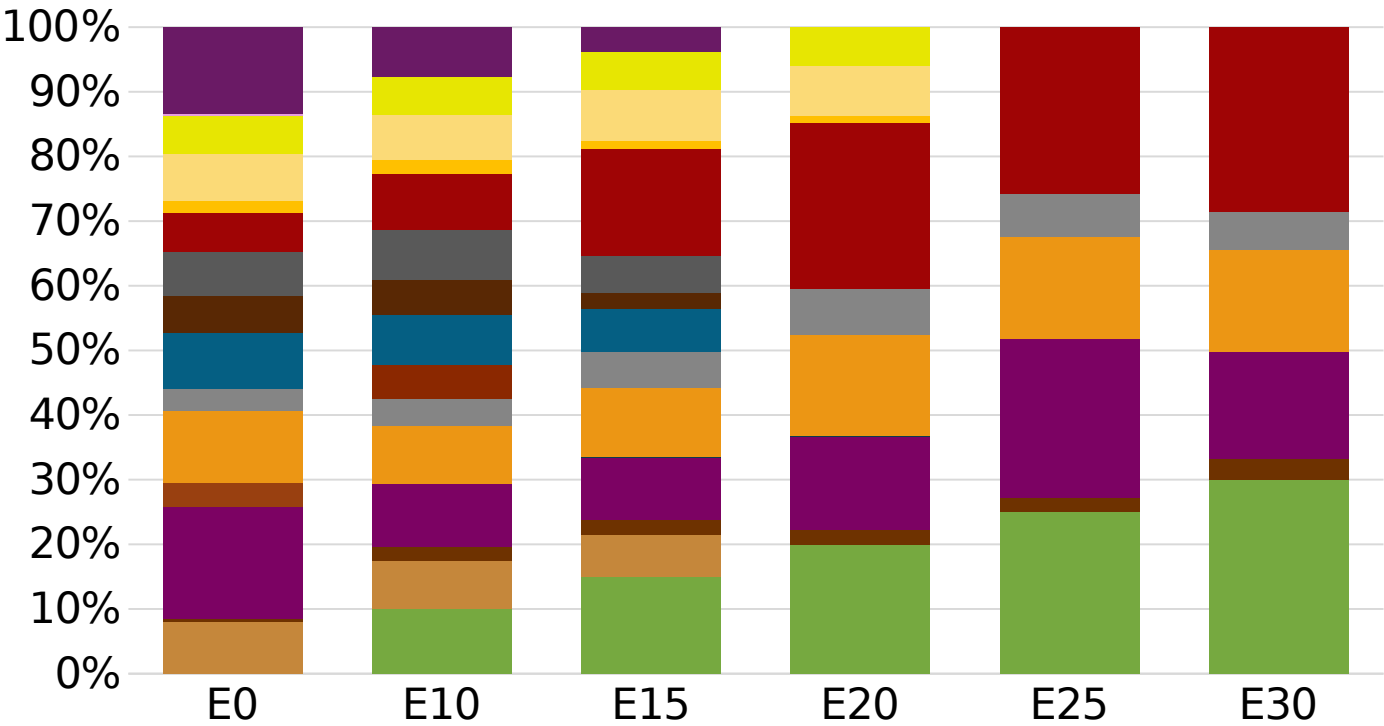
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Czechia Available Blending Components



Czechia – Gasoline 95 – Constant Octane

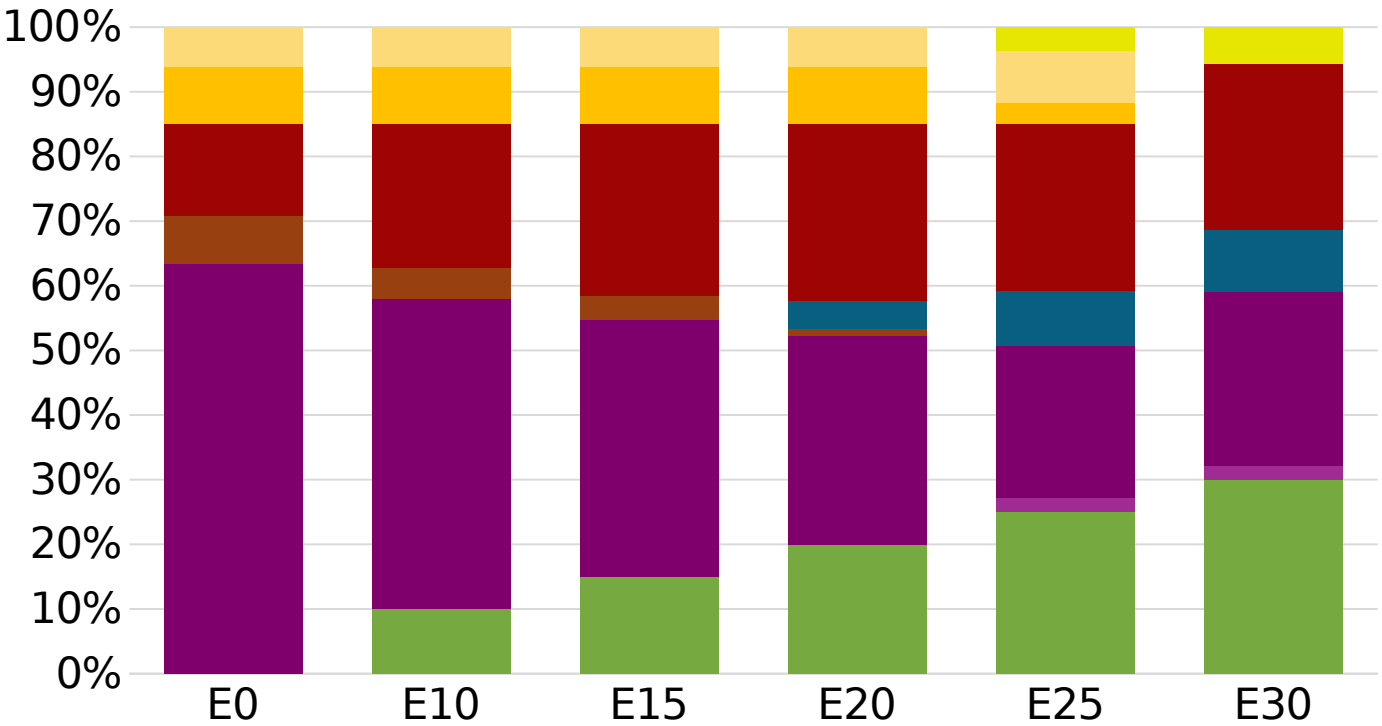
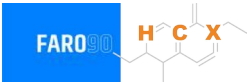


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.111	\$ 1.973	\$ 1.873	\$ 1.758	\$ 1.712	\$ 1.659

Source:
Faro90

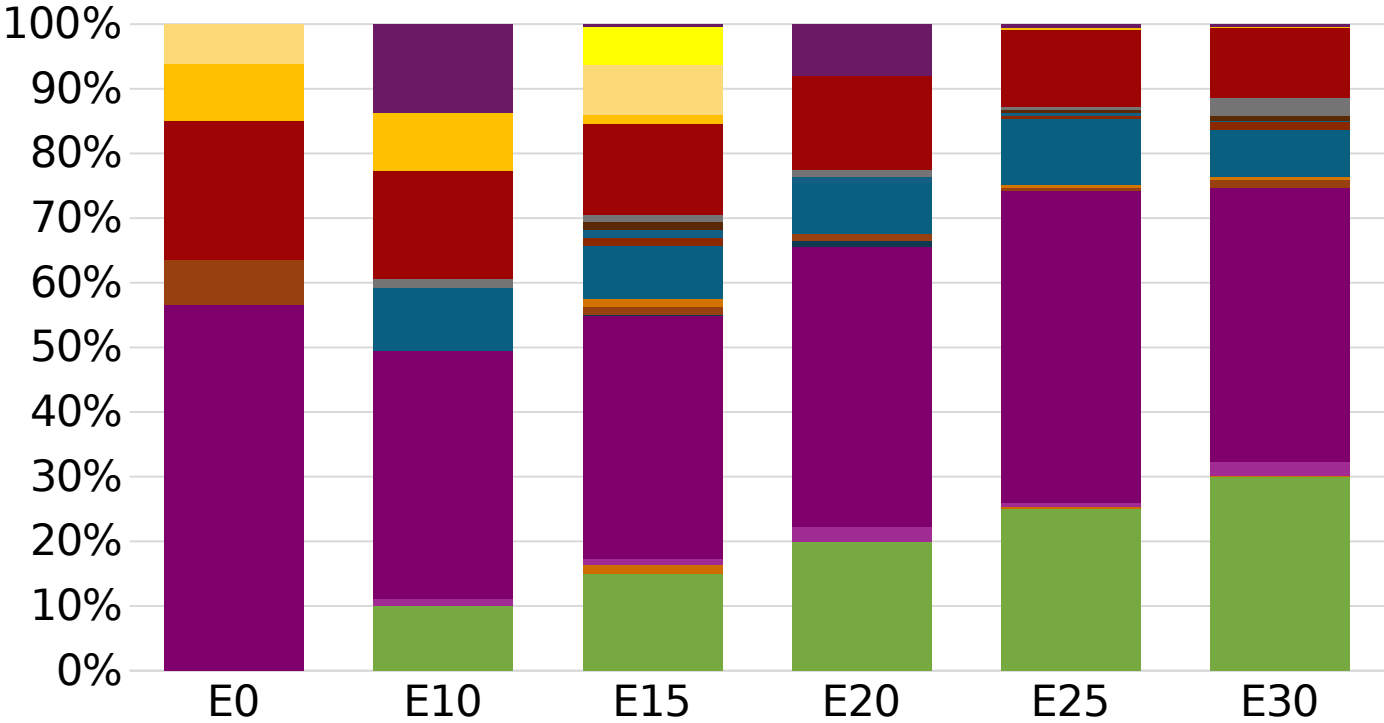
Czechia - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.131	\$ 2.014	\$ 1.945	\$ 1.873	\$ 1.808	\$ 1.758

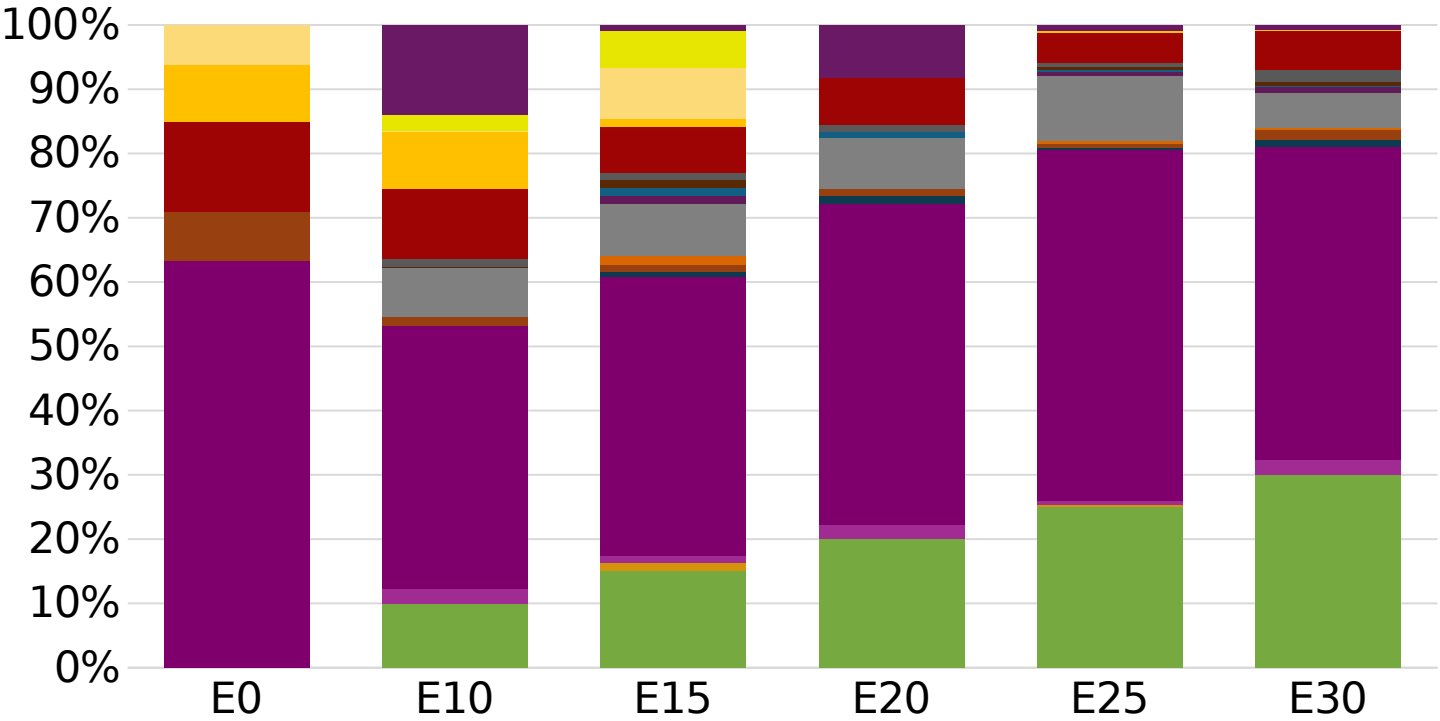
Average CIF 2025
Prices

Czechia – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

Czechia – Gasoline 98 – Increased Octane

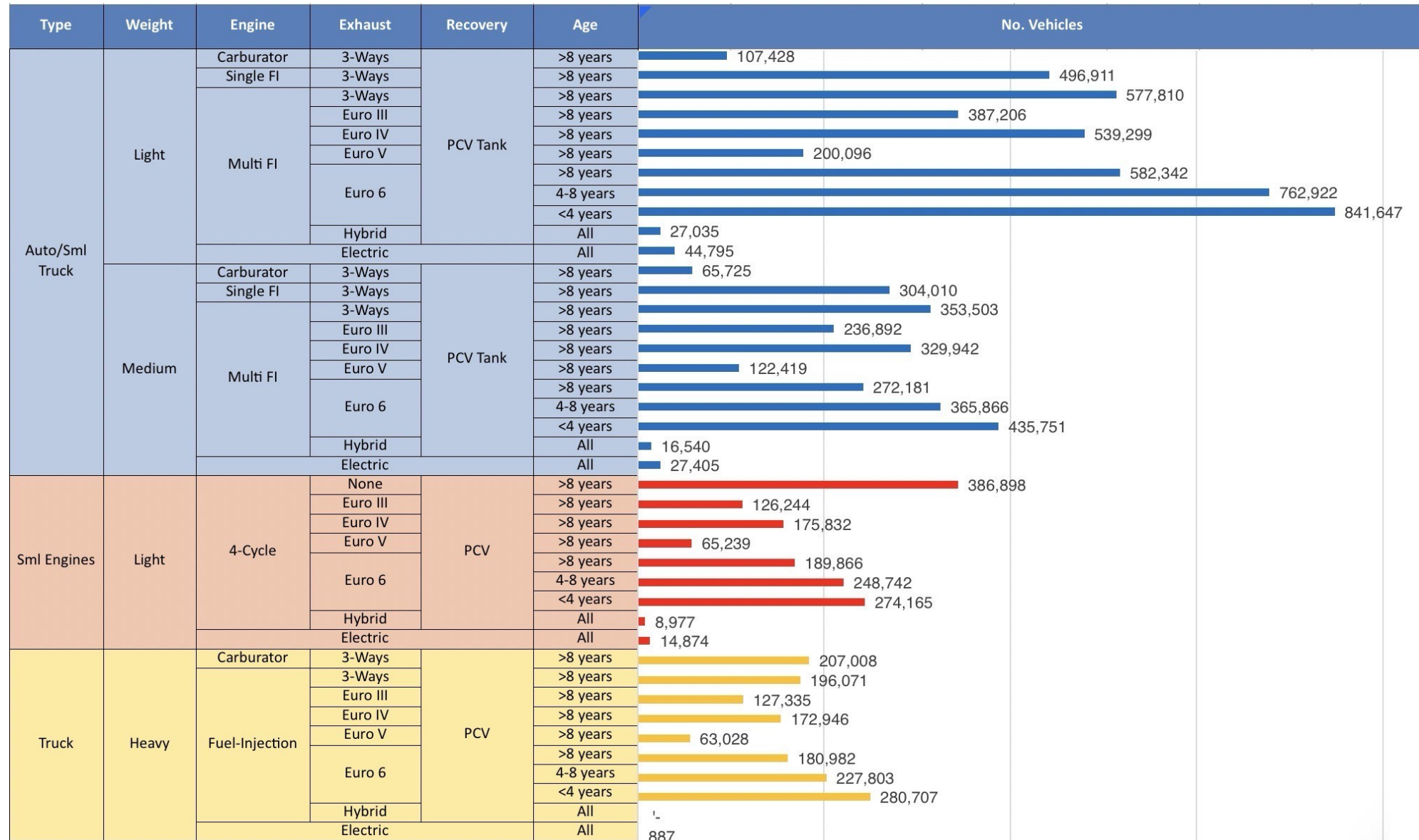


Average CIF 2025
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.131	\$ 2.131	\$ 2.131	\$ 2.131	\$ 2.131	\$ 2.131

Source:
Faro90

Gasoline Vehicle Fleet - Czechia

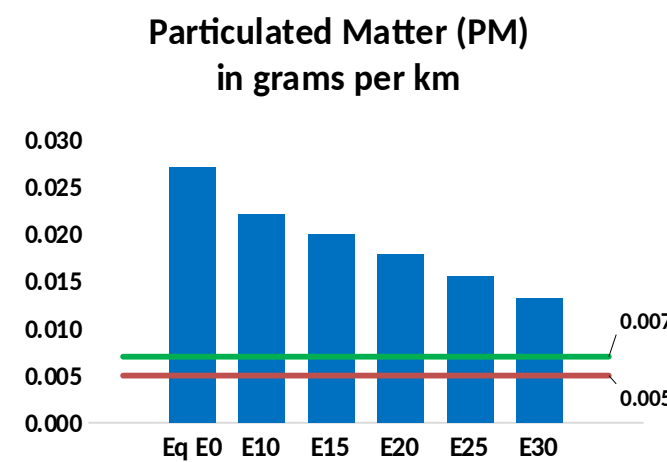
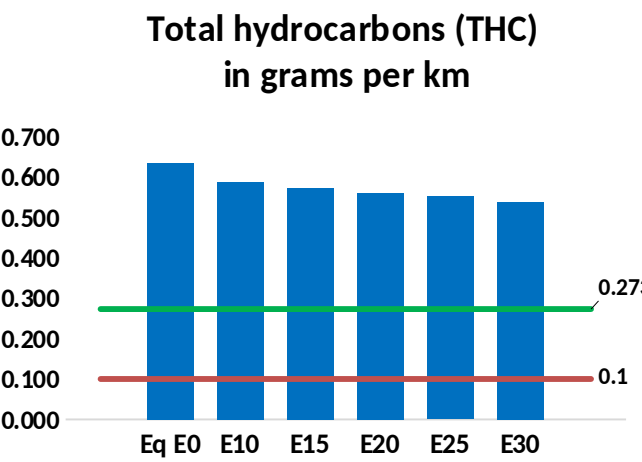
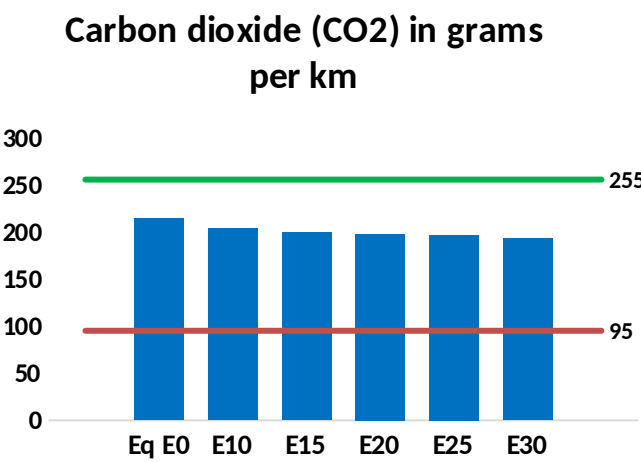
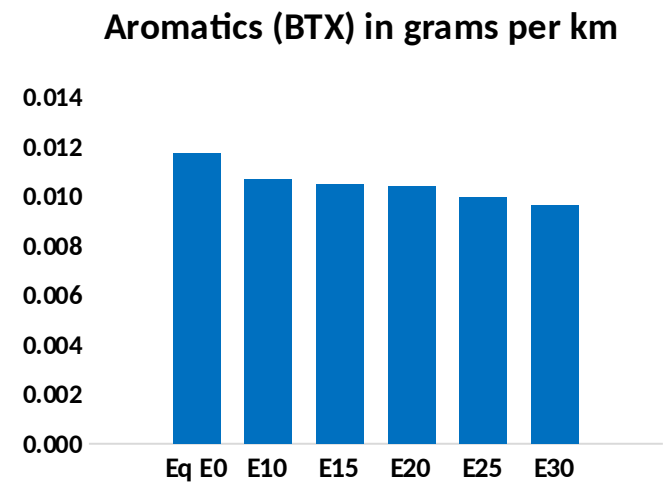
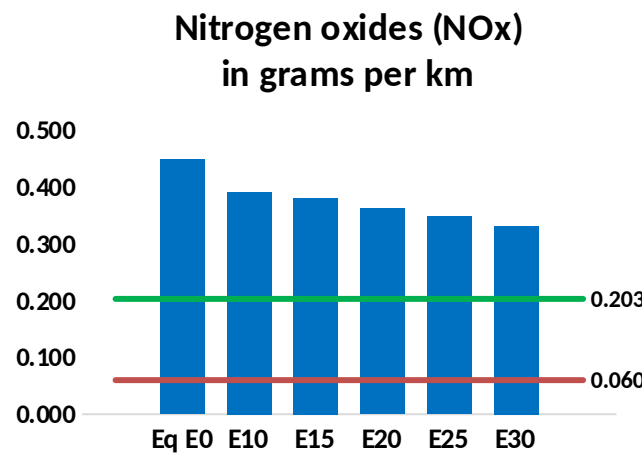
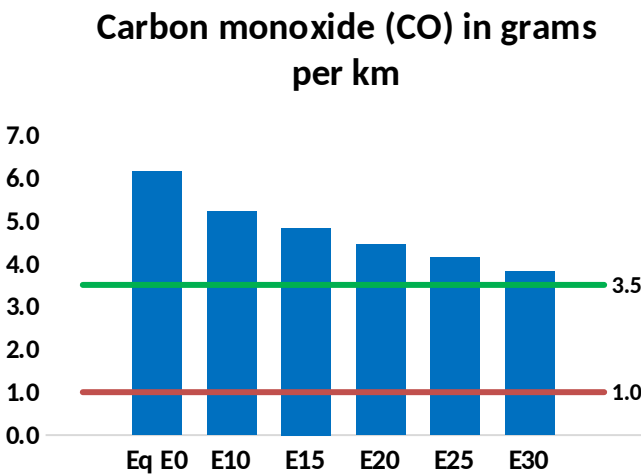
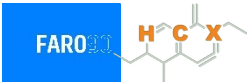


Vehicle Fleet: 10,045,330
Gasoline Vehicle Fleet: 3,692,443

Source: Eurostat, ACEA

Czechia – Vehicular Emissions

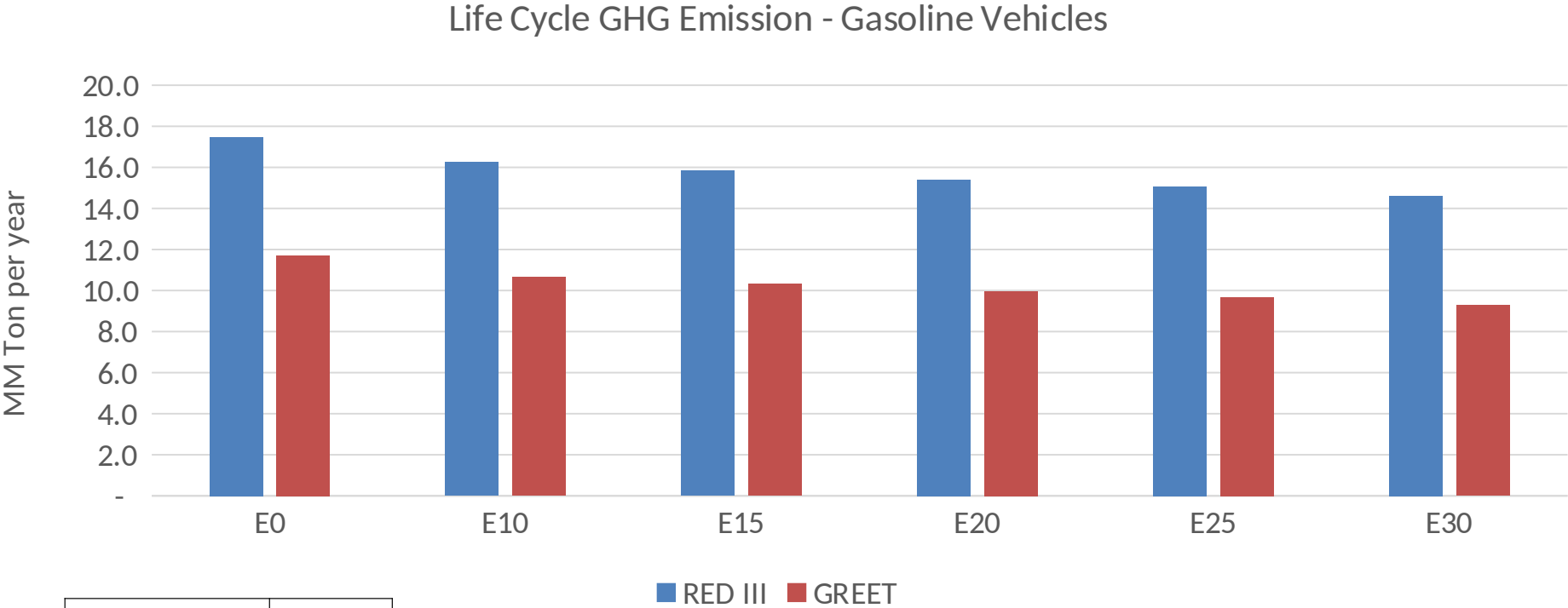
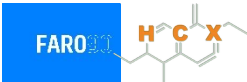
TIER III BIN 125 United States
Euro 6 Standard



Czechia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	521,318	481,576	441,834	409,332	376,655	351,537	322,496	-15%	-28%
CO2	18,149,222	17,682,528	17,241,761	16,895,111	16,723,575	16,642,885	16,336,109	-5%	-8%
VOC	53,728	51,740	49,753	48,543	47,517	46,786	45,607	-7%	-12%
NOx	38,011	35,097	33,189	32,265	30,723	29,538	28,129	-13%	-19%
SOx	208	193	177	164	151	137	122	-15%	-28%
NH3	6,238	6,176	6,115	6,144	6,213	6,261	6,302	-2%	0%
N2O	317	315	314	316	322	326	329	-1%	2%
CH4	11,854	11,854	11,854	12,032	12,292	12,518	12,731	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	362	344	326	313	301	292	279	-10%	-17%
Acetaldehyde	765	878	1,288	1,961	2,672	3,053	3,608	68%	249%
Formaldehyde	2,499	2,429	2,832	3,325	3,484	3,854	4,195	13%	39%
Benzene	994	955	905	891	884	845	818	-9%	-11%
PM2.5	1,040	945	851	768	686	599	508	-18%	-34%
PM10	1,254	1,140	1,026	926	828	722	612	-18%	-34%

Czechia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	108.0
Target @ 2035 (MMT/year)	60.8
Delta	47.2

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

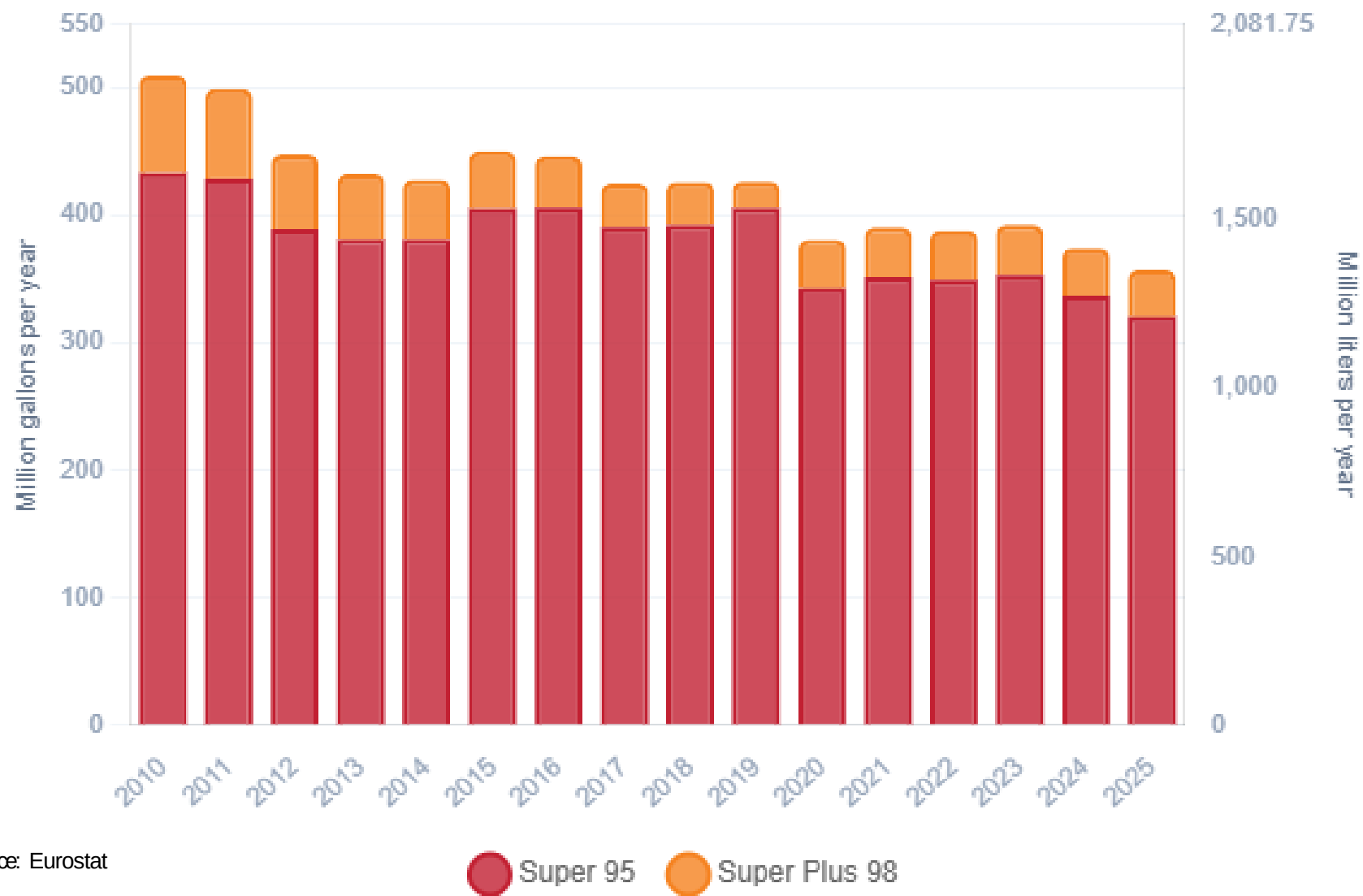
Ethanol Gasoline Blending in Denmark



In 2025, gasoline consumption in Denmark was 1,576 million liters (416 million gallons) and growth rate was -1.3%. Gasoline production and net imports were 2,633 and -1,062 million liters (696 and -280 million gallons) correspondingly. Denmark complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.3 RON.

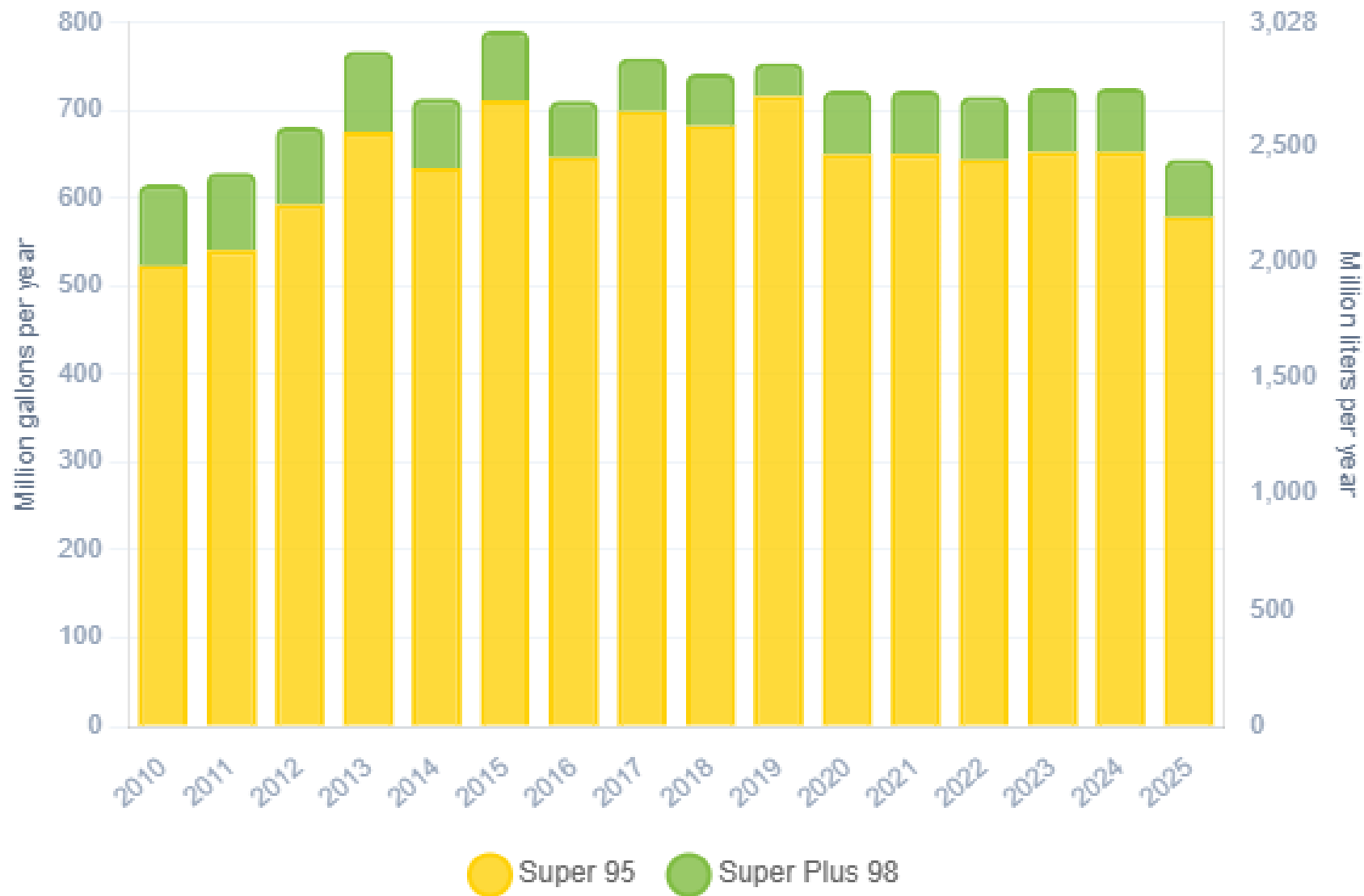
In 2025, ethanol consumption in Denmark was 242 million liters (64 million gallons) representing 18% of gasoline demand. Ethanol production and Net Imports were 206 and 270 million liters (54 and 72 million gallons) correspondingly.

Gasoline Demand in Denmark



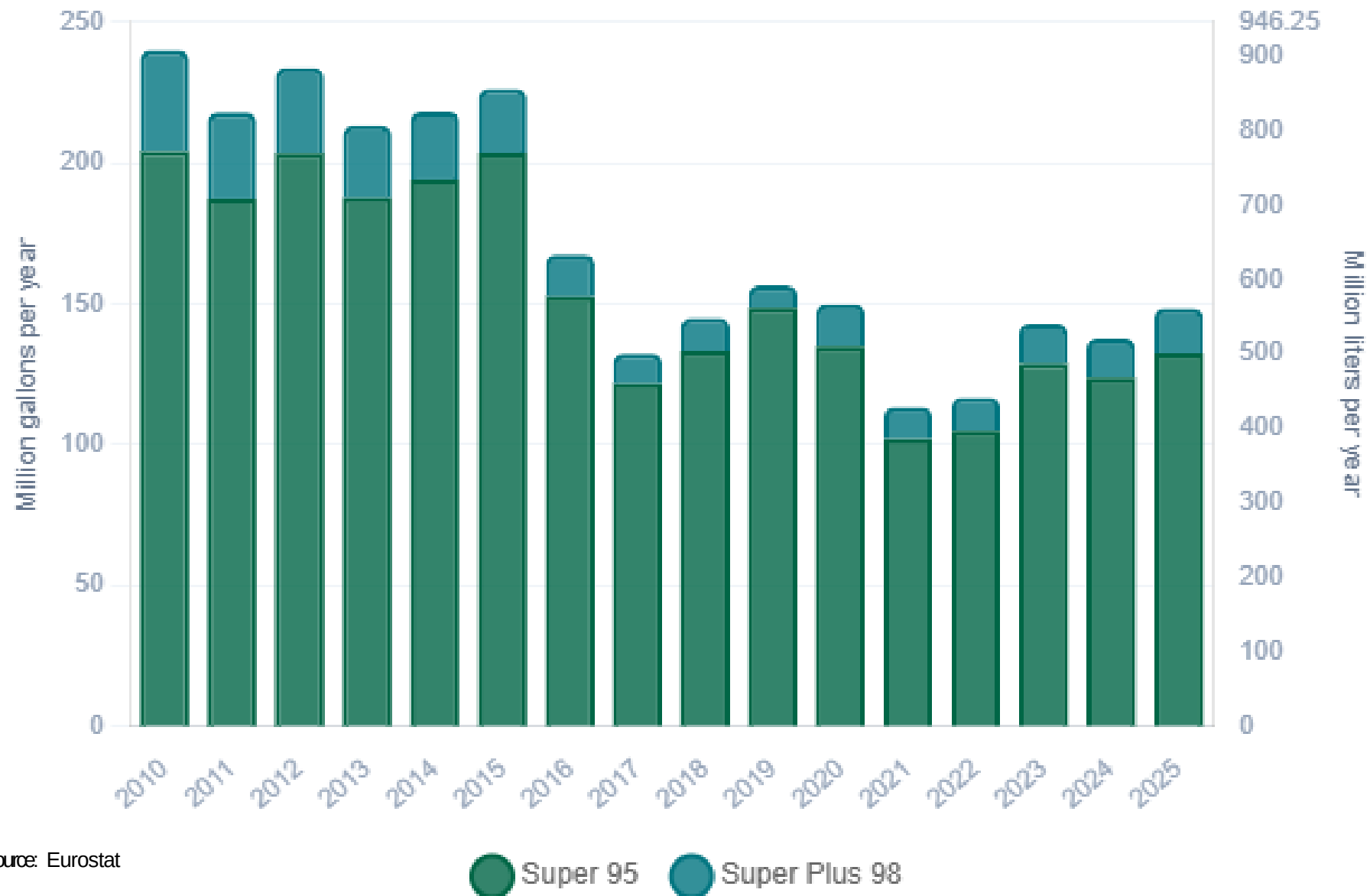
Source: Eurostat

Gasoline Production in Denmark



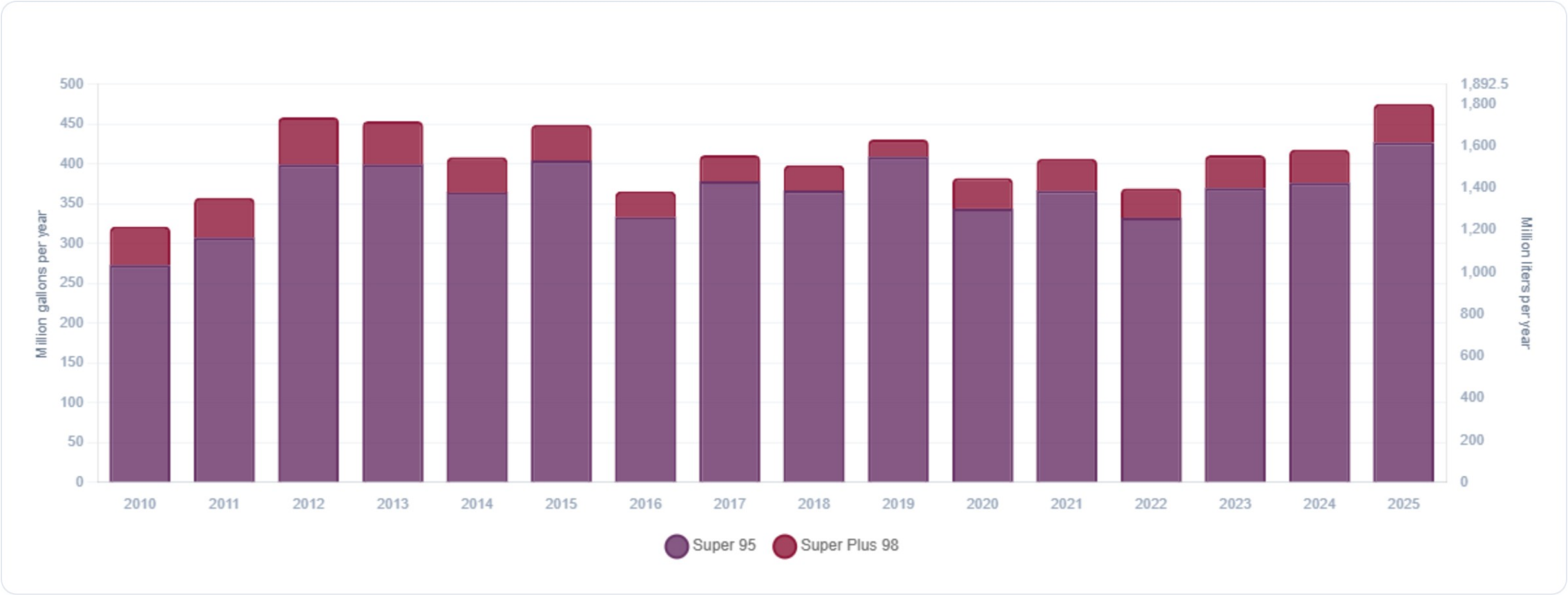
Source: Eurostat

Gasoline Imports in Denmark



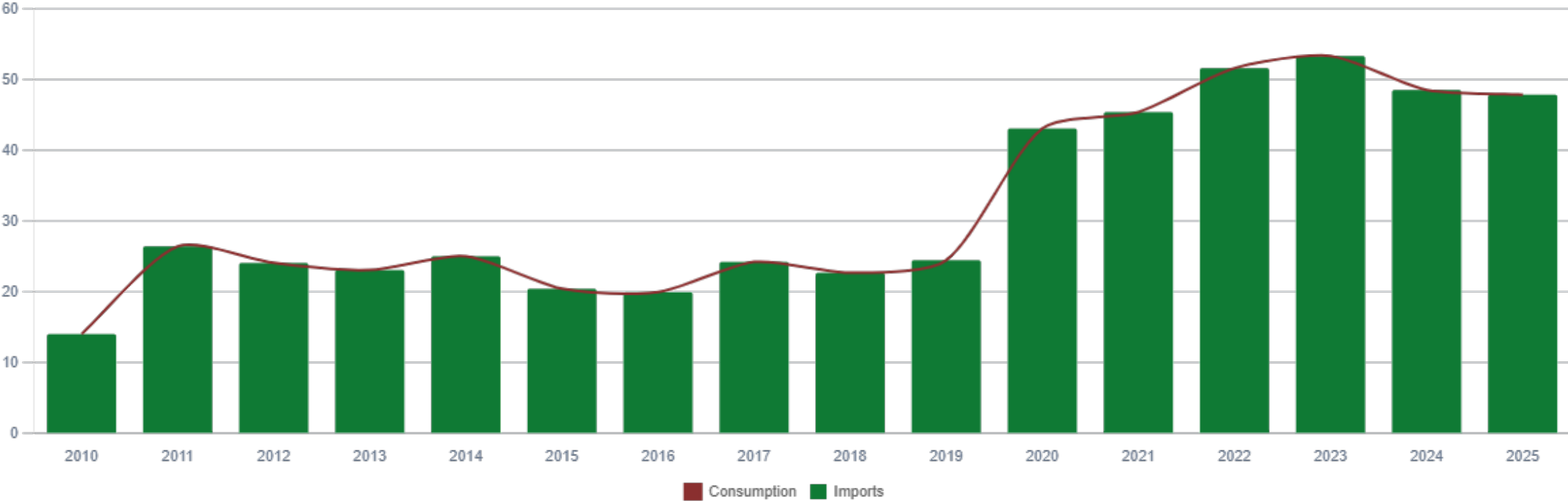
Source: Eurostat

Gasoline Exports in Denmark



Source: Eurostat

Ethanol Balance in Denmark



Source: Eurostat

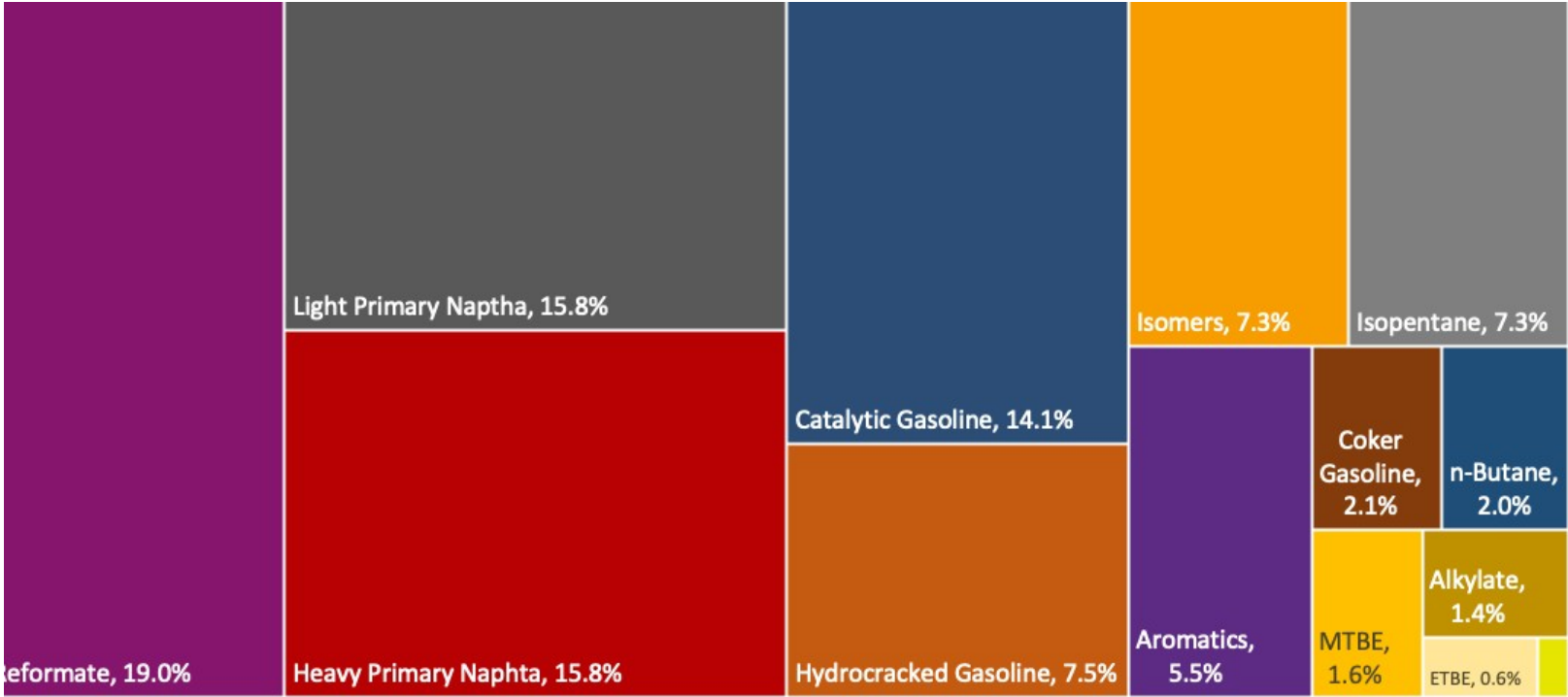
Gasoline Quality in Denmark

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	5.2							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

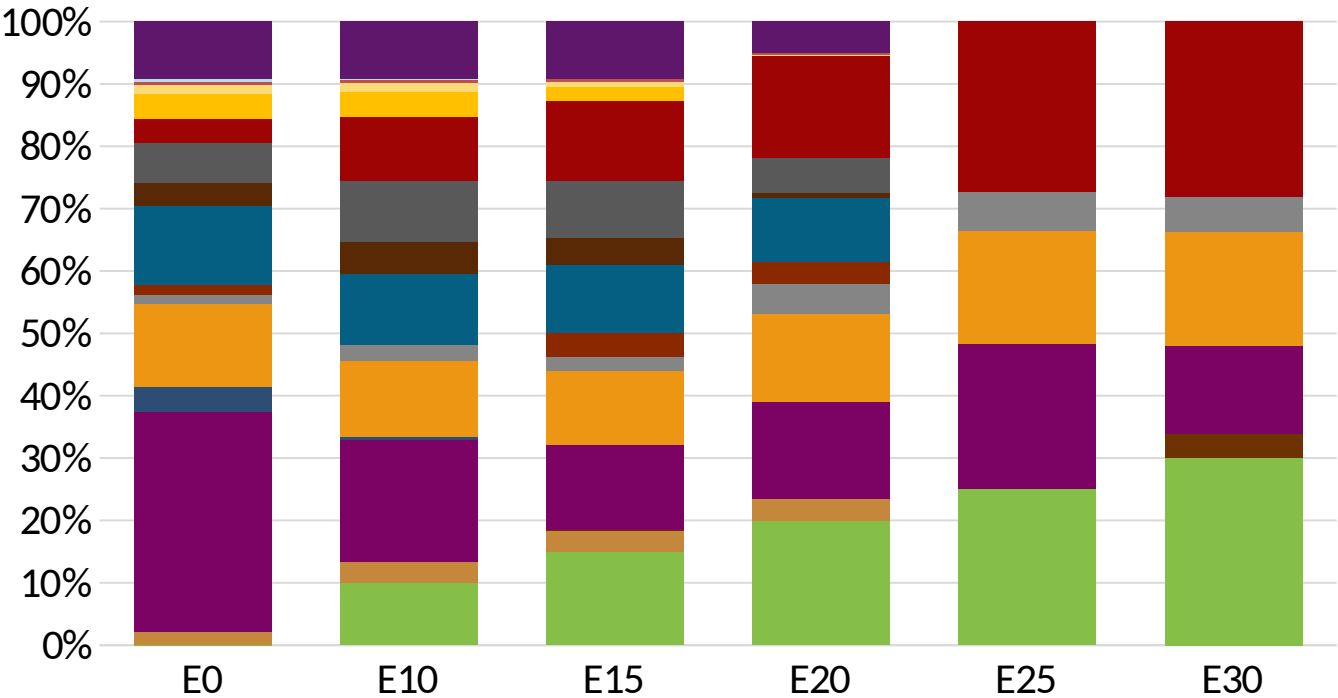
Source: European Commision

Denmark Available Blending Components



TAME,
0.2%

Denmark – Gasoline 95 – Constant Octane

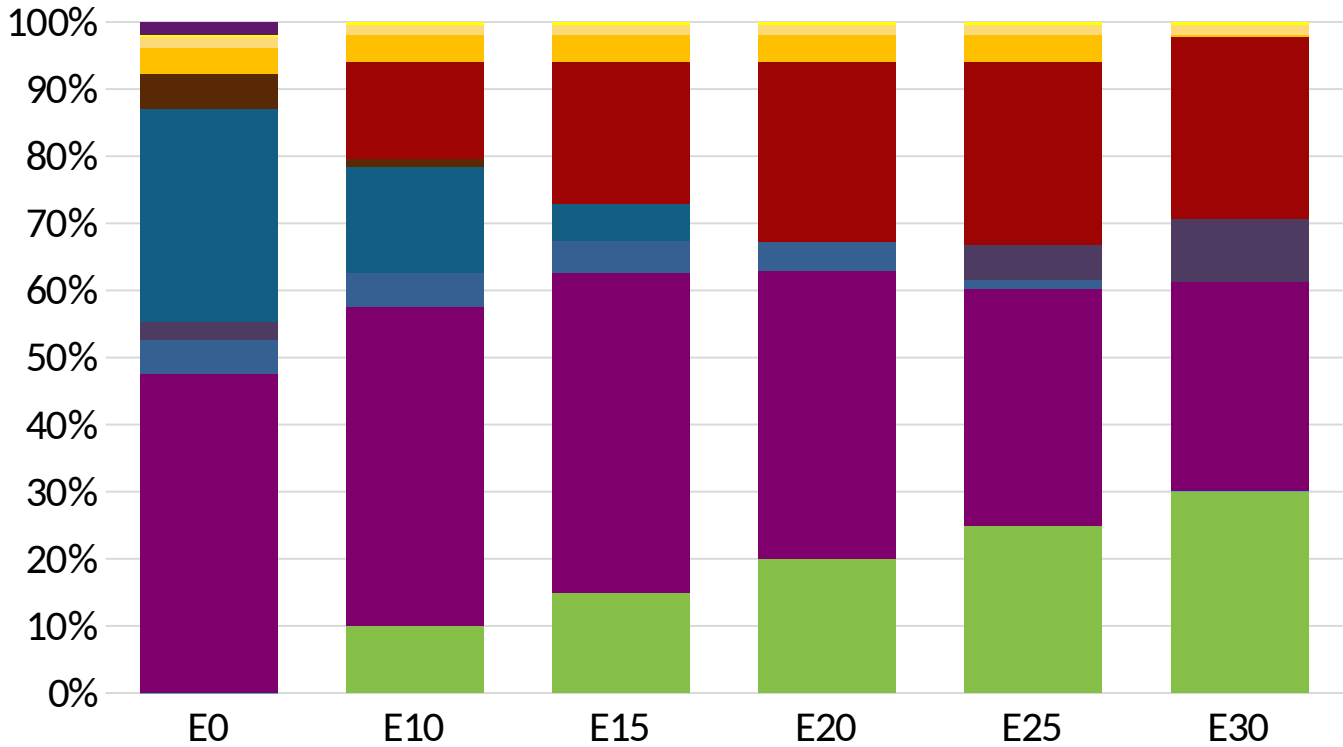


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.069	\$ 1.941	\$ 1.880	\$ 1.799	\$ 1.677	\$ 1.612

Source:
Faro90

Denmark - Gasoline 98 - Constant Octane

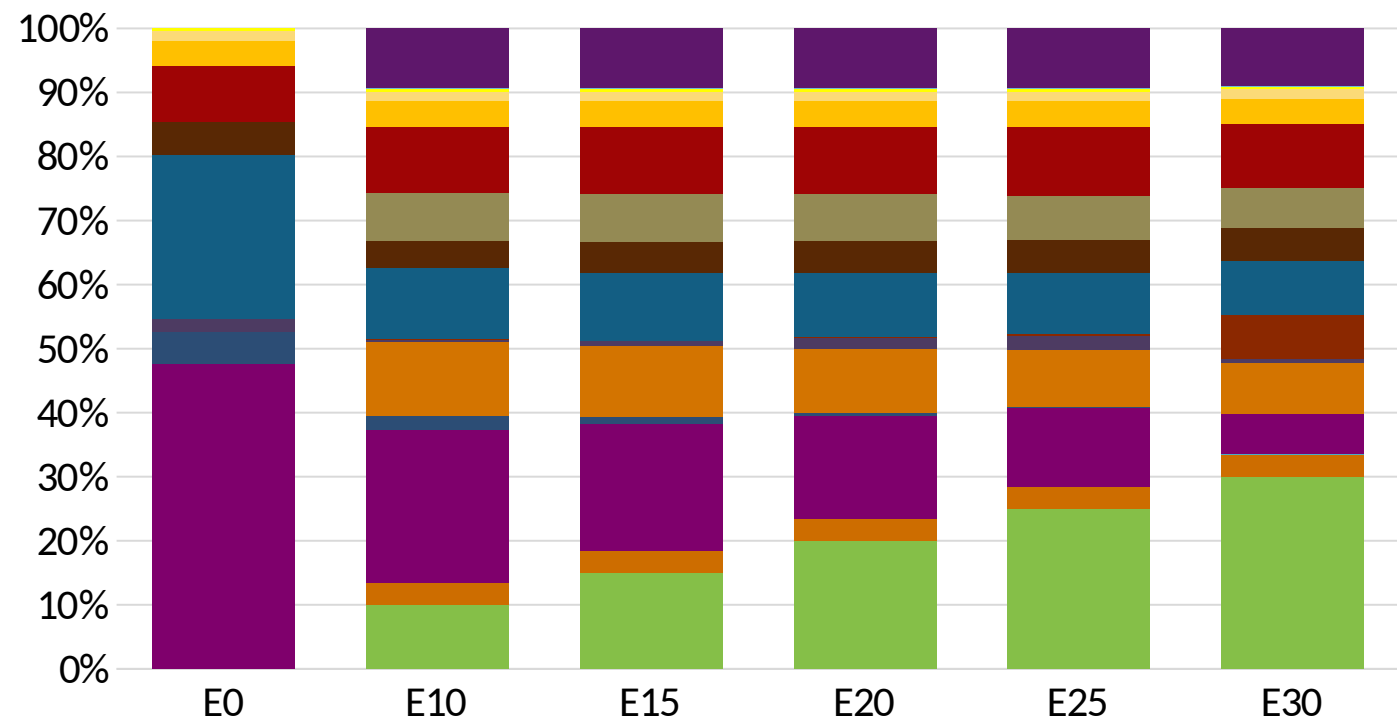
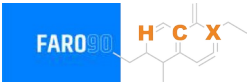


Average CIF 2025
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.144	\$ 2.015	\$ 1.943	\$ 1.866	\$ 1.789	\$ 1.722

Source:
Faro90

Denmark – Gasoline 95 – Increased Octane

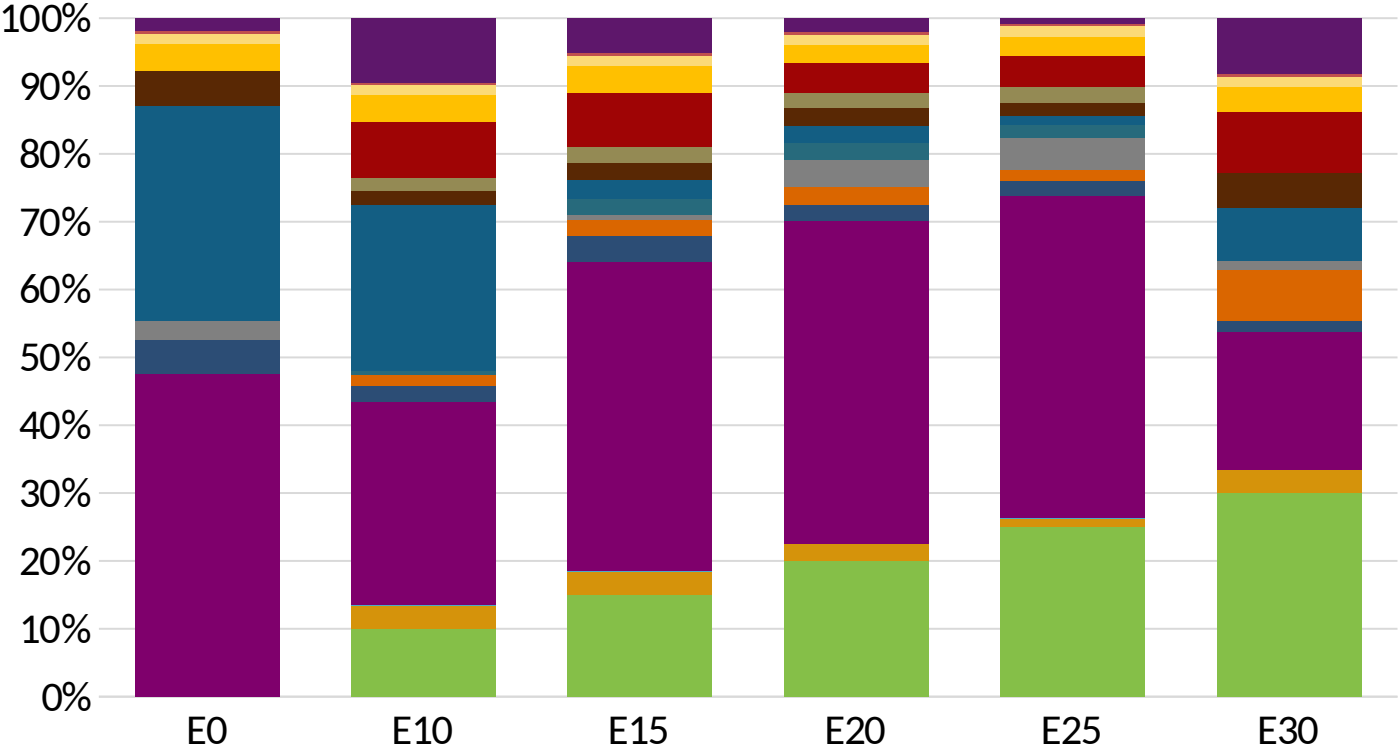


Average CIF 2025
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.006	\$ 2.006	\$ 2.006	\$ 2.006	\$ 2.006	\$ 2.006

Source:
Faro90

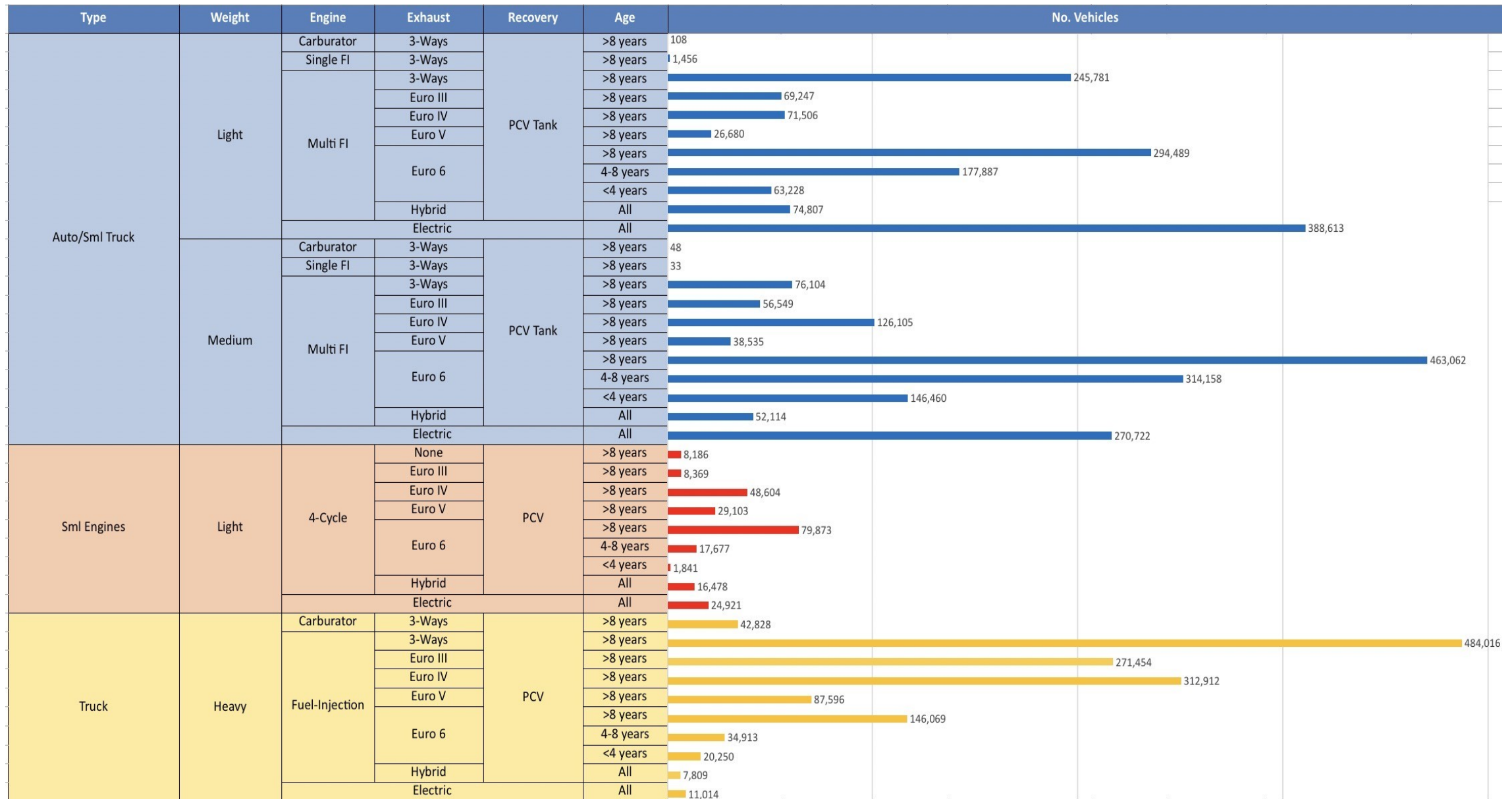
Denmark – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Source:
Faro90

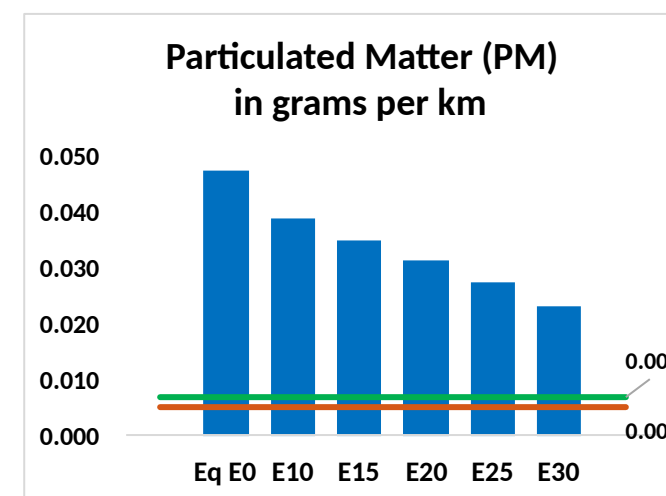
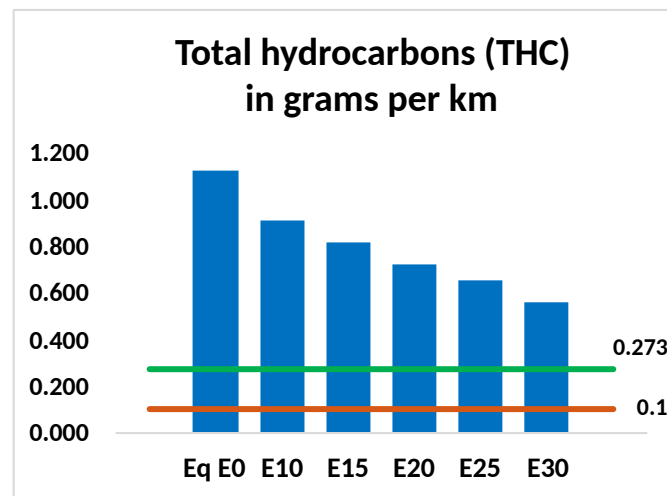
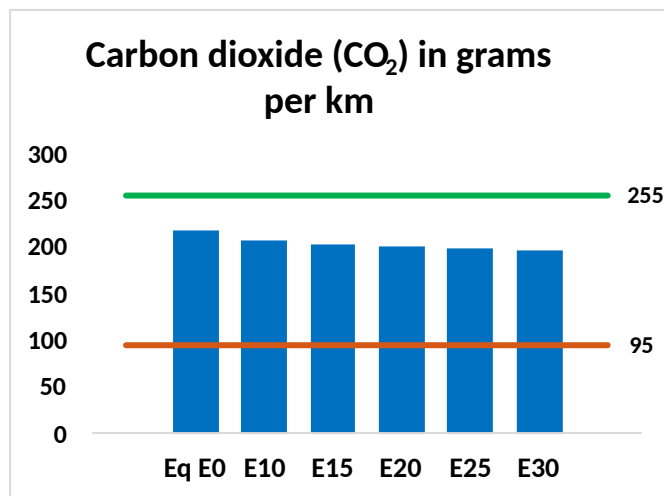
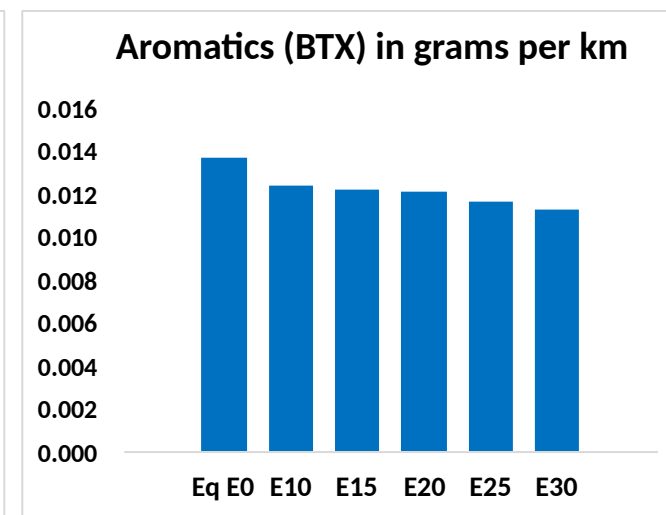
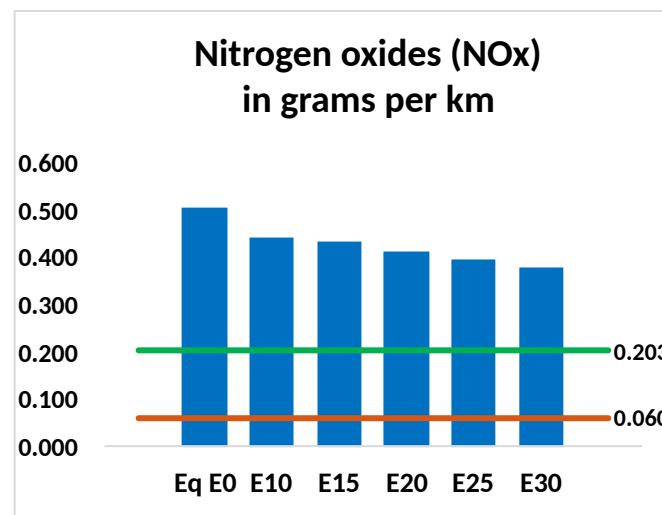
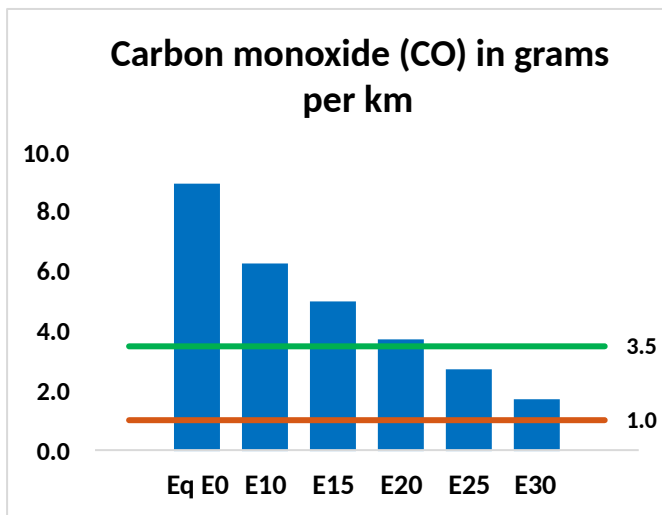
Gasoline Vehicle Fleet - Denmark



Vehicle Fleet: 4,611,606
Gasoline Vehicle Fleet: 2,006,235

Source: Eurostat, ACEA

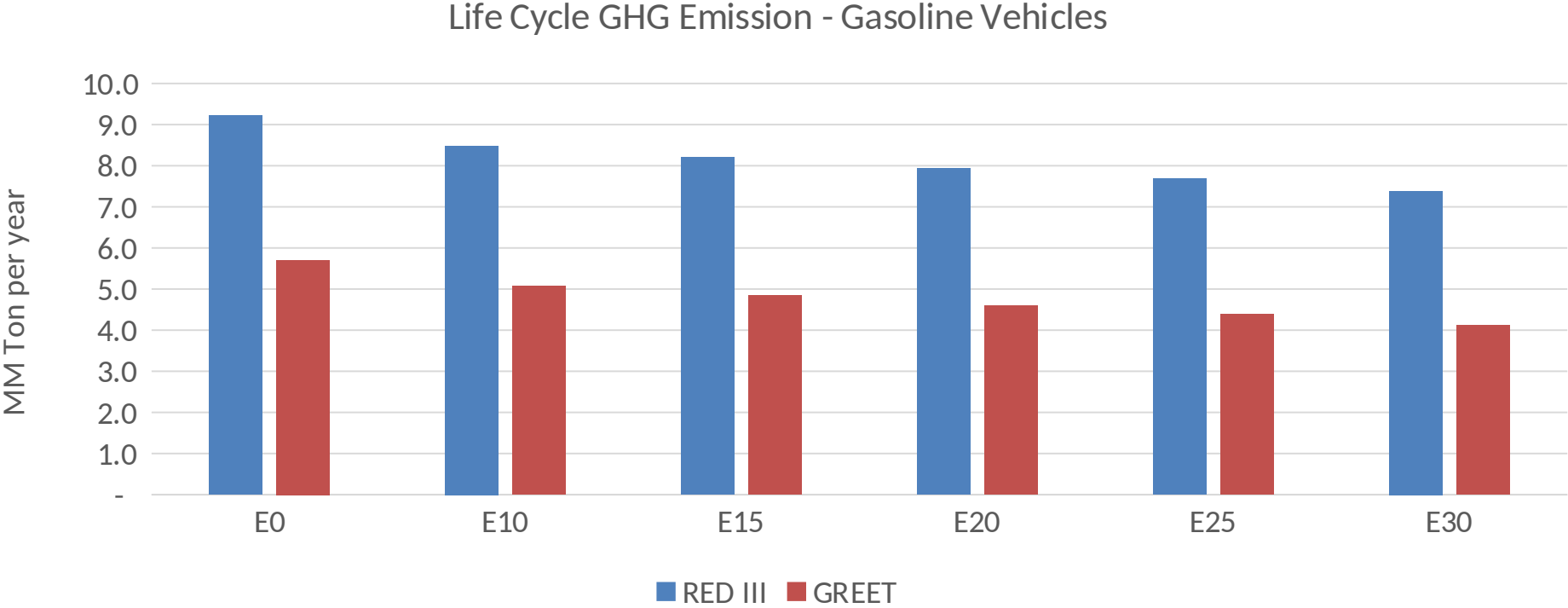
Denmark – Vehicular Emissions



Denmark – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	465,396	395,228	325,060	260,850	193,145	140,152	88,368	-30%	-58%
CO2	11,263,940	10,974,295	10,700,743	10,485,601	10,379,141	10,320,268	10,130,036	-5%	-8%
VOC	58,468	52,875	47,282	42,485	37,559	33,741	28,752	-19%	-36%
NOx	26,309	24,292	23,026	22,443	21,435	20,676	19,763	-12%	-19%
SOx	132	122	112	104	96	87	78	-15%	-28%
NH3	4,000	3,960	3,921	3,939	3,984	4,015	4,041	-2%	0%
N2O	262	260	259	260	266	269	272	-1%	2%
CH4	11,744	11,744	11,744	11,920	12,178	12,401	12,613	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	373	341	310	283	256	236	208	-17%	-31%
Acetaldehyde	978	1,122	1,647	2,508	3,417	3,904	4,613	68%	249%
Formaldehyde	3,638	3,537	4,123	4,841	5,072	5,611	6,108	13%	39%
Benzene	709	681	646	636	630	603	583	-9%	-11%
PM2.5	607	552	497	448	401	350	296	-18%	-34%
PM10	1,853	1,685	1,516	1,368	1,223	1,068	905	-18%	-34%

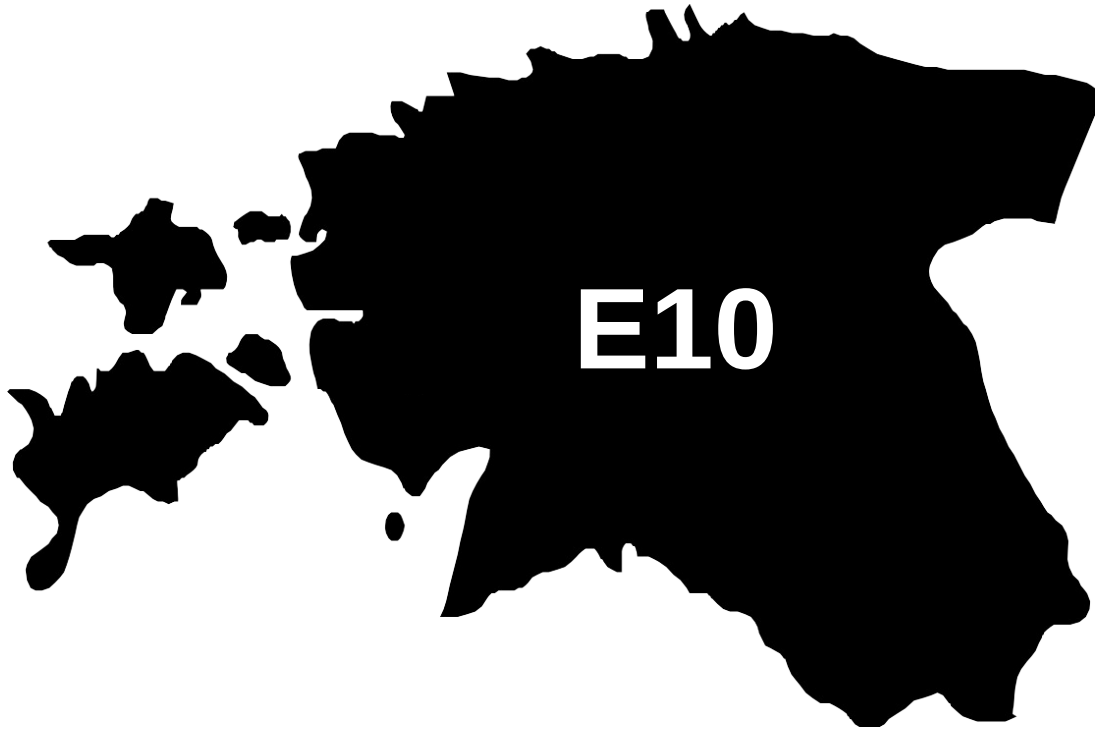
Denmark – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	45.5
Target @ 2035 (MMT/year)	25.6
Delta	19.9

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

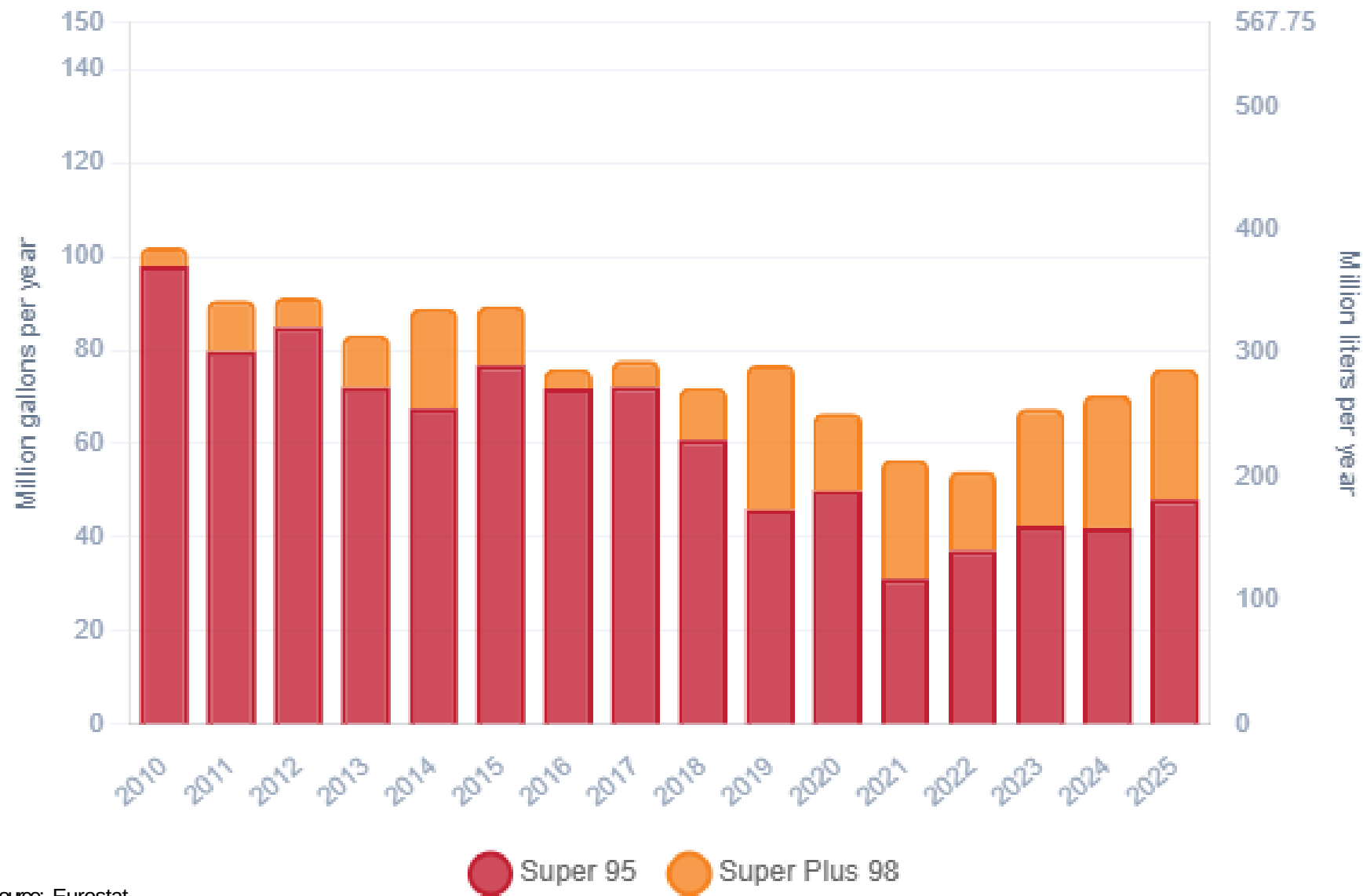
Ethanol Gasoline Blending in Estonia



In 2025, gasoline consumption in Estonia was 286 million liters (76 million gallons) and growth rate was -1.9%. There is no gasoline production. Gasoline net imports were 341 million liters (90 million gallons). Estonia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 96.1 RON.

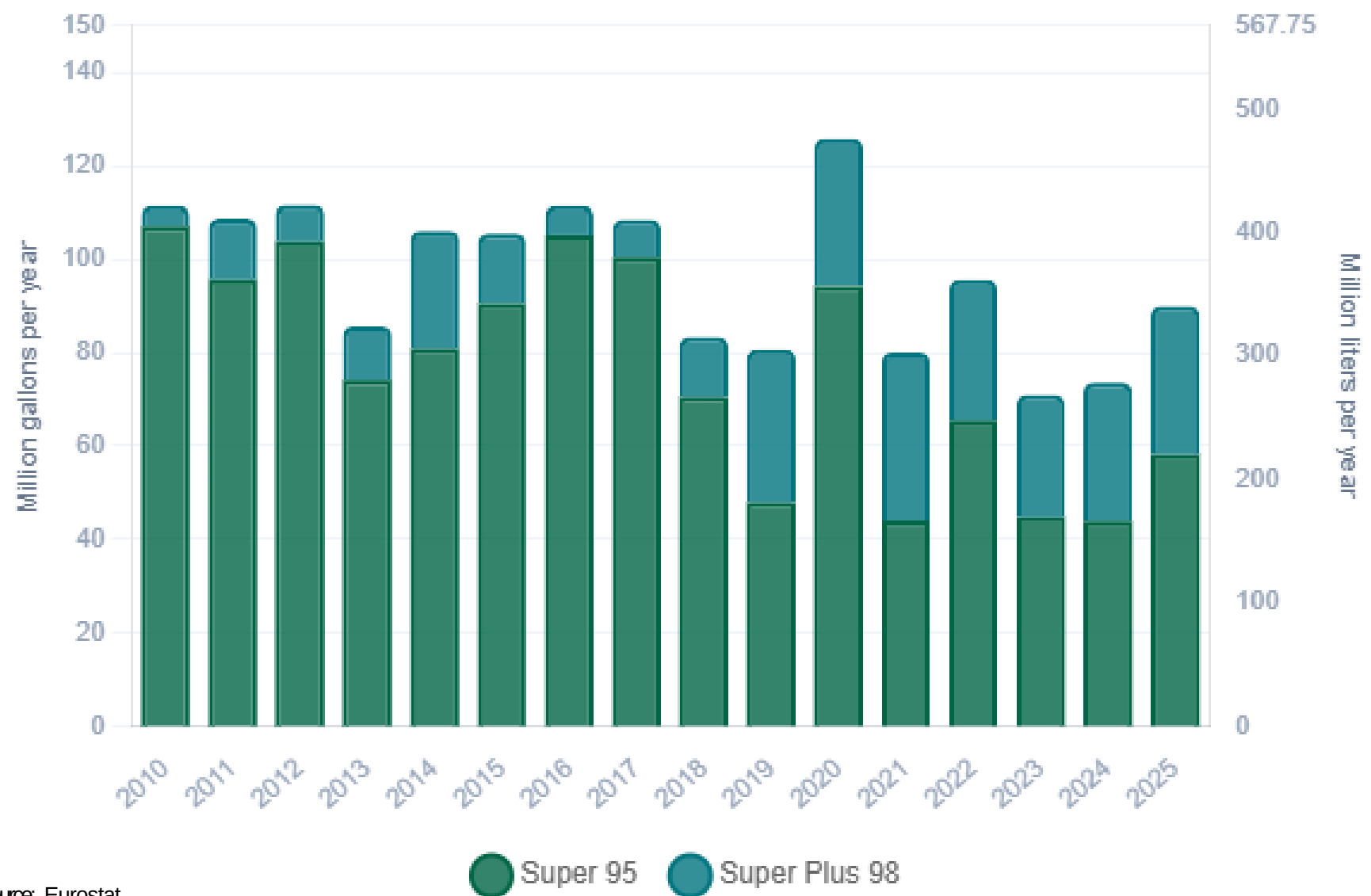
In 2025, ethanol consumption in Estonia was 5 million liters (1 million gallons) representing 2% of gasoline demand. Ethanol imports were 5 million liters (1 million gallons). Ethanol sources is Cereals 100%.

Gasoline Demand in Estonia



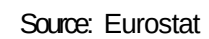
Source: Eurostat

Gasoline Imports in Estonia

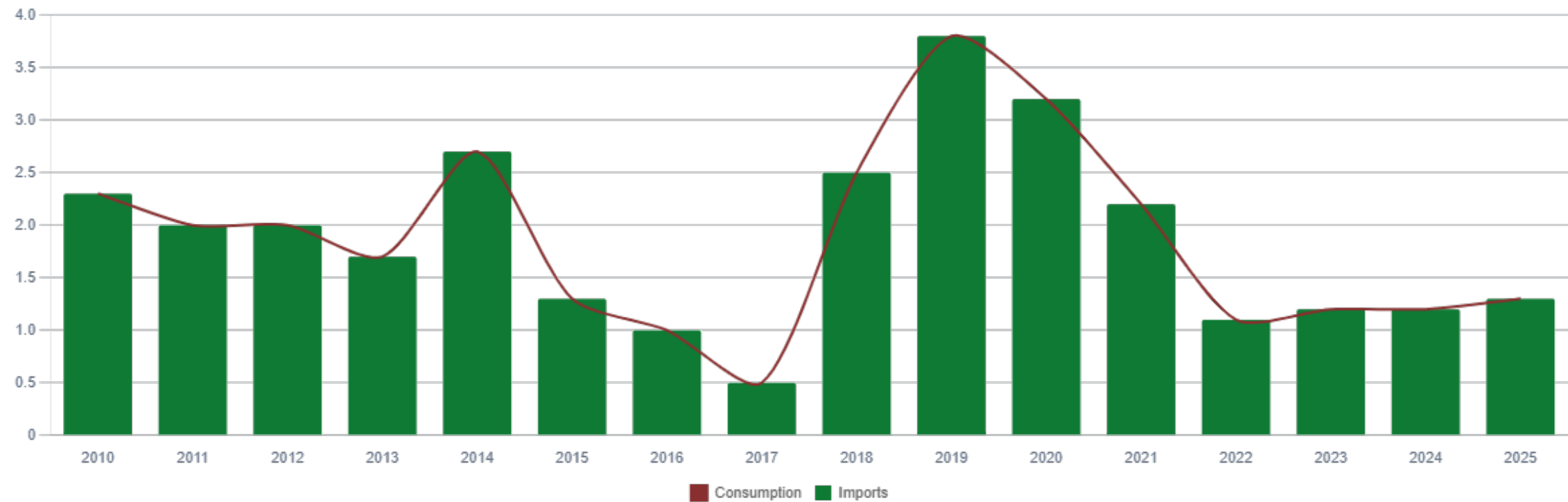


Source: Eurostat

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexene. The structure features a six-membered ring with a double bond. Substituents include a hydrogen atom (H), a carbon atom (C), and a group labeled 'X' (which is a vinyl group, -CH=CH₂). The ring also has methyl groups and a propyl group attached.

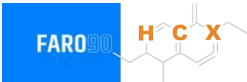


Ethanol Balance in Estonia



Source: Eurostat, Statistics Estonia

Gasoline Quality in Estonia

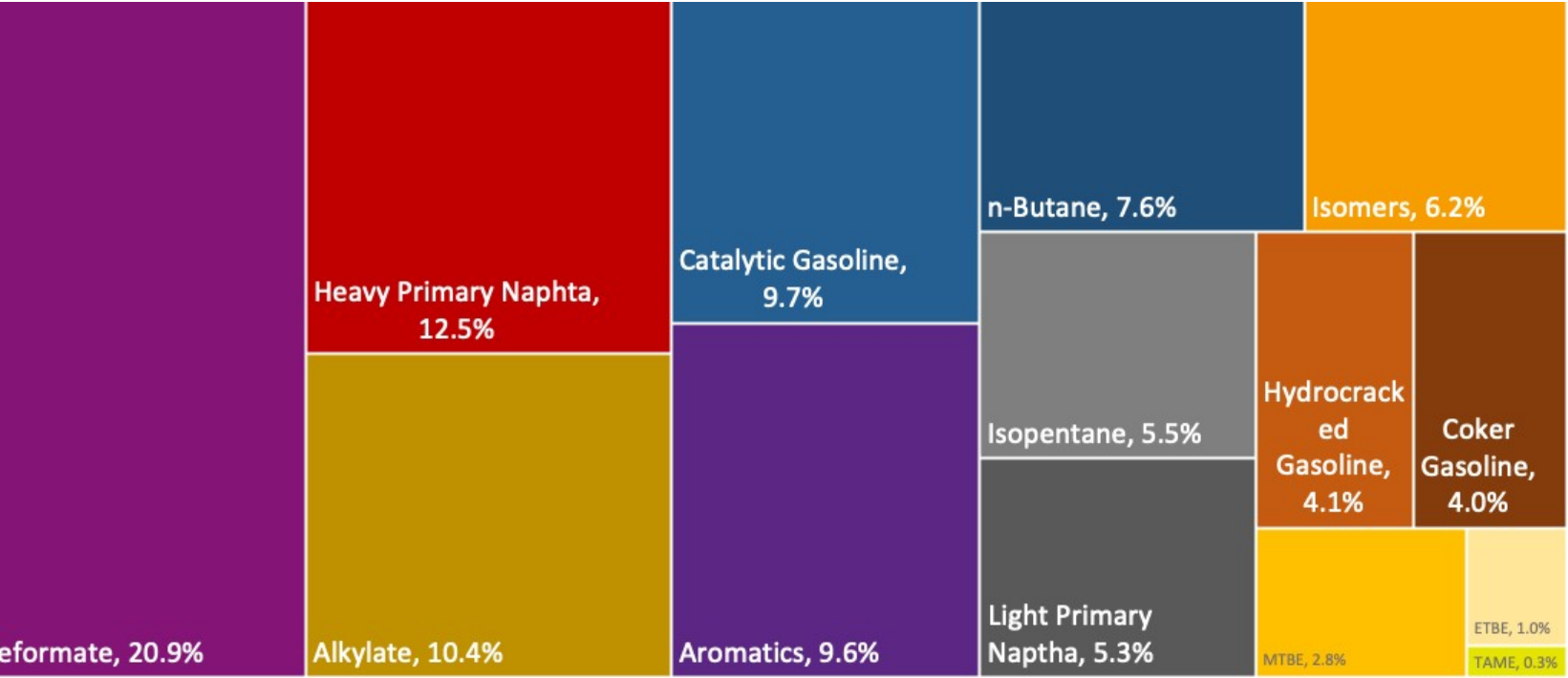


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	7.5							
Advanced Biofuels (%cal)	0.5							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

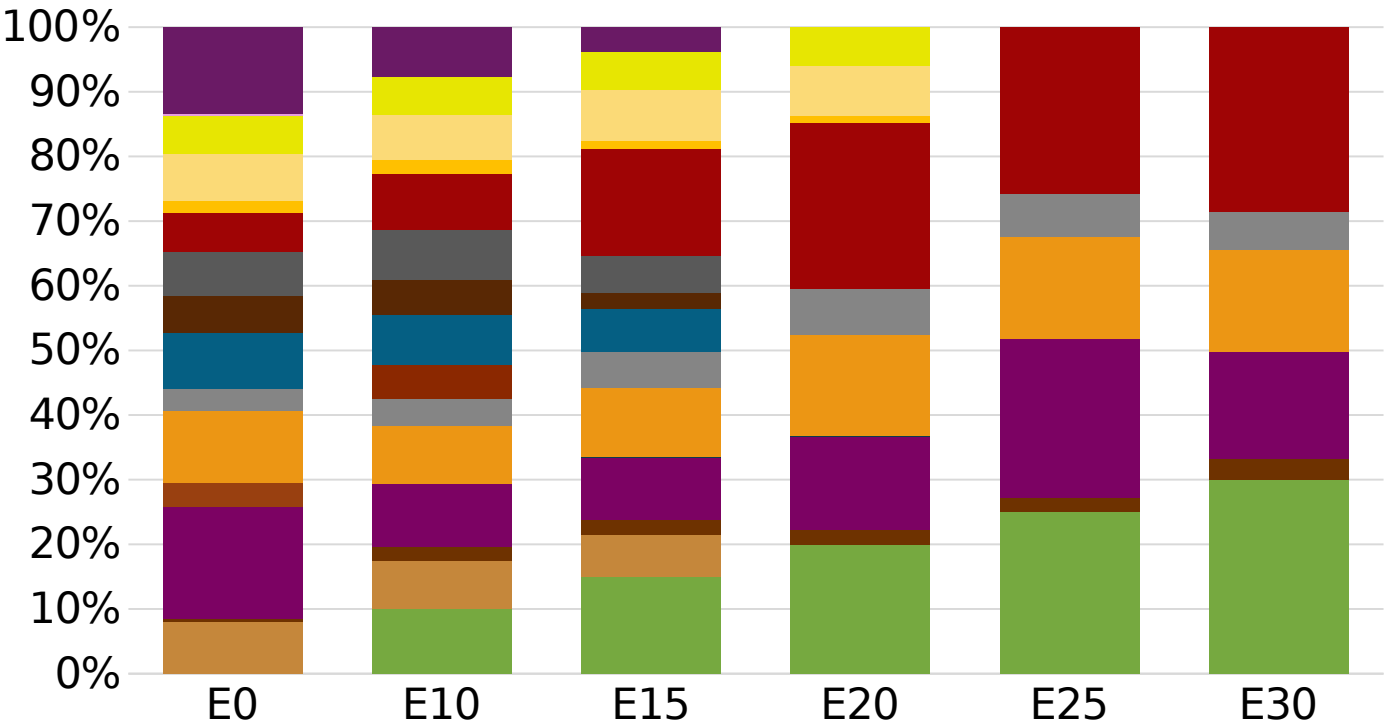
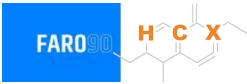
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Estonia Available Blending Components



Estonia – Gasoline 95 – Constant Octane

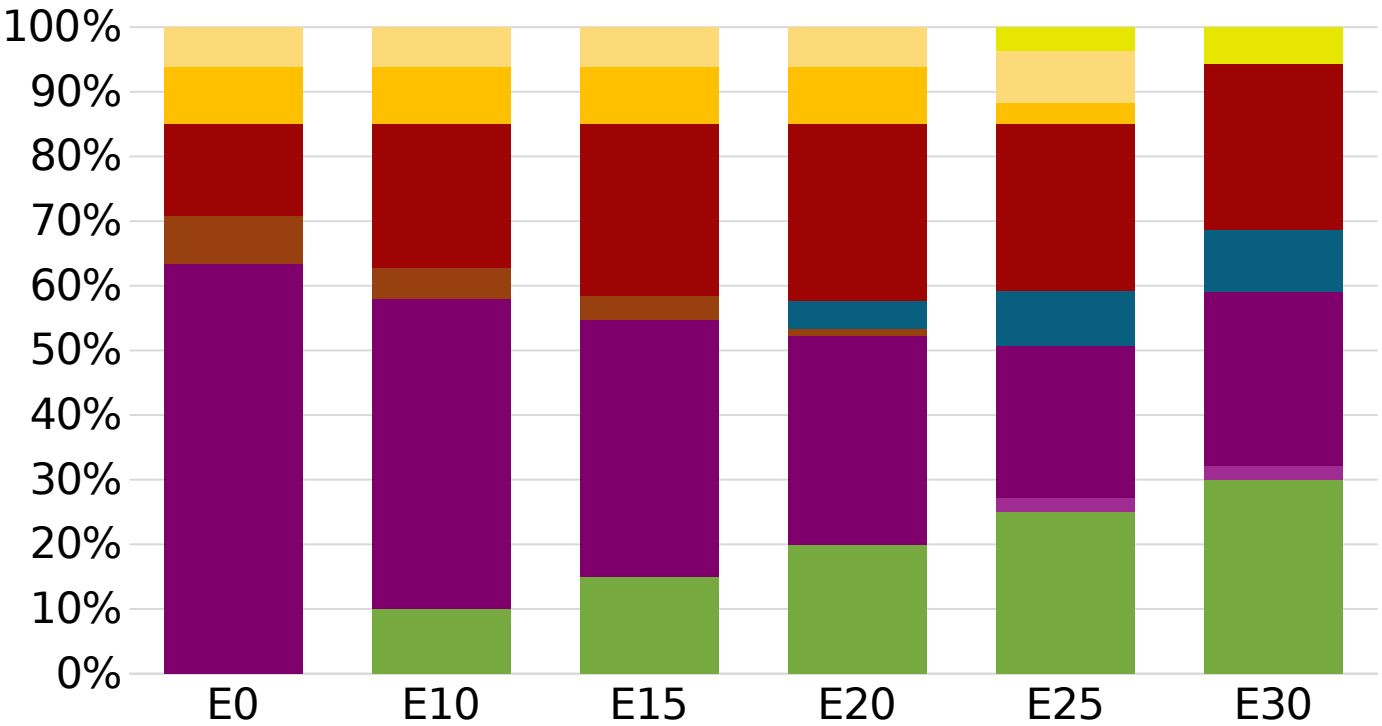
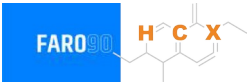


Average CIF 2025 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.071	\$ 1.933	\$ 1.833	\$ 1.718	\$ 1.672	\$ 1.619

Source:
Faro90

Estonia - Gasoline 98 - Constant Octane

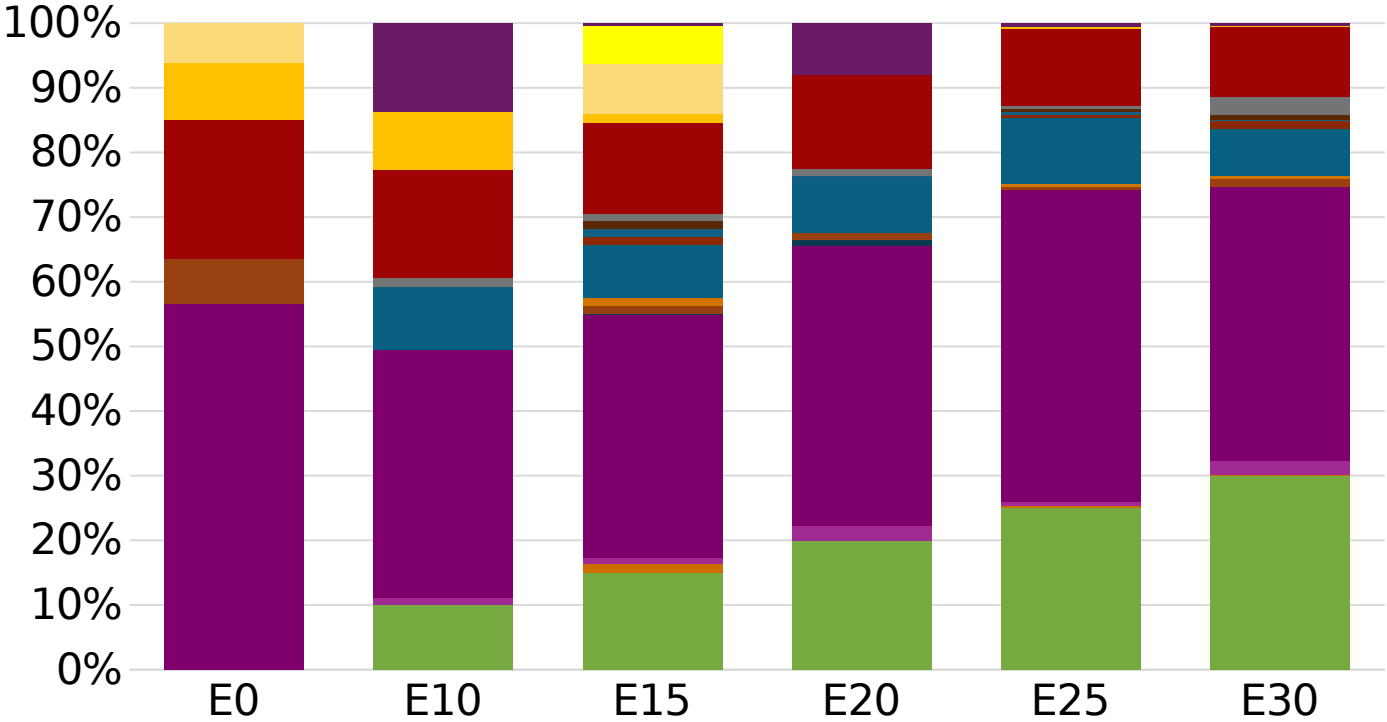


Average CIF 2025
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.091	\$ 1.974	\$ 1.905	\$ 1.833	\$ 1.768	\$ 1.718

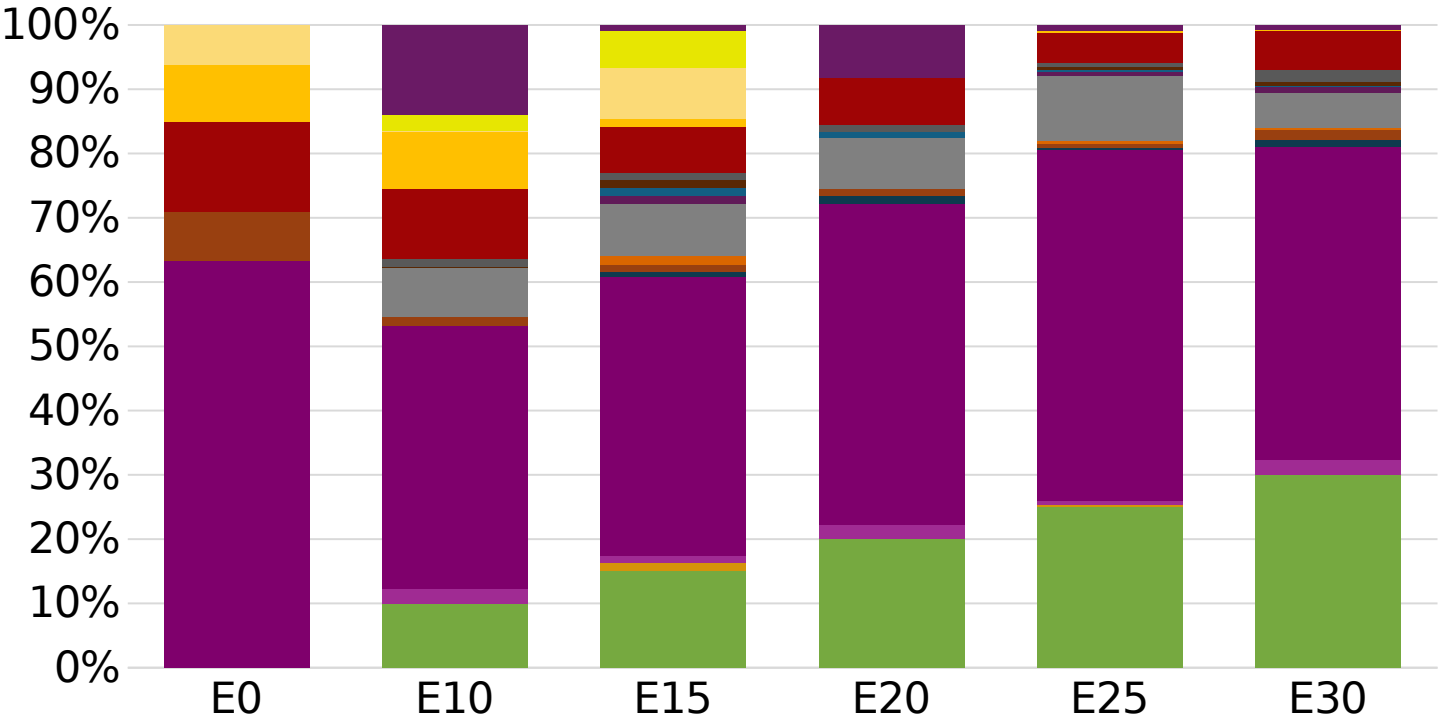
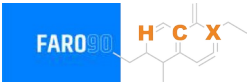
Source:
Faro90

Estonia – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

Estonia – Gasoline 98 – Increased Octane

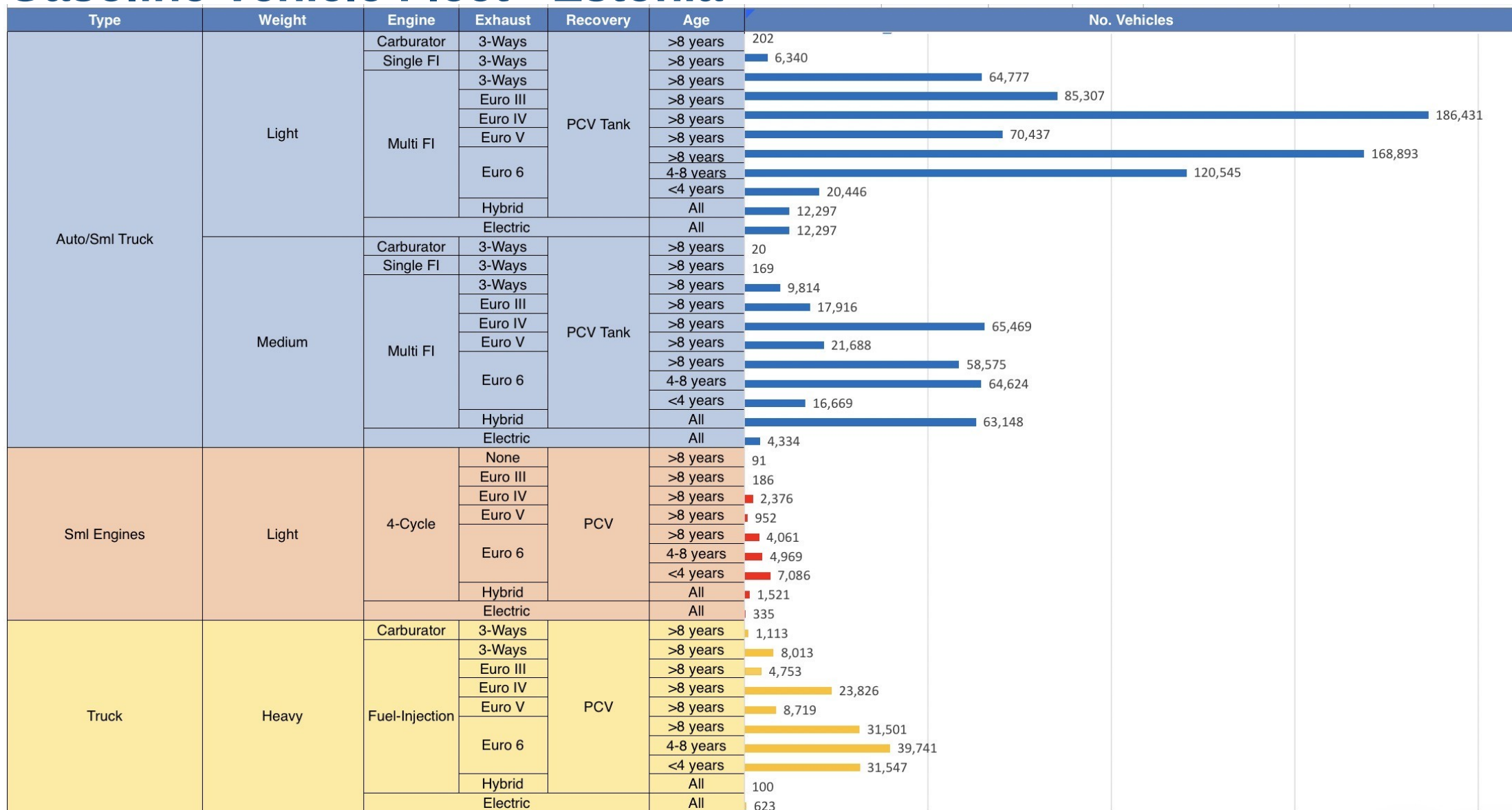


Average CIF 2025
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.091	\$ 2.091	\$ 2.091	\$ 2.091	\$ 2.091	\$ 2.091

Source:
Faro90

Gasoline Vehicle Fleet - Estonia

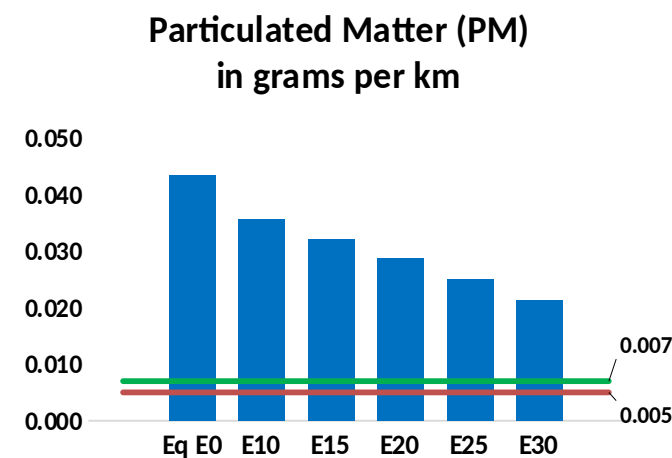
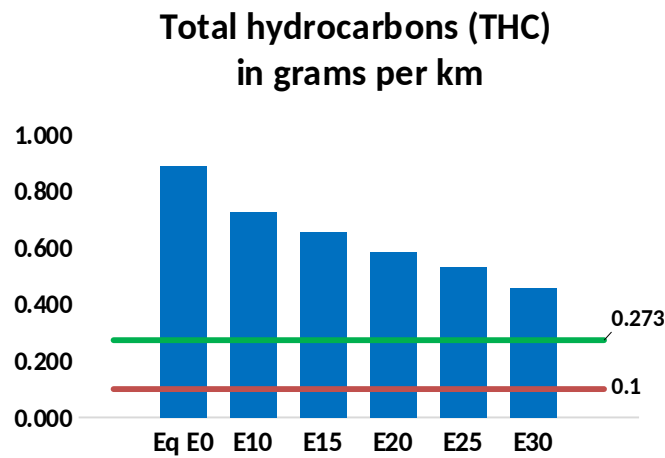
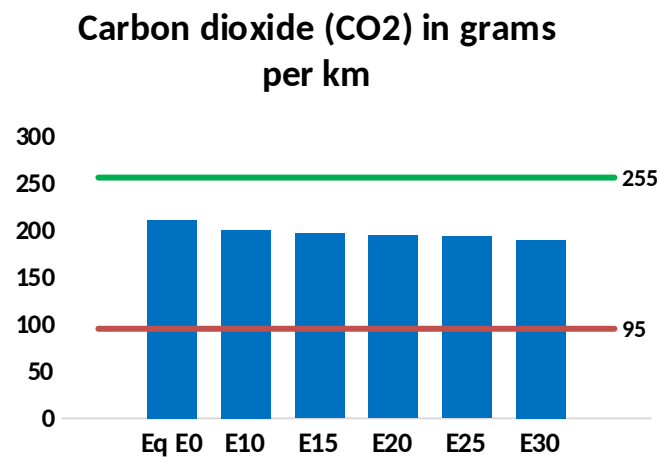
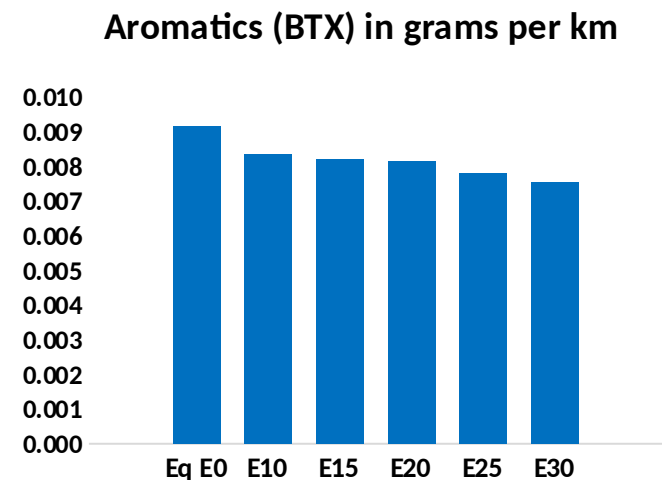
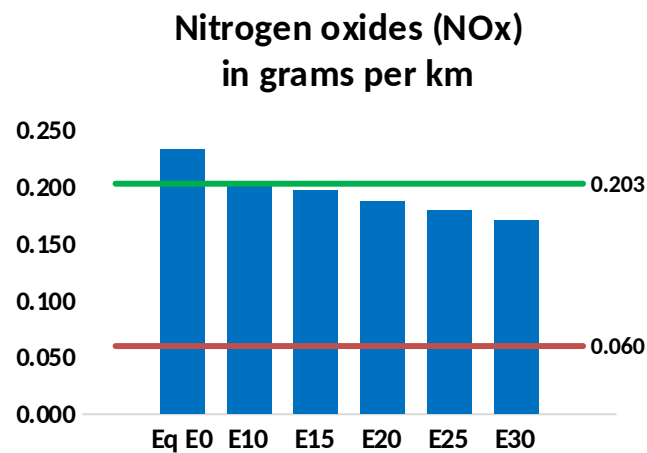
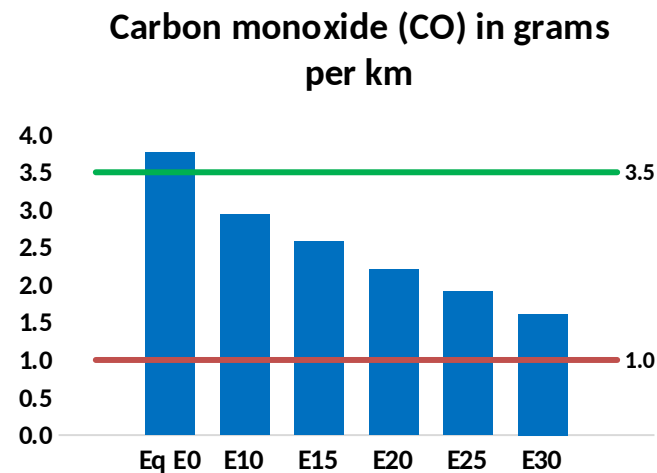
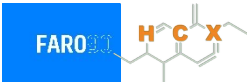


Vehicle Fleet: 1,241,909
Gasoline Vehicle Fleet: 589,169

Source: Eurostat, ACEA

Estonia – Vehicular Emissions

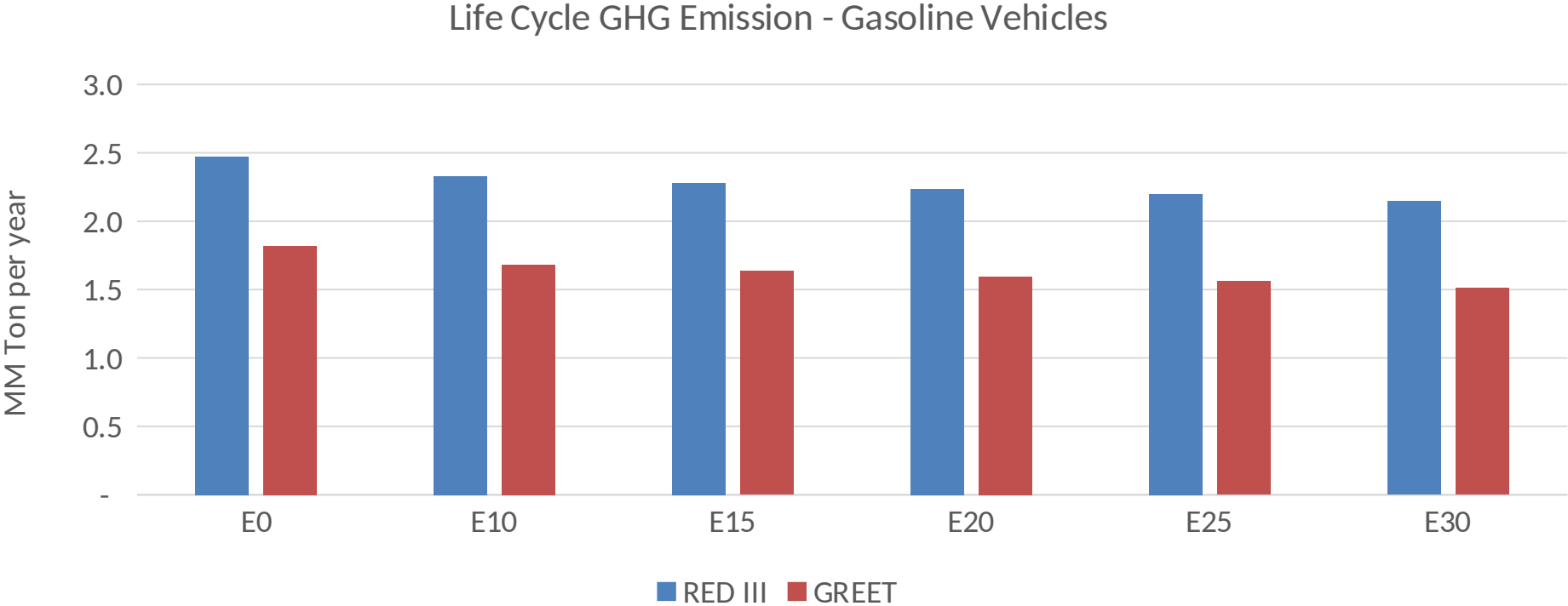
TIER III BIN 125 United States
Euro 6 Standard



Estonia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	56,434	50,255	44,075	38,652	33,029	28,656	24,190	-22%	-41%
CO2	3,158,152	3,076,942	3,000,244	2,939,923	2,910,074	2,892,954	2,839,628	-5%	-8%
VOC	13,317	12,098	10,880	9,843	8,783	7,962	6,885	-18%	-34%
NOx	3,497	3,229	3,048	2,956	2,808	2,693	2,556	-13%	-20%
SOx	37	34	31	29	26	24	21	-15%	-28%
NH3	1,220	1,208	1,196	1,201	1,215	1,224	1,232	-2%	0%
N2O	43	42	42	42	43	44	44	-1%	2%
CH4	2,643	2,643	2,643	2,683	2,741	2,791	2,839	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	55	52	49	47	45	44	41	-11%	-18%
Acetaldehyde	204	234	343	523	712	814	962	68%	249%
Formaldehyde	785	763	889	1,044	1,094	1,210	1,318	13%	39%
Benzene	138	132	125	123	122	117	113	-9%	-11%
PM2.5	187	170	153	138	123	108	91	-18%	-34%
PM10	466	423	381	344	307	268	227	-18%	-34%

Estonia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	15.6
Target @ 2035 (MMT/year)	8.8
Delta	6.8

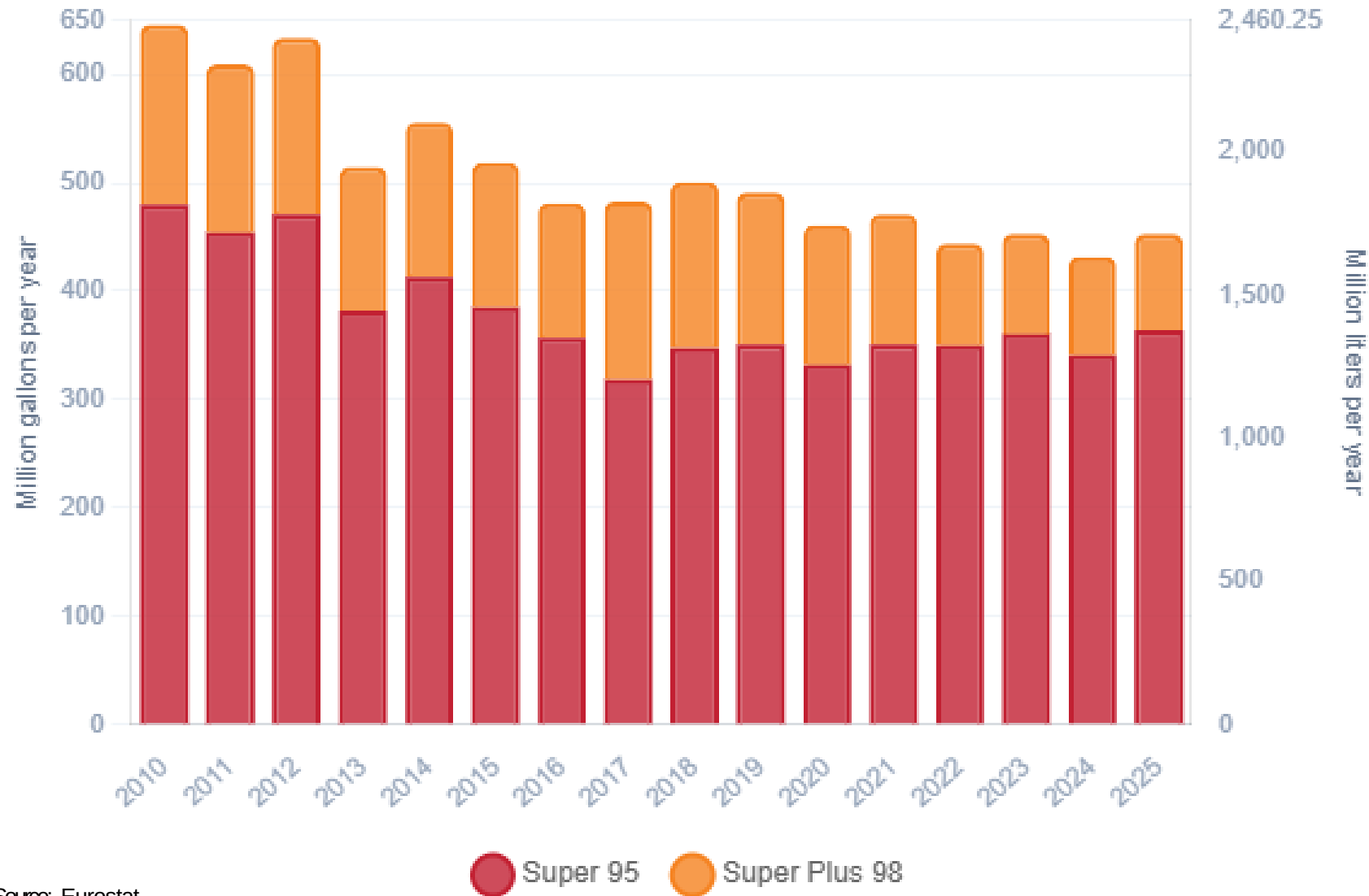
Ethanol Gasoline Blending in Finland



In 2025, gasoline consumption in Finland was 1,723 million liters (455 million gallons) and growth rate was -1.3%. Gasoline production and net imports were 5,696 and -3,919 million liters (1,505 and -1,035 million gallons) correspondingly. Finland complies with the European gasoline standards and offers two main grades: RON 95 and RON 98) with an average octane of 95.6 RON.

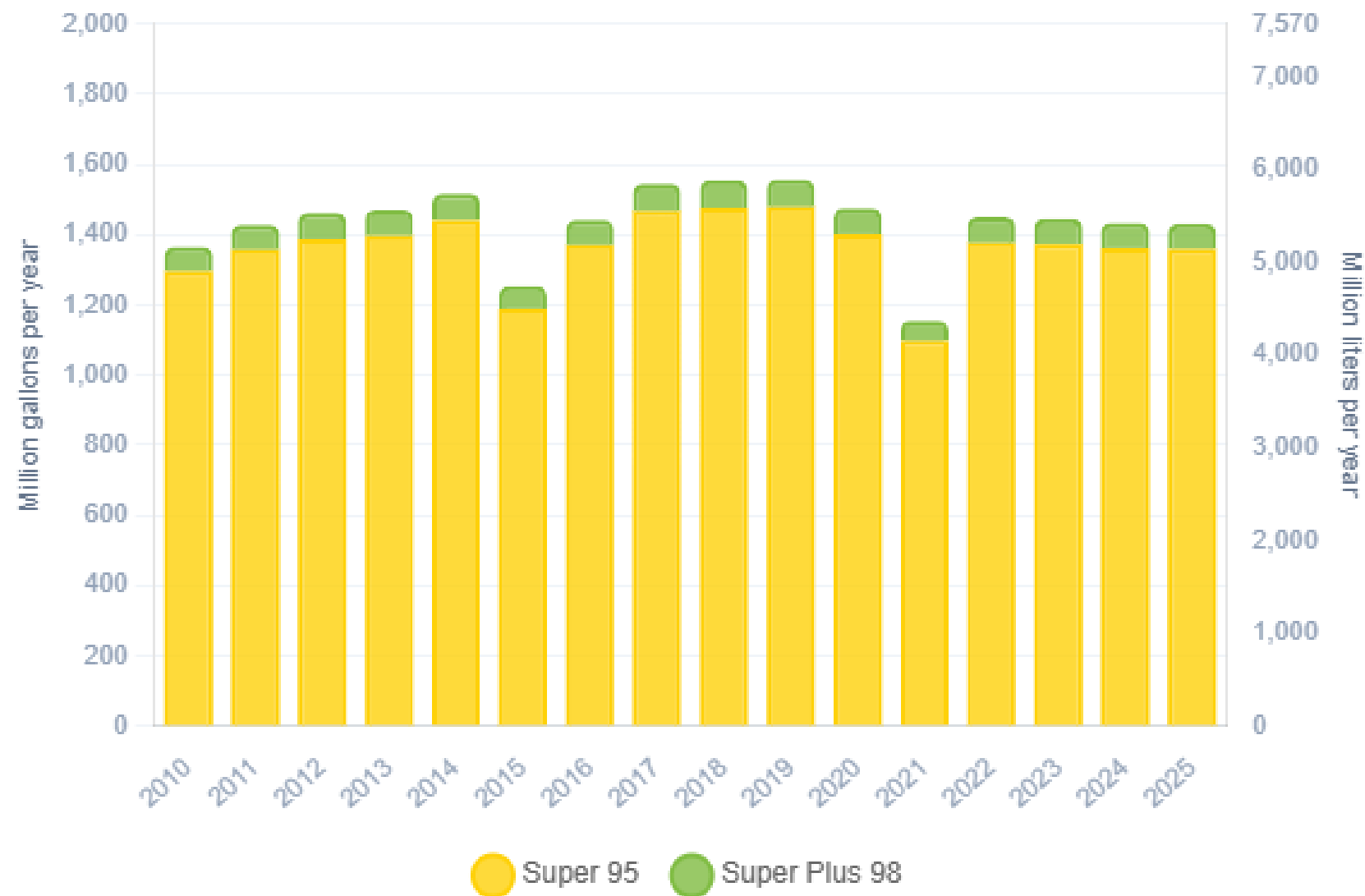
In 2025, ethanol consumption in Finland was 155 million liters (41 million gallons) representing 9% of gasoline demand. Ethanol production and Net Imports were 23 and 87 million liters (6 and 23 million gallons) correspondingly. Ethanol sources are Corn 5%, Cereals 85% and 2G lignocelulosic Ethanol 10%.

Gasoline Demand in Finland



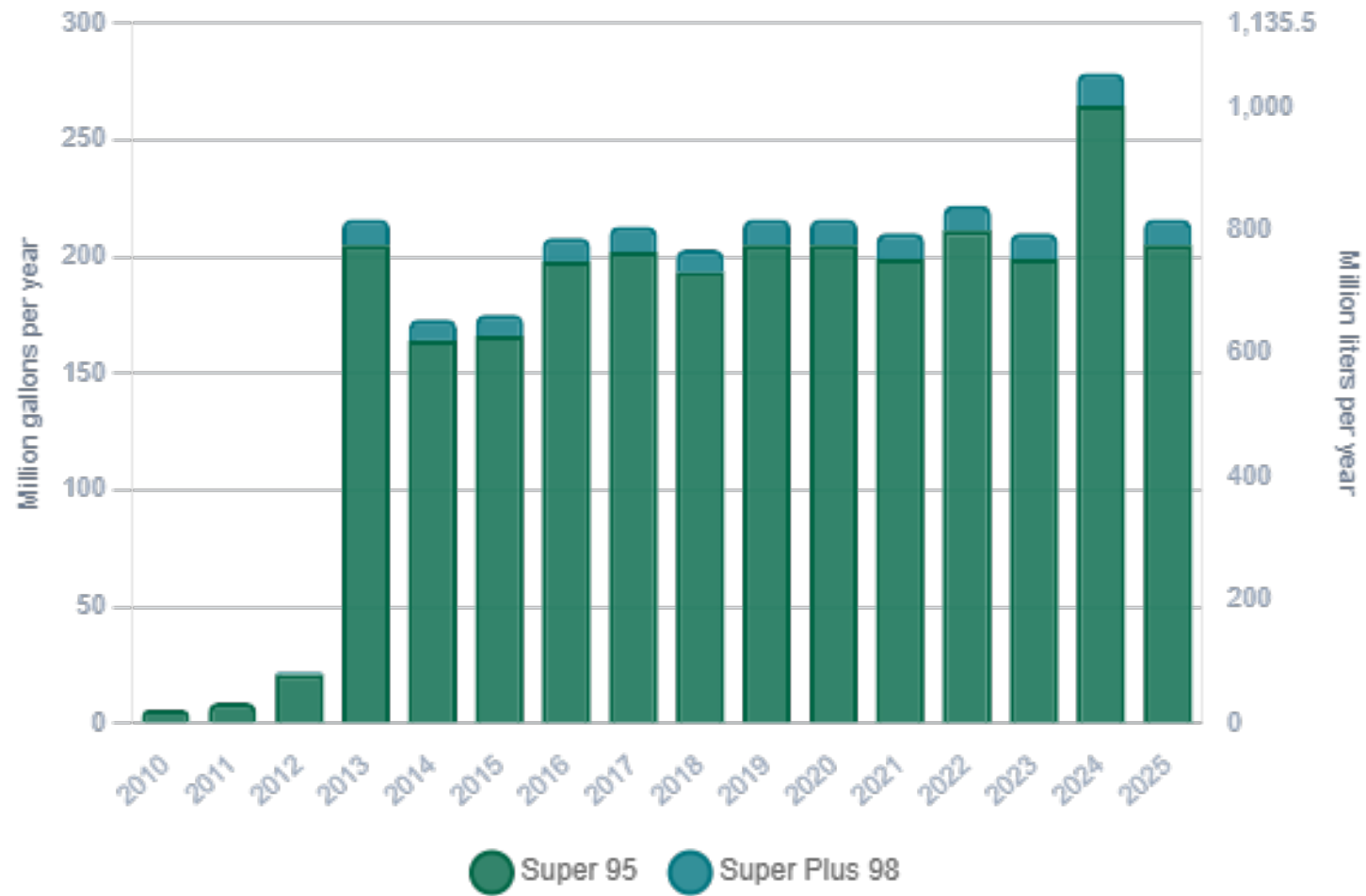
Source: Eurostat

Gasoline Production in Finland



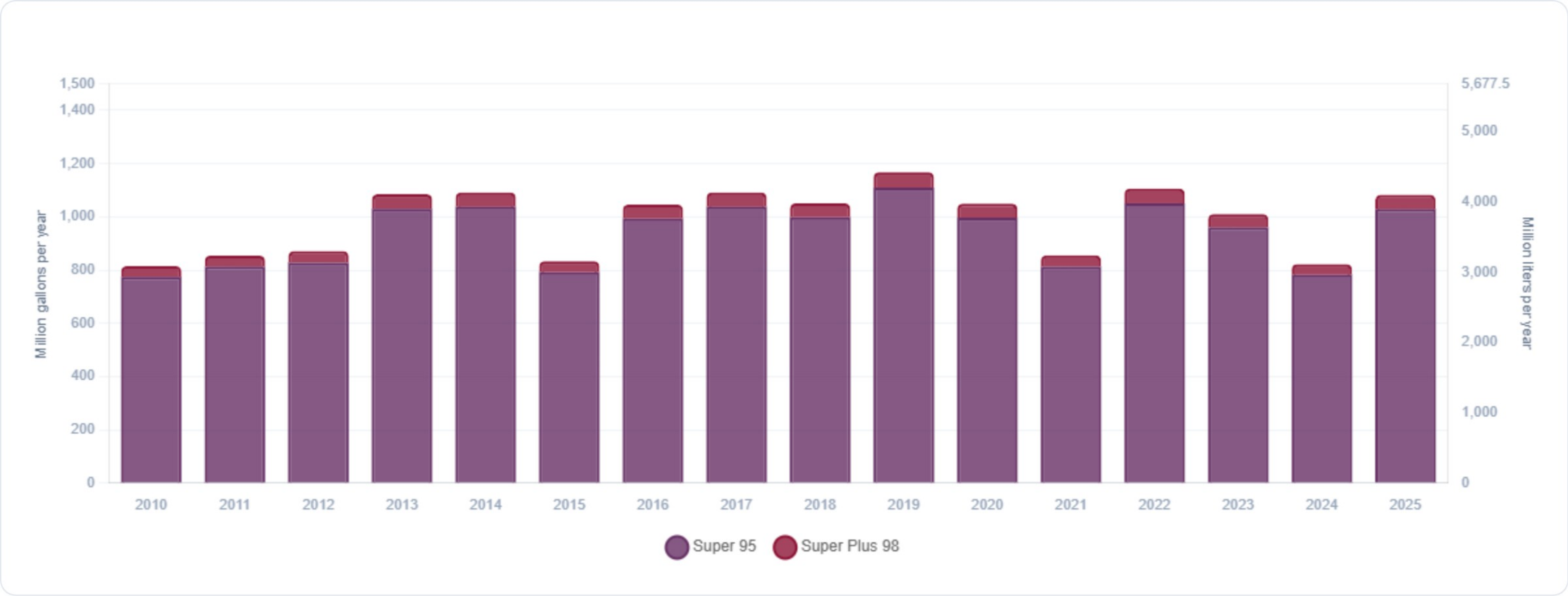
Source: Eurostat

Gasoline Imports in Finland



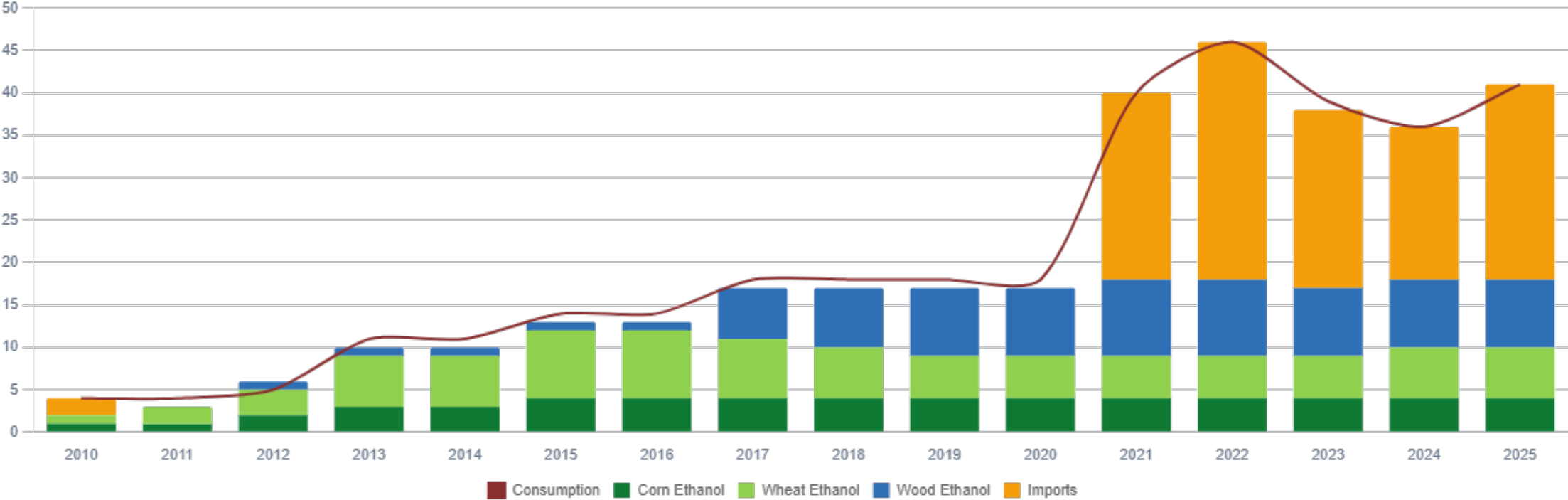
Source: Eurostat

Gasoline Exports in Finland



Source: Eurostat

Ethanol Balance in Finland



Source: Eurostat

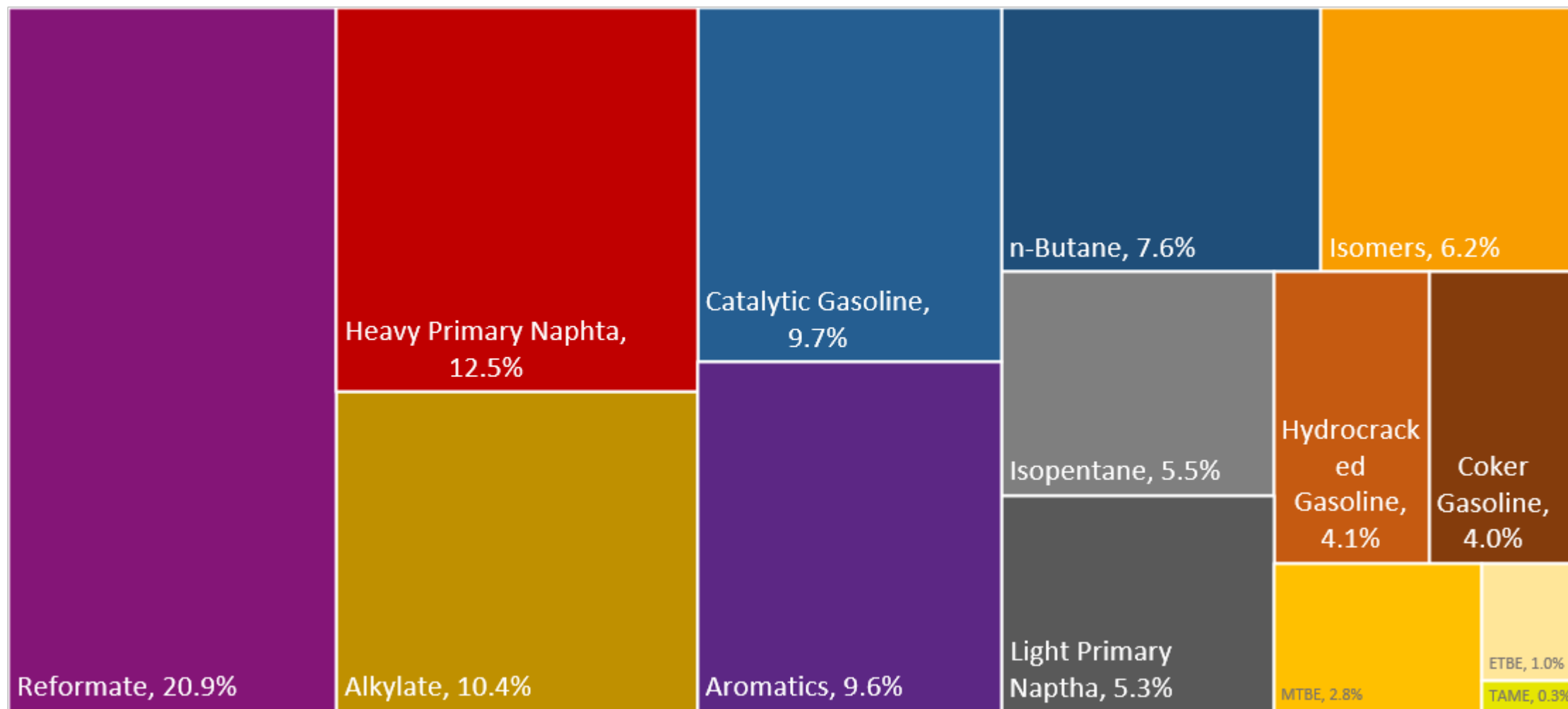
Gasoline Quality in Finland

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	< > 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	16.5							
Advanced Biofuels (%cal)	4.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

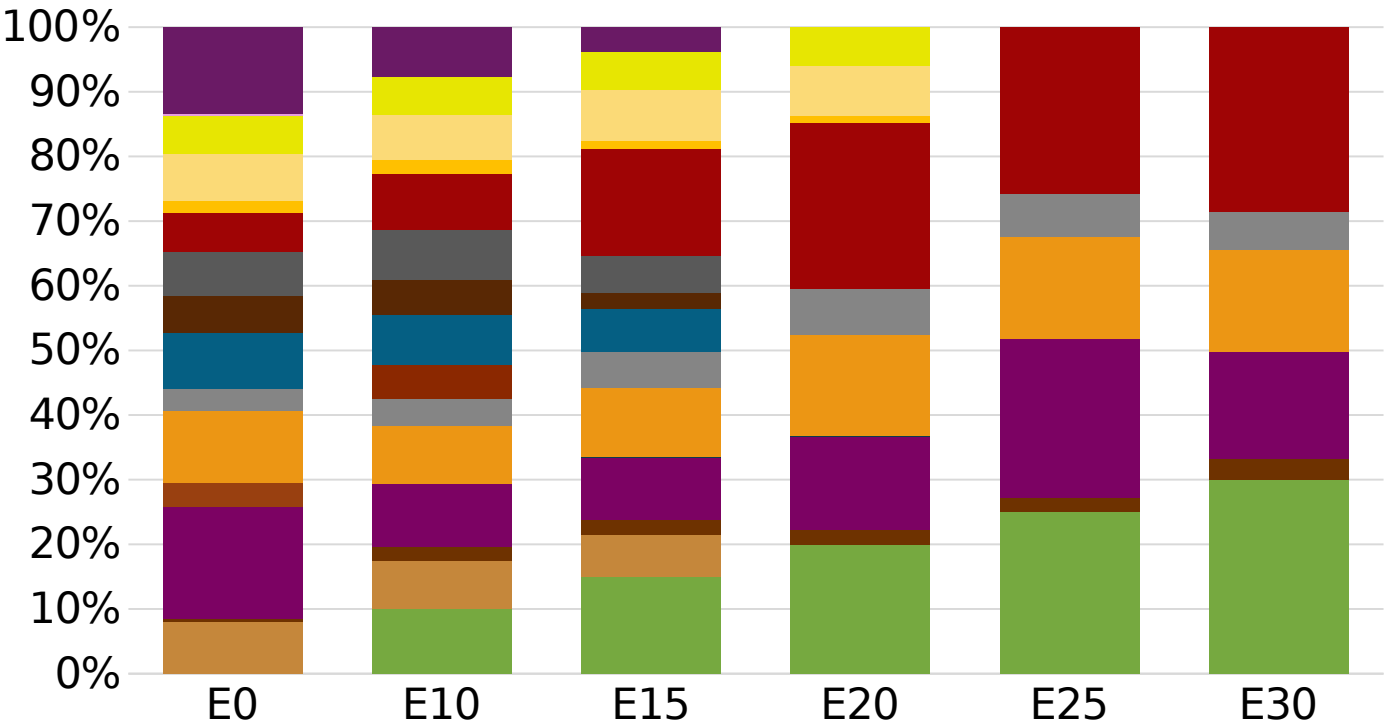
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Finland Available Blending Components



Finland – Gasoline 95 – Constant Octane

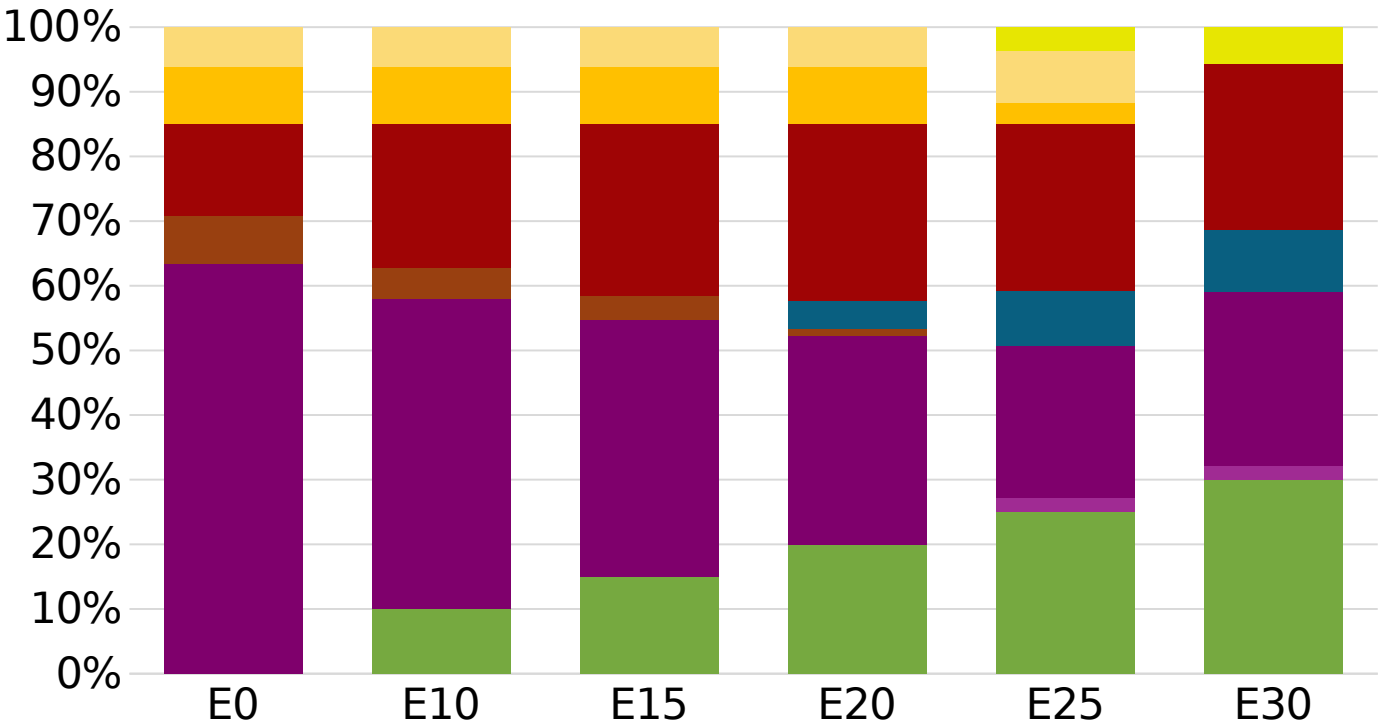
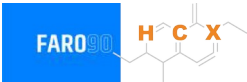


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.091	\$ 1.953	\$ 1.853	\$ 1.738	\$ 1.692	\$ 1.639

Source:
Faro90

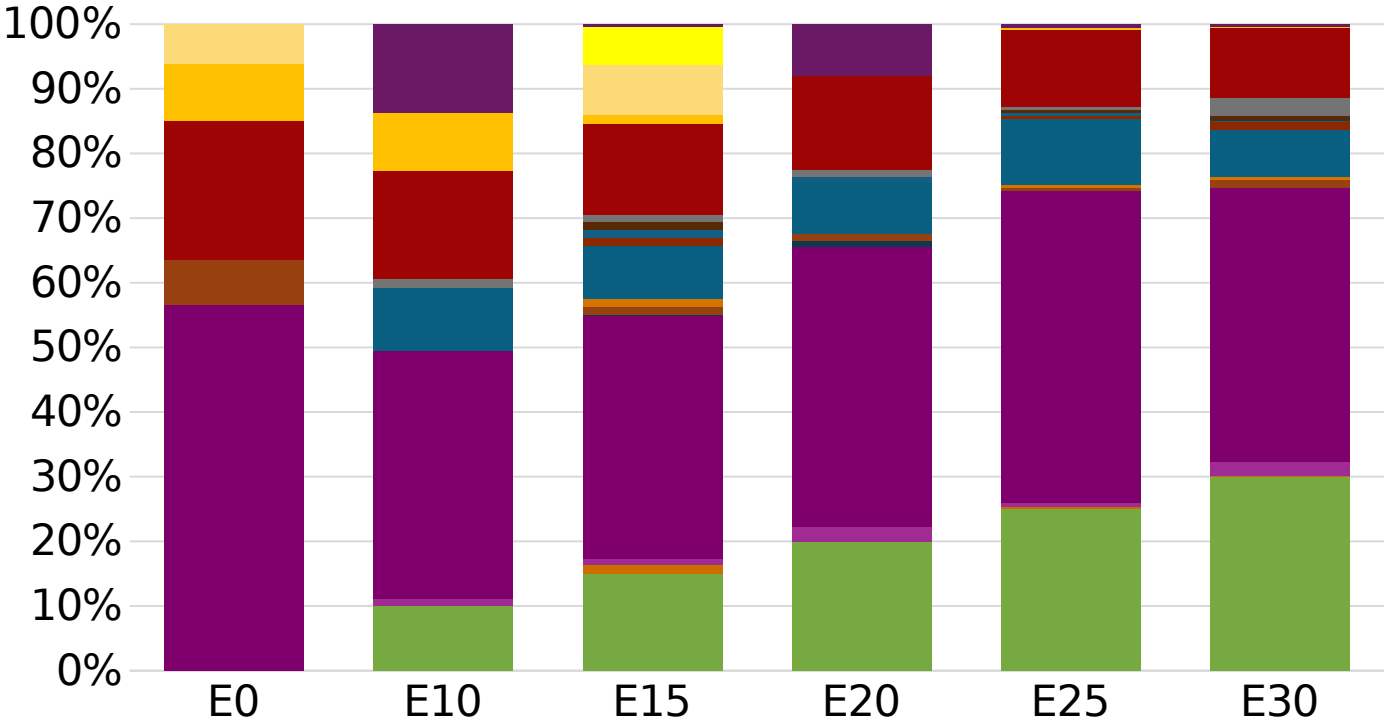
Finland - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.111	\$ 1.994	\$ 1.925	\$ 1.853	\$ 1.788	\$ 1.738

Average CIF 2025
Prices

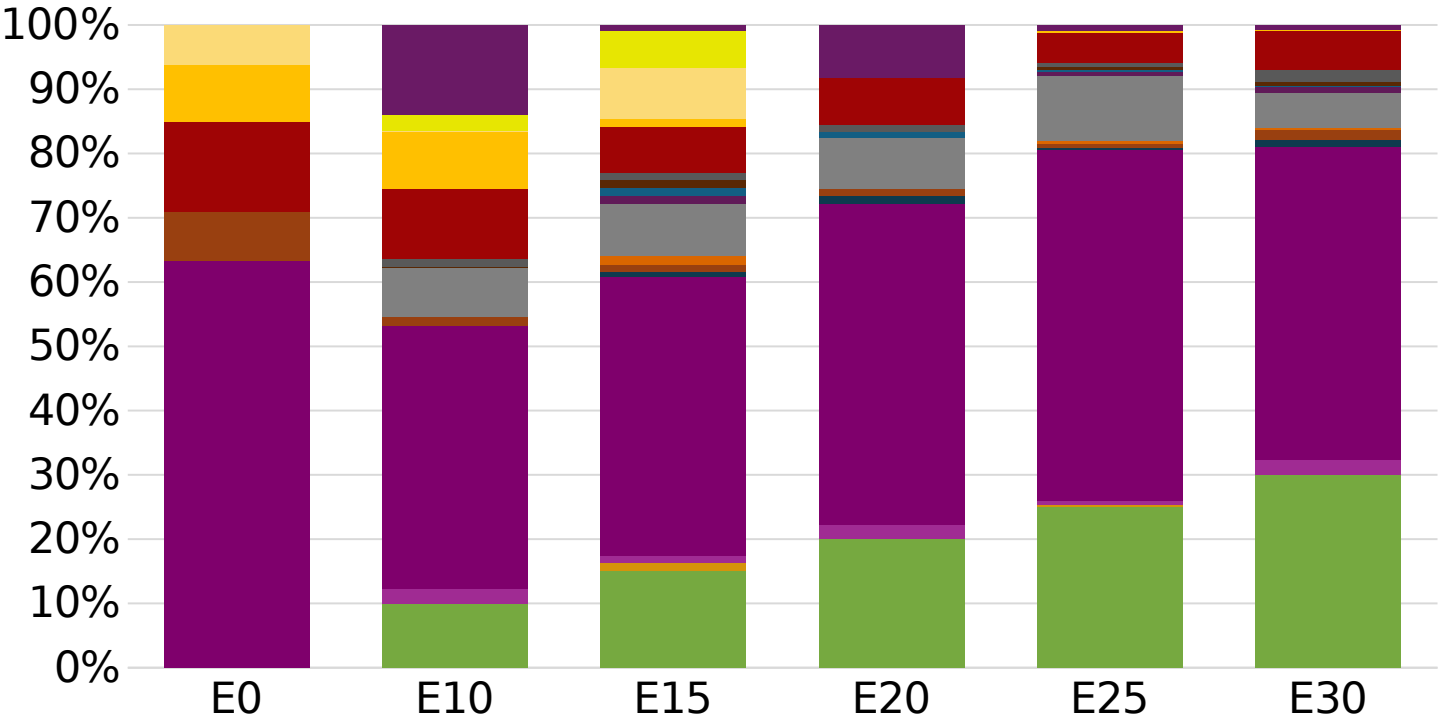
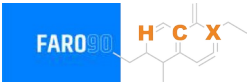
Finland – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

Source:
Faro90

Finland – Gasoline 98 – Increased Octane

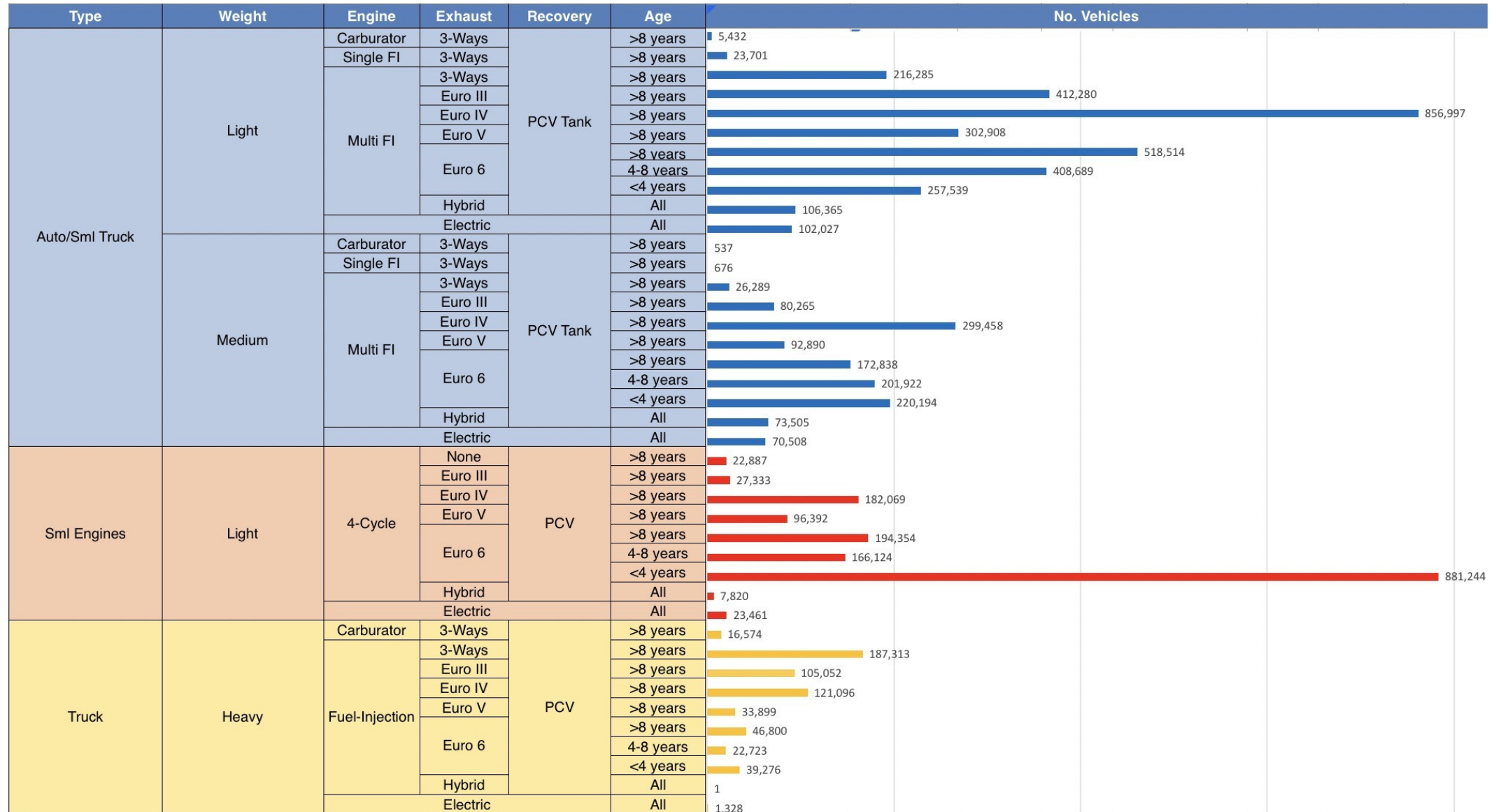


Average CIF 2025
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.111	\$ 2.111	\$ 2.111	\$ 2.111	\$ 2.111	\$ 2.111

Source:
Faro90

Gasoline Vehicle Fleet - Finland



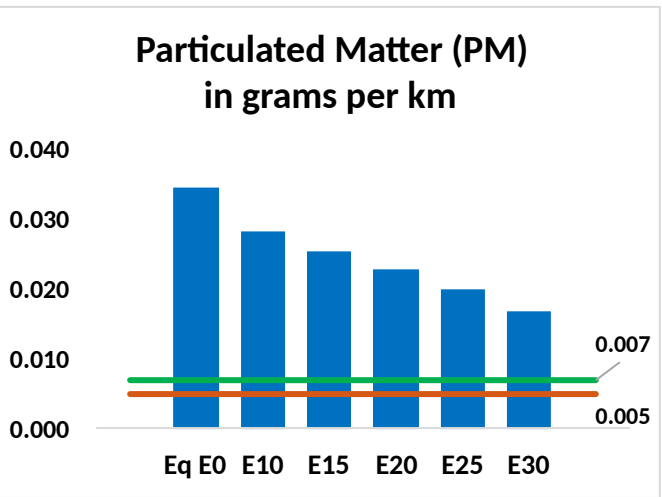
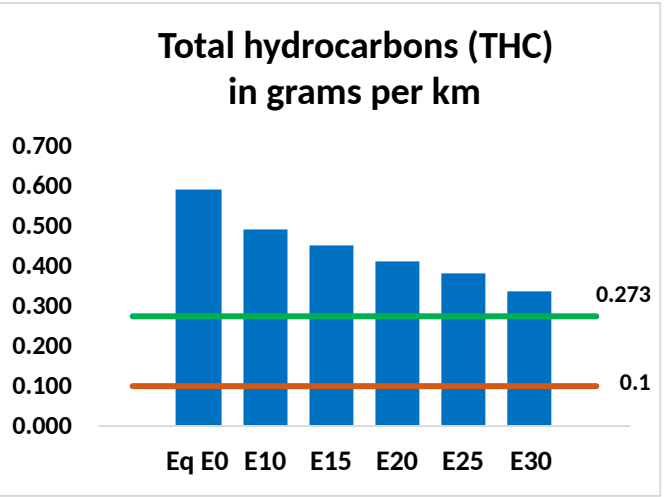
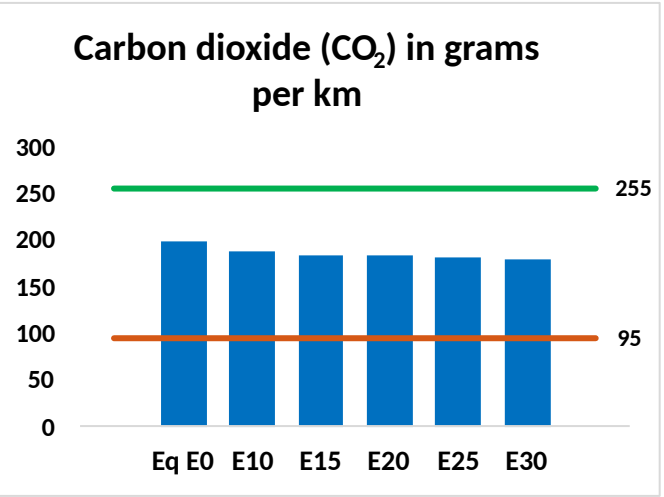
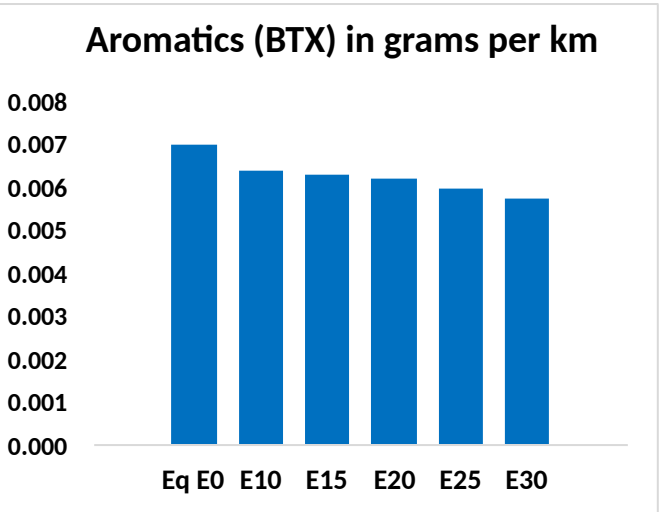
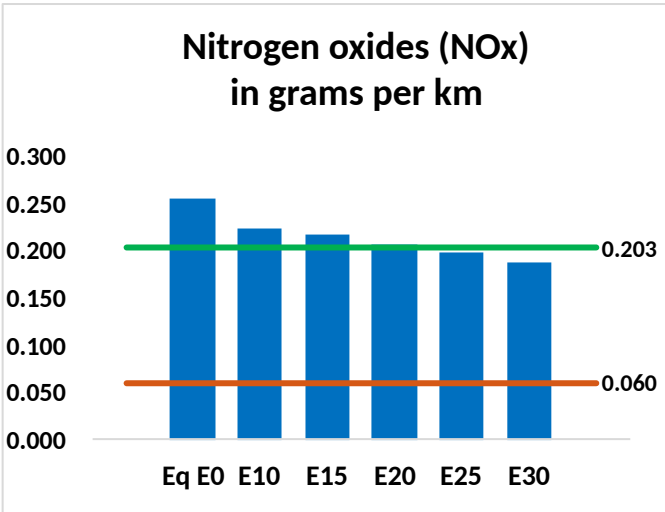
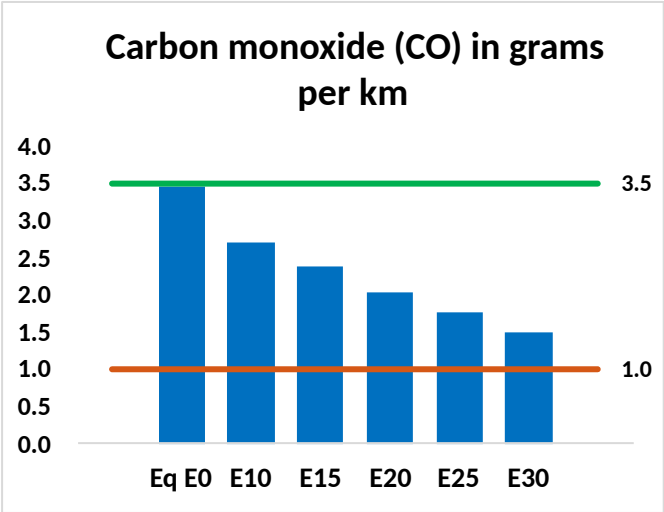
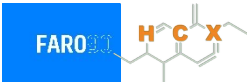
Vehicle Fleet: 6,625,564

Gasoline Vehicle Fleet: 3,473,949

Source: Eurostat, ACEA

Finland – Vehicular Emissions

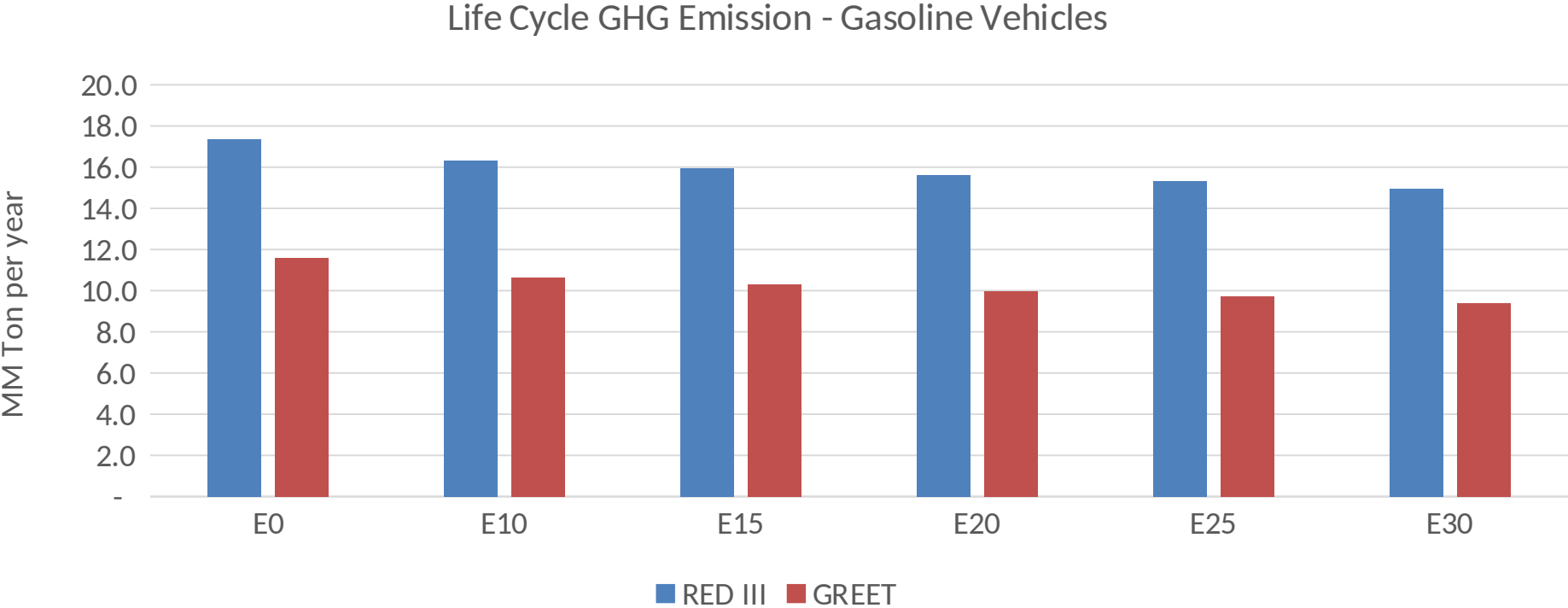
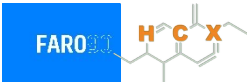
TIER III BIN 125 United States
Euro 6 Standard



Finland – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	312,839	278,880	244,922	215,160	184,321	160,339	135,854	-22%	-41%
CO2	17,985,650	17,523,162	17,086,367	16,742,841	16,572,851	16,474,128	16,170,462	-5%	-8%
VOC	53,535	49,130	44,726	41,060	37,345	34,481	30,686	-16%	-30%
NOx	23,215	21,435	20,243	19,650	18,679	17,926	17,034	-13%	-20%
SOx	205	189	174	161	148	135	120	-15%	-28%
NH3	6,546	6,481	6,417	6,447	6,520	6,571	6,613	-2%	0%
N2O	276	274	273	275	280	283	287	-1%	2%
CH4	11,529	11,529	11,529	11,702	11,956	12,175	12,382	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	317	300	282	269	257	247	234	-11%	-19%
Acetaldehyde	941	1,080	1,584	2,412	3,287	3,756	4,438	68%	249%
Formaldehyde	3,517	3,419	3,986	4,681	4,904	5,425	5,905	13%	39%
Benzene	636	611	579	570	565	540	523	-9%	-11%
PM2.5	1,075	978	880	794	710	619	525	-18%	-34%
PM10	2,045	1,859	1,673	1,510	1,349	1,178	998	-18%	-34%

Finland – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	64.9
Target @ 2035 (MMT/year)	36.6
Delta	28.4

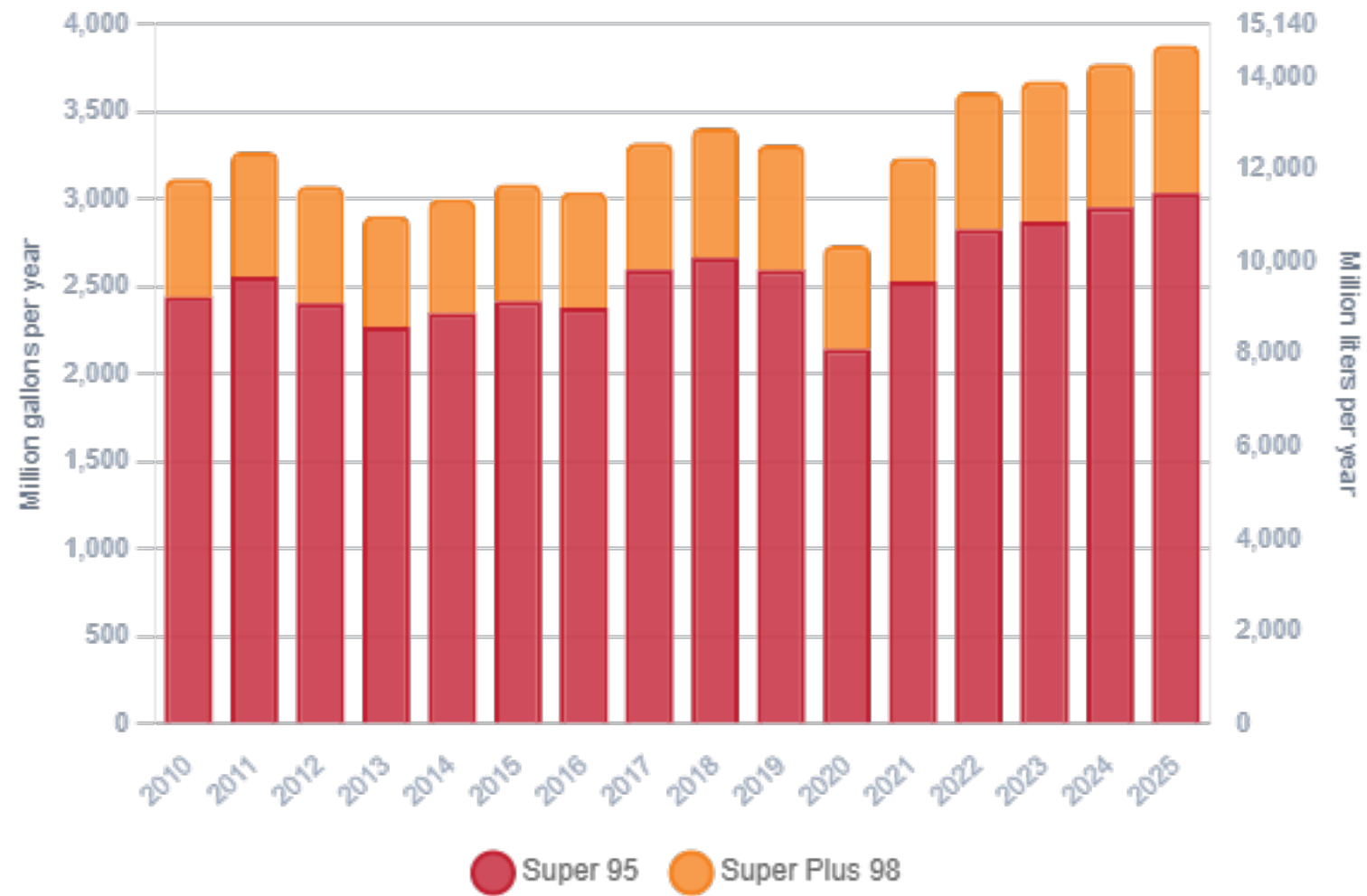
Ethanol Gasoline Blending in France



In 2025, gasoline consumption in France was 14,702 million liters (3,884 million gallons) and growth rate was 1.5%. Gasoline production and net imports were 11,926 and 1,072 million liters (3,150 and 283 million gallons) correspondingly. France complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.7 RON.

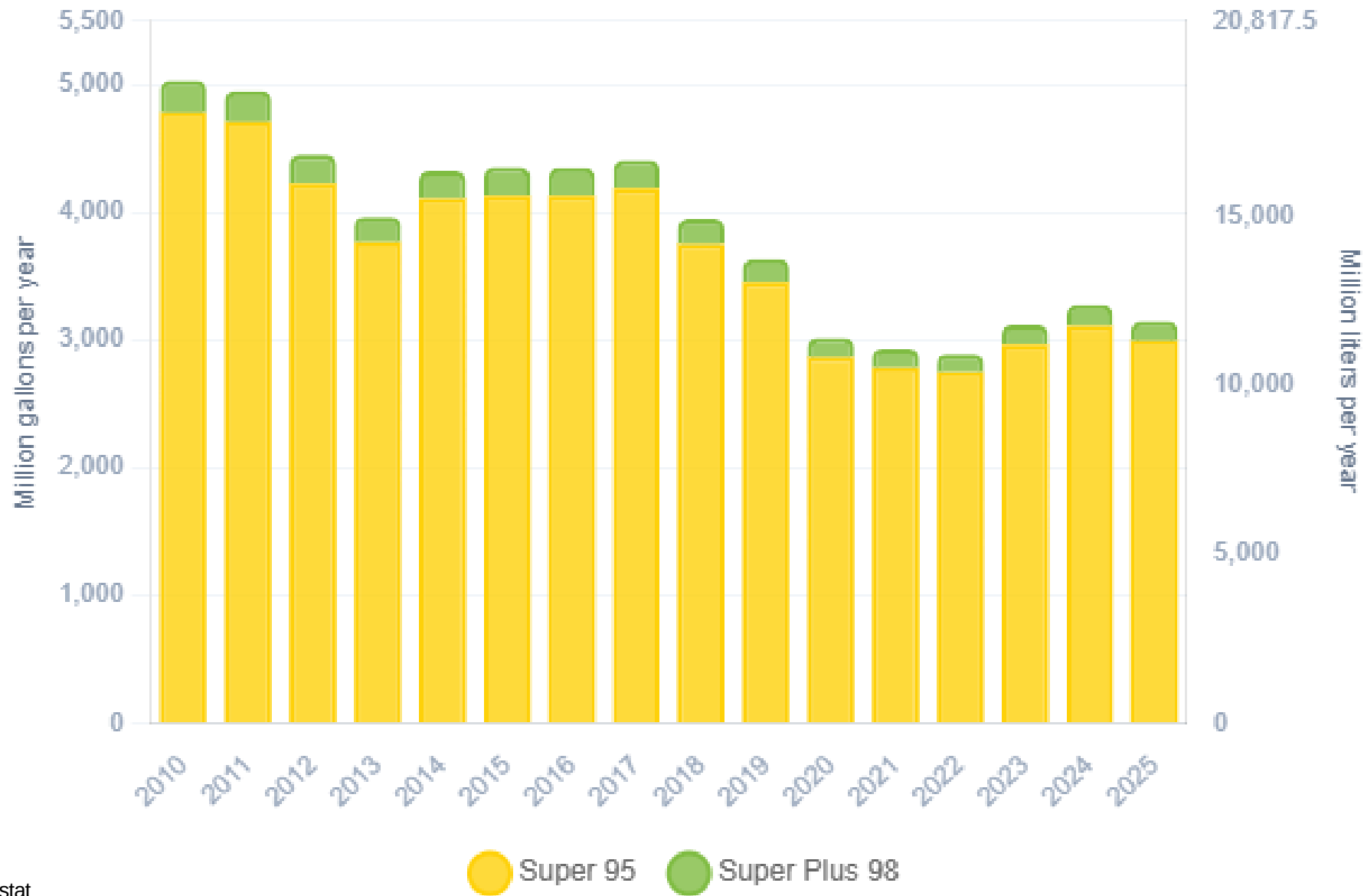
In 2025, ethanol consumption in France was 1,518 million liters (401 million gallons) representing 10% of gasoline demand. Ethanol production and Net Imports were 1,192 and 325 million liters (315 and 86 million gallons) correspondingly. Ethanol sources are Corn 32%, Cereals 21% and Sugar Beet 47%.

Gasoline Demand in France



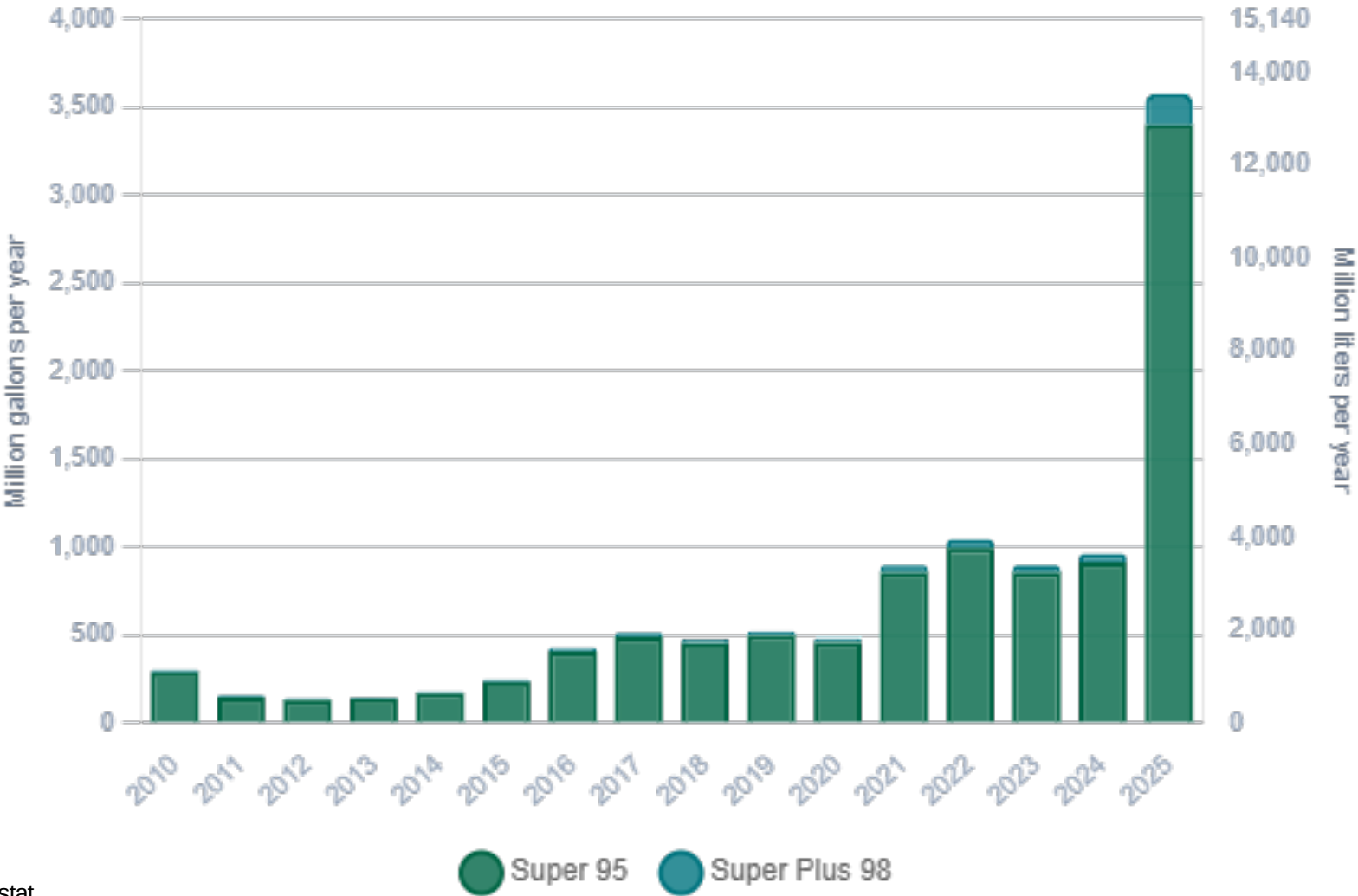
Source: Eurostat

Gasoline Production in France



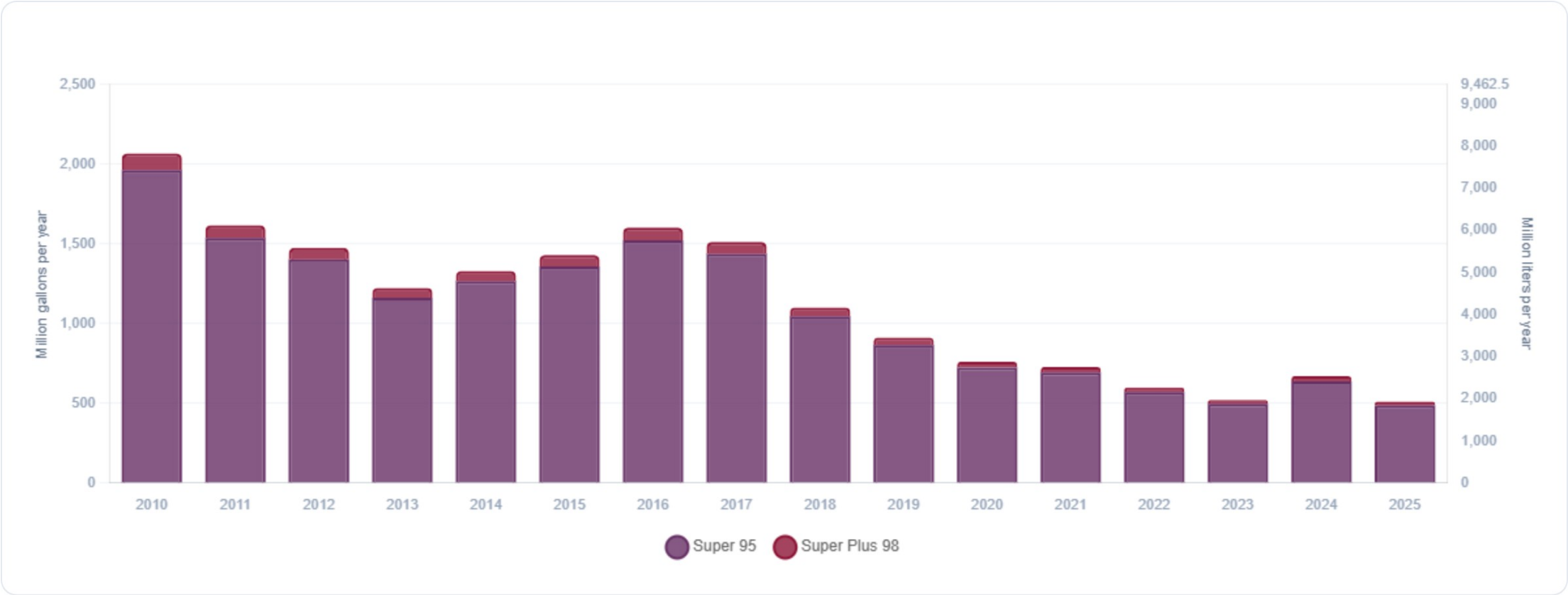
Source: Eurostat

Gasoline Imports in France



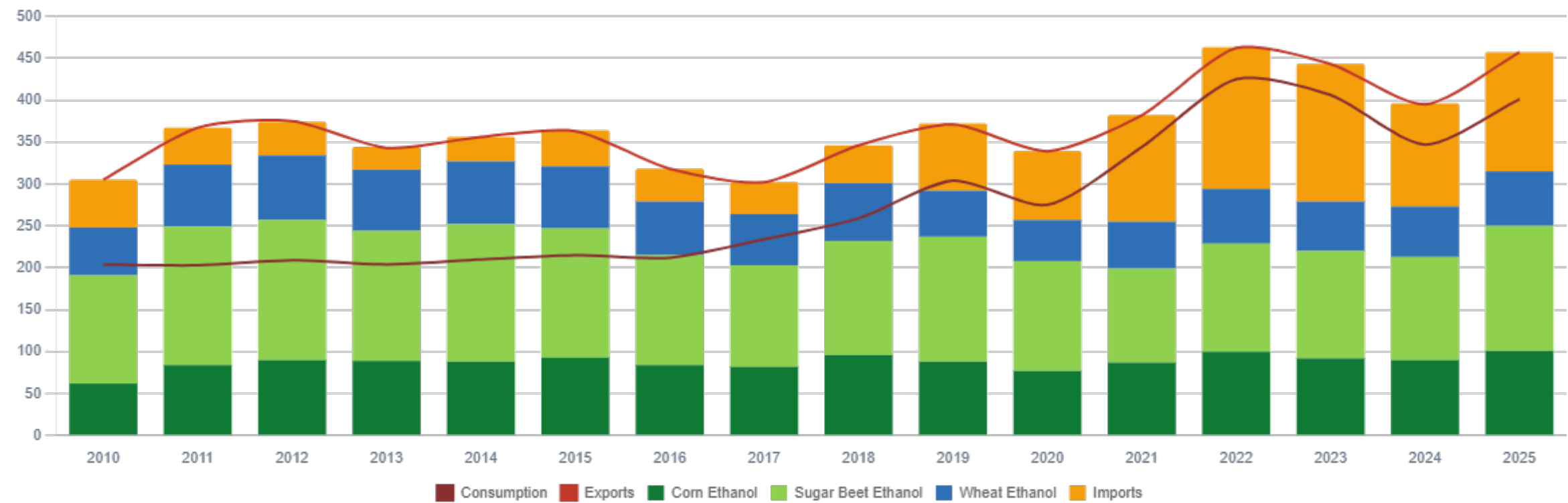
Source: Eurostat

Gasoline Exports in France



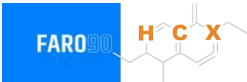
Source: Eurostat

Ethanol Balance in France



Source: Eurostat

Gasoline Quality in France

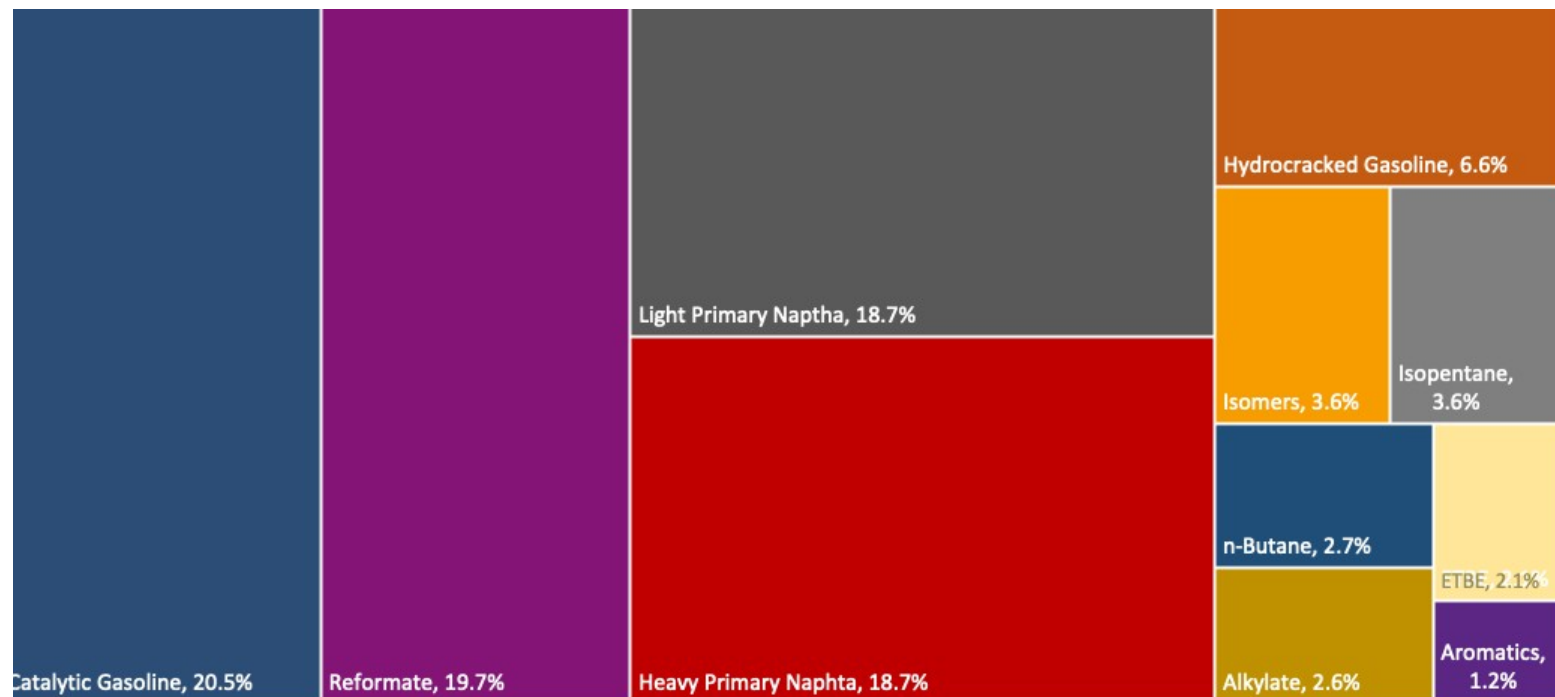


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	1.2							
Bioethanol in Gasoline Sales (%cal)	8.6							
GHG Emission Reduction (%)	-							

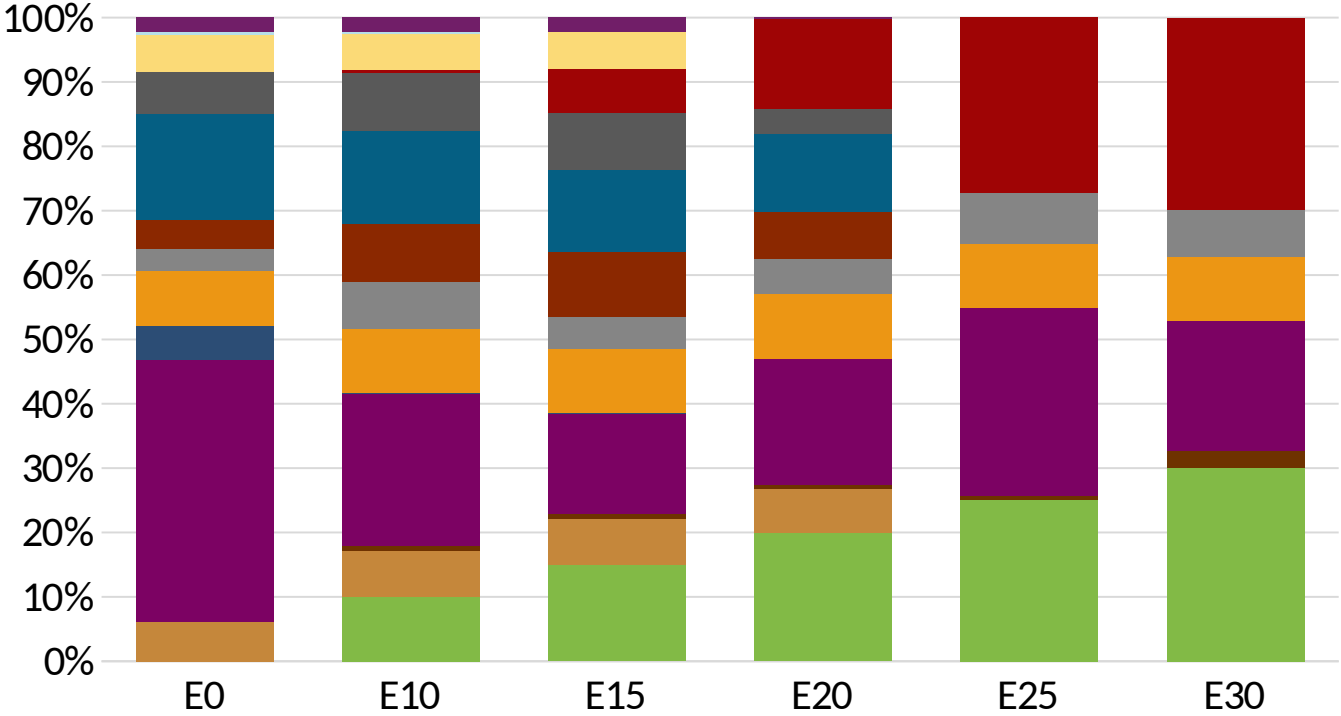
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

France Available Blending Components



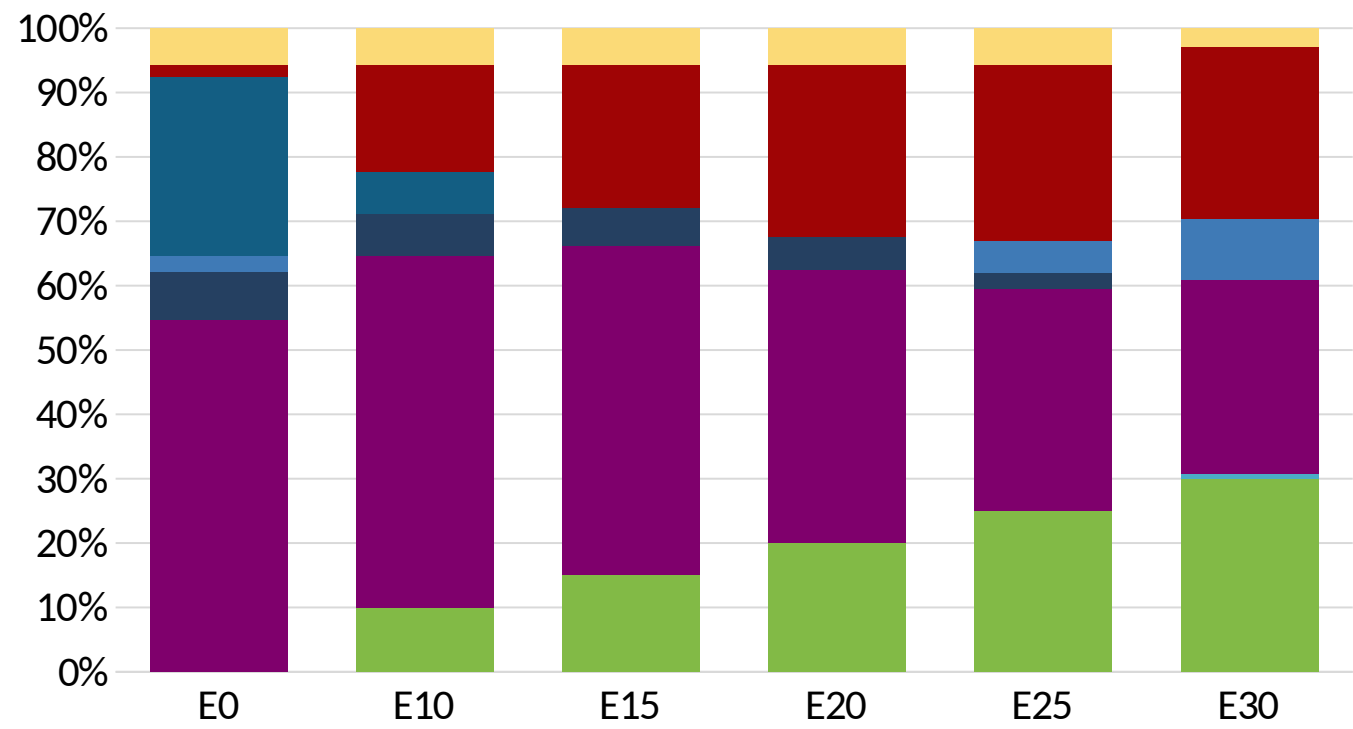
France – Gasoline 95 – Constant Octane



Average CIF 2025
Prices

Source:
Faro90

France - Gasoline 98 - Constant Octane

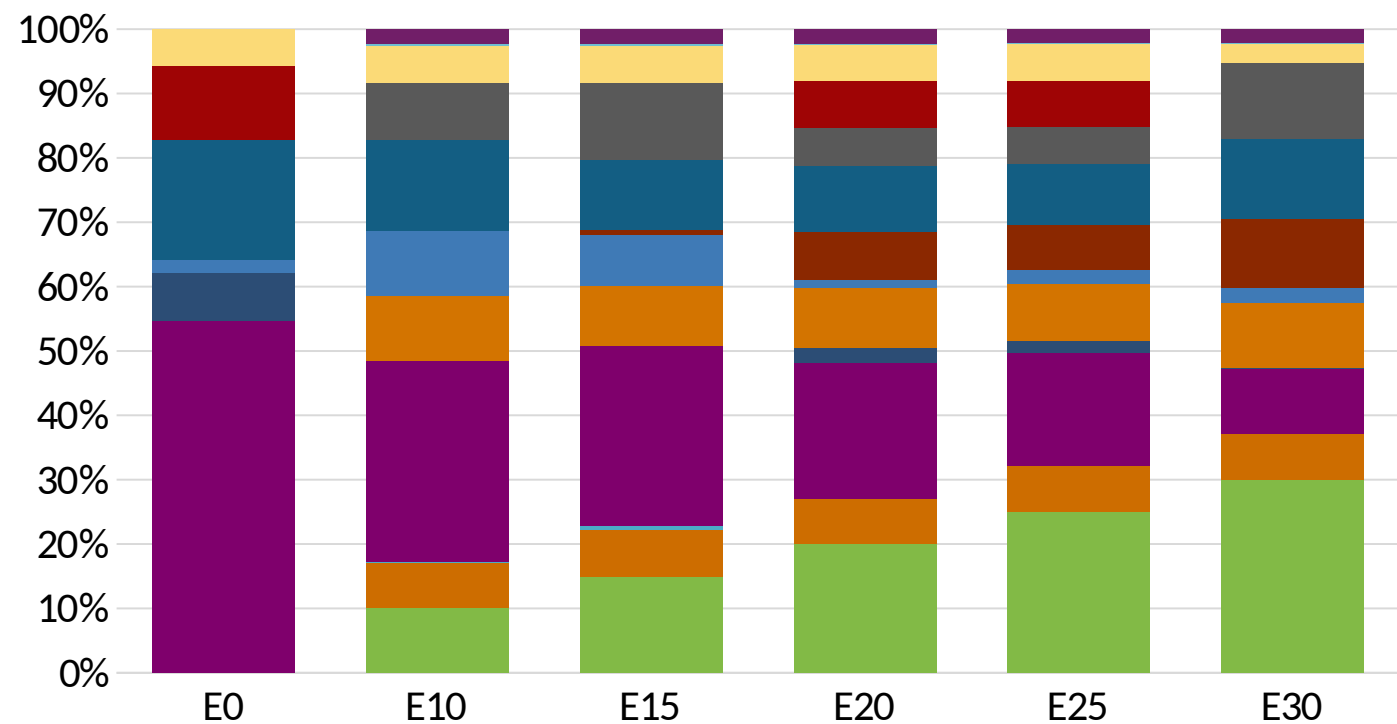


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.143	\$ 2.019	\$ 1.948	\$ 1.873	\$ 1.795	\$ 1.721

Average CIF 2025 Prices

Source:
Faro90

France – Gasoline 95 – Increased Octane

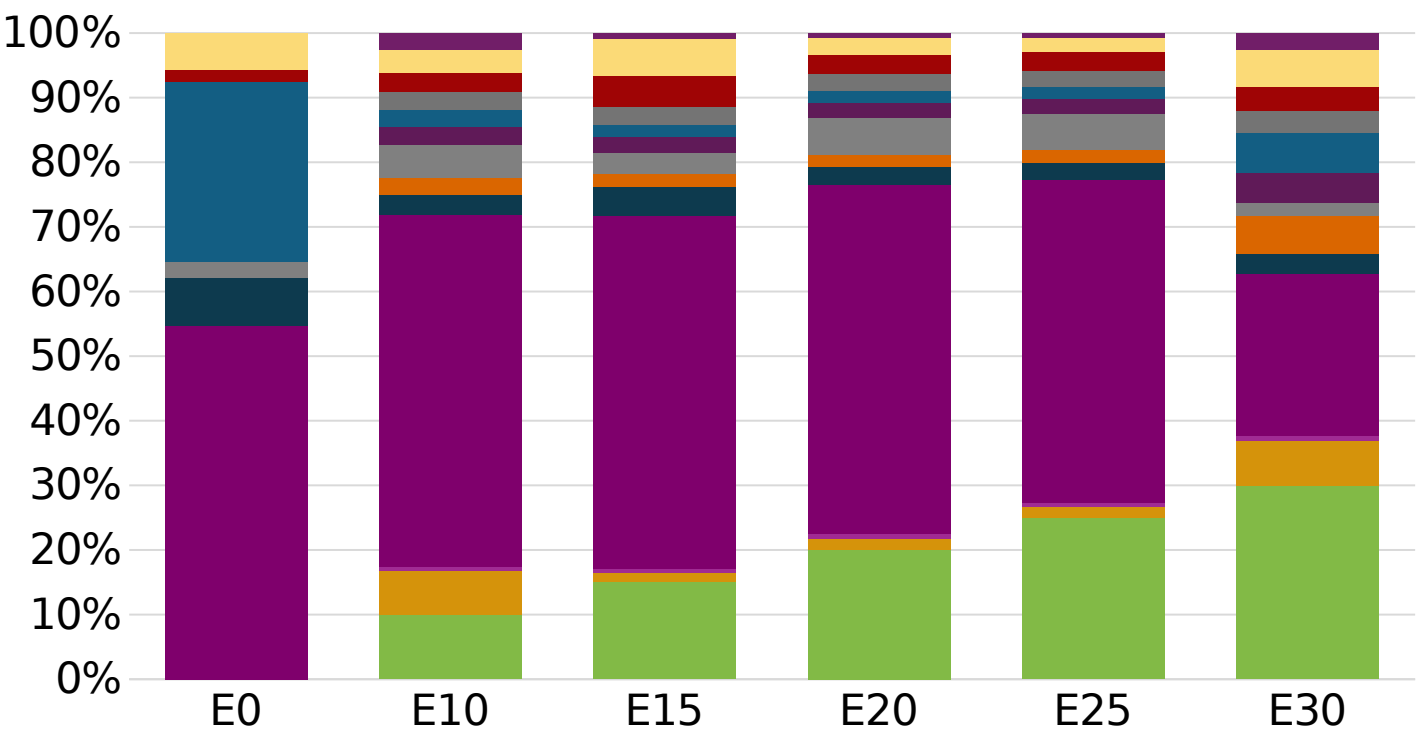


Average CIF 2025
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	101.7
Price (US\$/gal)	\$ 2.009	\$ 2.009	\$ 2.009	\$ 2.009	\$ 2.009	\$ 2.009

Source:
Faro90

France – Gasoline 98 – Increased Octane

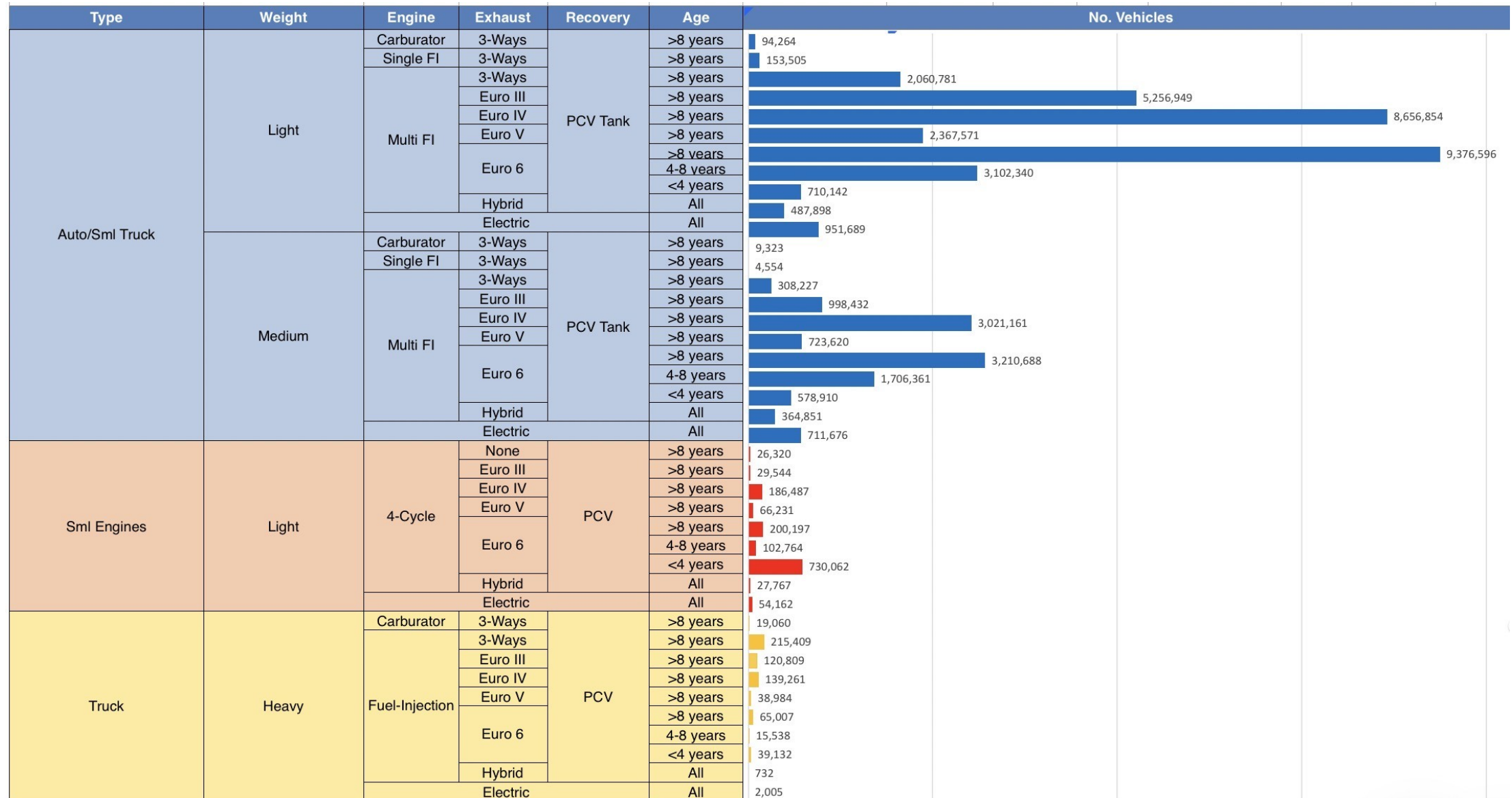


Average CIF 2025
Prices

Octane (RON)	98.0	99.3	101.4	102.4	103.7	104.6
Price (US\$/gal)	\$ 2.143	\$ 2.143	\$ 2.143	\$ 2.143	\$ 2.143	\$ 2.143

Source:
Faro90

Gasoline Vehicle Fleet - France

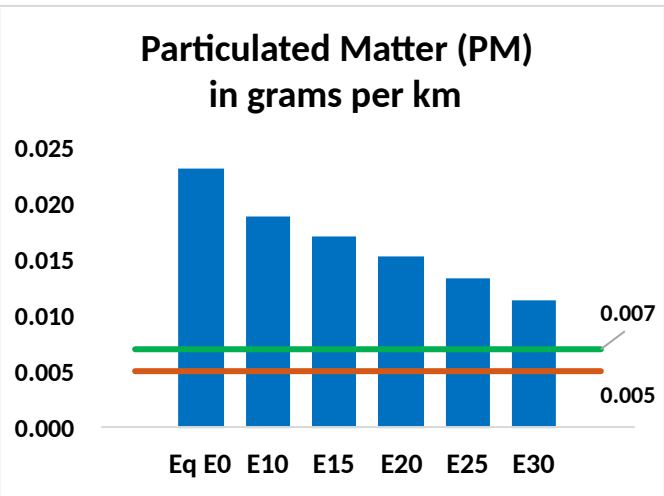
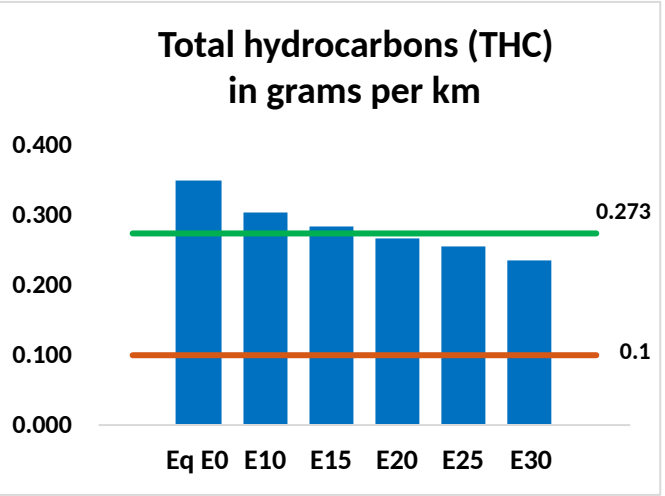
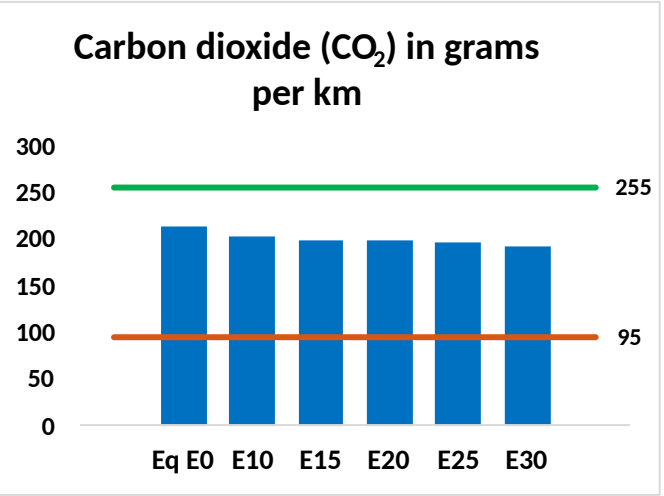
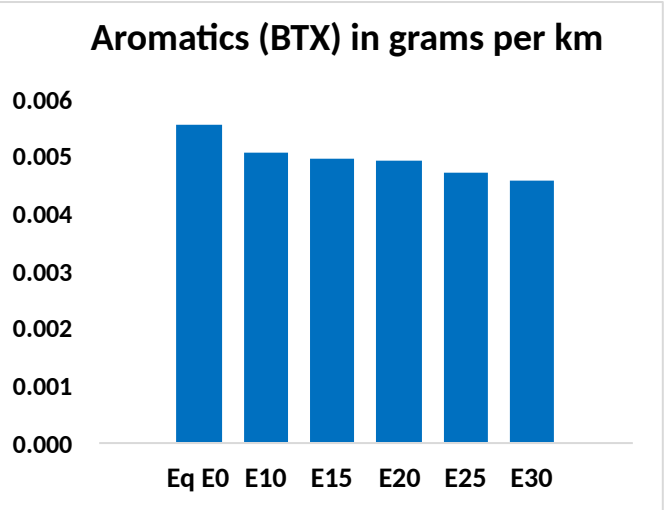
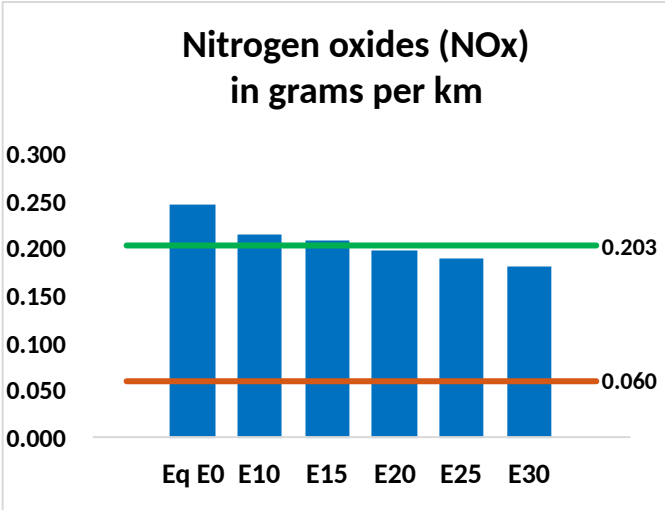
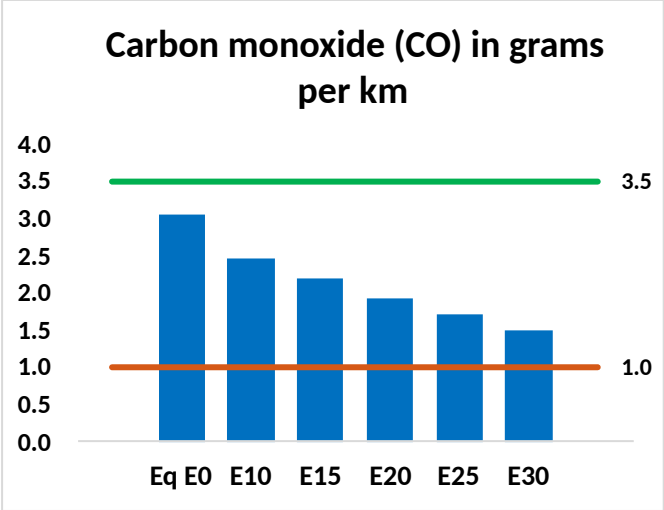
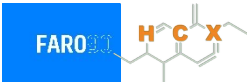


Vehicle Fleet: 46,878,458
 Gasoline Vehicle Fleet:
 22,915,839

Source: Eurostat, ACEA

France – Vehicular Emissions

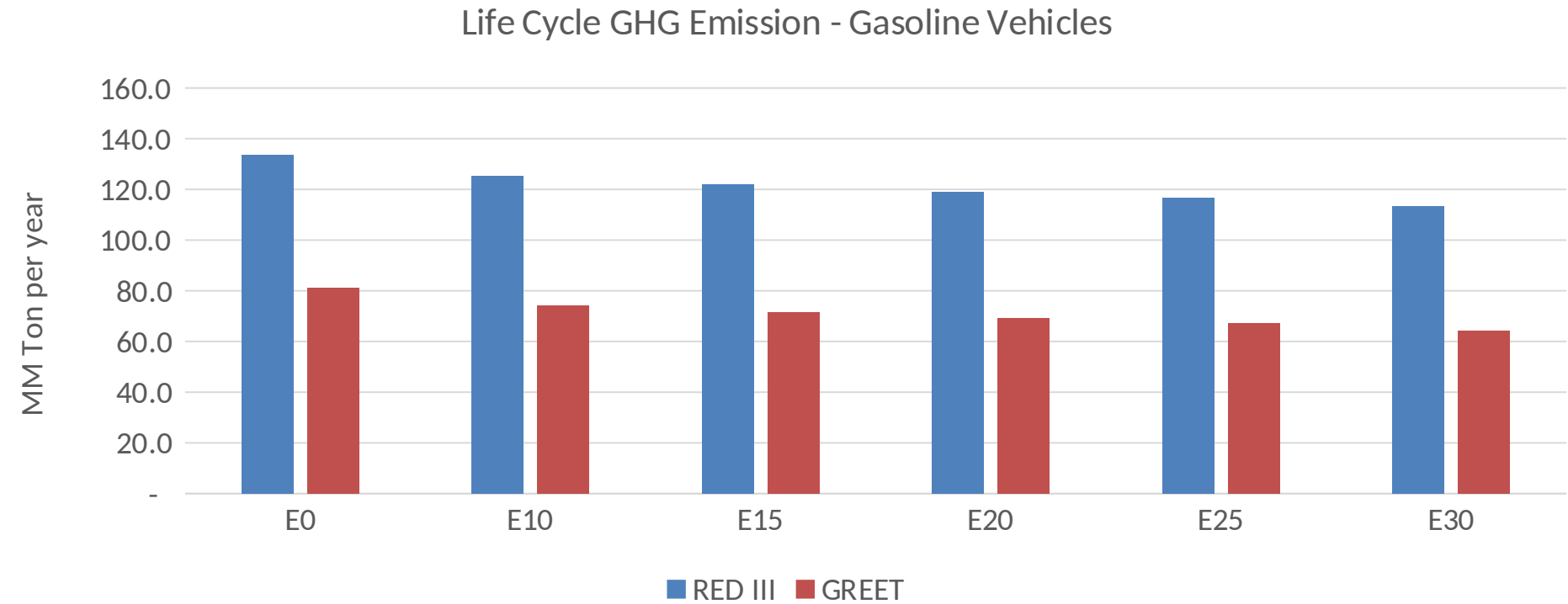
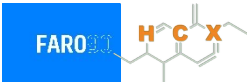
TIER III BIN 125 United States
Euro 6 Standard



France – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,829,288	1,646,298	1,463,308	1,305,089	1,142,065	1,015,561	884,729	-20%	-38%
CO2	128,233,648	124,936,211	121,821,966	119,372,703	118,160,712	117,444,650	115,279,808	-5%	-8%
VOC	208,099	194,530	180,961	170,315	159,820	151,816	140,903	-13%	-23%
NOx	147,567	136,254	128,697	124,955	118,807	114,044	108,400	-13%	-19%
SOx	1,484	1,372	1,260	1,165	1,074	975	871	-15%	-28%
NH3	44,156	43,719	43,282	43,487	43,976	44,320	44,606	-2%	0%
N2O	1,711	1,702	1,695	1,703	1,738	1,758	1,778	-1%	2%
CH4	44,539	44,539	44,539	45,207	46,187	47,033	47,835	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,783	1,709	1,634	1,586	1,542	1,511	1,463	-8%	-13%
Acetaldehyde	4,009	4,602	6,752	10,282	14,009	16,007	18,915	68%	249%
Formaldehyde	13,095	12,732	14,841	17,427	18,258	20,198	21,987	13%	39%
Benzene	3,327	3,197	3,029	2,981	2,957	2,828	2,736	-9%	-11%
PM2.5	7,507	6,825	6,142	5,542	4,954	4,324	3,665	-18%	-34%
PM10	6,379	5,799	5,219	4,709	4,209	3,674	3,114	-18%	-34%

France – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	331.1
Target @ 2035 (MMT/year)	186.4
Delta	144.7

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

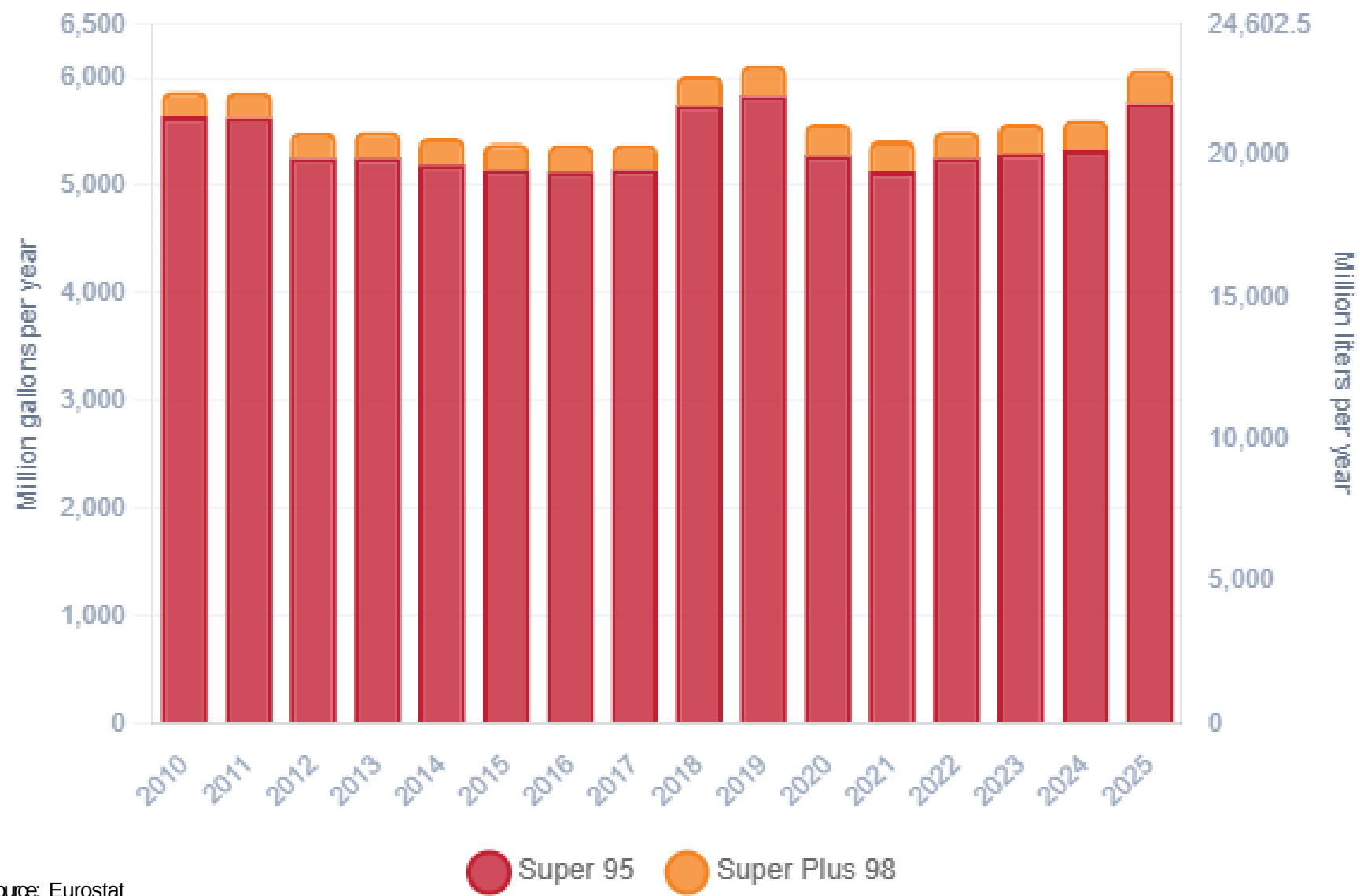
Ethanol Gasoline Blending in Germany



In 2025, gasoline consumption in Germany was 23,067 million liters (6,093 million gallons) and growth rate was 0.23%. Gasoline production and net imports were 25,530 and -2,754 million liters (6,744 and -727 million gallons) correspondingly. Germany complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

In 2025, ethanol consumption in Germany was 1,548 million liters (409 million gallons) representing 7% of gasoline demand. Ethanol production and Net Imports were 1,001 and 547 million liters (265 and 145 million gallons) correspondingly. Ethanol sources are Wheat 47%, Sugar Beet 20%, Corn 11% and Other Cereals 21%.

Gasoline Demand in Germany



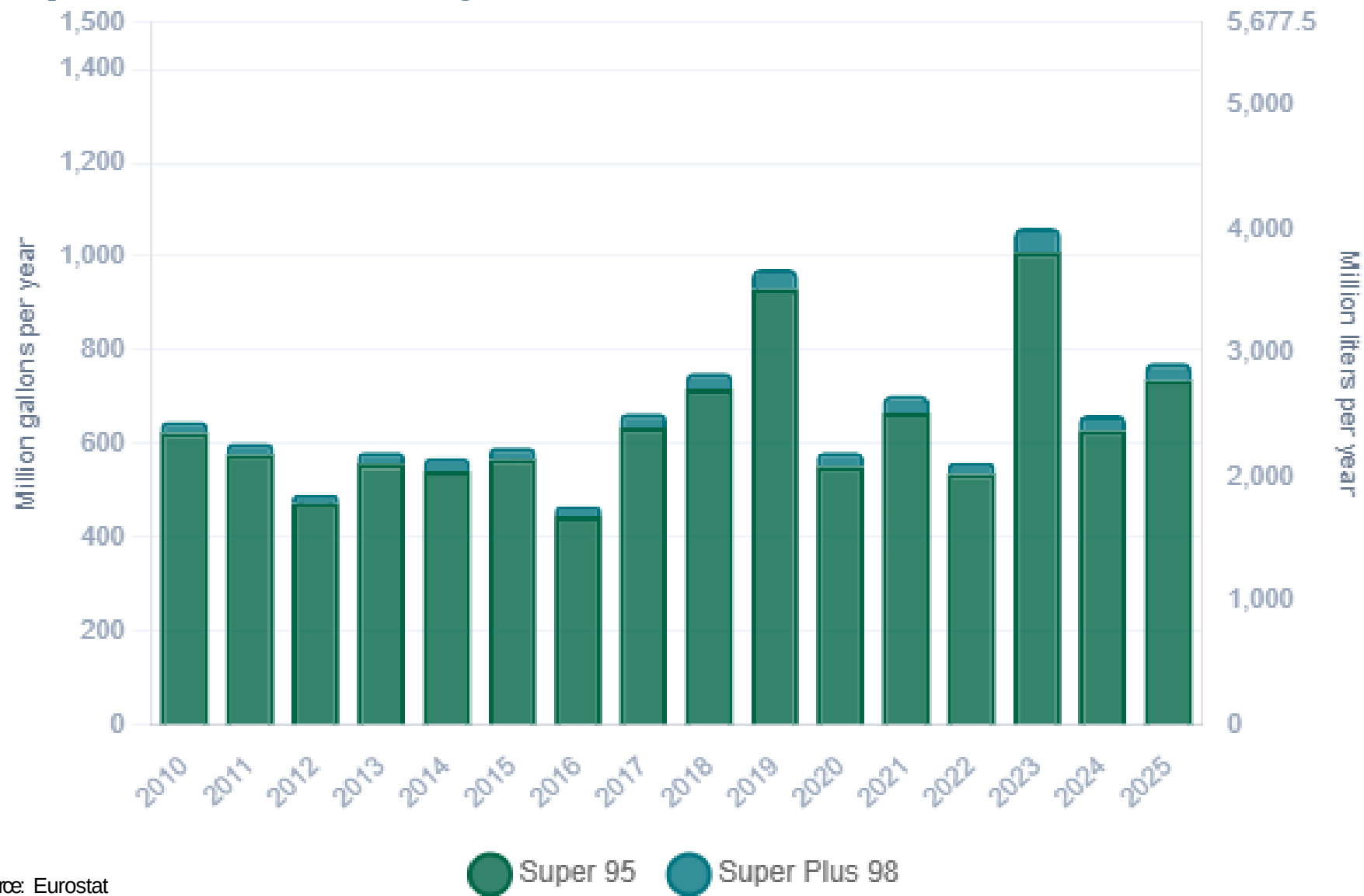
Source: Eurostat

Gasoline Production in Germany



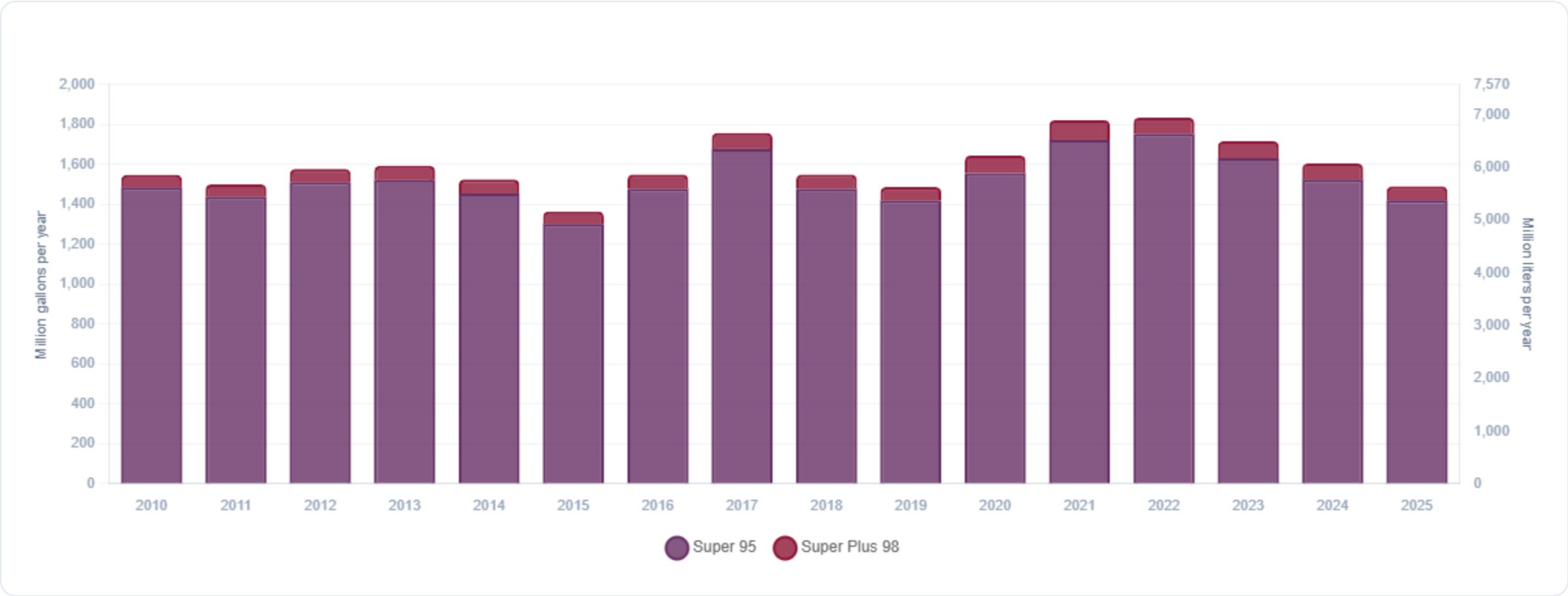
Source: Eurostat

Gasoline Imports in Germany



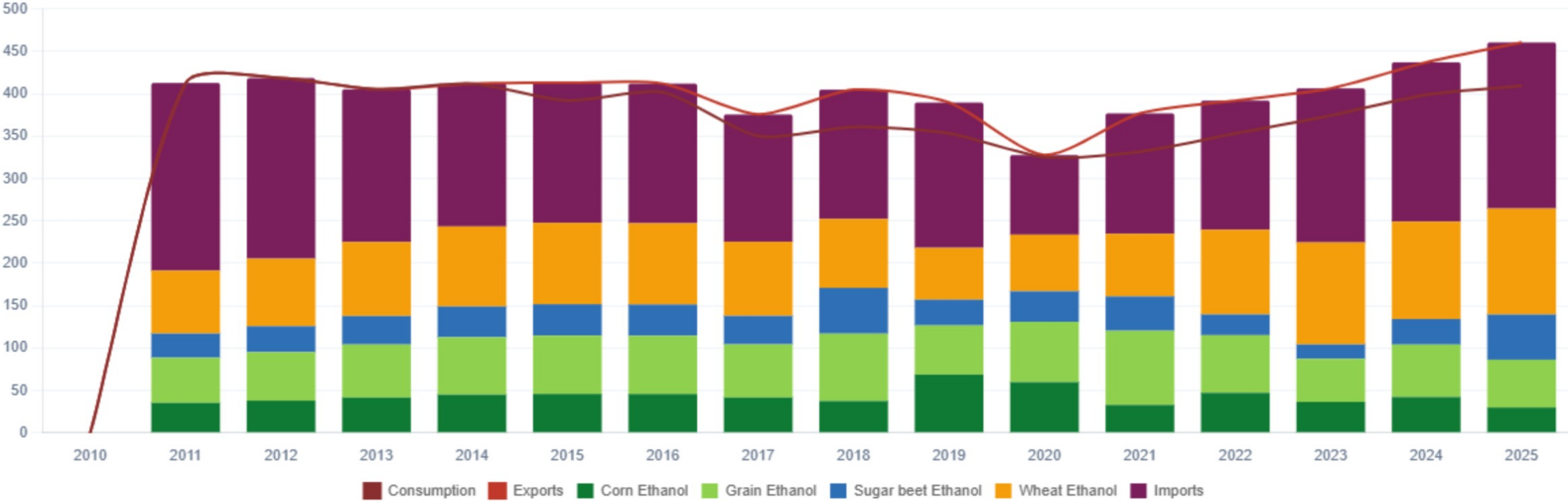
Source: Eurostat

Gasoline Exports in Germany



Source: Eurostat

Ethanol Balance in Germany



Source: Eurostat

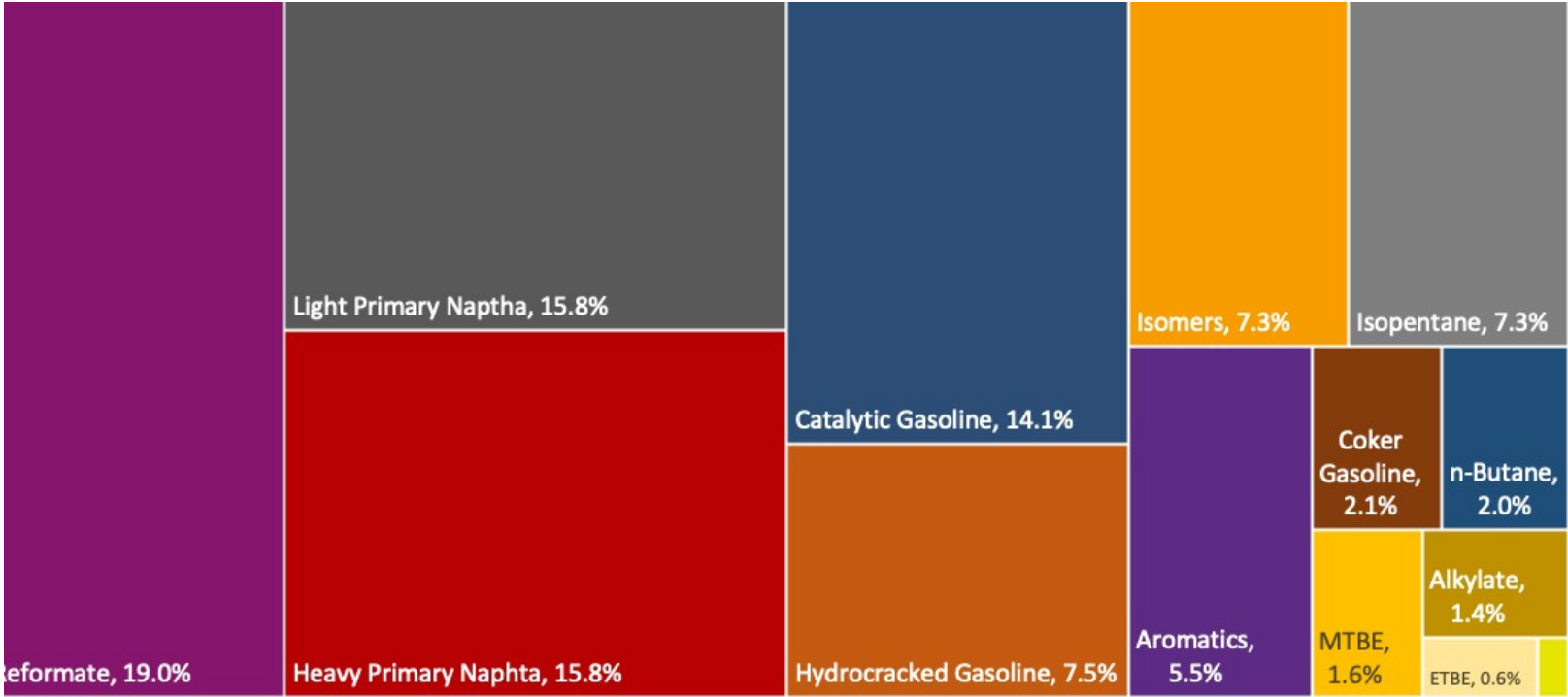
Gasoline Quality in Germany

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Countries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.7							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	10.5							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

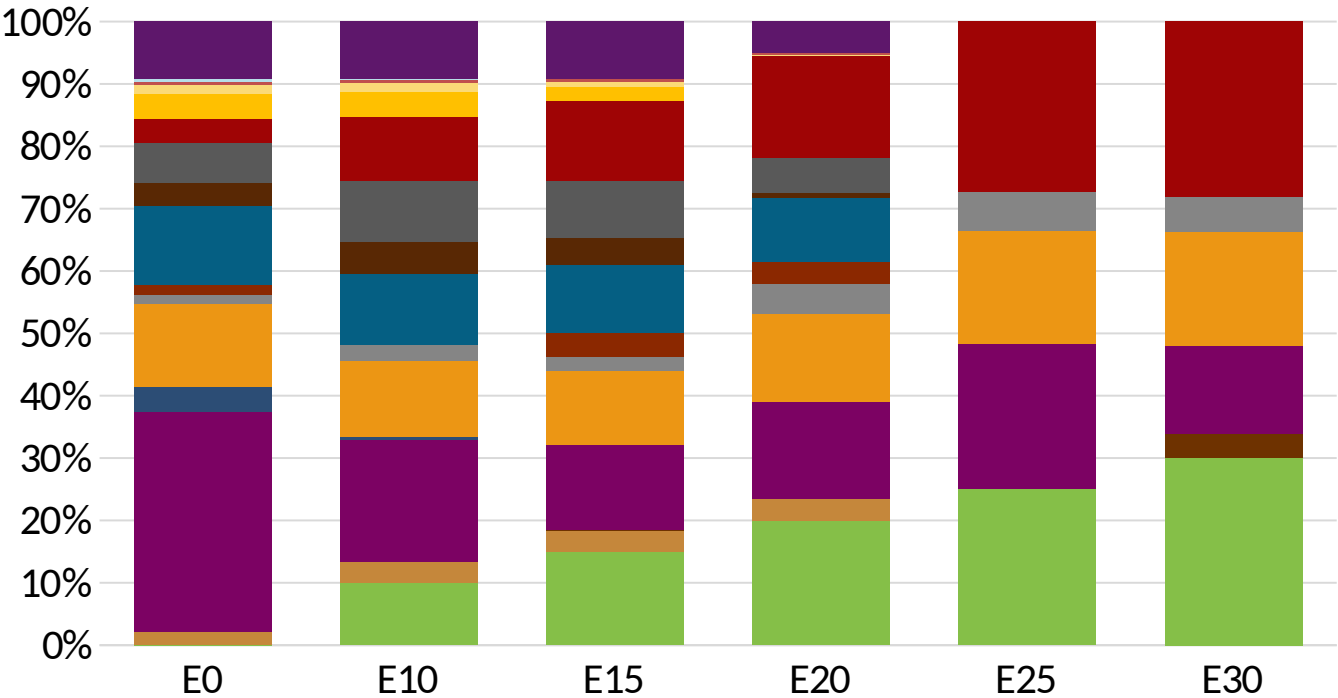
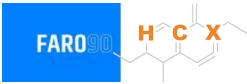
Source: European Commission

Germany Available Blending Components



TAME,
0.2%

Germany – Gasoline 95 – Constant Octane

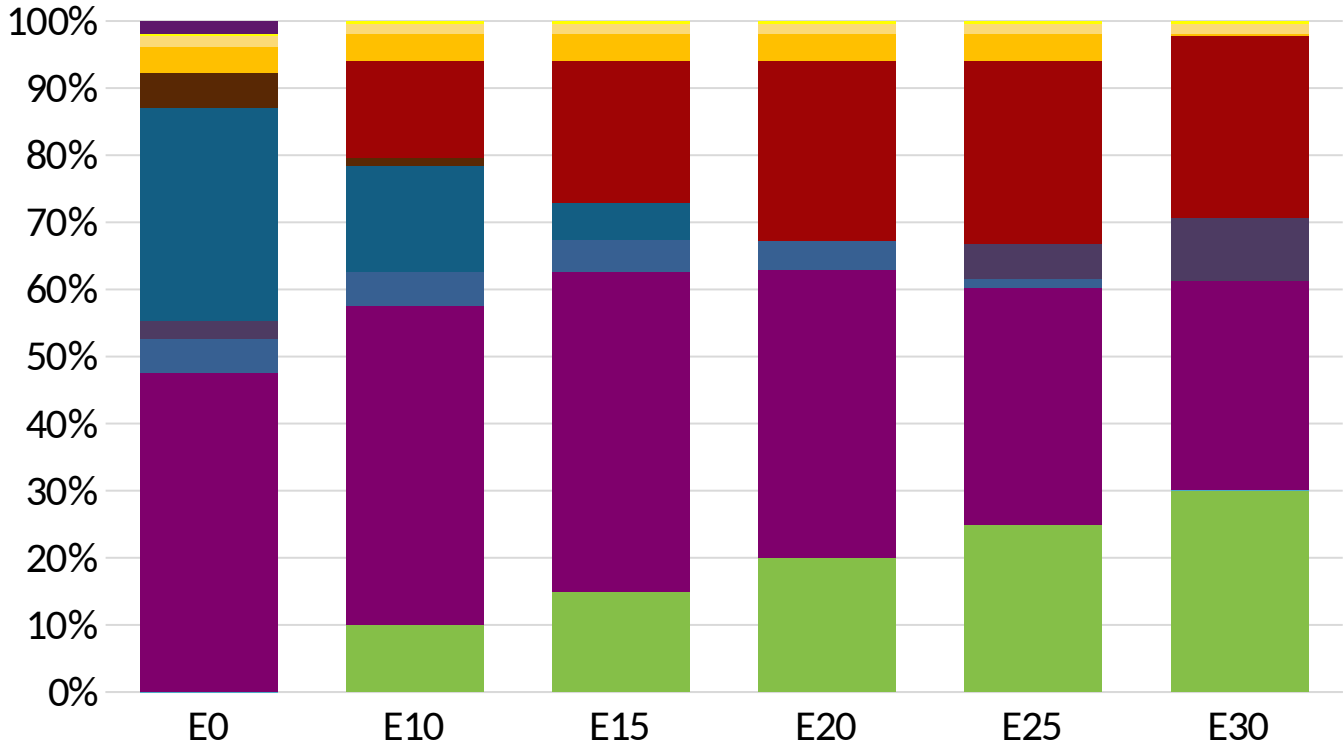


Average CIF 2025 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.069	\$ 1.941	\$ 1.880	\$ 1.799	\$ 1.677	\$ 1.612

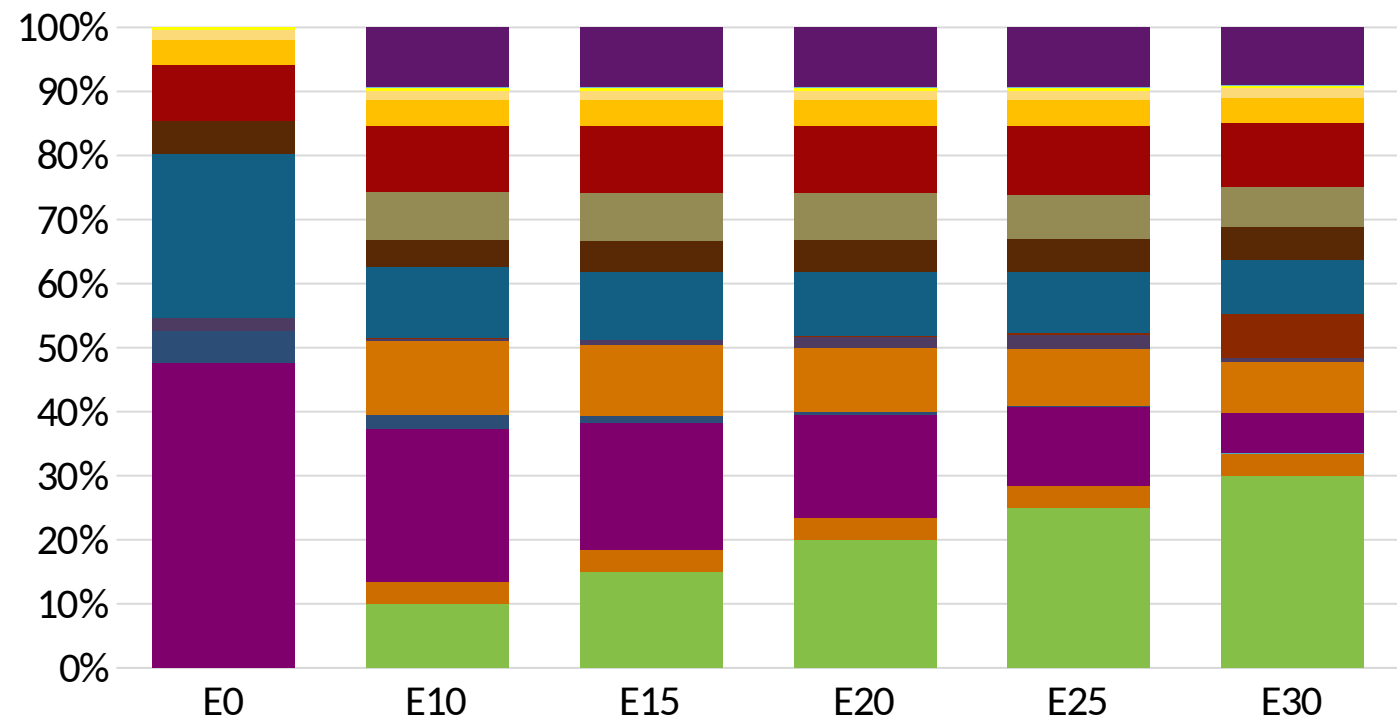
Source:
Faro90

Germany - Gasoline 98 - Constant Octane



Average CIF 2025
Prices

Germany – Gasoline 95 – Increased Octane

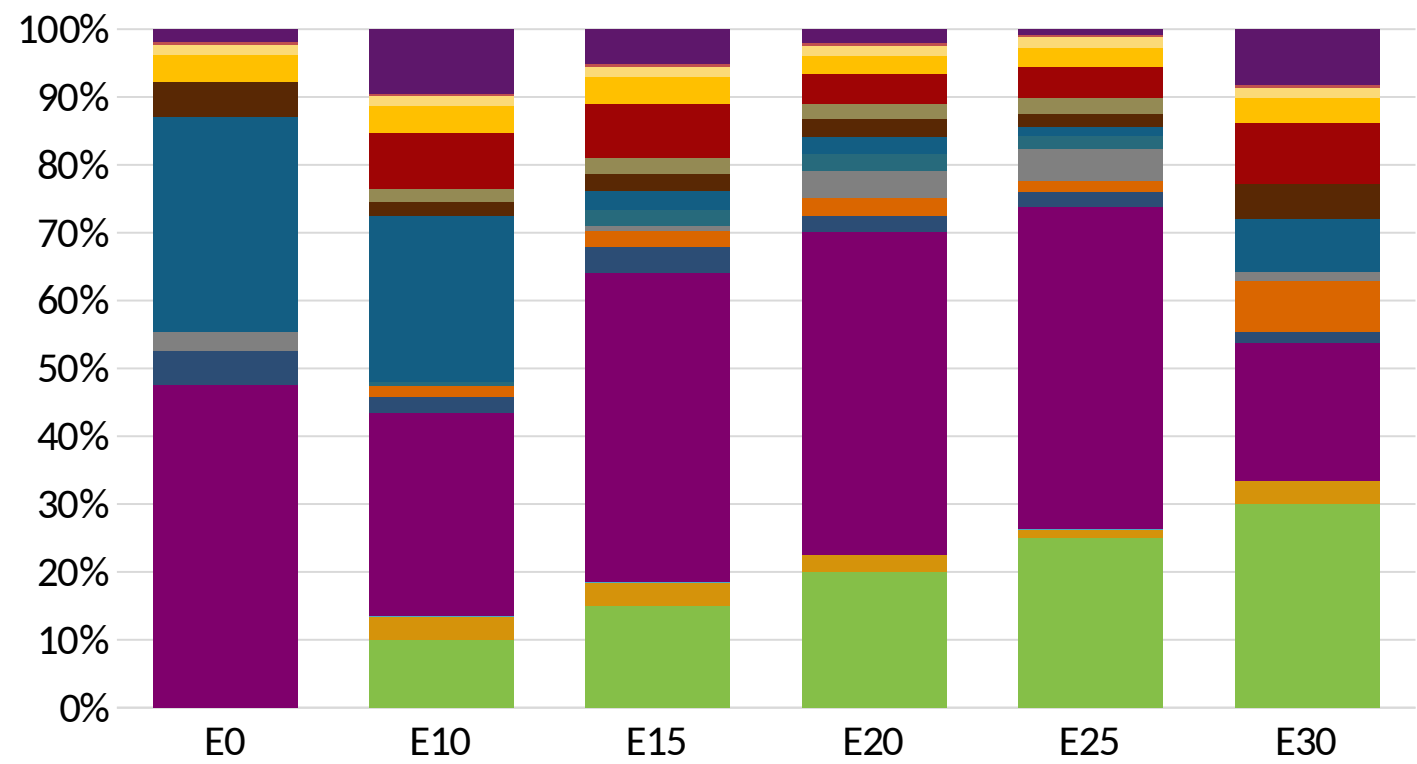


Average CIF 2025
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.006	\$ 2.006	\$ 2.006	\$ 2.006	\$ 2.006	\$ 2.006

Source:
Faro90

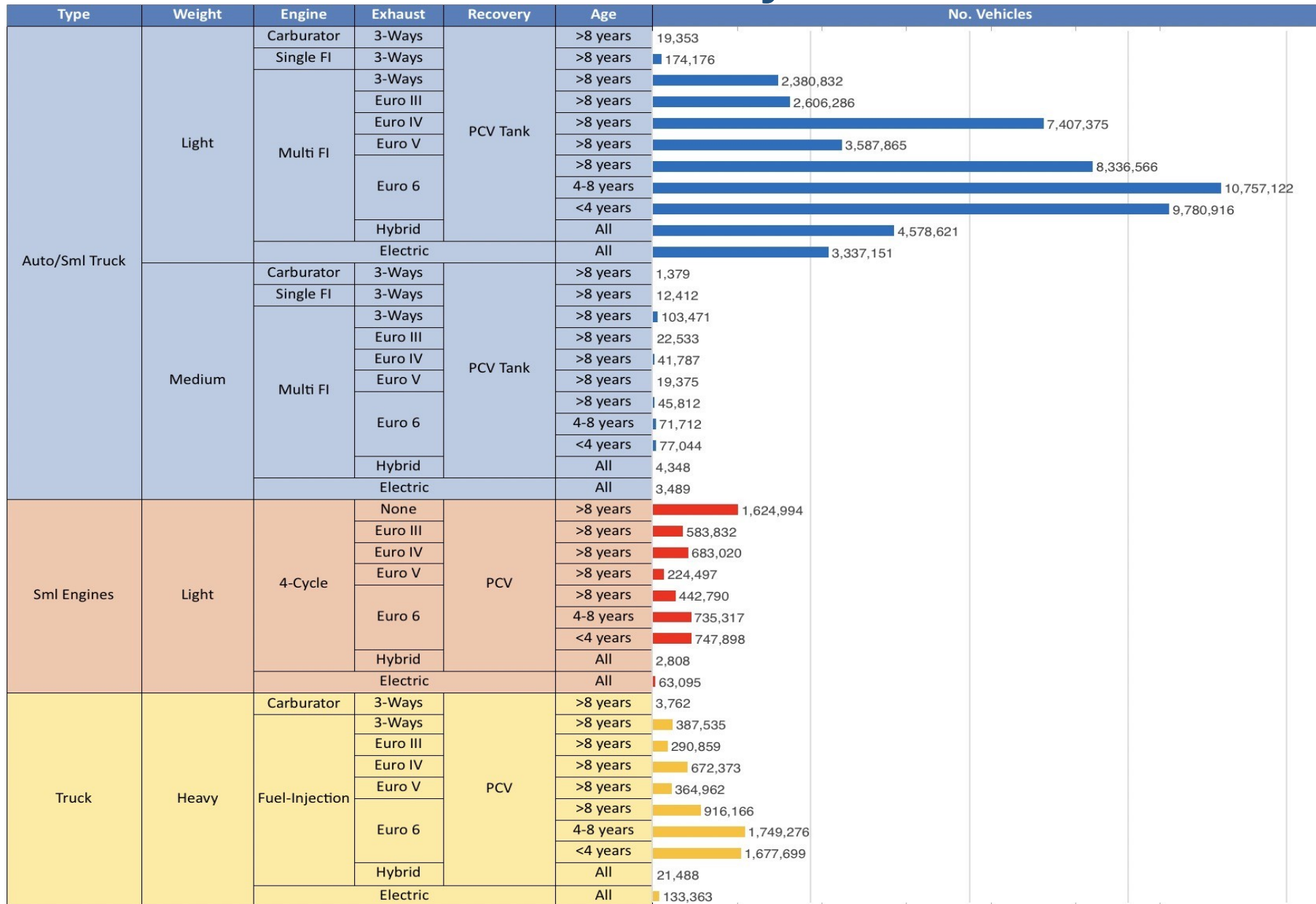
Germany – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Source:
Faro90

Gasoline Vehicle Fleet - Germany

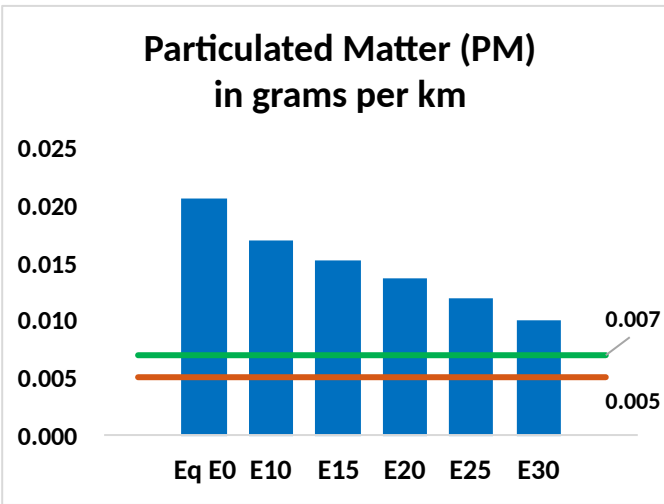
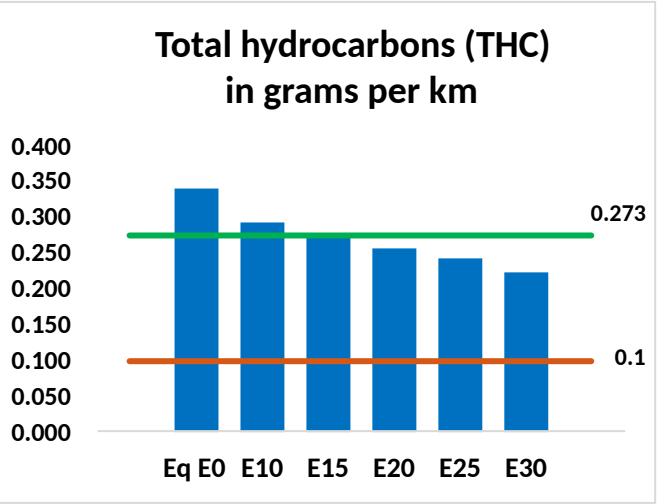
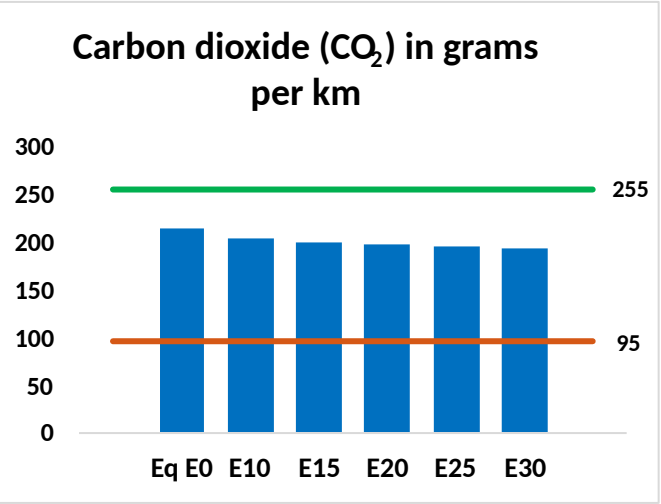
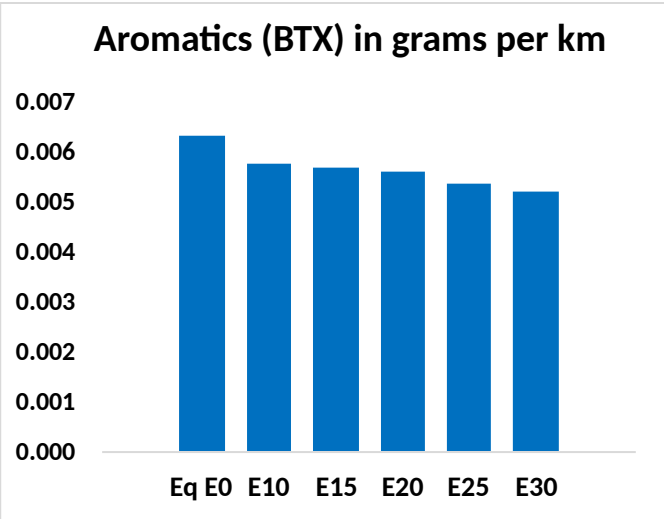
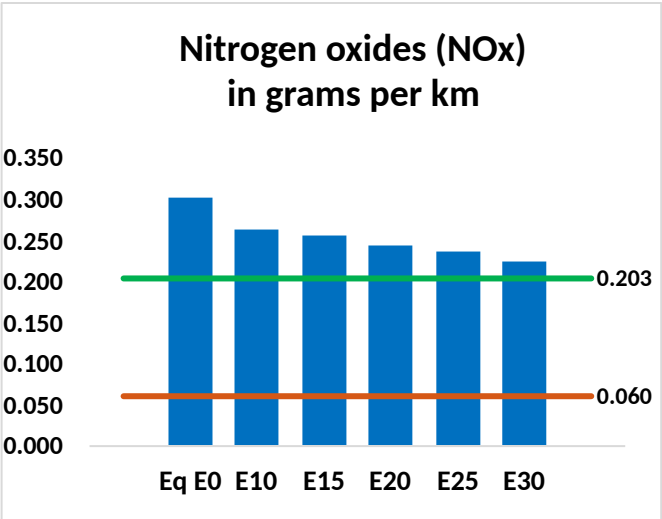
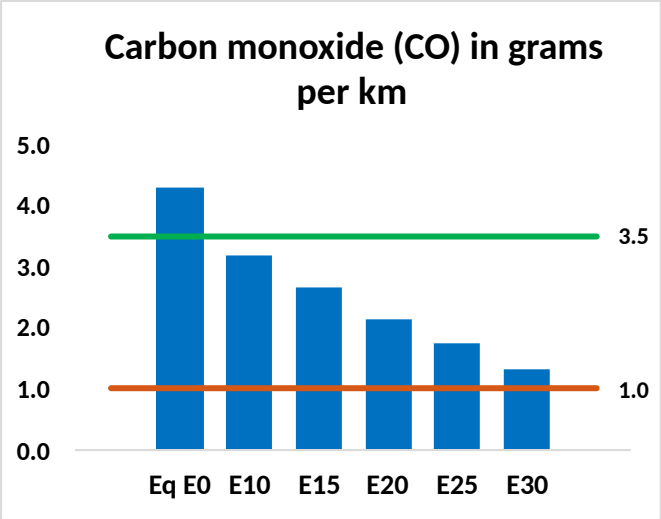
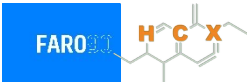


Vehicle Fleet: 64,695,361
Gasoline Vehicle Fleet: 36,155,376

Source: Eurostat, ACEA

Germany – Vehicular Emissions

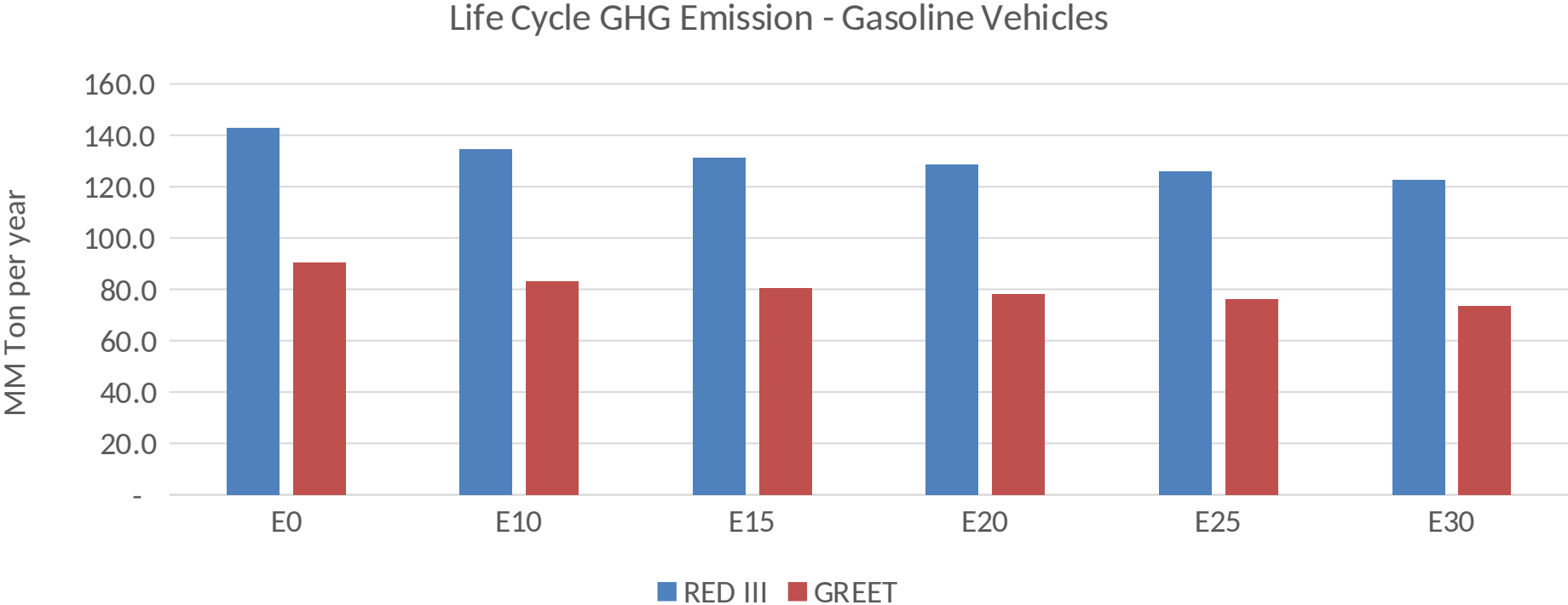
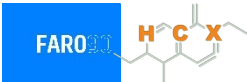
TIER III BIN 125 United States
Euro 6 Standard



Germany – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	3,965,507	3,448,118	2,930,729	2,465,280	1,977,796	1,597,190	1,219,777	-26%	-50%
CO2	197,042,612	191,975,802	187,190,481	183,426,967	181,564,633	180,472,100	177,145,480	-5%	-8%
VOC	313,299	291,576	269,852	252,511	235,273	222,082	204,253	-14%	-25%
NOx	277,237	255,982	242,400	236,002	225,116	216,850	206,951	-13%	-19%
SOx	2,209	2,042	1,875	1,734	1,598	1,451	1,297	-15%	-28%
NH3	66,276	65,619	64,963	65,272	66,006	66,522	66,951	-2%	0%
N2O	2,817	2,802	2,791	2,804	2,862	2,894	2,927	-1%	2%
CH4	66,970	66,970	66,970	67,975	69,448	70,720	71,926	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	3,152	2,955	2,759	2,606	2,457	2,344	2,188	-12%	-22%
Acetaldehyde	5,901	6,773	9,938	15,134	20,618	23,560	27,839	68%	249%
Formaldehyde	18,239	17,732	20,671	24,272	25,428	28,131	30,623	13%	39%
Benzene	5,806	5,579	5,286	5,203	5,160	4,934	4,775	-9%	-11%
PM2.5	11,476	10,432	9,389	8,471	7,573	6,610	5,602	-18%	-34%
PM10	7,588	6,898	6,208	5,601	5,007	4,371	3,704	-18%	-34%

Germany – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	717.1
Target @ 2035 (MMT/year)	403.6
Delta	313.4

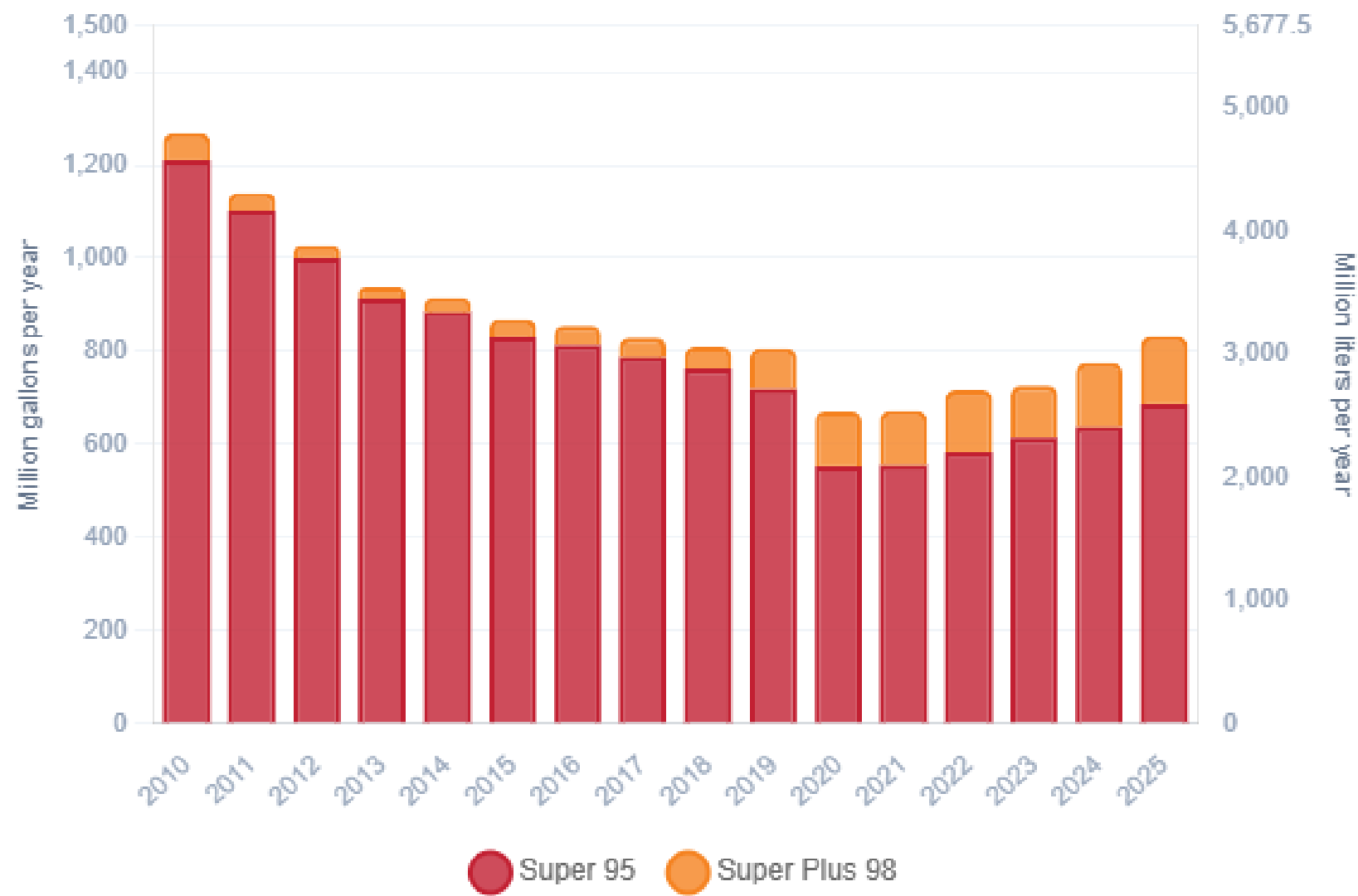
Ethanol Gasoline Blending in Greece



In 2025, gasoline consumption in Greece was 3,156 million liters (834 million gallons) and growth rate was -2.8%. Gasoline production and net imports were 7,983 and -6041 million liters (2,109 and -1,596 million gallons) correspondingly. Greece complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.5 RON.

Ethanol consumption in Greece is determined by 3.3% bioethanol content mandate (by energy). In 2025 demand was 164 million liters (43 million gallons) representing 5.0% of gasoline demand. Ethanol is supplied by imports.

Gasoline Demand in Greece



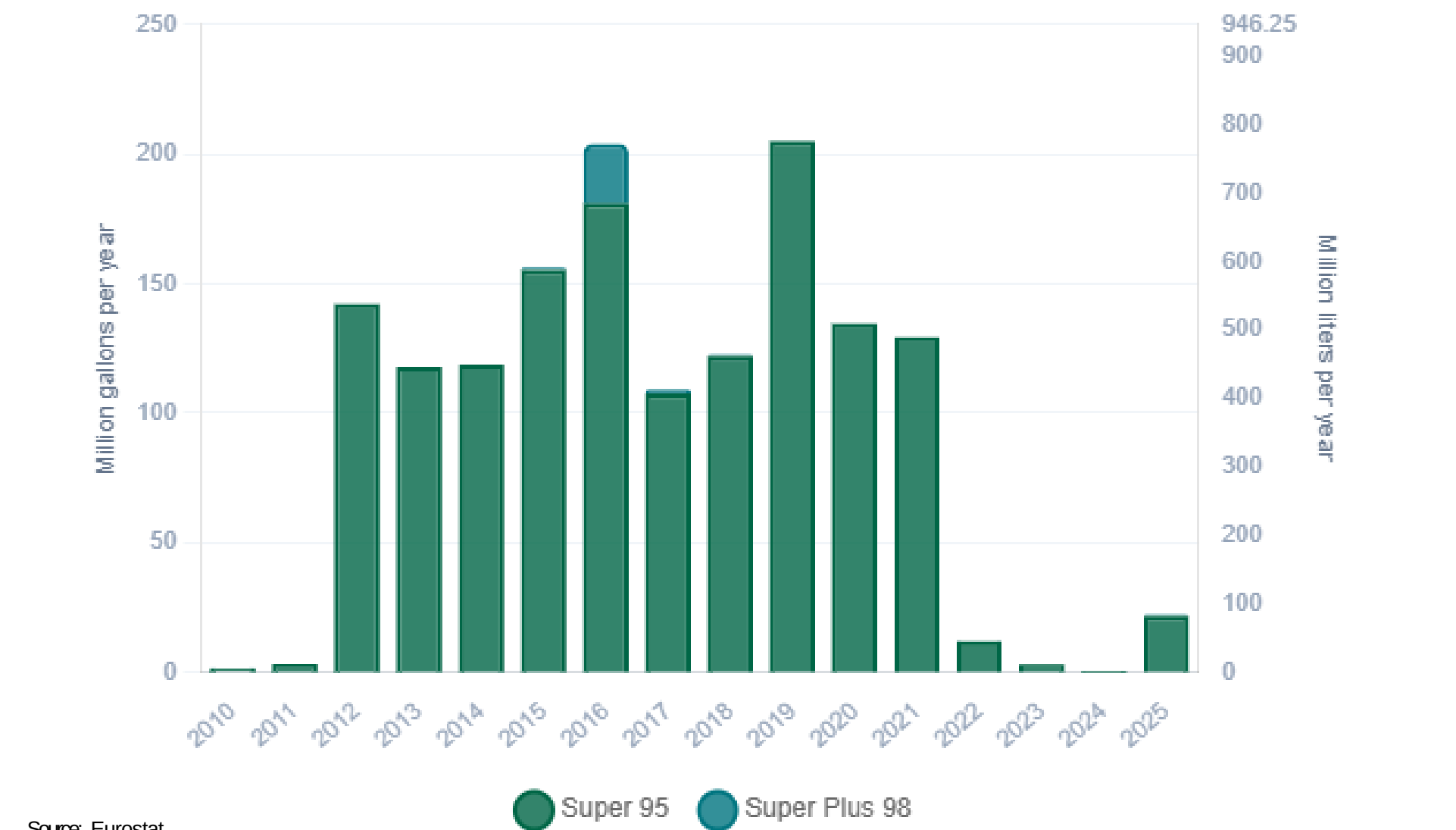
Source: Eurostat

Gasoline Production in Greece



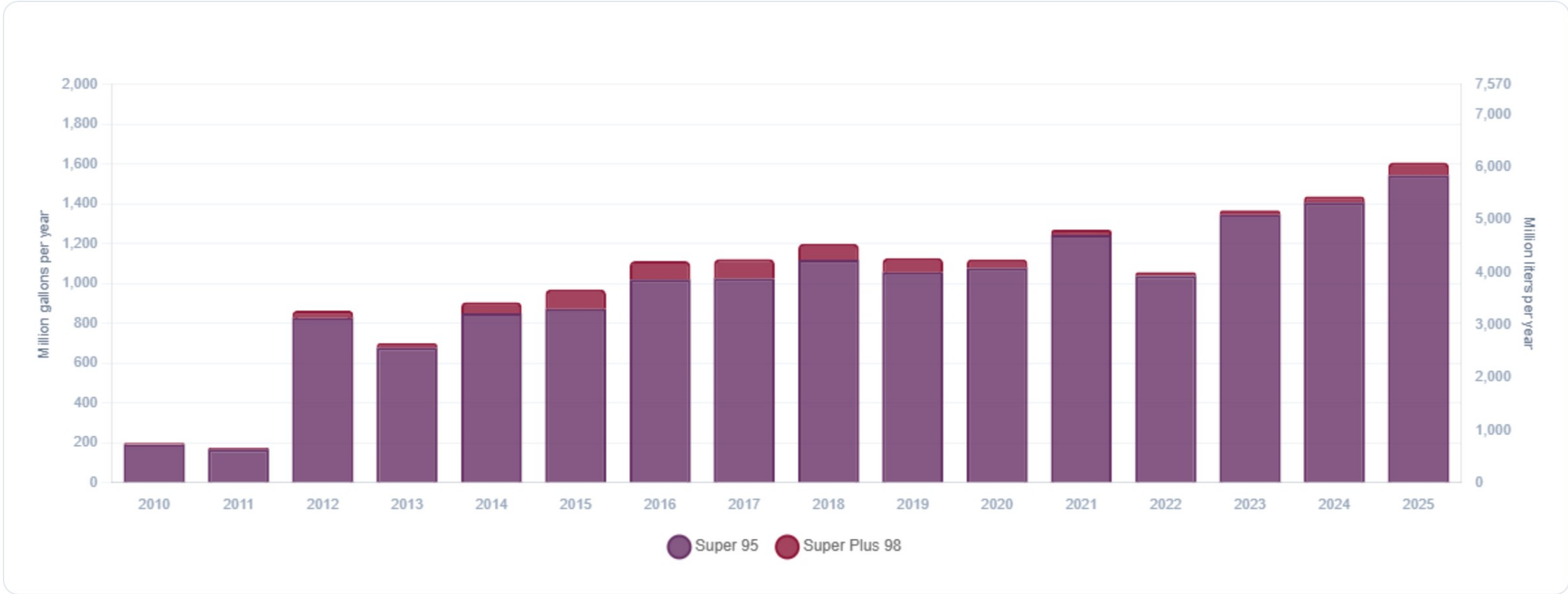
Source: Eurostat

Gasoline Imports in Greece



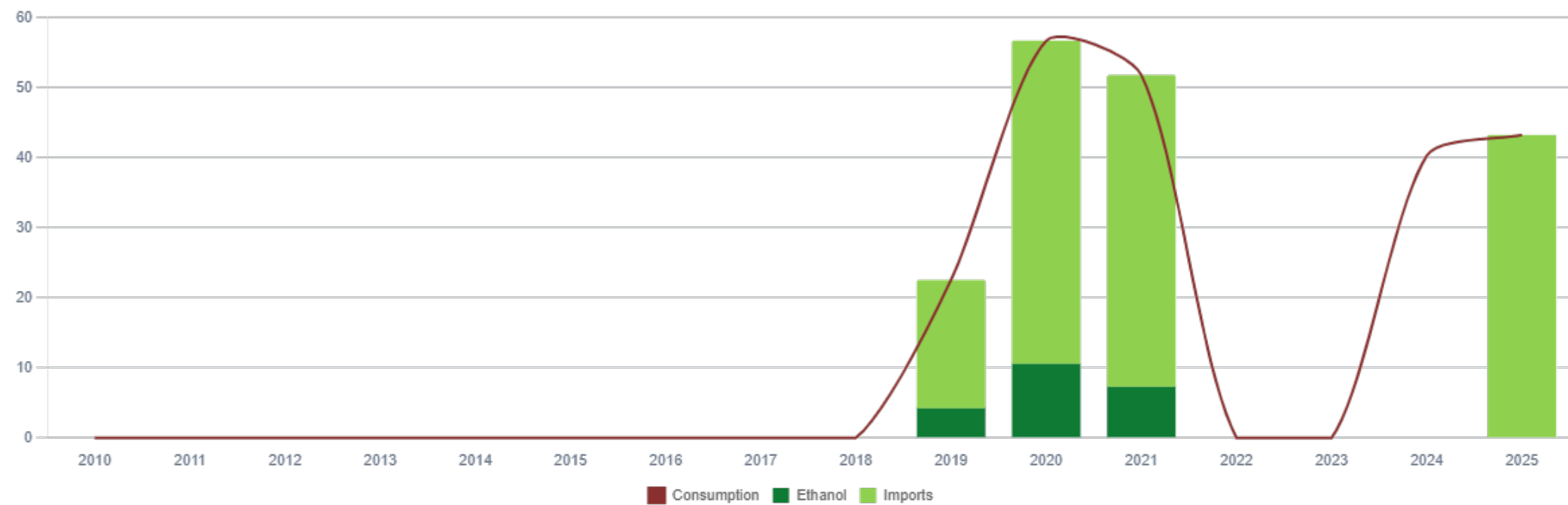
Source: Eurostat

Gasoline Exports in Greece



Source: Eurostat

Ethanol Balance in Greece



Source: Eurostat, USDA – Fuel Mandates in EU

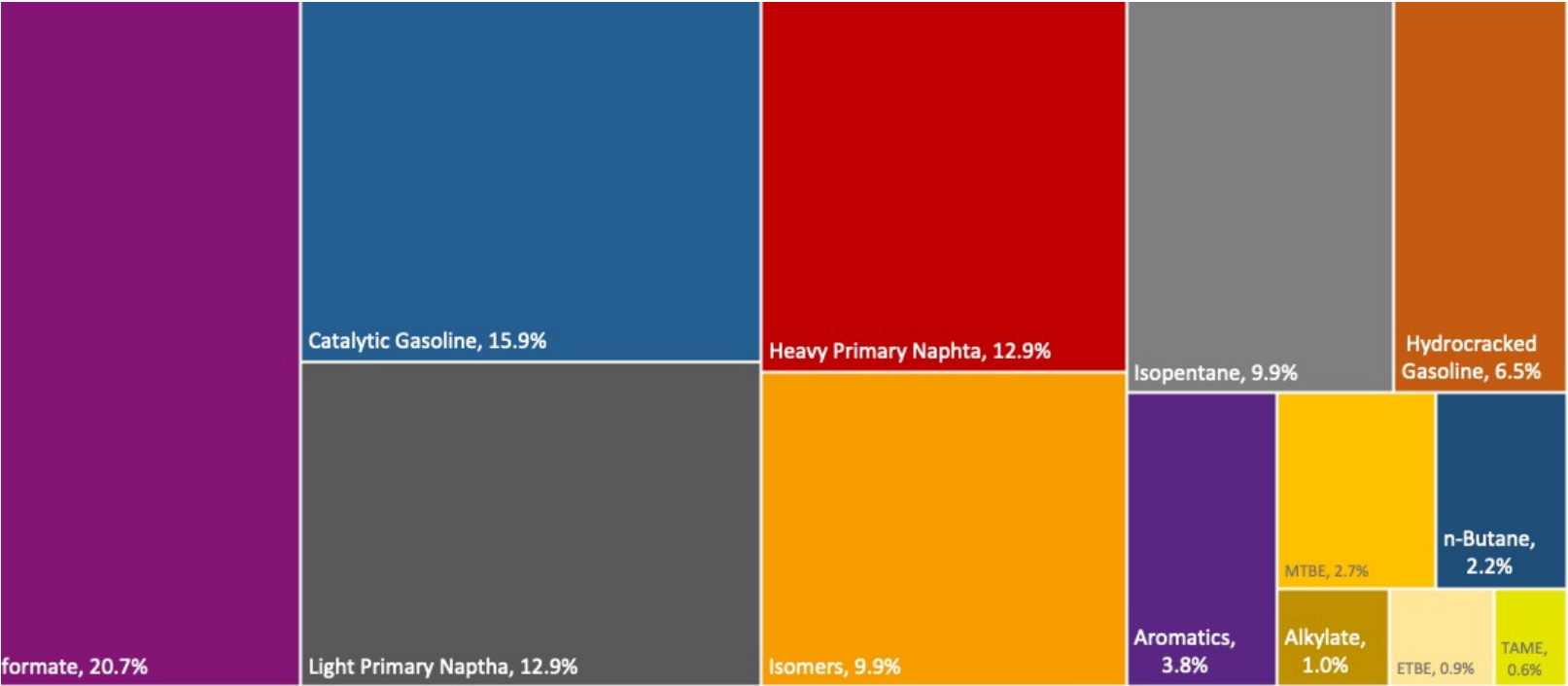
Gasoline Quality in Greece

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.0							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	3.3							
GHG Emission Reduction (%)	-							

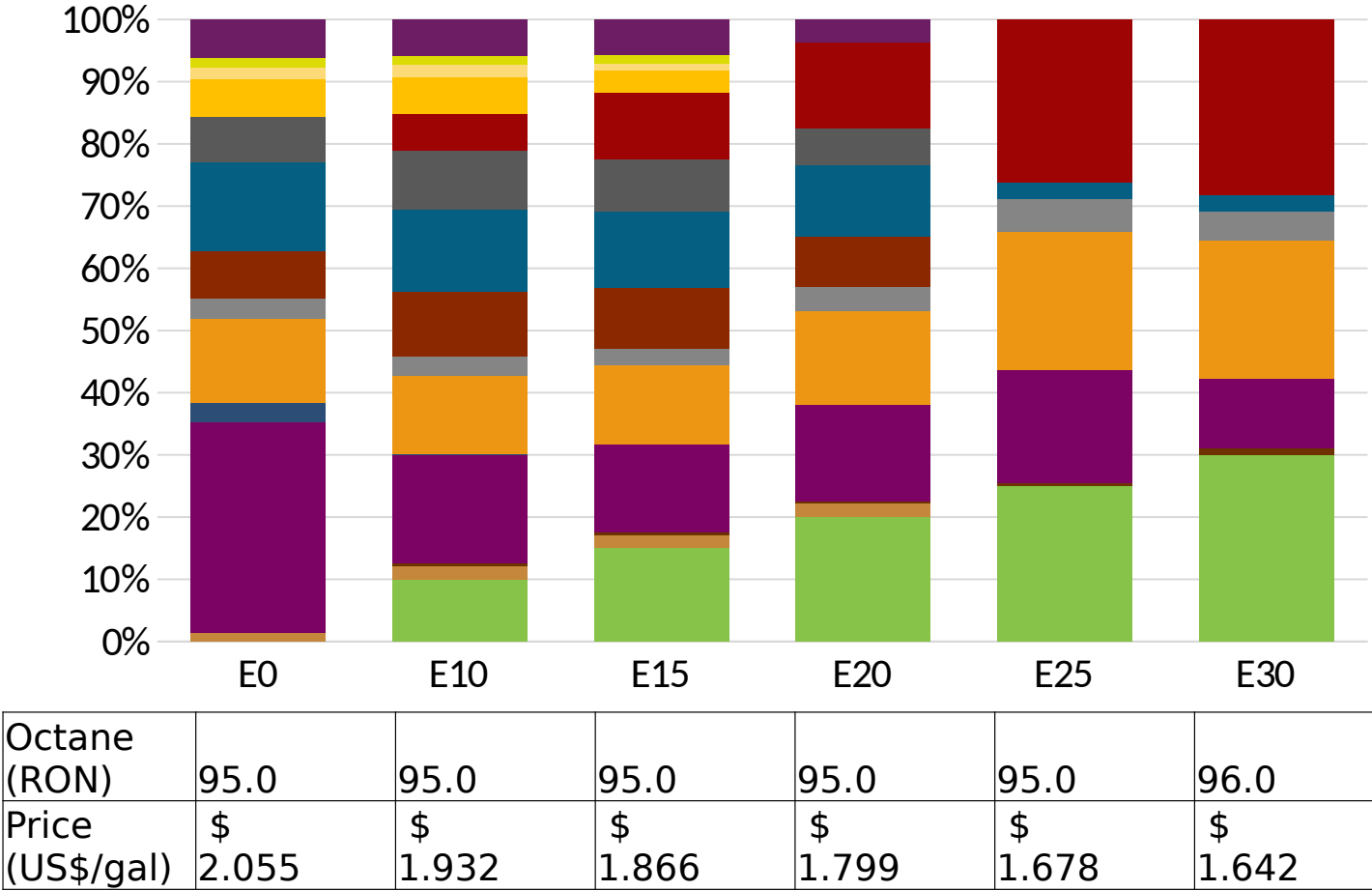
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Greece Available Blending Components



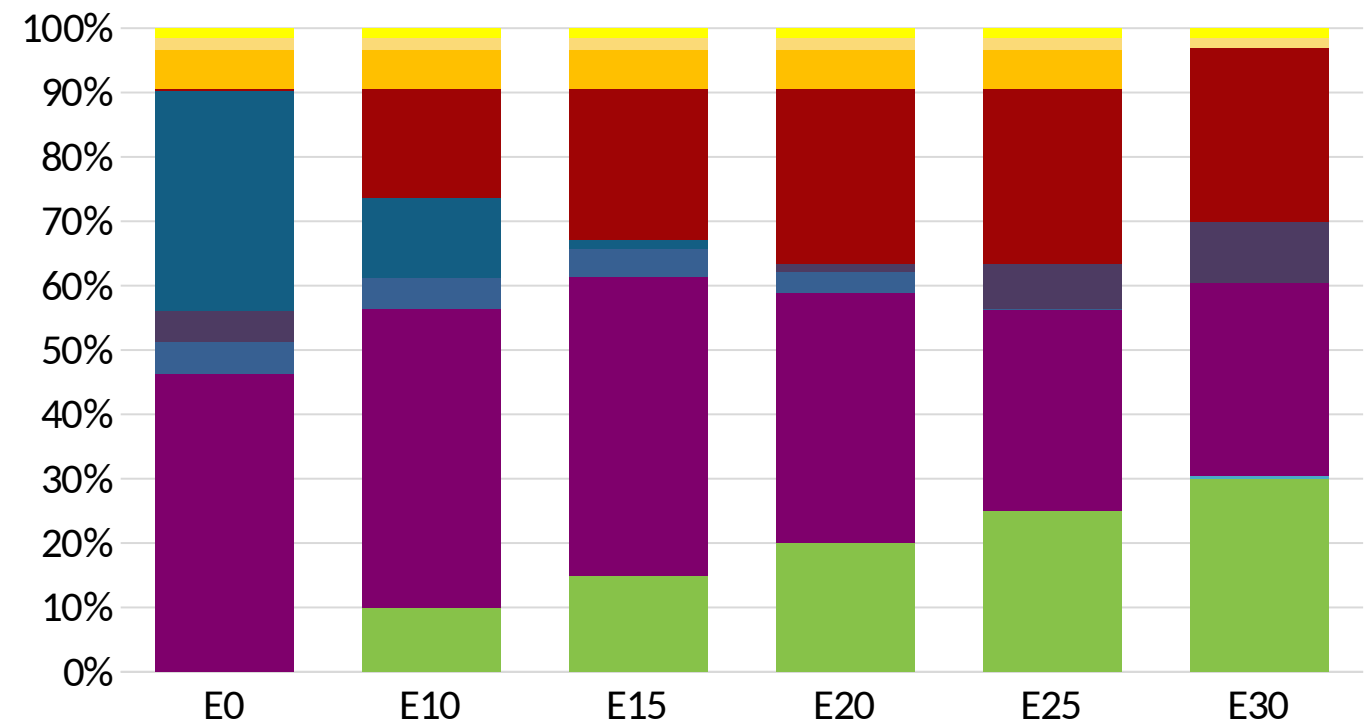
Greece – Gasoline 95 – Constant Octane



Average CIF 2025 Prices

Source:
Faro90

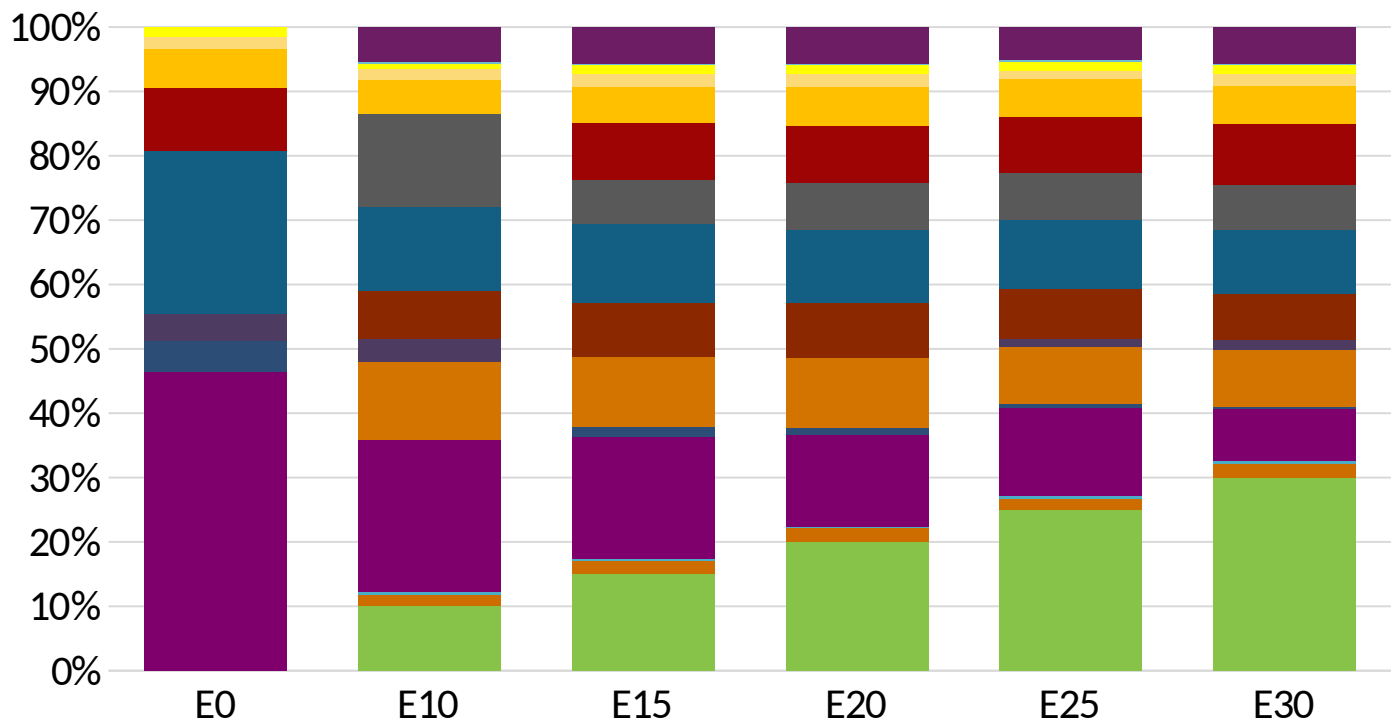
Greece - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.127	\$ 2.001	\$ 1.929	\$ 1.854	\$ 1.777	\$ 1.721

Average CIF 2025
Prices

Greece – Gasoline 95 – Increased Octane

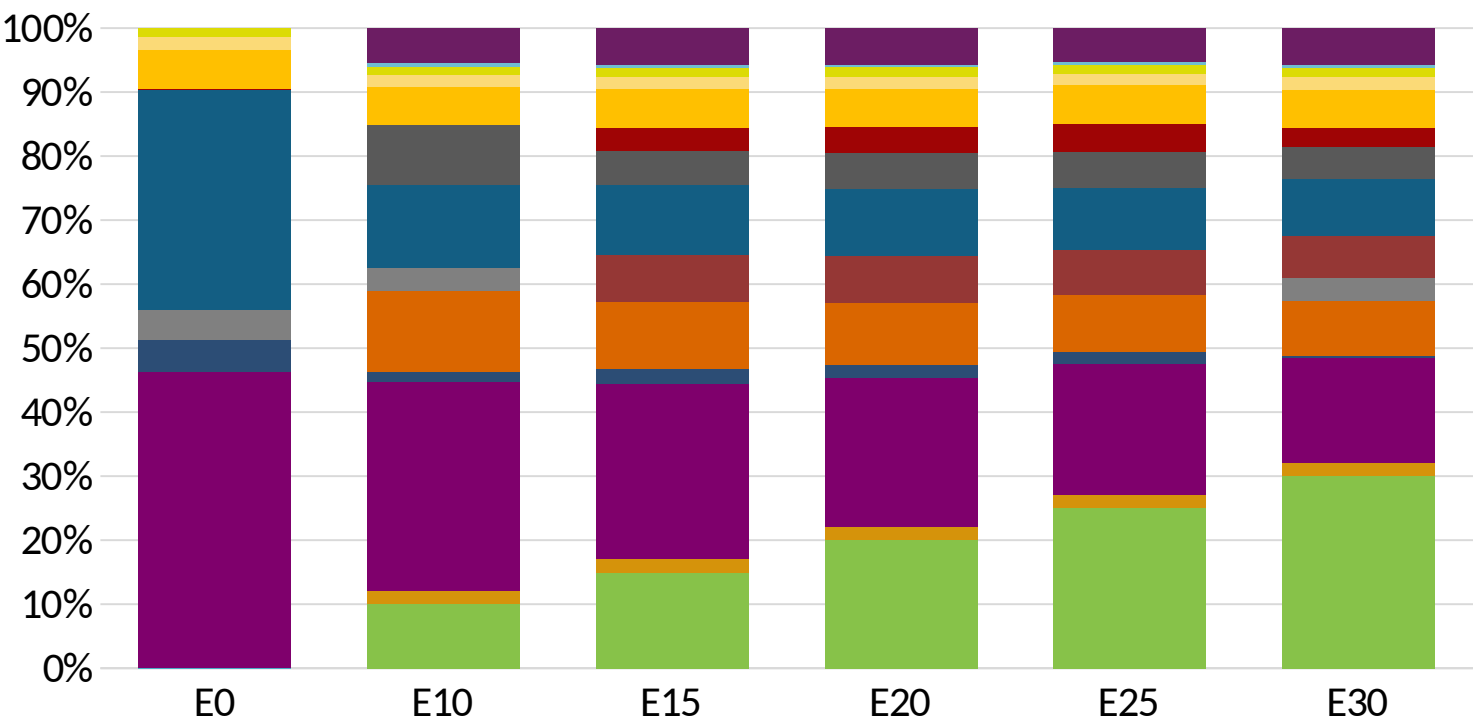


Average CIF 2025
Prices

Octane (RON)	95.0	96.2	97.9	99.4	100.8	102.2
Price (US\$/gal)	\$ 1.993	\$ 1.993	\$ 1.993	\$ 1.993	\$ 1.993	\$ 1.993

Source:
Faro90

Greece – Gasoline 98 – Increased Octane

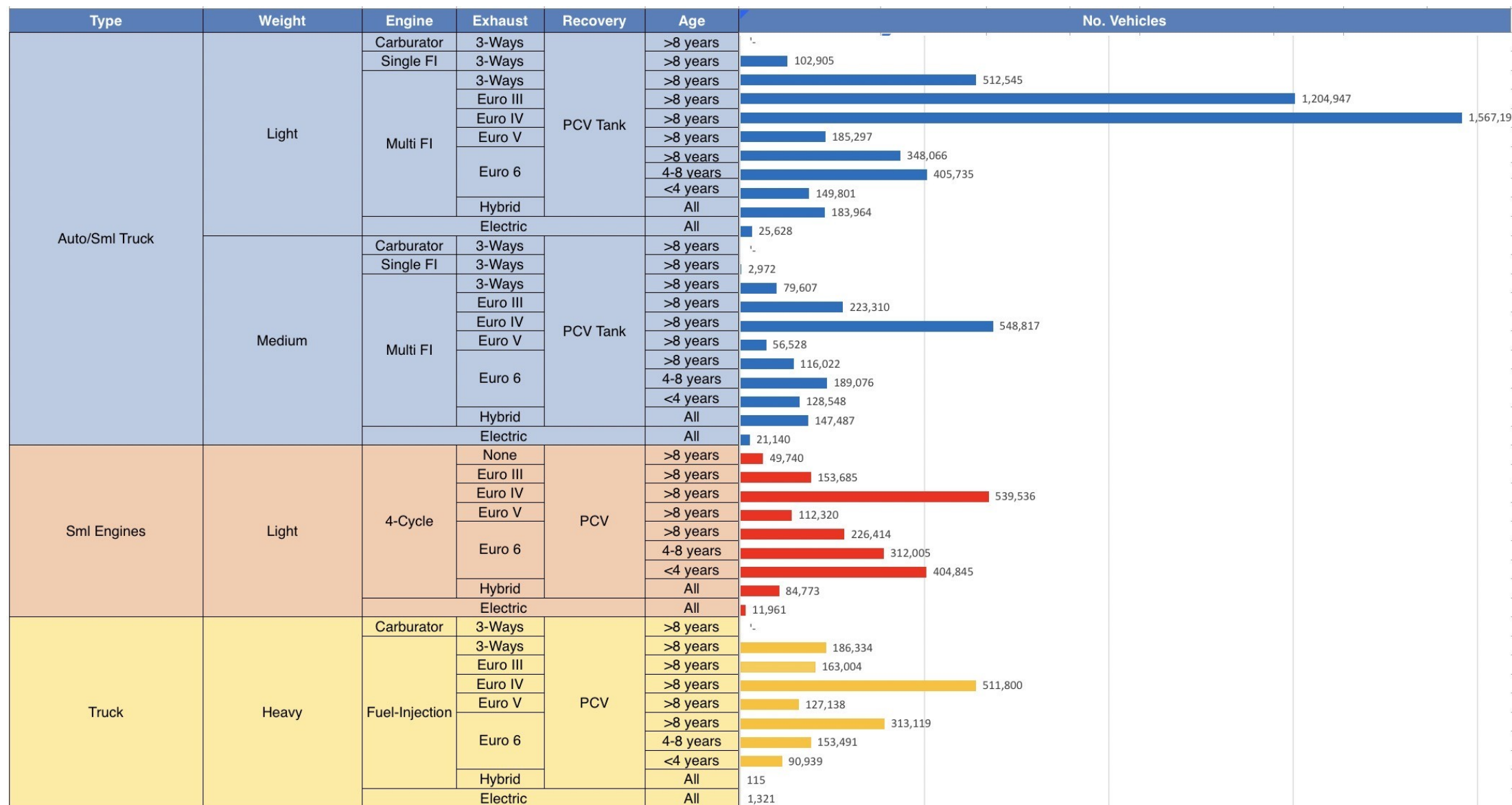


Average CIF 2025
Prices

Octane (RON)	98.0	99.4	100.7	102.0	103.3	104.5
Price (US\$/gal)	\$ 2.127	\$ 2.127	\$ 2.127	\$ 2.127	\$ 2.127	\$ 2.127

Source:
Faro90

Gasoline Vehicle Fleet - Greece

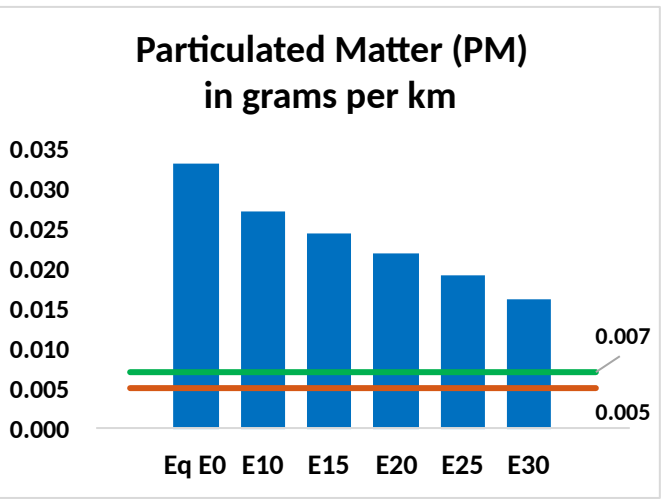
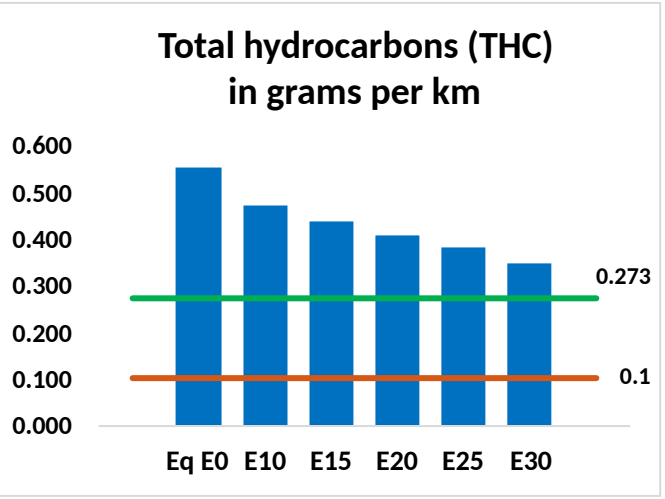
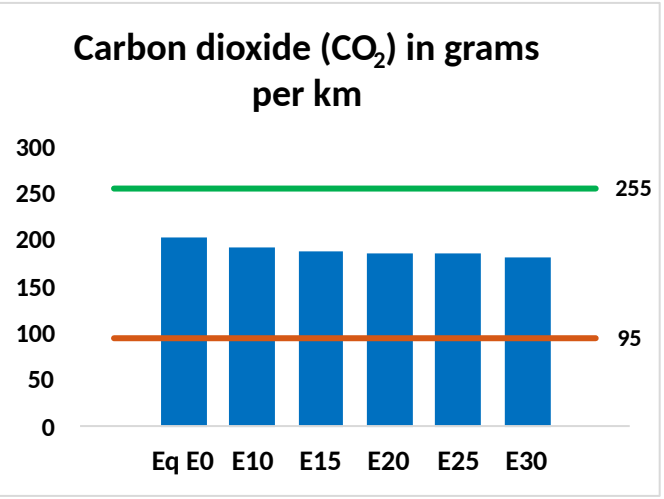
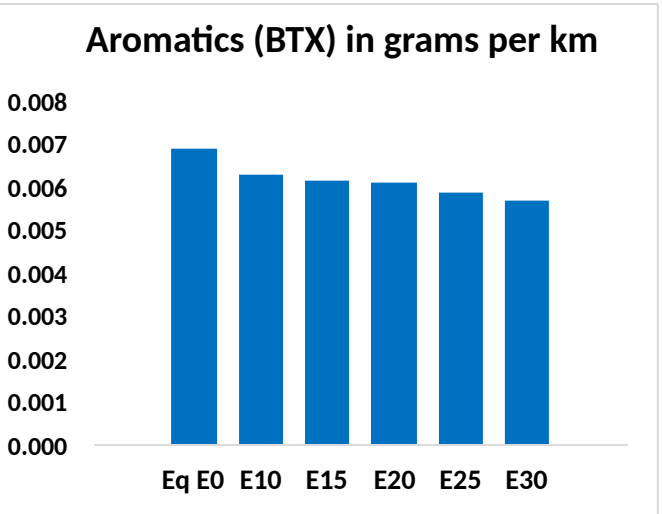
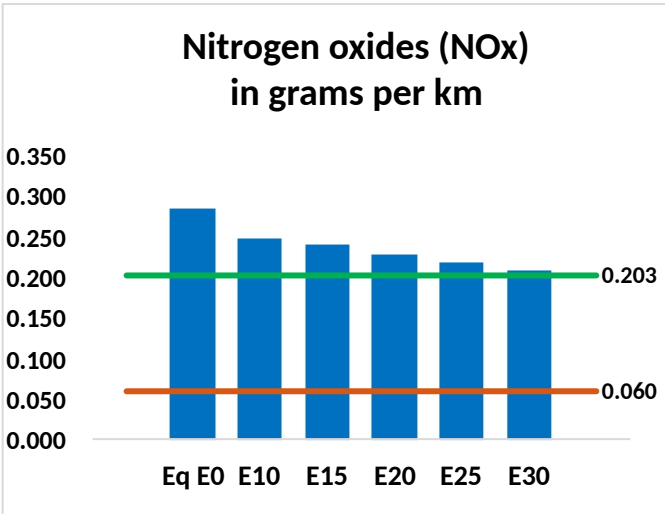
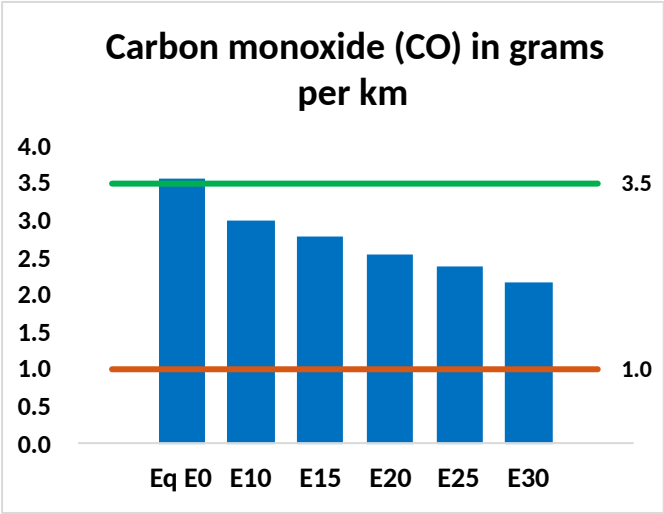
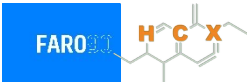


Vehicle Fleet: 9,642,126
Gasoline Vehicle Fleet: 6,833,958

Source: Eurostat, ACEA

Greece – Vehicular Emissions

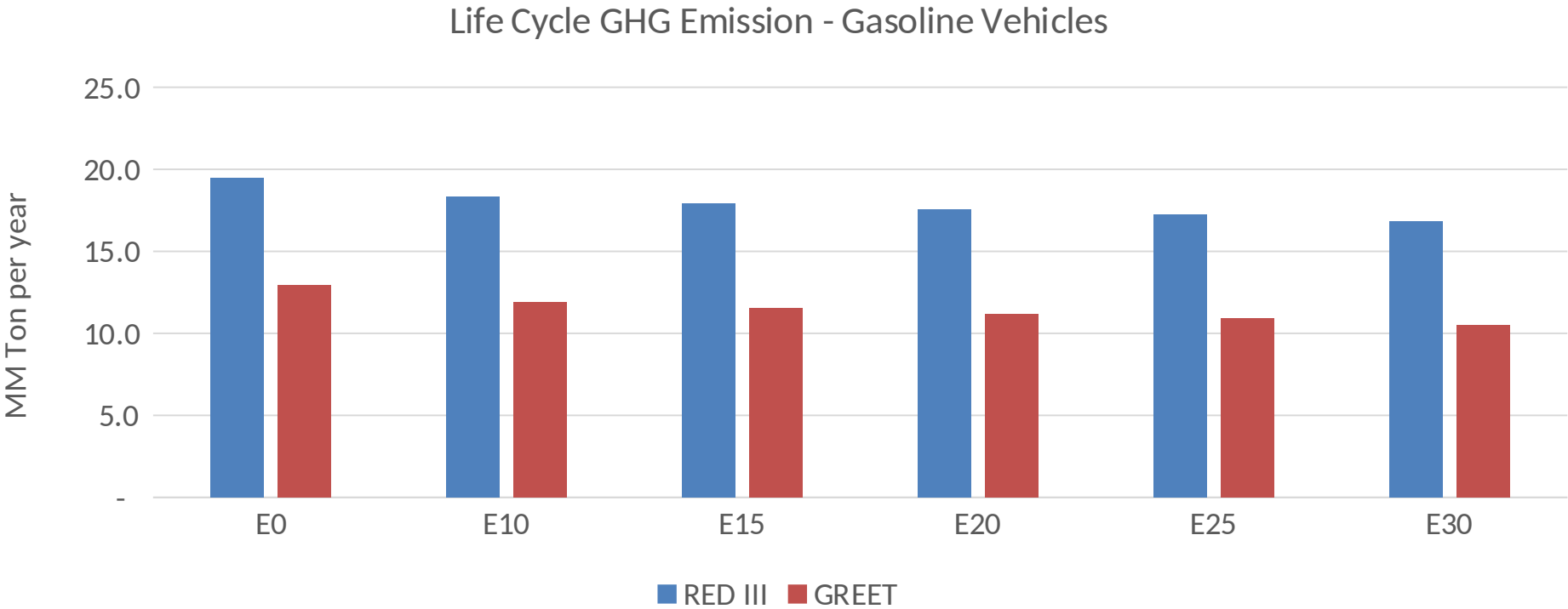
TIER III BIN 125 United States
Euro 6 Standard



Greece – Vehicular Emissions

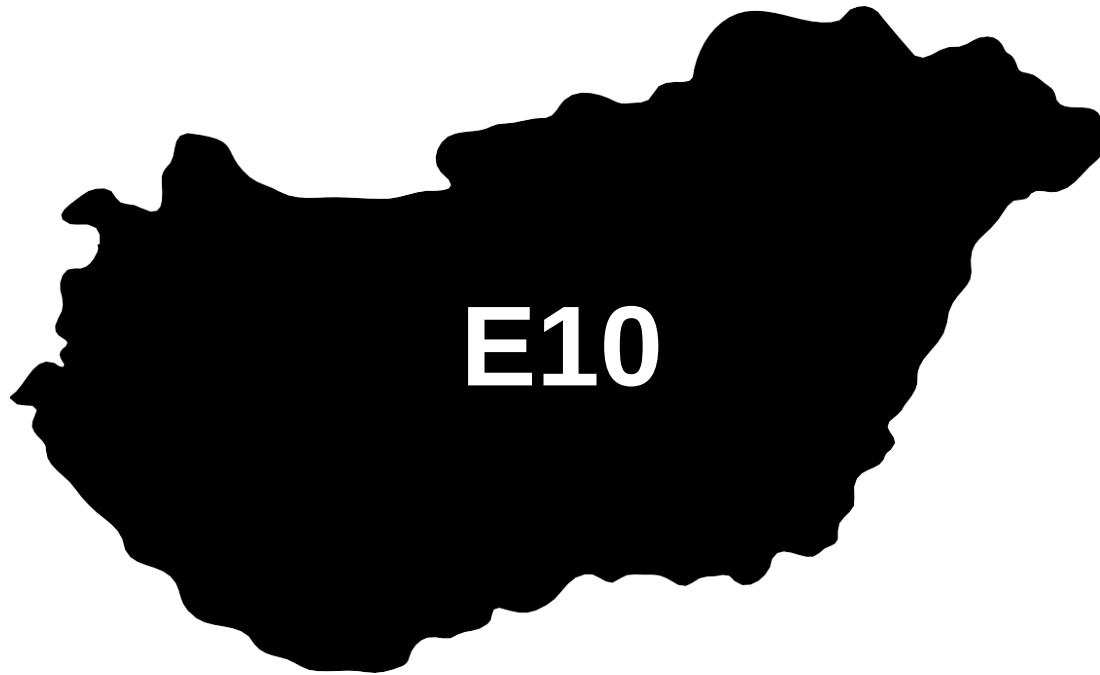
Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	566,219	522,014	477,809	441,453	404,806	376,609	345,253	-16%	-29%
CO2	32,196,010	31,368,113	30,586,209	29,971,266	29,666,967	29,499,709	28,955,945	-5%	-8%
VOC	88,038	81,630	75,221	70,040	64,858	60,883	55,544	-15%	-26%
NOx	45,556	42,063	39,681	38,476	36,525	35,000	33,202	-13%	-20%
SOx	365	338	310	287	264	240	214	-15%	-28%
NH3	11,790	11,674	11,557	11,612	11,742	11,834	11,911	-2%	0%
N2O	442	440	438	440	449	454	459	-1%	2%
CH4	19,149	19,149	19,149	19,436	19,857	20,221	20,566	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	579	556	533	518	505	496	481	-8%	-13%
Acetaldehyde	1,629	1,870	2,743	4,178	5,692	6,504	7,685	68%	249%
Formaldehyde	5,855	5,692	6,636	7,792	8,163	9,030	9,831	13%	39%
Benzene	1,099	1,056	1,001	985	977	934	904	-9%	-11%
PM2.5	1,932	1,756	1,581	1,426	1,275	1,113	943	-18%	-34%
PM10	3,366	3,060	2,754	2,484	2,221	1,939	1,643	-18%	-34%

Greece – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	74.3
Target @ 2035 (MMT/year)	41.8
Delta	32.5

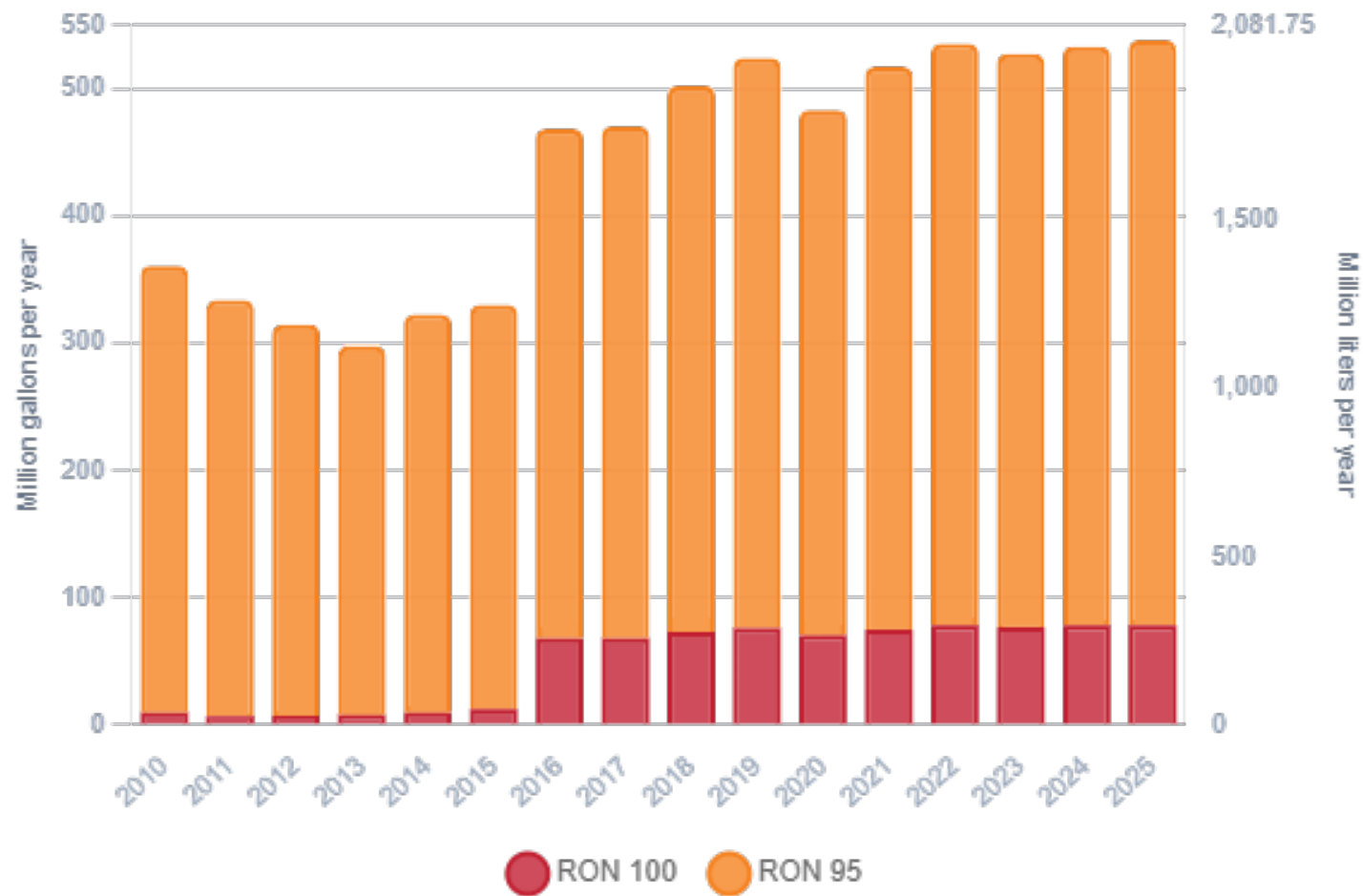
Ethanol Gasoline Blending in Hungary



In 2025, gasoline consumption in Hungary was 1,908 million liters (504 million gallons) and growth rate was 2.2%. Gasoline production and net imports were 1,438 and 473 million liters (380 and 125 million gallons) correspondingly. Hungary complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.4 RON.

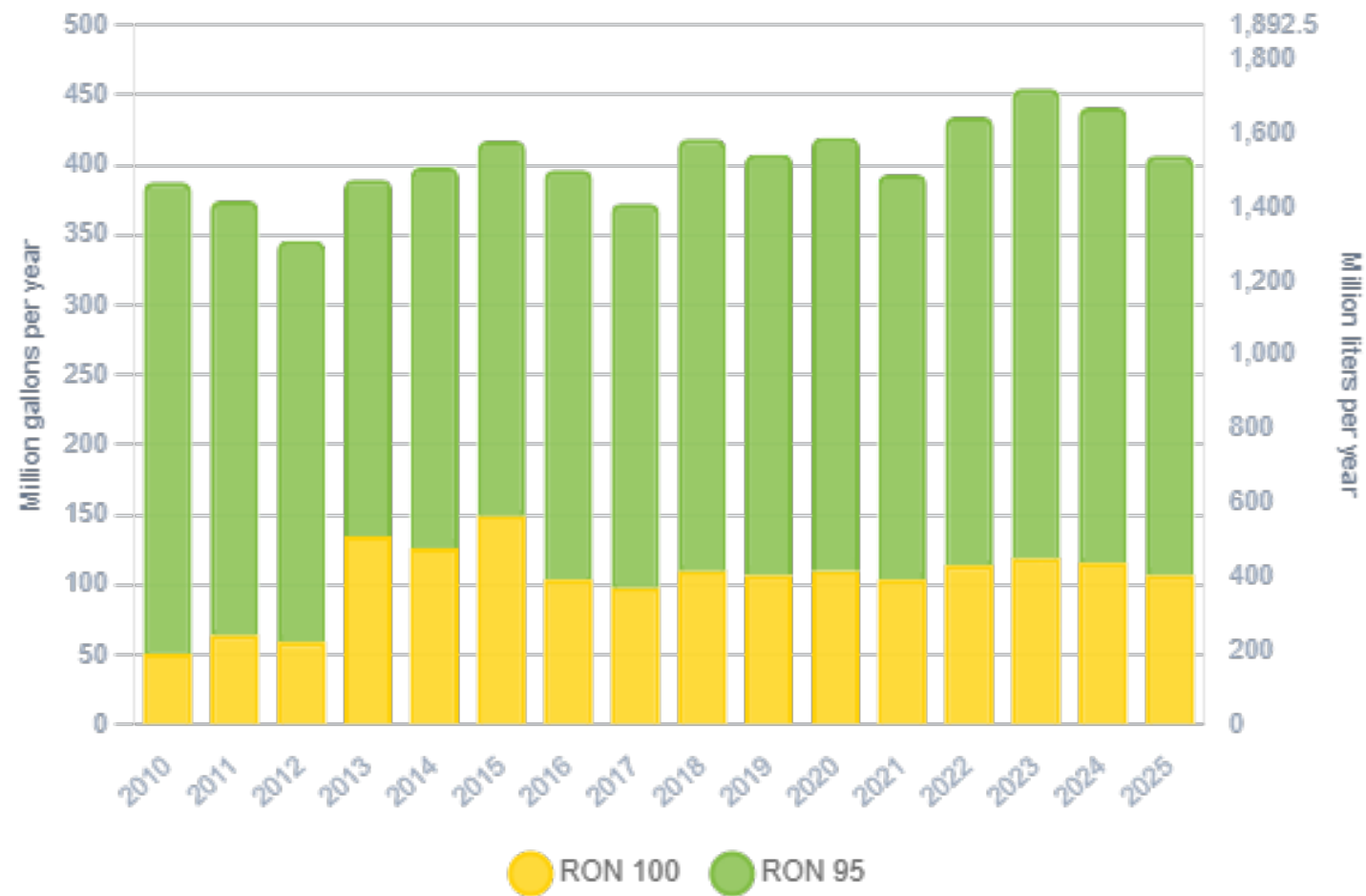
In 2025, ethanol consumption in Hungary was 113 million liters (30 million gallons) representing 6% of gasoline demand. Ethanol production and Net Imports were 531 and -418 million liters (140 and -110 million gallons) correspondingly. Ethanol source is mainly Corn based.

Gasoline Demand in Hungary



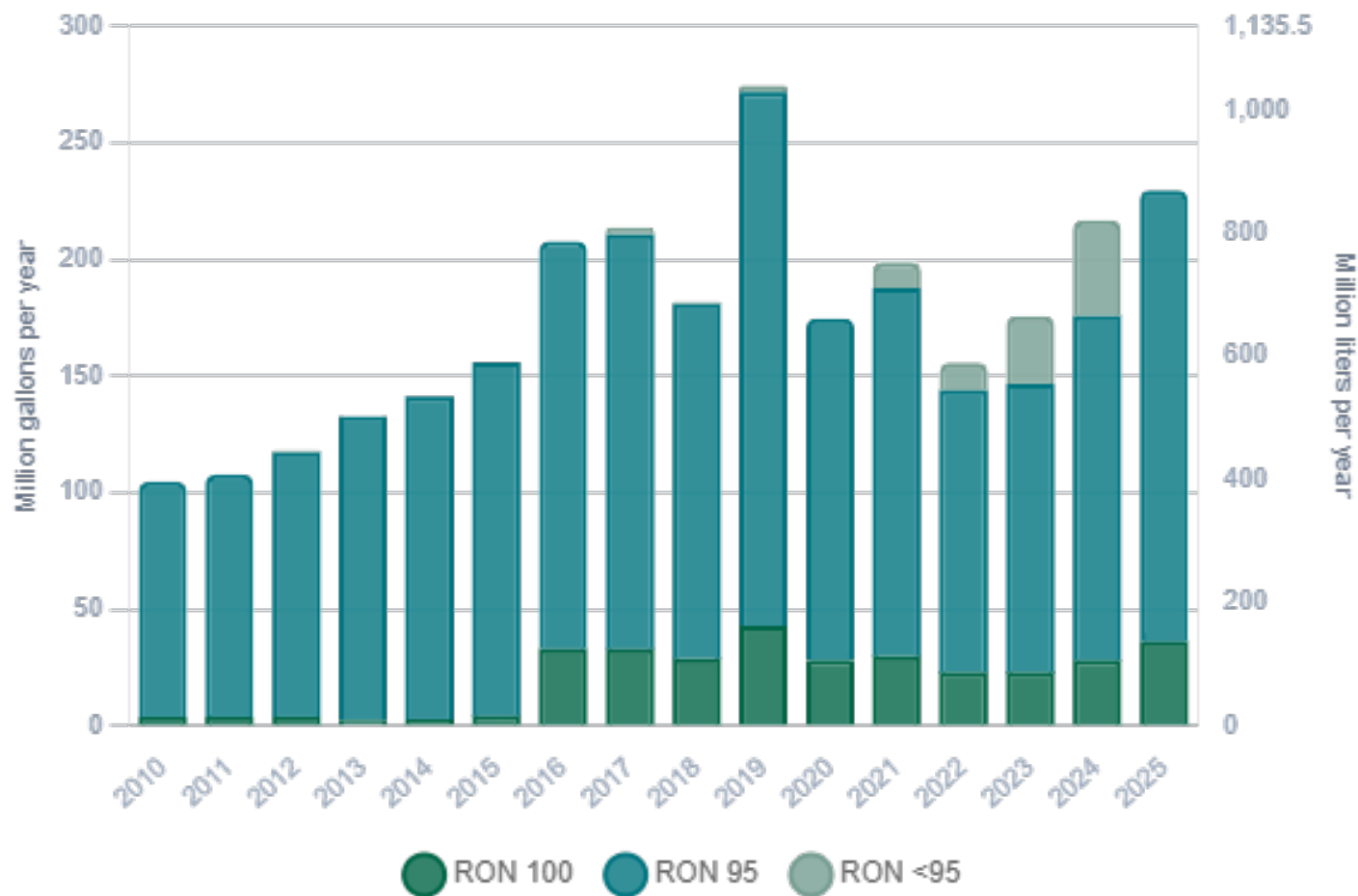
Source: Eurostat

Gasoline Production in Hungary



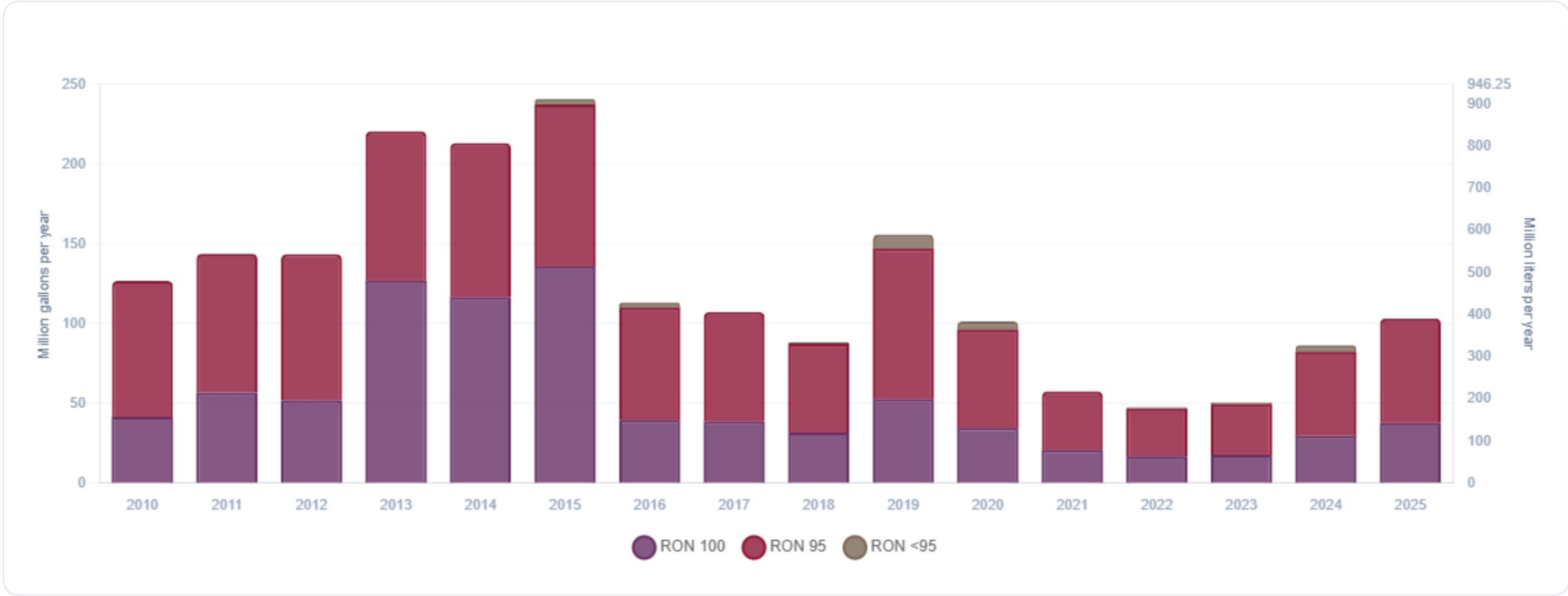
Source: Eurostat

Gasoline Imports in Hungary



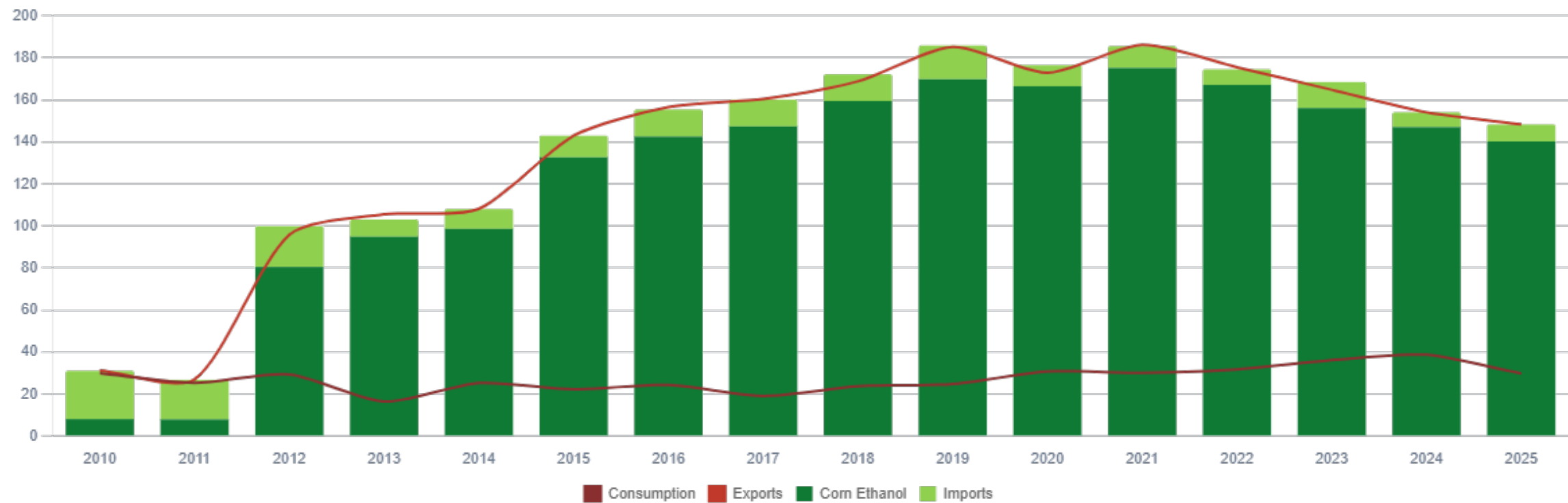
Source: Eurostat

Gasoline Exports in Hungary



Source: Eurostat

Ethanol Balance in Hungary



Source: Eurostat

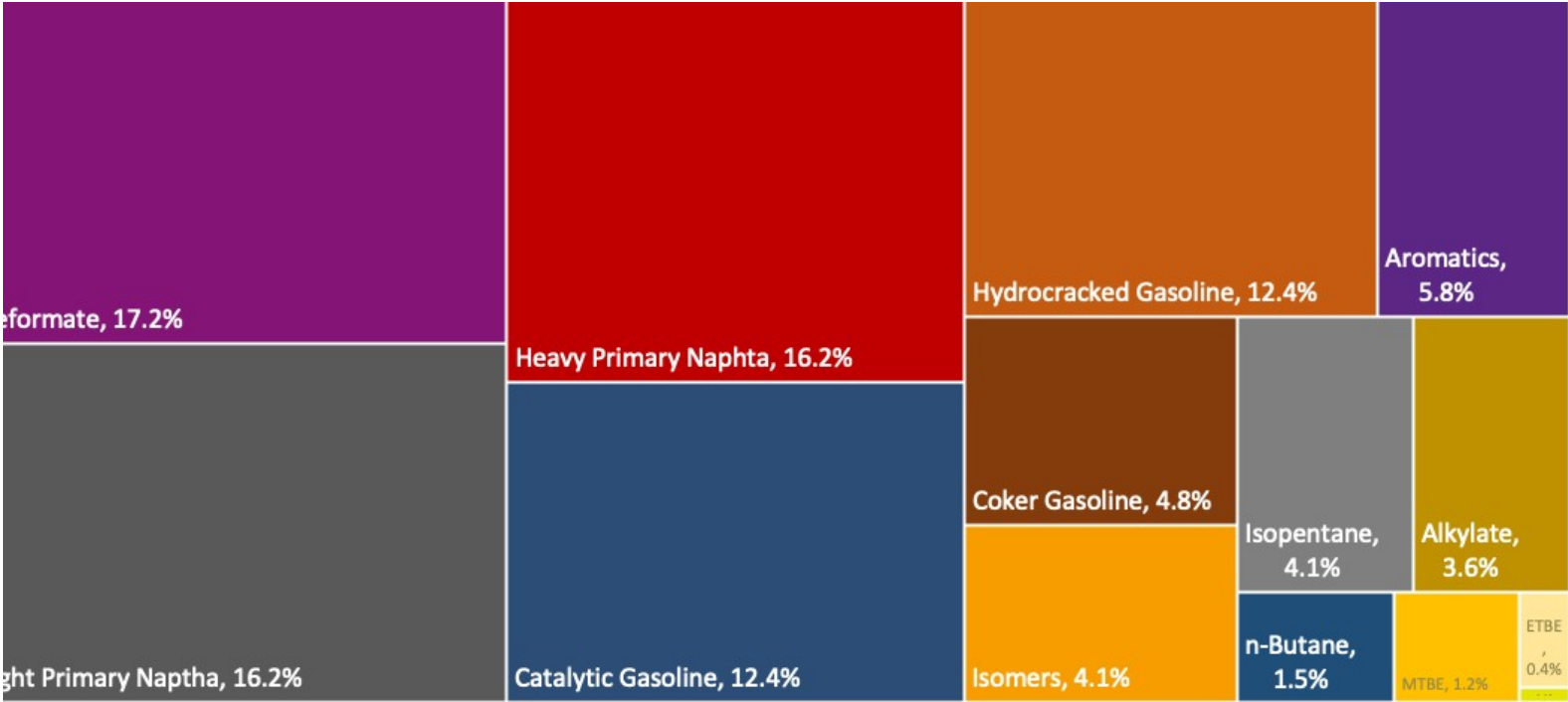
Gasoline Quality in Hungary

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.5							
Bioethanol in Gasoline Sales (%cal)	6.1							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

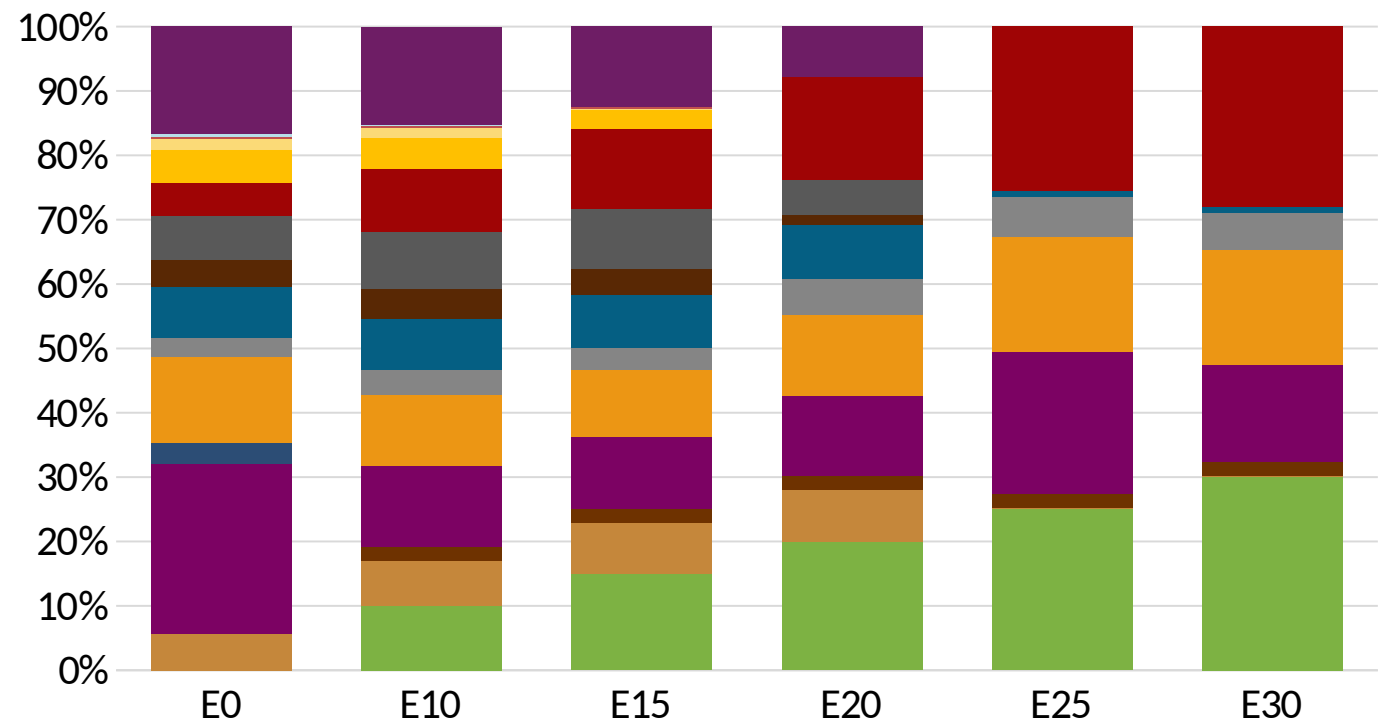
Source: European Commision

Hungary Available Blending Components



TAME,
0.1%

Hungary – Gasoline 95 – Constant Octane

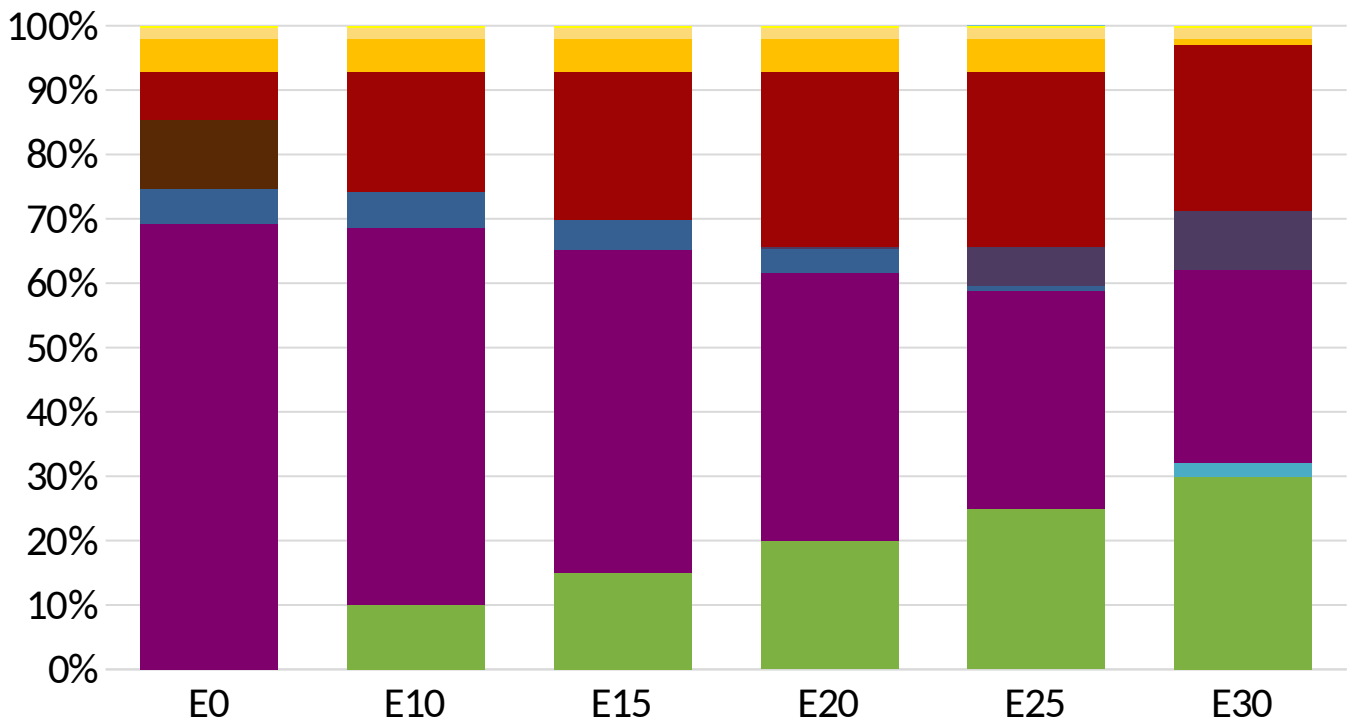
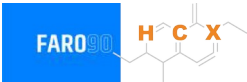


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	96.0
Price (US\$/gal)	\$ 2.094	\$ 1.965	\$ 1.896	\$ 1.818	\$ 1.675	\$ 1.638

Source:
Faro90

Hungary - Gasoline 98 - Constant Octane

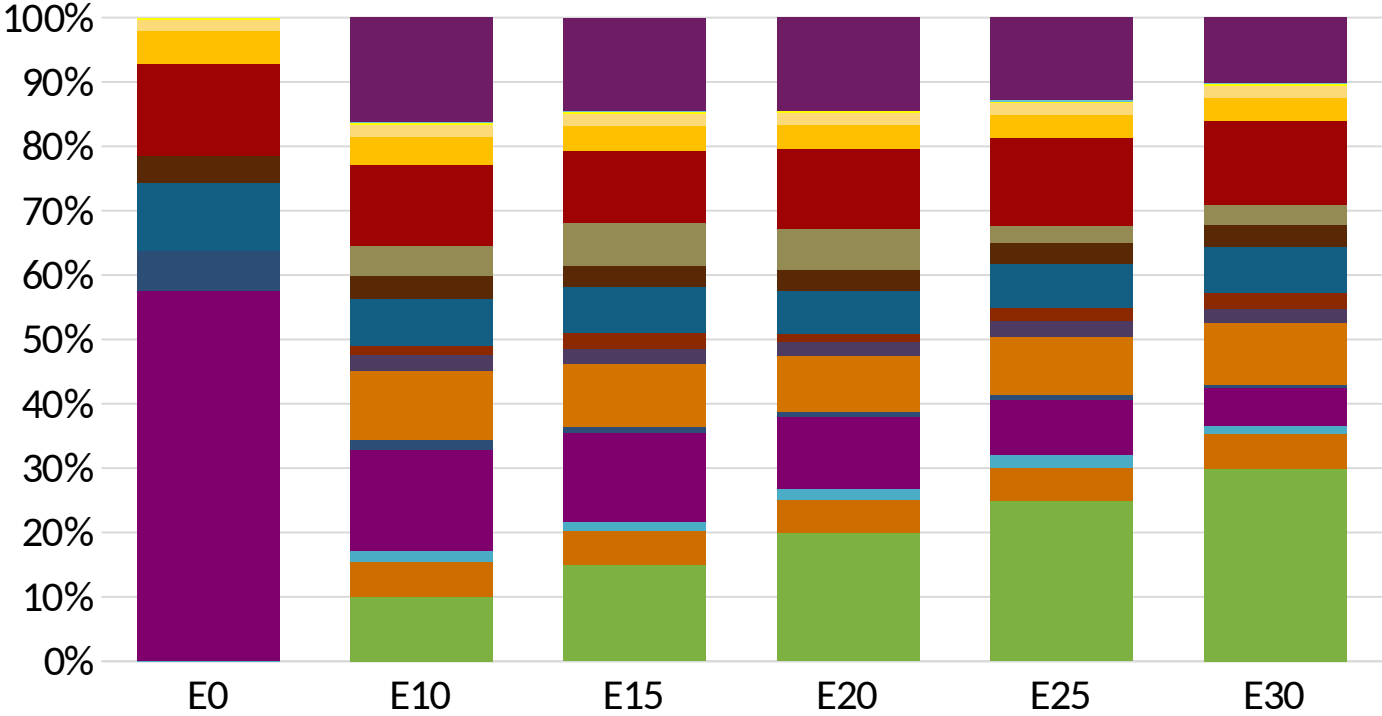
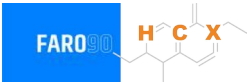


Average CIF 2025 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.126	\$ 2.005	\$ 1.936	\$ 1.861	\$ 1.784	\$ 1.720

Source:
Faro90

Hungary – Gasoline 95 – Increased Octane

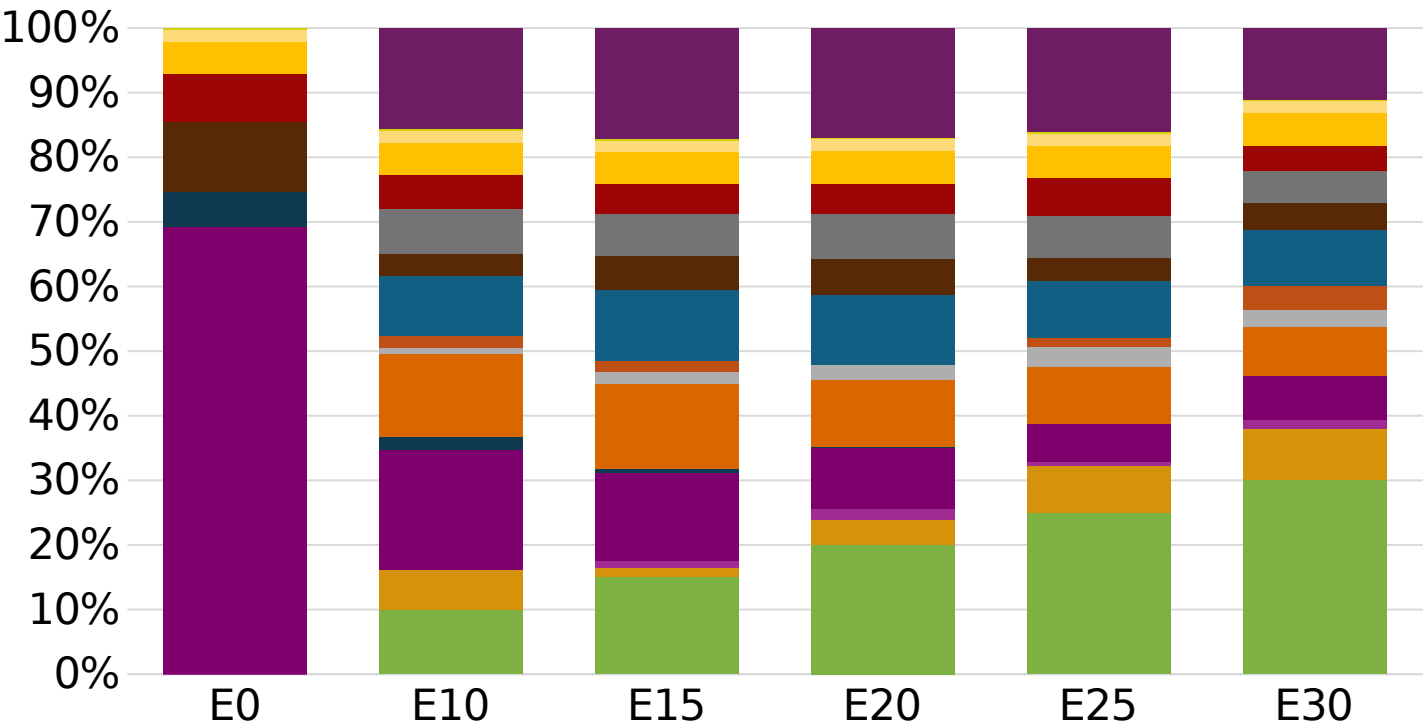


Average CIF 2025
Prices

Octane (RON)	95.0	95.7	97.2	98.8	100.3	101.8
Price (US\$/gal)	\$ 1.995	\$ 1.995	\$ 1.995	\$ 1.995	\$ 1.995	\$ 1.995

Source:
Faro90

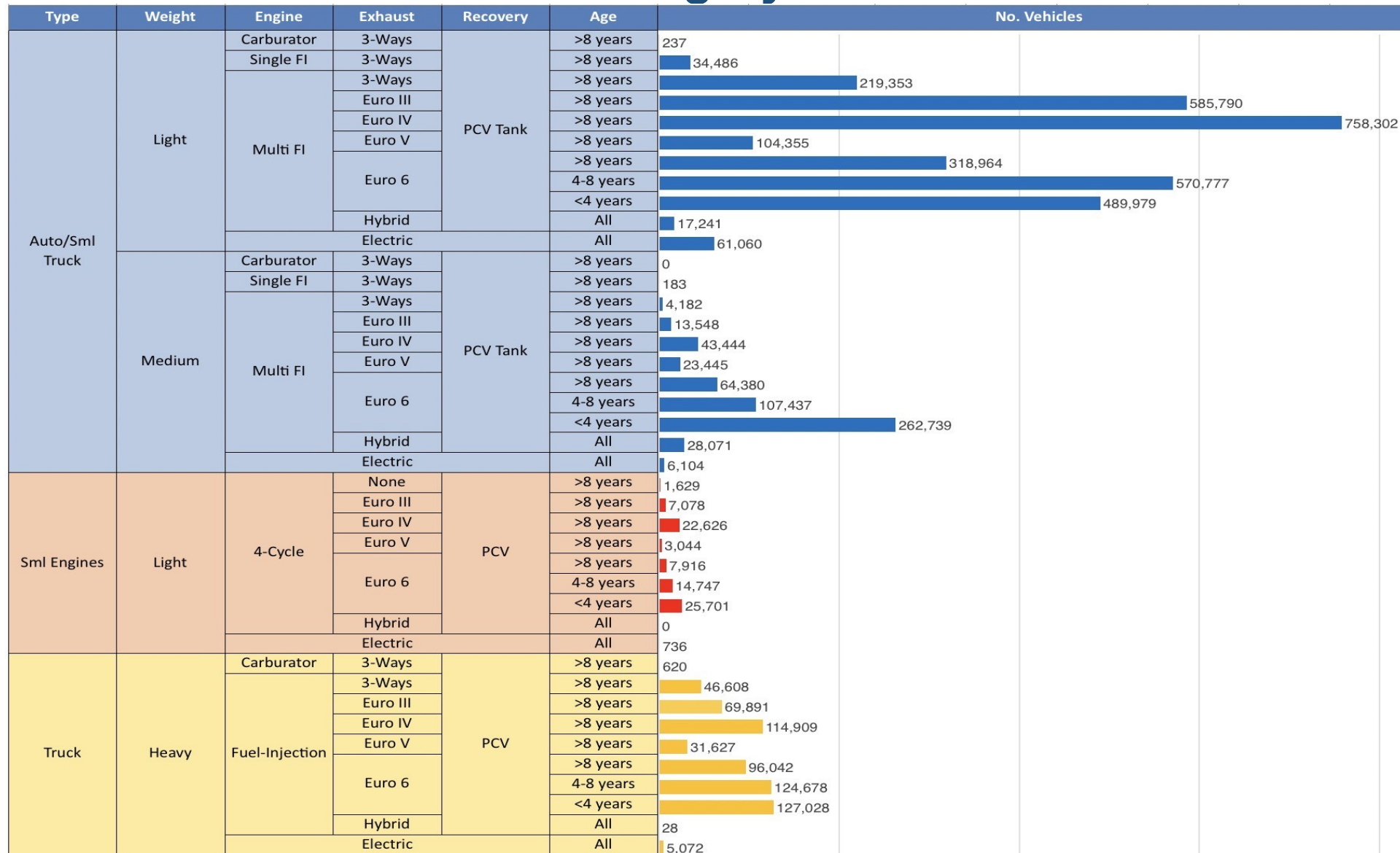
Hungary – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Source:
Faro90

Gasoline Vehicle Fleet - Hungary

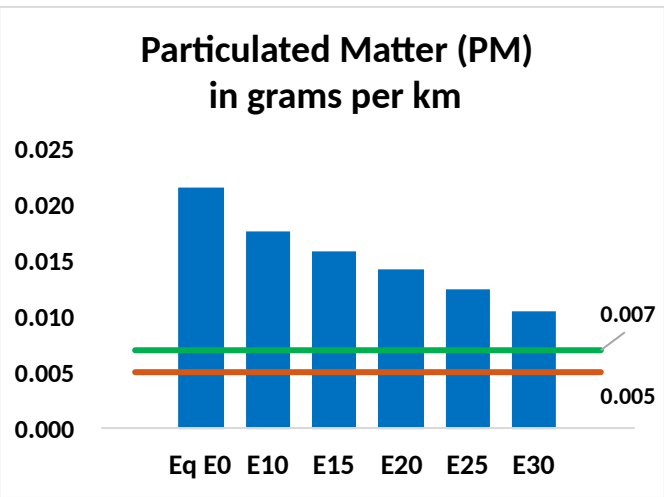
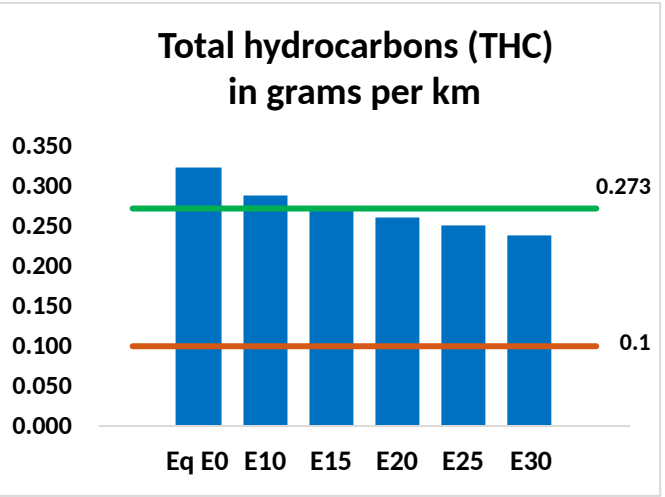
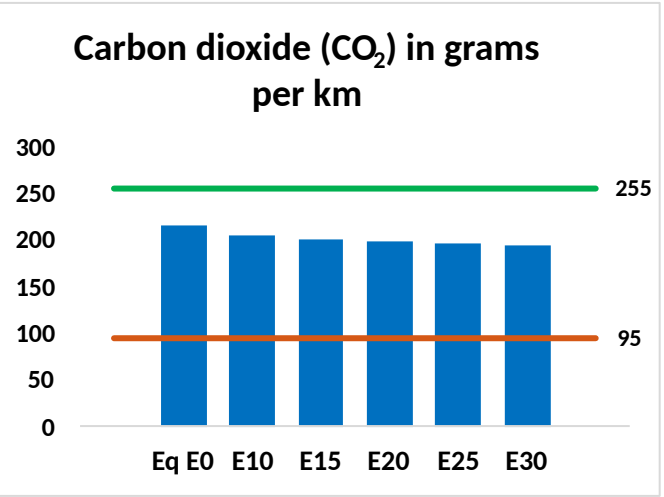
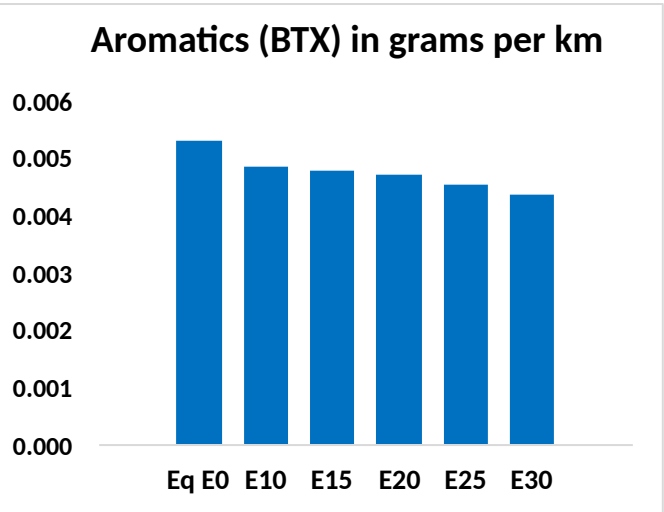
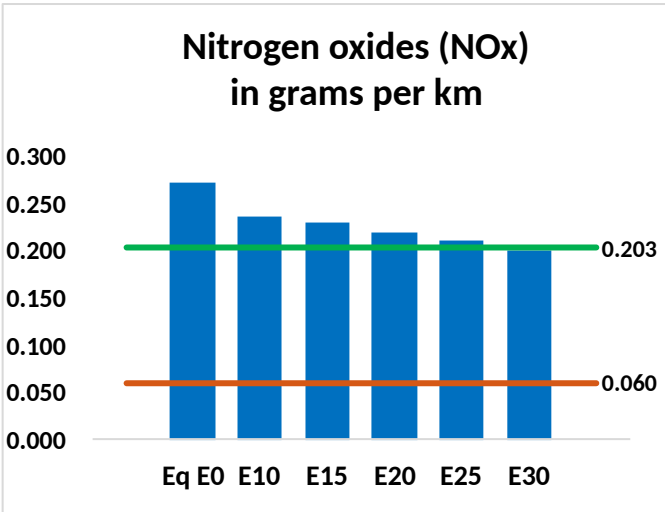
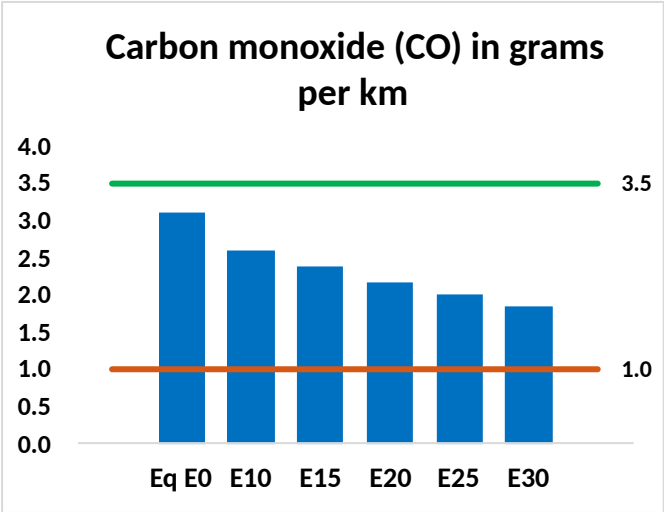
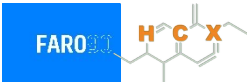


Vehicle Fleet: 4,414,058
Gasoline Vehicle Fleet: 2,379,575

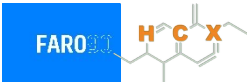
Source: Eurostat, ACEA

Hungary – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard

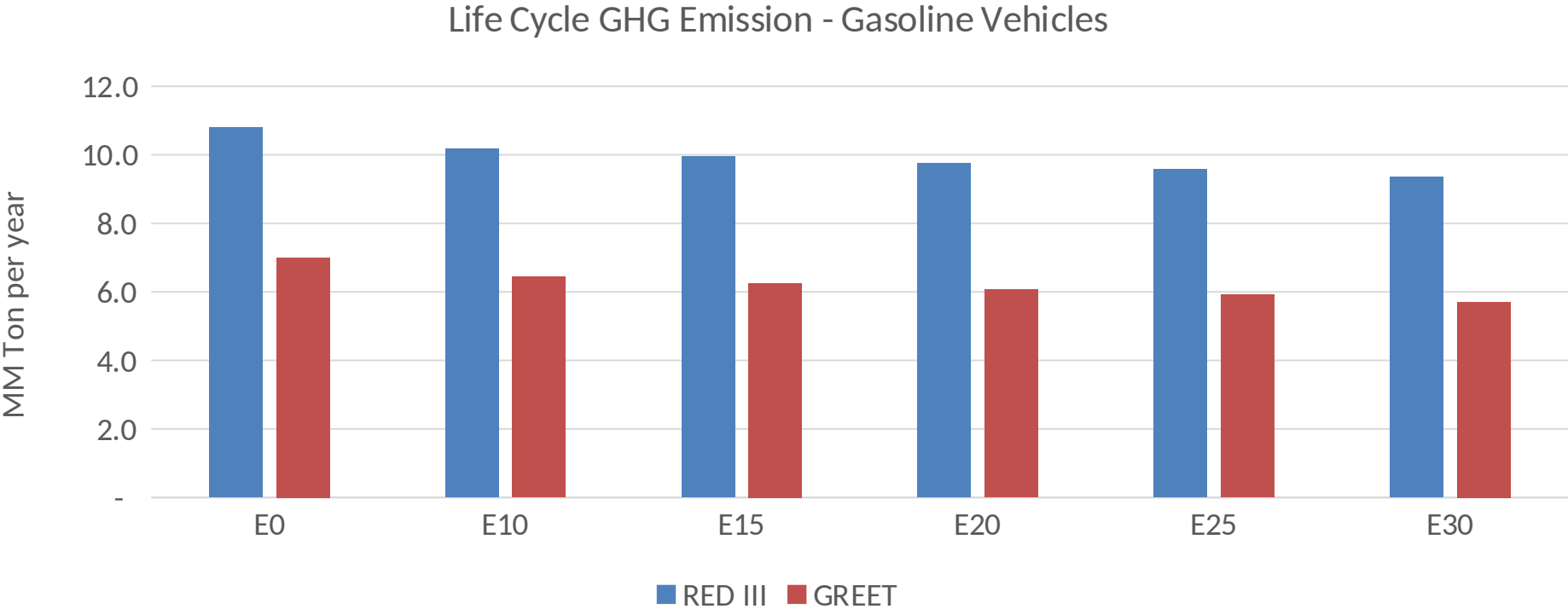
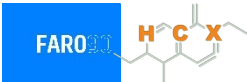


Hungary – Vehicular Emissions



Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	179,554	164,907	150,260	138,093	125,773	116,277	105,990	-16%	-30%
CO2	12,474,387	12,153,617	11,850,668	11,612,407	11,494,506	11,431,684	11,220,966	-5%	-8%
VOC	18,773	17,724	16,676	15,894	15,142	14,575	13,780	-11%	-19%
NOx	15,810	14,598	13,783	13,376	12,711	12,195	11,584	-13%	-20%
SOx	141	131	120	111	102	93	83	-15%	-28%
NH3	4,231	4,189	4,147	4,167	4,214	4,247	4,274	-2%	0%
N2O	193	192	191	192	196	198	200	-1%	2%
CH4	4,091	4,091	4,091	4,152	4,242	4,320	4,394	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	163	157	152	149	146	144	141	-7%	-10%
Acetaldehyde	351	403	592	901	1,227	1,403	1,657	68%	249%
Formaldehyde	1,103	1,073	1,250	1,468	1,538	1,702	1,852	13%	39%
Benzene	310	297	282	277	275	263	255	-9%	-11%
PM2.5	730	664	598	539	482	421	357	-18%	-34%
PM10	523	476	428	386	345	301	255	-18%	-34%

Hungary – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	64.6
Target @ 2035 (MMT/year)	36.4
Delta	28.2

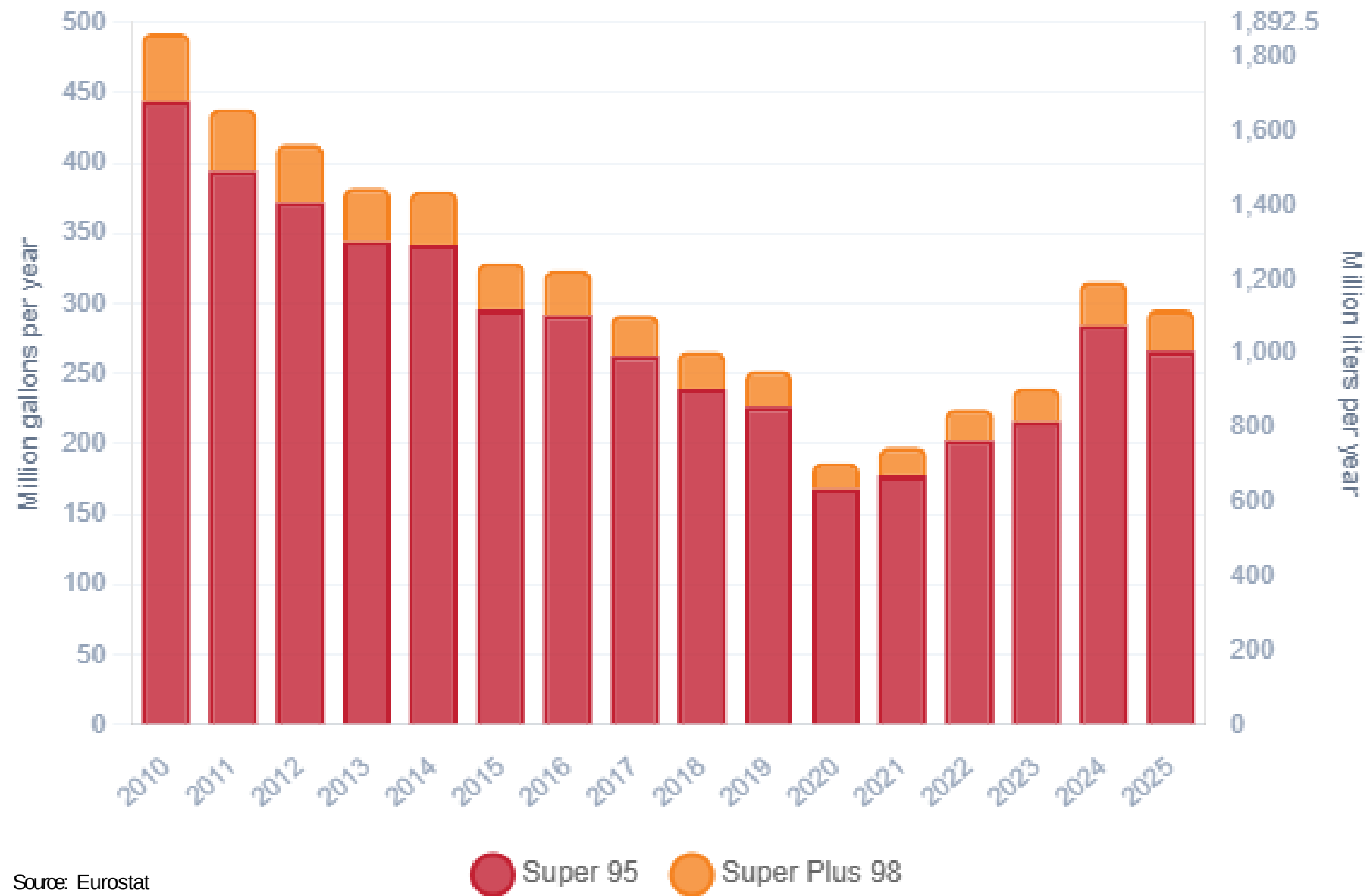
Ethanol Gasoline Blending in Ireland



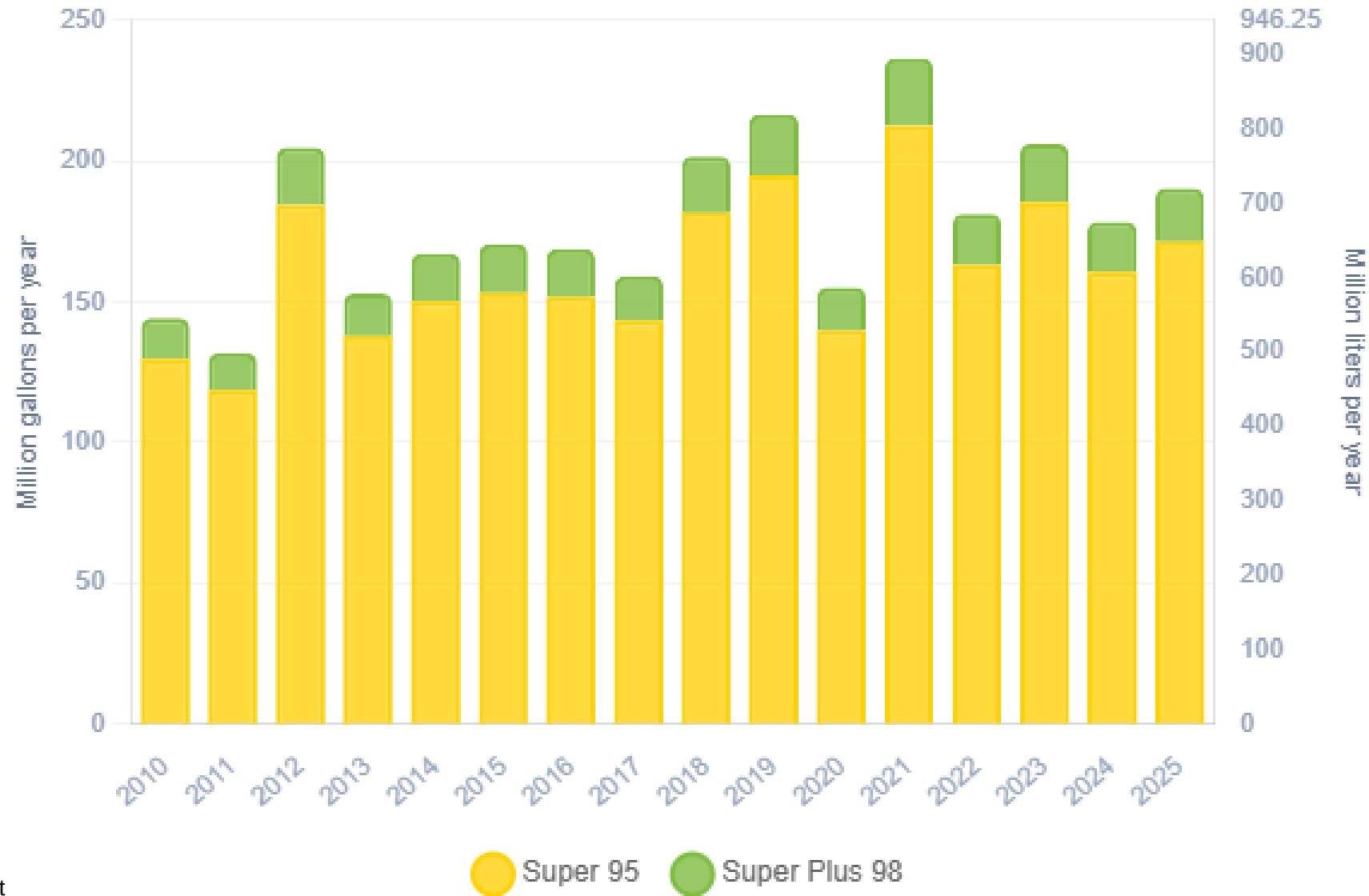
In 2025, gasoline consumption in Ireland was 1,122 million liters (296 million gallons) and growth rate was -3.3%. Gasoline production and net imports were 721 and 246 million liters (191 and 65 million gallons) correspondingly. Ireland complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.3 RON.

In 2025, ethanol consumption in Ireland was 107 million liters (28 million gallons) representing 10% of gasoline demand. Ethanol production and Net Imports were 10 and 95 million liters (3 and 25 million gallons) correspondingly. Ethanol source is Cereals 100%.

Gasoline Demand in Ireland

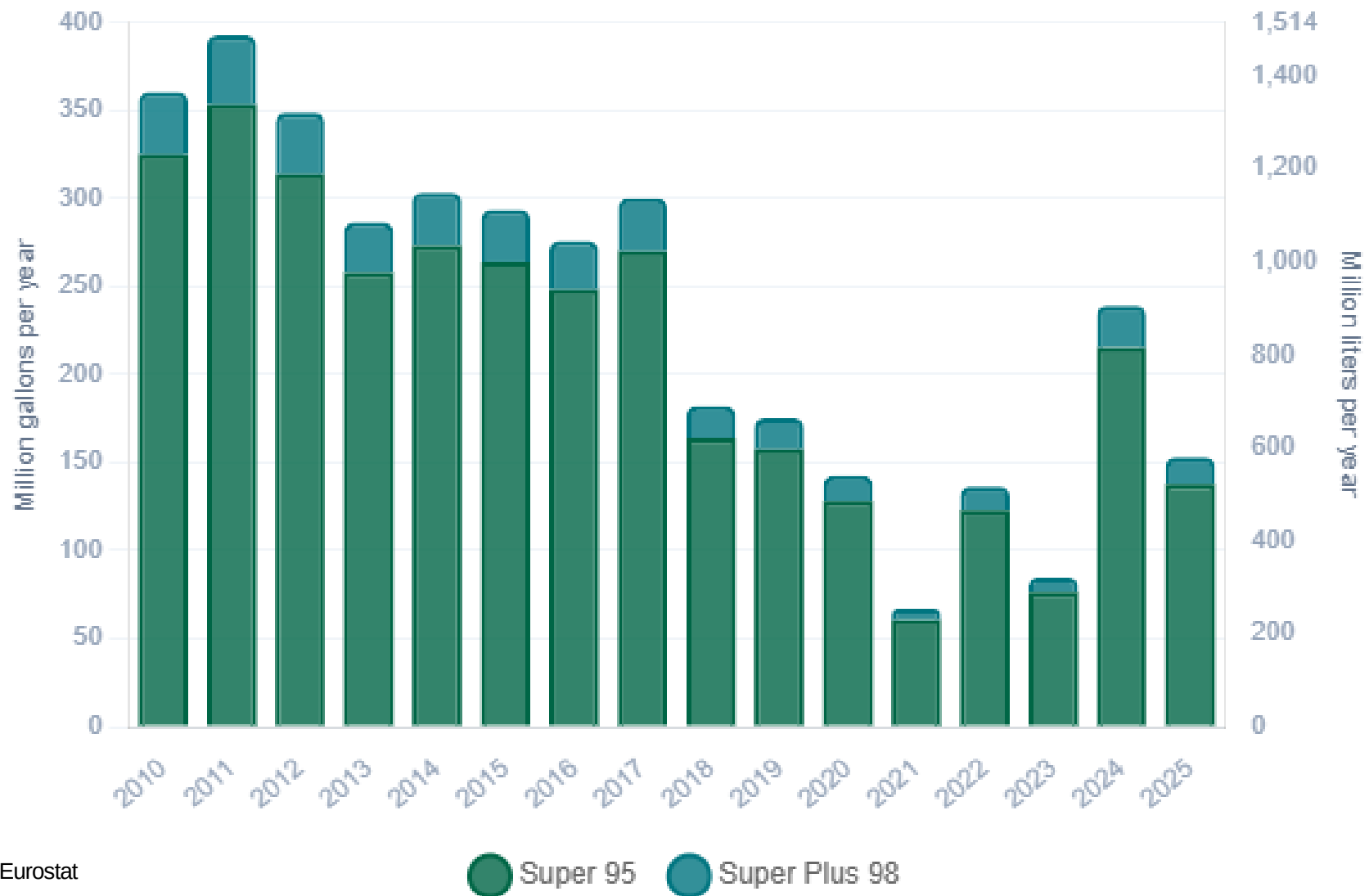


Gasoline Production in Ireland



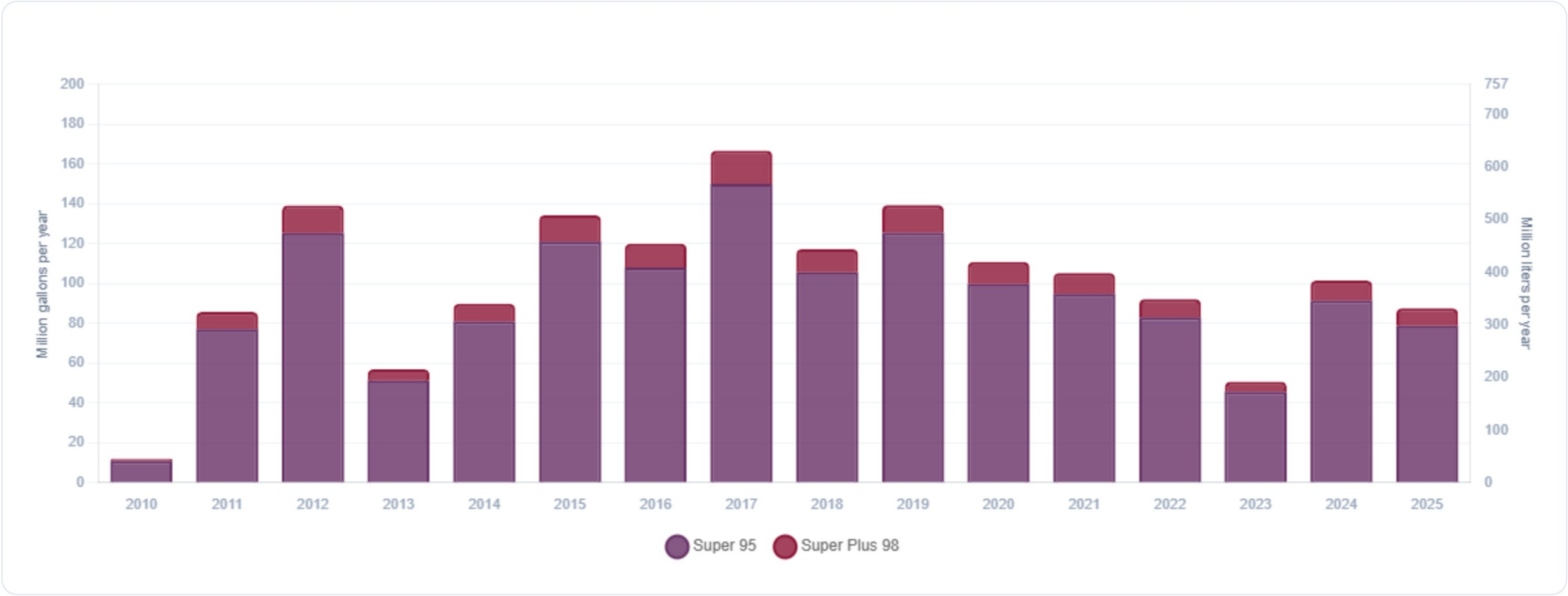
Source: Eurostat

Gasoline Imports in Ireland



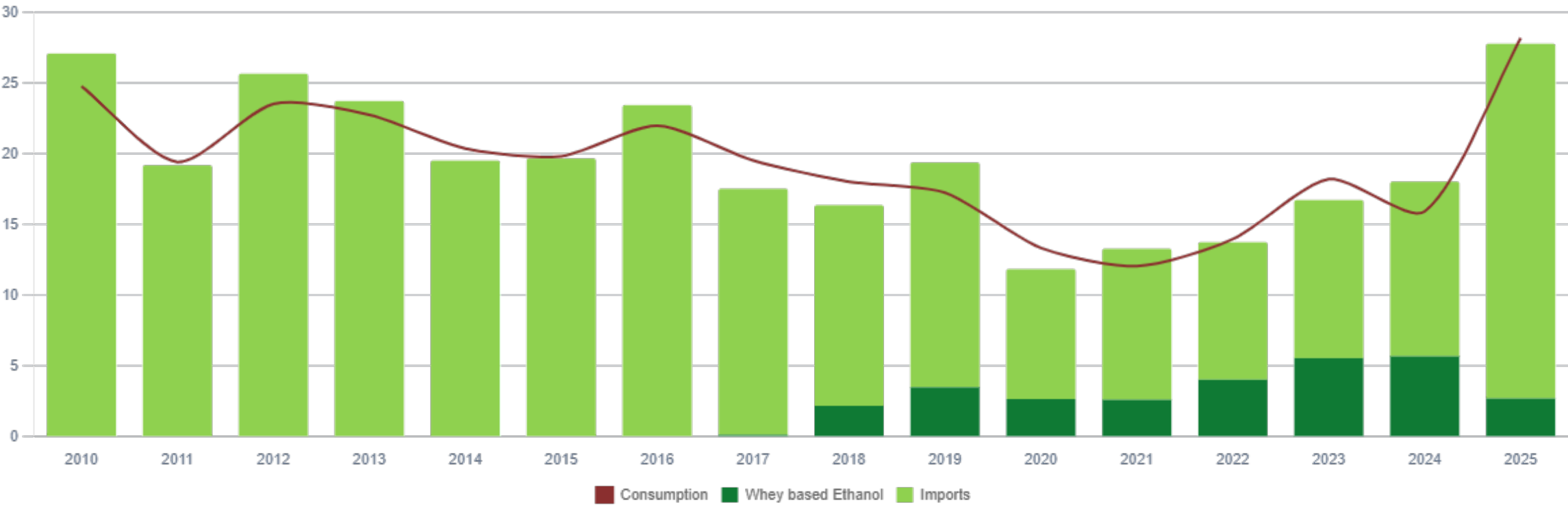
Source: Eurostat

Gasoline Exports in Ireland



Source: Eurostat

Ethanol Balance in Ireland



Source: Eurostat

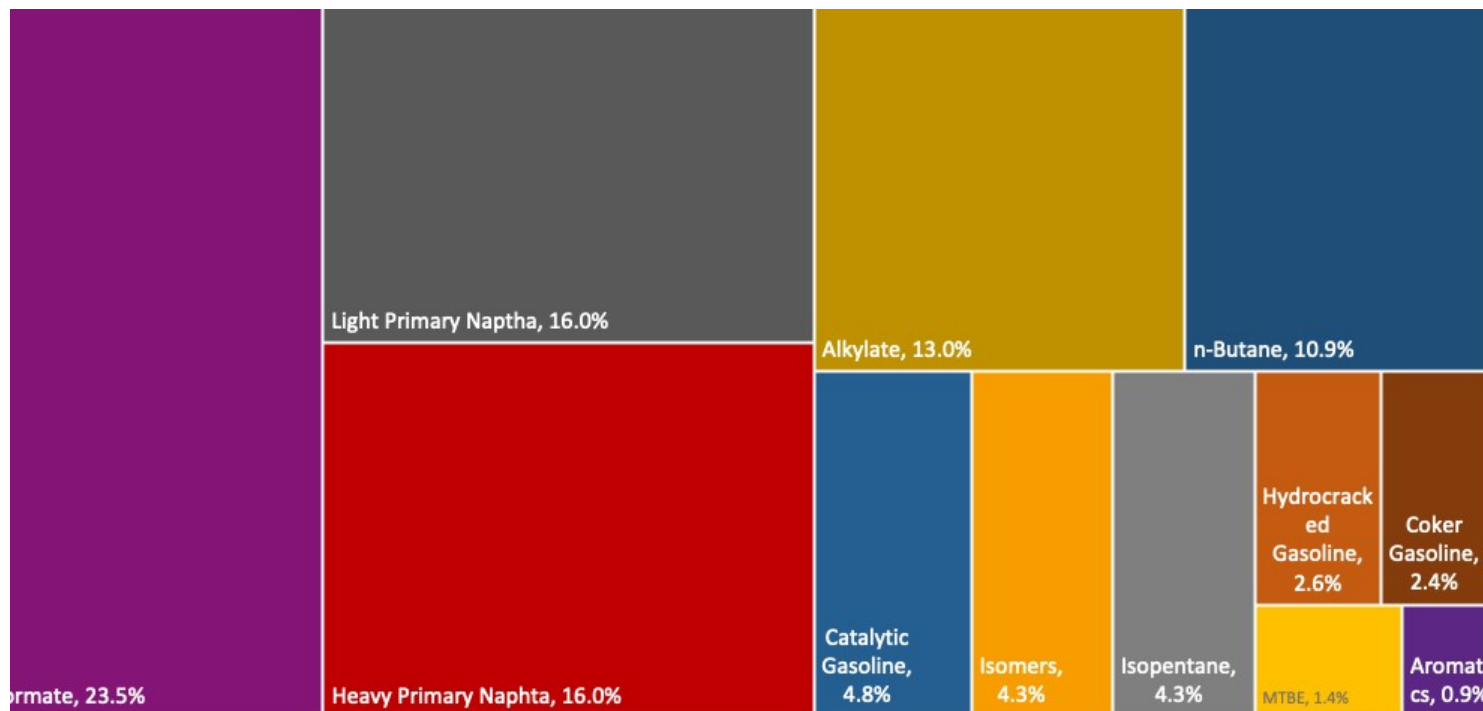
Gasoline Quality in Ireland

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Countries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	< 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	21.0							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

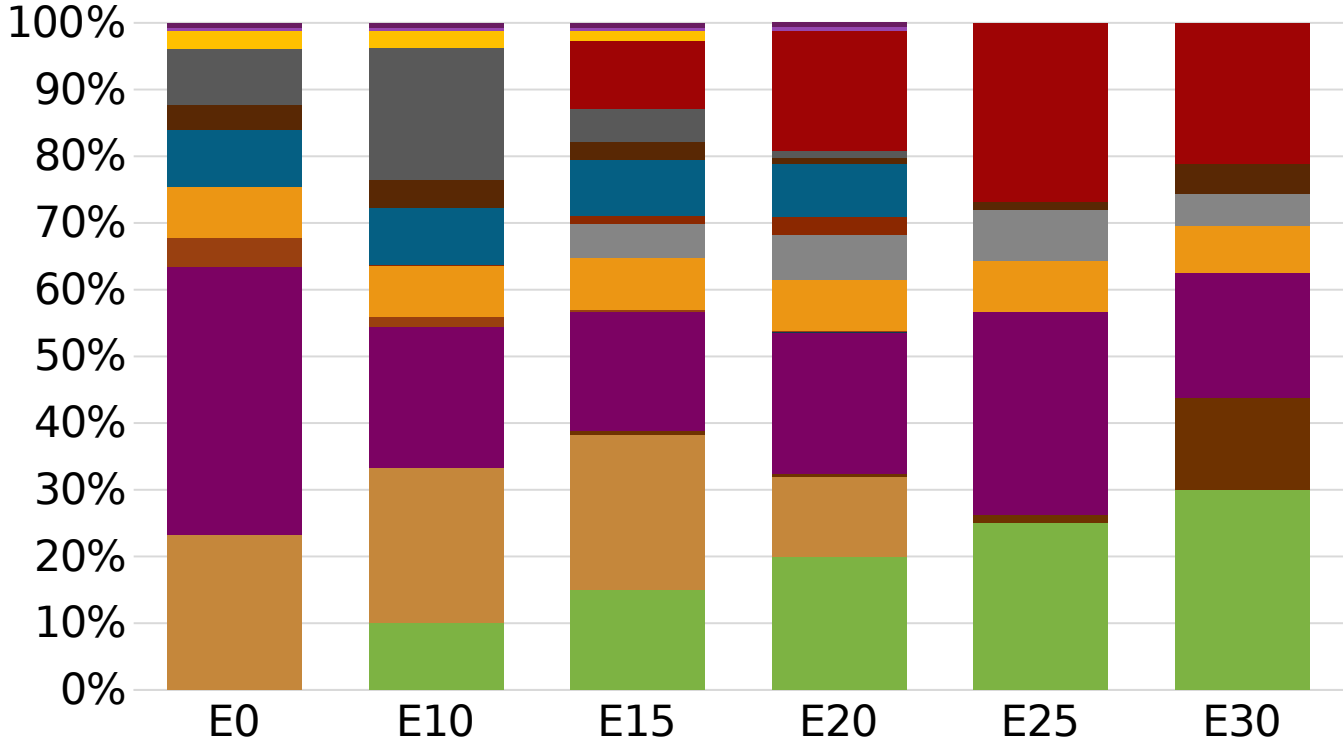
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commission

Ireland Available Blending Components



Ireland – Gasoline 95 – Constant Octane

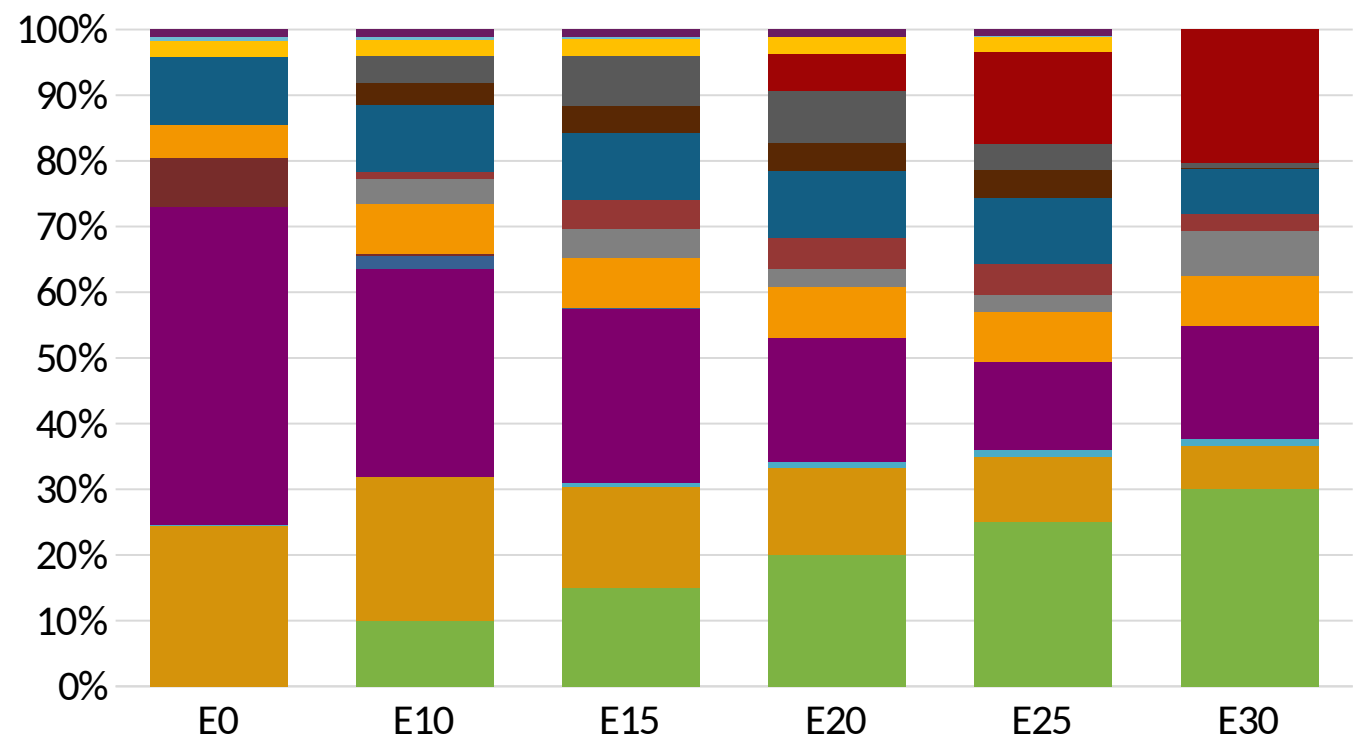


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US\$/gal)	\$ 2.114	\$ 1.984	\$ 1.902	\$ 1.793	\$ 1.683	\$ 1.618

Source:
Faro90

Ireland - Gasoline 98 - Constant Octane

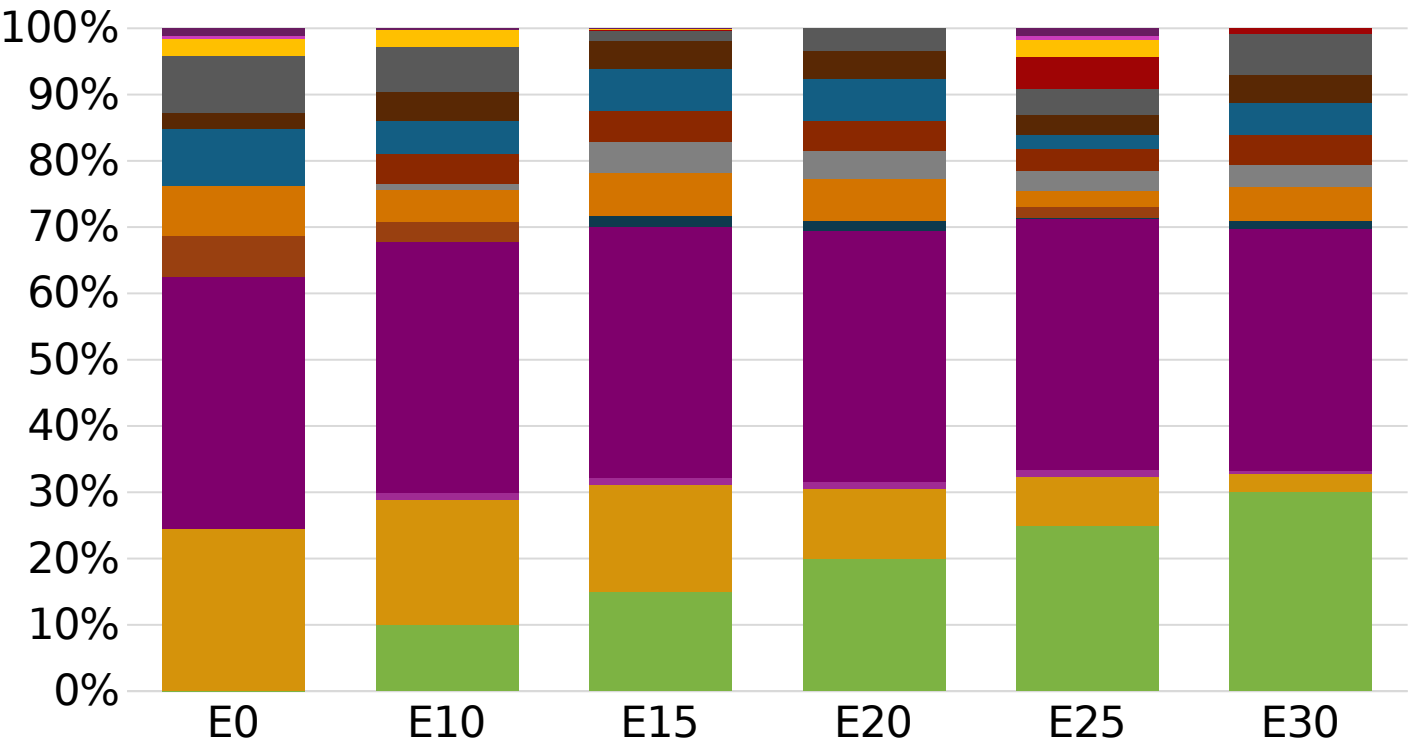


Average CIF 2025
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.237	\$ 2.114	\$ 2.037	\$ 1.954	\$ 1.859	\$ 1.764

Source:
Faro90

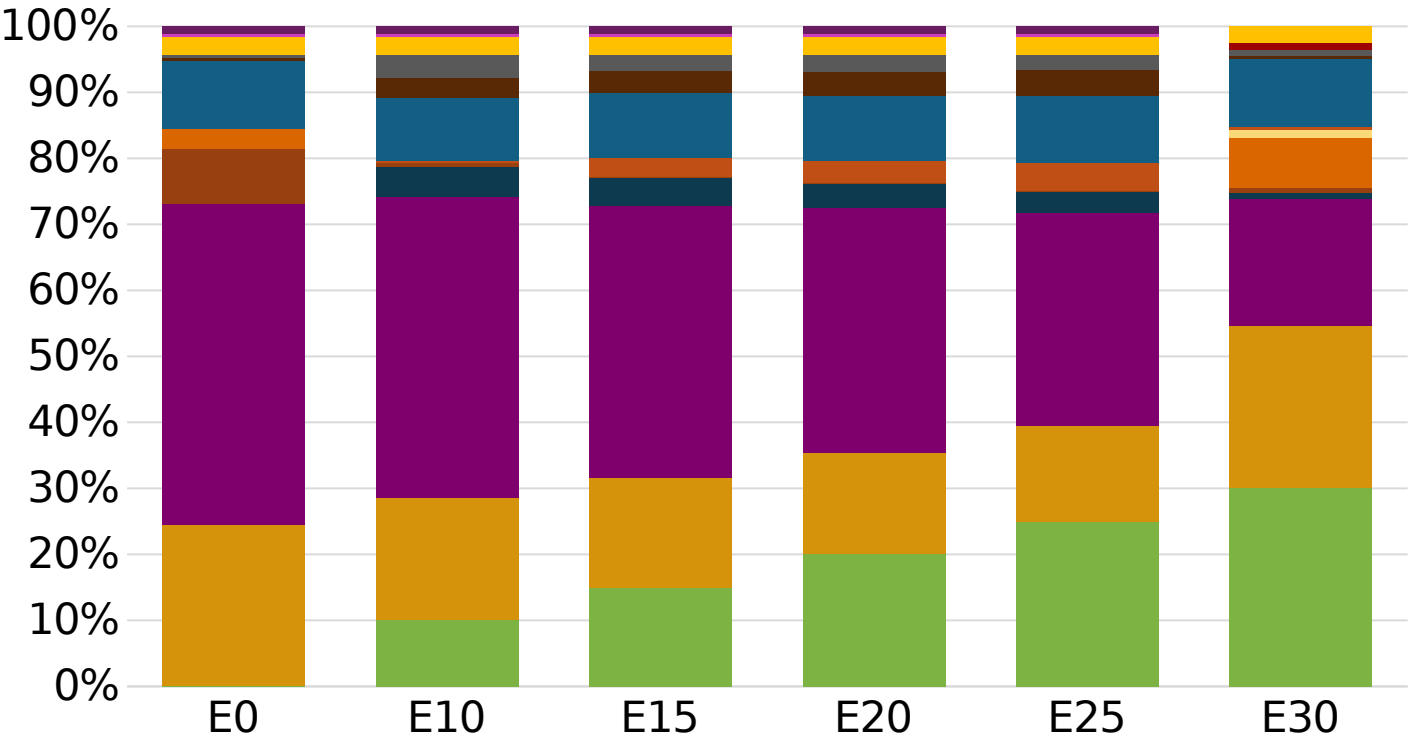
Ireland – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

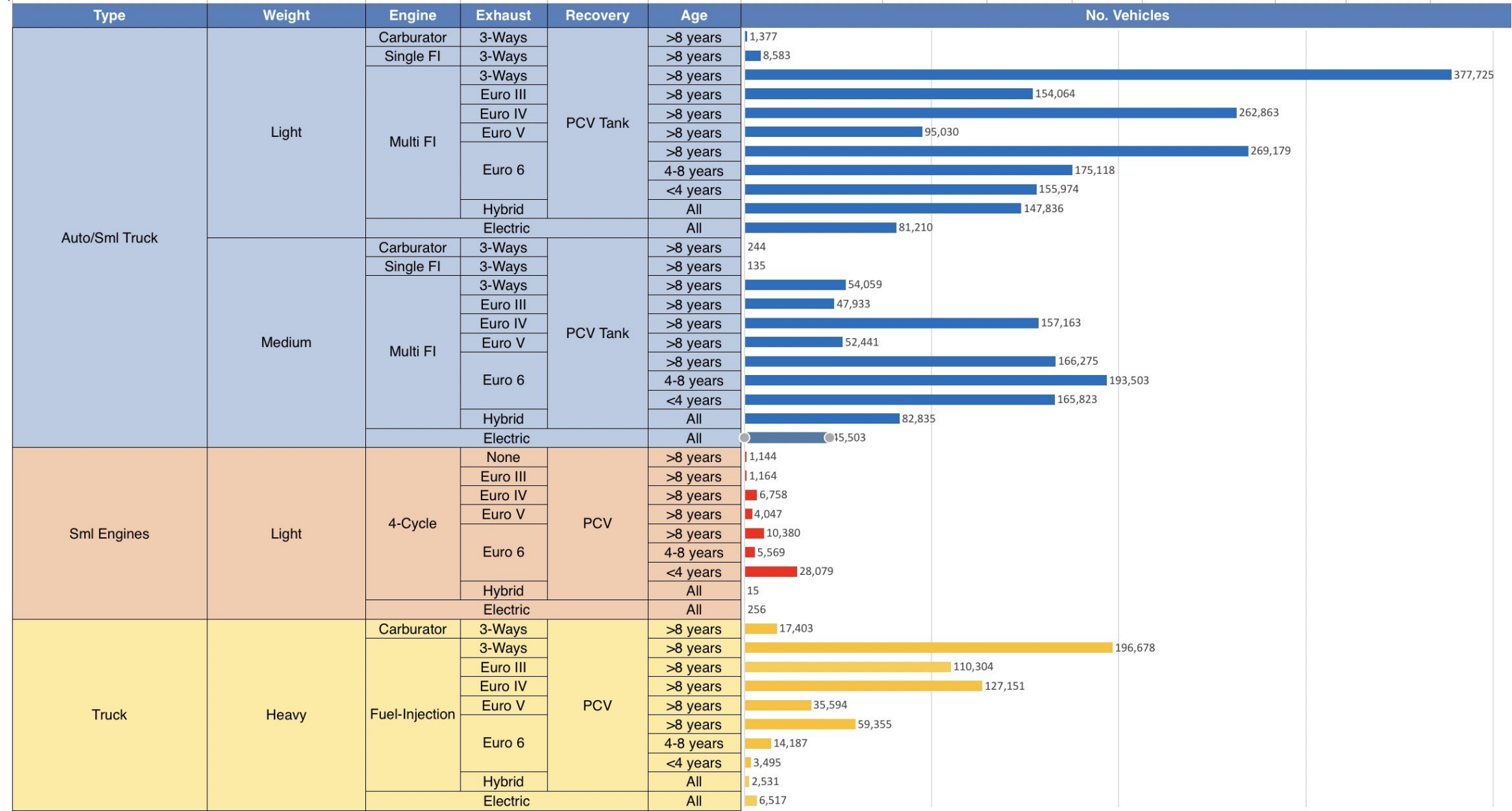
Source:
Faro90

Ireland – Gasoline 98 – Increased Octane



Average CIF 2025 Prices

Gasoline Vehicle Fleet - Ireland

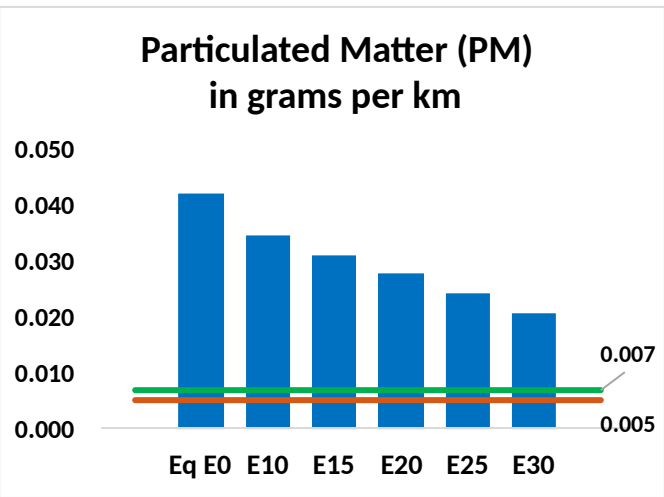
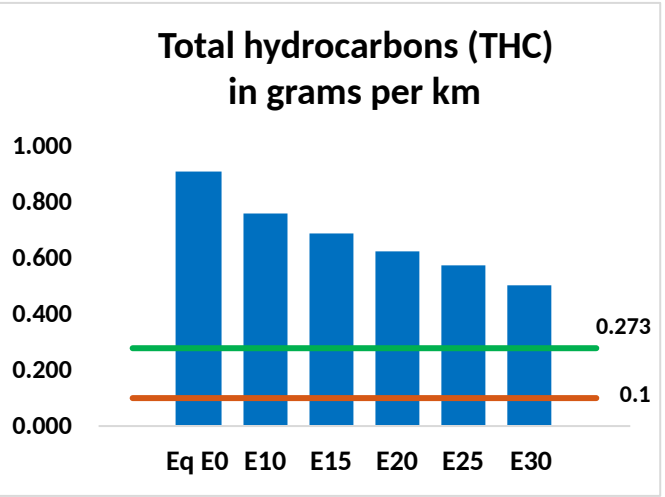
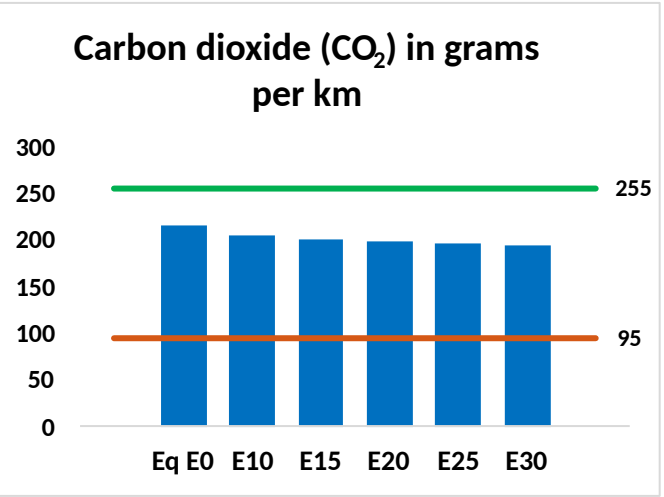
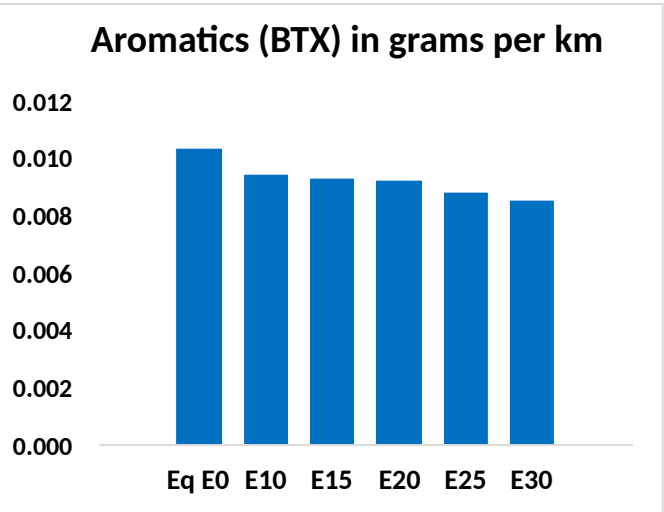
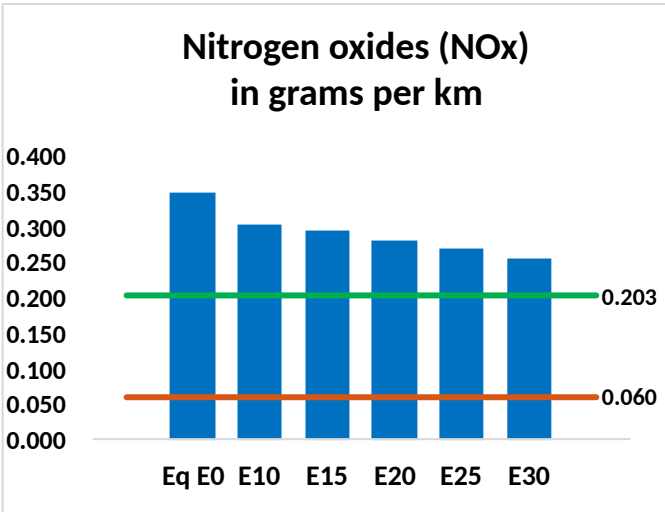
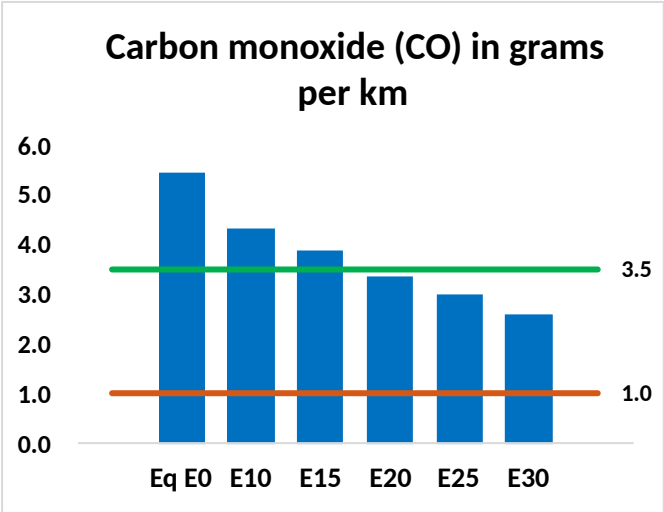
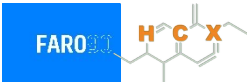


Vehicle Fleet: 3,325,499
Gasoline Vehicle Fleet: 966,996

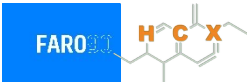
Source: Eurostat, ACEA

Ireland – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard

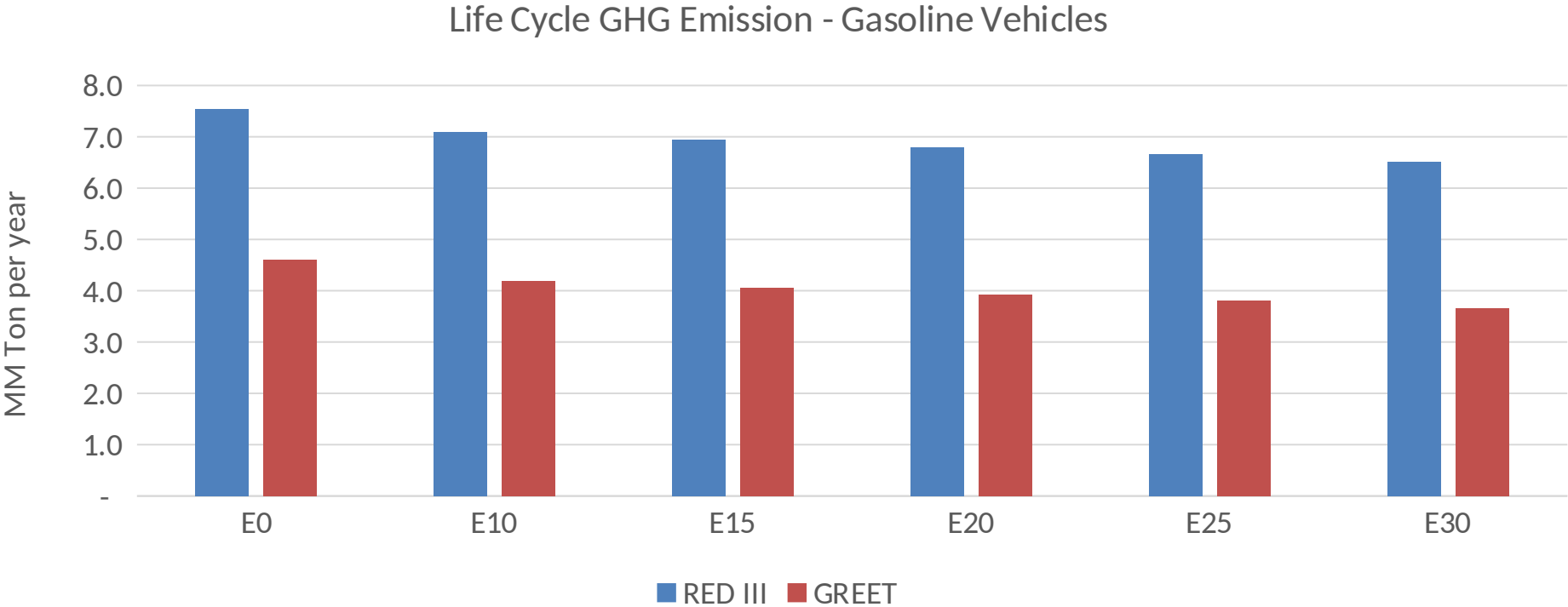
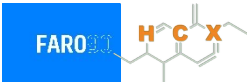


Ireland – Vehicular Emissions



Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	139,173	125,025	110,877	98,610	85,956	76,132	66,042	-20%	-38%
CO2	5,492,683	5,351,443	5,218,049	5,113,139	5,061,225	5,031,145	4,938,406	-5%	-8%
VOC	23,262	21,268	19,274	17,599	15,896	14,580	12,845	-17%	-32%
NOx	8,888	8,207	7,750	7,523	7,151	6,862	6,521	-13%	-20%
SOx	63	58	54	50	46	42	37	-15%	-28%
NH3	2,022	2,002	1,982	1,991	2,013	2,029	2,042	-2%	0%
N2O	107	106	106	106	108	109	111	-1%	2%
CH4	4,627	4,627	4,627	4,696	4,798	4,886	4,969	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	126	119	113	108	103	100	95	-10%	-18%
Acetaldehyde	385	442	648	987	1,345	1,536	1,816	68%	249%
Formaldehyde	1,422	1,383	1,612	1,893	1,983	2,194	2,388	13%	39%
Benzene	264	254	241	237	235	225	217	-9%	-11%
PM2.5	314	286	257	232	207	181	153	-18%	-34%
PM10	759	690	621	560	501	437	370	-18%	-34%

Ireland – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	70.0
Target @ 2035 (MMT/year)	39.4
Delta	30.6

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

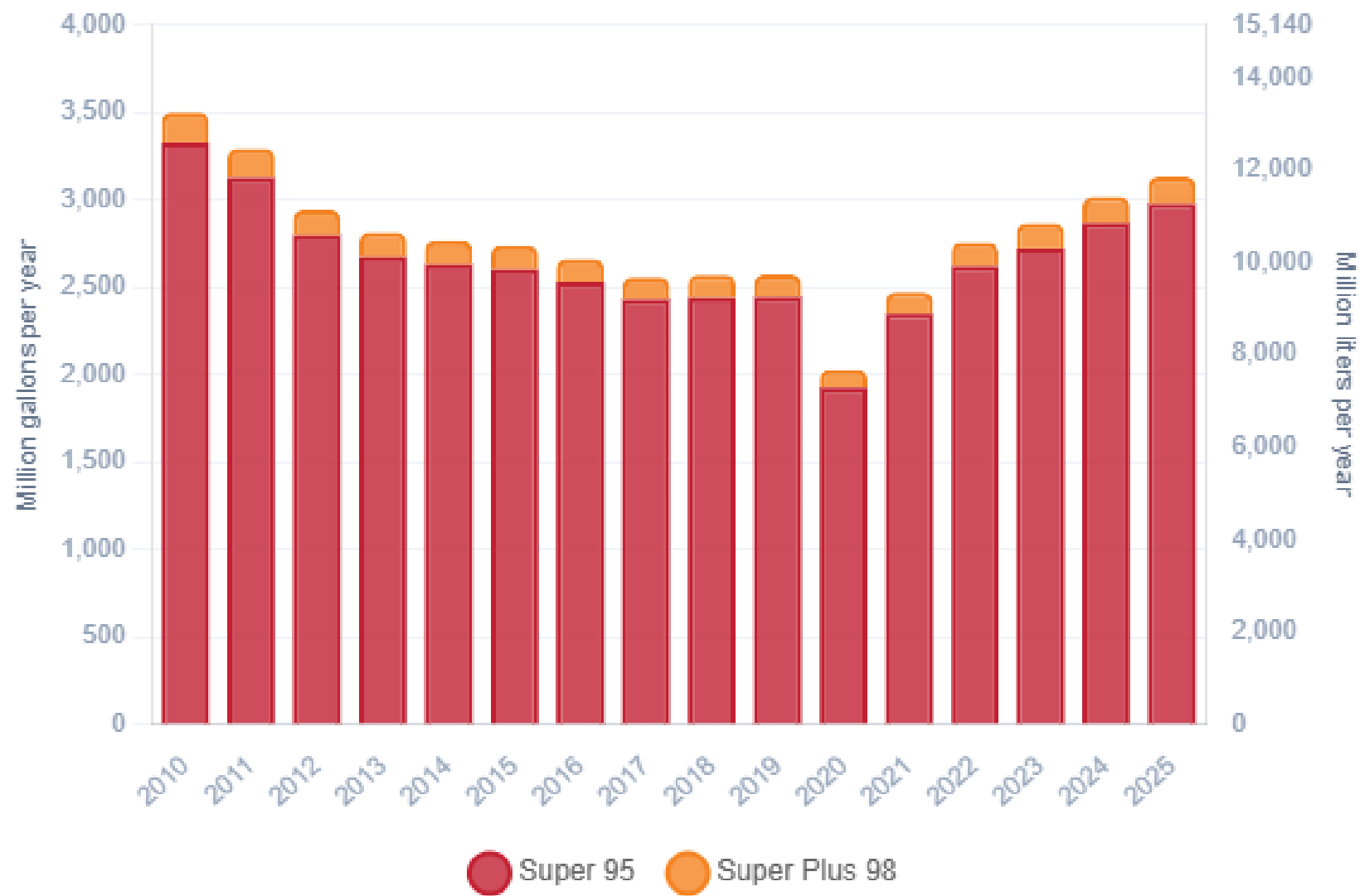
Ethanol Gasoline Blending in Italy



In 2025, gasoline consumption in Italy was 11,871 million liters (3,136 million gallons) and growth rate was 3.2%. Gasoline production and net imports were 17,503 and -5,636 million liters (4,624 and -1,489 million gallons) correspondingly. Italy complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

In 2025, ethanol consumption in Italy was 277 million liters (73 million gallons) representing 2% of gasoline demand. Ethanol production and Net Imports were 71 and 206 million liters (19 and 54 million gallons) correspondingly. Ethanol sources are Wine and Second Generation Lignocelulosic ethanol.

Gasoline Demand in Italy



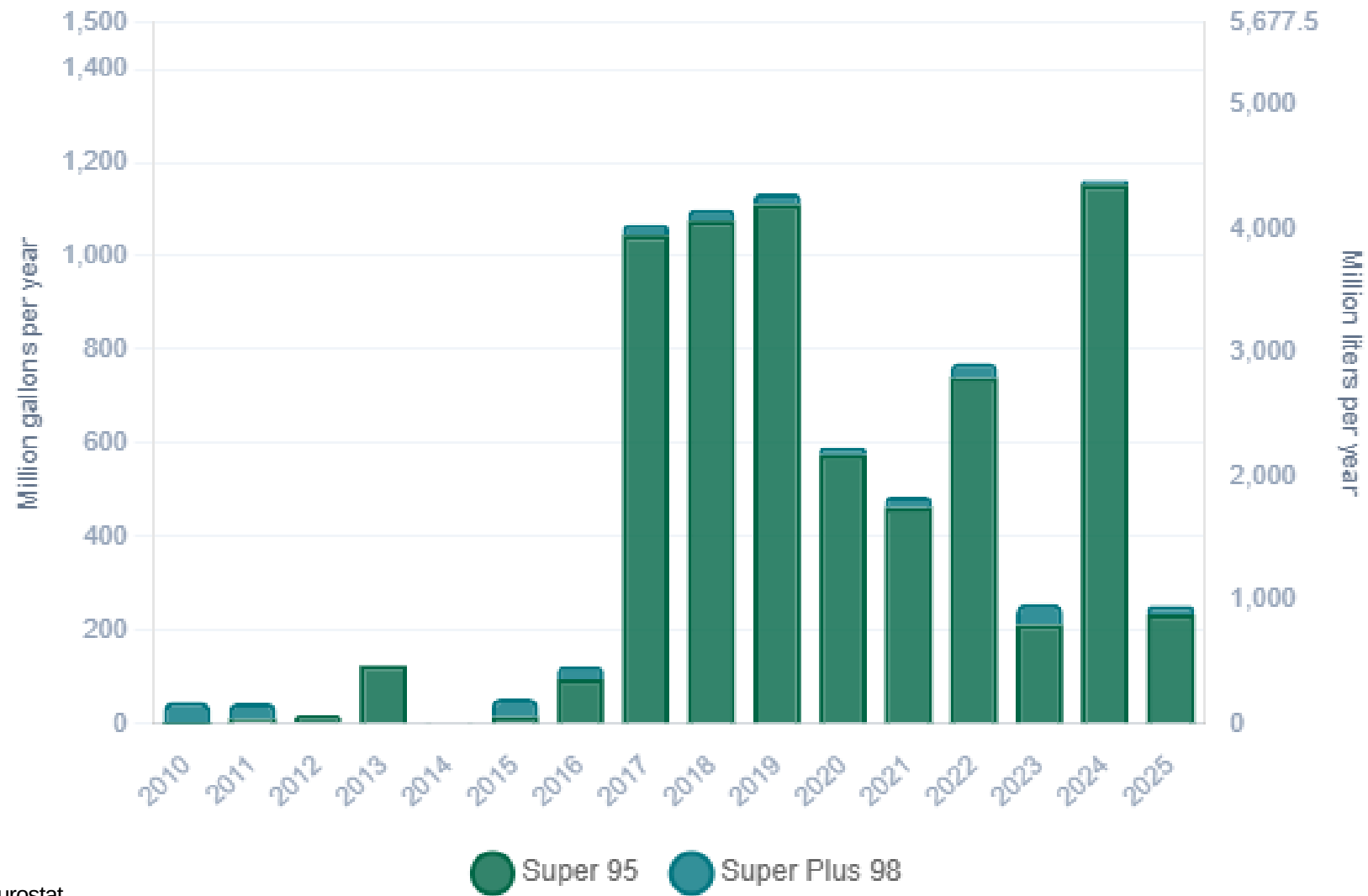
Source: Eurostat

Gasoline Production in Italy



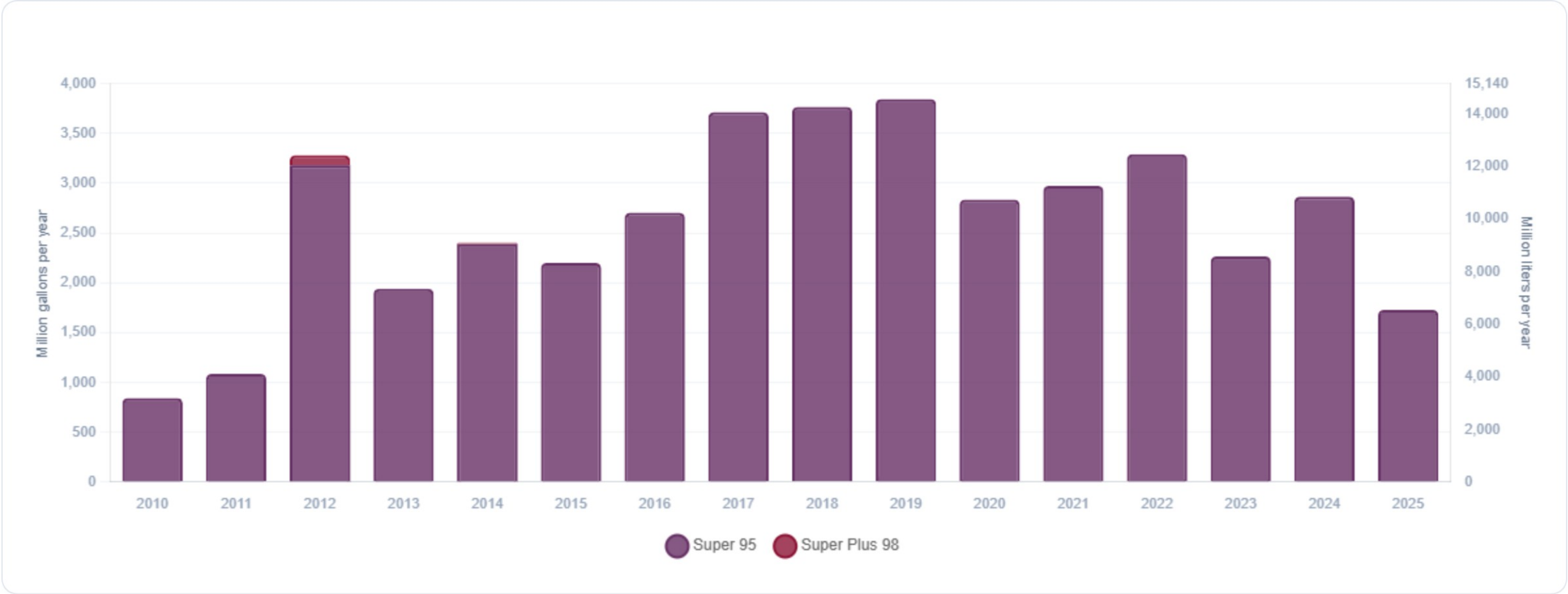
Source: Eurostat

Gasoline Imports in Italy



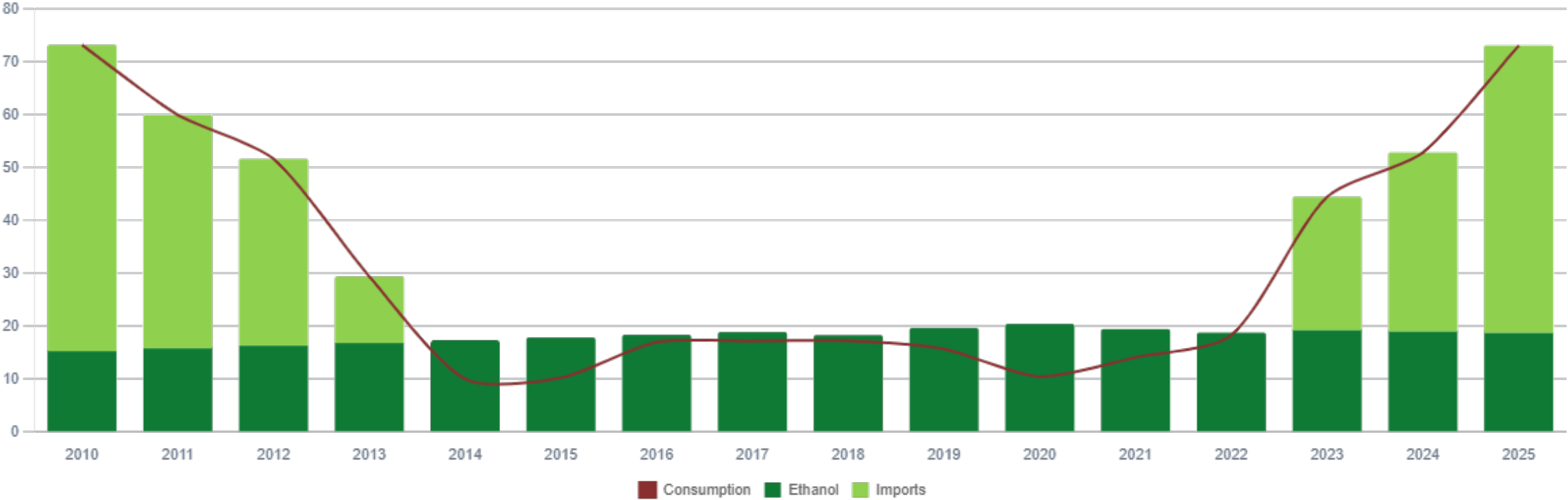
Source: Eurostat

Gasoline Exports in Italy



Source: Eurostat

Ethanol Balance in Italy



Source: Eurostat

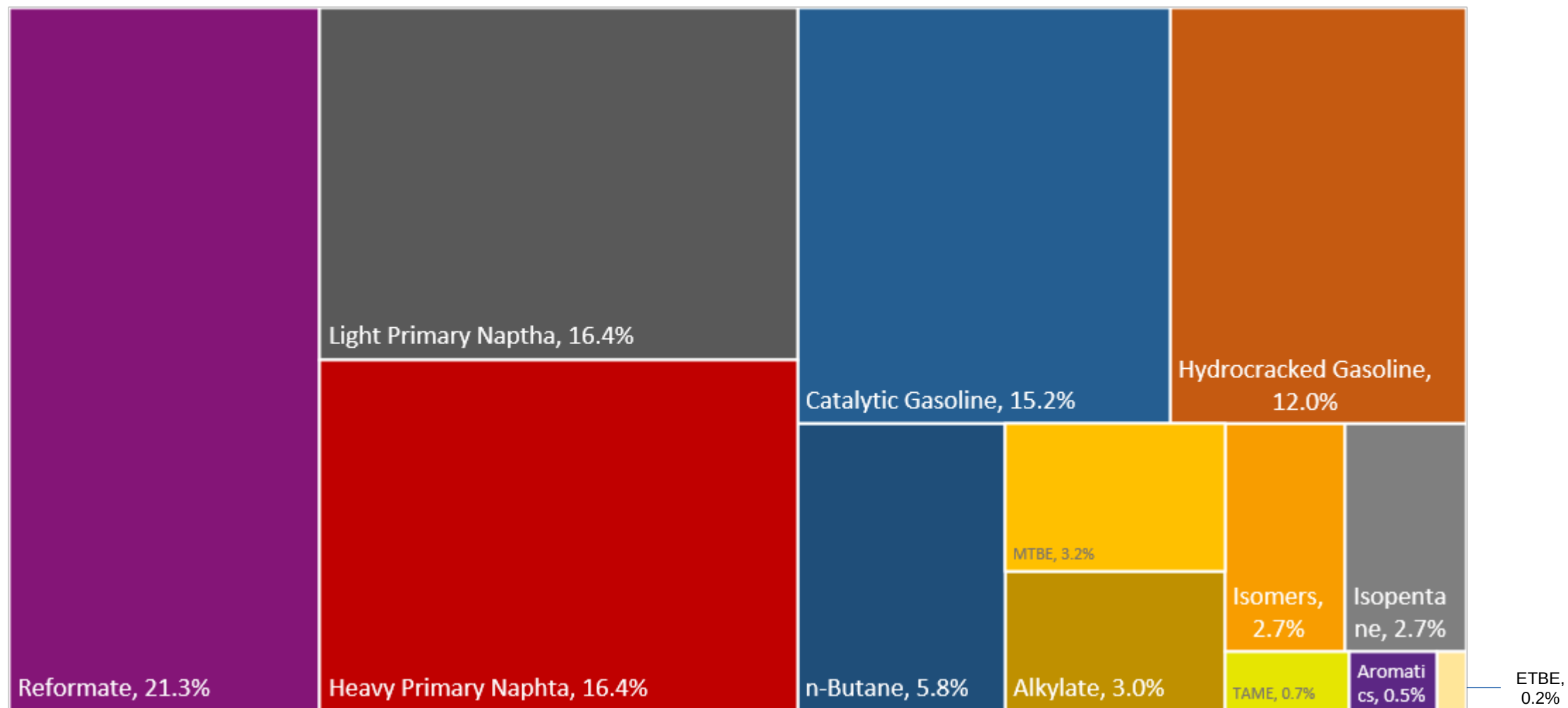
Gasoline Quality in Italy

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	11.7							
Advanced Biofuels (%cal)	4.9							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

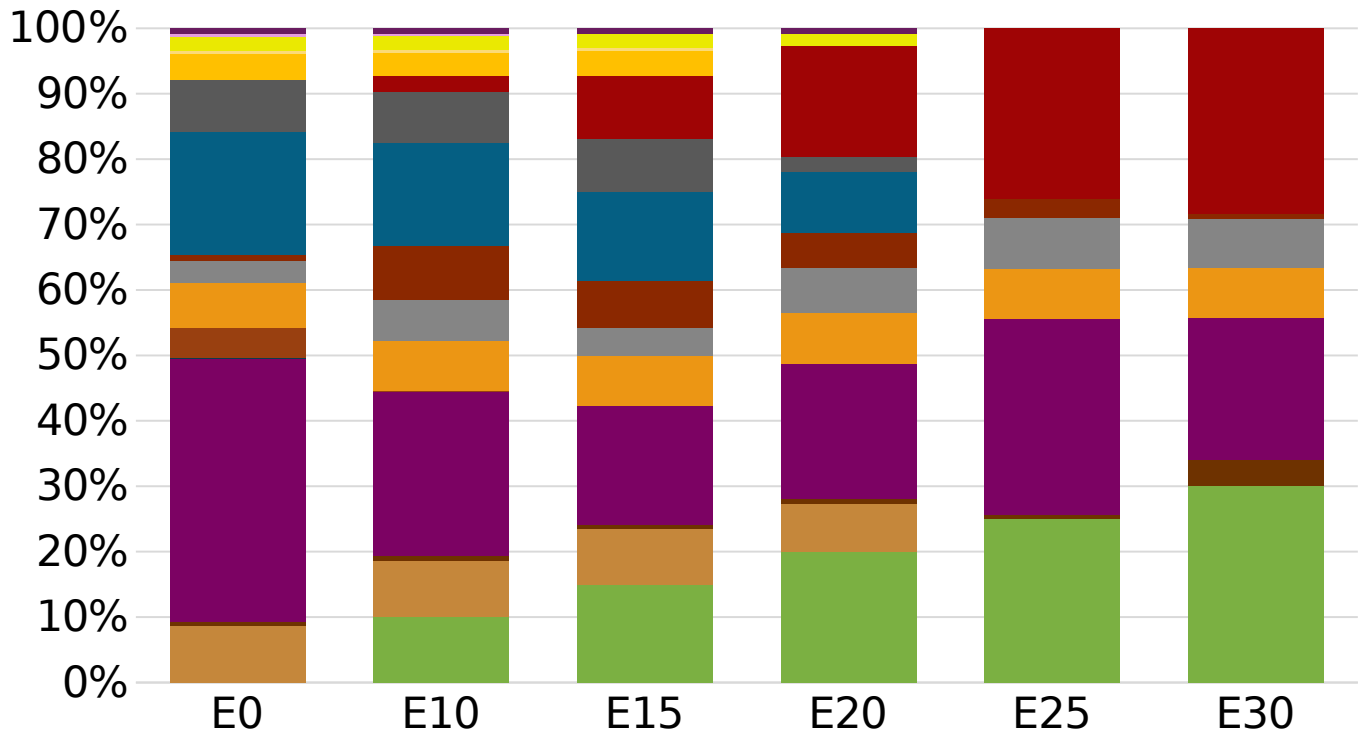
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Italy Available Blending Components



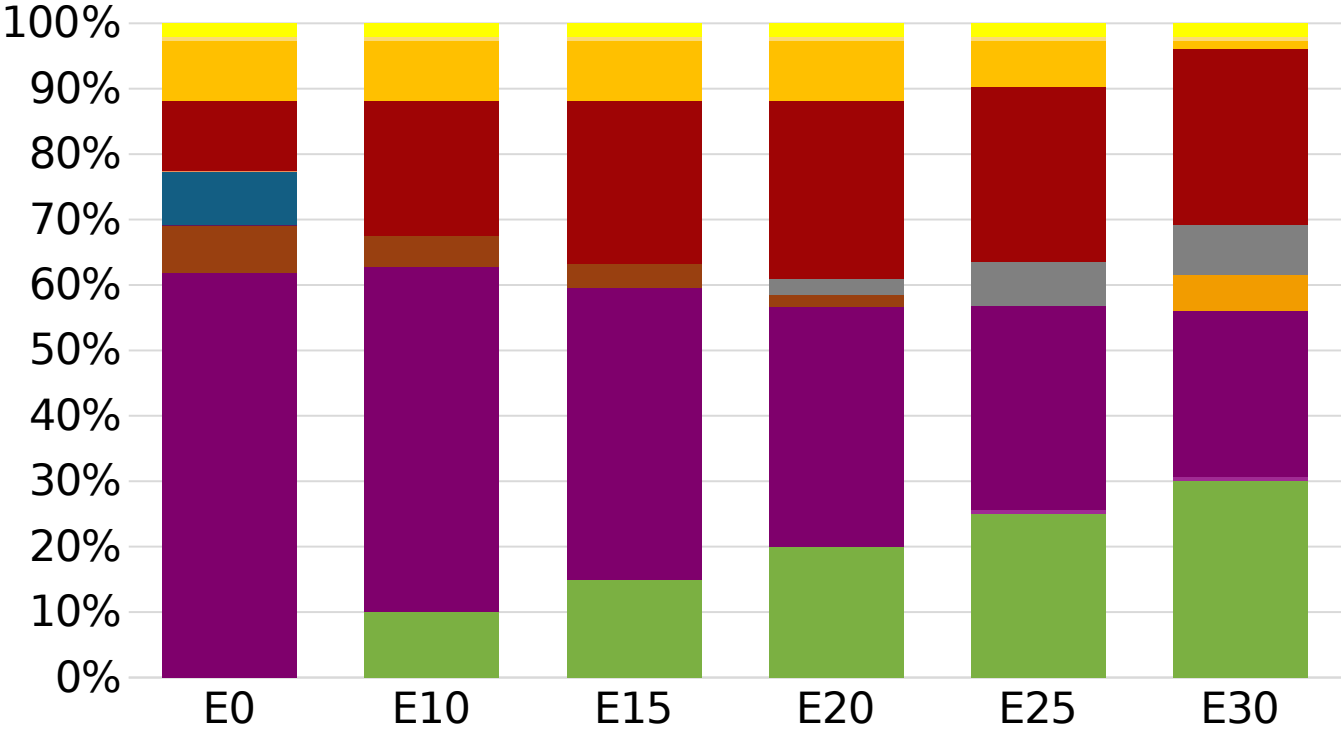
Italy – Gasoline 95 – Constant Octane



Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US\$/gal)	\$ 2.055	\$ 1.942	\$ 1.859	\$ 1.783	\$ 1.688	\$ 1.614

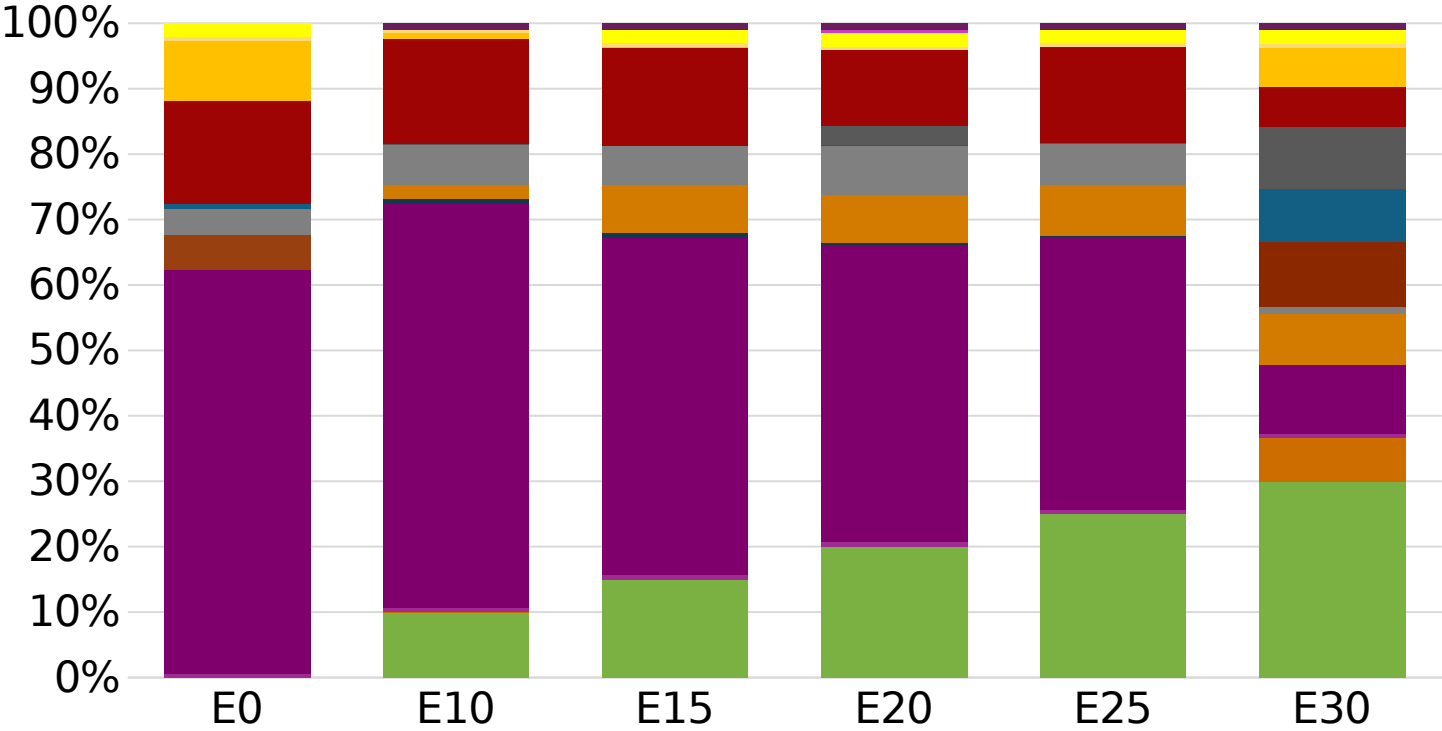
Italy - Gasoline 98 - Constant Octane



Average CIF 2025
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.104	\$ 1.985	\$ 1.916	\$ 1.846	\$ 1.776	\$ 1.723

Italy – Gasoline 95 – Increased Octane

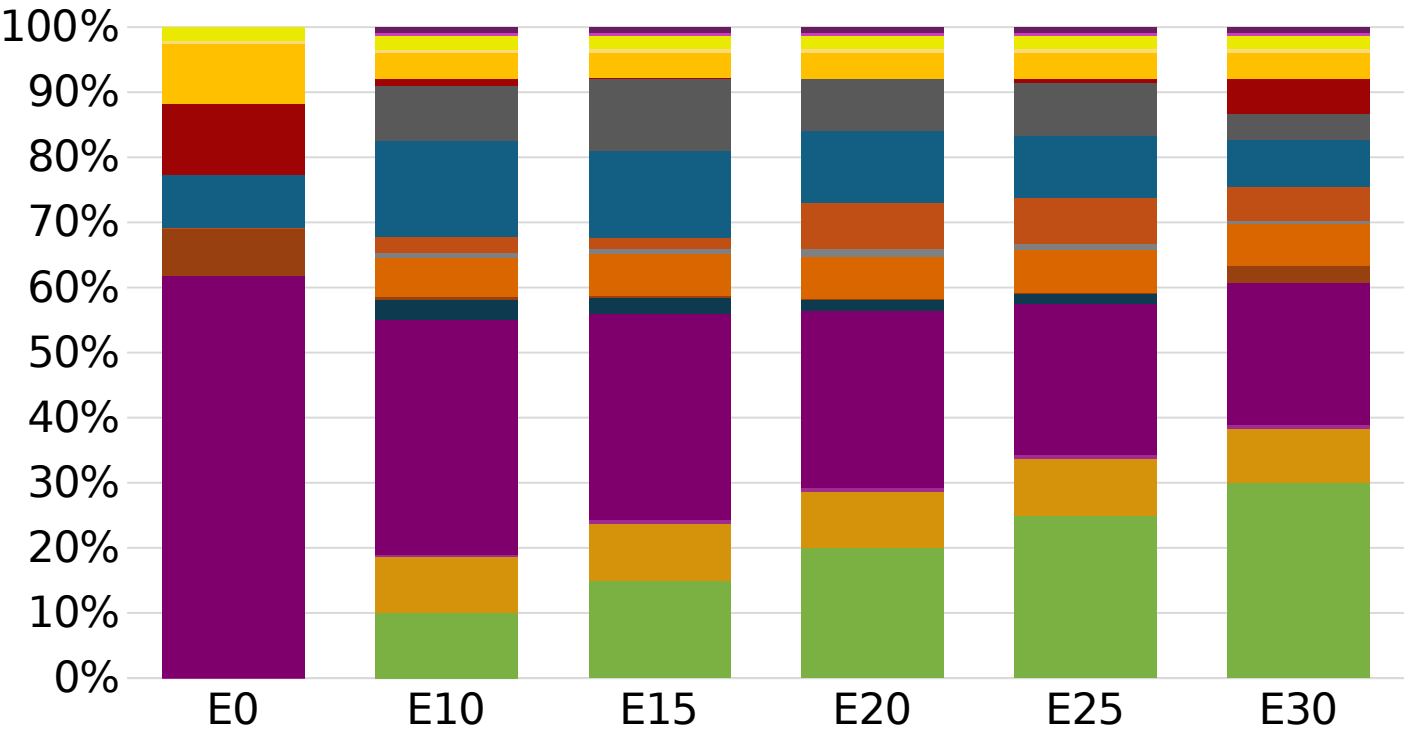


Average CIF 2025
Prices

Octane (RON)	95.0	95.9	97.5	99.0	100.5	101.8
Price (US\$/gal)	\$ 1.974	\$ 1.974	\$ 1.974	\$ 1.974	\$ 1.974	\$ 1.974

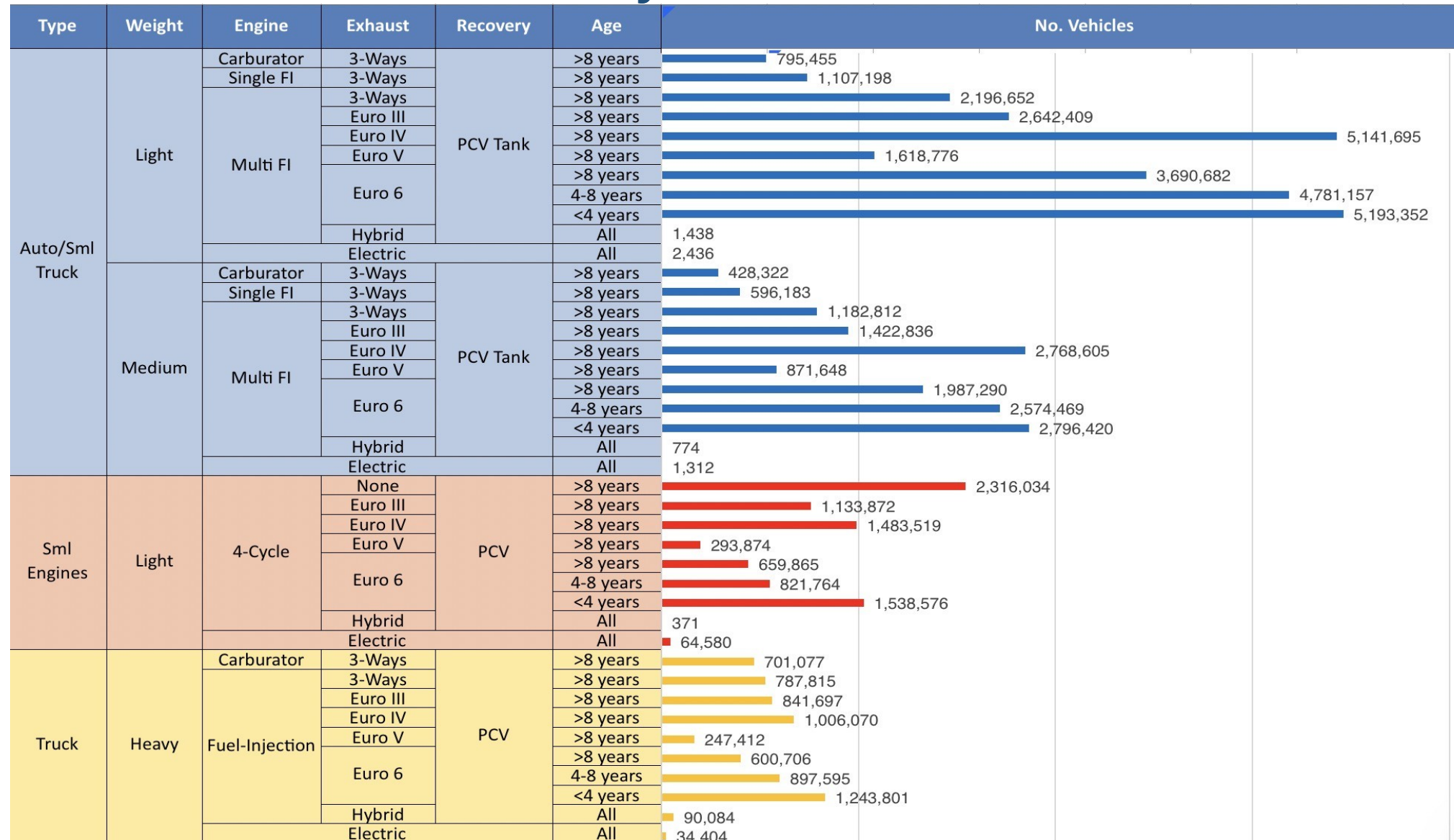
Source:
Faro90

Italy – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Gasoline Vehicle Fleet - Italy

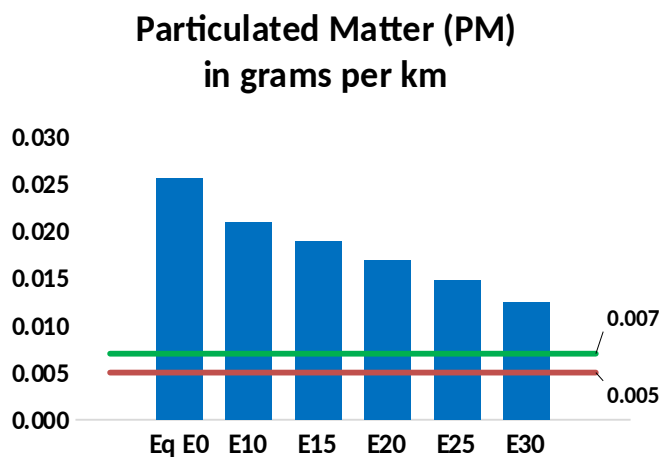
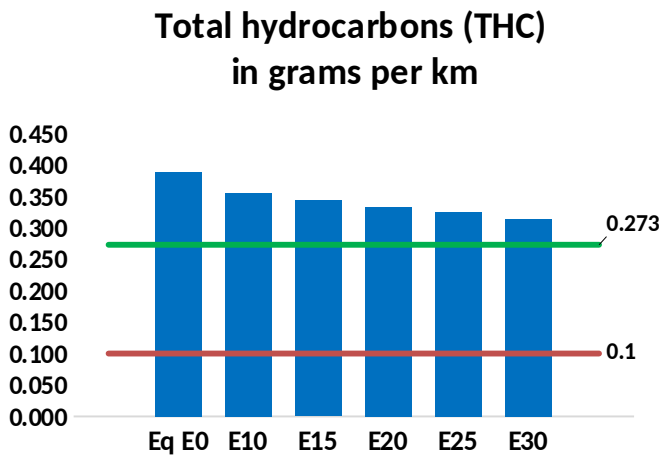
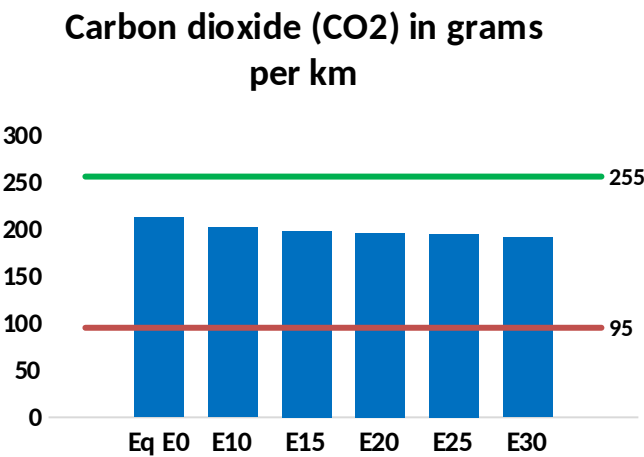
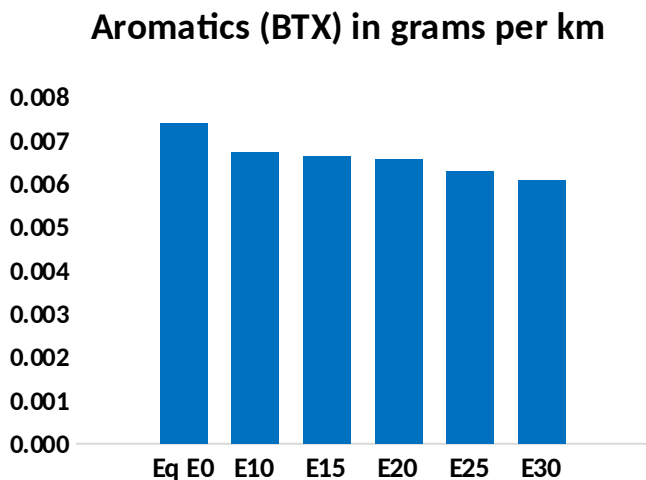
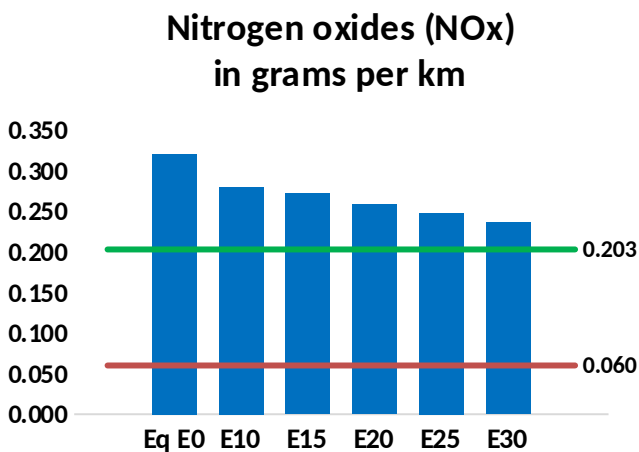
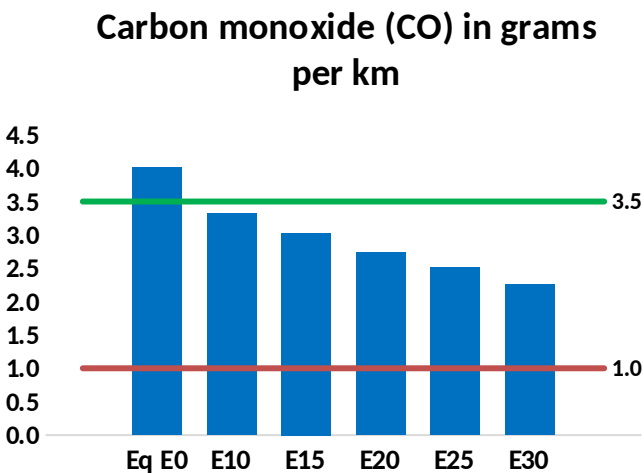
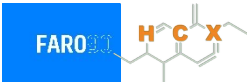


Vehicle Fleet: 56,565,039
Gasoline Vehicle Fleet: 20,088,725

Source: Eurostat, ACEA

Italy – Vehicular Emissions

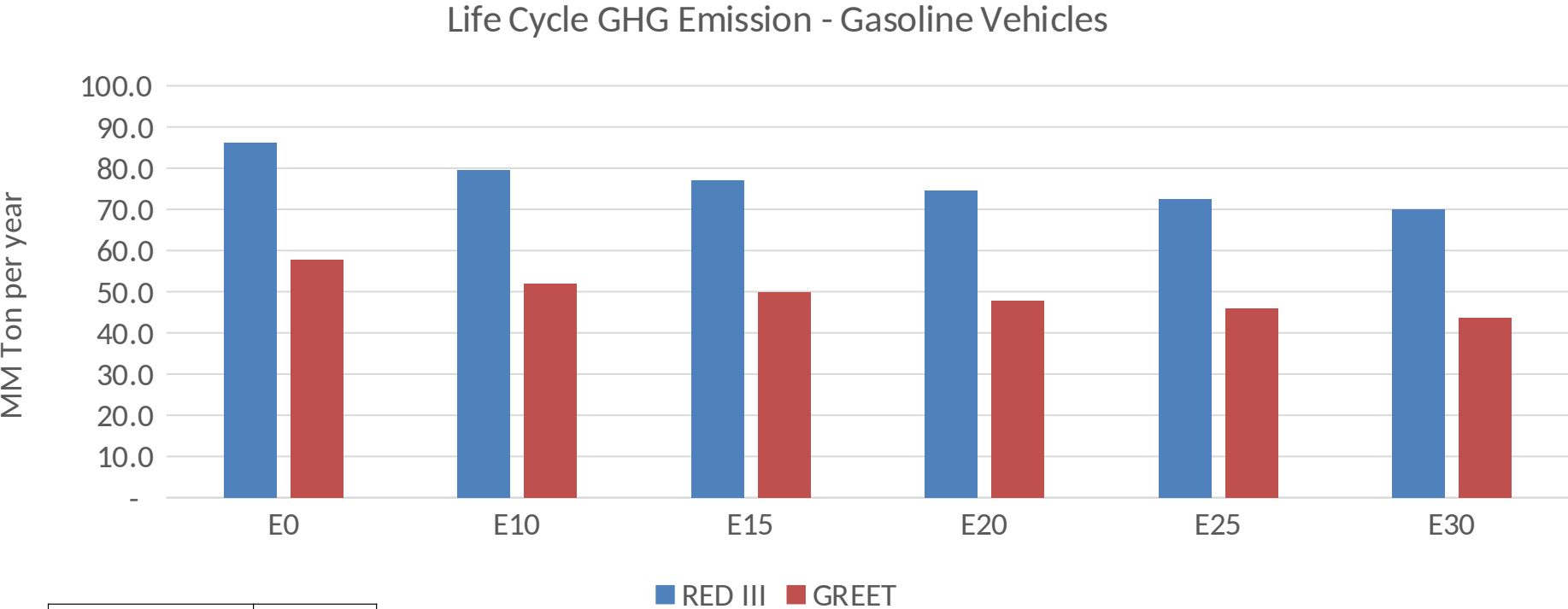
TIER III BIN 125 United States
Euro 6 Standard



Italy – Vehicular Emissions

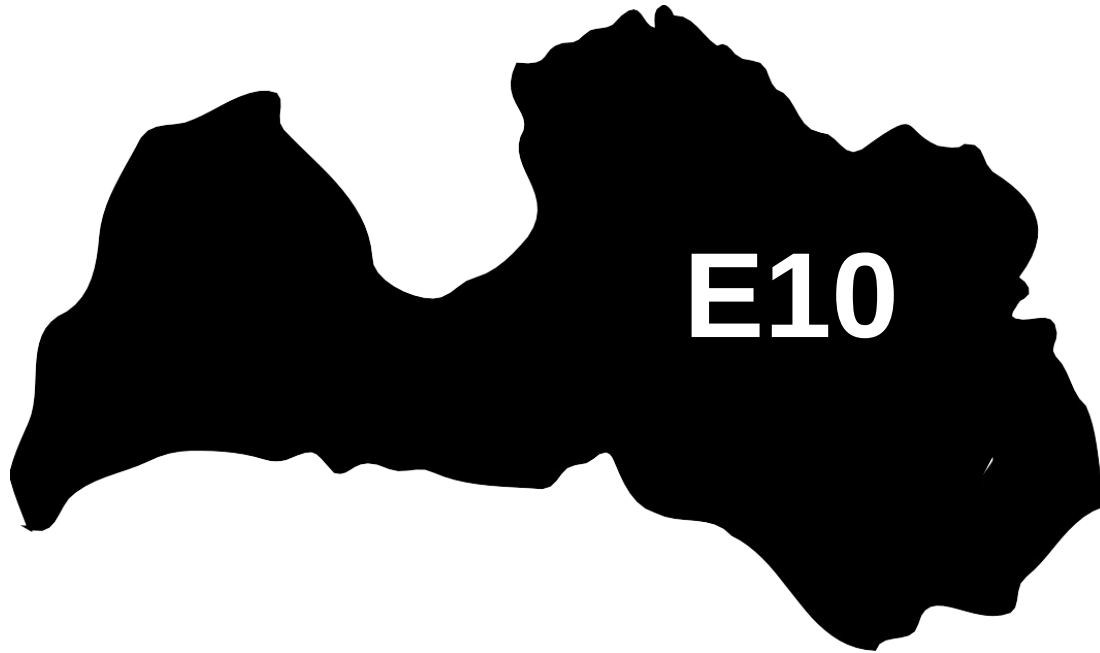
Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	1,846,146	1,687,397	1,528,648	1,395,276	1,259,550	1,154,728	1,038,885	-17%	-32%
CO2	97,604,300	95,094,475	92,724,085	90,859,843	89,937,343	89,487,006	87,837,504	-5%	-8%
VOC	179,403	171,498	163,594	158,265	153,431	149,874	144,564	-9%	-14%
NOx	147,812	136,479	128,951	125,244	119,129	114,402	108,795	-13%	-19%
SOx	1,120	1,035	951	879	810	736	657	-15%	-28%
NH3	33,389	33,058	32,728	32,883	33,253	33,513	33,729	-2%	0%
N2O	1,638	1,629	1,623	1,631	1,664	1,683	1,702	-1%	2%
CH4	41,617	41,617	41,617	42,241	43,156	43,947	44,696	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	1,523	1,448	1,373	1,319	1,269	1,231	1,177	-10%	-17%
Acetaldehyde	3,252	3,732	5,477	8,340	11,363	12,984	15,342	68%	249%
Formaldehyde	10,843	10,542	12,289	14,430	15,118	16,724	18,206	13%	39%
Benzene	3,408	3,275	3,103	3,054	3,028	2,896	2,803	-9%	-11%
PM2.5	5,647	5,134	4,620	4,168	3,726	3,253	2,757	-18%	-34%
PM10	6,142	5,584	5,025	4,534	4,053	3,538	2,998	-18%	-34%

Italy – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	357.1
Target @ 2035 (MMT/year)	201.0
Delta	156.1

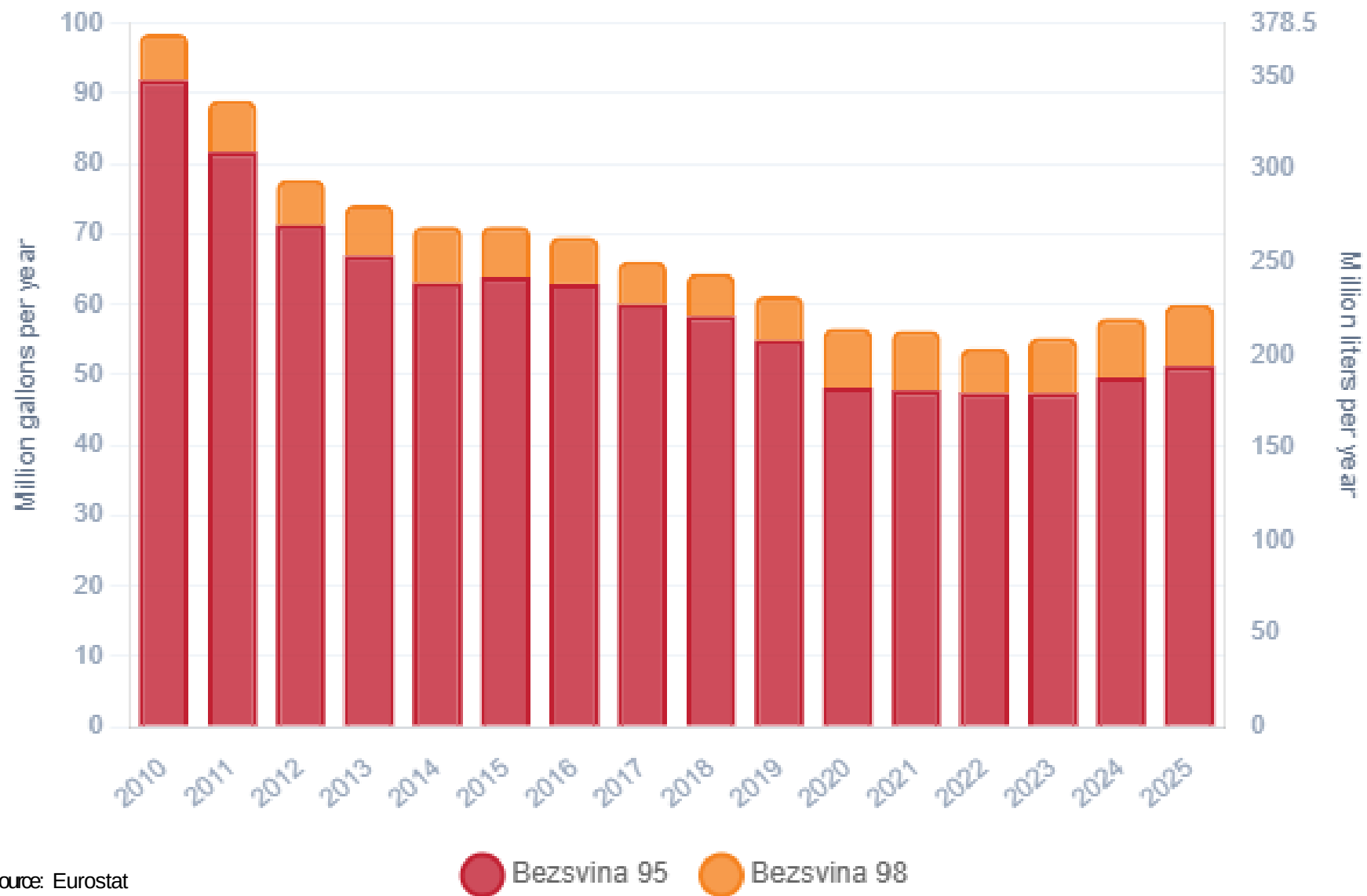
Ethanol Gasoline Blending in Latvia



In 2025, gasoline consumption in Latvia was 227 million liters (60 million gallons) and growth rate was -3.3%. There is no gasoline production. Gasoline net imports were 123 million liters (33 million gallons). Latvia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.3 RON.

In 2025, ethanol consumption in Latvia was 13 million liters (3 million gallons) representing 6% of gasoline demand. Net Imports were 13 million liters (3 million gallons) correspondingly. Ethanol source Cereals 100%.

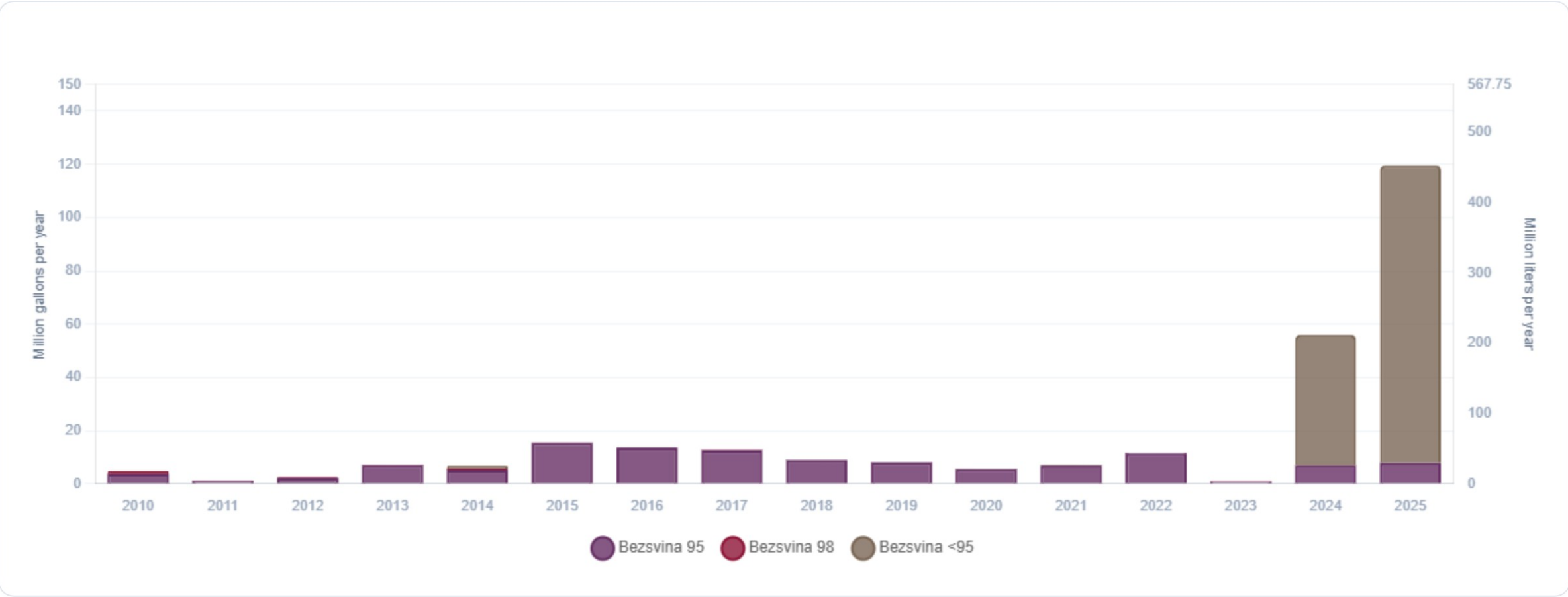
Gasoline Demand in Latvia



Gasoline Imports in Latvia

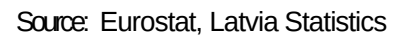


Gasoline Exports in Latvia



Source: Eurostat

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexene. The structure features a six-membered ring with a double bond. Substituents include a hydrogen atom (H), a carbon atom (C), and a group labeled 'X' (which is a vinyl group, -CH=CH₂). The ring also has methyl groups and a propyl group attached.



Source: Eurostat, Latvia Statistics

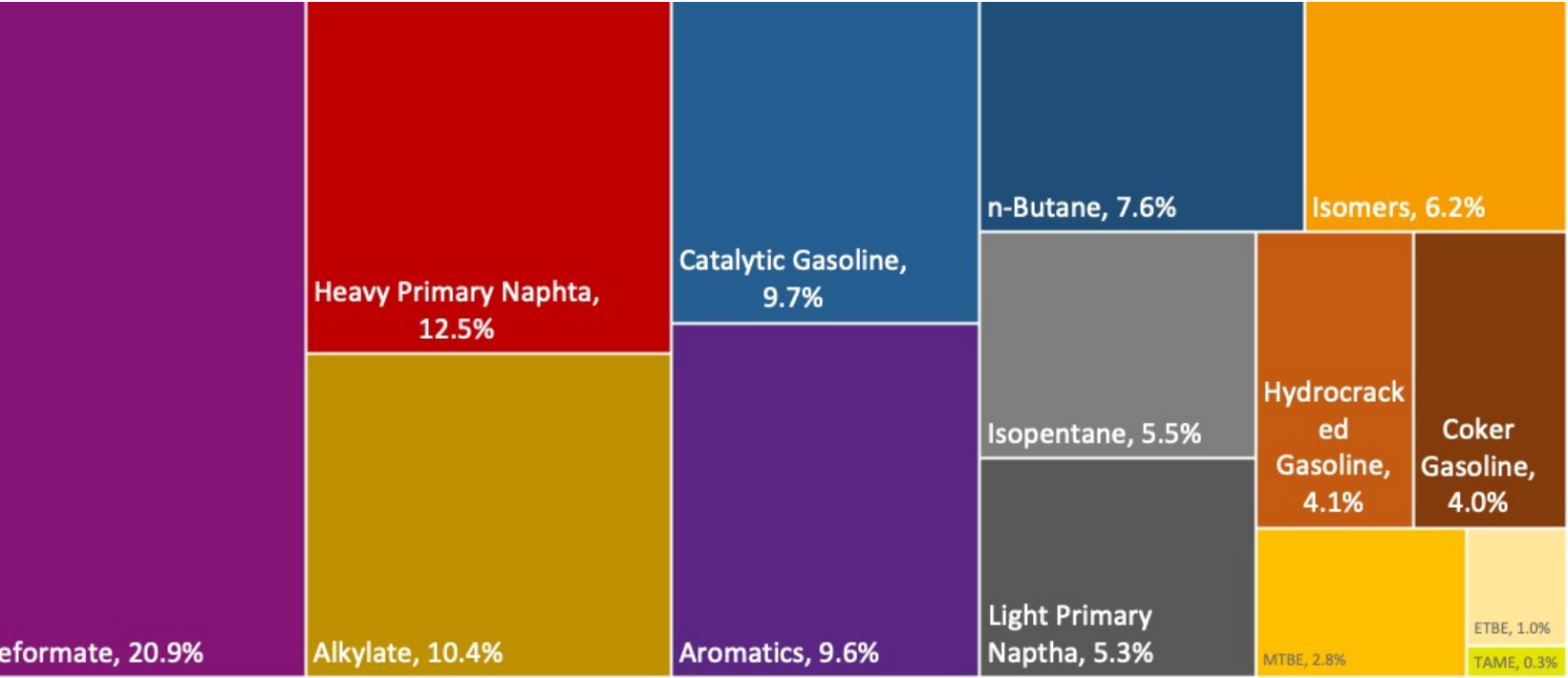
Gasoline Quality in Latvia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Countries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

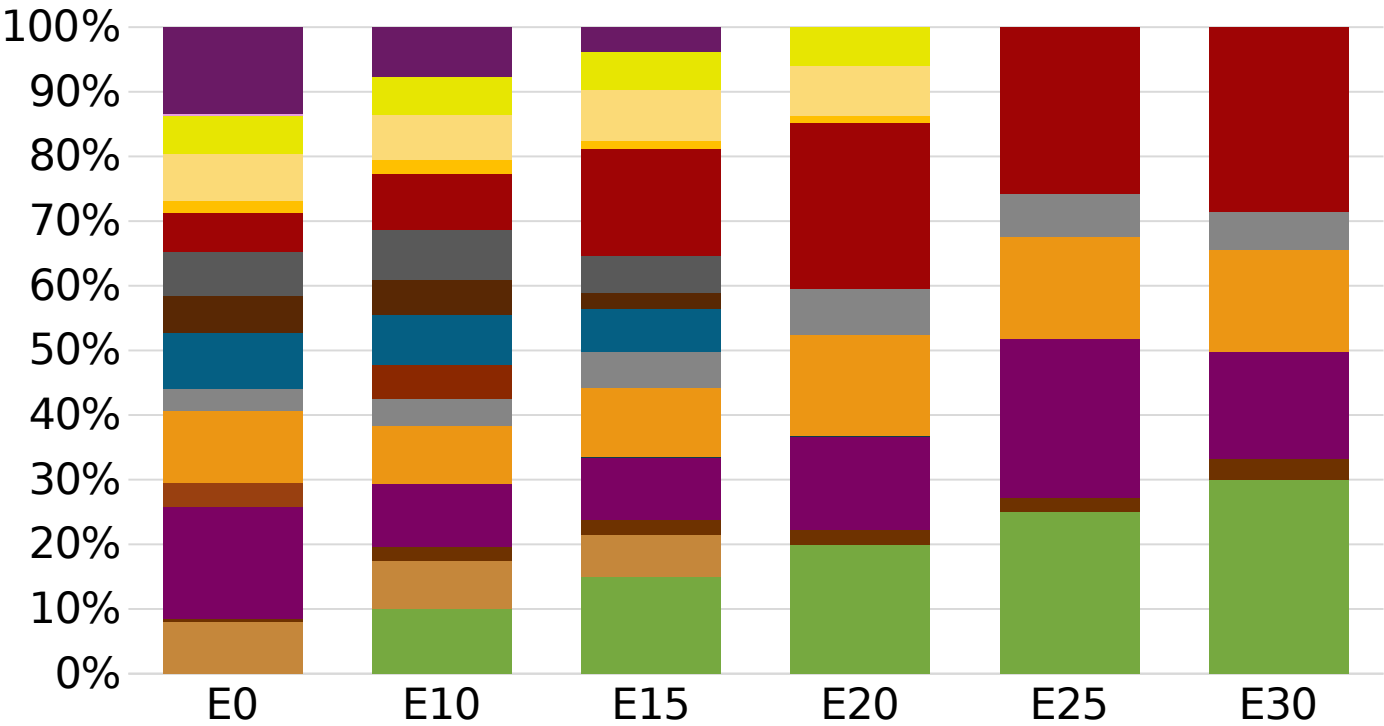
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commission

Latvia Available Blending Components



Latvia – Gasoline 95 – Constant Octane

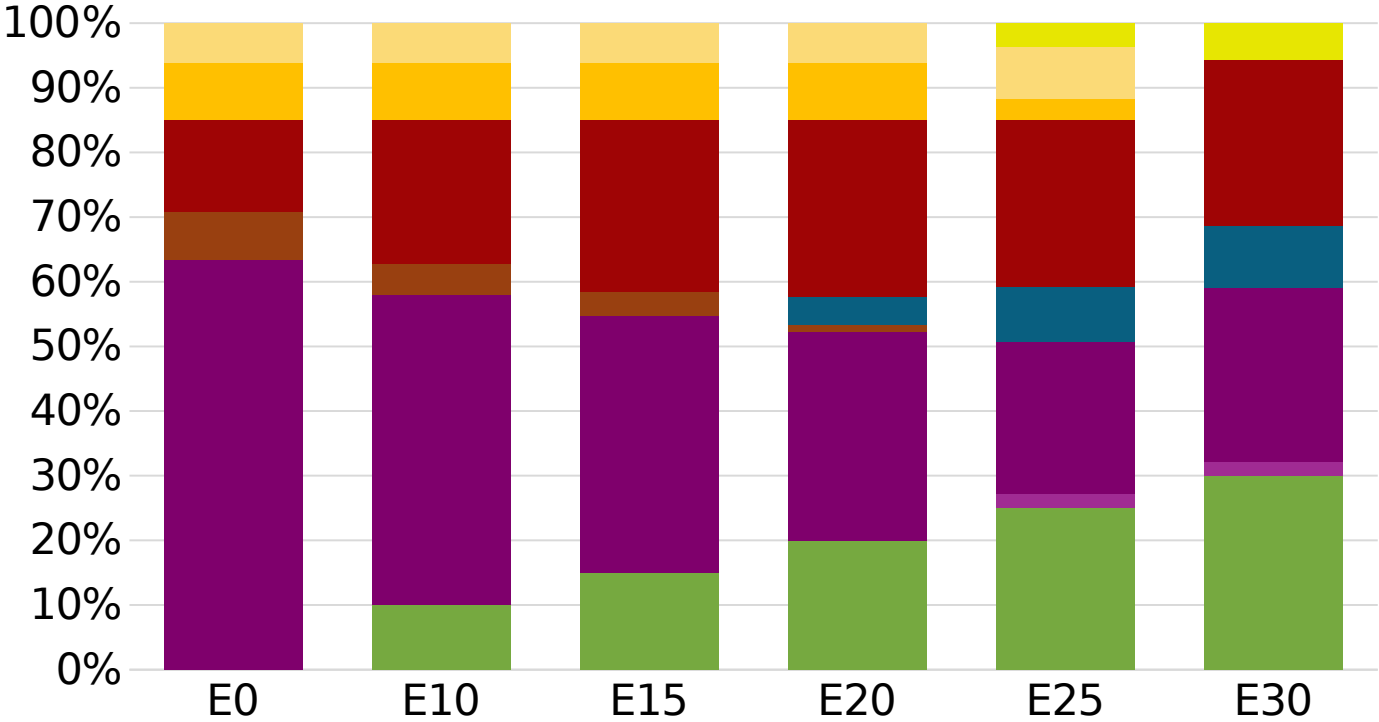


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.071	\$ 1.933	\$ 1.833	\$ 1.718	\$ 1.672	\$ 1.619

Source:
Faro90

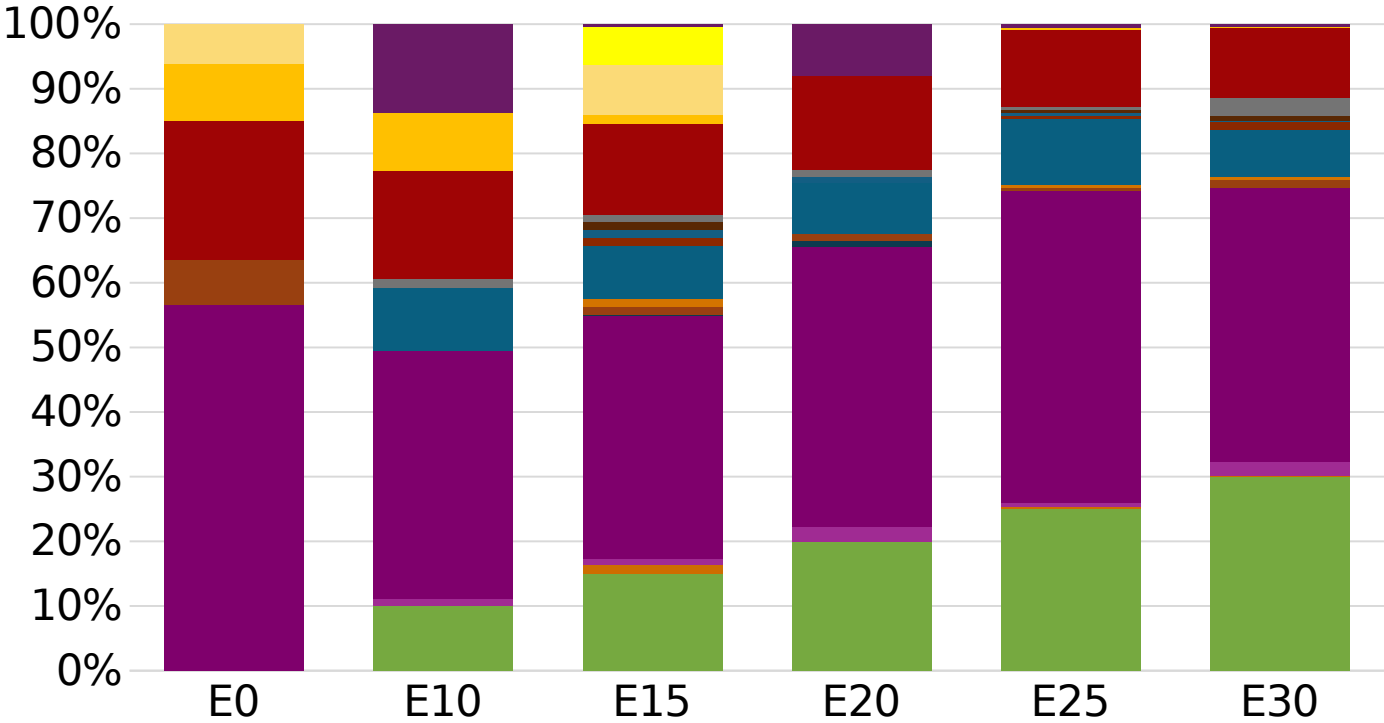
Latvia - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.091	\$ 1.974	\$ 1.905	\$ 1.833	\$ 1.768	\$ 1.718

Average CIF 2025
Prices

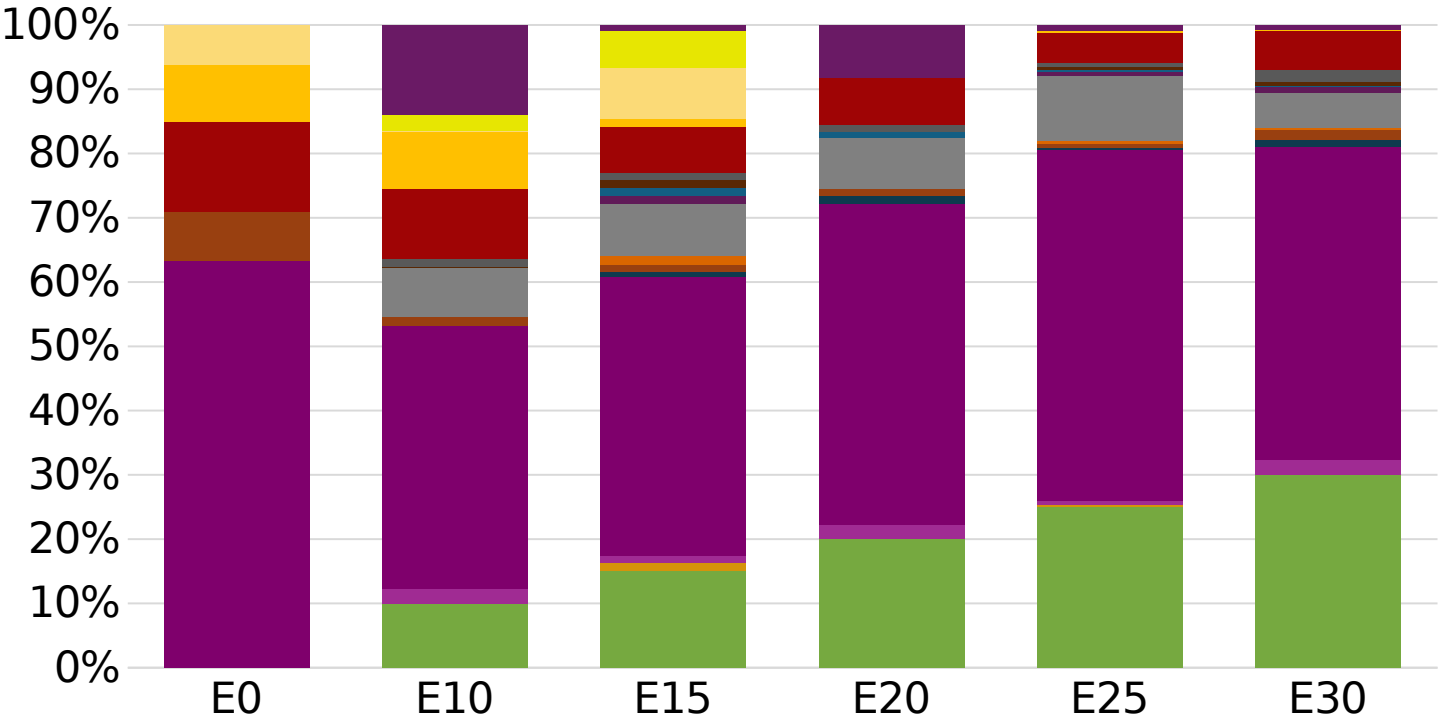
Latvia – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

Source:
Faro90

Latvia – Gasoline 98 – Increased Octane

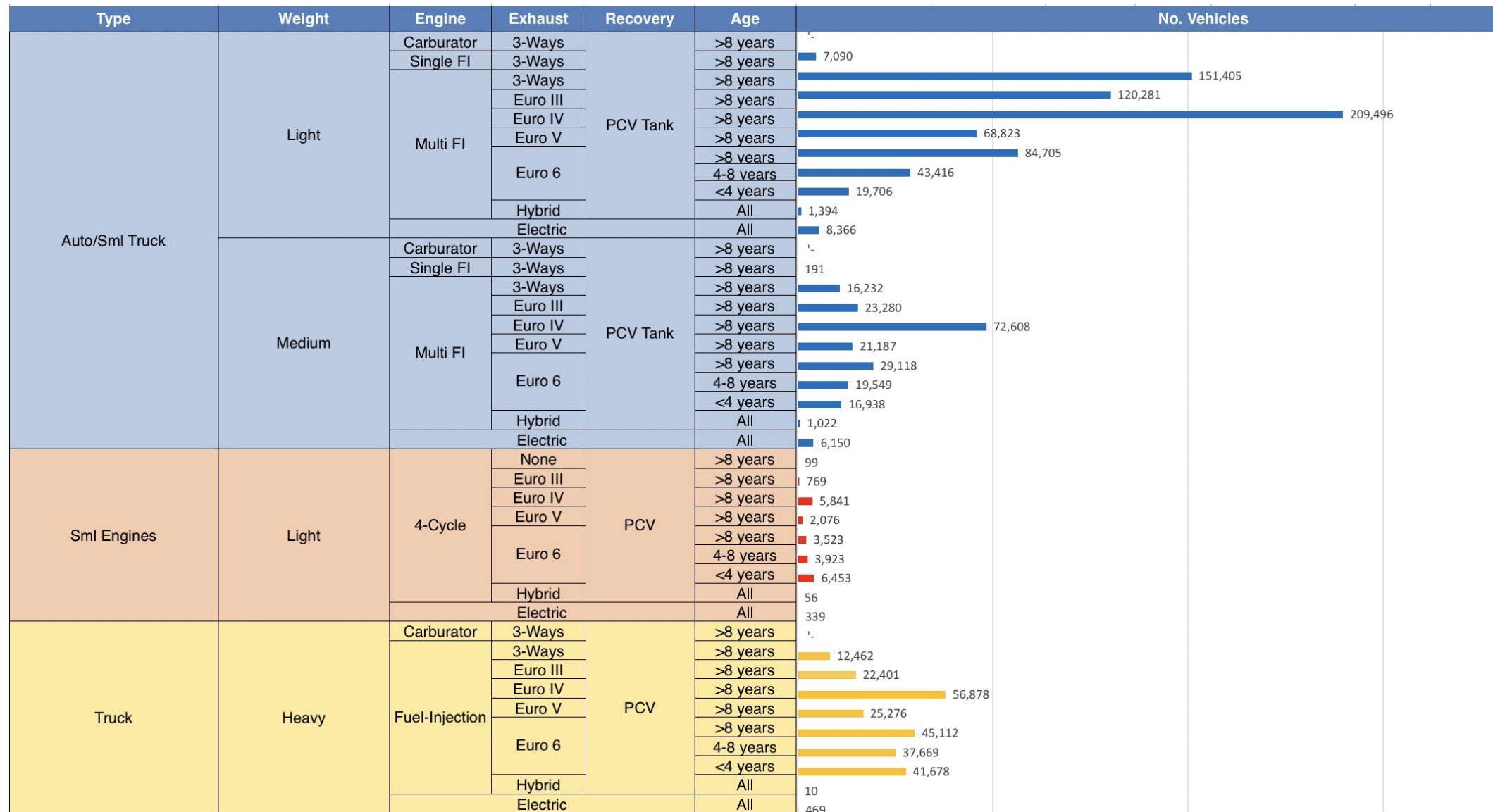


Average CIF 2025
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.091	\$ 2.091	\$ 2.091	\$ 2.091	\$ 2.091	\$ 2.091

Source:
Faro90

Gasoline Vehicle Fleet - Latvia



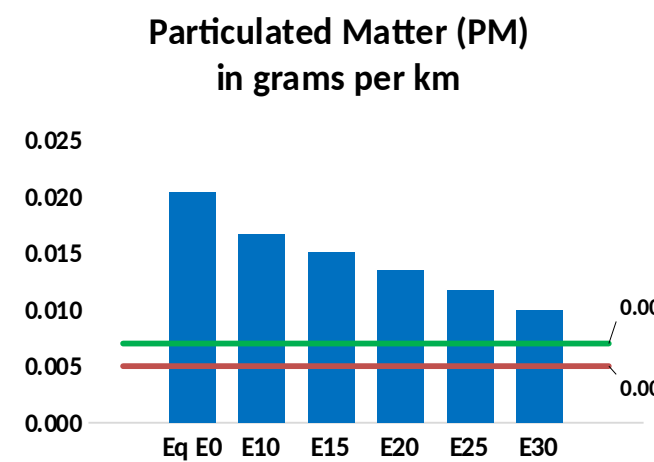
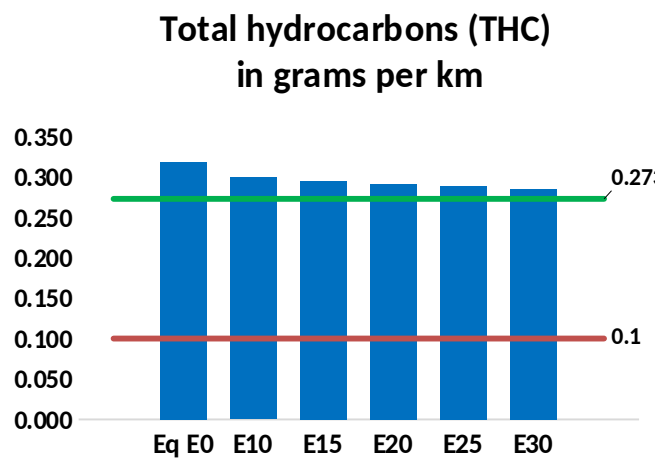
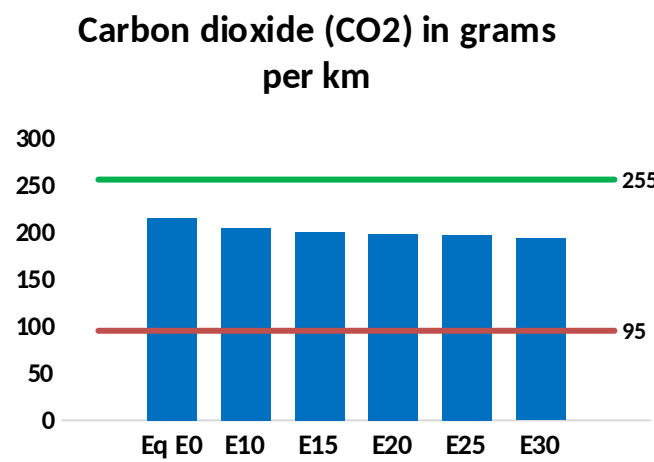
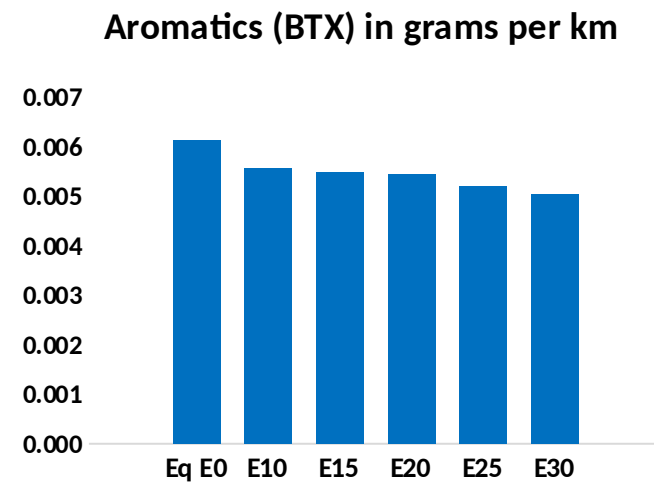
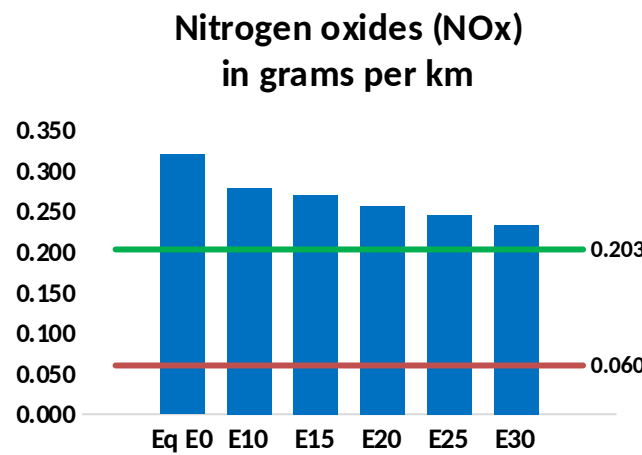
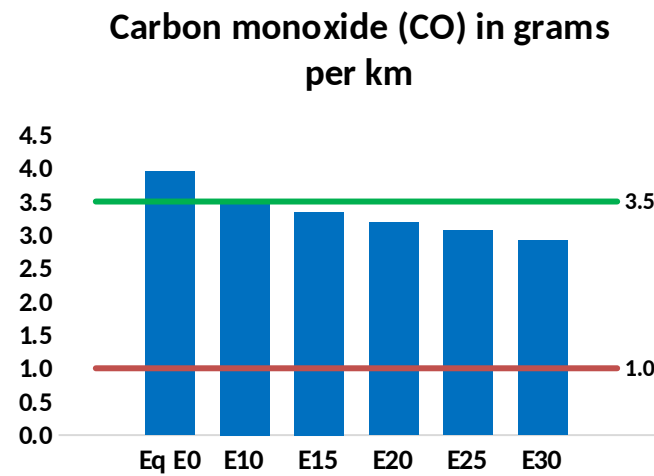
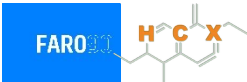
Vehicle Fleet: 1,185,994

Gasoline Vehicle Fleet: 293,341

Source: Eurostat, ACEA

Latvia – Vehicular Emissions

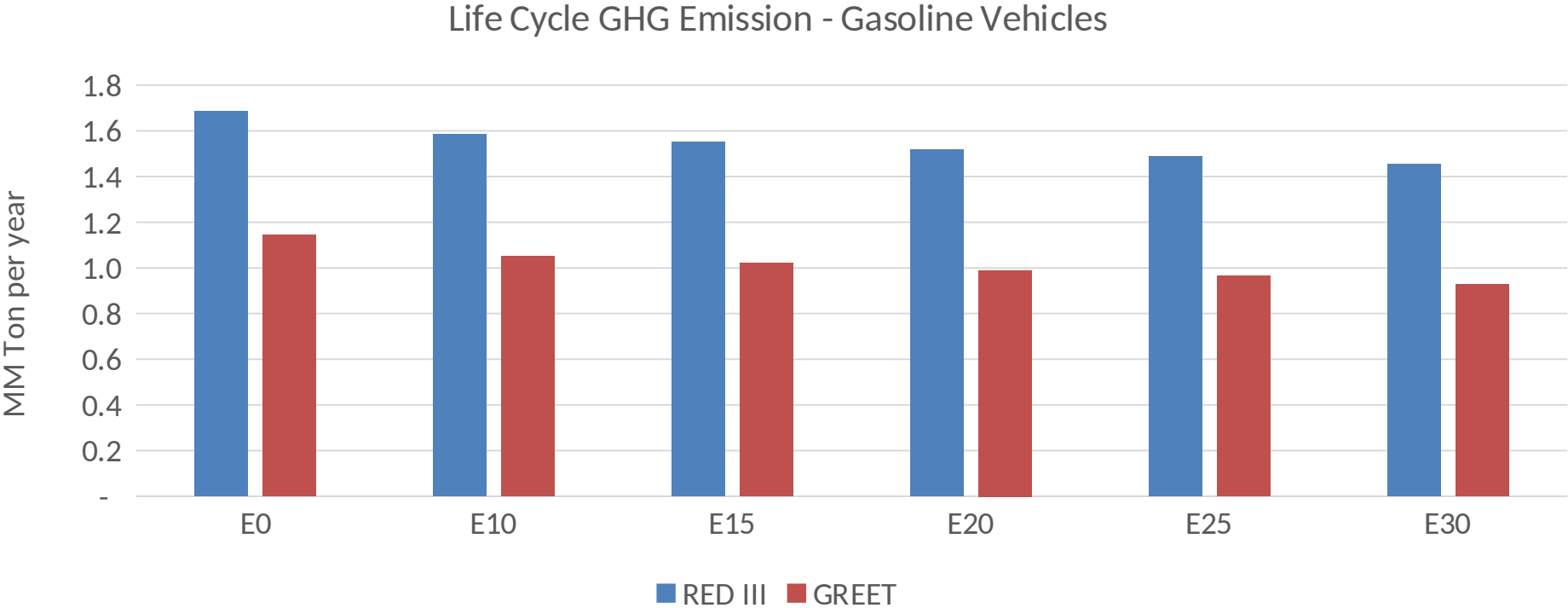
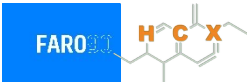
TIER III BIN 125 United States
Euro 6 Standard



Latvia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	30,410	28,711	27,012	25,745	24,527	23,608	22,478	-11%	-19%
CO2	1,653,068	1,610,561	1,570,415	1,538,841	1,523,218	1,513,827	1,485,923	-5%	-8%
VOC	2,450	2,378	2,307	2,271	2,245	2,228	2,195	-6%	-8%
NOx	2,466	2,277	2,147	2,082	1,976	1,893	1,795	-13%	-20%
SOx	19	18	16	15	14	12	11	-15%	-28%
NH3	560	555	549	552	558	562	566	-2%	0%
N2O	22	22	22	22	23	23	23	-1%	2%
CH4	542	542	542	551	562	573	583	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	30	29	29	28	28	28	28	-4%	-5%
Acetaldehyde	59	67	99	150	205	234	277	68%	249%
Formaldehyde	172	167	194	228	239	265	288	13%	39%
Benzene	47	45	43	42	42	40	39	-9%	-11%
PM2.5	97	88	79	72	64	56	47	-18%	-34%
PM10	60	55	49	44	40	35	29	-18%	-34%

Latvia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



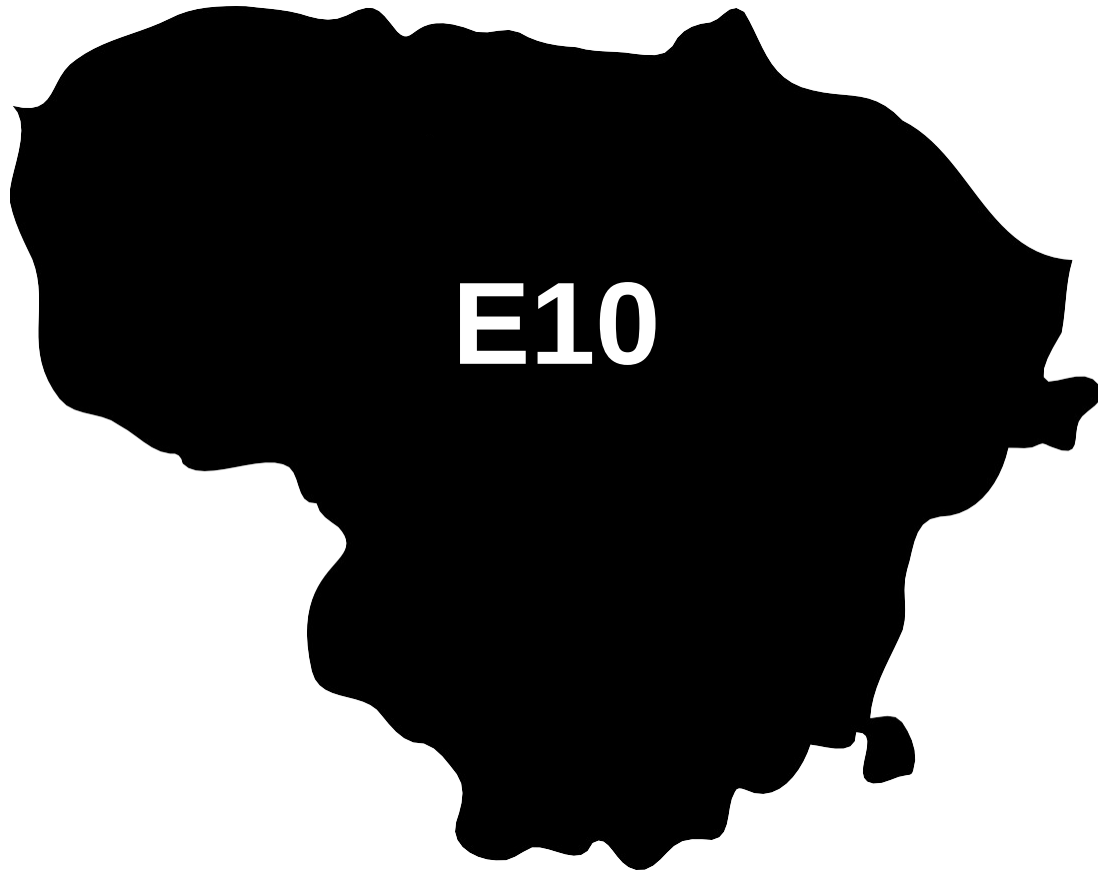
Country Emissions 2025 (MMT/año)	12.1
Target @ 2035 (MMT/year)	6.8
Delta	5.3

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.3%	11.0%	13.7%	15.9%	18.9%
RED II/III	0.0%	6.0%	8.1%	10.0%	11.6%	13.8%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	1.8%	2.4%	3.0%	3.4%	4.1%
RED II/III	0.0%	1.9%	2.6%	3.2%	3.7%	4.4%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

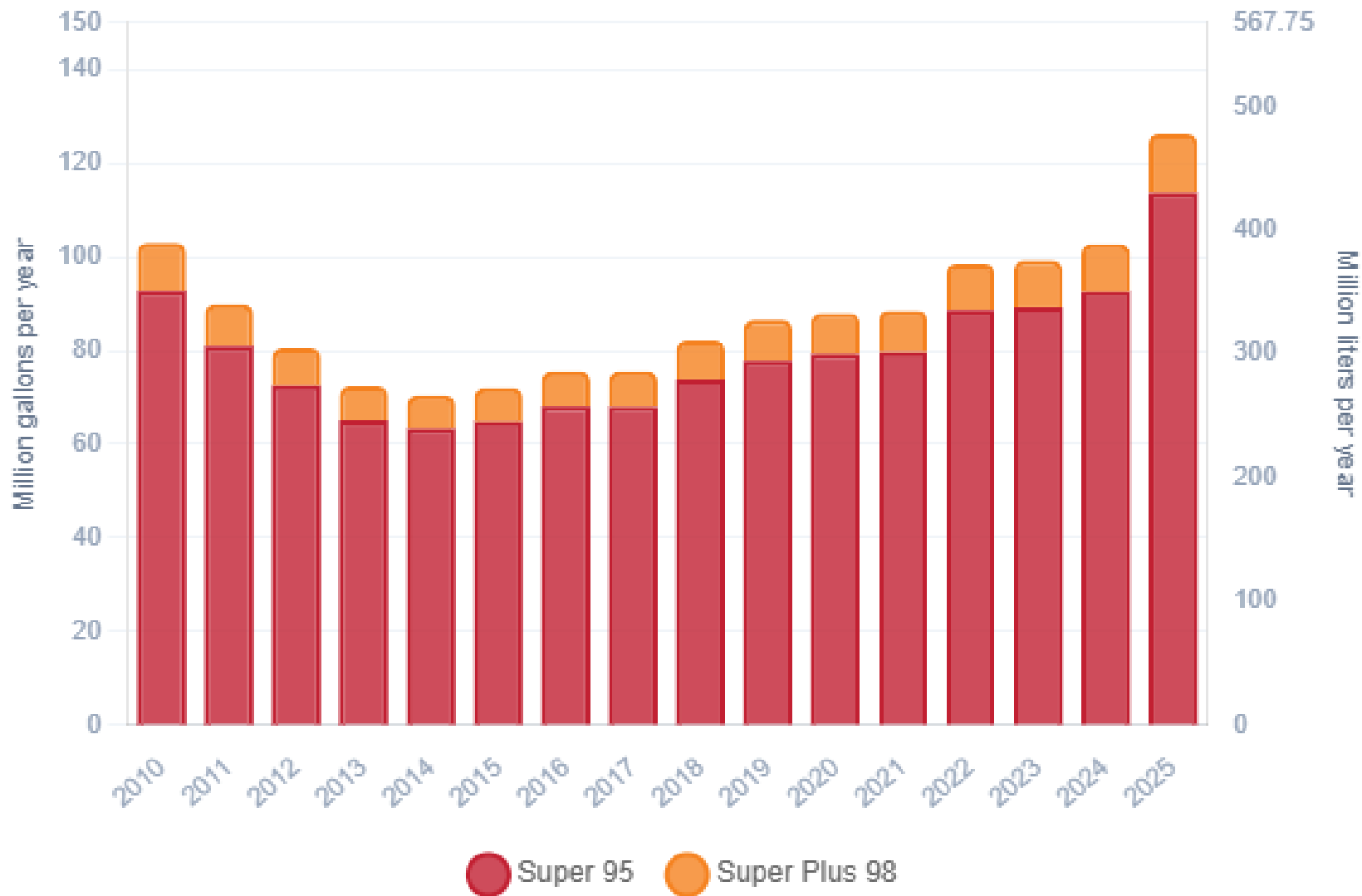
Ethanol Gasoline Blending in Lithuania



In 2025, gasoline consumption in Lithuania was 479 million liters (127 million gallons) and growth rate was 1.4%. Gasoline production and net imports were 4,056 and -3,577 million liters (1,071 and -944 million gallons) correspondingly. Lithuania complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.3 RON.

In 2025, ethanol consumption in Lithuania was 200 million liters (53 million gallons) representing 42% of gasoline demand. Ethanol production and Net Imports were 96 and 101 million liters (25 and 27 million gallons) correspondingly. Ethanol sources are Corn 15% and Cereals 85%.

Gasoline Demand in Lithuania



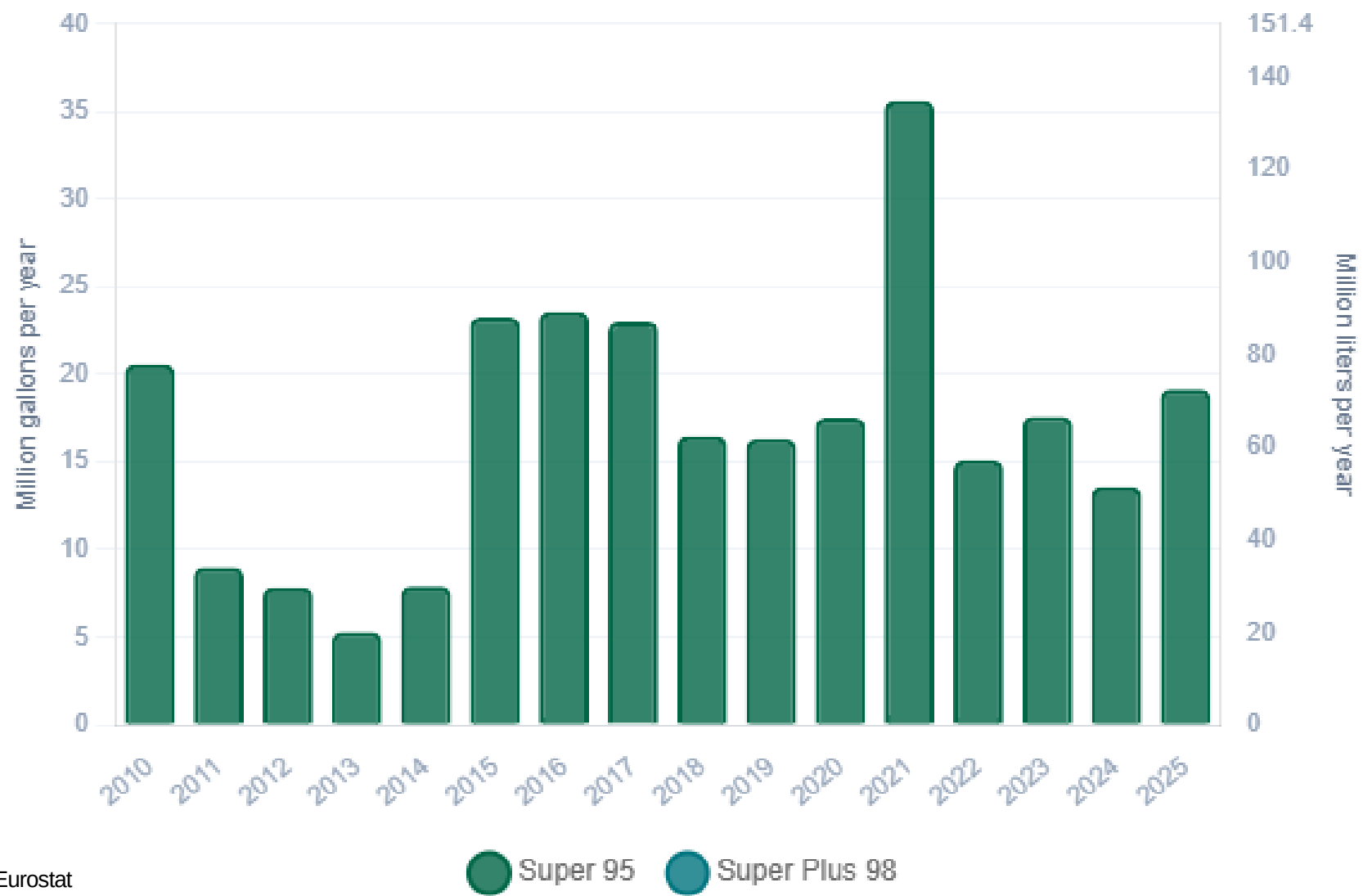
Source: Eurostat

Gasoline Production in Lithuania



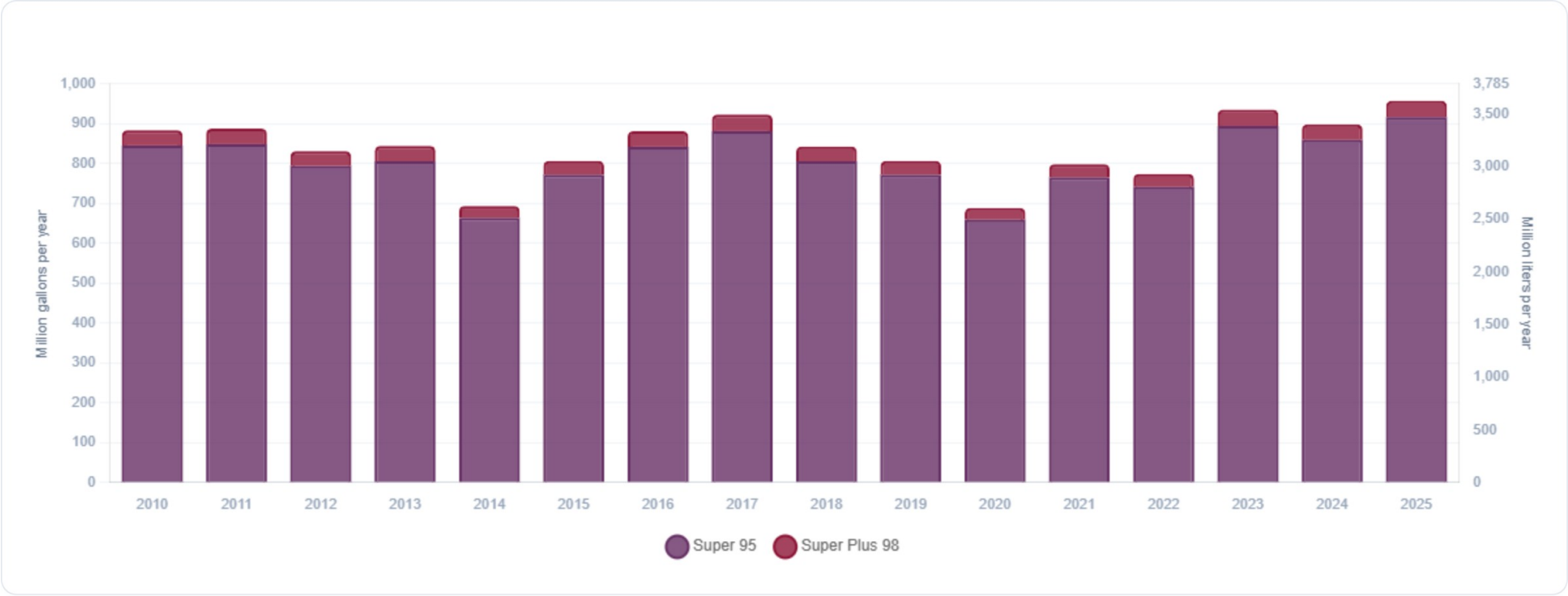
Source: Eurostat

Gasoline Imports in Lithuania



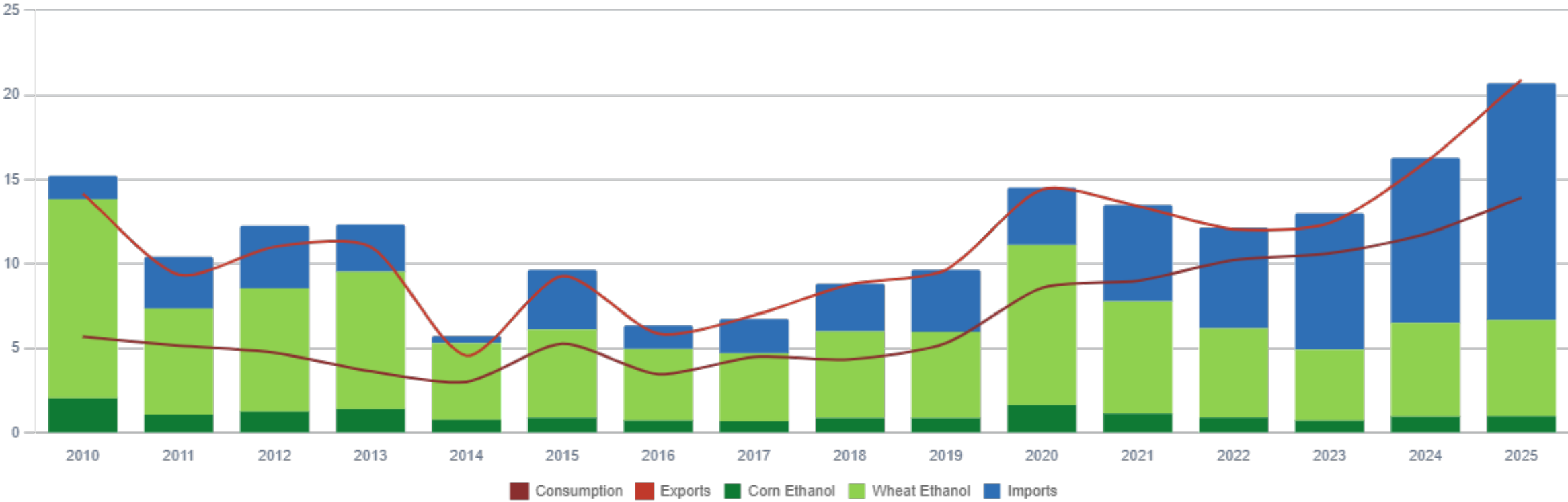
Source: Eurostat

Gasoline Exports in Lithuania



Source: Eurostat

Ethanol Balance in Lithuania



Source: Eurostat

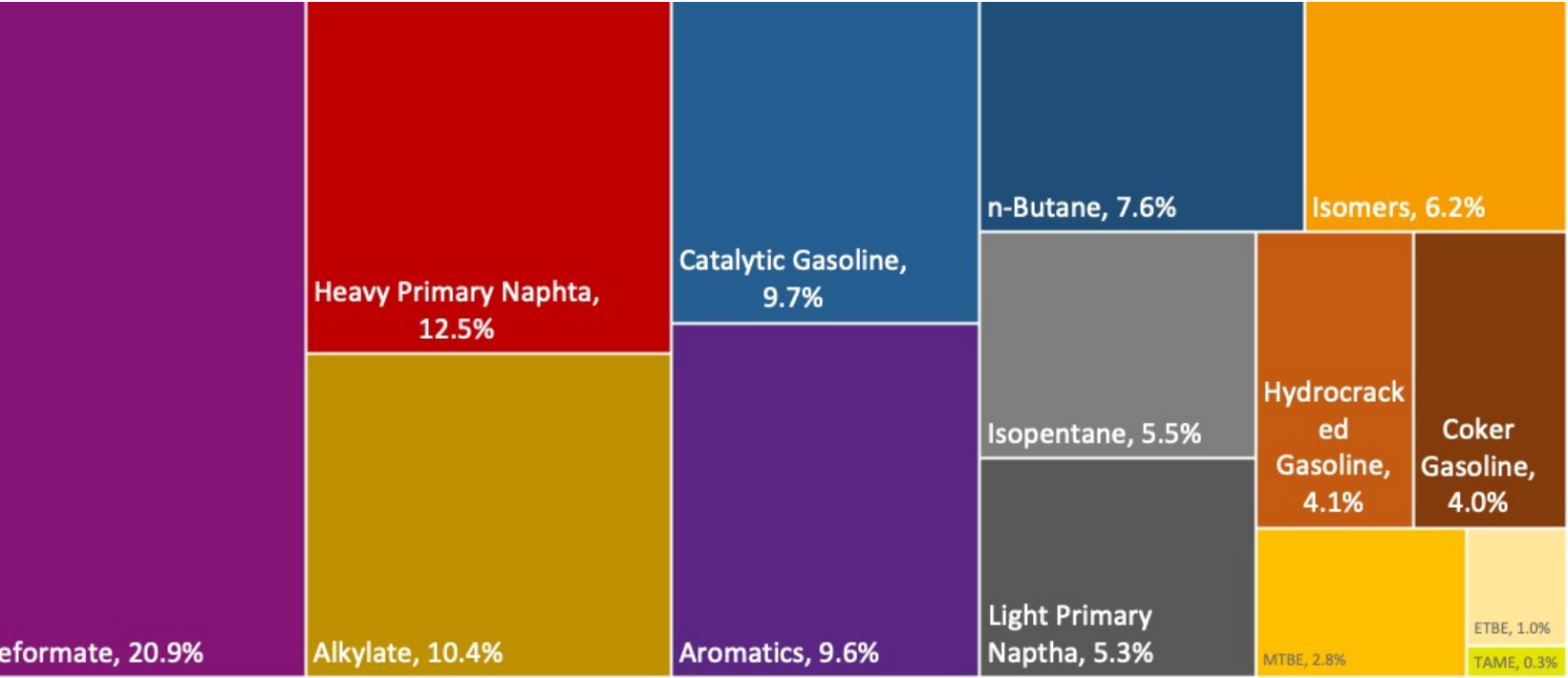
Gasoline Quality in Lithuania

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	8.6							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

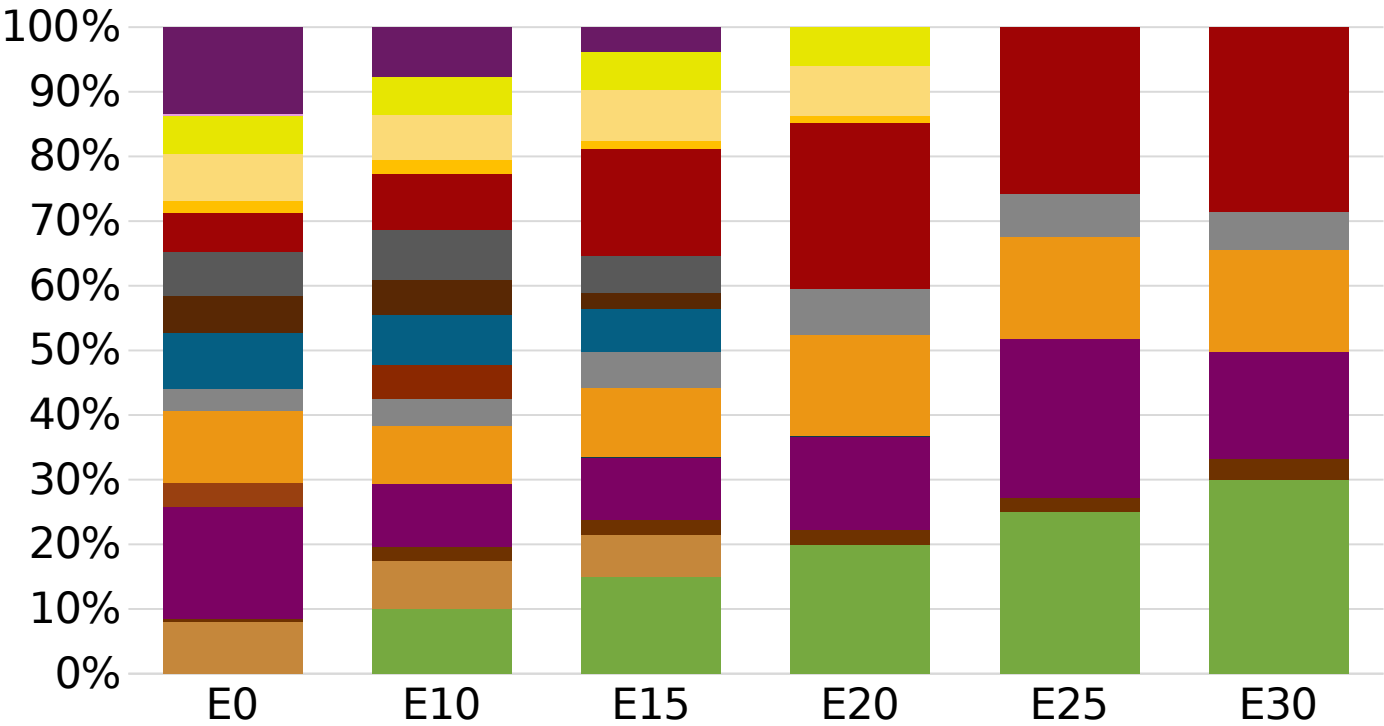
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Lithuania Available Blending Components



Lithuania – Gasoline 95 – Constant Octane

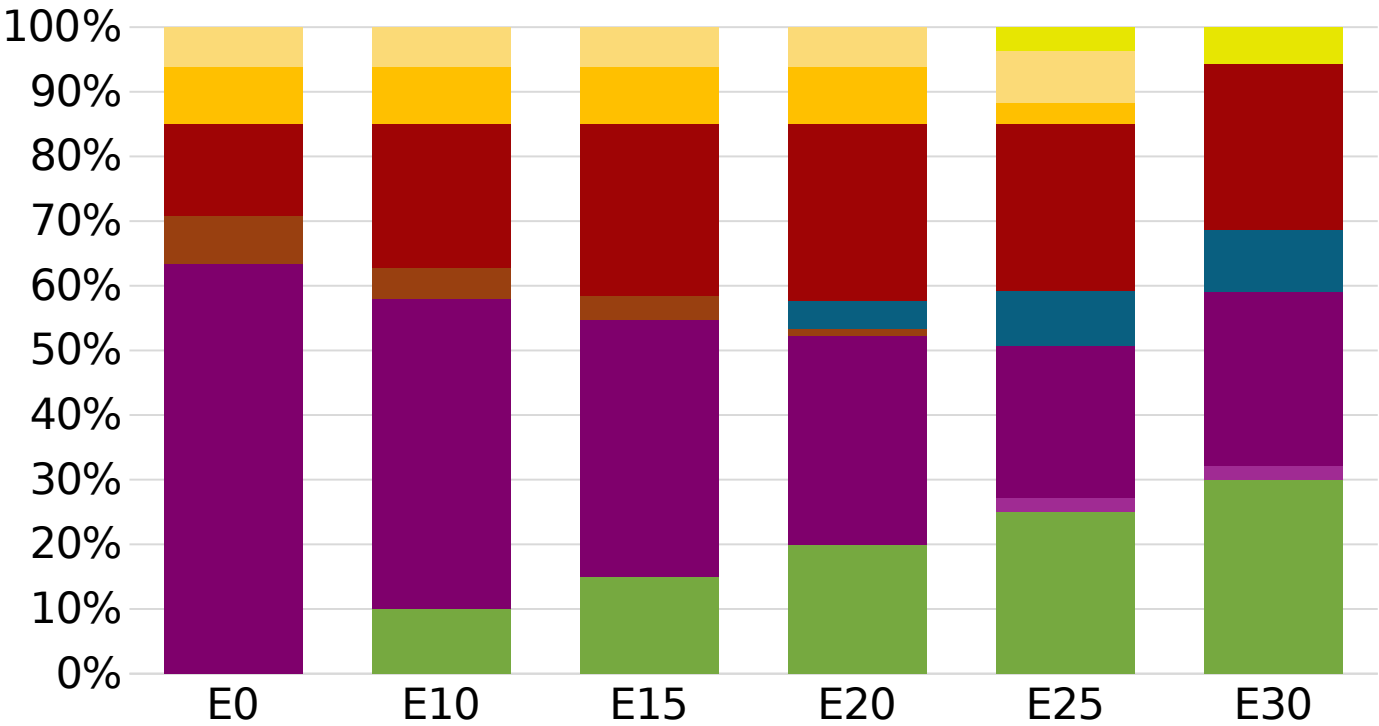
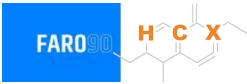


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.071	\$ 1.933	\$ 1.833	\$ 1.718	\$ 1.672	\$ 1.619

Source:
Faro90

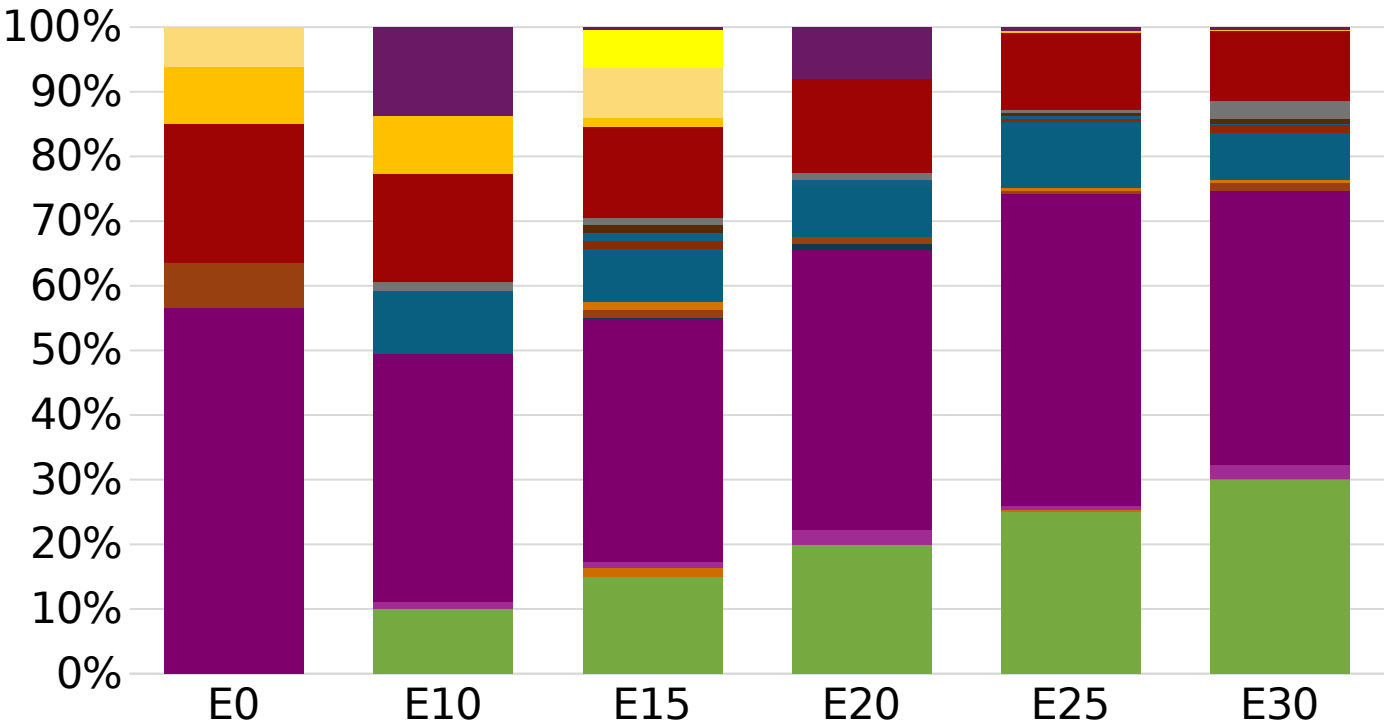
Lithuania - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.091	\$ 1.974	\$ 1.905	\$ 1.833	\$ 1.768	\$ 1.718

Average CIF 2025
Prices

Lithuania – Gasoline 95 – Increased Octane

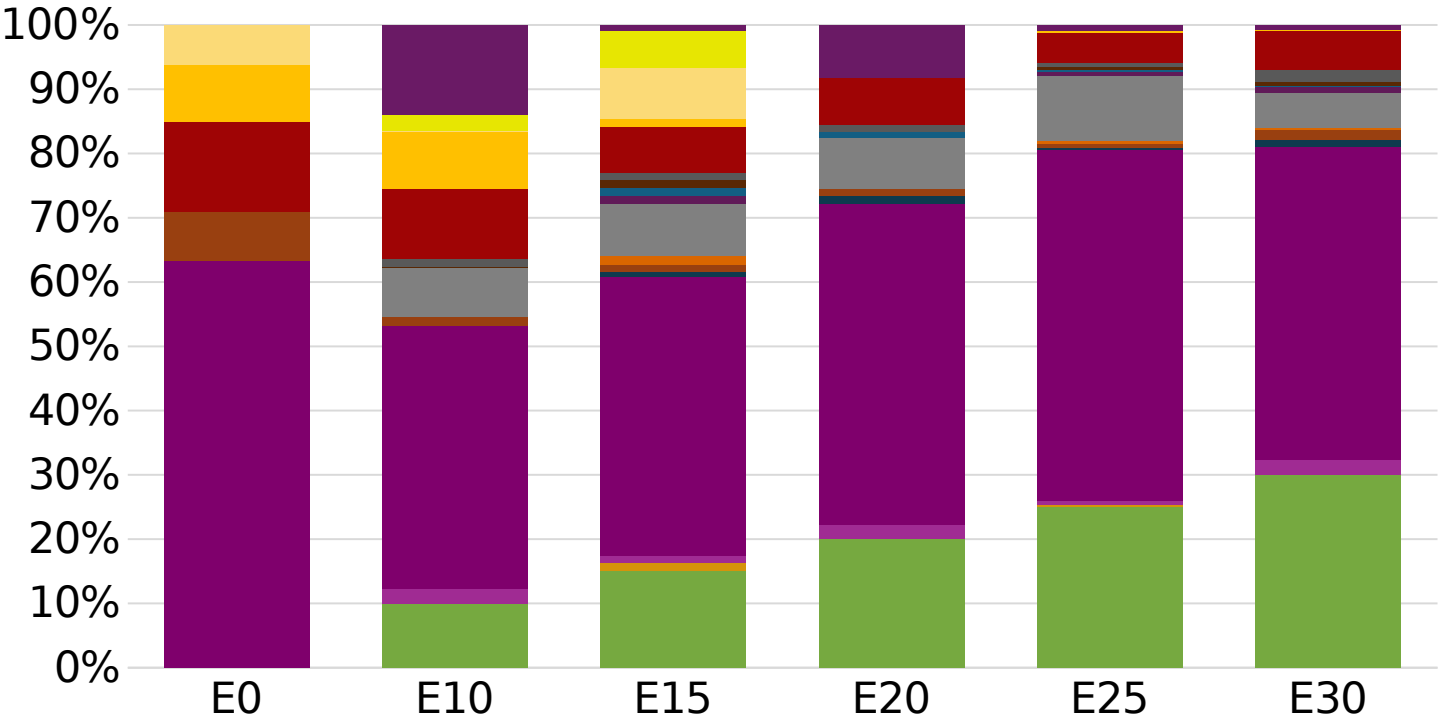
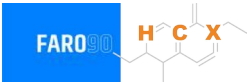


Average CIF 2025
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 1.959	\$ 1.959	\$ 1.959	\$ 1.959	\$ 1.959	\$ 1.959

Source:
Faro90

Lithuania – Gasoline 98 – Increased Octane

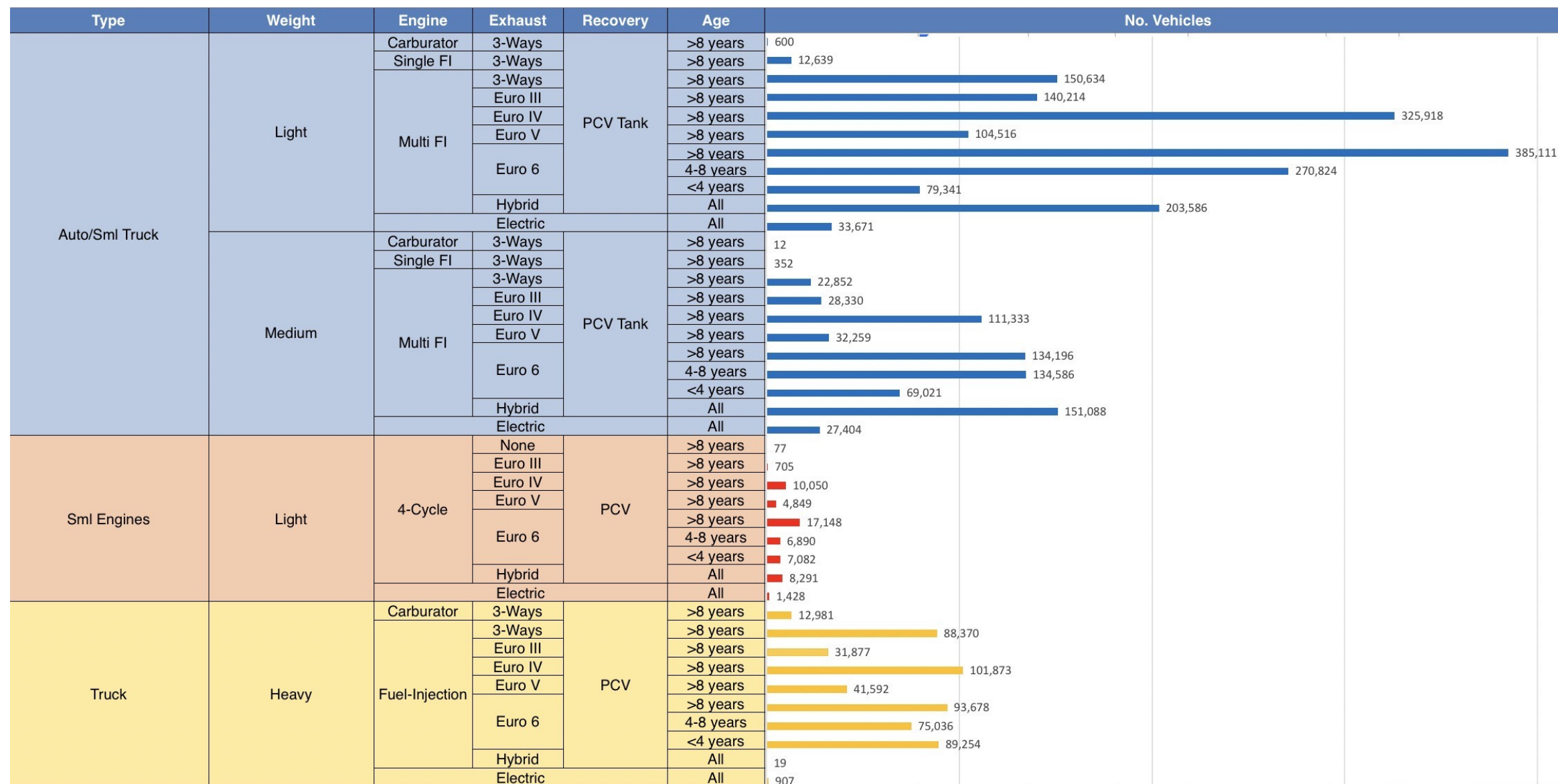


Average CIF 2025 Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.091	\$ 2.091	\$ 2.091	\$ 2.091	\$ 2.091	\$ 2.091

Source:
Faro90

Gasoline Vehicle Fleet - Lithuania

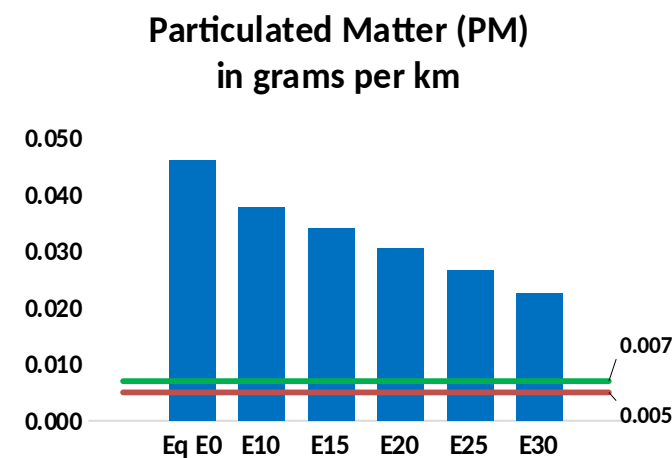
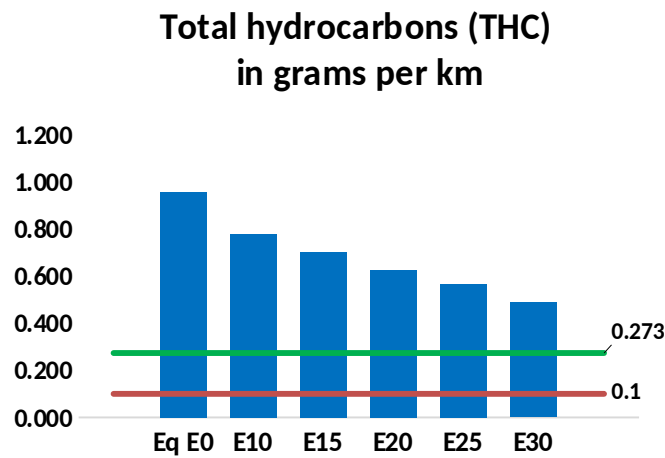
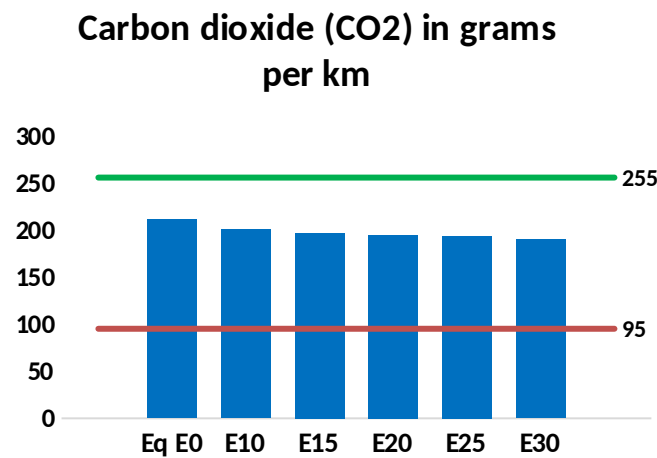
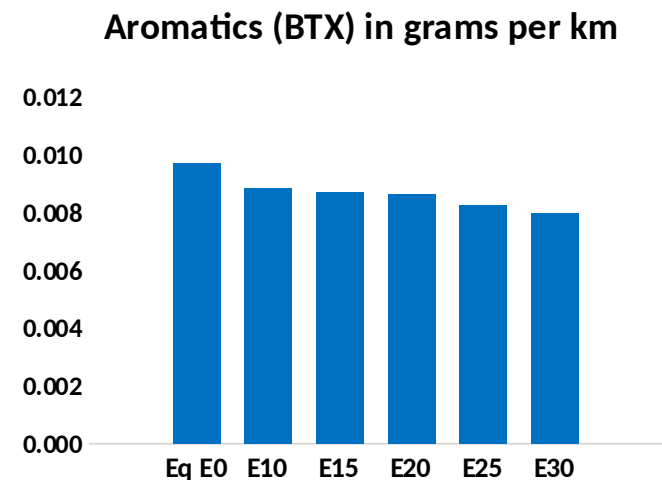
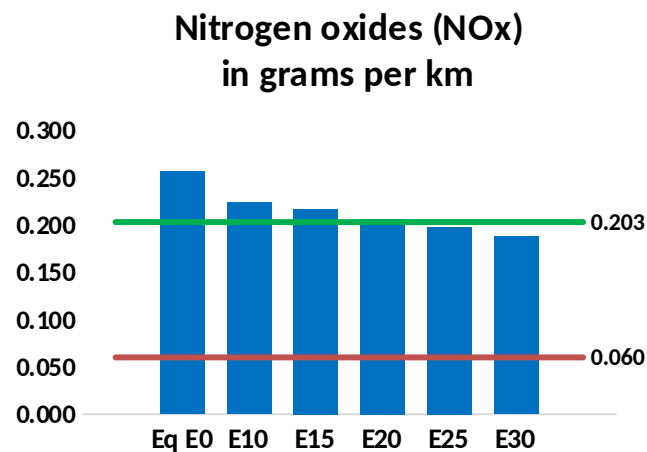
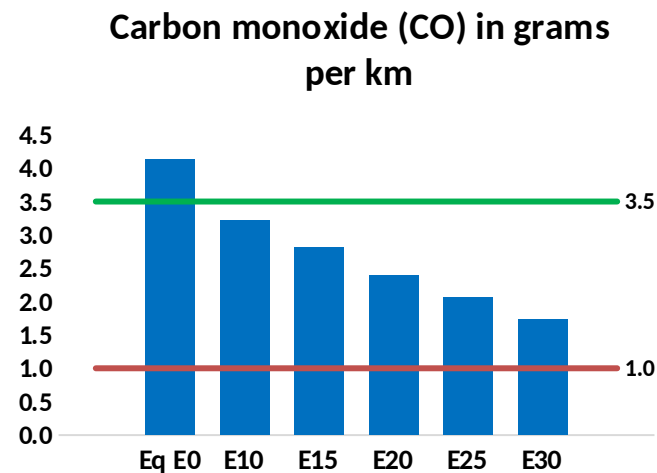
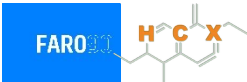


Vehicle Fleet: 3,010,595
Gasoline Vehicle Fleet 609,635

Source: Eurostat, ACEA

Lithuania – Vehicular Emissions

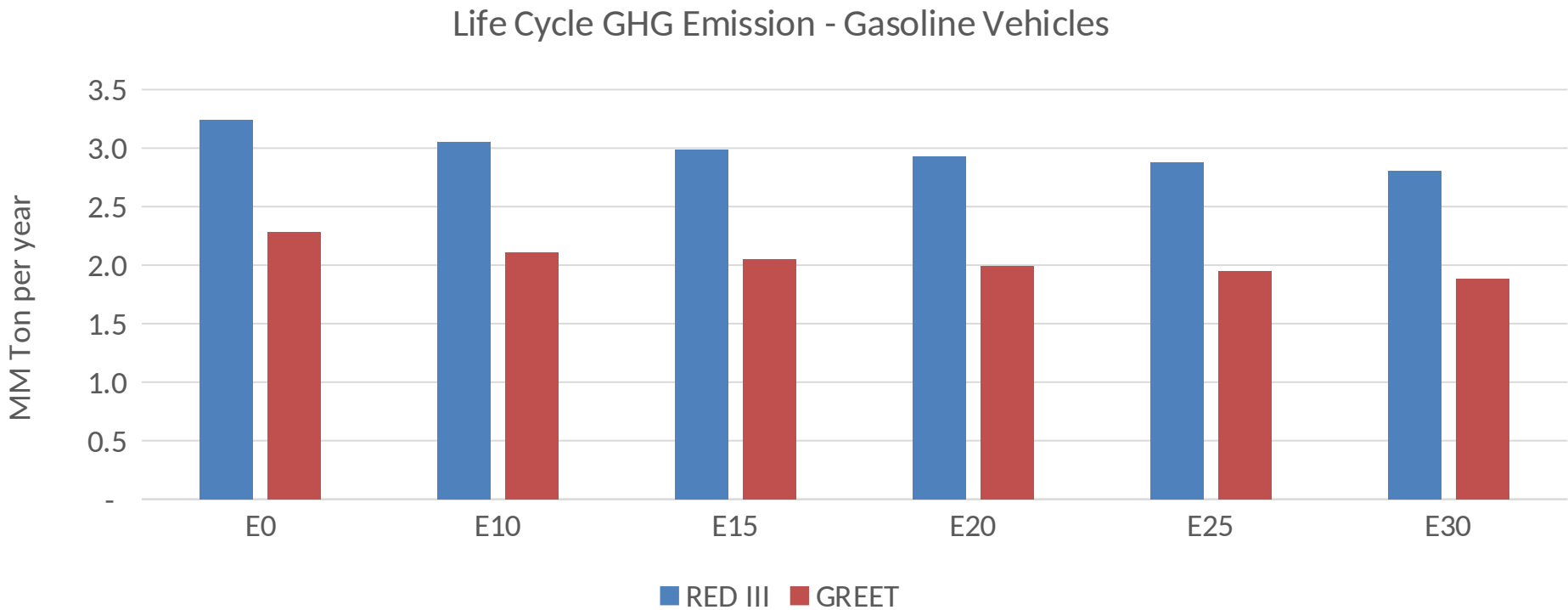
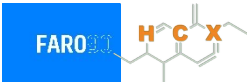
TIER III BIN 125 United States
Euro 6 Standard



Lithuania – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	63,460	56,418	49,376	43,184	36,758	31,758	26,685	-22%	-42%
CO2	3,242,116	3,158,748	3,080,010	3,018,086	2,987,443	2,969,963	2,915,218	-5%	-8%
VOC	14,706	13,343	11,981	10,819	9,629	8,707	7,500	-19%	-35%
NOx	3,944	3,642	3,438	3,335	3,168	3,038	2,884	-13%	-20%
SOx	37	34	32	29	27	24	22	-15%	-28%
NH3	1,265	1,252	1,240	1,246	1,260	1,270	1,278	-2%	0%
N2O	54	54	54	54	55	55	56	-1%	2%
CH4	2,920	2,920	2,920	2,964	3,028	3,084	3,136	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	60	56	53	51	48	46	44	-11%	-20%
Acetaldehyde	222	255	374	570	776	887	1,048	68%	249%
Formaldehyde	862	838	977	1,147	1,202	1,330	1,447	13%	39%
Benzene	150	144	136	134	133	127	123	-9%	-11%
PM2.5	191	173	156	141	126	110	93	-18%	-34%
PM10	519	471	424	383	342	299	253	-18%	-34%

Lithuania – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	25.4
Target @ 2035 (MMT/year)	14.3
Delta	11.1

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	7.8%	10.4%	12.8%	14.8%	17.6%
RED II/III	0.0%	5.9%	7.9%	9.7%	11.2%	13.3%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	1.6%	2.1%	2.6%	3.0%	3.6%
RED II/III	0.0%	1.7%	2.3%	2.8%	3.3%	3.9%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

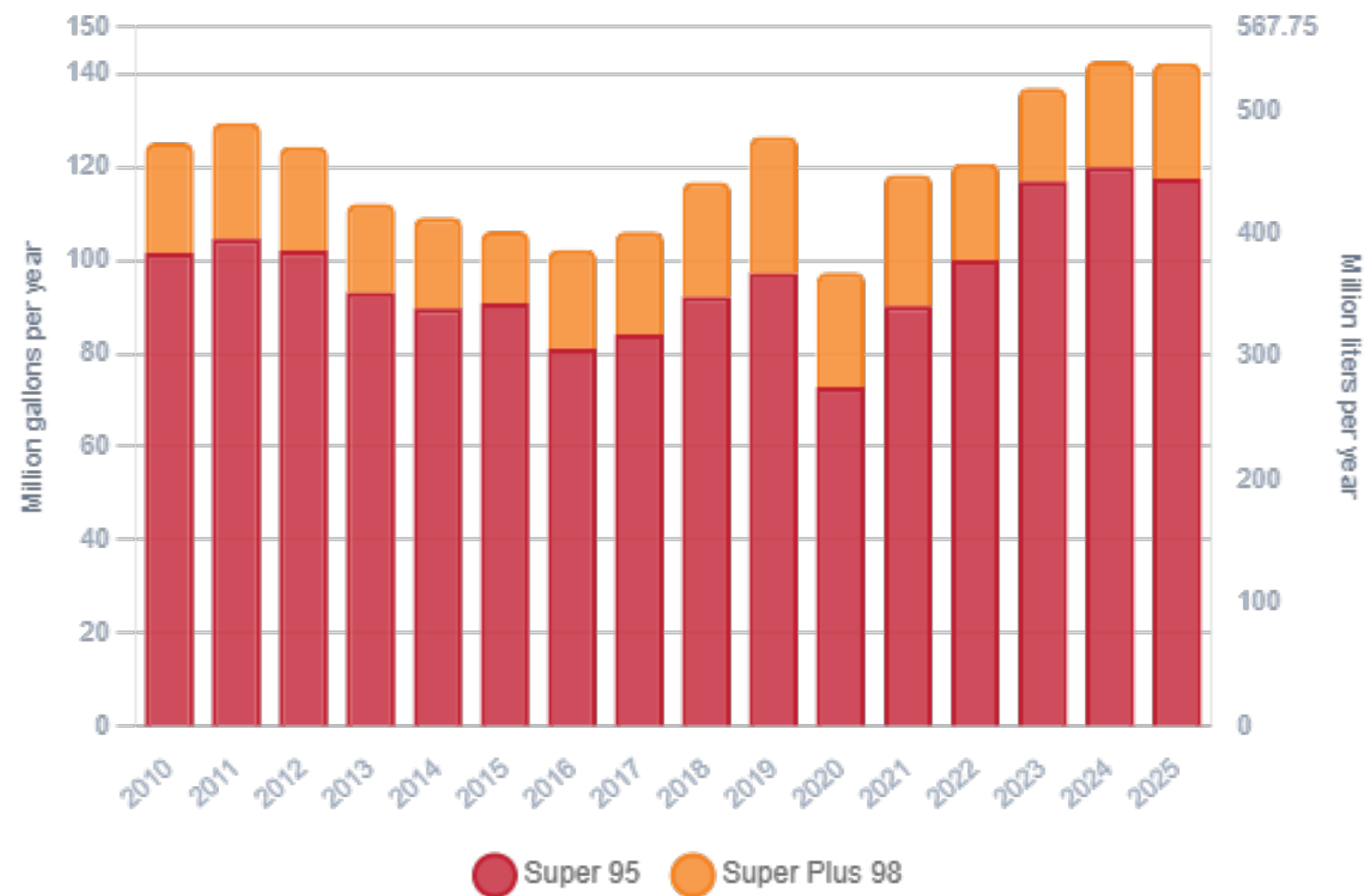
Ethanol Gasoline Blending in Luxembourg



In 2025, gasoline consumption in Luxembourg was 539 million liters (142 million gallons) and growth rate was 0.86%. There is no gasoline production. Gasoline net imports were 485 million liters (128 million gallons) correspondingly. Luxembourg complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.6 RON.

In 2025, ethanol consumption in Luxembourg was 24 million liters (6 million gallons) representing 4% of gasoline demand. Net Imports were 27 million liters (7 million gallons). Ethanol sources are Sugar Cane 80% and Corn 20%.

Gasoline Demand in Luxembourg



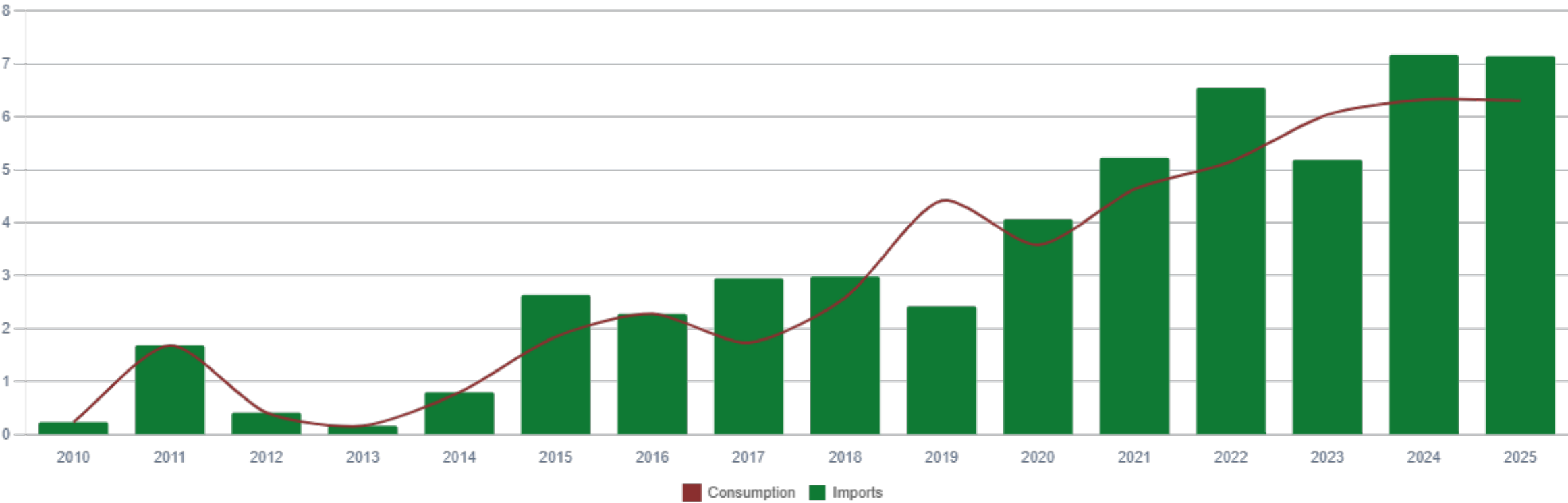
Source: Eurostat

Gasoline Imports in Luxembourg



Source: Eurostat

Ethanol Balance in Luxembourg



Ethanol is supplied only by imports

Source: Eurostat

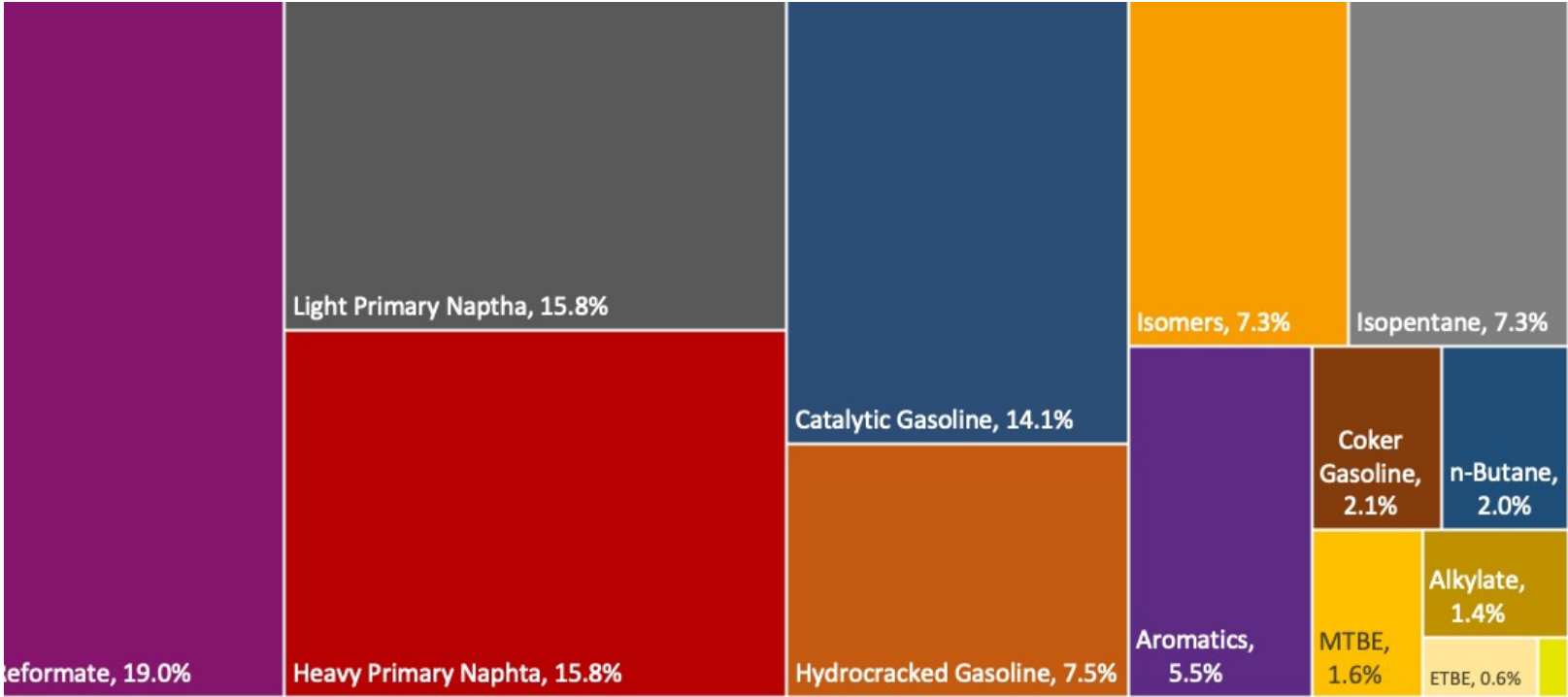
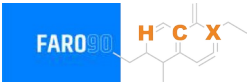
Gasoline Quality in Luxembourg

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	7.7							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	6.0							

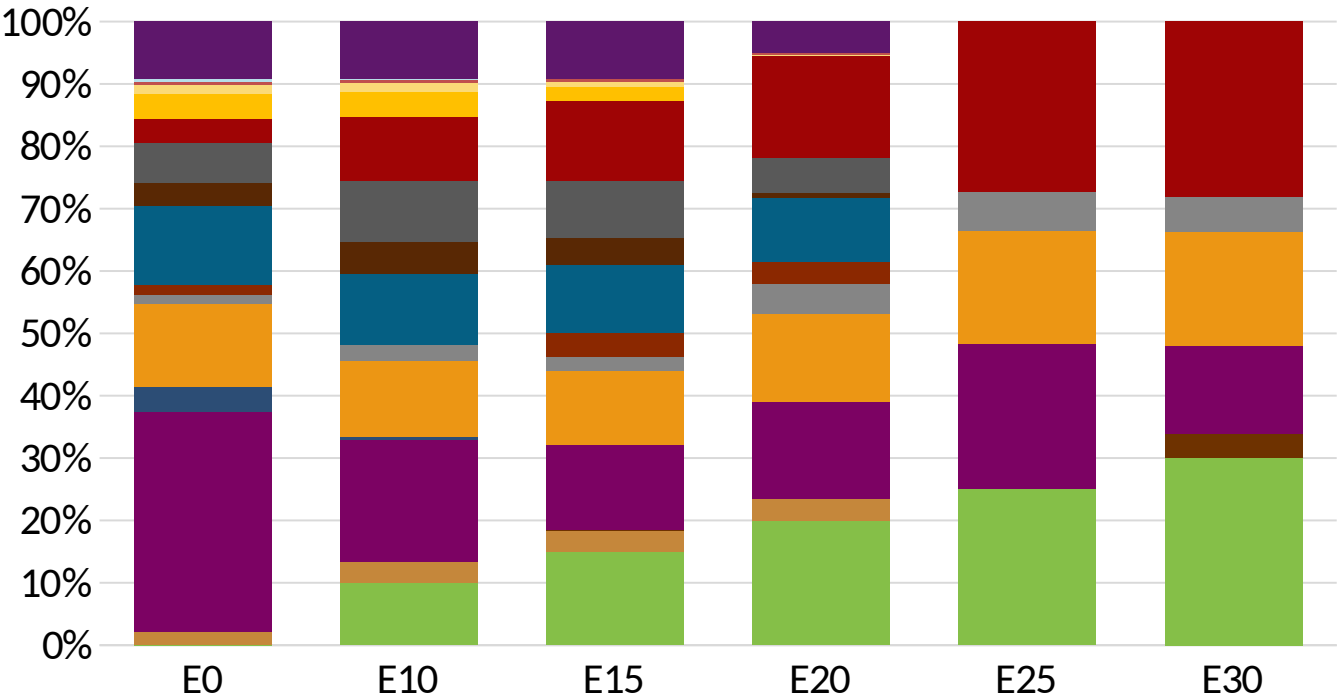
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Luxembourg Available Blending Components



Luxembourg – Gasoline 95 – Constant Octane

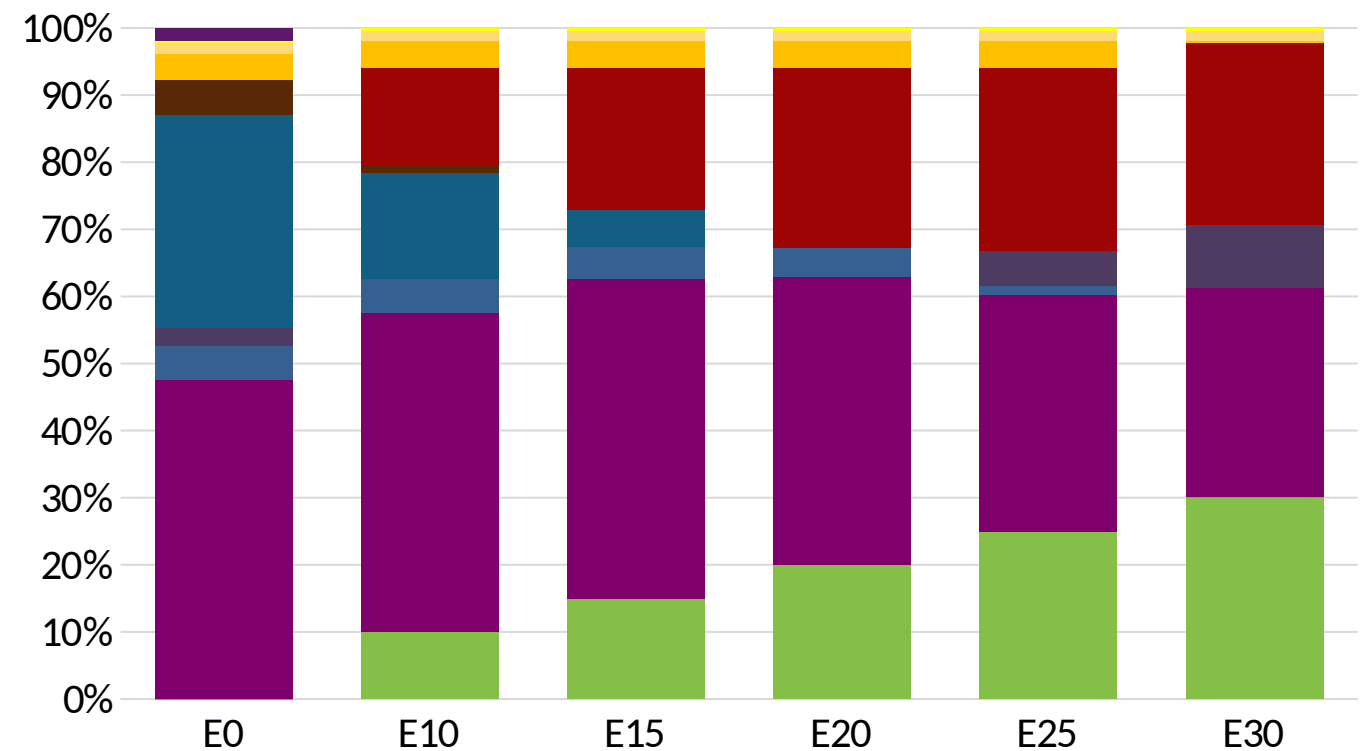


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.069	\$ 1.941	\$ 1.880	\$ 1.799	\$ 1.677	\$ 1.612

Source:
Faro90

Luxembourg - Gasoline 98 - Constant Octane

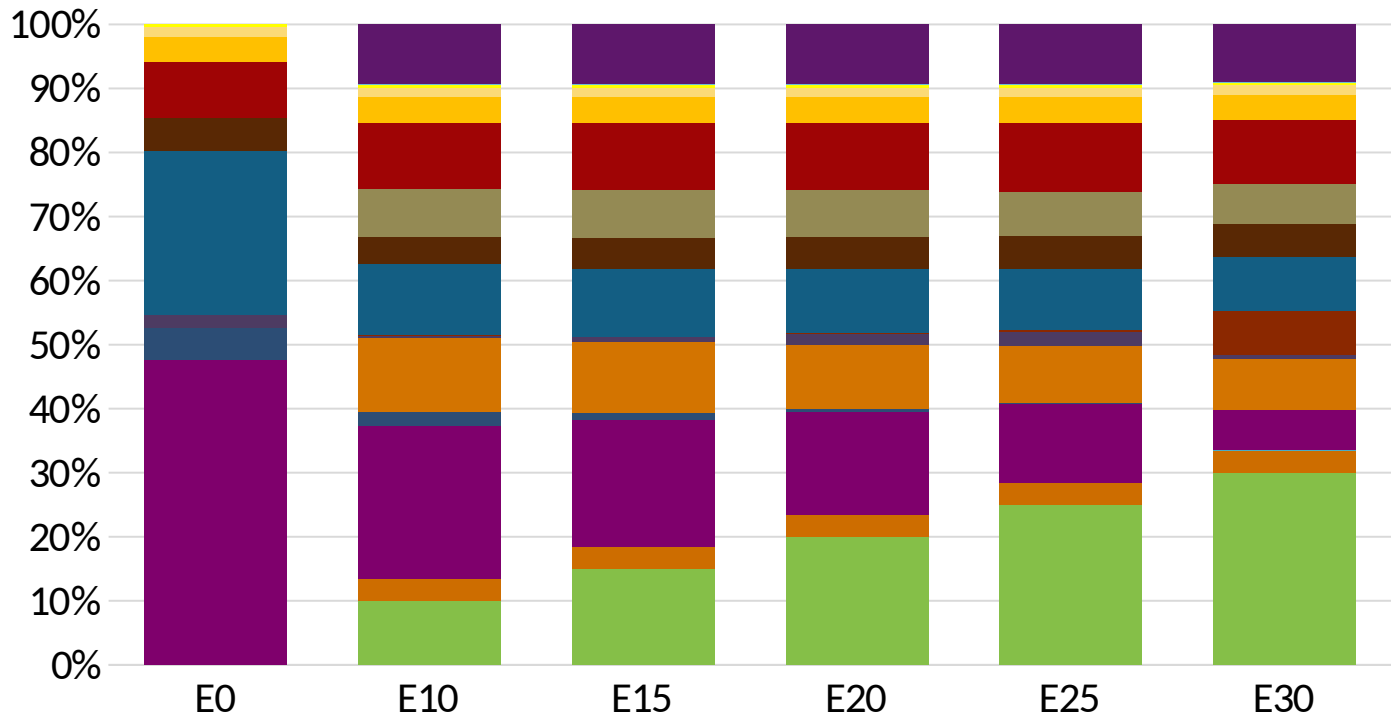


Average CIF 2025
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.144	\$ 2.015	\$ 1.943	\$ 1.866	\$ 1.789	\$ 1.722

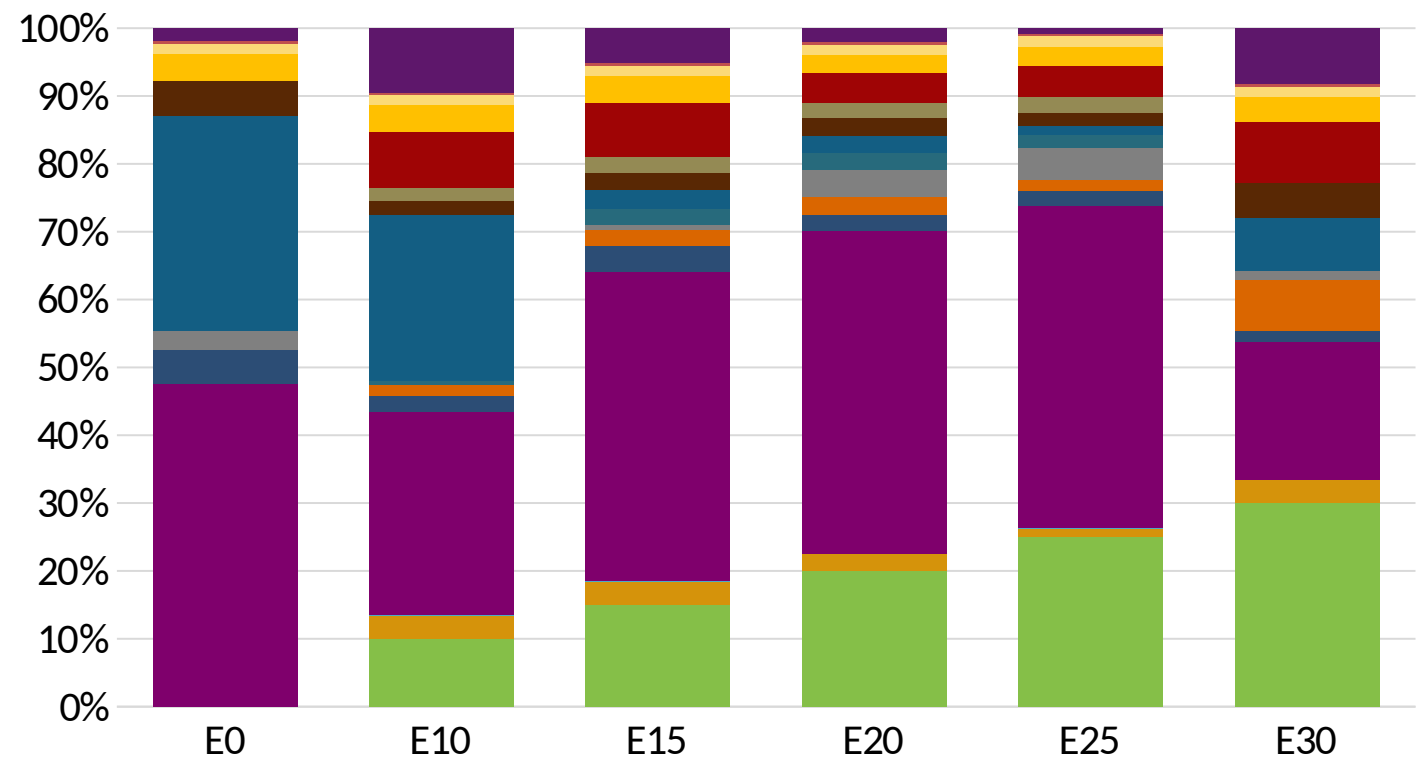
Source:
Faro90

Luxembourg – Gasoline 95 – Increased Octane



Average CIF 2025 Prices

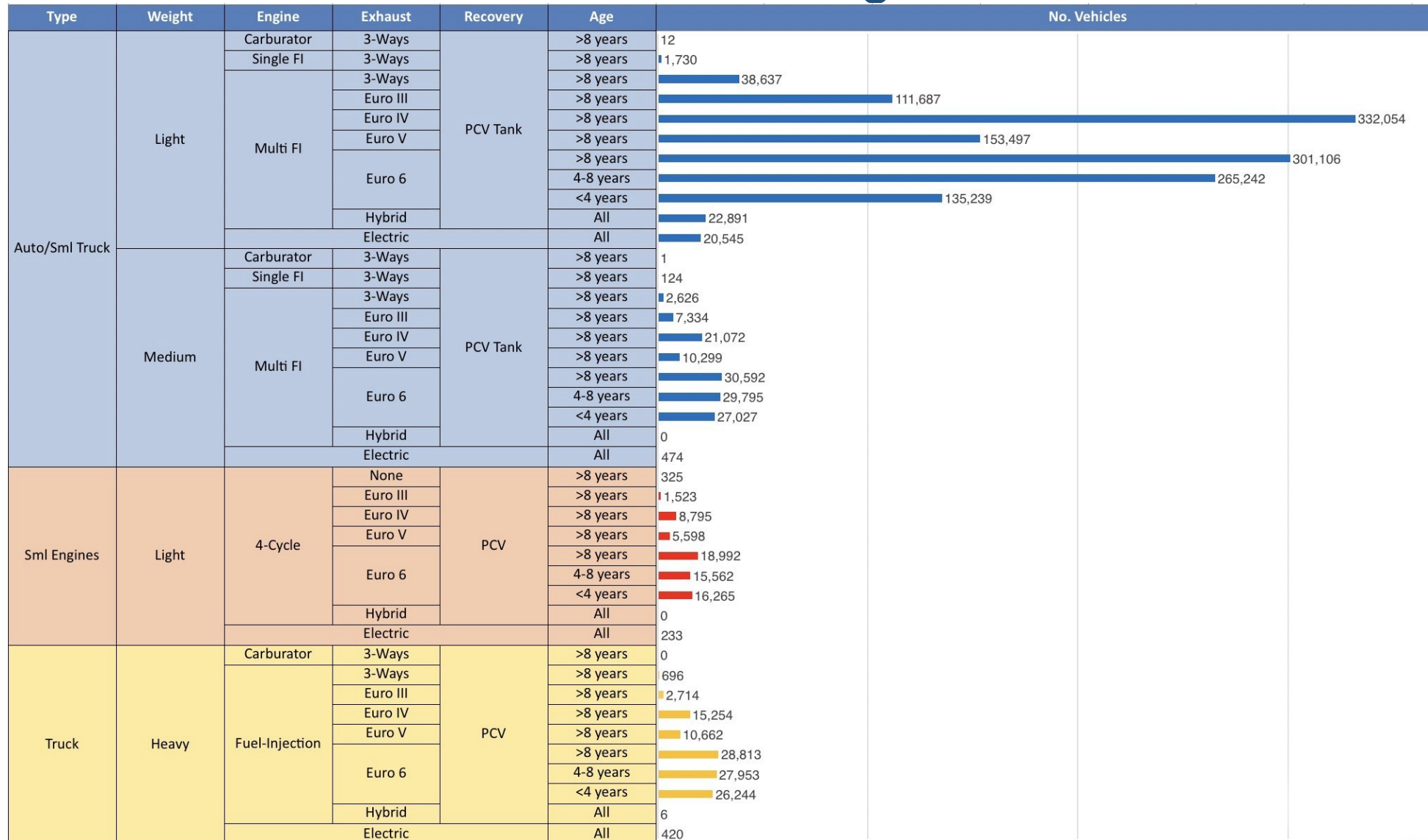
Luxembourg – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Source:
Faro90

Gasoline Vehicle Fleet - Luxembourg

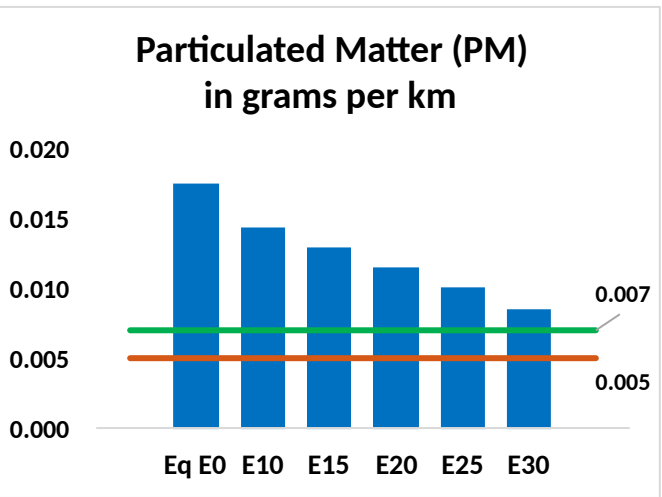
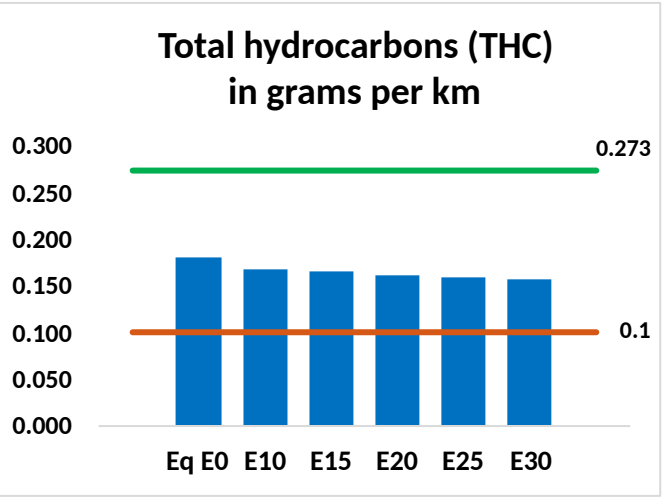
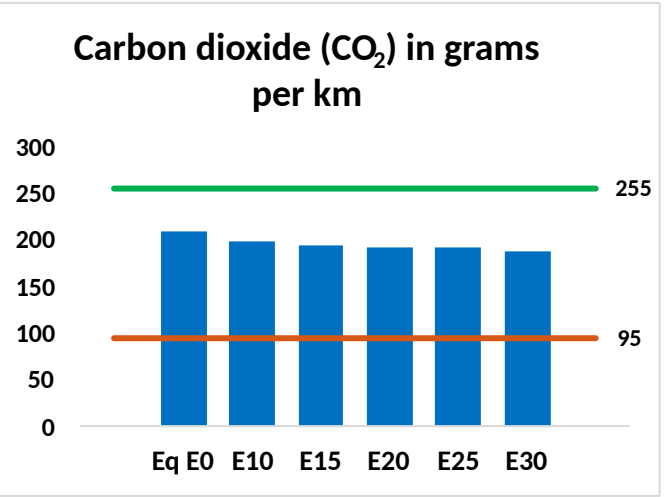
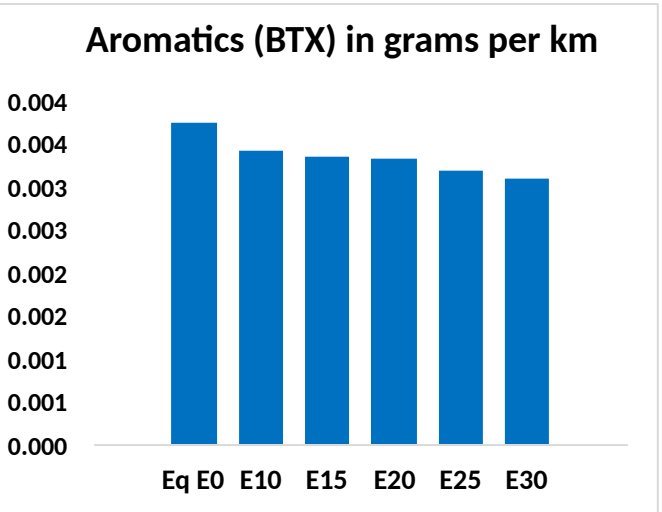
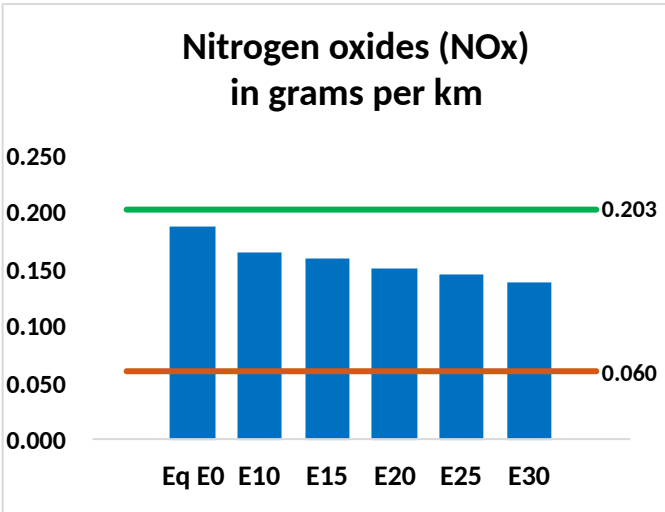
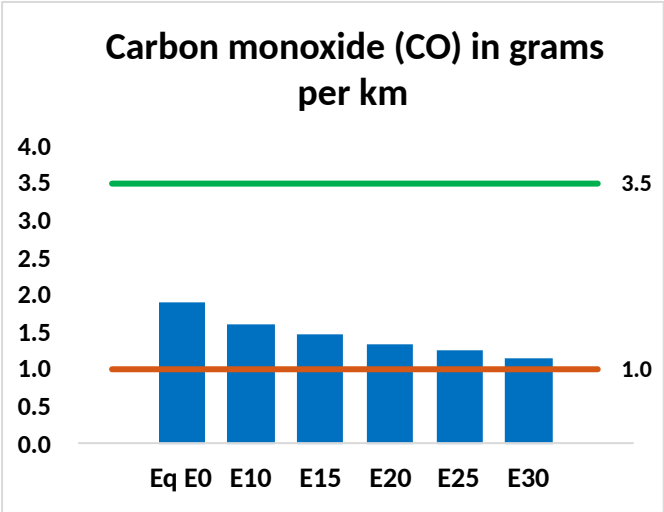
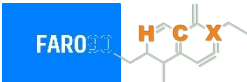


Vehicle Fleet: 1,692,040
Gasoline Vehicle Fleet: 1,150,362

Source: Eurostat, ACEA

Luxembourg – Vehicular Emissions

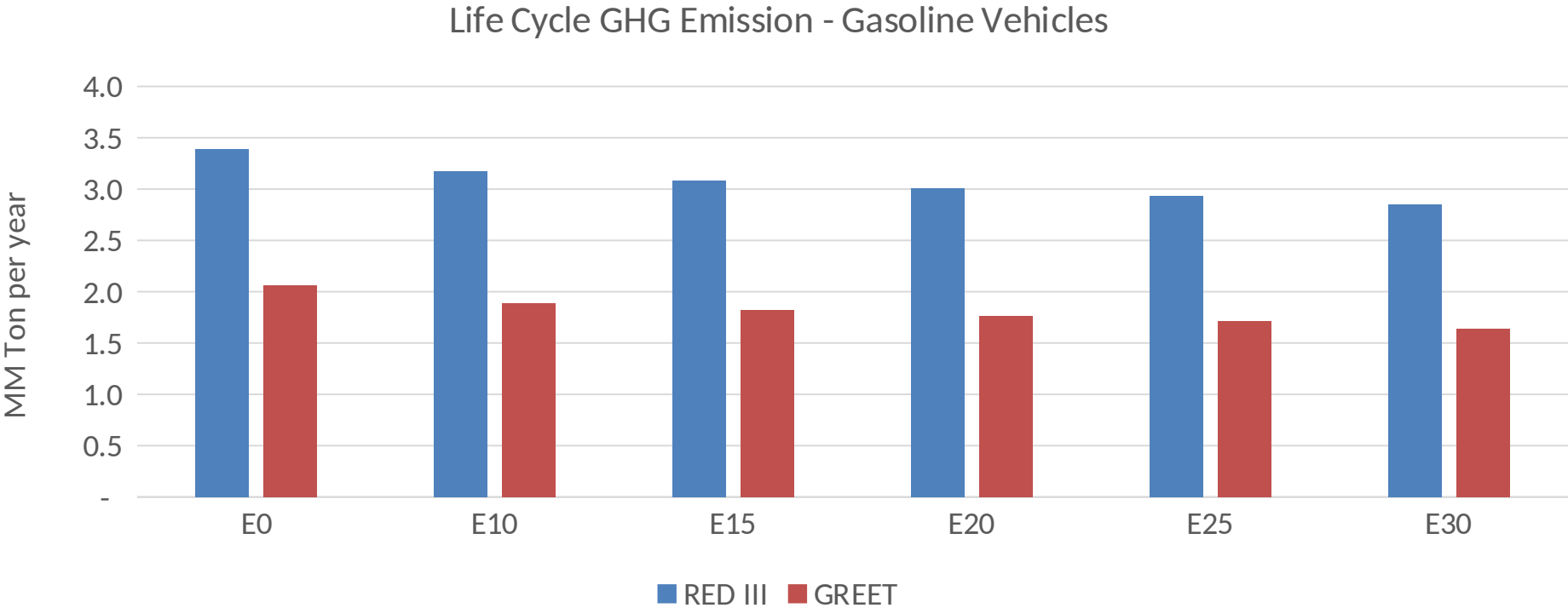
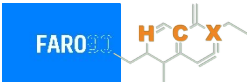
TIER III BIN 125 United States
Euro 6 Standard



Luxembourg – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	58,155	53,465	48,776	44,890	40,960	37,933	34,697	-16%	-30%
CO2	6,391,079	6,226,737	6,071,525	5,949,455	5,889,050	5,852,719	5,744,836	-5%	-8%
VOC	5,518	5,330	5,142	5,033	4,946	4,885	4,781	-7%	-10%
NOx	5,779	5,336	5,037	4,887	4,642	4,452	4,227	-13%	-20%
SOx	73	67	62	57	53	48	43	-15%	-28%
NH3	2,172	2,150	2,129	2,139	2,163	2,180	2,194	-2%	0%
N2O	73	72	72	72	74	75	76	-1%	2%
CH4	1,324	1,324	1,324	1,344	1,373	1,398	1,422	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	63	61	60	59	58	58	57	-6%	-8%
Acetaldehyde	120	137	201	307	418	477	564	68%	249%
Formaldehyde	349	339	395	464	486	538	585	13%	39%
Benzene	115	110	105	103	102	98	95	-9%	-11%
PM2.5	382	348	313	282	252	220	187	-18%	-34%
PM10	153	139	125	113	101	88	75	-18%	-34%

Luxembourg – Gasoline Vehicle Fleet Life Cycle GHG Emissions



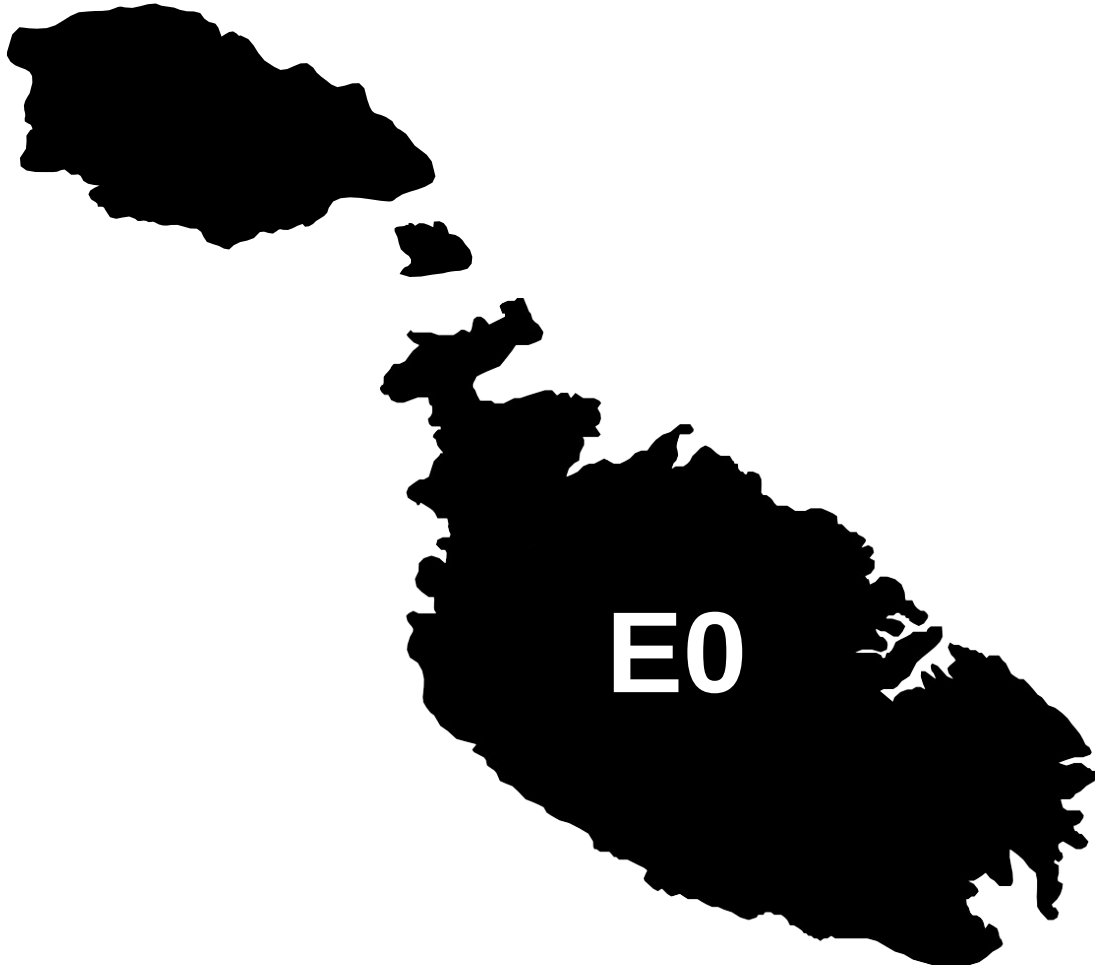
Country Emissions 2025 (MMT/año)	9.0
Target @ 2035 (MMT/year)	5.1
Delta	3.9

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.4%	11.6%	14.5%	17.1%	20.5%
RED II/III	0.0%	6.4%	8.9%	11.2%	13.3%	15.8%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.4%	6.1%	7.6%	8.9%	10.8%
RED II/III	0.0%	5.5%	7.7%	9.7%	11.4%	13.7%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

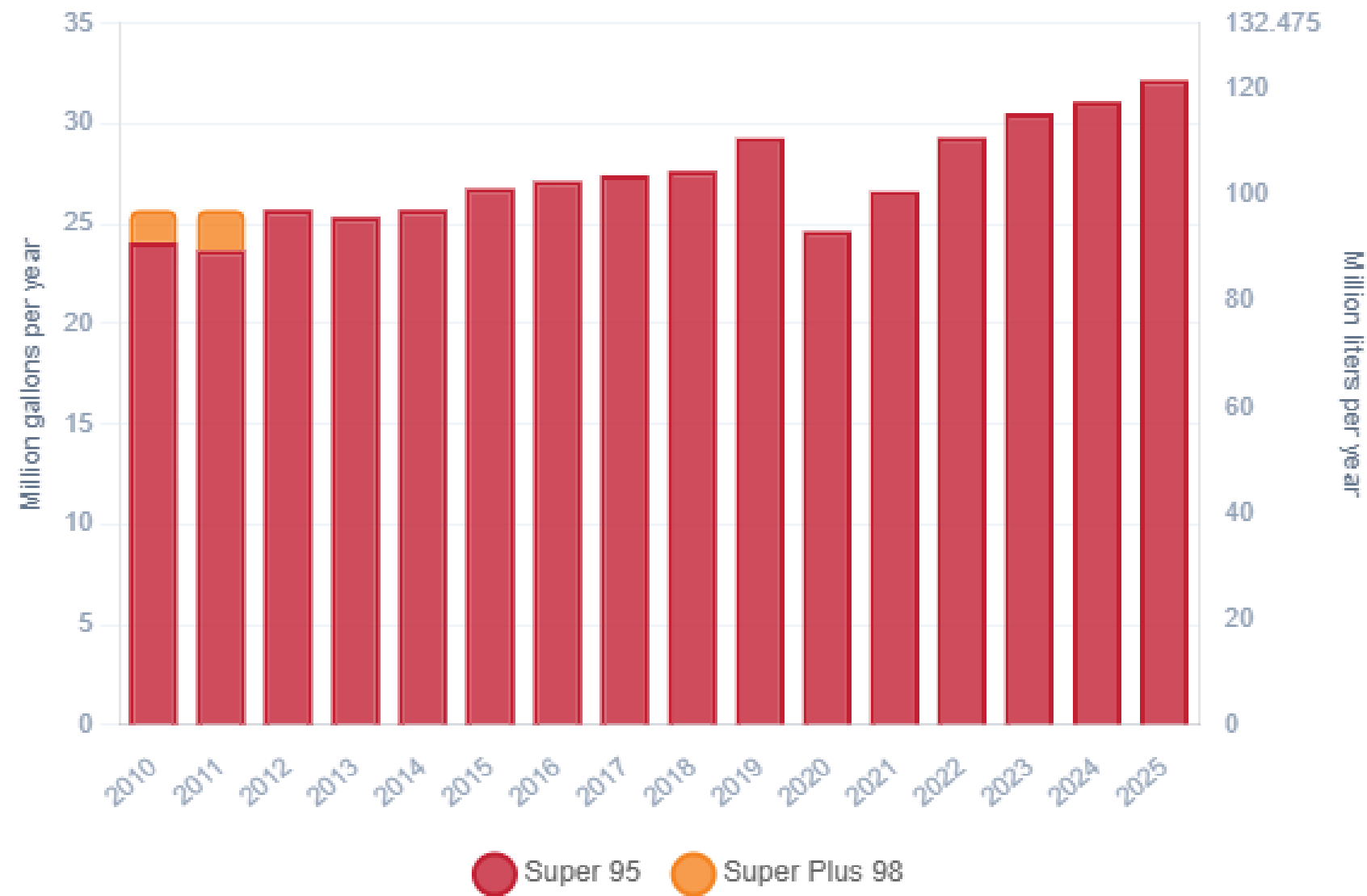
Ethanol Gasoline Blending in Malta



In 2025, gasoline consumption in Malta was 122 million liters (32 million gallons) and growth rate was 1.5%. There is no gasoline production. Gasoline imports were and 74 million liters (20 million gallons) . Malta complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.0 RON.

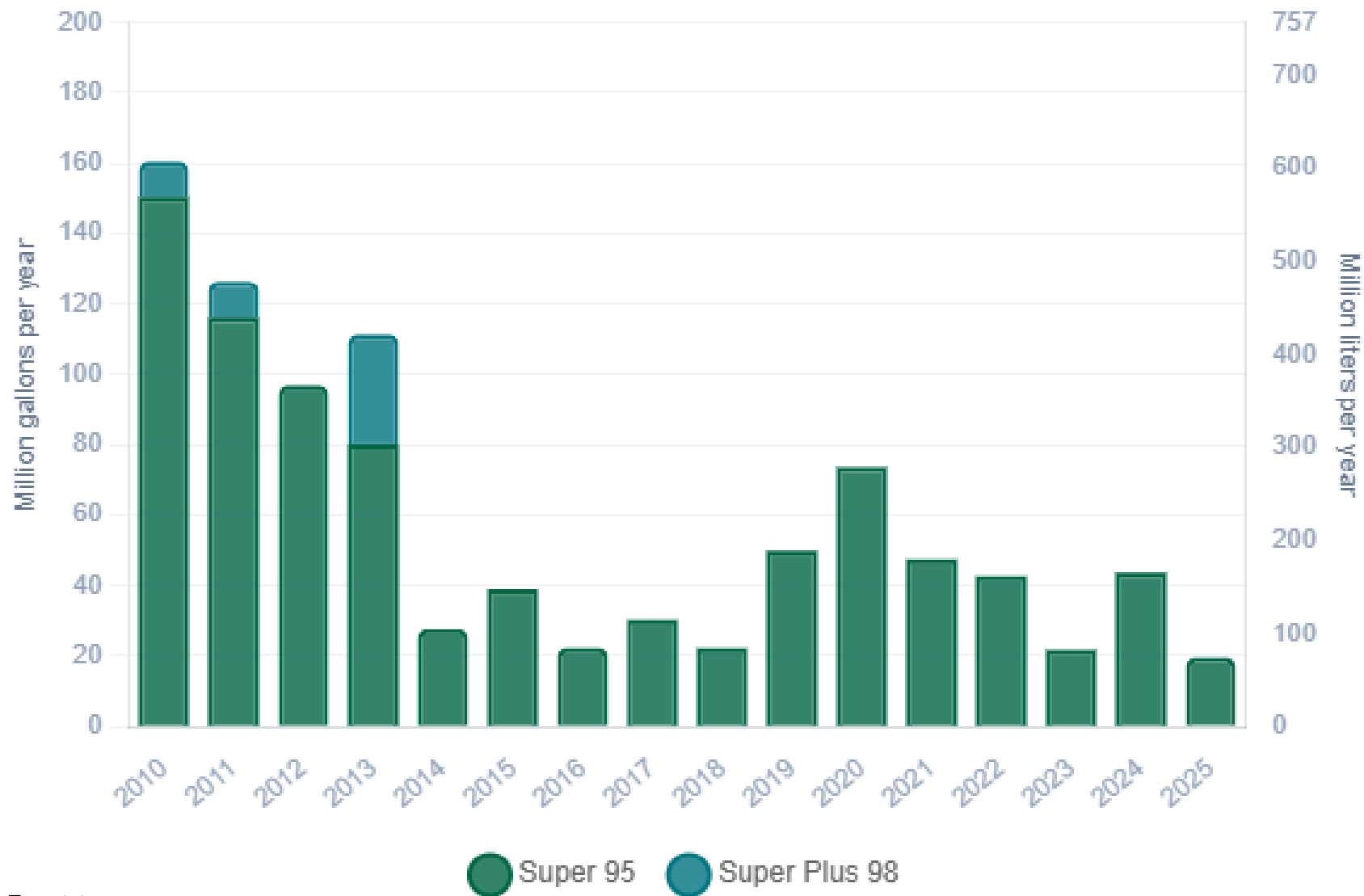
E10 is allowed but Malta does not produce or consume bioethanol at a commercial level

Gasoline Demand in Malta



Source: Eurostat

Gasoline Imports in Malta



Source: Eurostat

The FARO90 logo is shown in a blue box, followed by a chemical structure diagram of a substituted cyclohexadiene derivative. The structure features a six-membered ring with two double bonds. Substituents include a methyl group, a vinyl group, and a group labeled 'X' in orange. The letters 'H', 'C', and 'X' are also highlighted in orange in the diagram.

Source: Eurostat

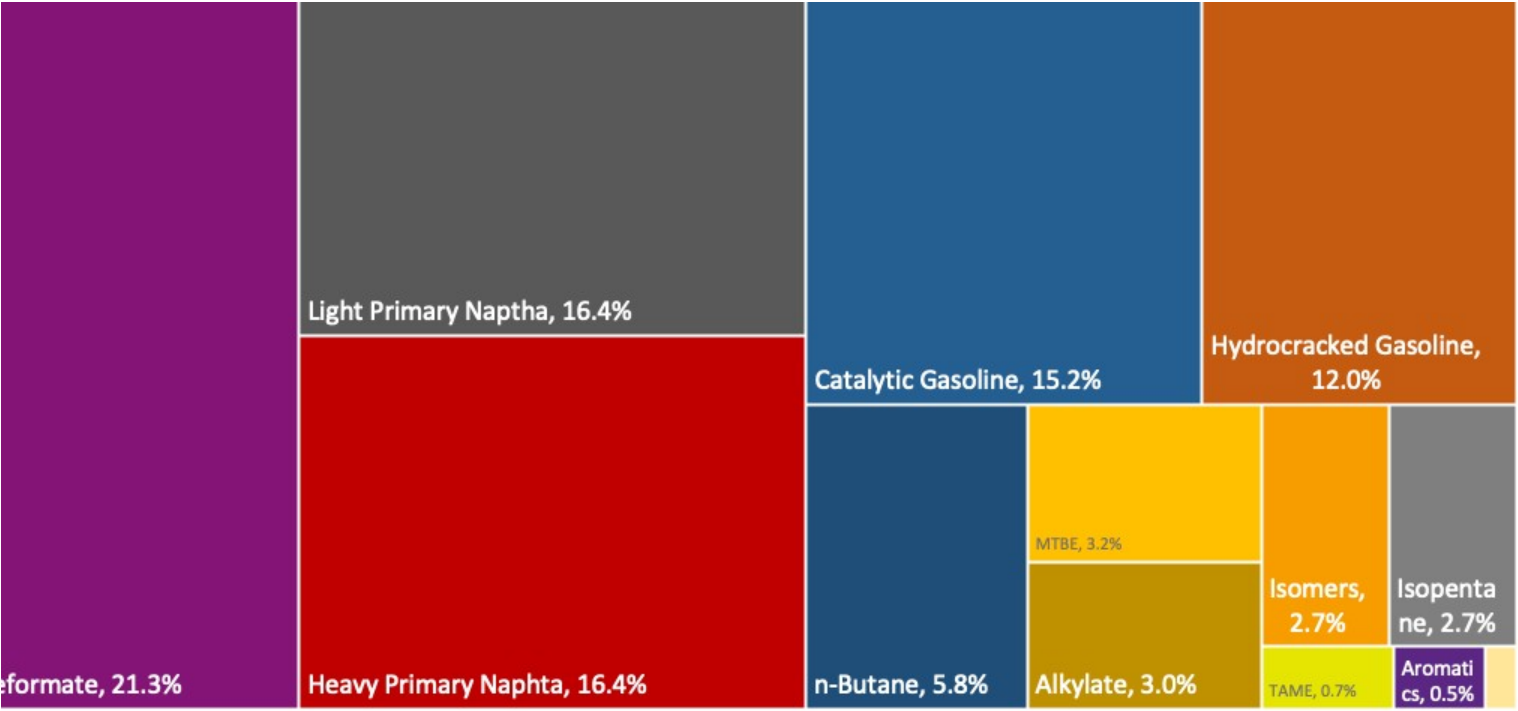
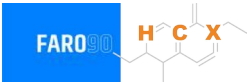
Gasoline Quality in Malta

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

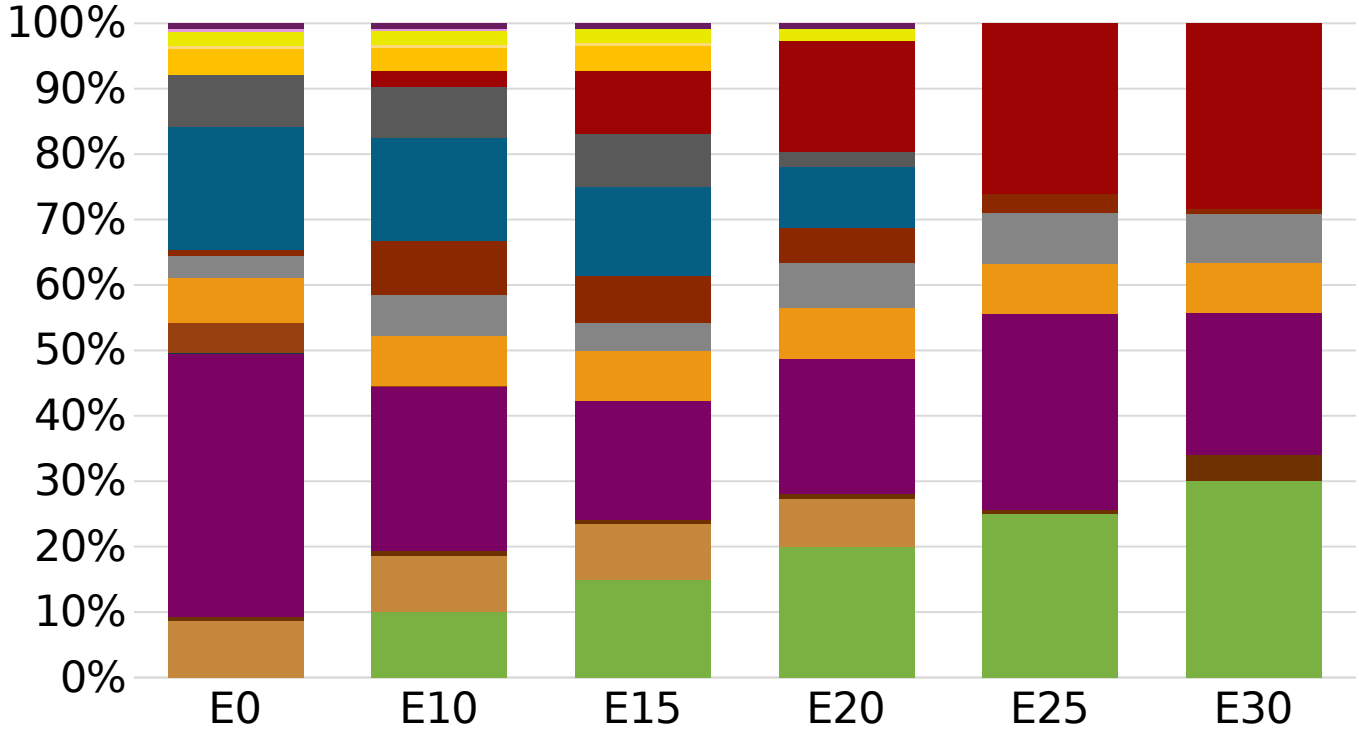
Source: European Commision

Malta Available Blending Components



ETBE,
0.2%

Malta – Gasoline 95 – Constant Octane

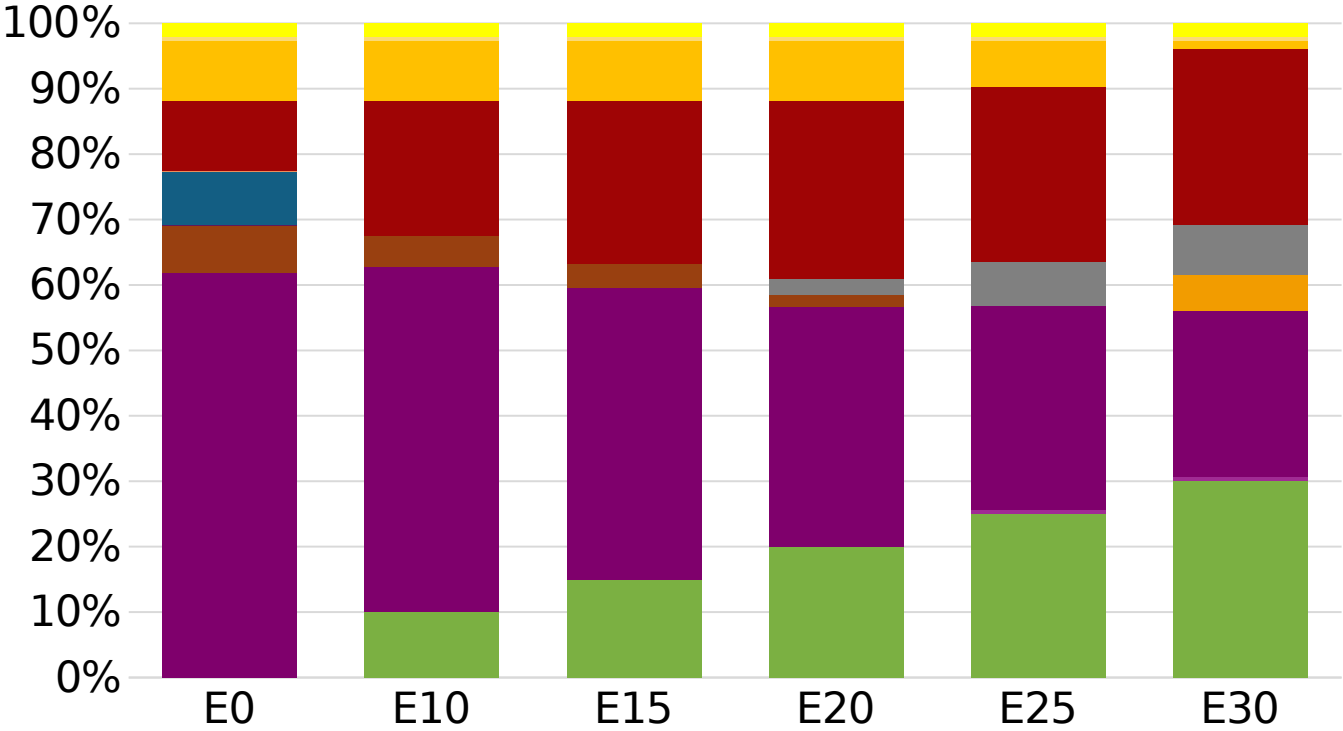


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.4	95.8
Price (US\$/gal)	\$ 2.055	\$ 1.942	\$ 1.859	\$ 1.783	\$ 1.688	\$ 1.614

Source:
Faro90

Malta - Gasoline 98 - Constant Octane

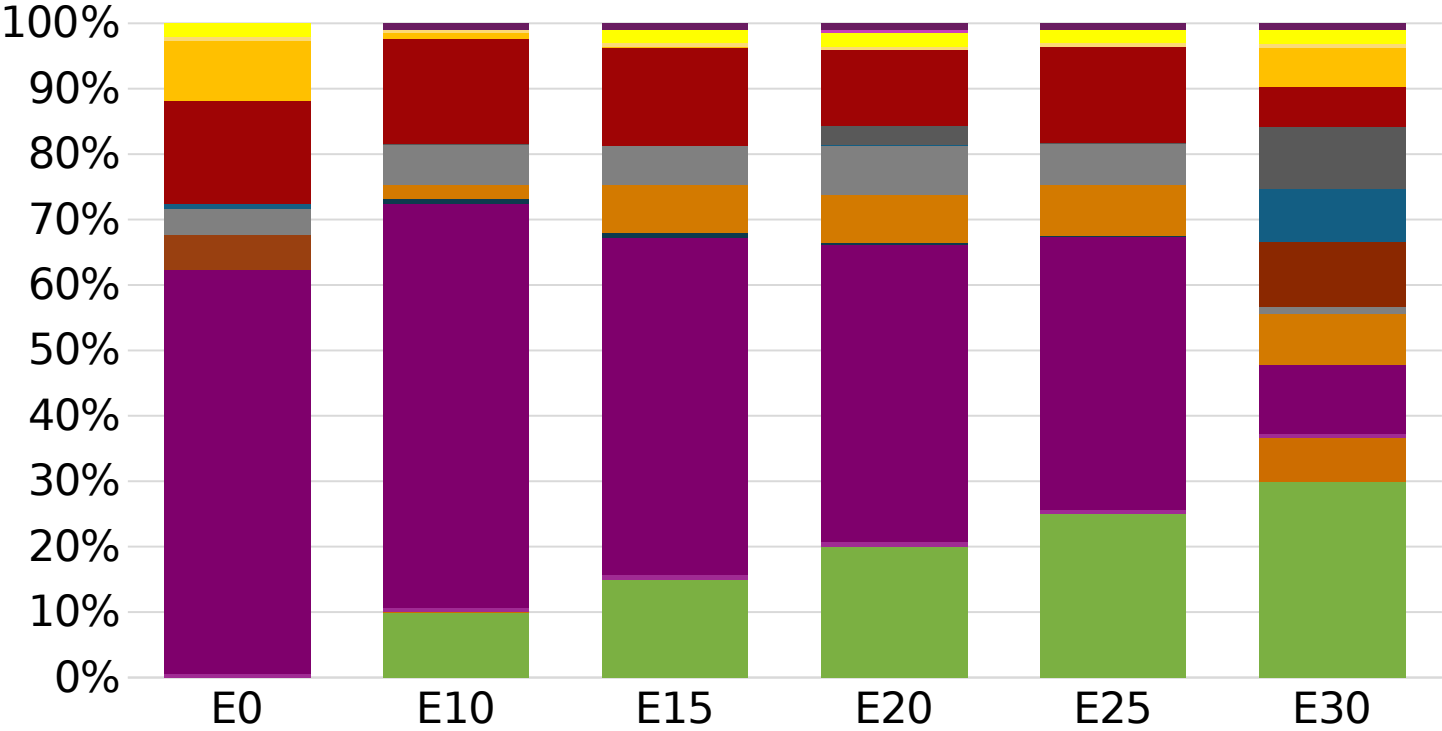


Average CIF 2025
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.104	\$ 1.985	\$ 1.916	\$ 1.846	\$ 1.776	\$ 1.723

Source:
Faro90

Malta – Gasoline 95 – Increased Octane

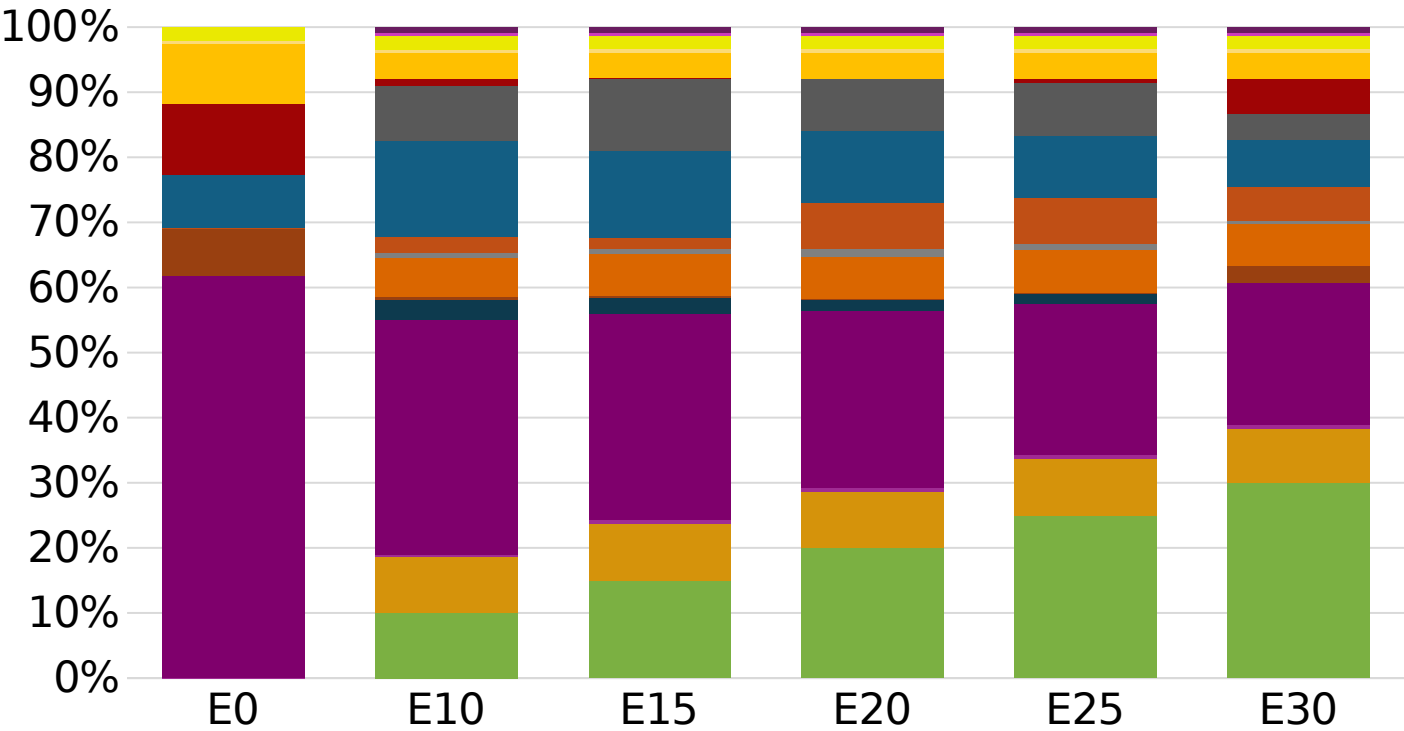
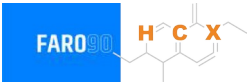


Average CIF 2025
Prices

Octane (RON)	95.0	95.9	97.5	99.0	100.5	101.8
Price (US\$/gal)	\$ 1.974	\$ 1.974	\$ 1.974	\$ 1.974	\$ 1.974	\$ 1.974

Source:
Faro90

Malta – Gasoline 98 – Increased Octane

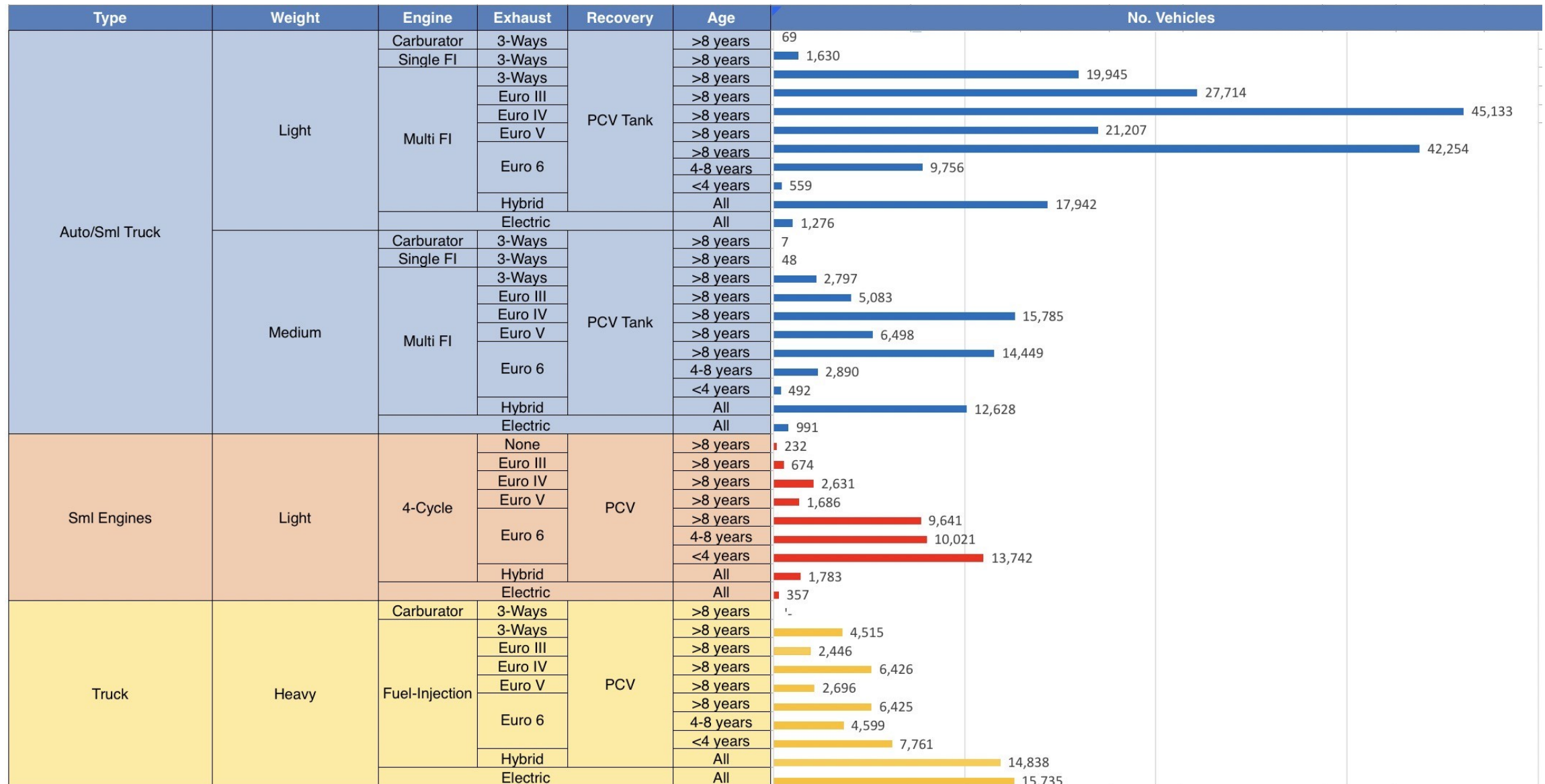


Average CIF 2025
Prices

Octane (RON)	98.0	98.8	100.1	101.3	102.6	104.1
Price (US\$/gal)	\$ 2.104	\$ 2.104	\$ 2.104	\$ 2.104	\$ 2.104	\$ 2.104

Source:
Faro90

Gasoline Vehicle Fleet - Malta

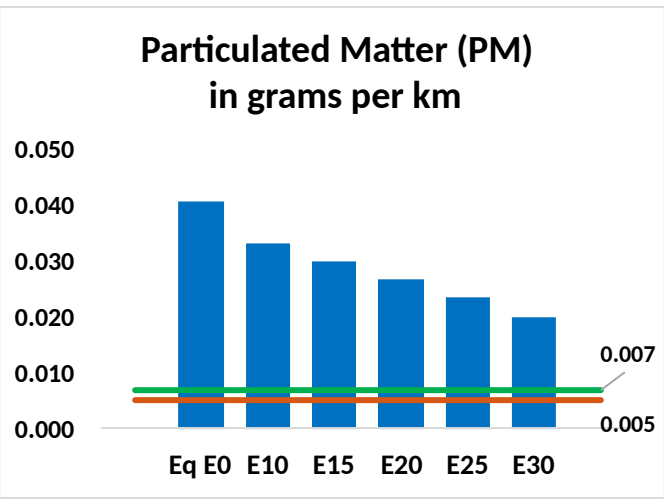
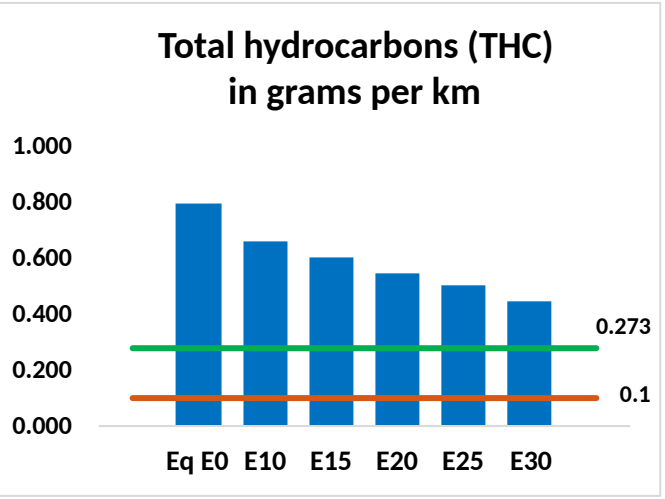
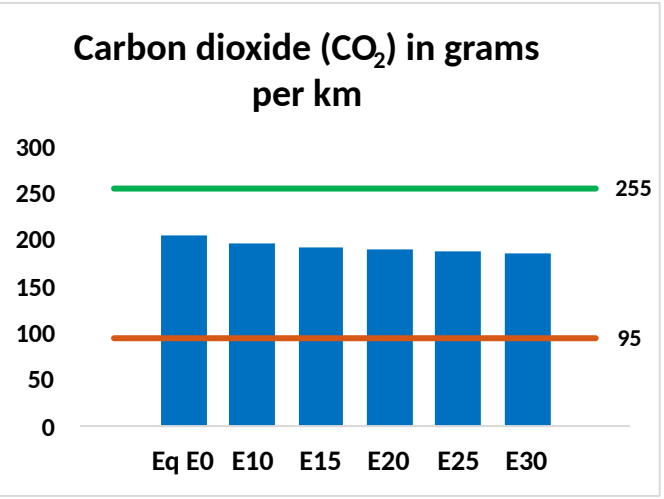
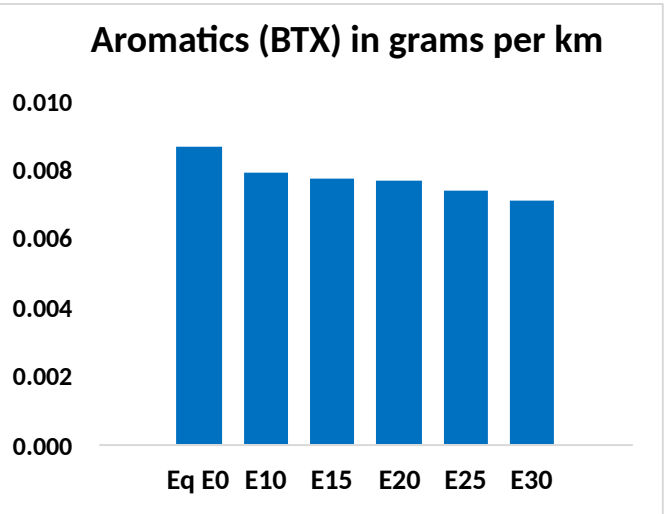
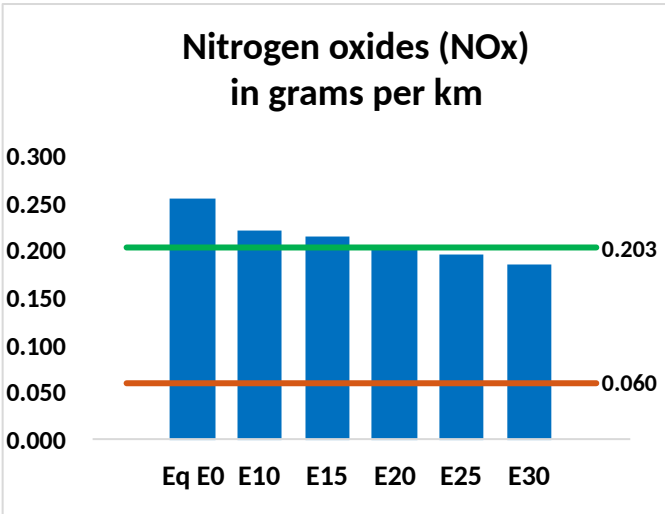
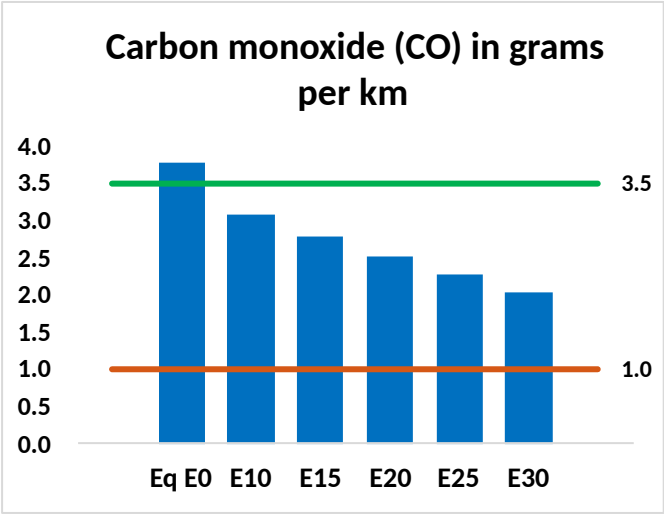
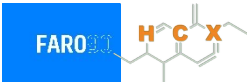


Vehicle Fleet: 355,360
Gasoline Vehicle Fleet 200,048

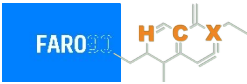
Source: Eurostat, ACEA

Malta – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard

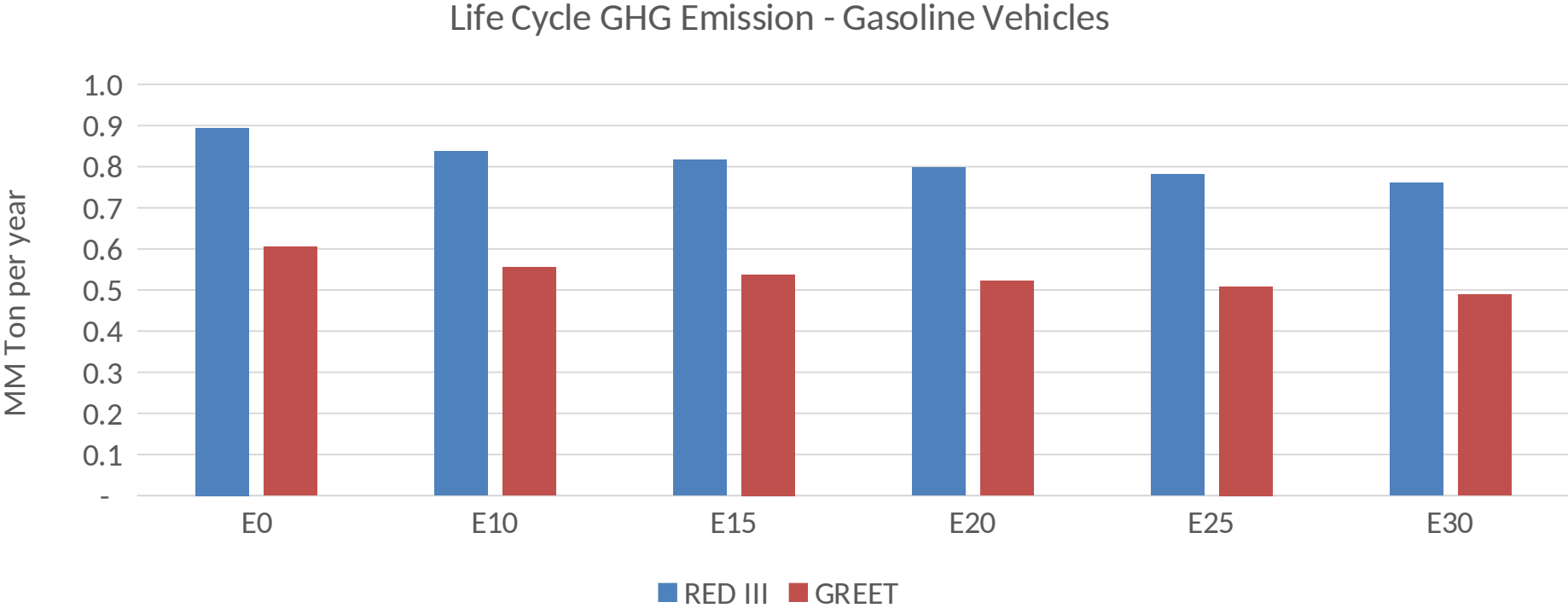
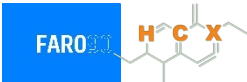


Malta – Vehicular Emissions



Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	18,330	16,673	15,017	13,612	12,175	11,064	9,890	-18%	-34%
CO2	1,003,265	977,467	953,102	933,940	924,458	919,120	902,178	-5%	-8%
VOC	3,880	3,550	3,220	2,943	2,662	2,445	2,158	-17%	-31%
NOx	1,243	1,148	1,083	1,049	996	954	904	-13%	-20%
SOx	12	11	10	9	8	8	7	-15%	-28%
NH3	388	384	380	382	386	389	392	-2%	0%
N2O	14	14	14	14	15	15	15	-1%	2%
CH4	786	786	786	797	815	830	844	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	19	18	17	17	16	16	15	-9%	-14%
Acetaldehyde	63	73	107	163	222	253	299	68%	249%
Formaldehyde	238	232	270	317	332	367	400	13%	39%
Benzene	42	41	39	38	38	36	35	-9%	-11%
PM2.5	60	55	49	44	40	35	29	-18%	-34%
PM10	138	126	113	102	91	80	68	-18%	-34%

Malta – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	2.2
Target @ 2035 (MMT/year)	1.2
Delta	1.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.2%	11.1%	13.7%	15.9%	19.1%
RED II/III	0.0%	6.3%	8.6%	10.7%	12.5%	14.9%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.2%	7.0%	8.7%	10.1%	12.0%
RED II/III	0.0%	5.9%	8.0%	10.0%	11.7%	13.9%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

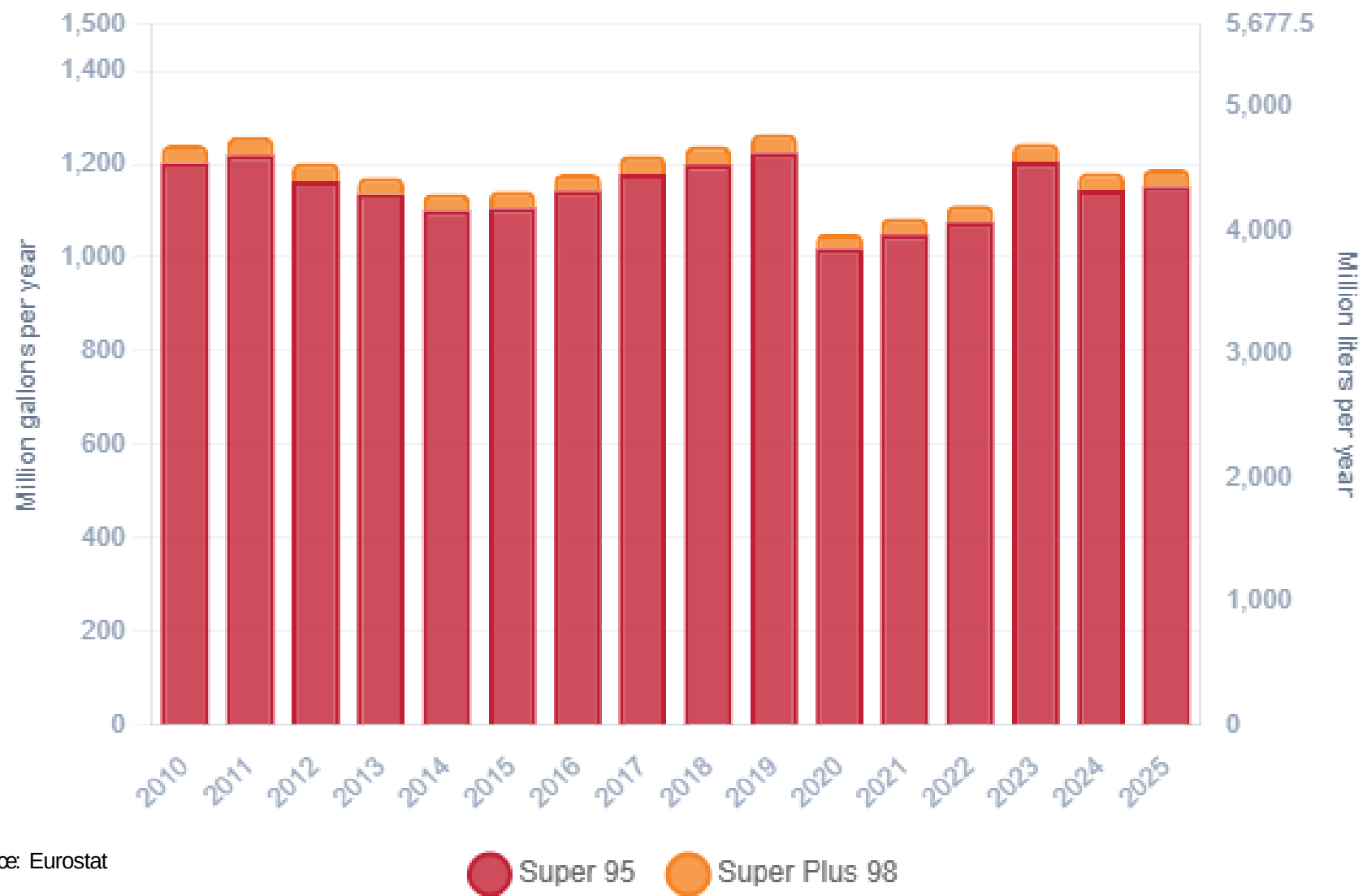
Ethanol Gasoline Blending in Netherlands



In 2025, gasoline consumption in Netherlands was 4,514 million liters (1,192 million gallons) and growth rate was -0.28%. Gasoline production and net imports were 18,087 and -13,573 million liters (4,778 and -3,586 million gallons) correspondingly. Netherlands complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.1 RON.

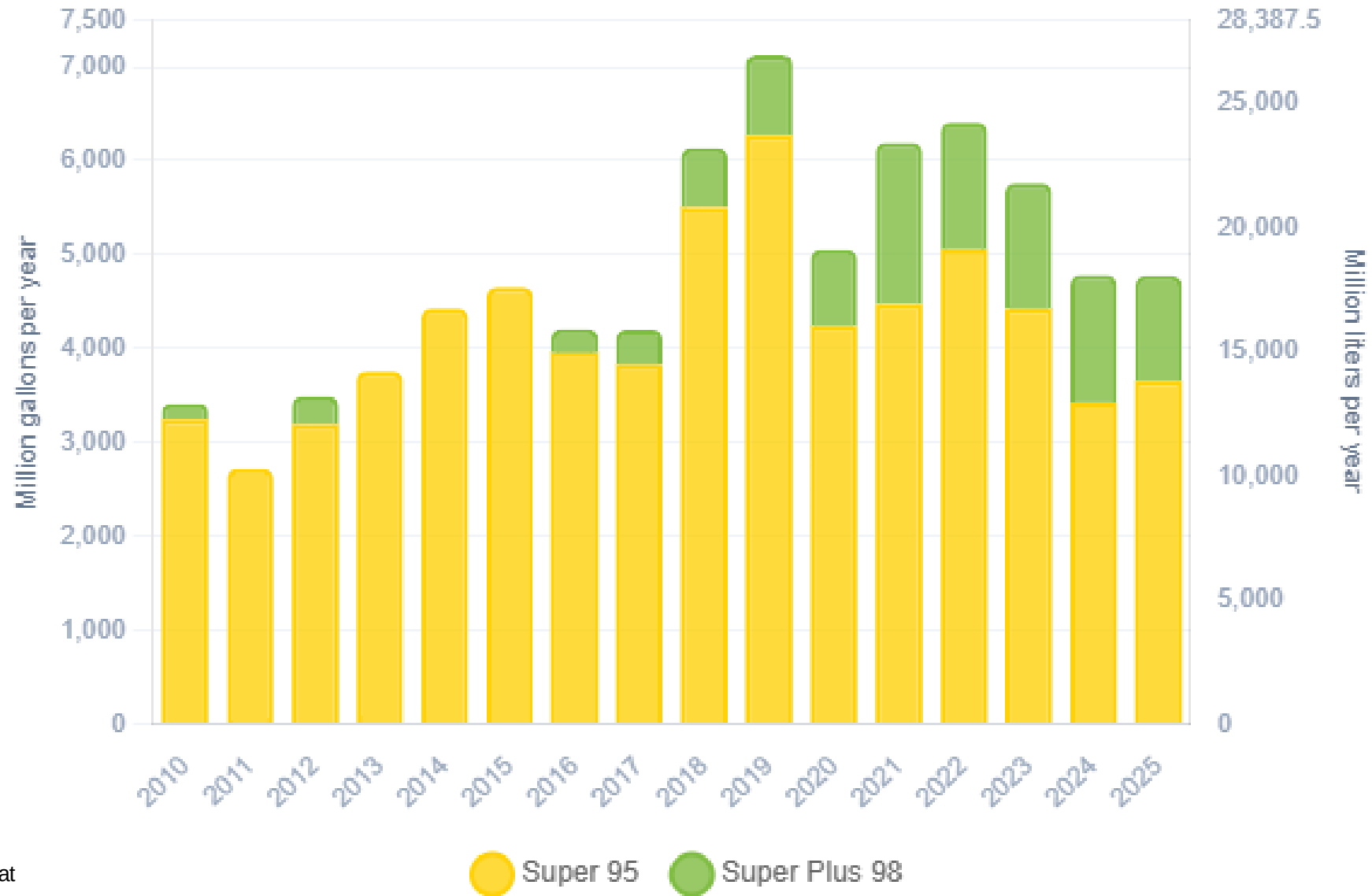
In 2025, ethanol consumption in Netherlands was 500 million liters (132 million gallons) representing 11% of gasoline demand. Ethanol Net Imports were 695 million liters (0 and 184 million gallons) correspondingly. Ethanol sources are Sugar Cane 5% and Cereals 95%.

Gasoline Demand in Netherlands



Source: Eurostat

Gasoline Production in Netherlands



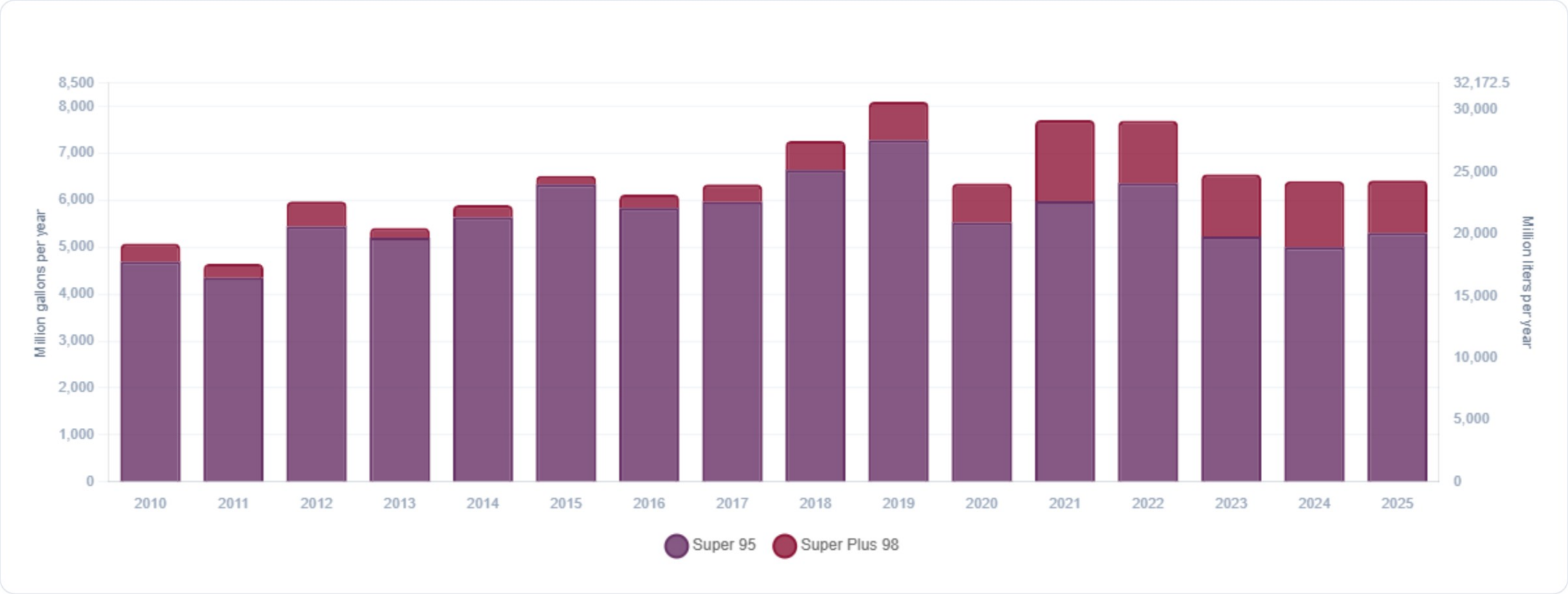
Source: Eurostat

Gasoline Imports in Netherlands



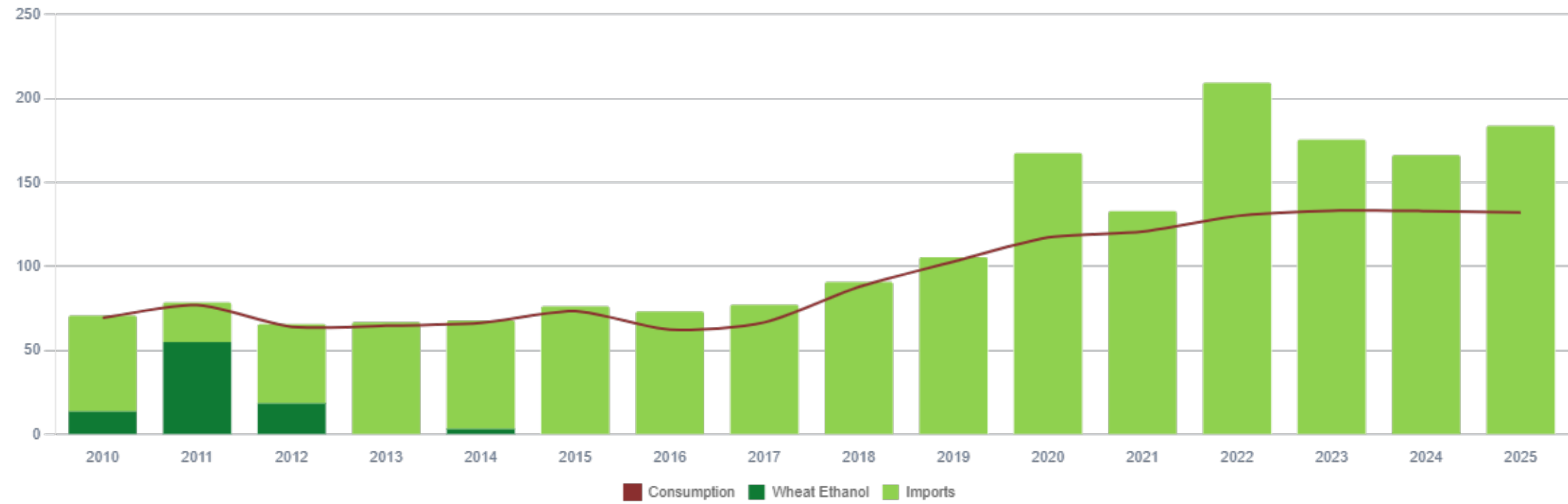
Source: Eurostat

Gasoline Exports in Netherlands



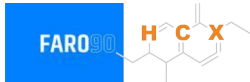
Source: Eurostat

Ethanol Balance in Netherlands



Source: Eurostat, Statistics Netherlands

Gasoline Quality in Netherlands

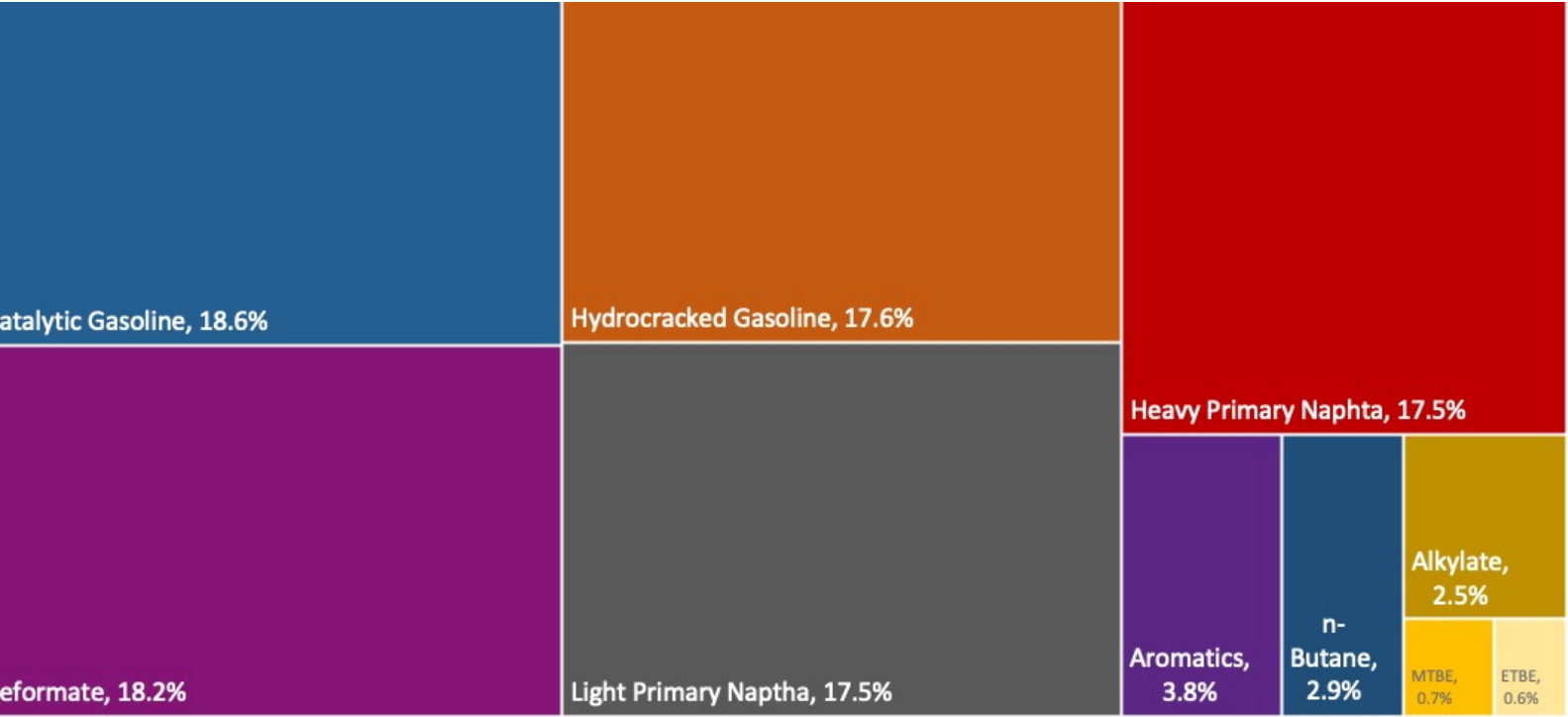


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	28.4							
Advanced Biofuels (%cal)	3.6							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

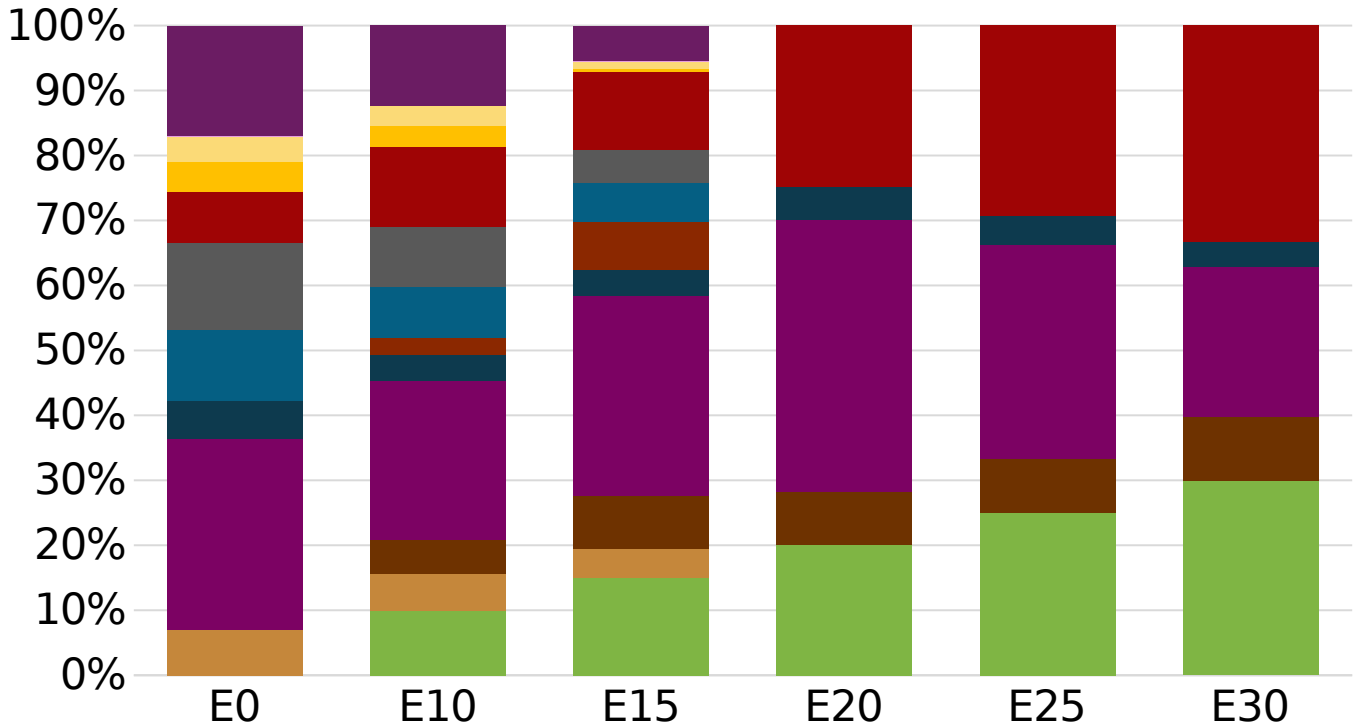
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Netherlands Available Blending Components



Netherlands – Gasoline 95 – Constant Octane

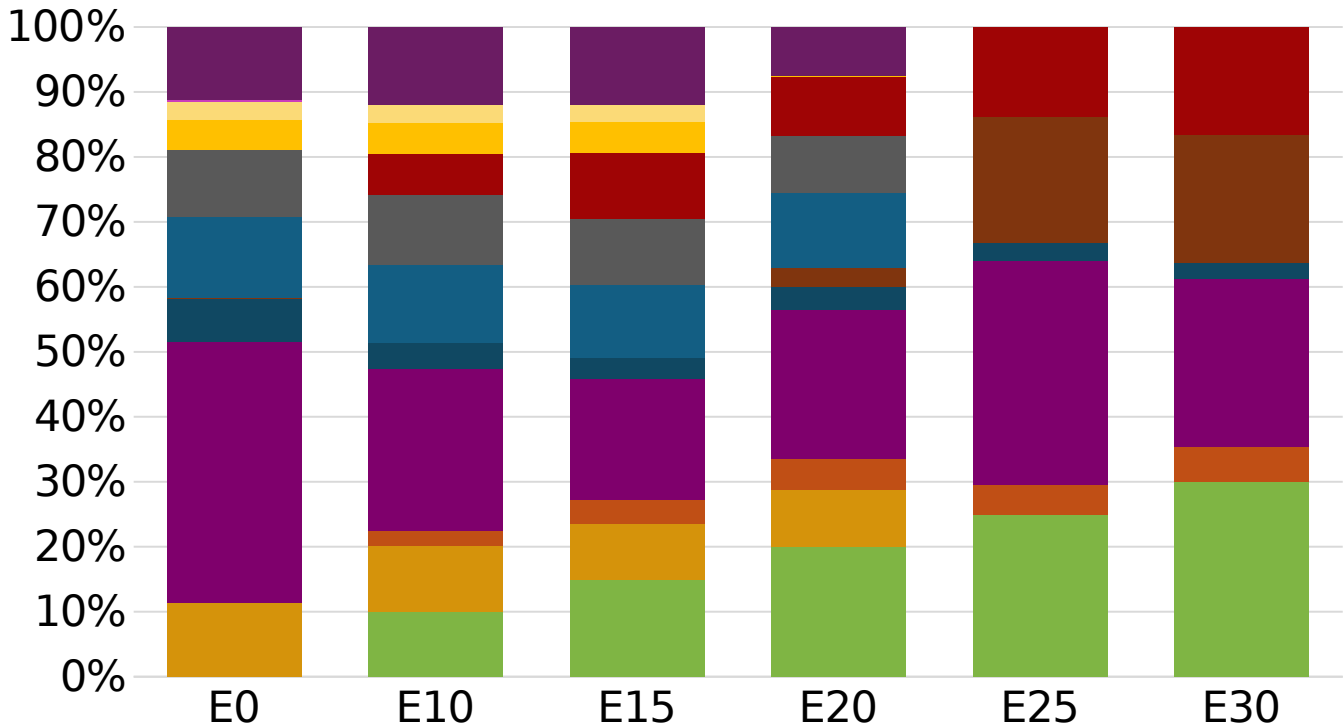


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.1	95.3	95.3
Price (US\$/gal)	\$ 2.078	\$ 1.944	\$ 1.869	\$ 1.748	\$ 1.672	\$ 1.577

Source:
Faro90

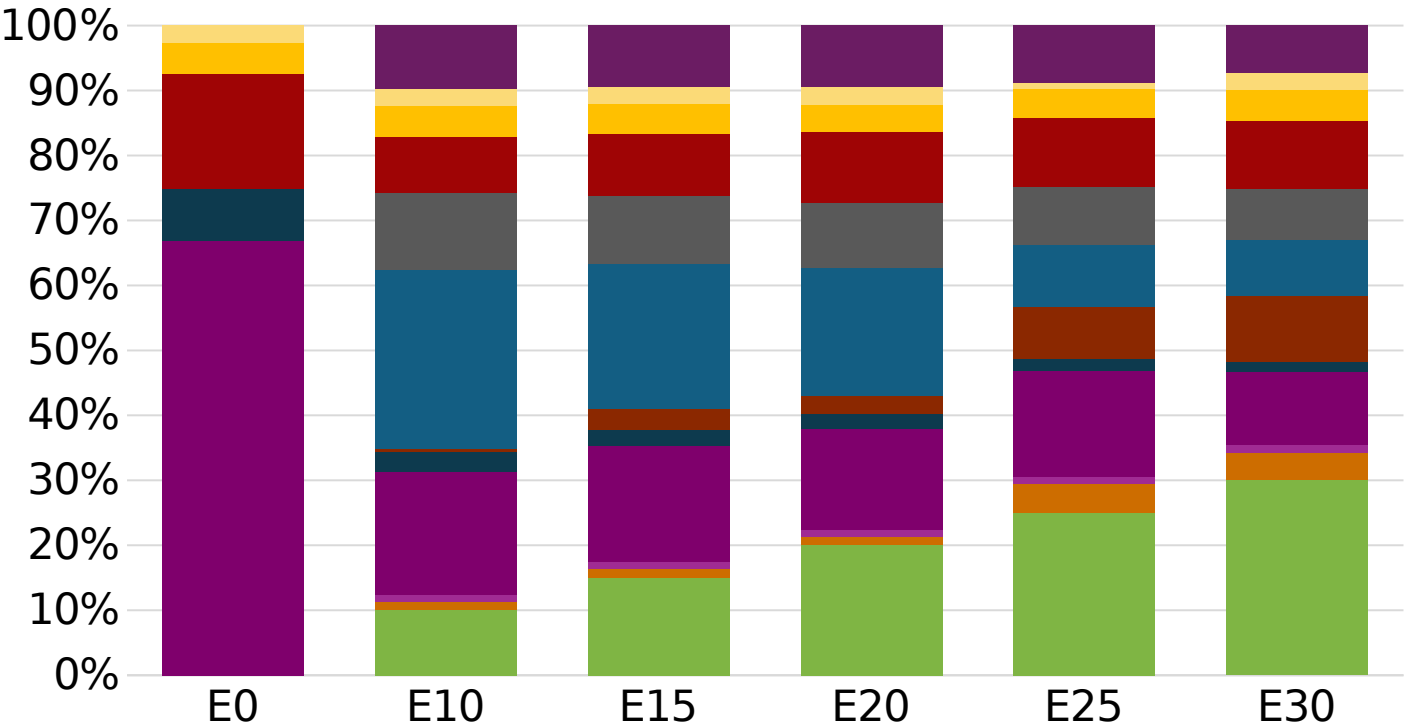
Netherlands - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.1	98.5
Price (US\$/gal)	\$ 2.206	\$ 2.087	\$ 2.013	\$ 1.961	\$ 1.854	\$ 1.791

Average CIF 2025
Prices

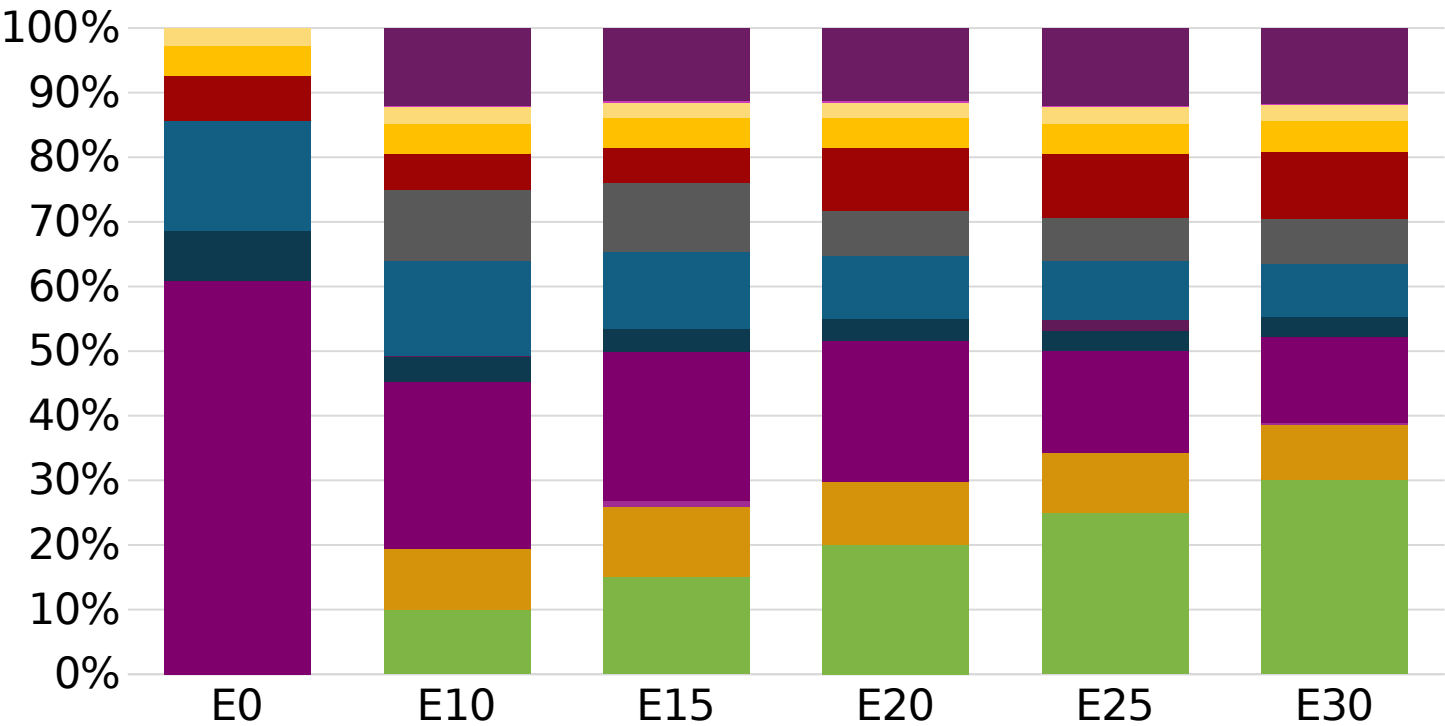
Netherlands – Gasoline 95 – Increased Octane



Average CIF 2025
Prices

Source:
Faro90

Netherlands – Gasoline 98 – Increased Octane

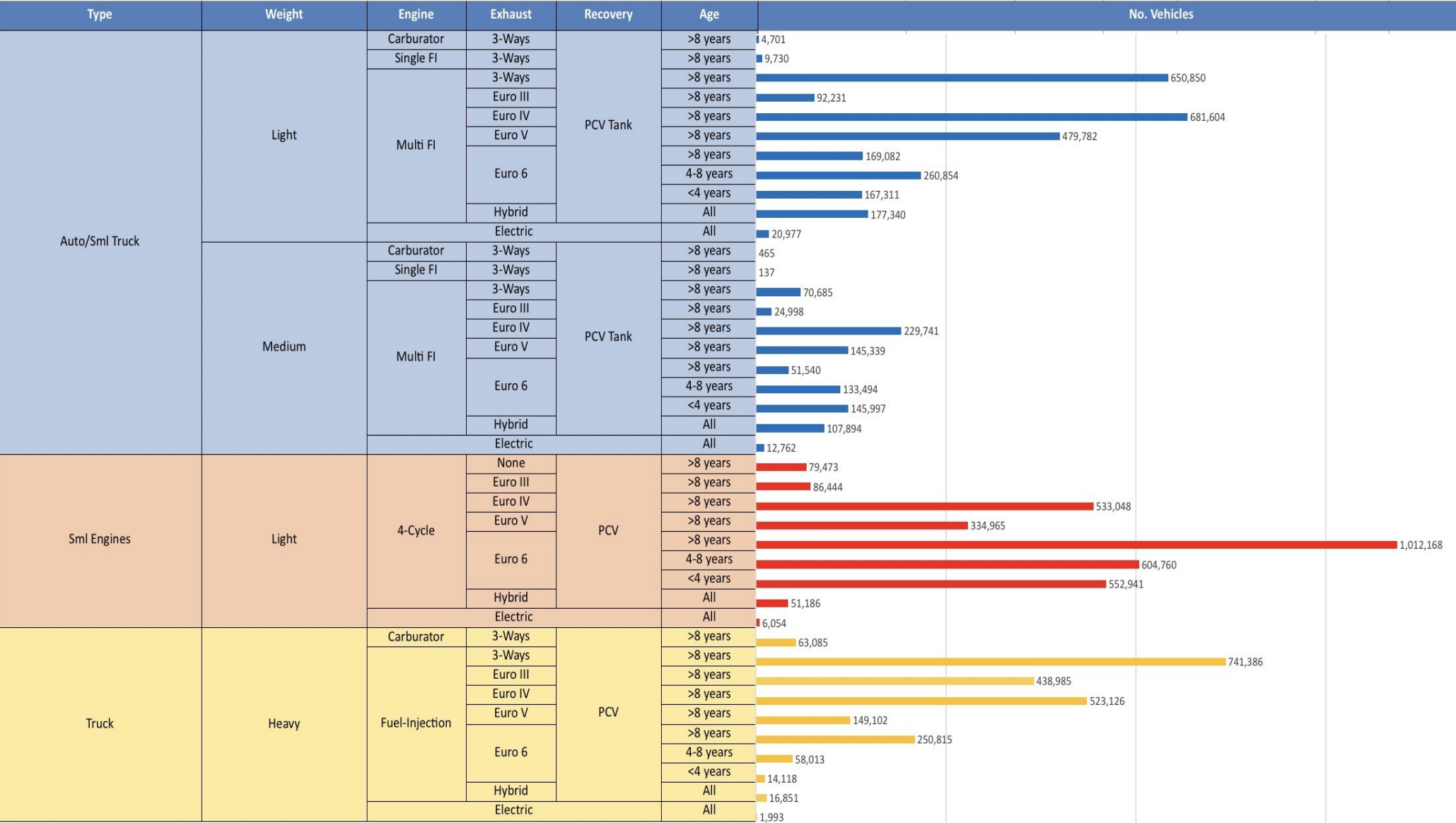


Average CIF 2025
Prices

Octane (RON)	98.0	98.8	100.1	101.6	102.9	104.2
Price (US\$/gal)	\$ 2.126	\$ 2.126	\$ 2.126	\$ 2.126	\$ 2.126	\$ 2.126

Source:
Faro90

Gasoline Vehicle Fleet - Netherlands

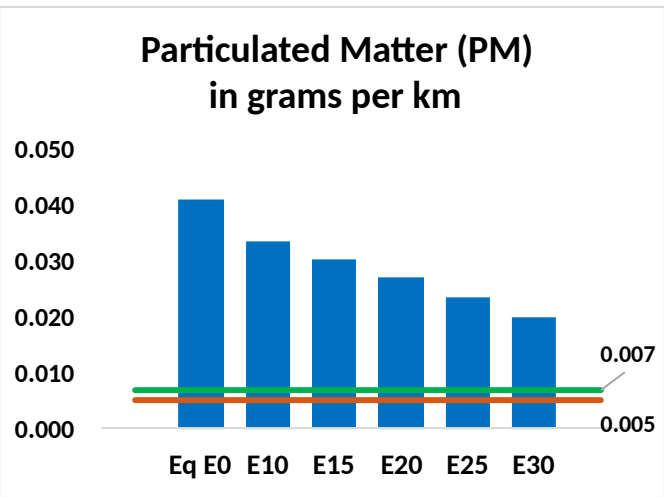
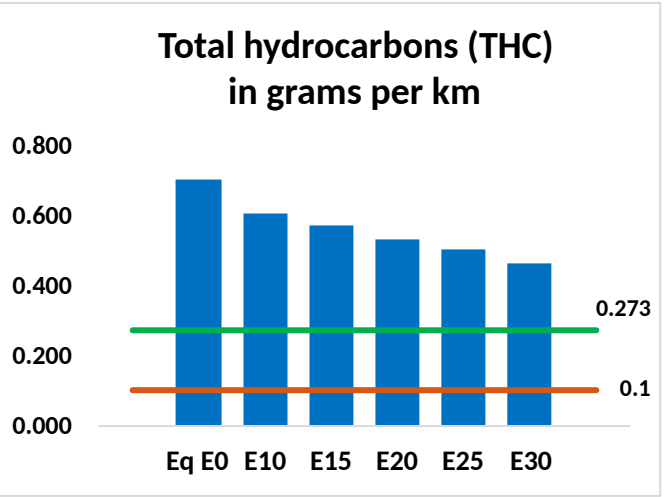
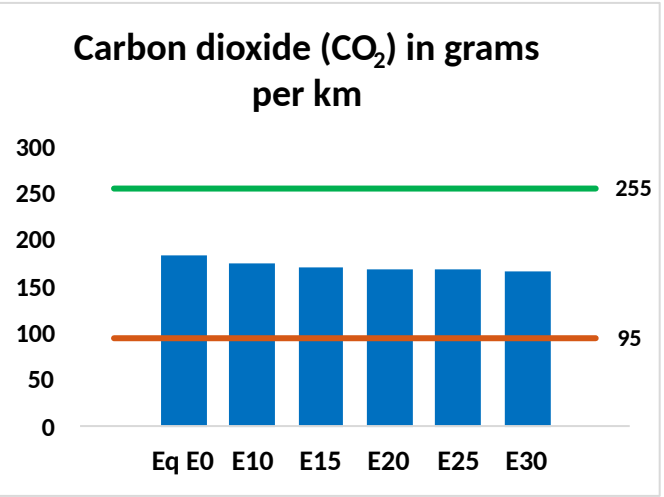
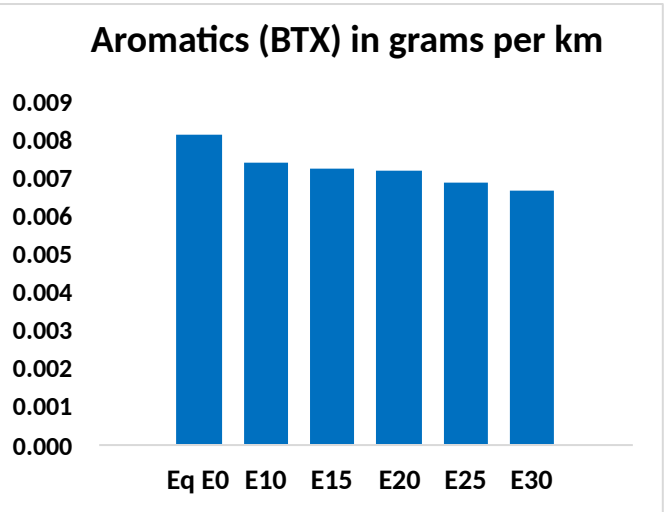
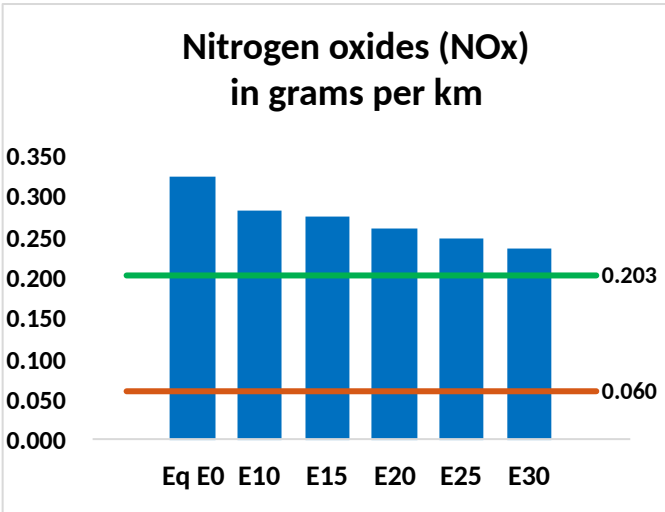
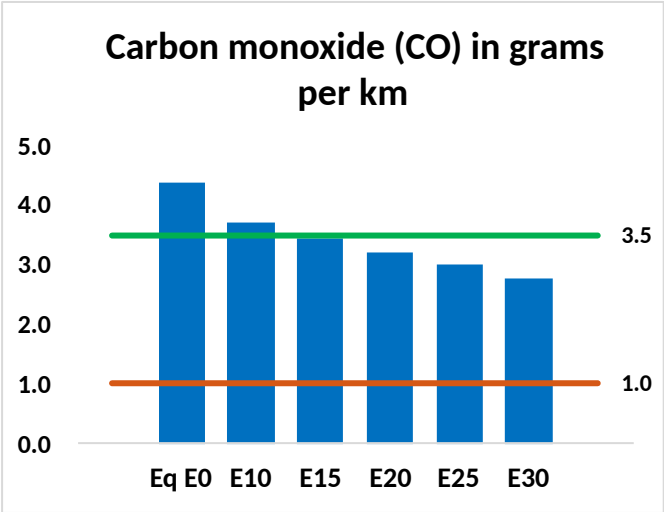
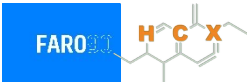


Vehicle Fleet: 9,156,027
Gasoline Vehicle Fleet: 7,160,013

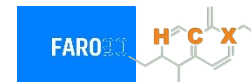
Source: Eurostat, ACEA

Netherlands – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard

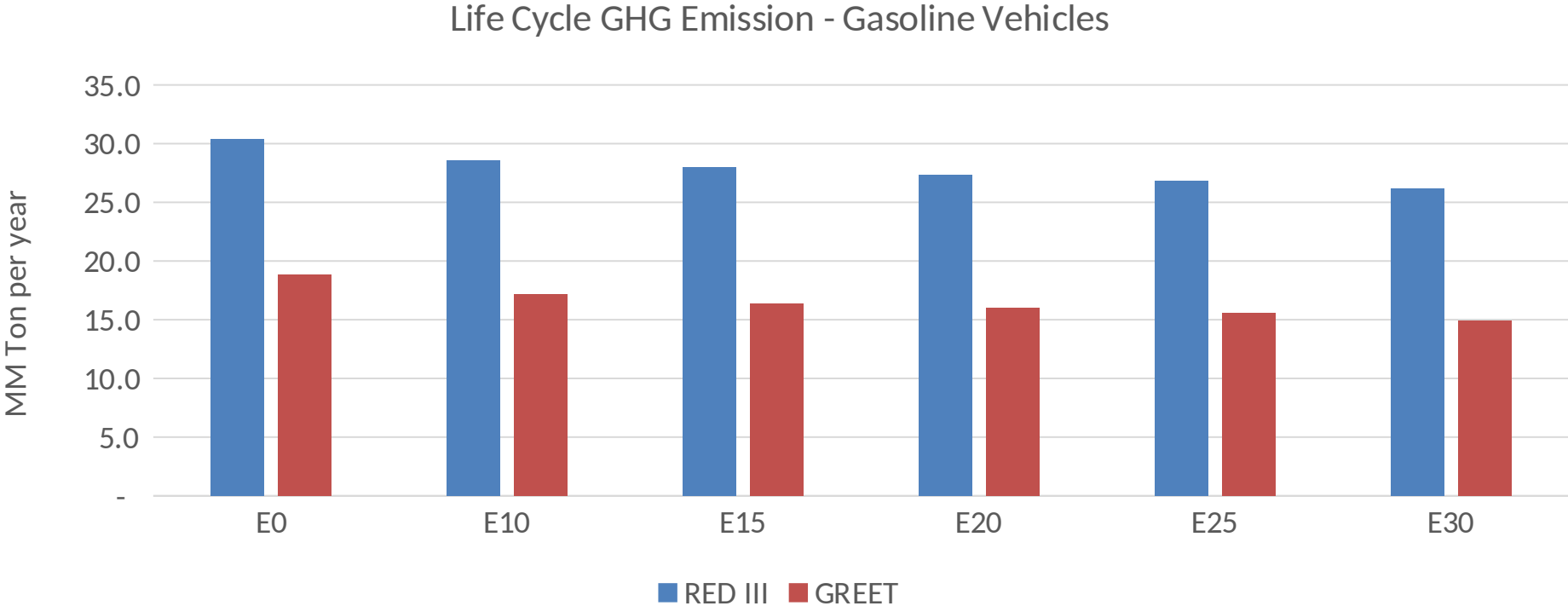
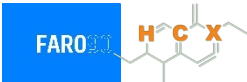


Netherlands – Vehicular Emissions



Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	641,117	593,663	546,209	507,680	469,072	439,434	406,683	-15%	-27%
CO2	27,000,283	26,305,990	25,650,269	25,134,563	24,879,372	24,767,928	24,311,384	-5%	-8%
VOC	103,136	96,195	89,254	83,759	78,317	74,160	68,518	-13%	-24%
NOx	47,938	44,263	41,727	40,428	38,343	36,705	34,778	-13%	-20%
SOx	306	283	260	240	222	201	180	-15%	-28%
NH3	10,439	10,335	10,232	10,280	10,396	10,477	10,545	-2%	0%
N2O	450	447	445	448	457	462	467	-1%	2%
CH4	22,880	22,880	22,880	23,223	23,727	24,161	24,573	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	671	645	619	603	589	579	563	-8%	-12%
Acetaldehyde	1,960	2,249	3,300	5,025	6,846	7,823	9,244	68%	249%
Formaldehyde	7,275	7,073	8,245	9,682	10,143	11,221	12,215	13%	39%
Benzene	1,193	1,146	1,086	1,069	1,060	1,014	981	-9%	-11%
PM2.5	1,667	1,515	1,364	1,230	1,100	960	814	-18%	-34%
PM10	4,343	3,948	3,553	3,206	2,866	2,502	2,120	-18%	-34%

Netherlands – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	171.1
Target @ 2035 (MMT/year)	96.3
Delta	74.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.8%	13.0%	14.8%	17.3%	20.7%
RED II/III	0.0%	6.0%	8.0%	10.1%	11.8%	14.0%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.2%	3.3%	3.7%	4.3%	5.2%
RED II/III	0.0%	2.4%	3.3%	4.1%	4.8%	5.7%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

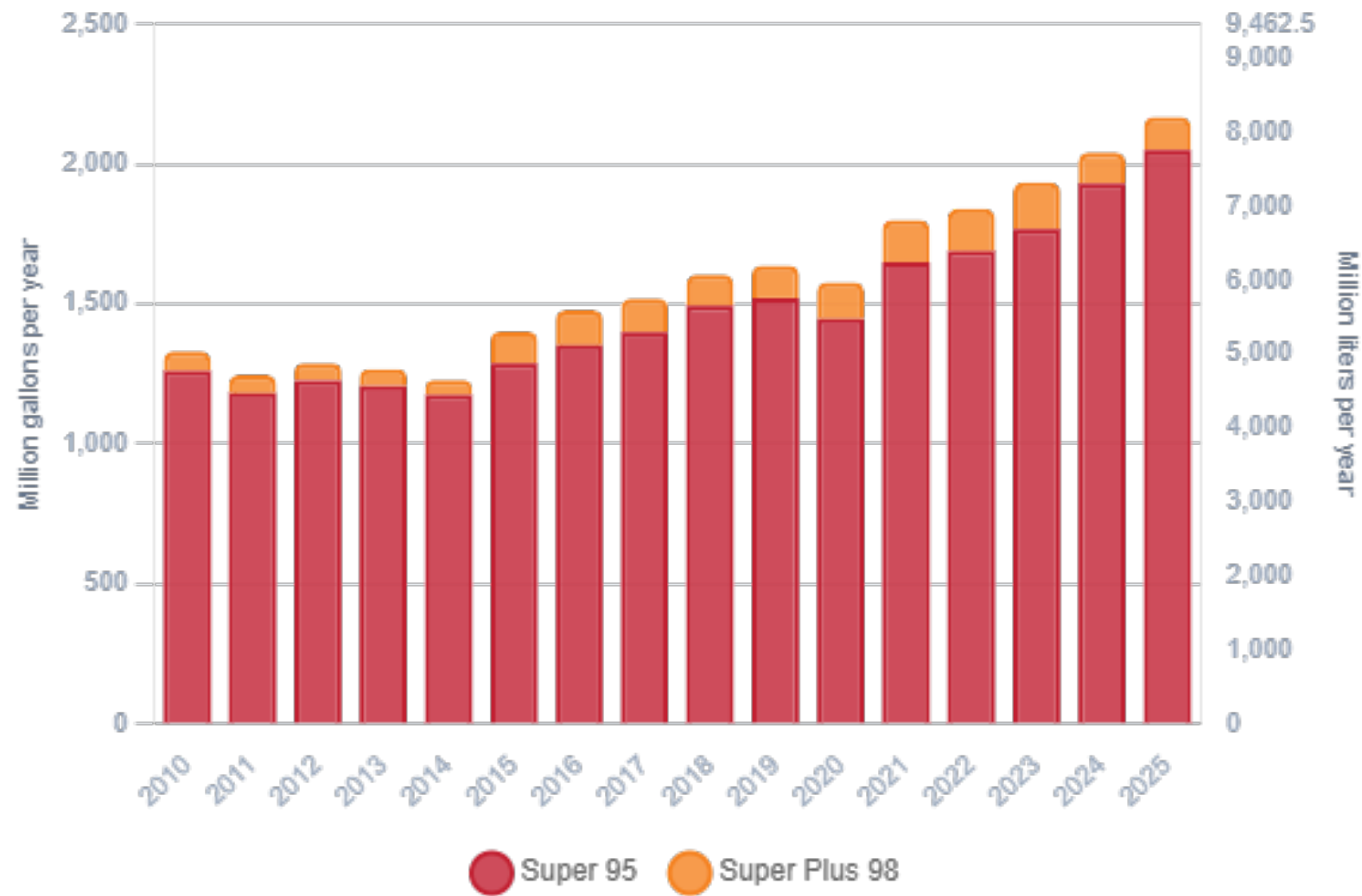
Ethanol Gasoline Blending in Poland



In 2025, gasoline consumption in Poland was 8,220 million liters (2,172 million gallons) and growth rate was 3.31%. Gasoline production and net imports were 6,532 and 1,599 million liters (1,726 and 422 million gallons) correspondingly. Poland complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

In 2025, ethanol consumption in Poland was 659 million liters (174 million gallons) representing 8% of gasoline demand. Ethanol production and Net Imports were 560 and 99 million liters (148 and 26 million gallons) correspondingly. Ethanol sources are Corn 57% and Cereals 43%.

Gasoline Demand in Poland



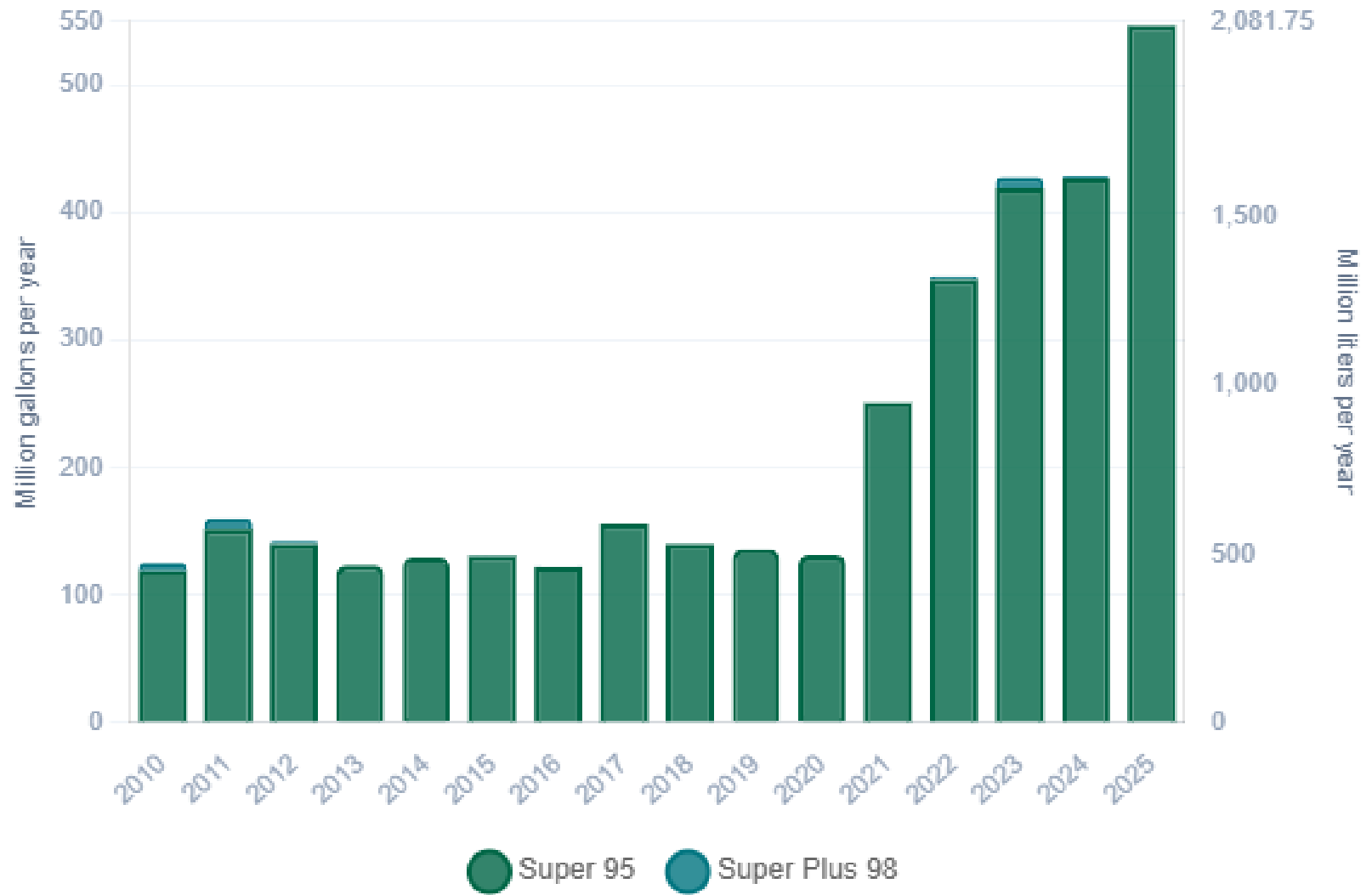
Source: Eurostat

Gasoline Production in Poland



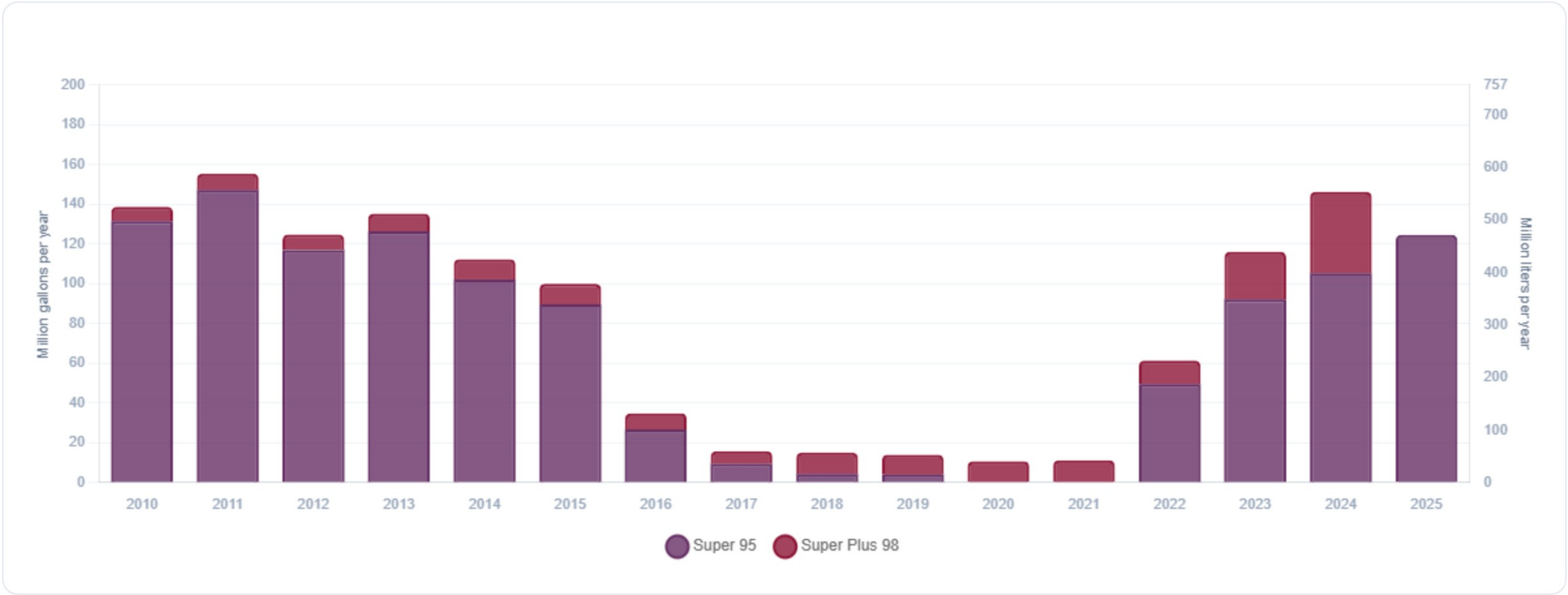
Source: Eurostat

Gasoline Imports in Poland



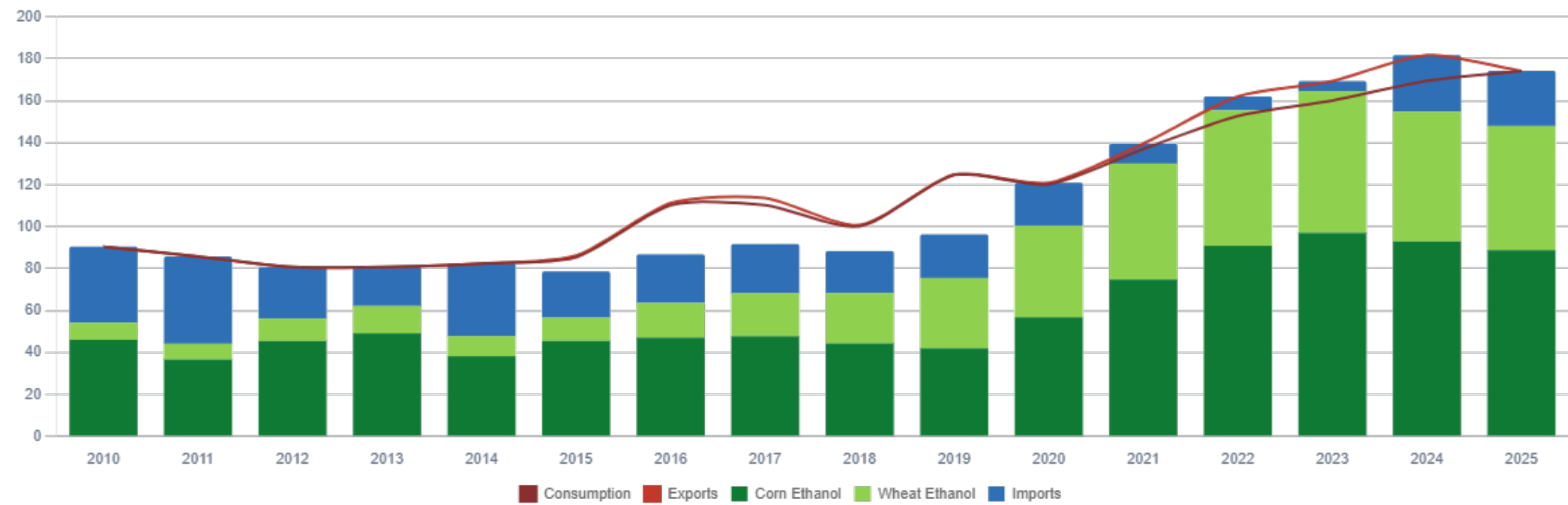
Source: Eurostat

Gasoline Exports in Poland



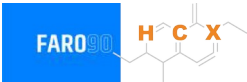
Source: Eurostat

Ethanol Balance in Poland



Source: Eurostat

Gasoline Quality in Poland

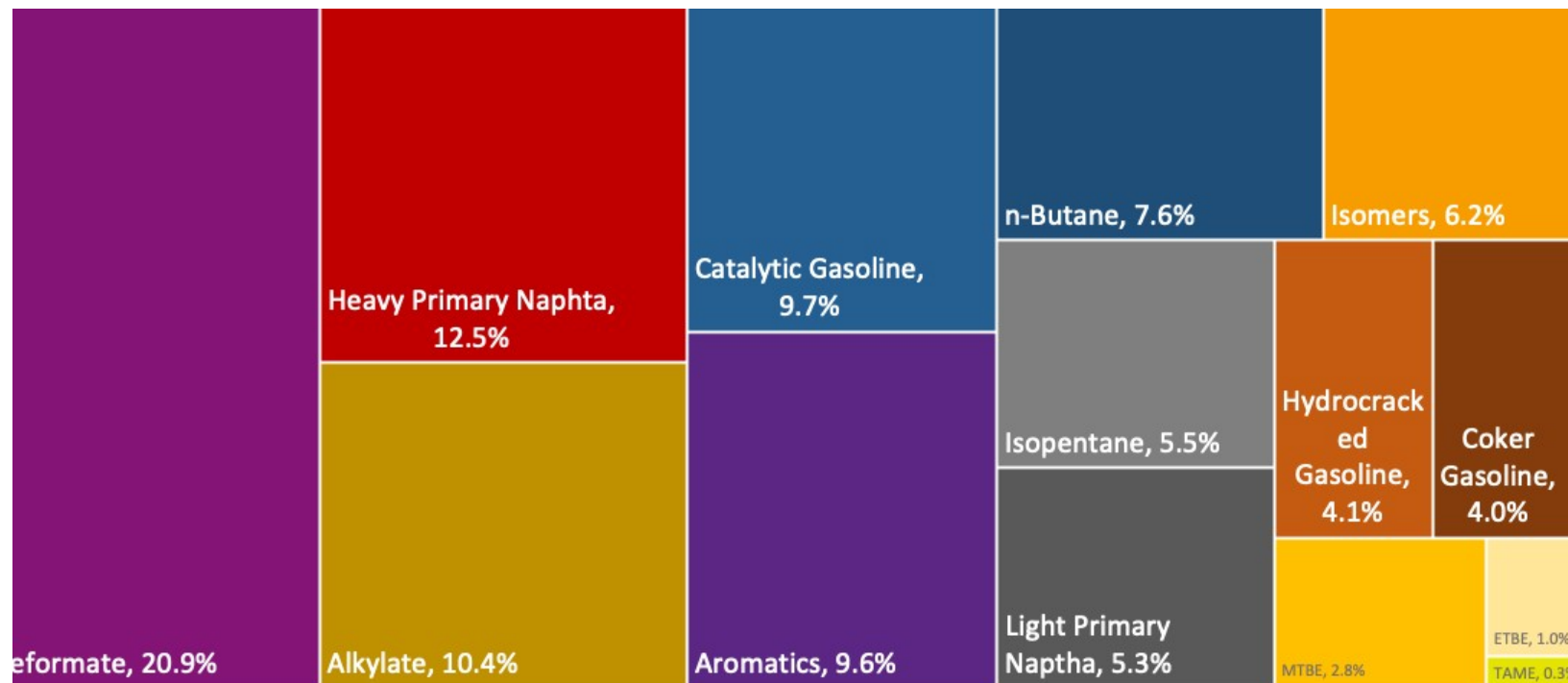


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	9.2							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	4.6							
GHG Emission Reduction (%)	-							

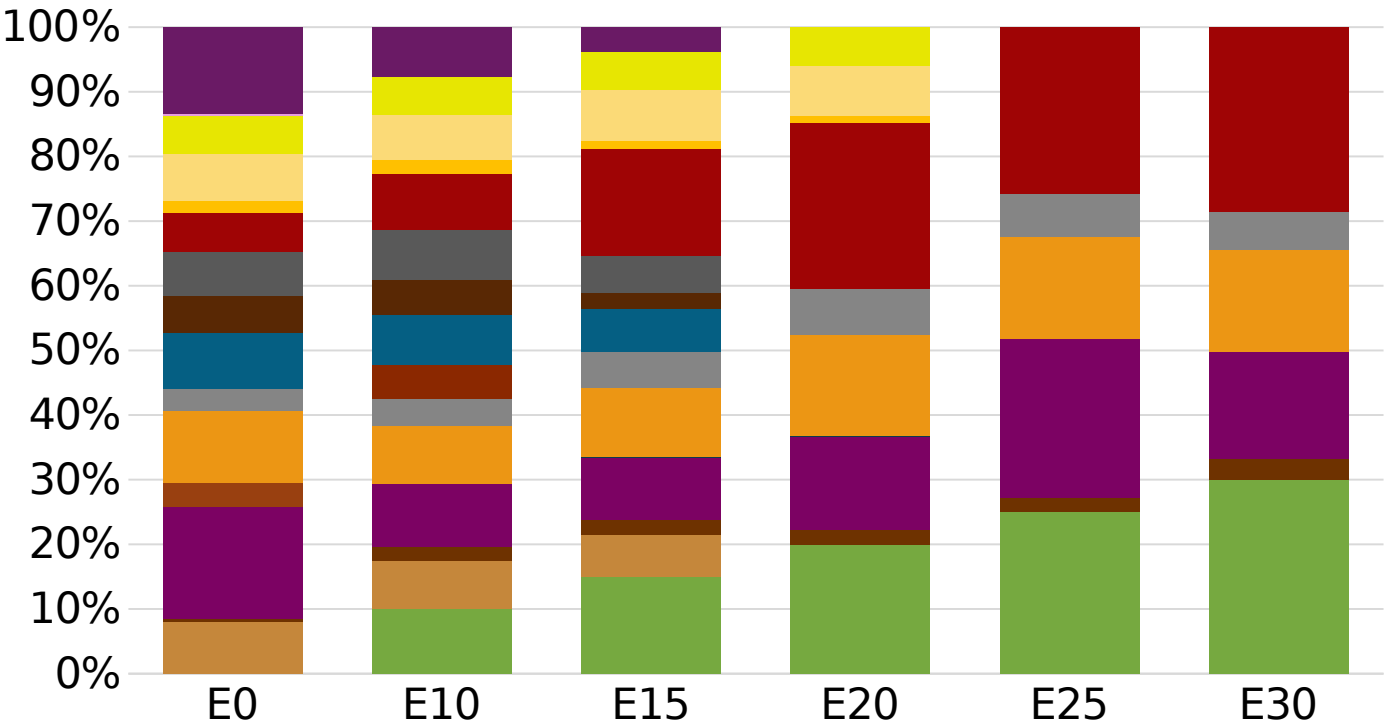
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Poland Available Blending Components



Poland – Gasoline 95 – Constant Octane

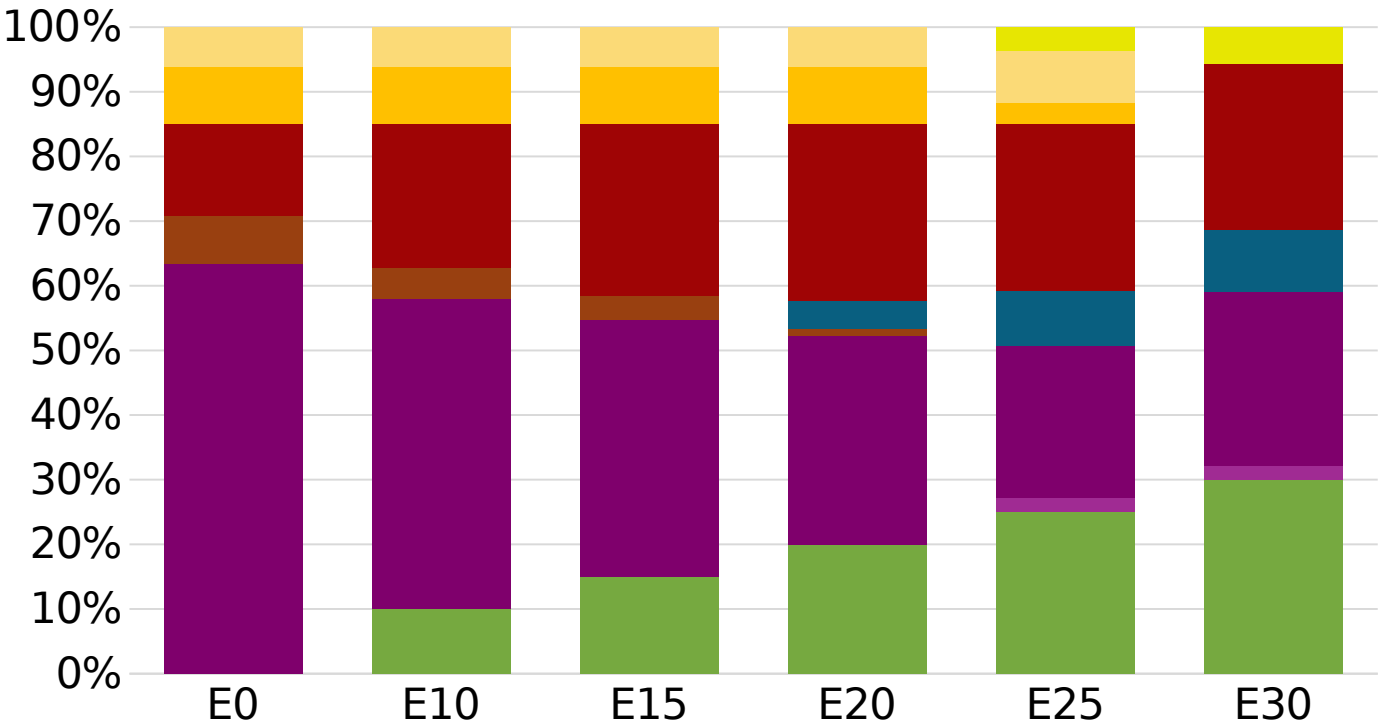
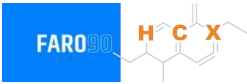


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.081	\$ 1.943	\$ 1.843	\$ 1.728	\$ 1.682	\$ 1.629

Source:
Faro90

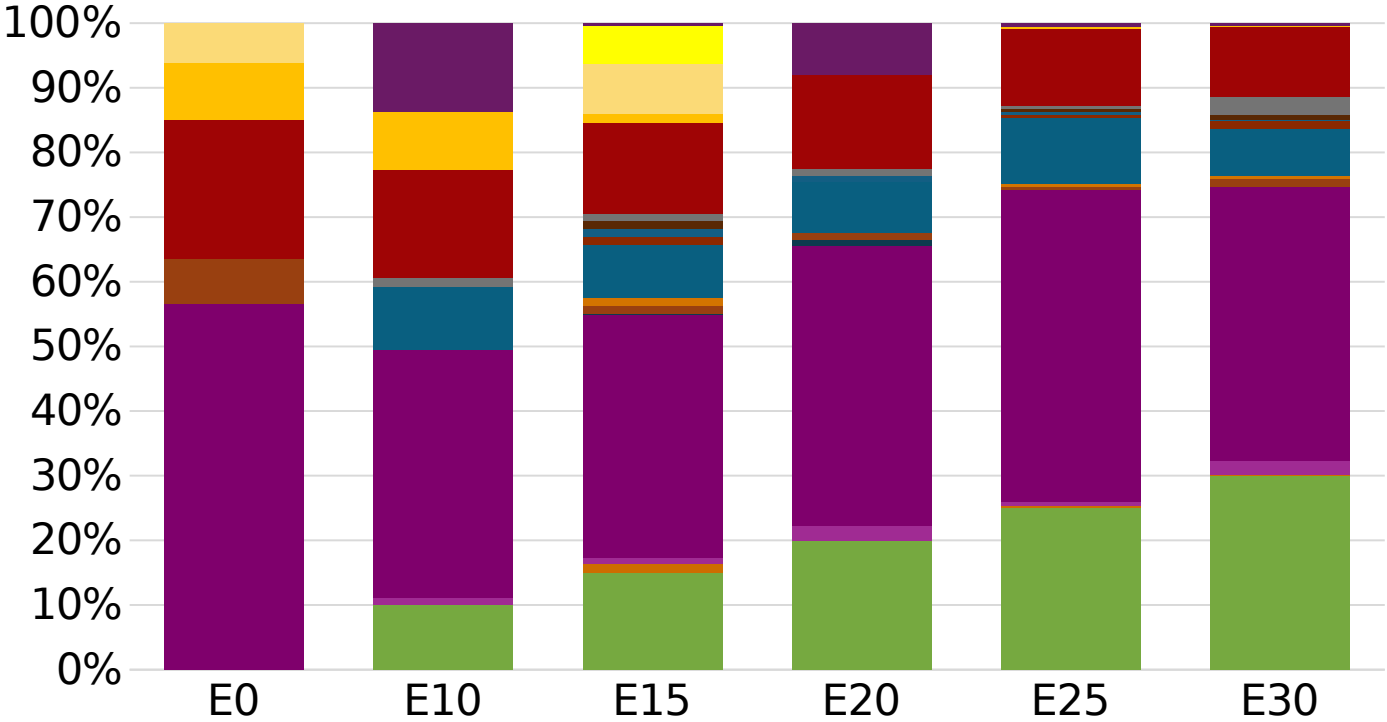
Poland - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.101	\$ 1.984	\$ 1.915	\$ 1.843	\$ 1.778	\$ 1.728

Average CIF 2025
Prices

Poland – Gasoline 95 – Increased Octane

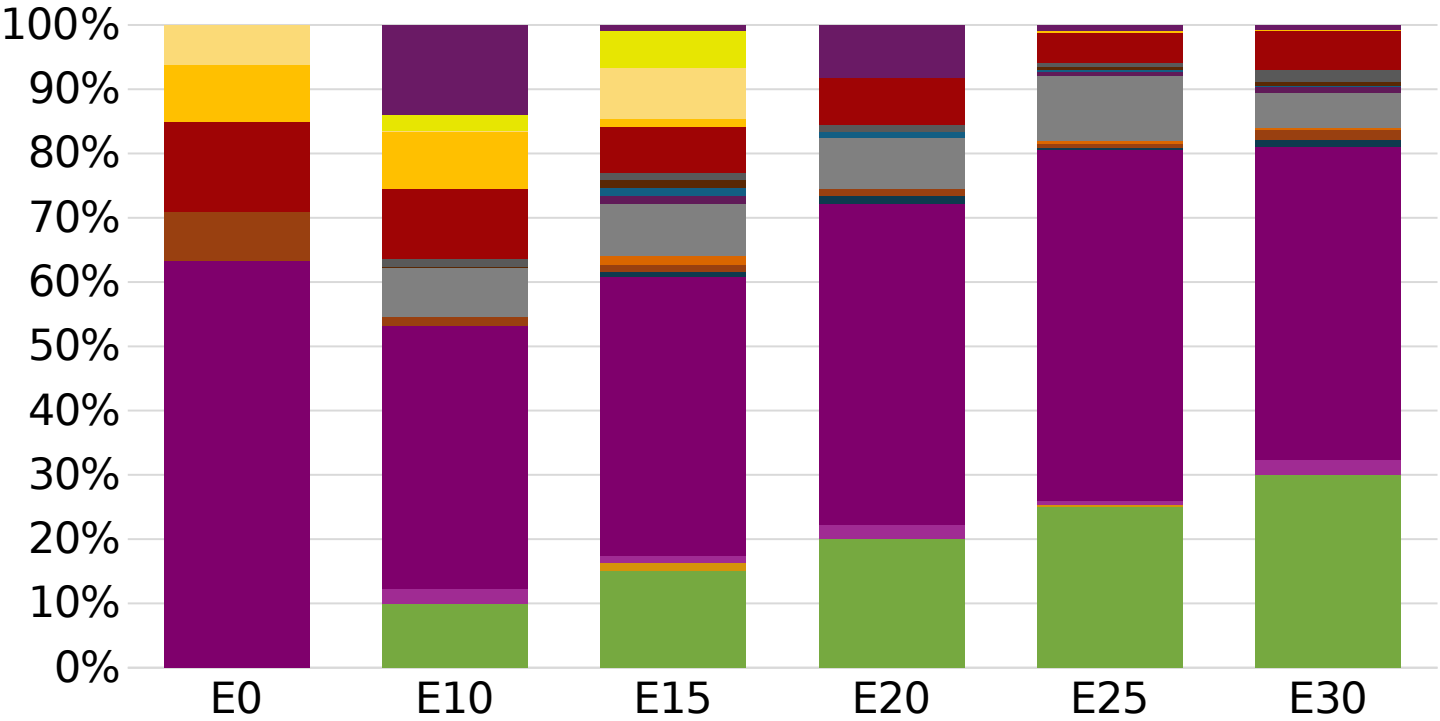
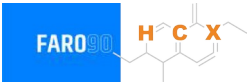


Average CIF 2025
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 1.969	\$ 1.969	\$ 1.969	\$ 1.969	\$ 1.969	\$ 1.969

Source:
Faro90

Poland – Gasoline 98 – Increased Octane

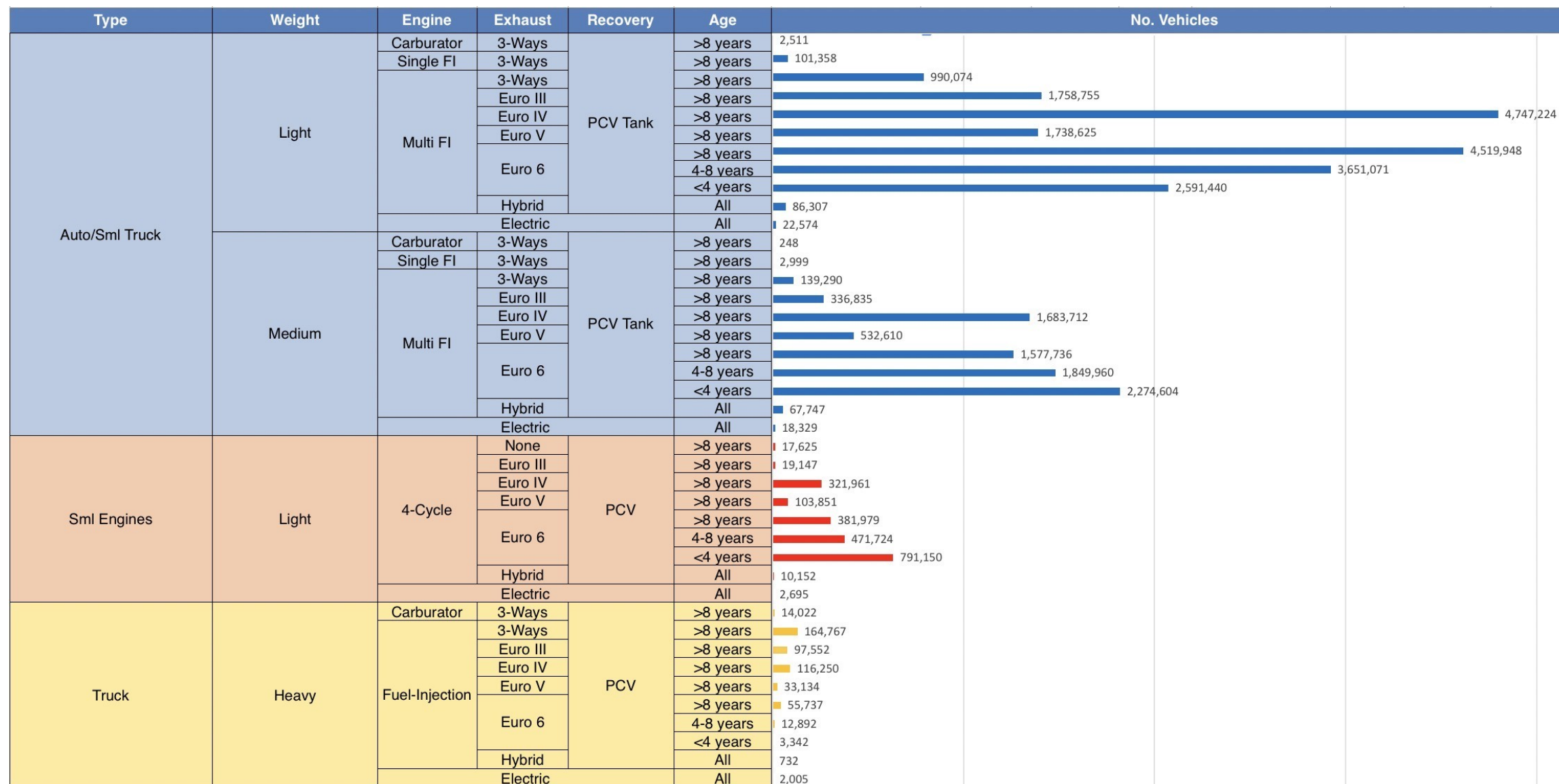


Average CIF 2025
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.101	\$ 2.101	\$ 2.101	\$ 2.101	\$ 2.101	\$ 2.101

Source:
Faro90

Gasoline Vehicle Fleet - Poland

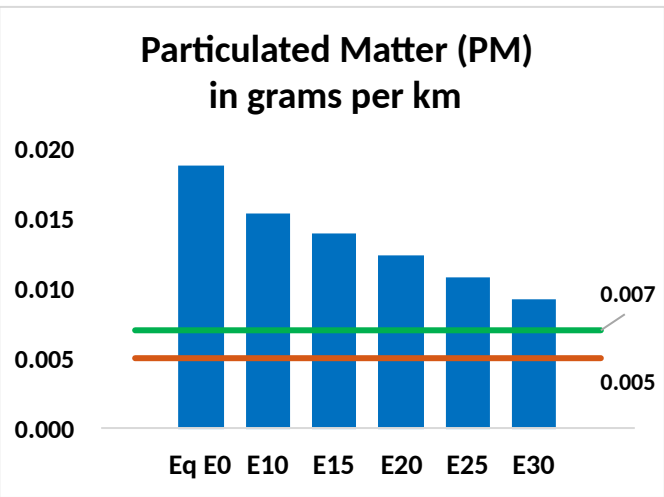
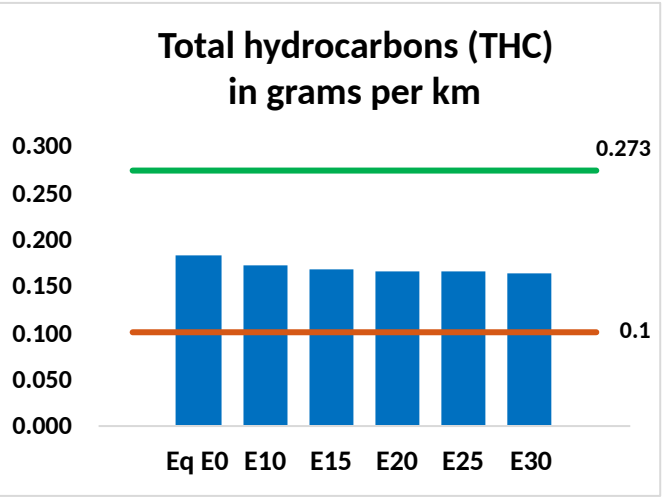
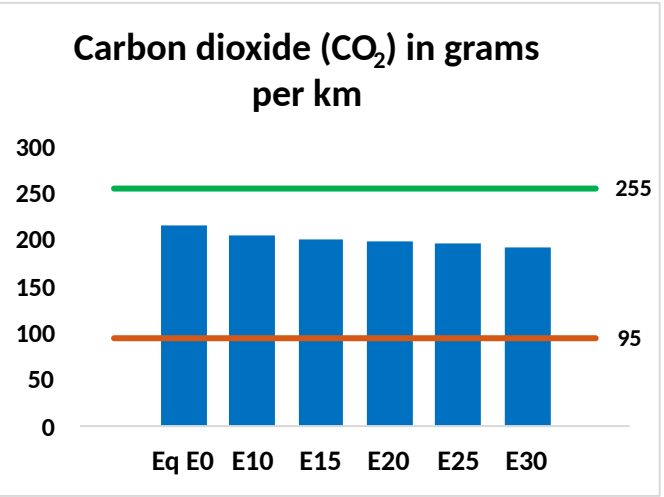
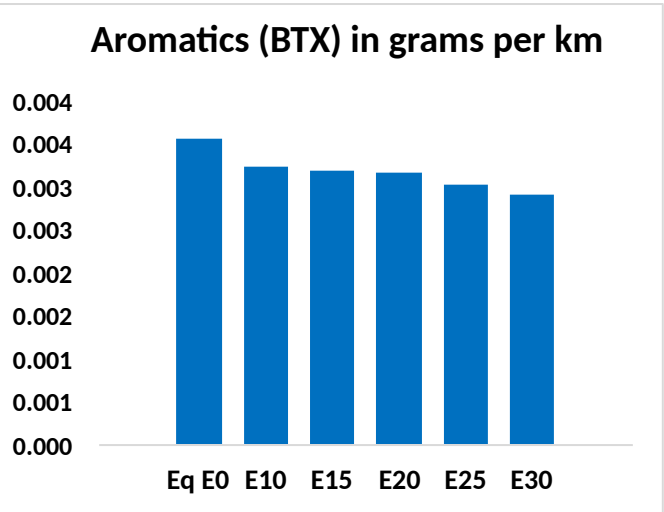
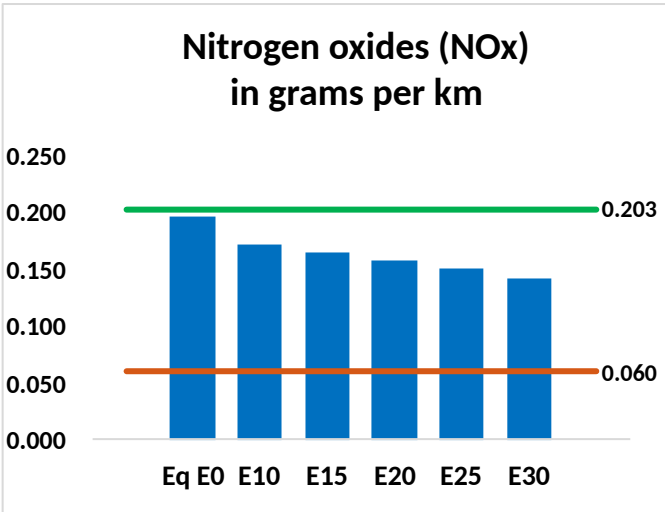
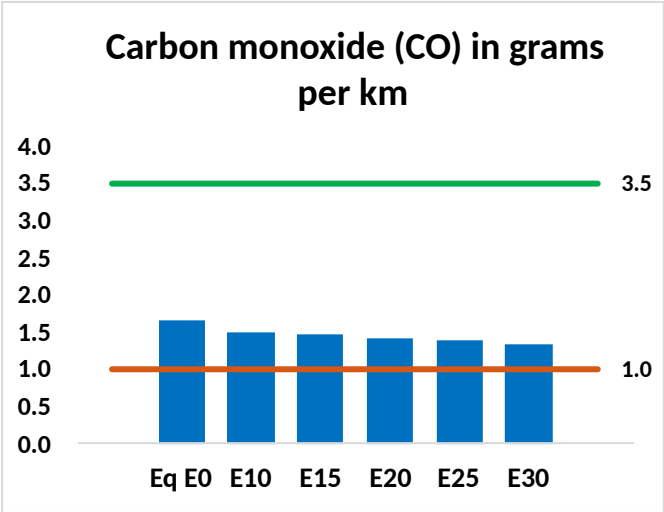
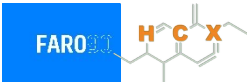


Vehicle Fleet: 31,314,673
Gasoline Vehicle Fleet: 11,384,568

Source: Eurostat, ACEA

Poland – Vehicular Emissions

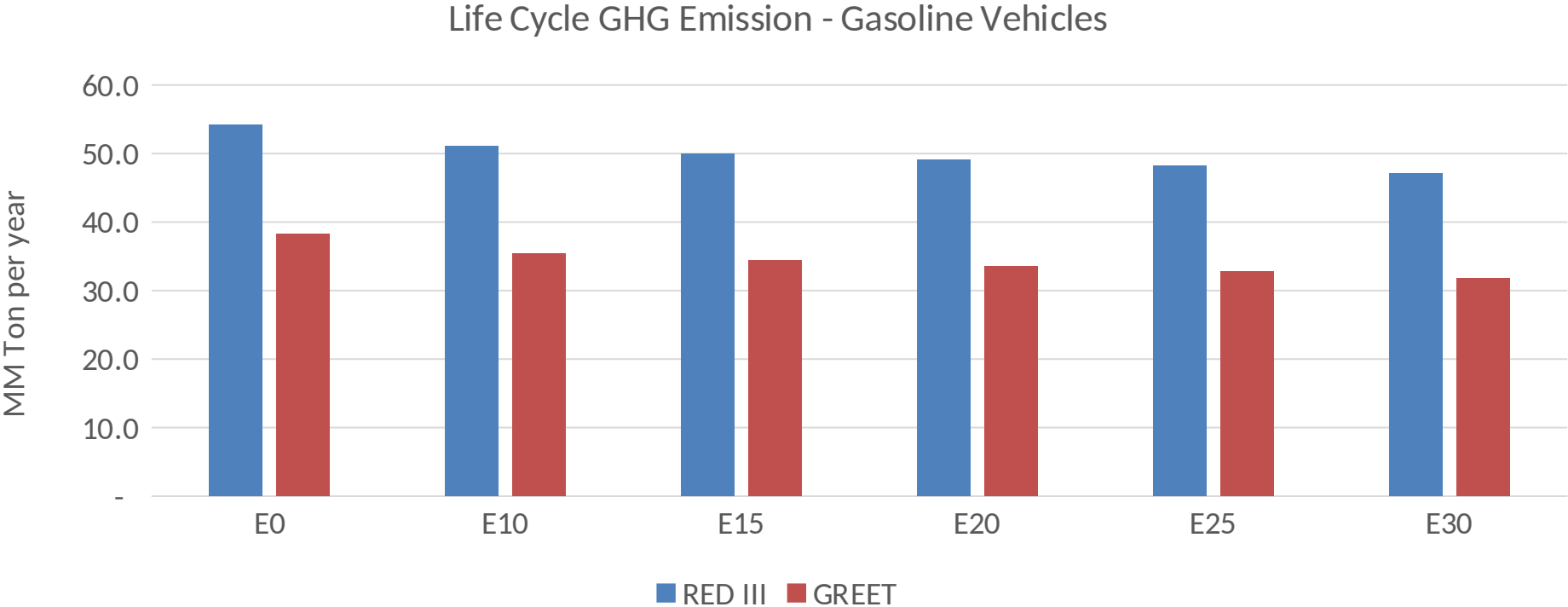
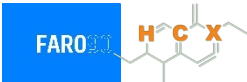
TIER III BIN 125 United States
Euro 6 Standard



Poland – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	571,190	545,994	520,797	503,802	488,379	477,027	461,847	-9%	-14%
CO2	74,492,027	72,576,518	70,767,426	69,344,628	68,640,572	68,234,765	66,977,002	-5%	-8%
VOC	63,270	61,351	59,432	58,434	57,694	57,205	56,271	-6%	-9%
NOx	68,371	63,129	59,506	57,647	54,666	52,323	49,568	-13%	-20%
SOx	854	789	725	670	618	561	501	-15%	-28%
NH3	24,969	24,722	24,475	24,591	24,868	25,062	25,224	-2%	0%
N2O	1,159	1,152	1,148	1,153	1,177	1,190	1,204	-1%	2%
CH4	15,302	15,302	15,302	15,531	15,868	16,158	16,434	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	646	635	624	623	625	627	627	-3%	-3%
Acetaldehyde	1,296	1,487	2,182	3,324	4,528	5,174	6,114	68%	249%
Formaldehyde	3,913	3,804	4,435	5,207	5,455	6,035	6,570	13%	39%
Benzene	1,234	1,186	1,124	1,106	1,097	1,049	1,015	-9%	-11%
PM2.5	4,356	3,960	3,564	3,215	2,874	2,509	2,126	-18%	-34%
PM10	2,198	1,998	1,798	1,622	1,450	1,266	1,073	-18%	-34%

Poland – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	337.8
Target @ 2035 (MMT/year)	190.2
Delta	147.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	7.5%	10.1%	12.4%	14.3%	17.1%
RED II/III	0.0%	5.7%	7.7%	9.5%	11.0%	13.1%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	1.9%	2.6%	3.2%	3.7%	4.4%
RED II/III	0.0%	2.1%	2.8%	3.5%	4.0%	4.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

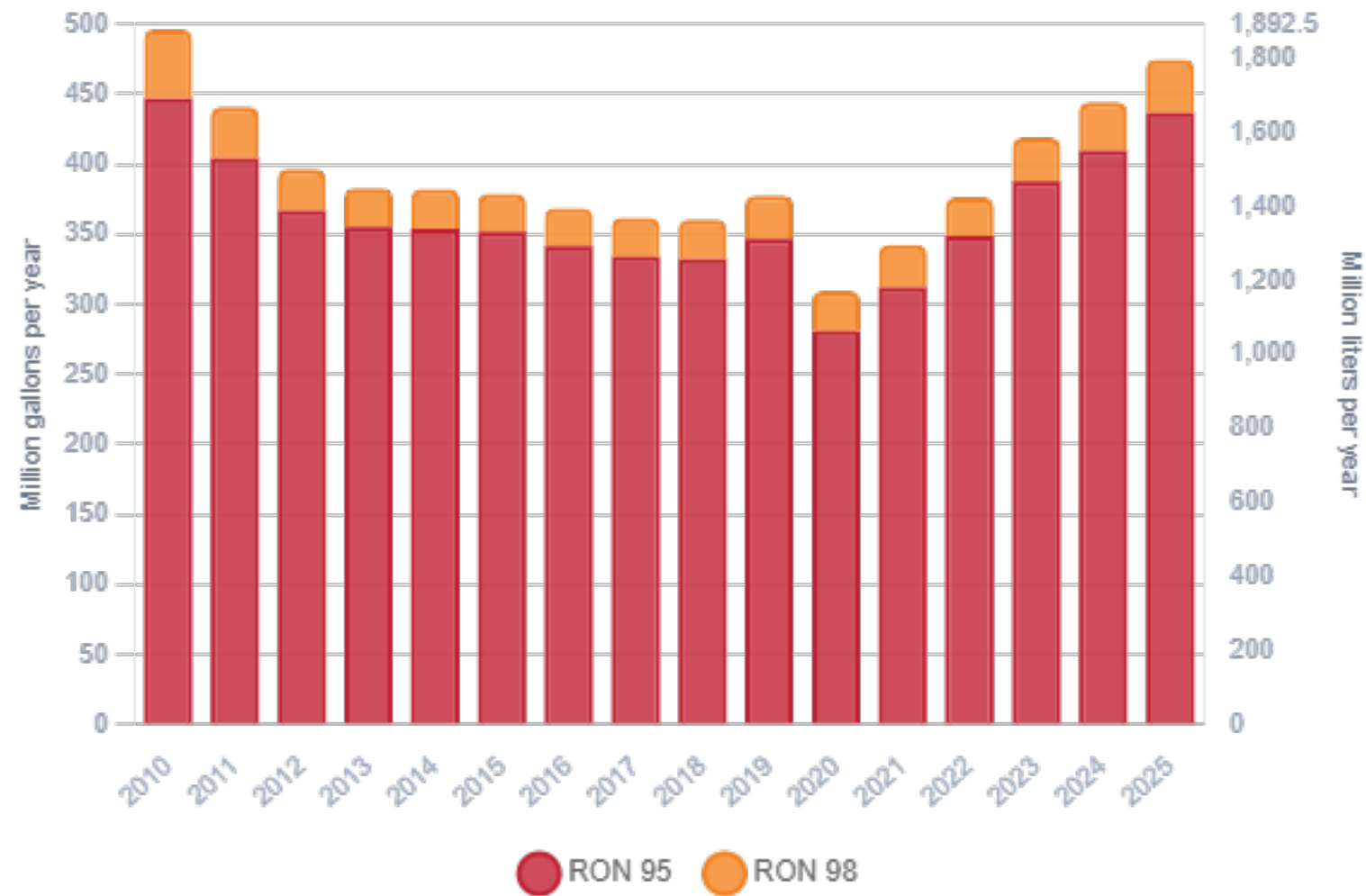
Ethanol Gasoline Blending in Portugal



In 2025, gasoline consumption in Portugal was 1,797 million liters (475 million gallons) and growth rate was - 0.3%. Gasoline production and net imports were 2,464 and -667 million liters (651 and -176 million gallons) correspondingly. Portugal complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

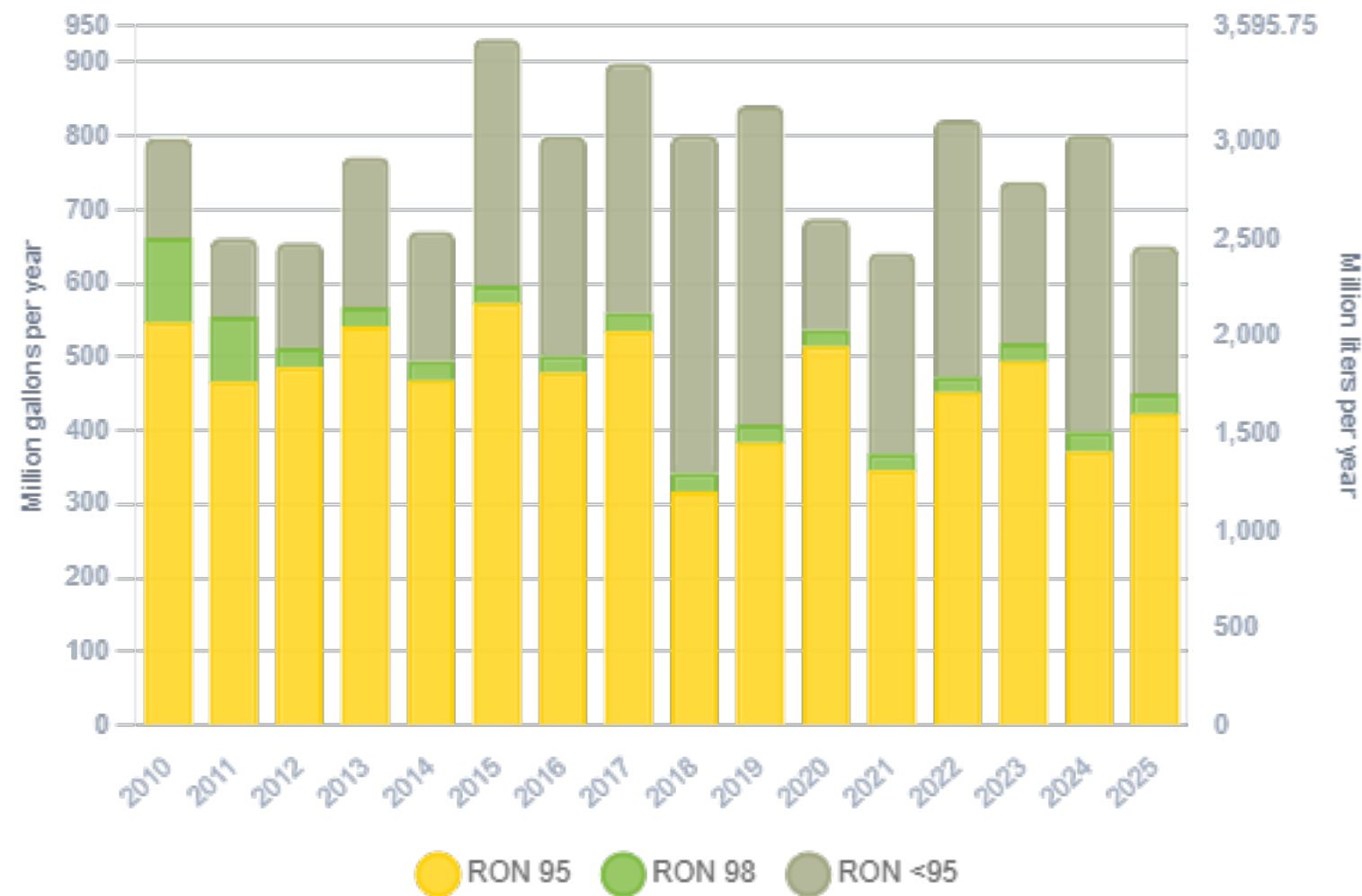
In 2025, ethanol consumption in Portugal was 263 million liters (69 million gallons) representing 15% of gasoline demand. Portugal does not produce ethanol, supply is done through imports. Local production plants are being built.

Gasoline Demand in Portugal



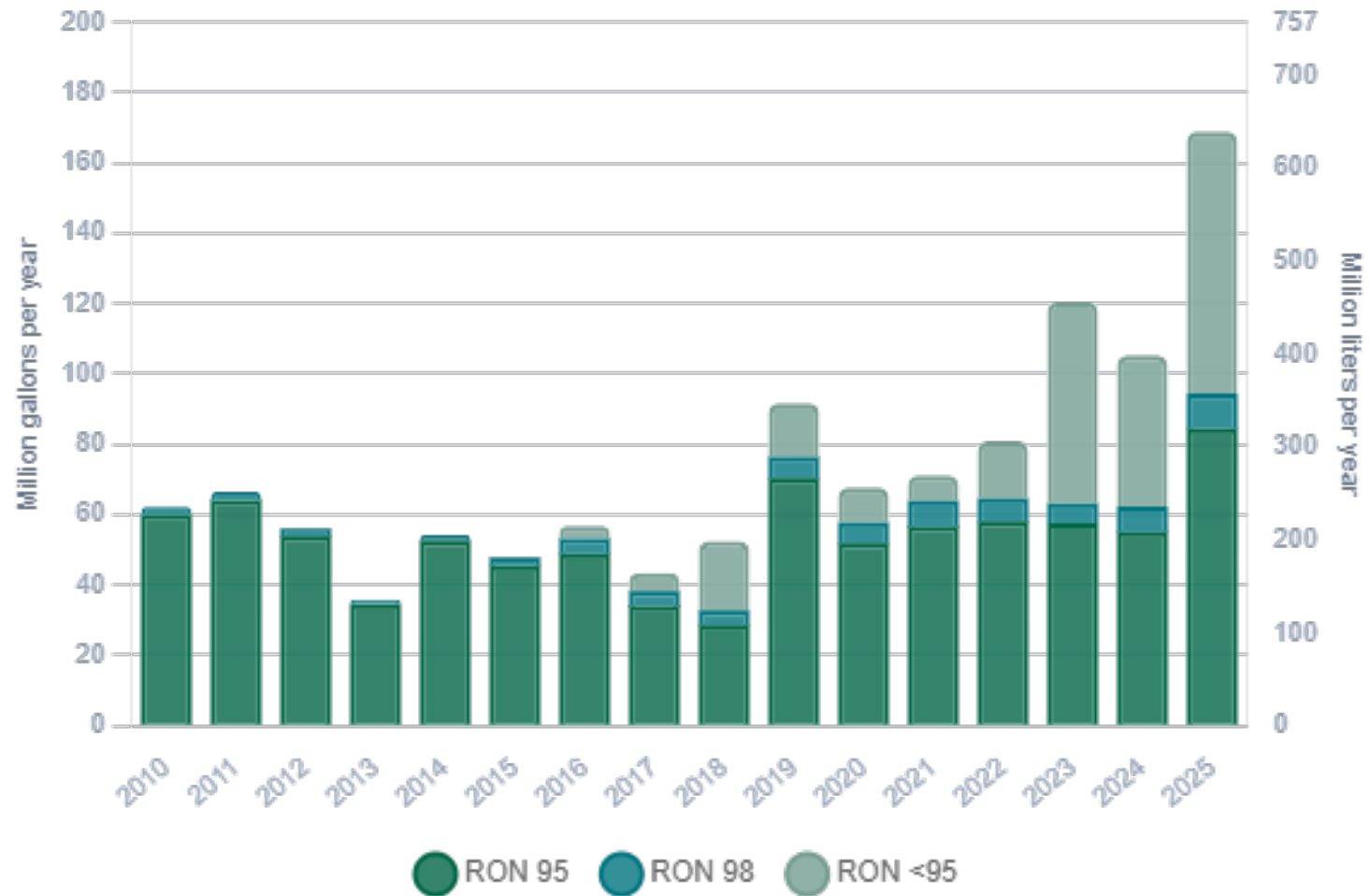
Source: Eurostat

Gasoline Production in Portugal



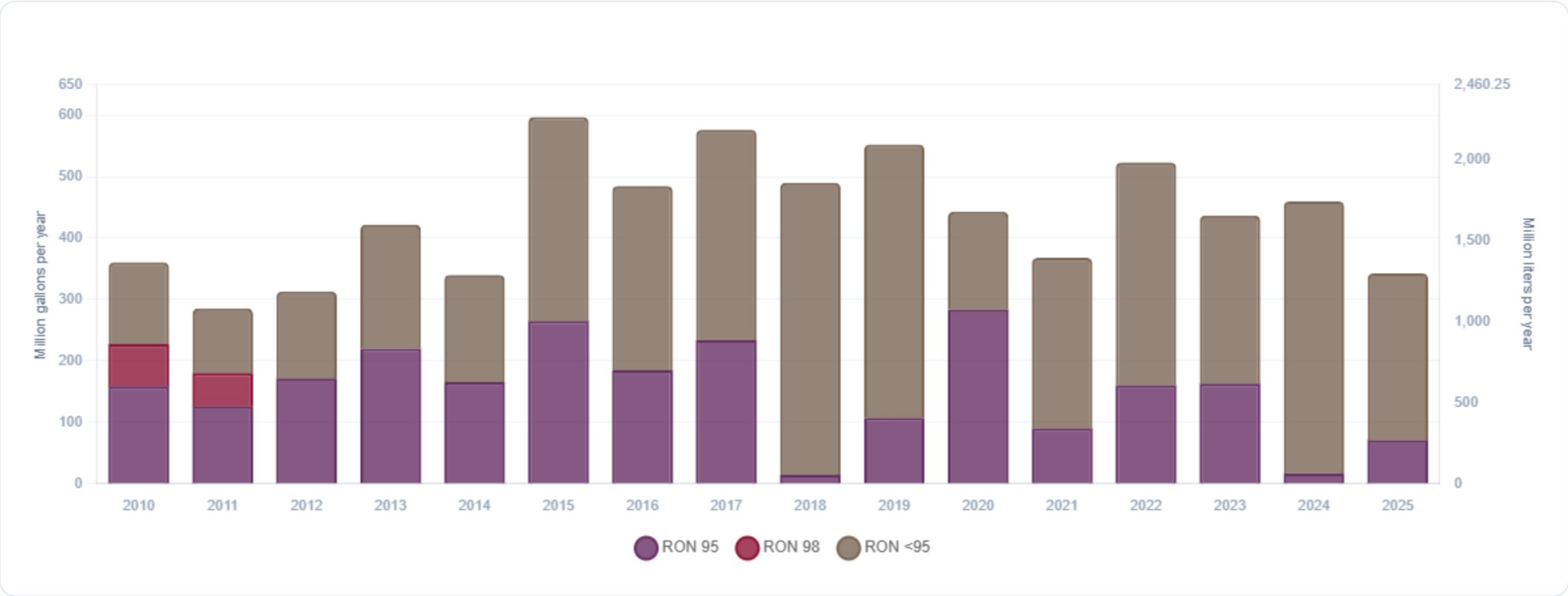
Source: Eurostat

Gasoline Imports in Portugal



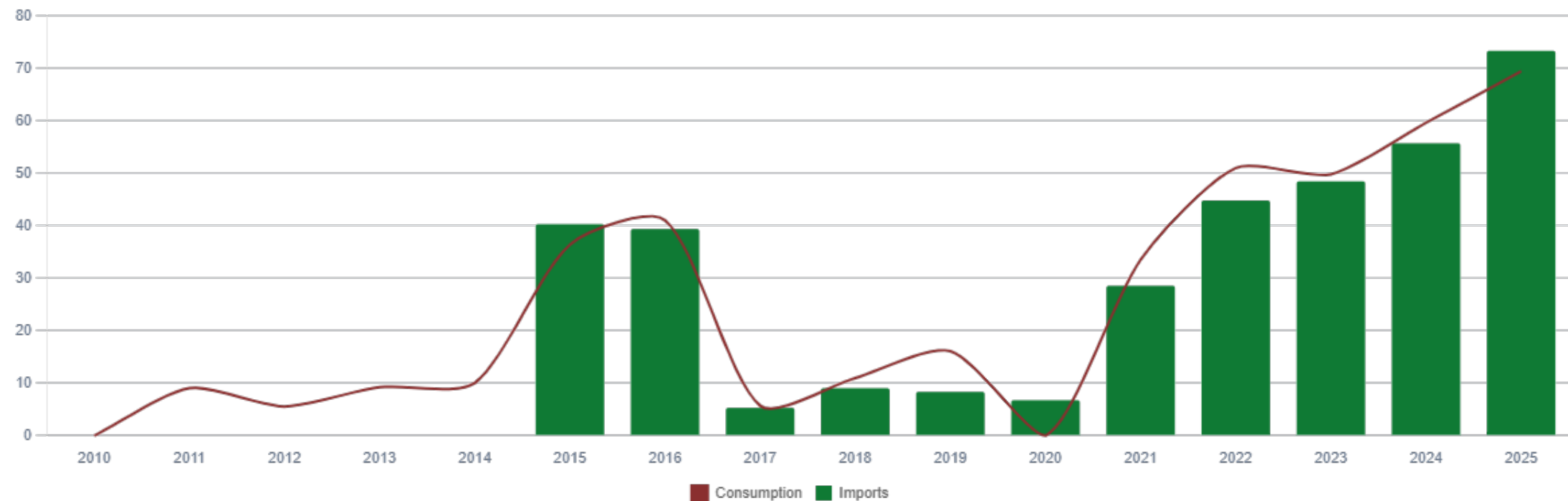
Source: Eurostat

Gasoline Exports in Portugal



Source: Eurostat

Ethanol Balance in Portugal



Ethanol is supplied by imports

Source: Eurostat

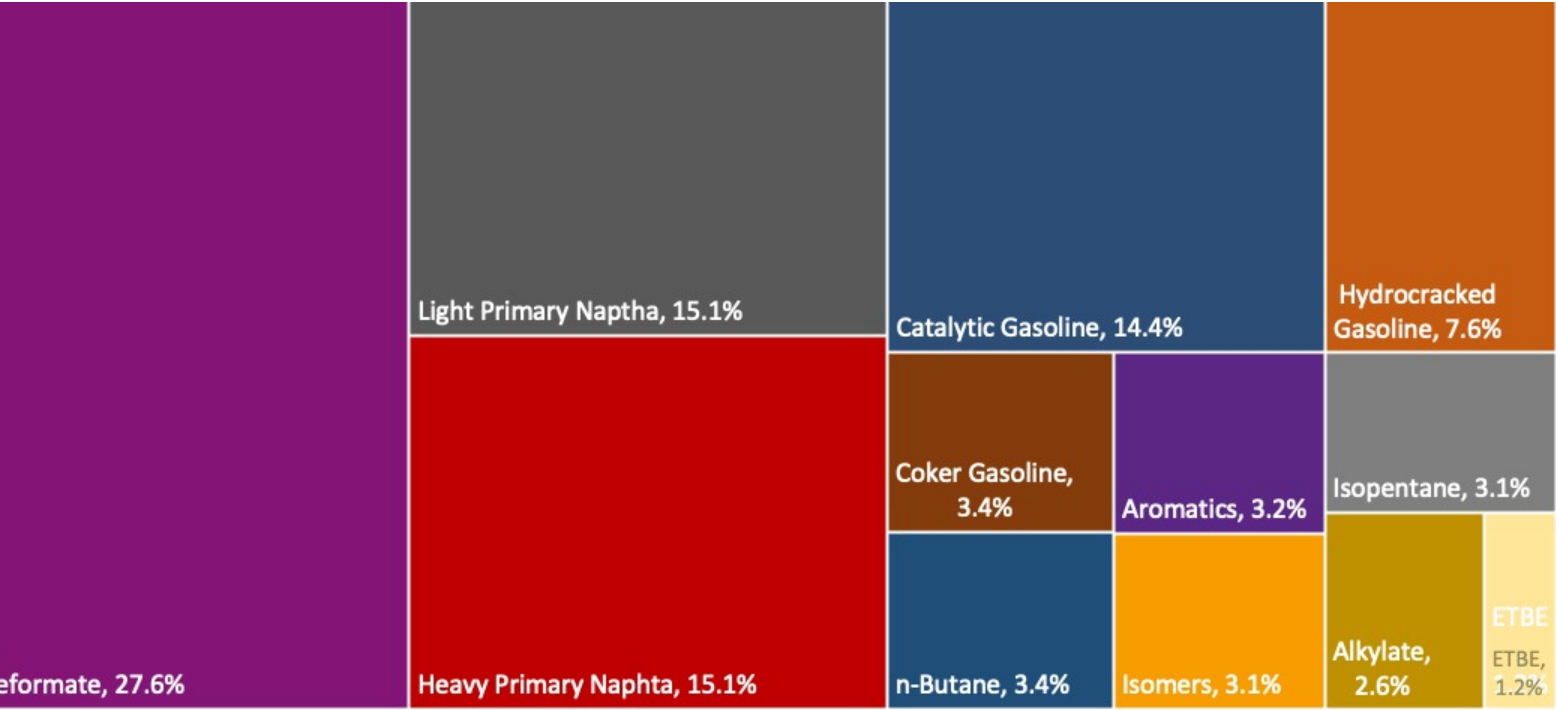
Gasoline Quality in Portugal

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	13.0							
Advanced Biofuels (%cal)	2.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

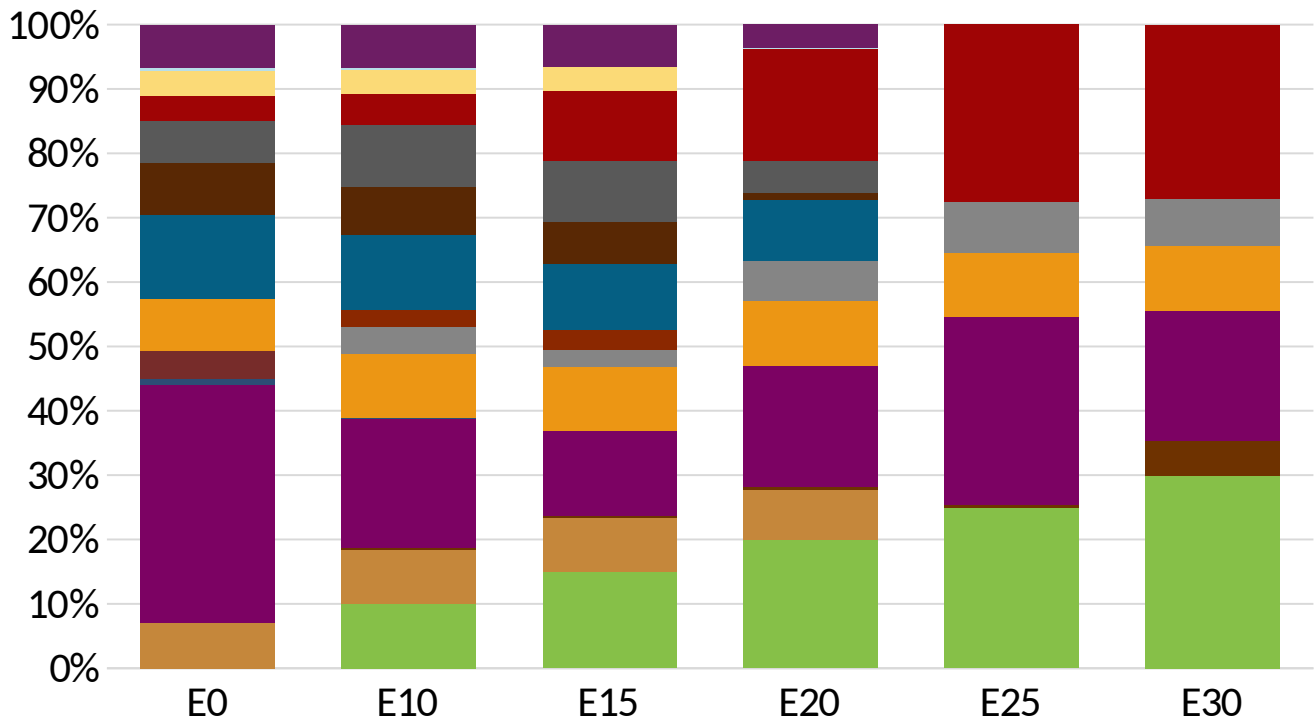
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Portugal Available Blending Components



Portugal – Gasoline 95 – Constant Octane

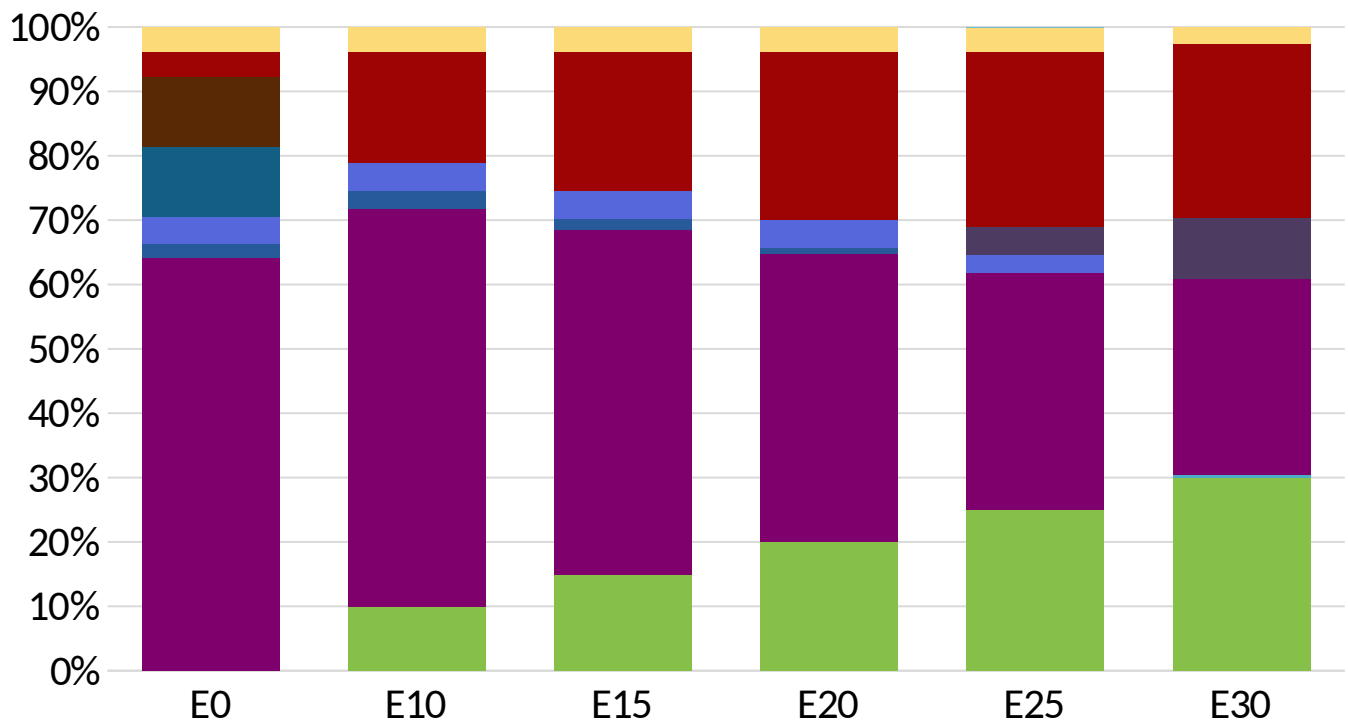


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.3	95.8
Price (US\$/gal)	\$ 2.087	\$ 1.975	\$ 1.896	\$ 1.806	\$ 1.690	\$ 1.627

Source:
Faro90

Portugal - Gasoline 98 - Constant Octane

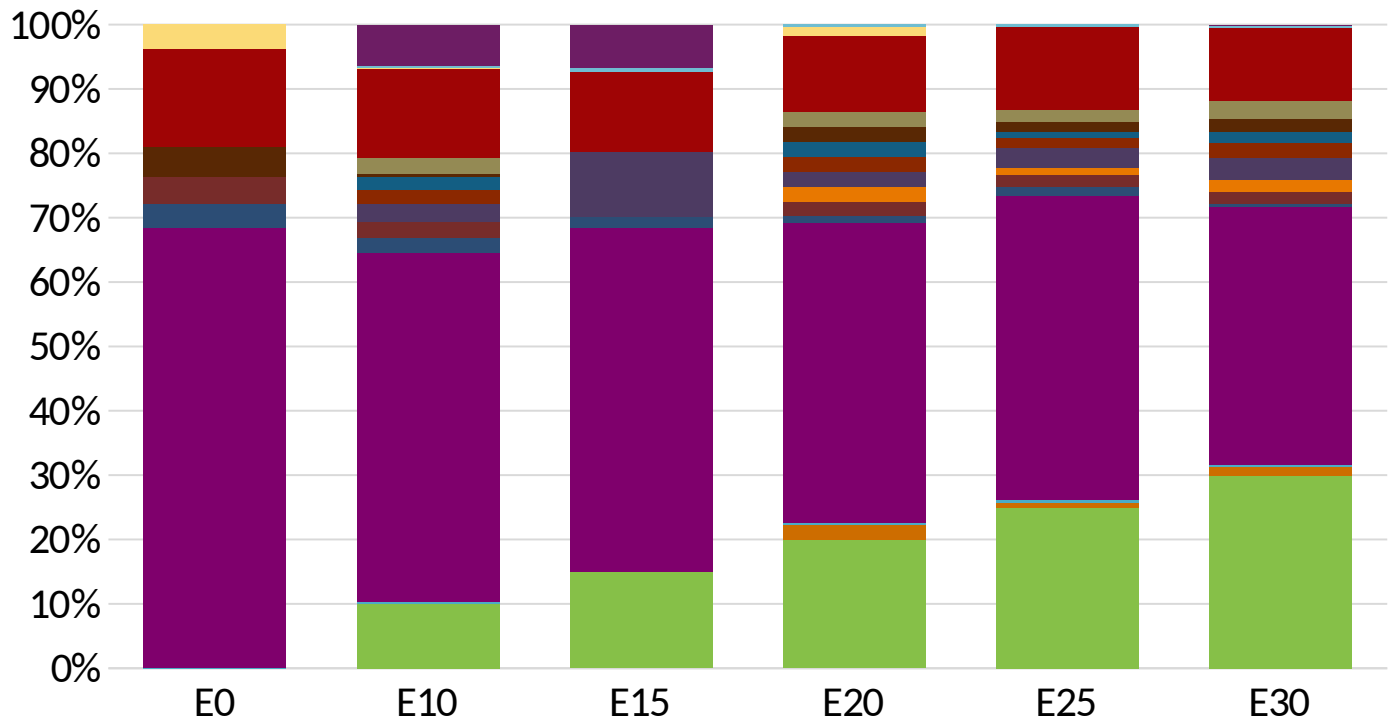


Average CIF 2025
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.156	\$ 2.032	\$ 1.963	\$ 1.888	\$ 1.810	\$ 1.731

Source:
Faro90

Portugal – Gasoline 95 – Increased Octane

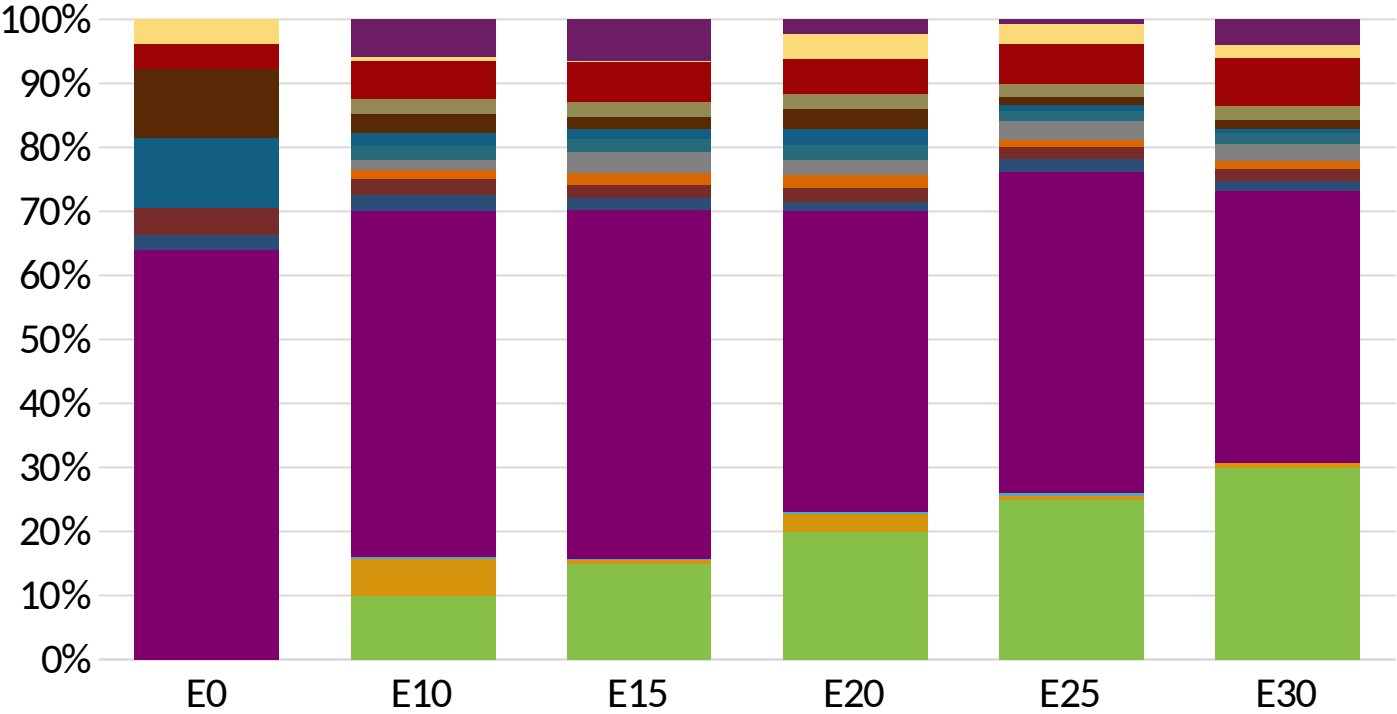


Average CIF 2025
Prices

Octane (RON)	95.0	96.8	98.2	99.9	101.4	102.6
Price (US\$/gal)	\$ 2.019	\$ 2.019	\$ 2.019	\$ 2.019	\$ 2.019	\$ 2.019

Source:
Faro90

Portugal – Gasoline 98 – Increased Octane

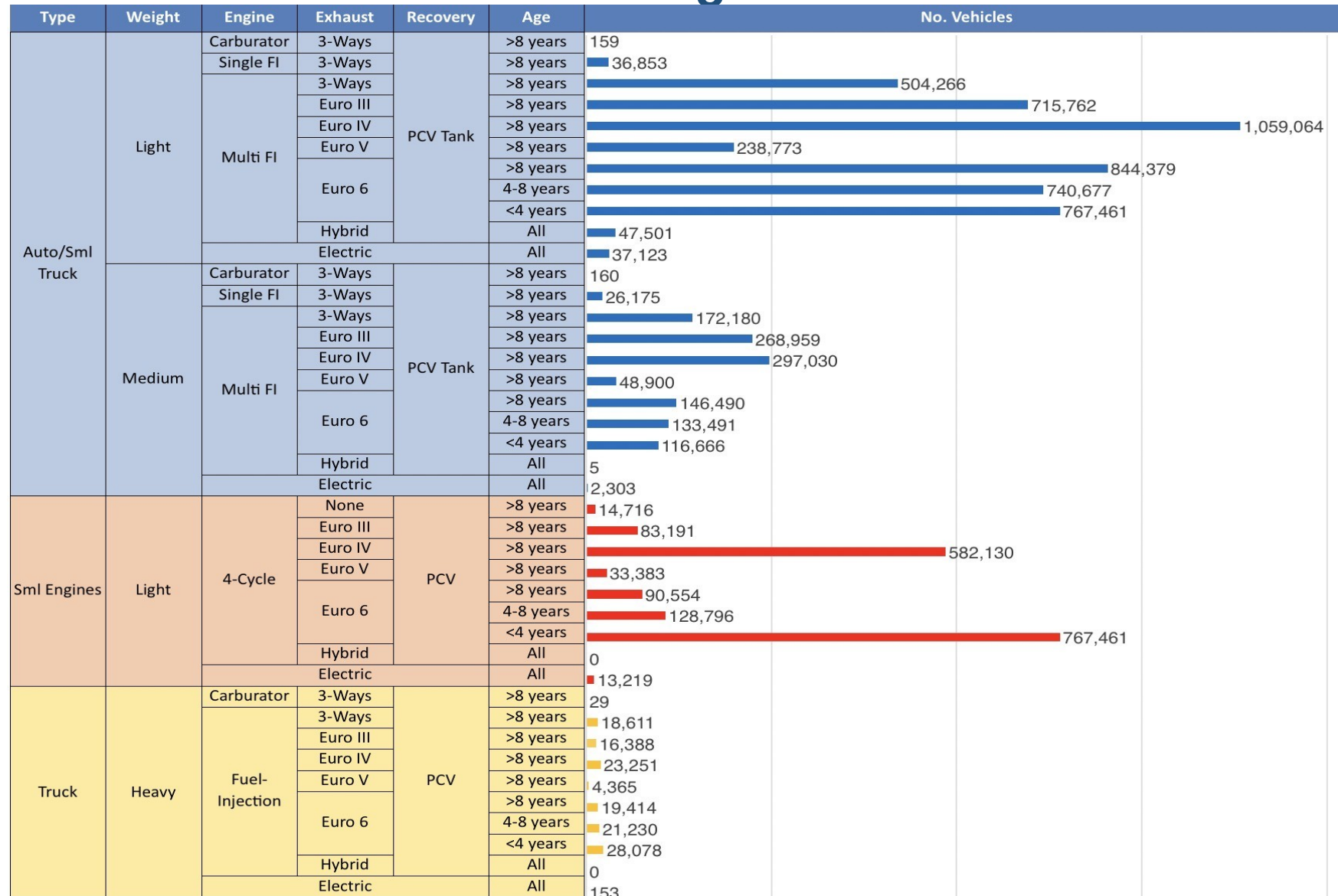


Average CIF 2025
Prices

Octane (RON)	98.0	99.3	100.9	102.5	104.0	105.0
Price (US\$/gal)	\$ 2.156	\$ 2.156	\$ 2.156	\$ 2.156	\$ 2.156	\$ 2.156

Source:
Faro90

Gasoline Vehicle Fleet - Portugal

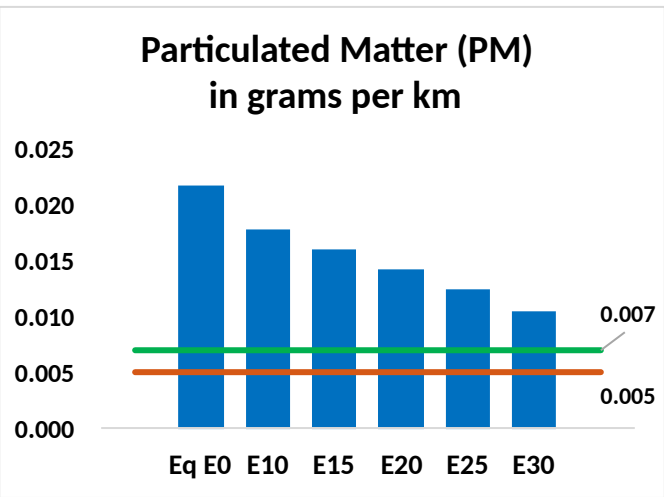
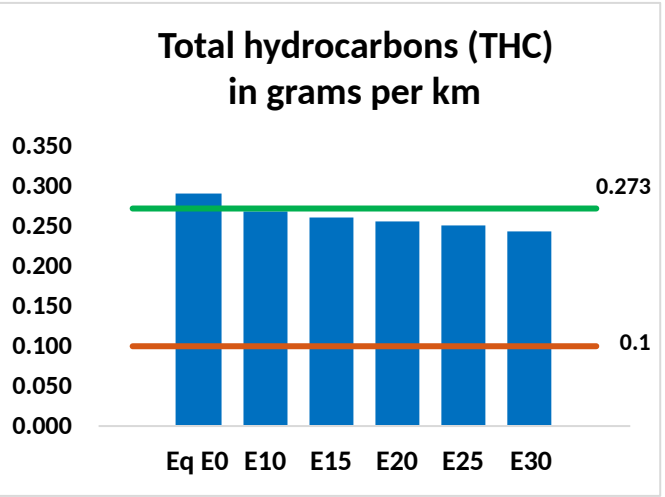
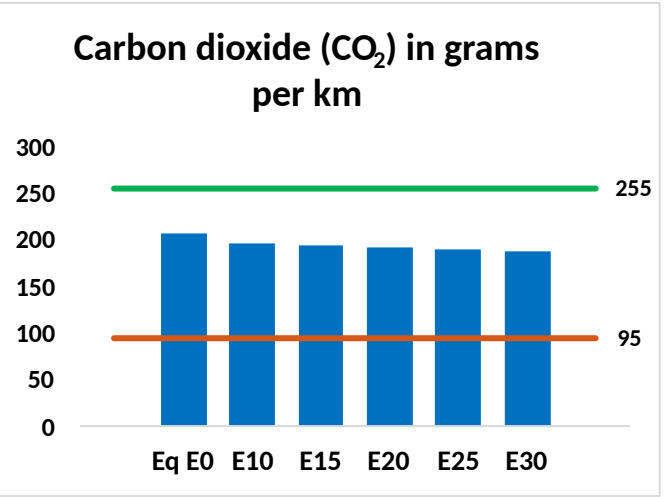
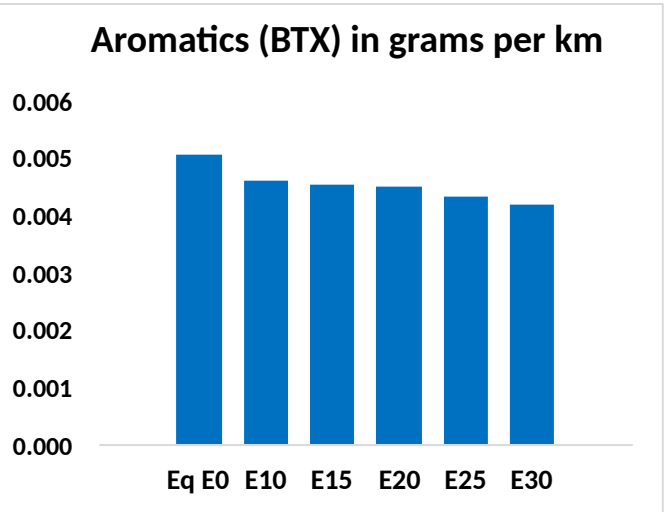
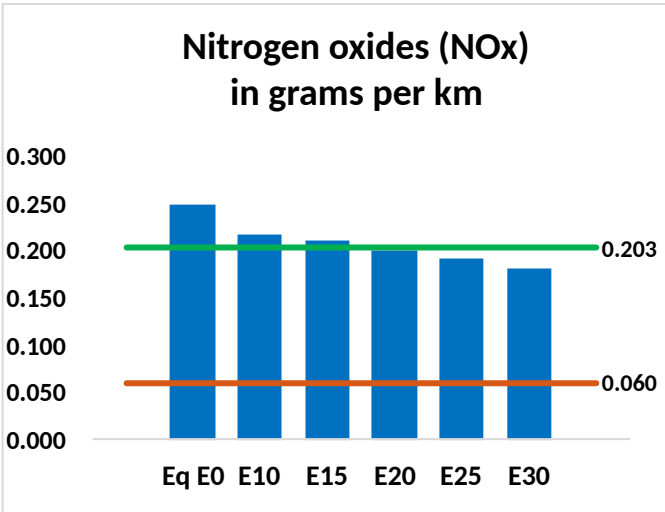
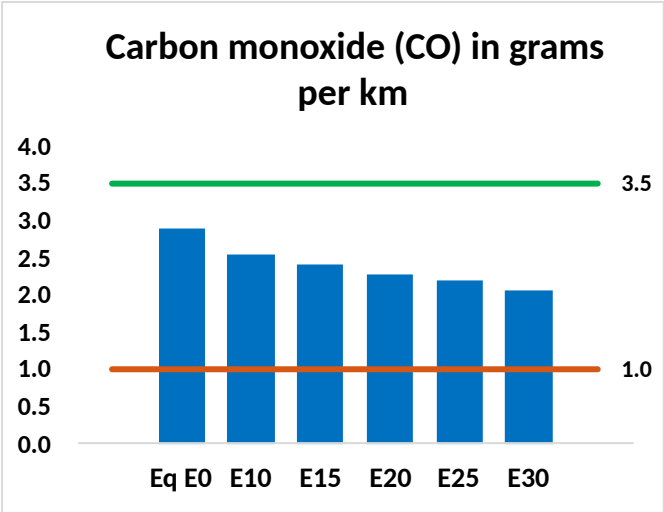
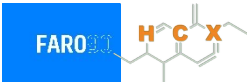


Vehicle Fleet: 8,049,349
Gasoline Vehicle Fleet: 2,605,060

Source: Eurostat, ACEA

Portugal – Vehicular Emissions

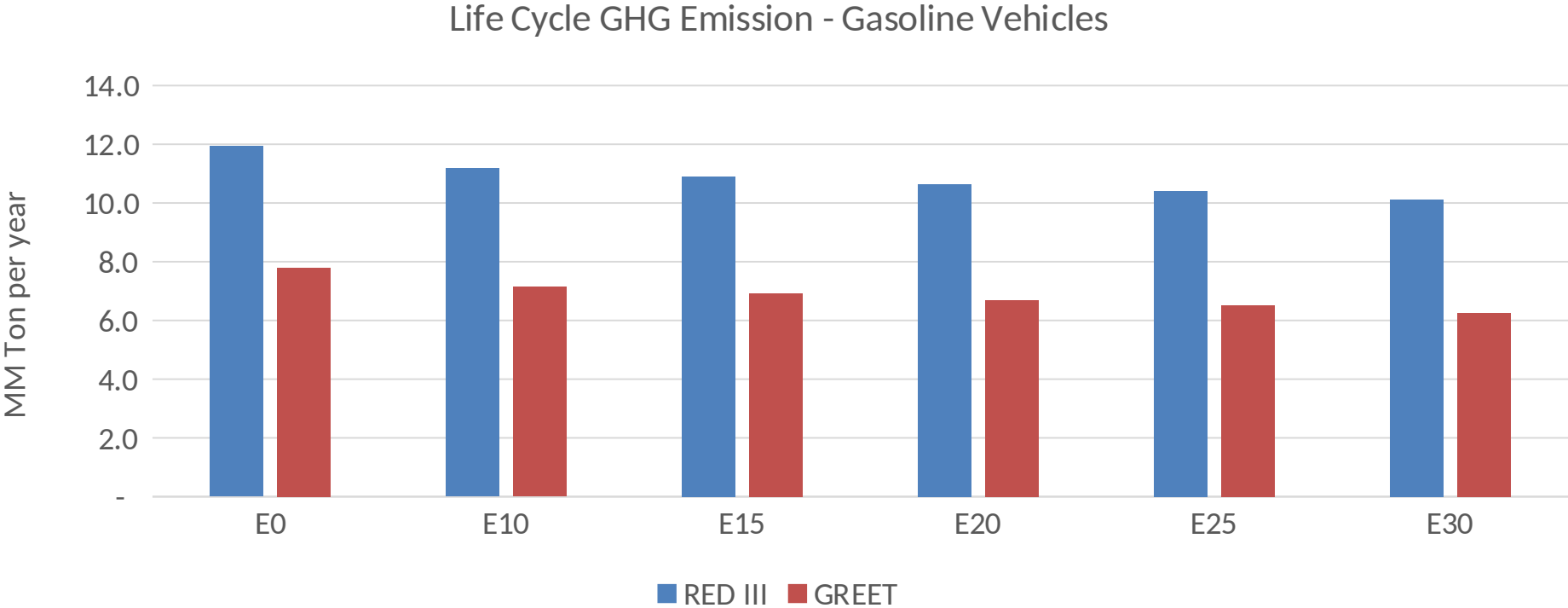
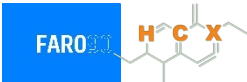
TIER III BIN 125 United States
Euro 6 Standard



Portugal – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	184,660	173,546	162,431	153,925	145,641	139,355	131,969	-12%	-21%
CO2	13,276,939	12,935,532	12,613,092	12,359,502	12,234,016	12,162,548	11,938,357	-5%	-8%
VOC	18,529	17,824	17,118	16,680	16,304	16,034	15,605	-8%	-12%
NOx	15,941	14,718	13,881	13,455	12,768	12,230	11,596	-13%	-20%
SOx	151	139	128	118	109	99	88	-15%	-28%
NH3	4,531	4,486	4,441	4,462	4,512	4,548	4,577	-2%	0%
N2O	169	168	168	168	172	174	176	-1%	2%
CH4	4,428	4,428	4,428	4,495	4,592	4,676	4,756	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	180	175	170	168	166	165	163	-6%	-8%
Acetaldehyde	381	437	642	977	1,331	1,521	1,798	68%	249%
Formaldehyde	1,209	1,175	1,370	1,608	1,685	1,864	2,029	13%	39%
Benzene	325	313	296	291	289	276	267	-9%	-11%
PM2.5	787	715	644	581	519	453	384	-18%	-34%
PM10	600	545	491	443	396	346	293	-18%	-34%

Portugal – Gasoline Vehicle Fleet Life Cycle GHG Emissions



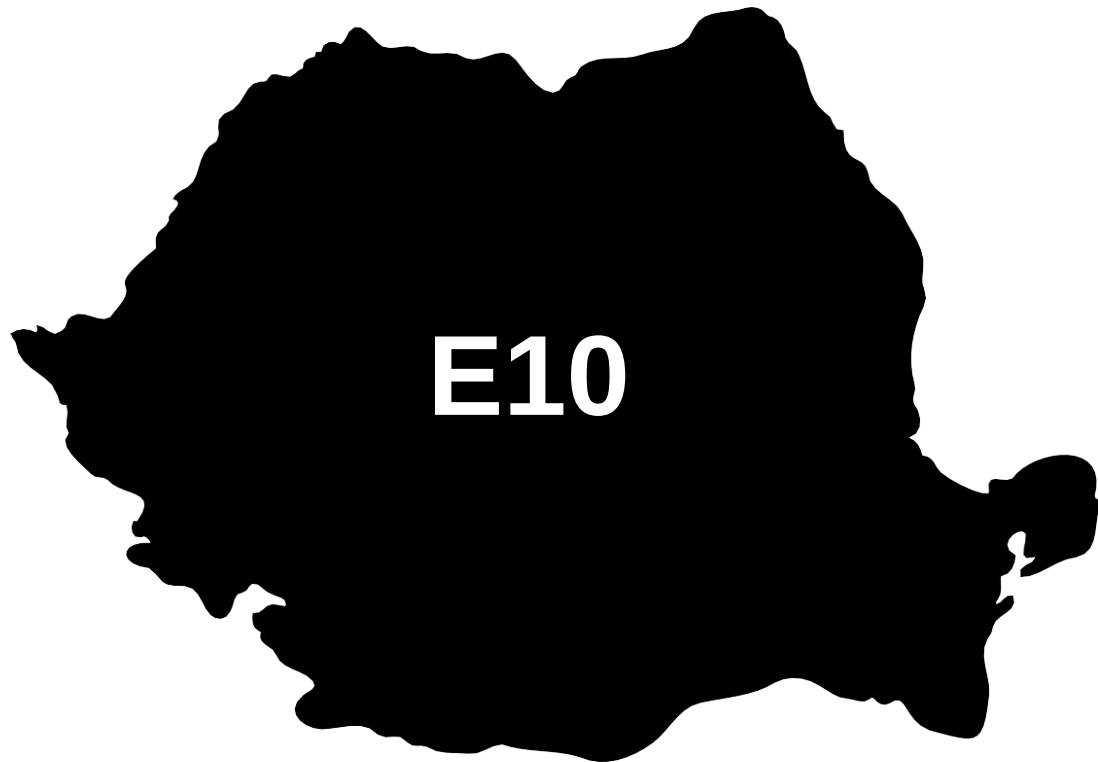
Country Emissions 2025 (MMT/año)	59.0
Target @ 2035 (MMT/year)	33.2
Delta	25.8

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.4%	11.4%	14.1%	16.5%	19.7%
RED II/III	0.0%	6.3%	8.7%	10.9%	12.7%	15.2%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.5%	3.4%	4.3%	5.0%	6.0%
RED II/III	0.0%	2.9%	4.0%	5.0%	5.9%	7.0%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

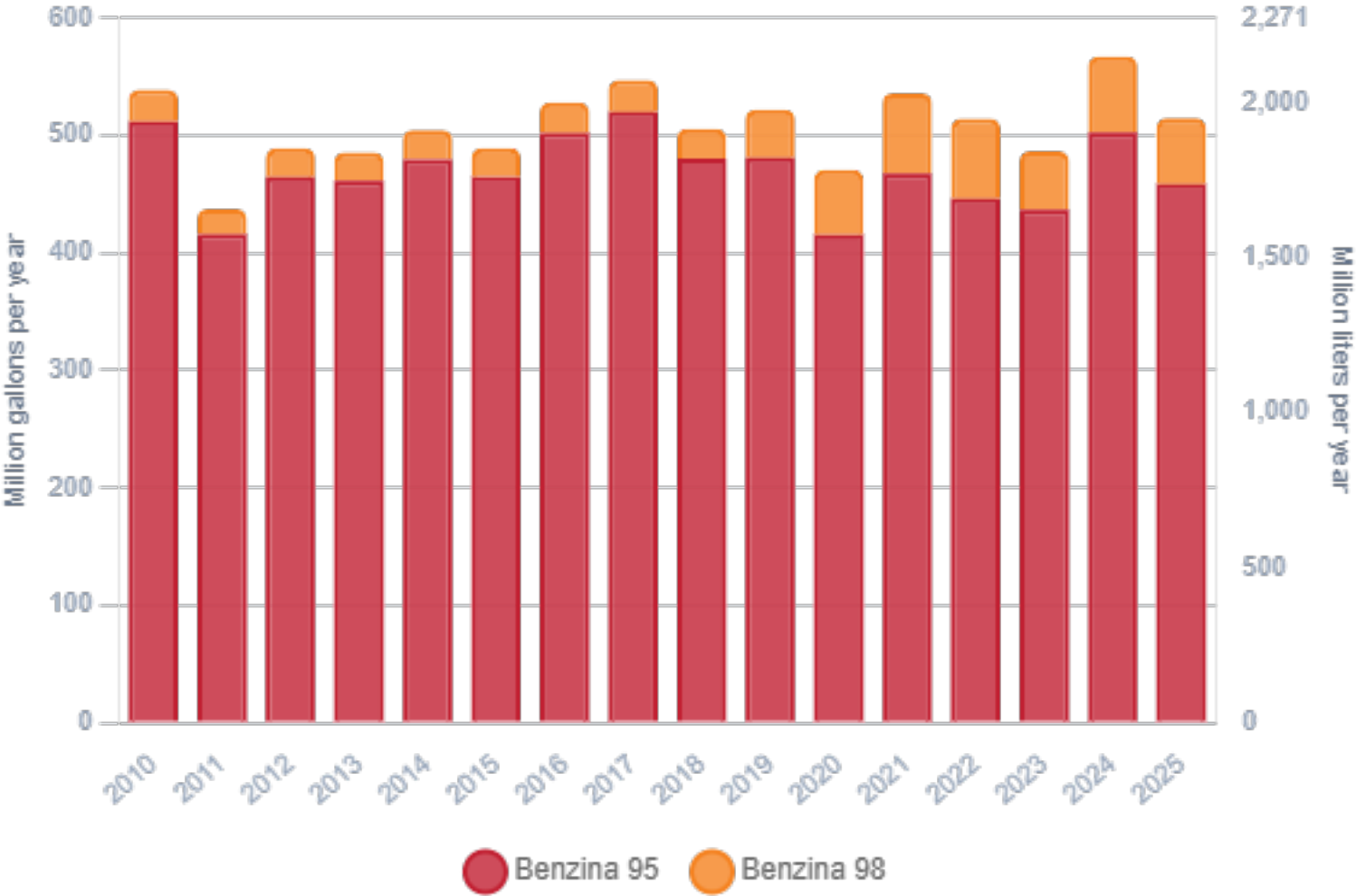
Ethanol Gasoline Blending in Romania



In 2025, gasoline consumption in Romania was 1,948 million liters (515 million gallons) and growth rate was -0.3%. Gasoline production and net imports were 4,115 and -2,167 million liters (1,087 and -573 million gallons) correspondingly. Romania complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

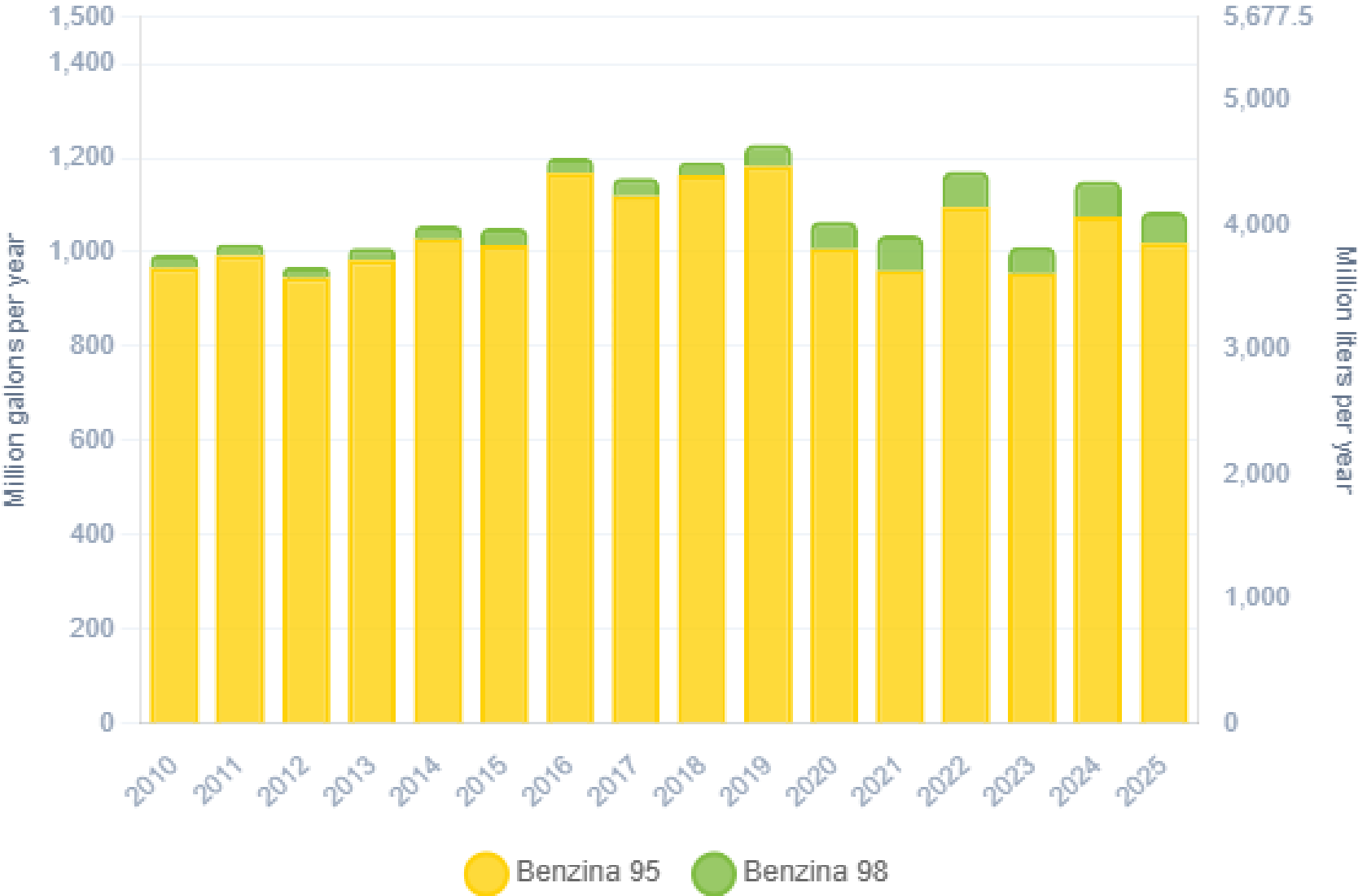
In 2025, ethanol consumption in Romania was 9 million liters (2.4 million gallons) representing 0.5% of gasoline demand. Ethanol Net Imports were 9 million liters (0 and 2.4 million gallons) correspondingly. Ethanol sources are Corn 28% and Cereals 72%.

Gasoline Demand in Romania



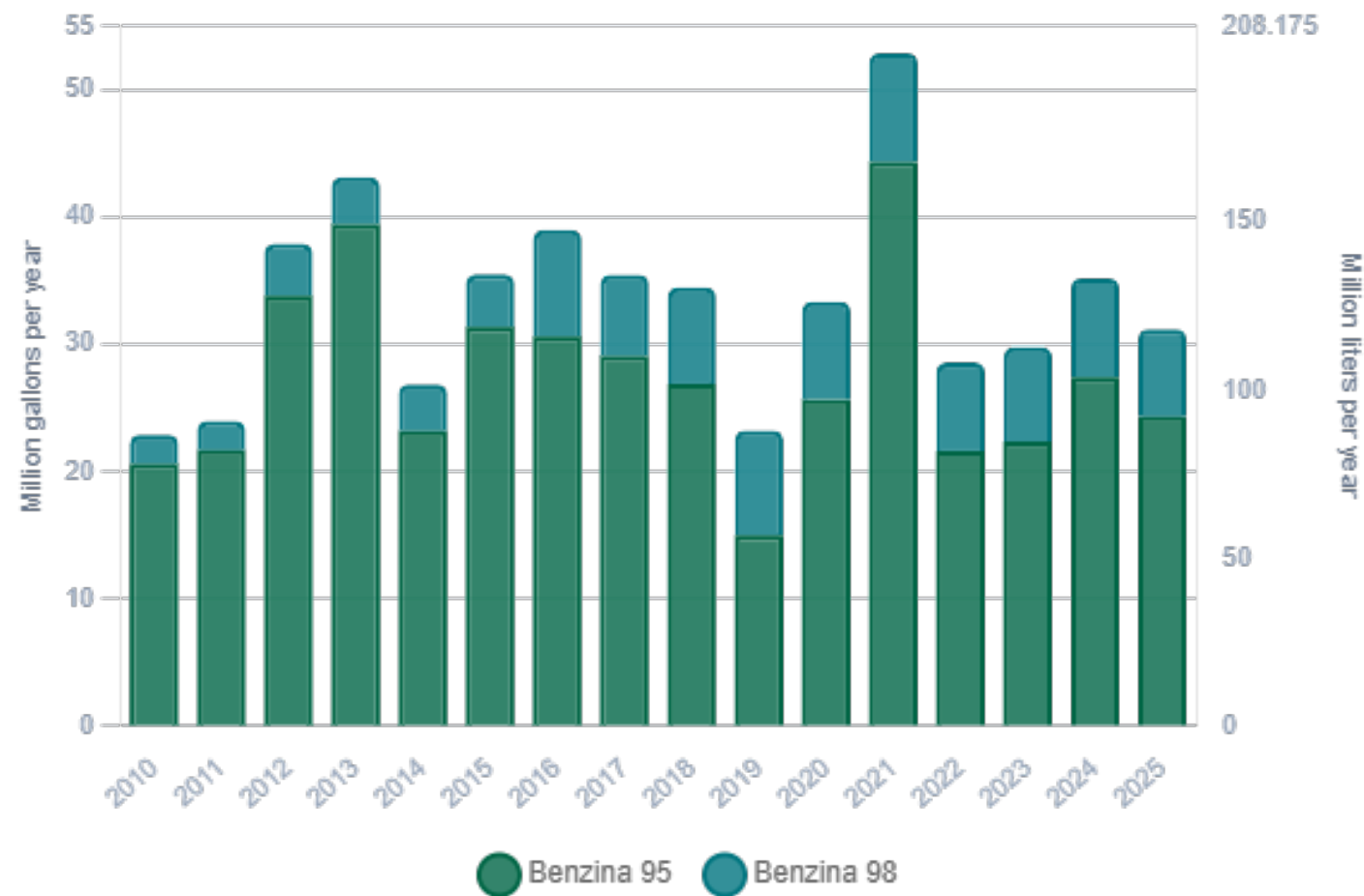
Source: Eurostat

Gasoline Production in Romania



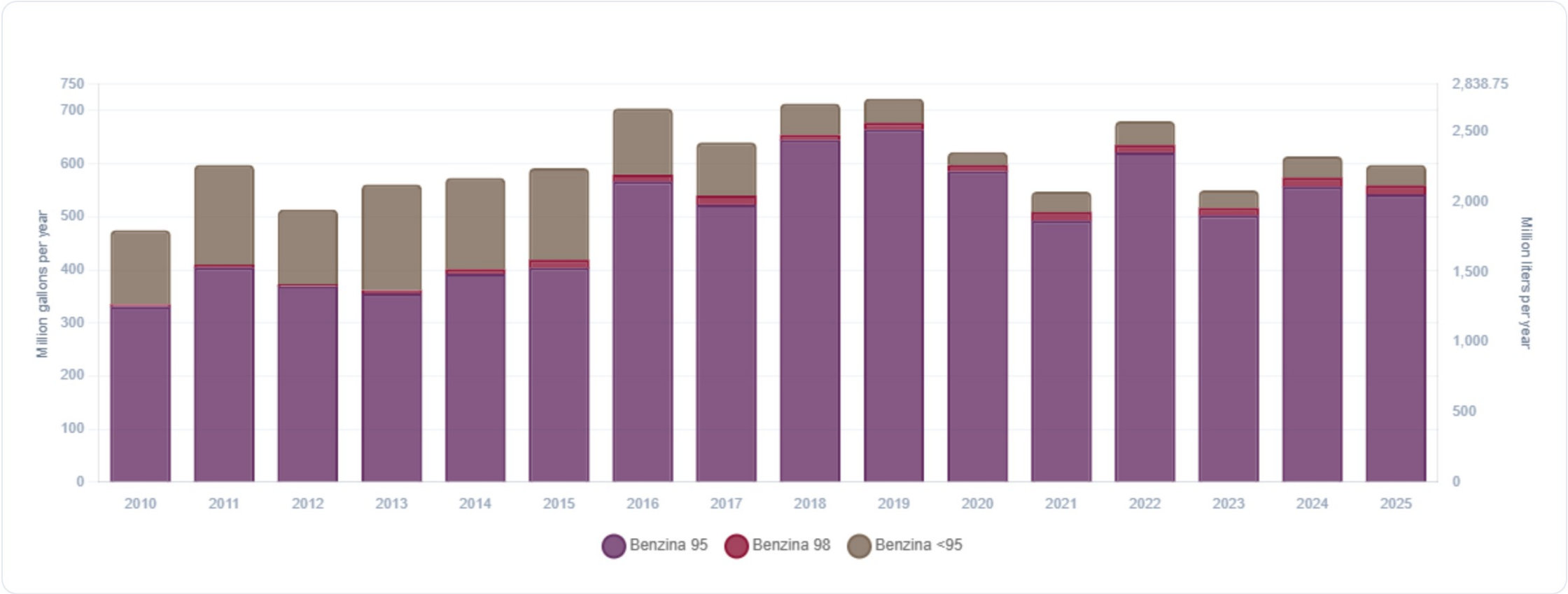
Source: Eurostat

Gasoline Imports in Romania



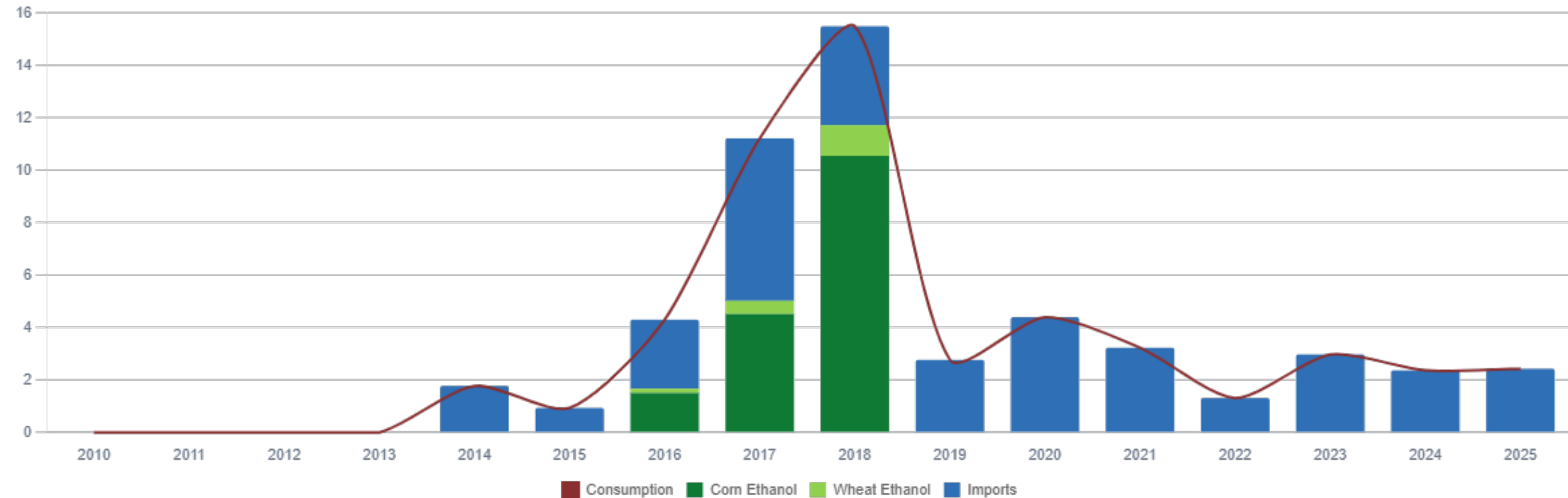
Source: Eurostat

Gasoline Exports in Romania



Source: Eurostat

Ethanol Balance in Romania



Source: Eurostat

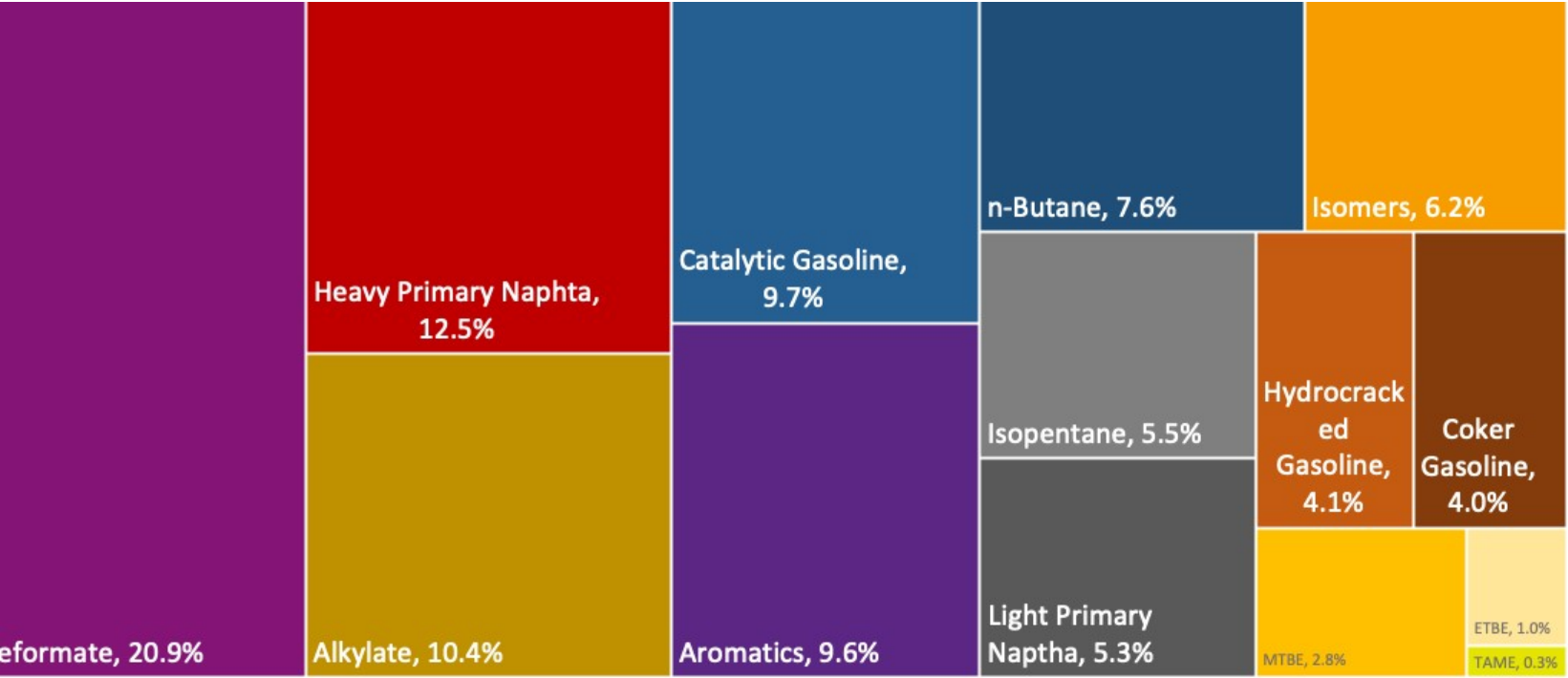
Gasoline Quality in Romania

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	10.0							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

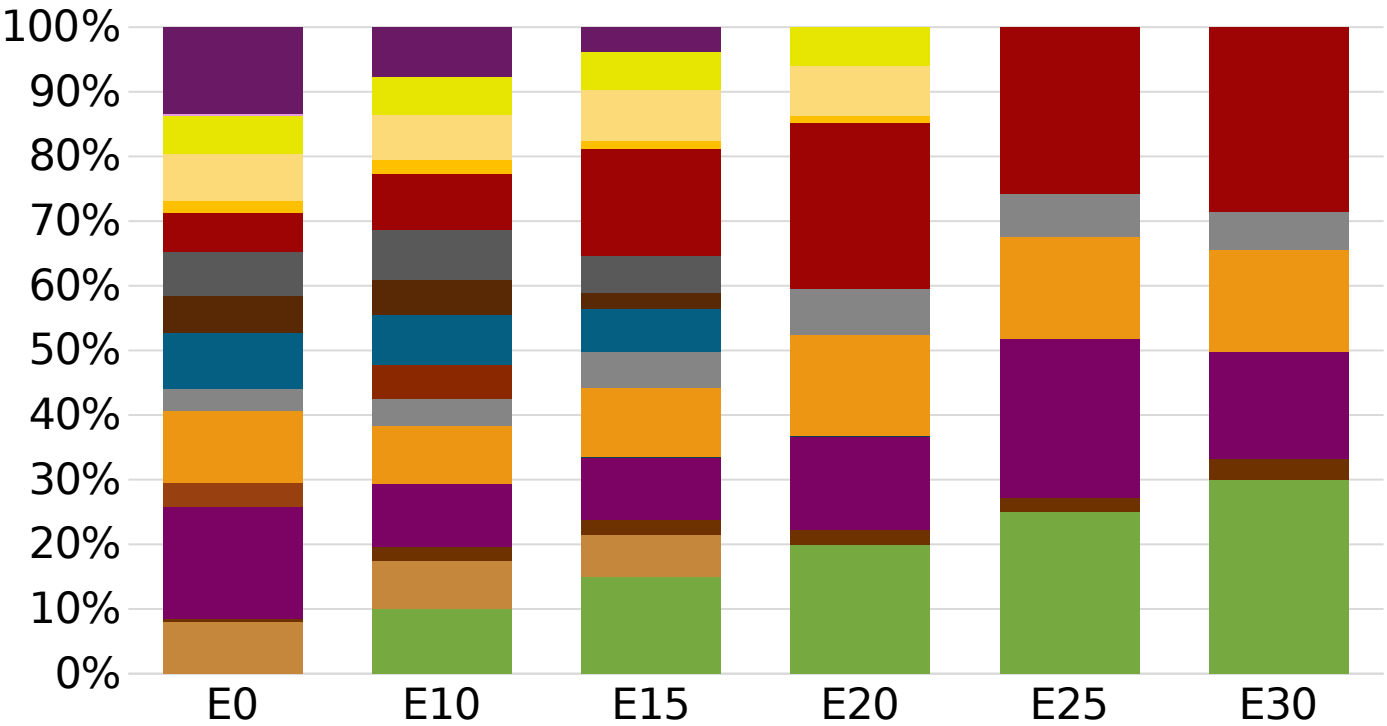
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Romania Available Blending Components



Romania – Gasoline 95 – Constant Octane

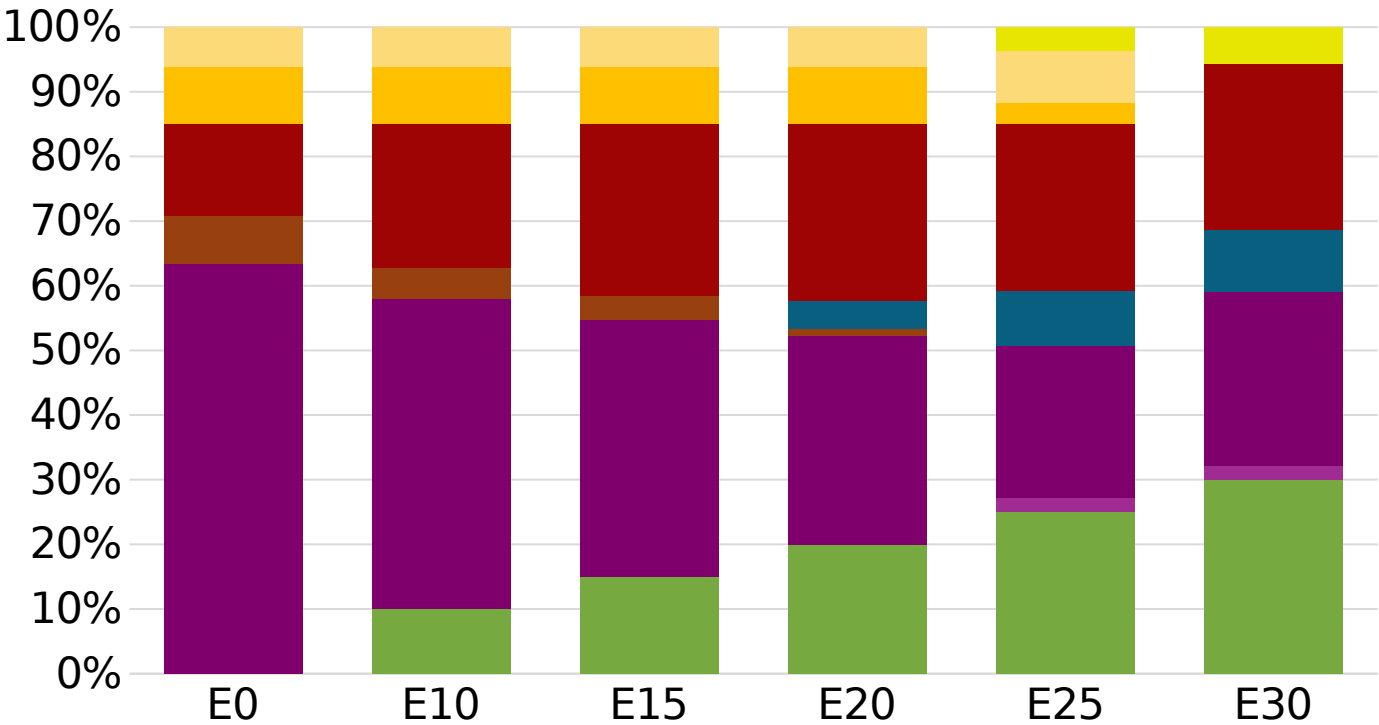
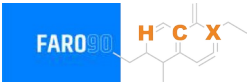


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.081	\$ 1.943	\$ 1.843	\$ 1.728	\$ 1.682	\$ 1.629

Source:
Faro90

Romania - Gasoline 98 - Constant Octane

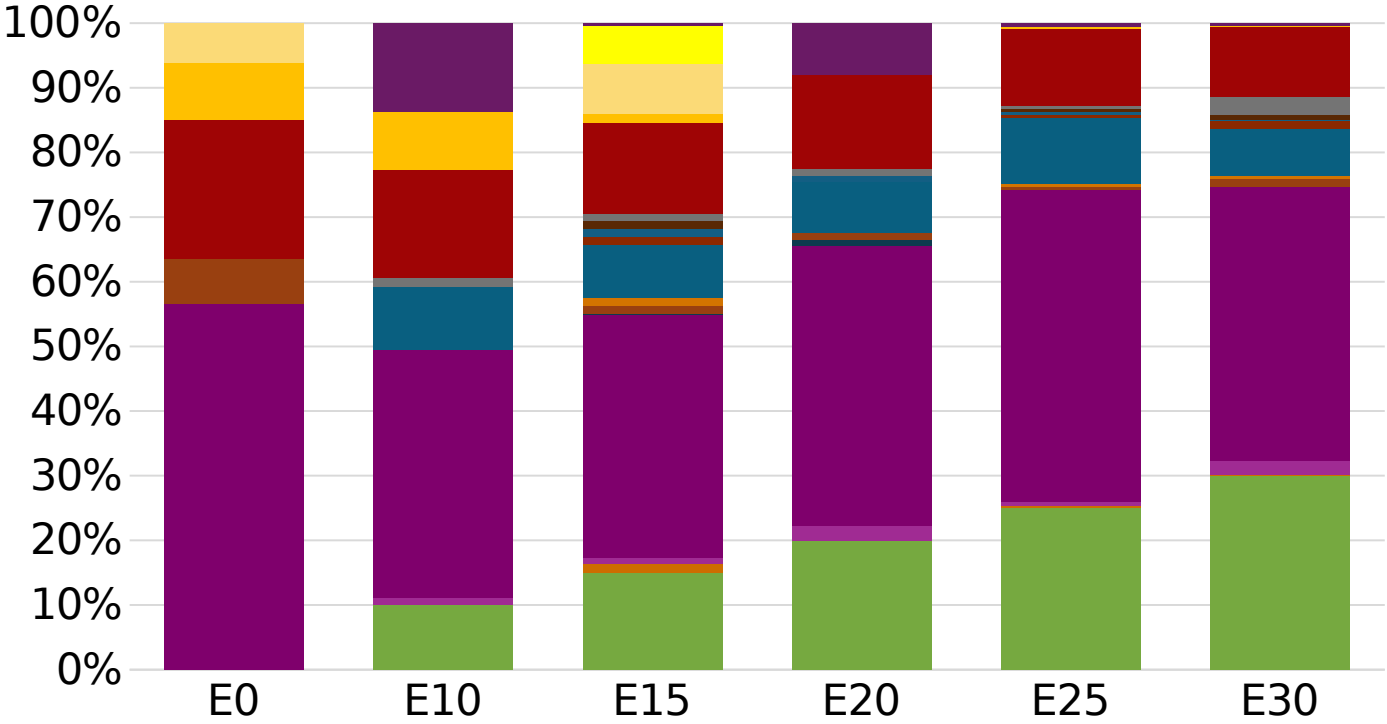


Average CIF 2025 Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.101	\$ 1.984	\$ 1.915	\$ 1.843	\$ 1.778	\$ 1.728

Source:
Faro90

Romania – Gasoline 95 – Increased Octane

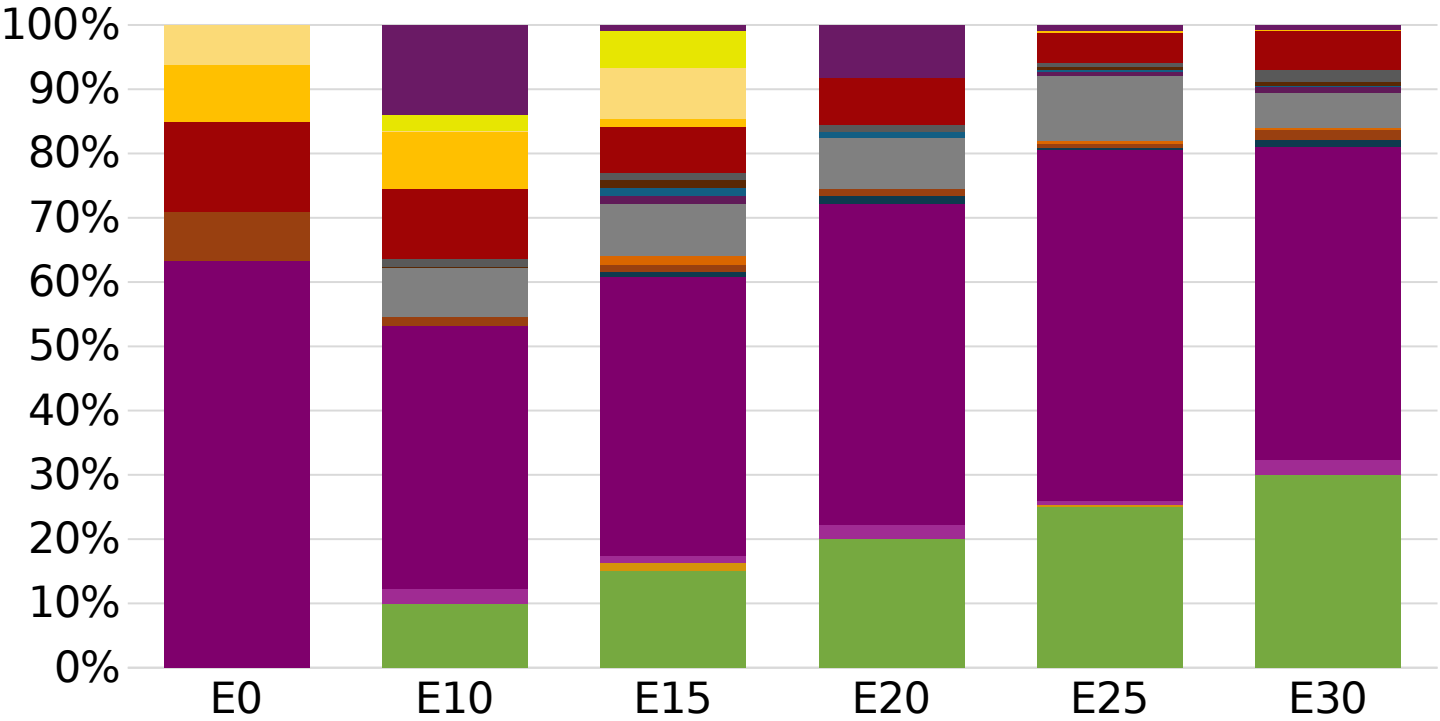


Average CIF 2025
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 1.969	\$ 1.969	\$ 1.969	\$ 1.969	\$ 1.969	\$ 1.969

Source:
Faro90

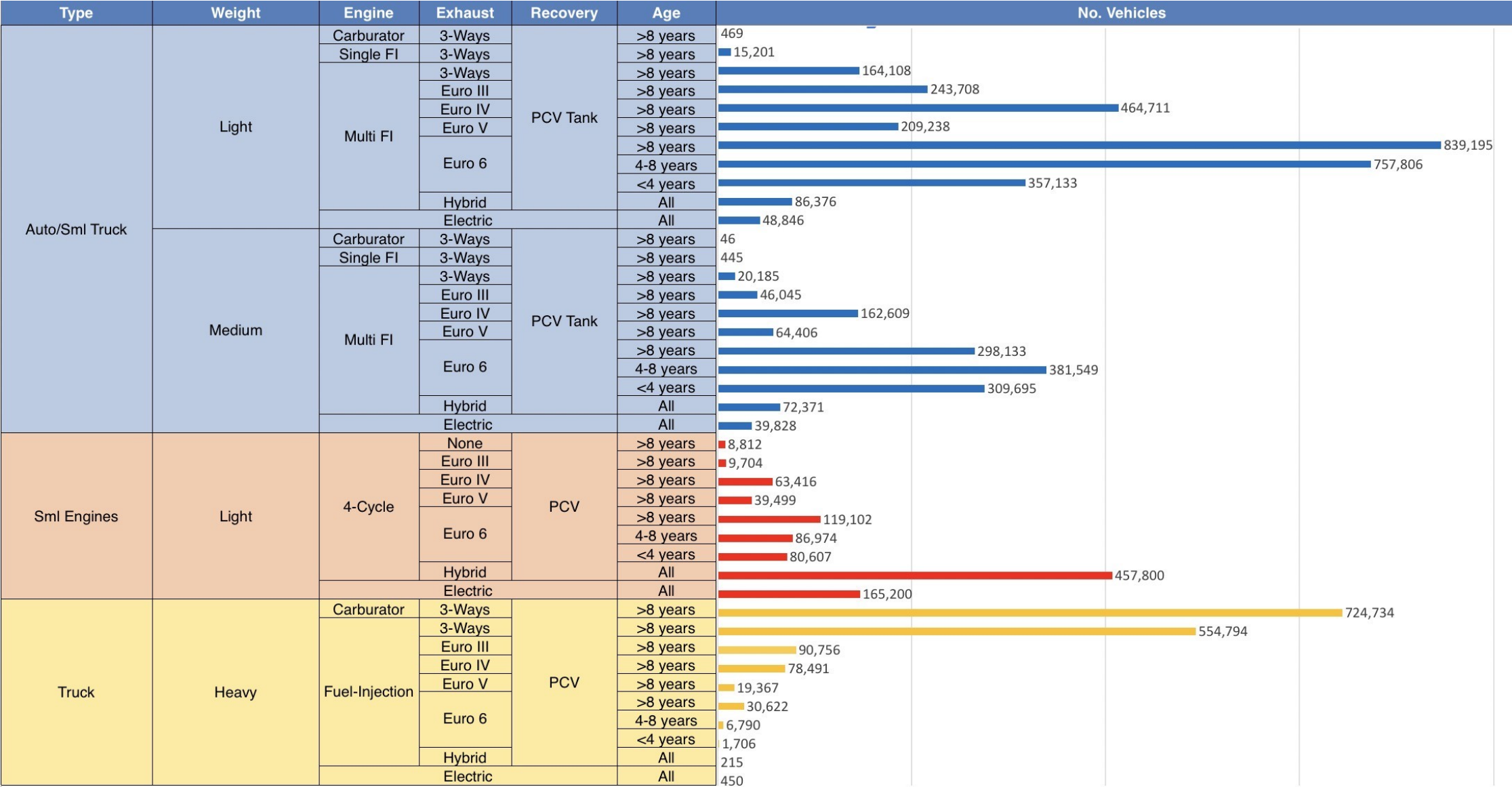
Romania – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Octane (RON)	98.0	99.0	100.9	101.2	102.8	104.1
Price (US\$/gal)	\$ 2.101	\$ 2.101	\$ 2.101	\$ 2.101	\$ 2.101	\$ 2.101

Gasoline Vehicle Fleet - Romania



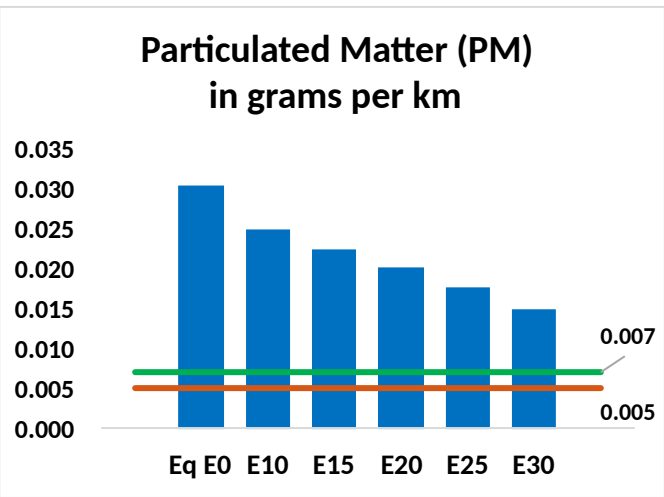
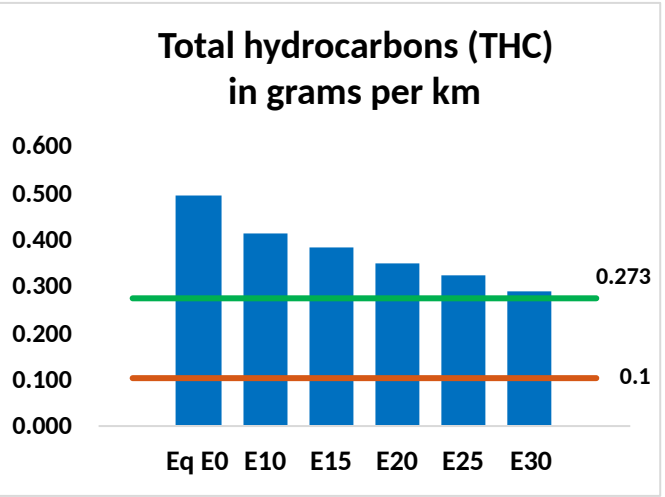
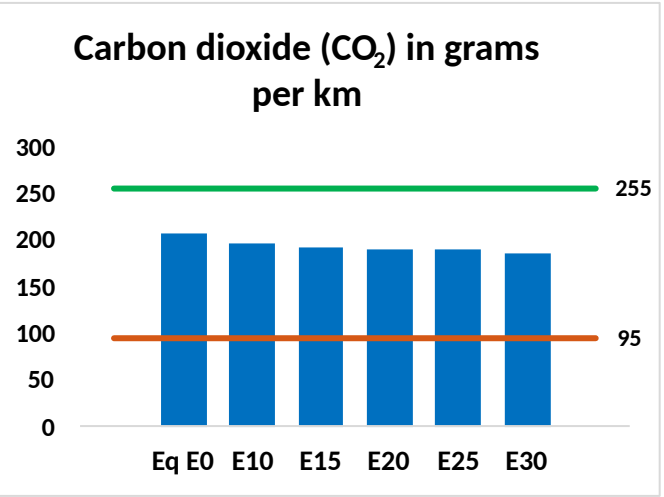
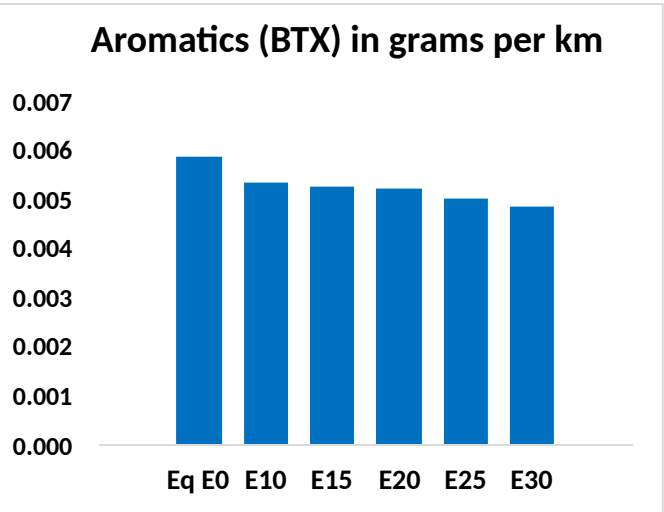
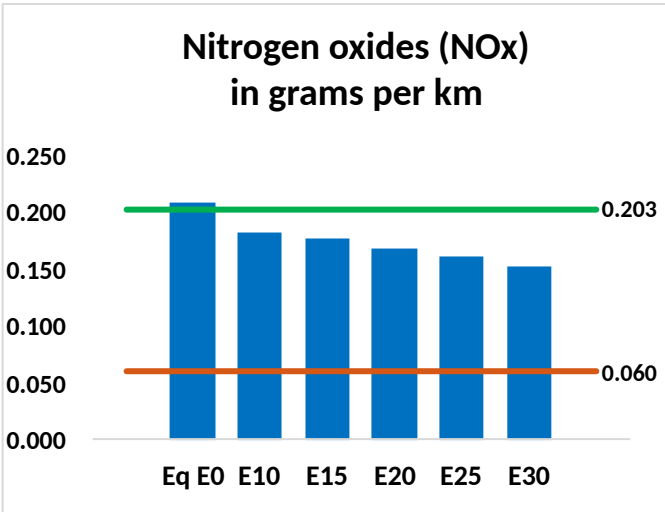
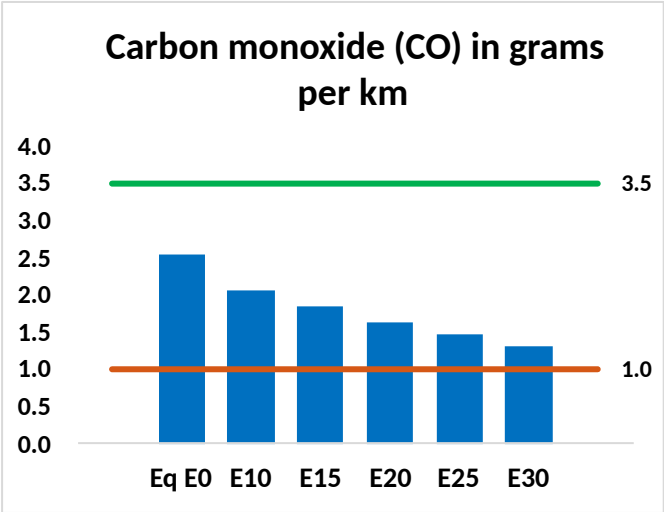
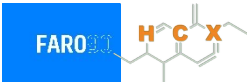
Vehicle Fleet: 7,121,143

Gasoline Vehicle Fleet: 2,695,377

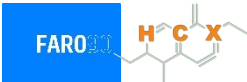
Source: Eurostat, ACEA

Romania – Vehicular Emissions

TIER III BIN 125 United States
Euro 6 Standard

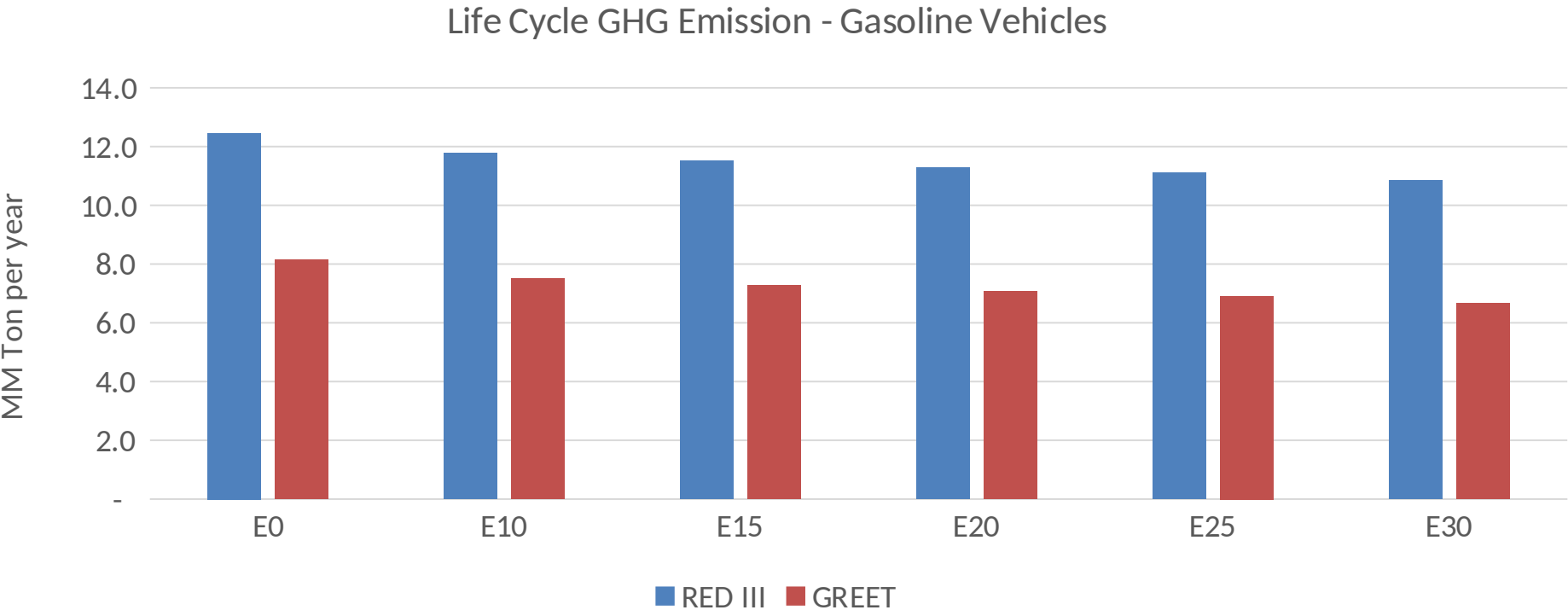
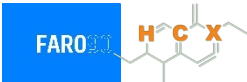


Romania – Vehicular Emissions



Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	180,118	162,910	145,702	130,945	115,793	104,051	91,912	-19%	-36%
CO2	14,700,751	14,322,732	13,965,714	13,684,930	13,545,986	13,466,227	13,218,006	-5%	-8%
VOC	35,069	32,240	29,410	27,066	24,695	22,868	20,442	-16%	-30%
NOx	14,919	13,776	12,998	12,606	11,969	11,473	10,886	-13%	-20%
SOx	169	156	143	133	122	111	99	-15%	-28%
NH3	5,233	5,181	5,129	5,154	5,212	5,252	5,286	-2%	0%
N2O	237	236	235	236	241	244	246	-1%	2%
CH4	7,432	7,432	7,432	7,544	7,707	7,848	7,982	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	188	179	170	163	157	153	146	-10%	-16%
Acetaldehyde	581	667	979	1,491	2,031	2,321	2,743	68%	249%
Formaldehyde	2,175	2,115	2,465	2,895	3,033	3,355	3,652	13%	39%
Benzene	419	403	382	376	372	356	345	-9%	-11%
PM2.5	868	789	710	641	573	500	424	-18%	-34%
PM10	1,305	1,186	1,067	963	861	751	637	-18%	-34%

Romania – Gasoline Vehicle Fleet Life Cycle GHG Emissions



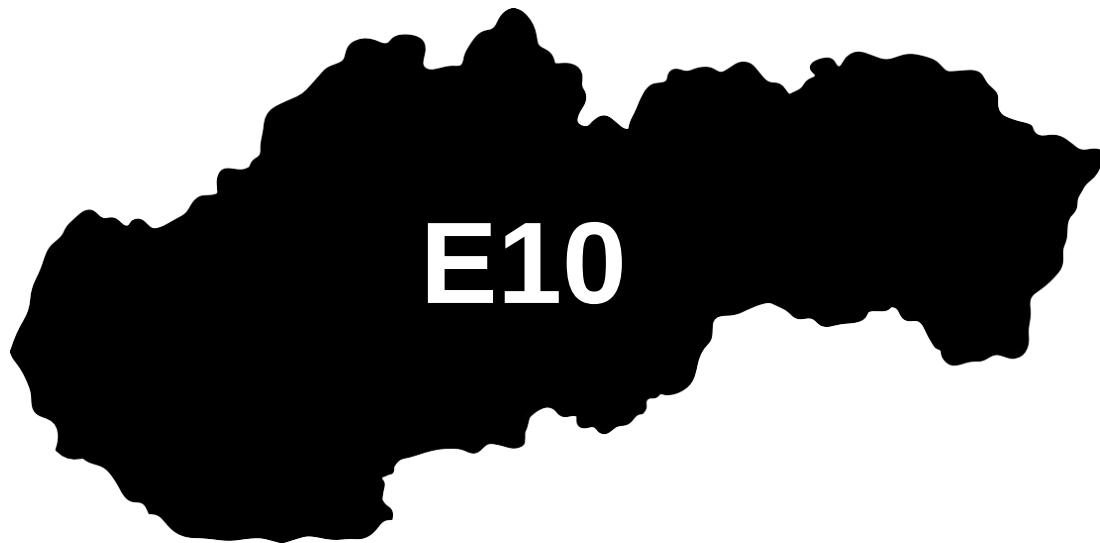
Country Emissions 2025 (MMT/año)	76.7
Target @ 2035 (MMT/year)	43.2
Delta	33.5

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	7.8%	10.6%	13.0%	15.1%	18.1%
RED II/III	0.0%	5.5%	7.5%	9.3%	10.8%	12.9%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	1.9%	2.6%	3.2%	3.7%	4.4%
RED II/III	0.0%	2.1%	2.8%	3.5%	4.0%	4.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

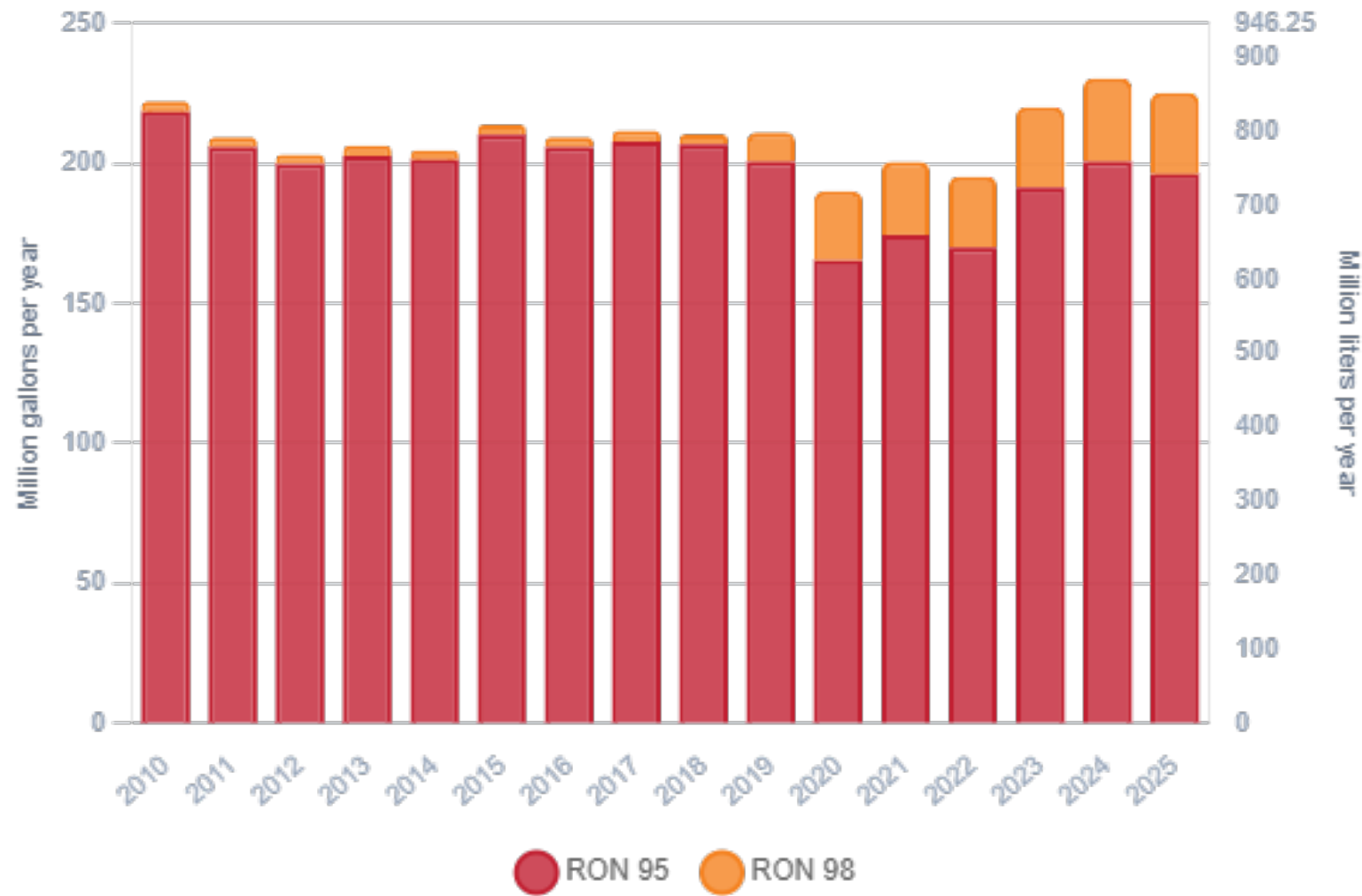
Ethanol Gasoline Blending in Slovakia



In 2025, gasoline consumption in Slovakia was 852 million liters (225 million gallons) and growth rate was 0.1%. Gasoline production and net imports were 1,887 and -1,035 million liters (498 and -273 million gallons) correspondingly. Slovakia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

In 2025, ethanol consumption in Slovakia was 80 million liters (21 million gallons) representing 9.3% of gasoline demand. Ethanol production and Net Imports were 166 and -87 million liters (44 and -23 million gallons) correspondingly. Ethanol source is Cereals 100%.

Gasoline Demand in Slovakia



Source: Eurostat

Gasoline Production in Slovakia



Source: Eurostat

Gasoline Imports in Slovakia



Source: Eurostat

The FARO90 logo is shown in a blue box, followed by a chemical structure of a substituted cyclohexene. The structure features a six-membered ring with a double bond. Substituents include a hydrogen atom (H), a carbon atom (C), and a group labeled 'X' (which is a vinyl group, -CH=CH₂). The ring also has methyl groups and a propyl group attached.



Source: Eurostat

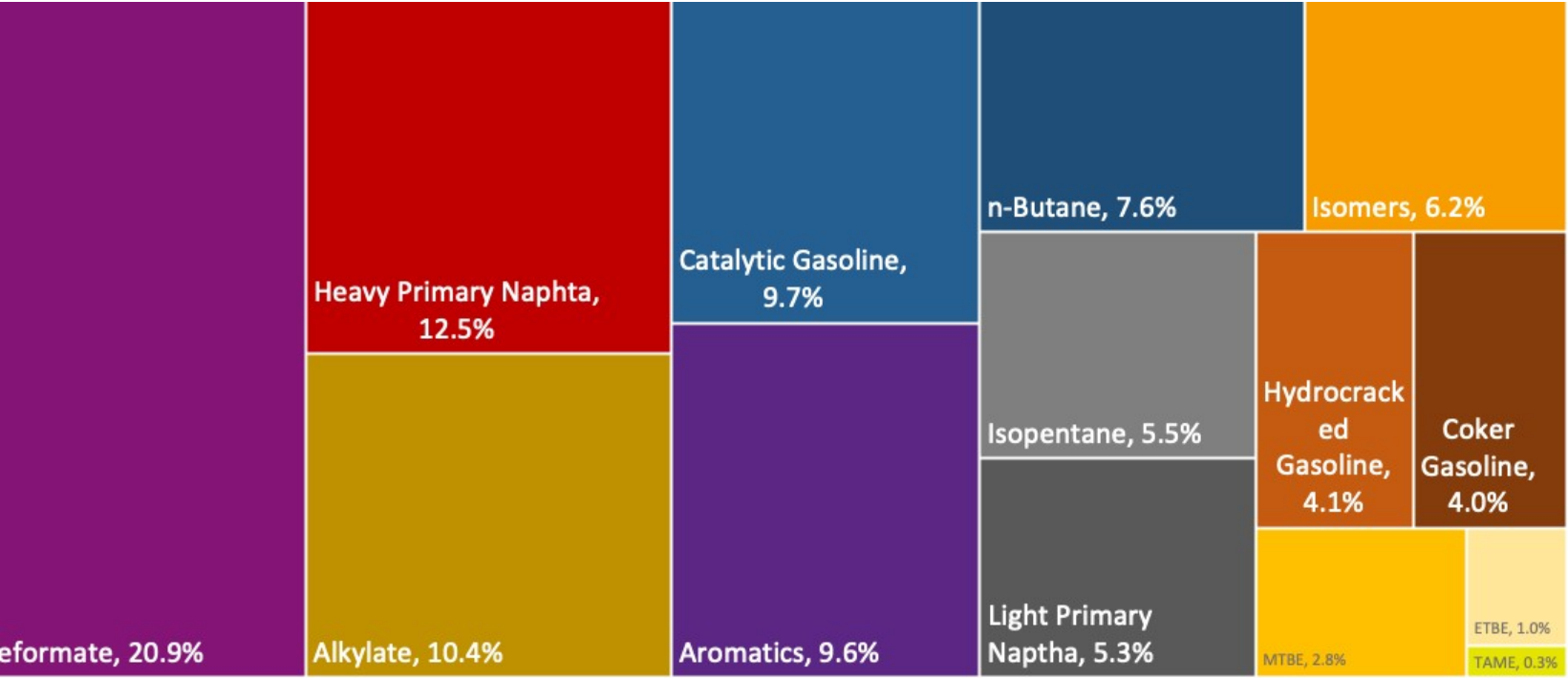
Gasoline Quality in Slovakia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	9.2							
Advanced Biofuels (%cal)	1.1							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

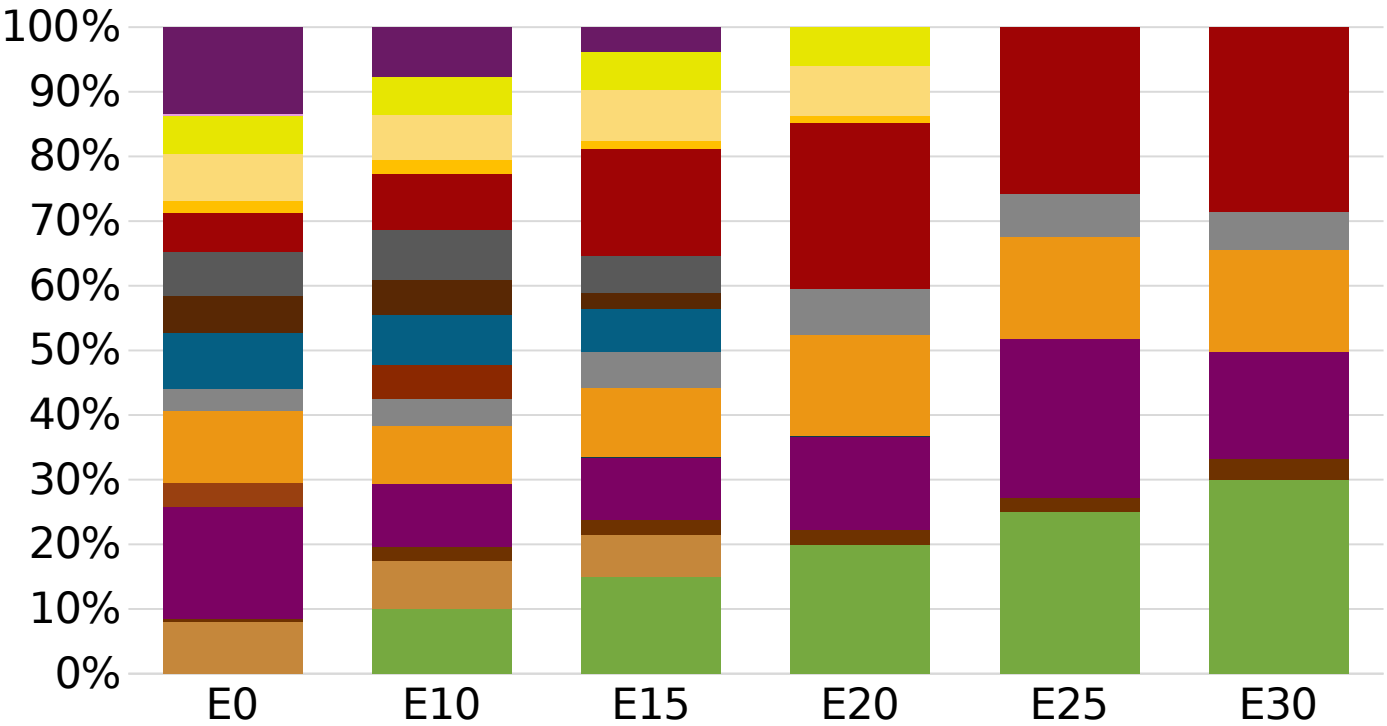
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Slovakia Available Blending Components



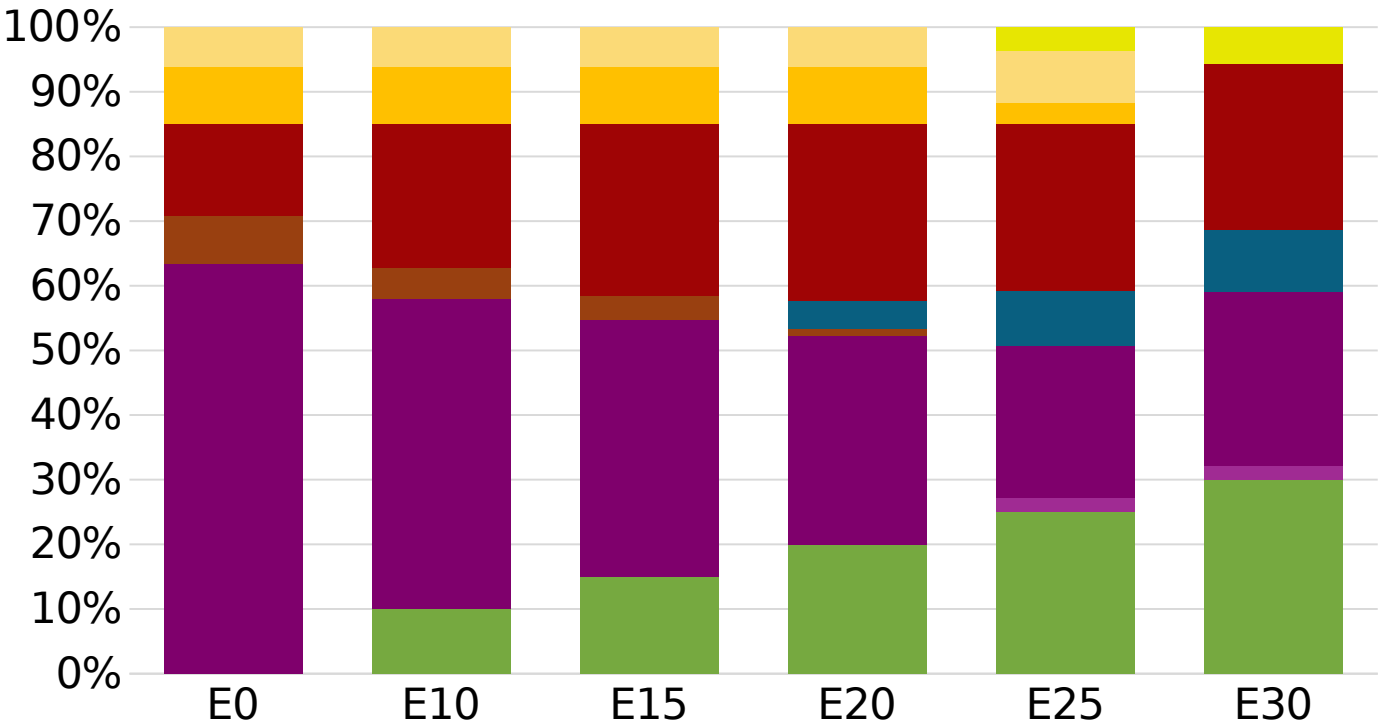
Slovakia – Gasoline 95 – Constant Octane



Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.0	95.8
Price (US\$/gal)	\$ 2.101	\$ 1.963	\$ 1.863	\$ 1.748	\$ 1.702	\$ 1.649

Slovakia - Gasoline 98 - Constant Octane

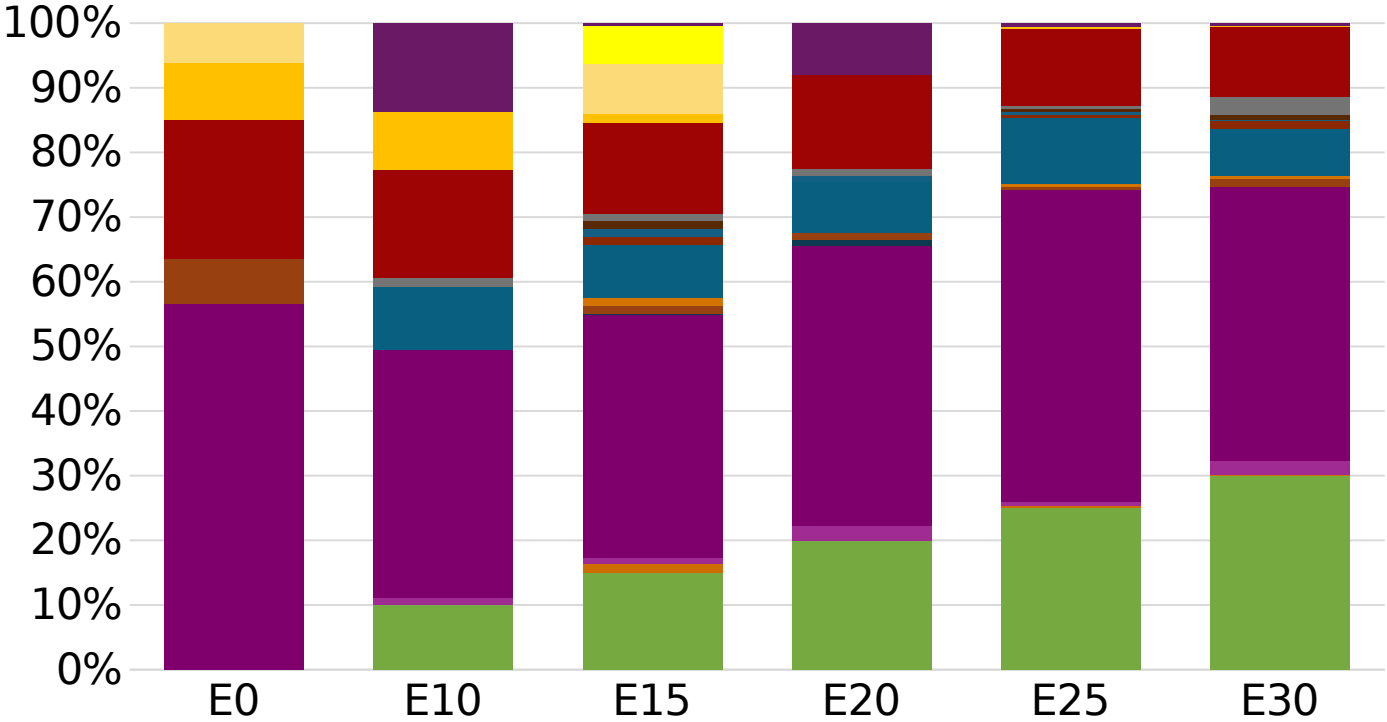


Average CIF 2025
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.121	\$ 2.004	\$ 1.935	\$ 1.863	\$ 1.798	\$ 1.748

Source:
Faro90

Slovakia – Gasoline 95 – Increased Octane

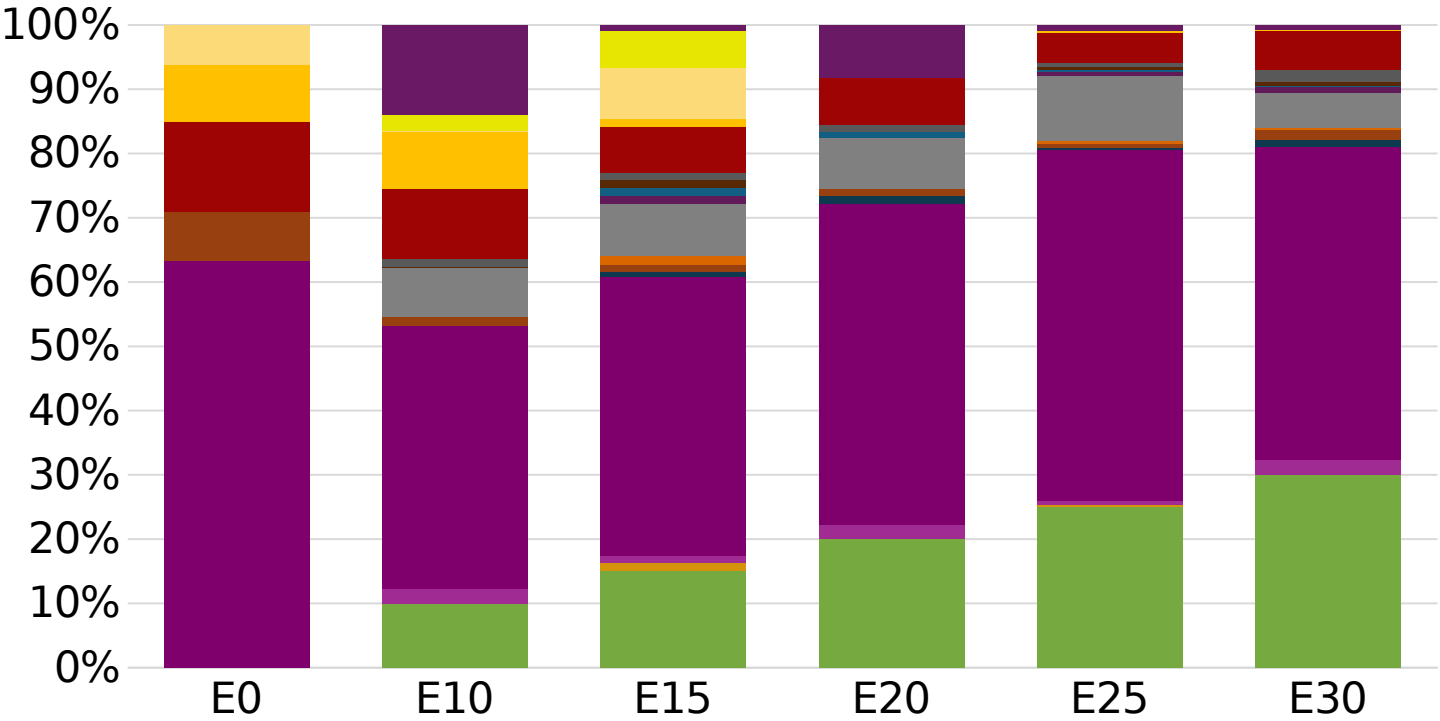


Average CIF 2025
Prices

Octane (RON)	95.0	96.2	98.3	98.7	100.5	101.8
Price (US\$/gal)	\$ 1.989	\$ 1.989	\$ 1.989	\$ 1.989	\$ 1.989	\$ 1.989

Source:
Faro90

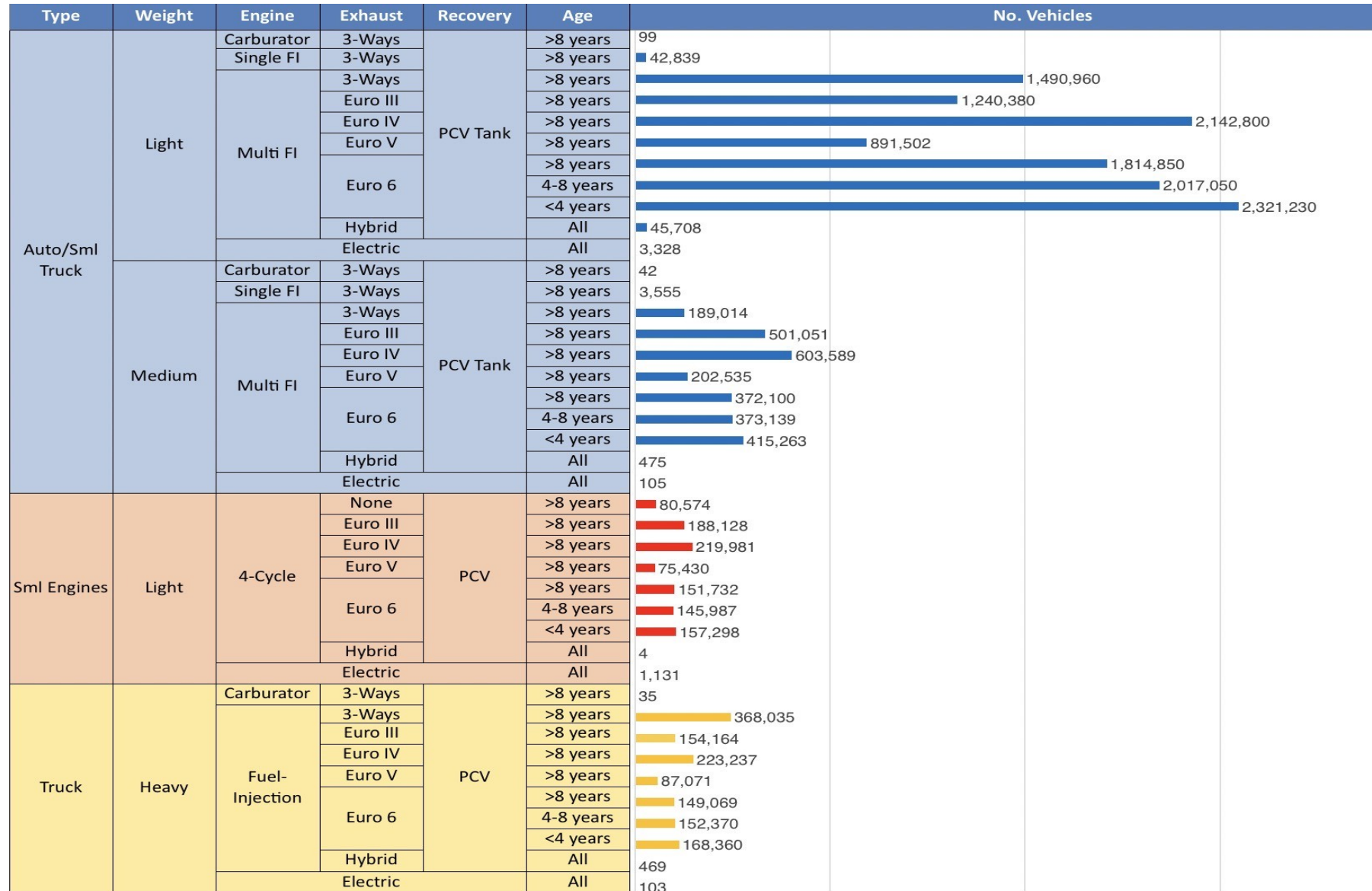
Slovakia – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Source:
Faro90

Gasoline Vehicle Fleet - Slovakia

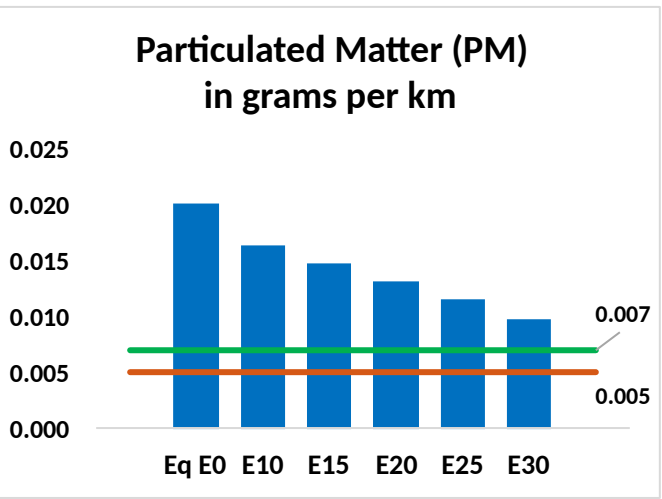
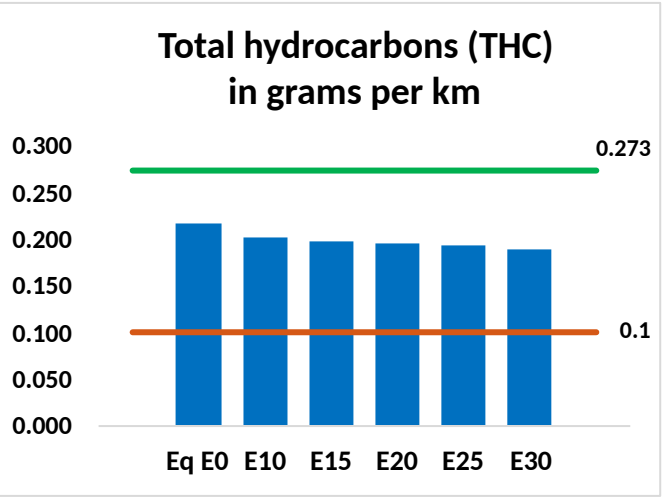
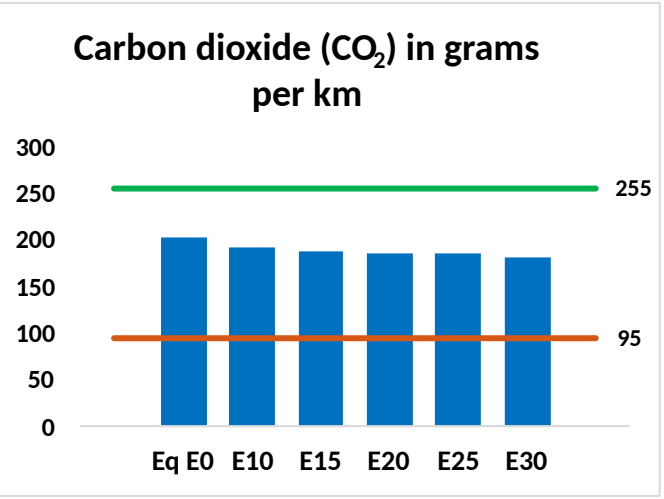
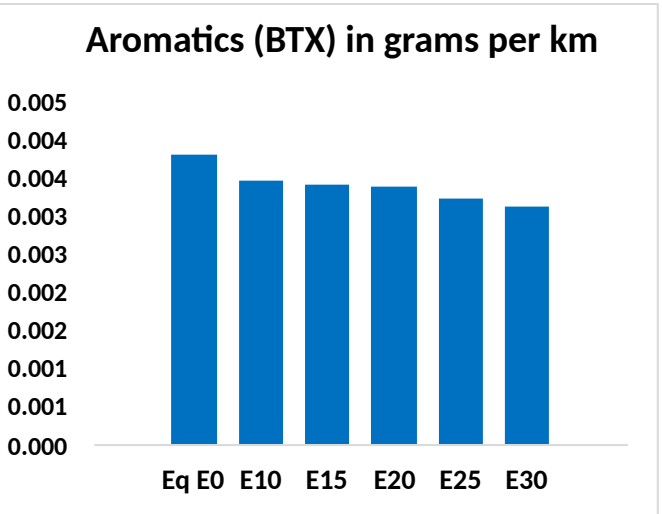
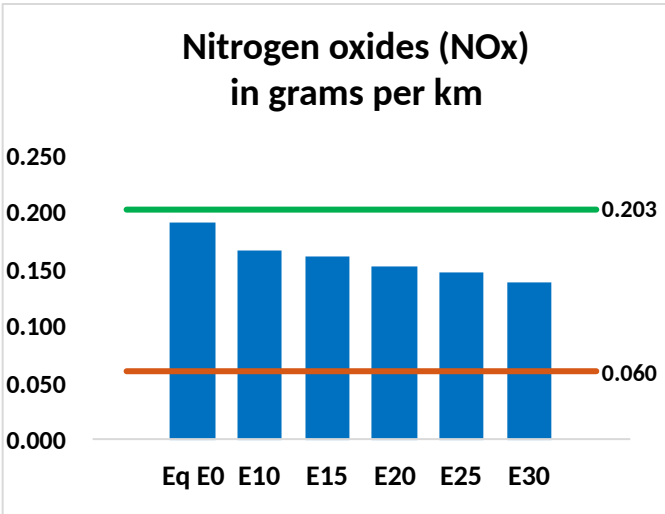
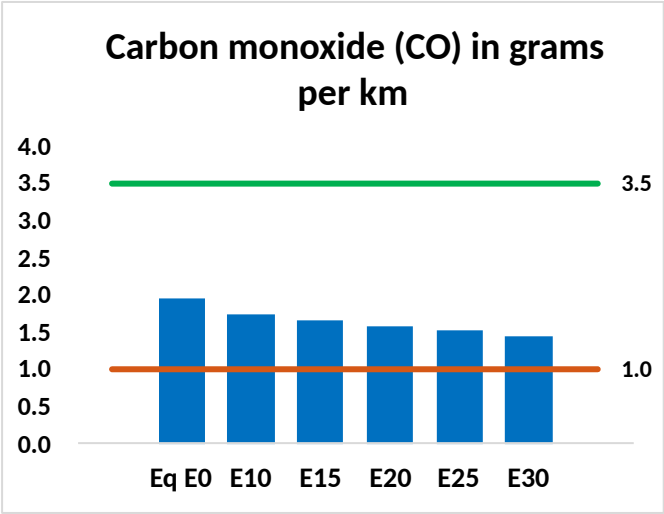
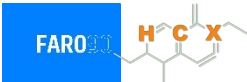


Vehicle Fleet: 5,707,470
Gasoline Vehicle Fleet: 1,816,846

Source: Eurostat, ACEA

Slovakia – Vehicular Emissions

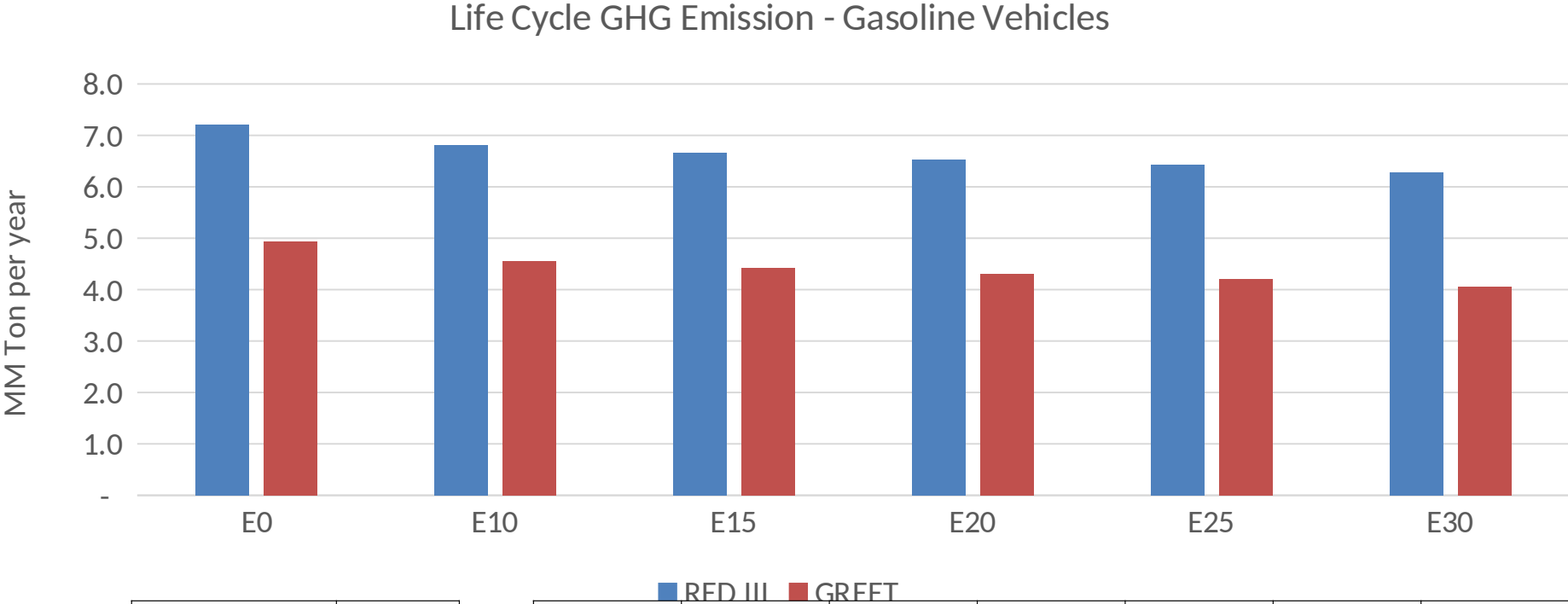
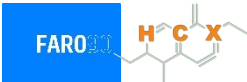
TIER III BIN 125 United States
Euro 6 Standard



Slovakia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	87,546	82,678	77,810	74,185	70,704	68,077	64,914	-11%	-19%
CO2	9,095,946	8,862,050	8,641,149	8,467,416	8,381,447	8,332,055	8,178,472	-5%	-8%
VOC	9,706	9,396	9,086	8,916	8,785	8,696	8,535	-6%	-9%
NOx	8,626	7,965	7,510	7,278	6,904	6,611	6,265	-13%	-20%
SOx	103	95	87	81	74	67	60	-15%	-28%
NH3	3,158	3,127	3,096	3,110	3,145	3,170	3,190	-2%	0%
N2O	101	101	100	101	103	104	105	-1%	2%
CH4	2,389	2,389	2,389	2,424	2,477	2,522	2,565	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	99	97	94	93	93	93	92	-5%	-6%
Acetaldehyde	212	243	356	543	739	845	998	68%	249%
Formaldehyde	675	657	765	899	941	1,041	1,134	13%	39%
Benzene	171	165	156	154	152	146	141	-9%	-11%
PM2.5	552	502	452	408	364	318	270	-18%	-34%
PM10	351	319	287	259	231	202	171	-18%	-34%

Slovakia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	36.8
Target @ 2035 (MMT/year)	20.7
Delta	16.1

%Reduction vs E0	GREET					
	E0	E10	E15	E20	E25	E30
	0.0%	7.7%	10.4%	12.8%	14.7%	17.6%
RED II/III	0.0%	5.6%	7.6%	9.4%	10.9%	13.0%

% 2035 Target	GREET					
	E0	E10	E15	E20	E25	E30
	0.0%	2.4%	3.2%	3.9%	4.5%	5.4%
RED II/III	0.0%	2.5%	3.4%	4.2%	4.9%	5.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

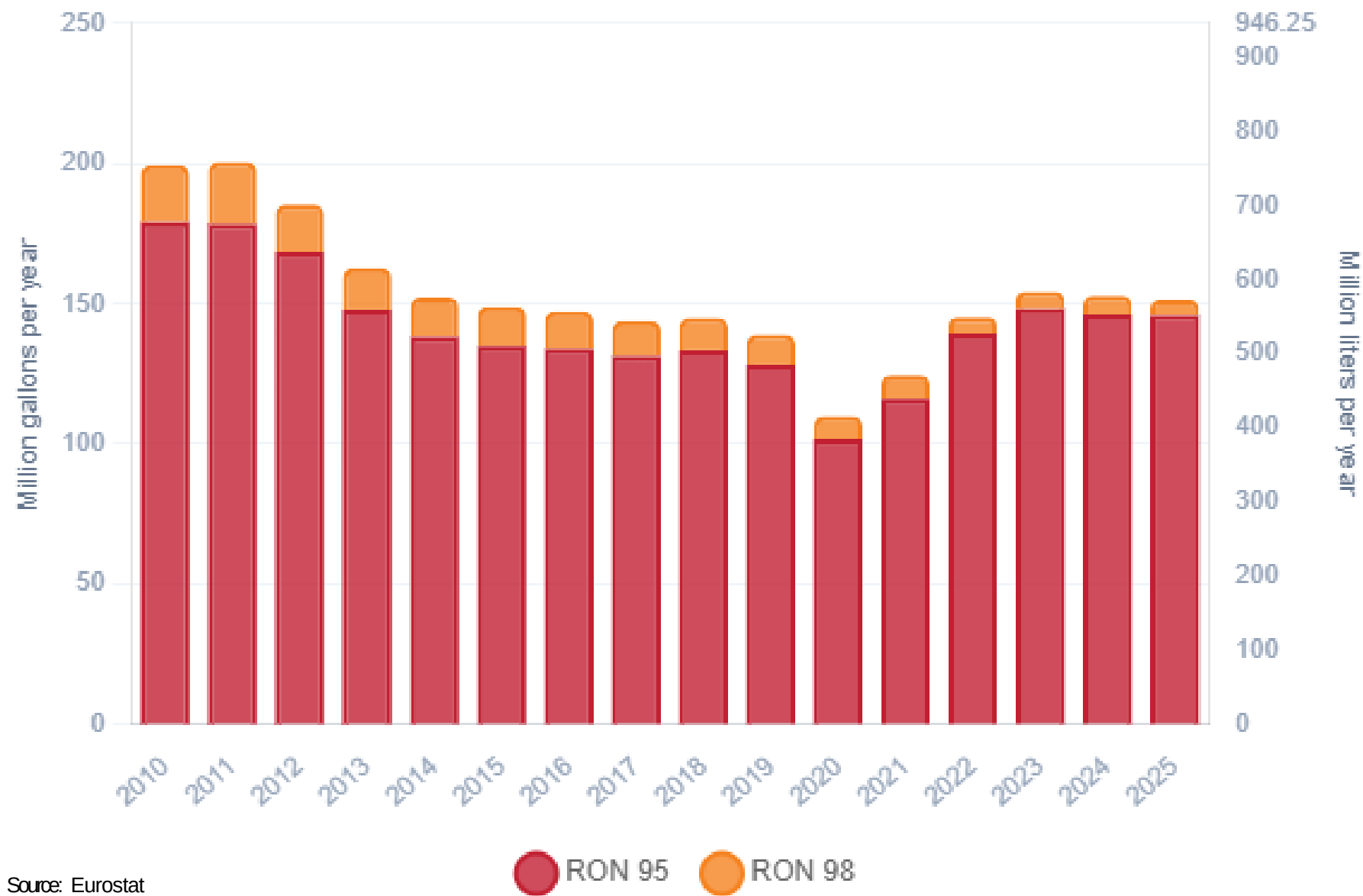
Ethanol Gasoline Blending in Slovenia



In 2025, gasoline consumption in Slovenia was 572 million liters (151 million gallons) and growth rate was -1.8%. Gasoline production and net imports were 0 and 572 million liters (0 and 151 million gallons) correspondingly. Slovenia complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

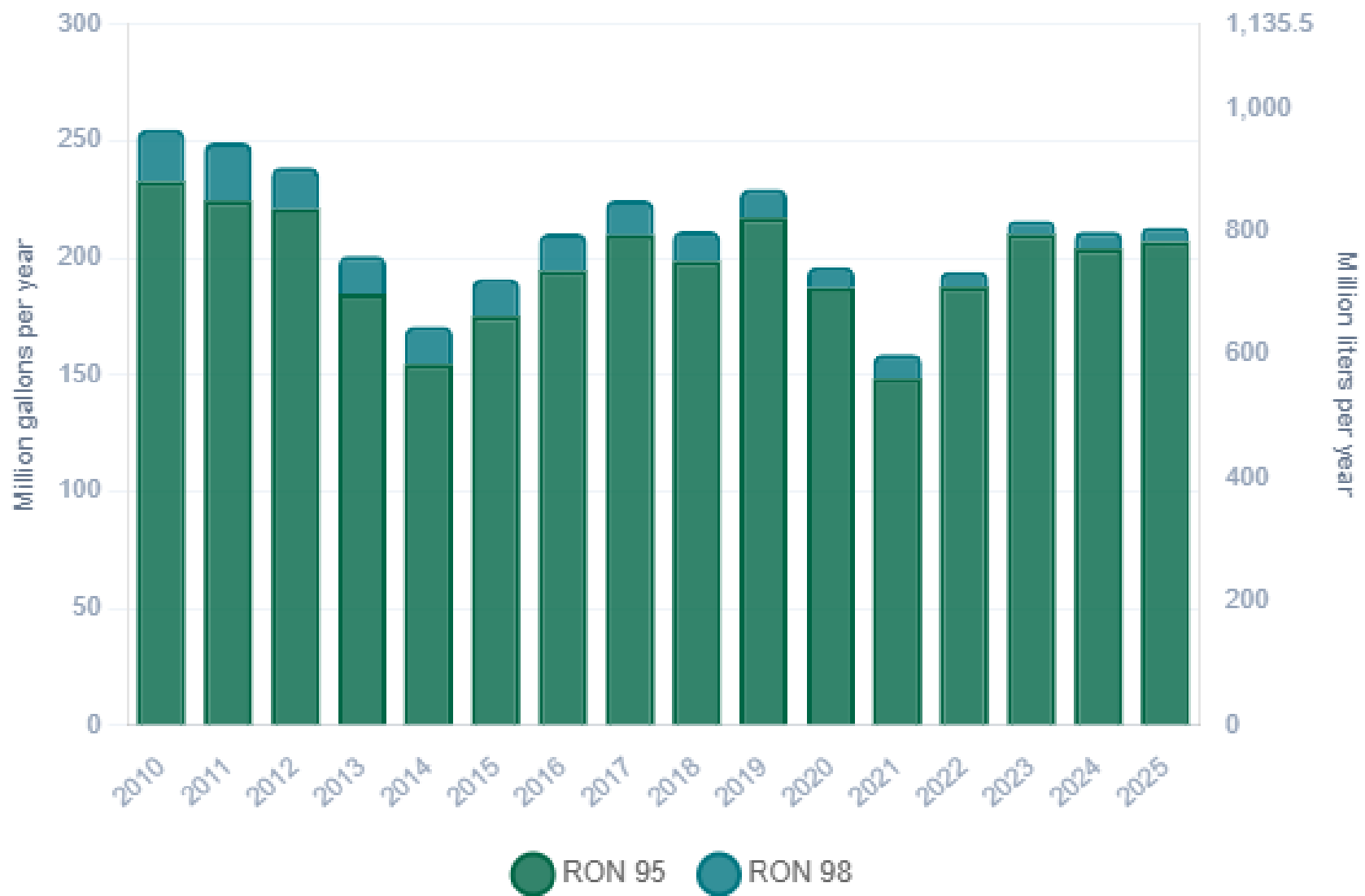
In 2025, ethanol consumption in Slovenia was 14 million liters (4 million gallons) representing 2.4% of gasoline demand. Ethanol production and Net Imports were 0 and 14 million liters (0 and 4 million gallons) correspondingly. Ethanol source is Cereals 100%.

Gasoline Demand in Slovenia



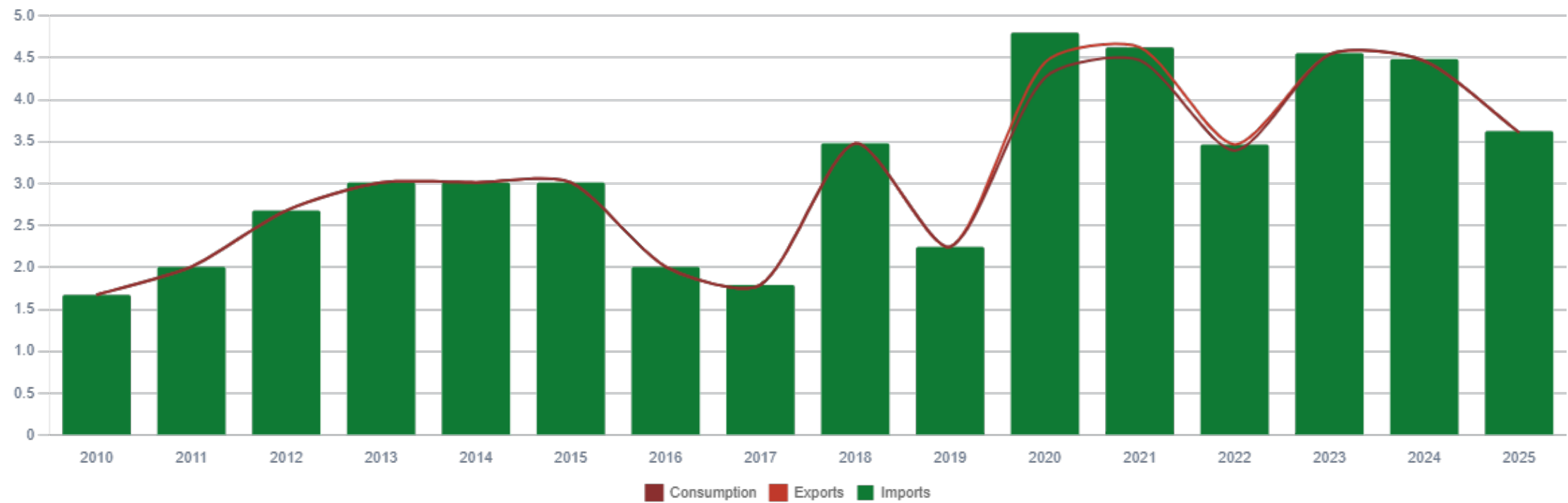
Source: Eurostat

Gasoline Imports in Slovenia



Source: Eurostat

Ethanol Balance in Slovenia



Source: Eurostat

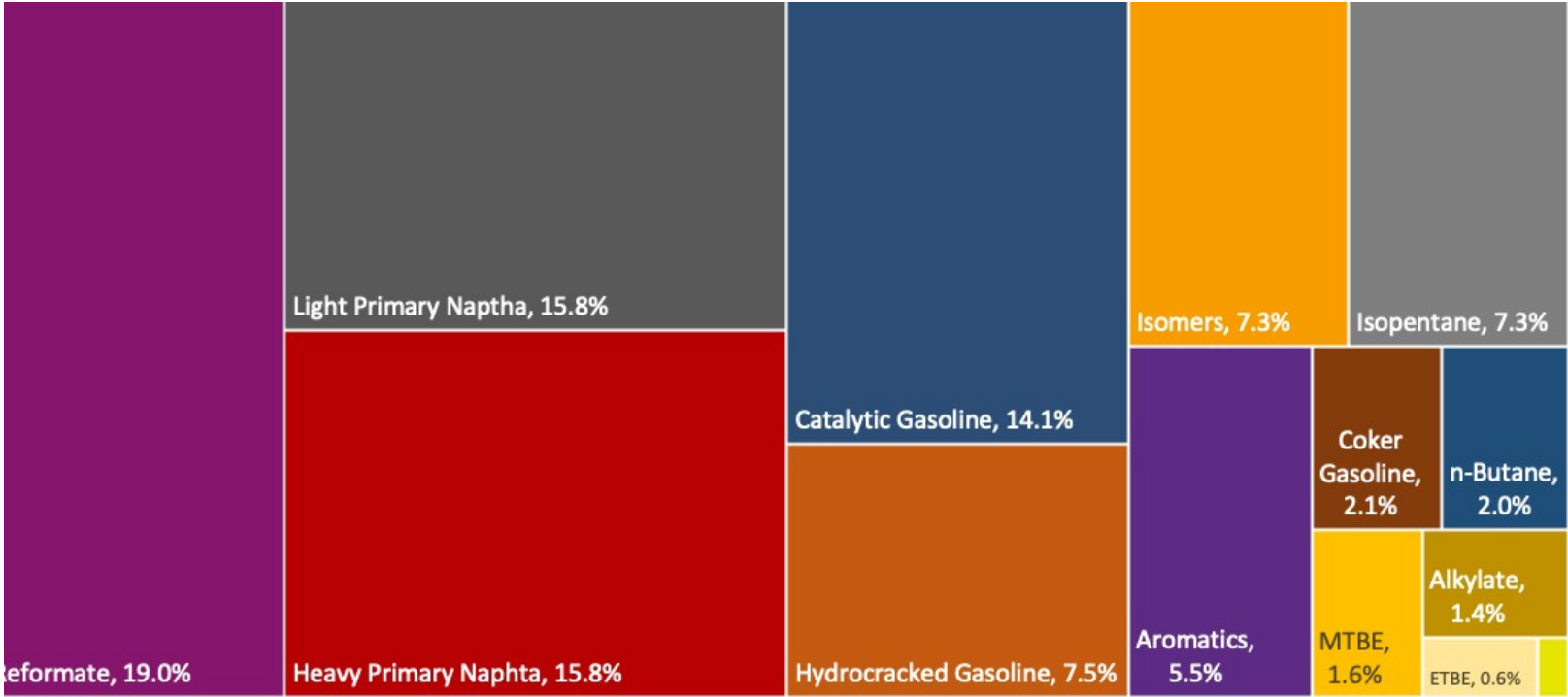
Gasoline Quality in Slovenia

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	11.2							
Advanced Biofuels (%cal)	0.2							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

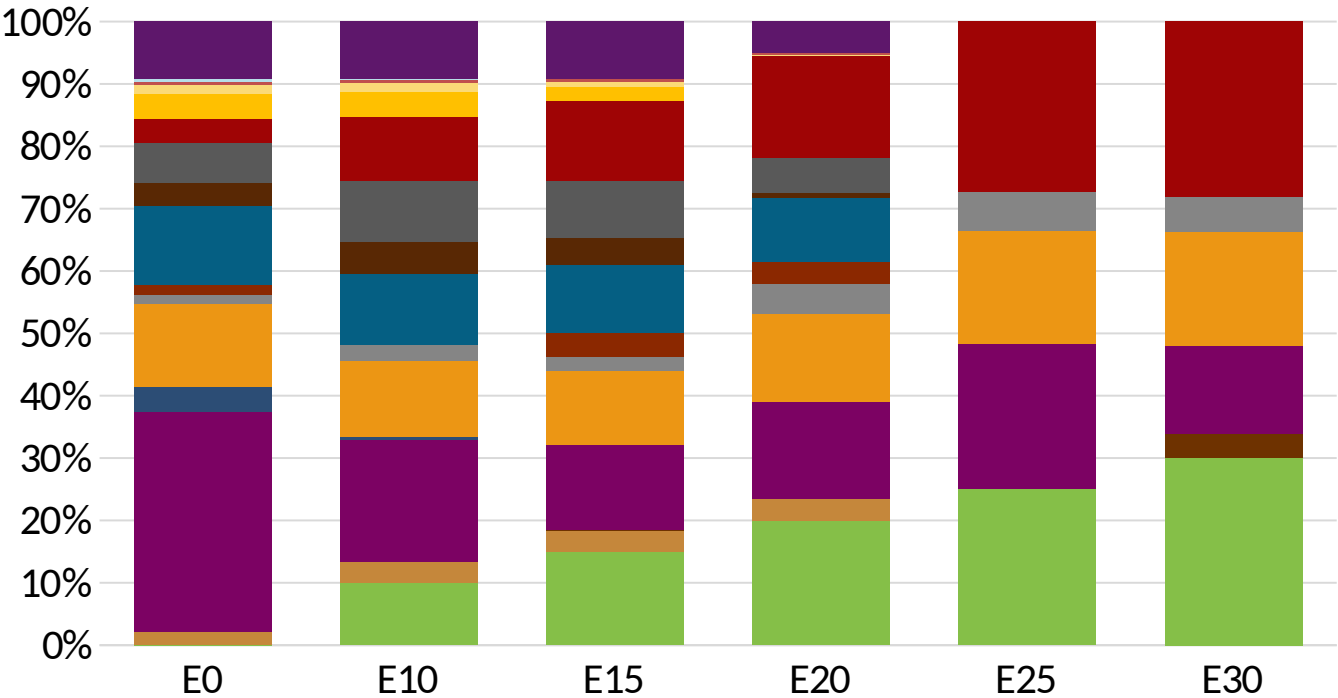
Source: European Commision

Slovenia Available Blending Components



TAME,
0.2%

Slovenia – Gasoline 95 – Constant Octane

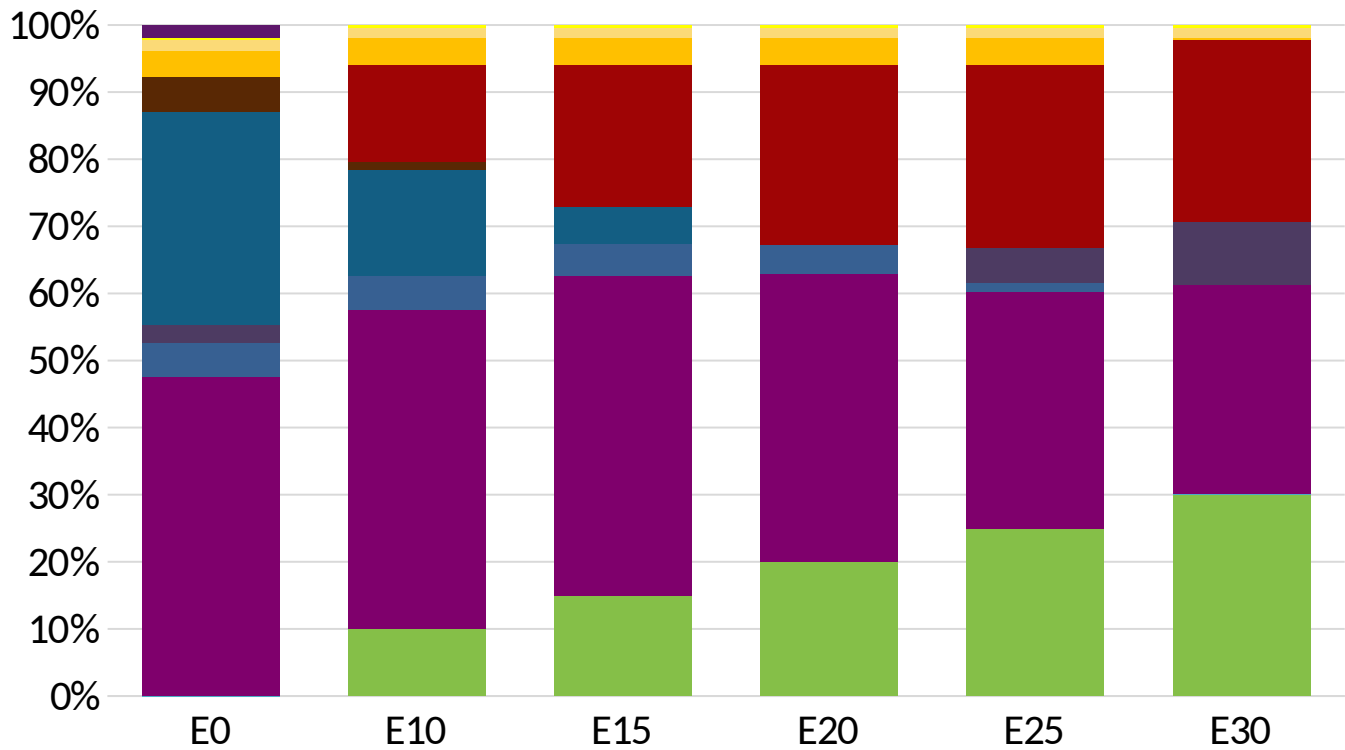


Average CIF 2025
Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.089	\$ 1.961	\$ 1.900	\$ 1.819	\$ 1.697	\$ 1.632

Source:
Faro90

Slovenia - Gasoline 98 - Constant Octane

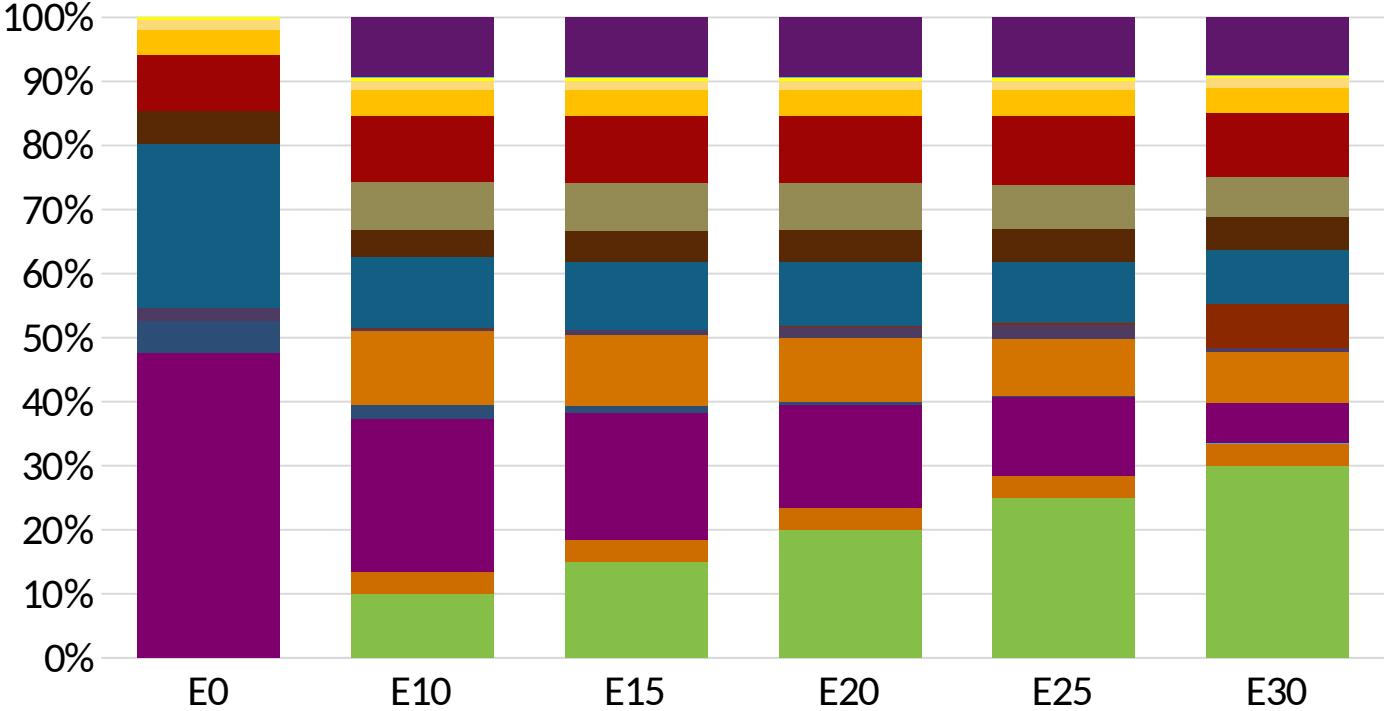


Average CIF 2025
Prices

Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.164	\$ 2.035	\$ 1.963	\$ 1.886	\$ 1.809	\$ 1.742

Source:
Faro90

Slovenia – Gasoline 95 – Increased Octane

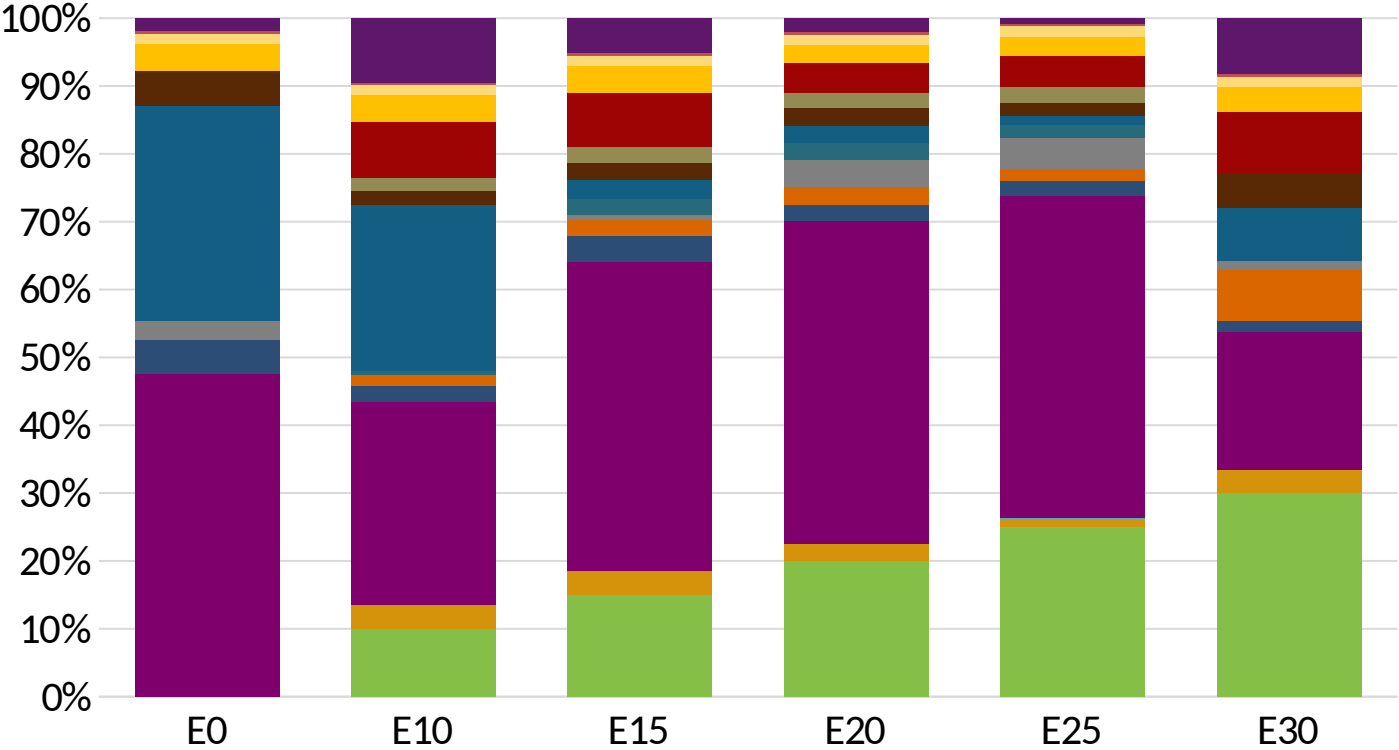


Average CIF 2025
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.026	\$ 2.026	\$ 2.026	\$ 2.026	\$ 2.026	\$ 2.026

Source:
Faro90

Slovenia – Gasoline 98 – Increased Octane

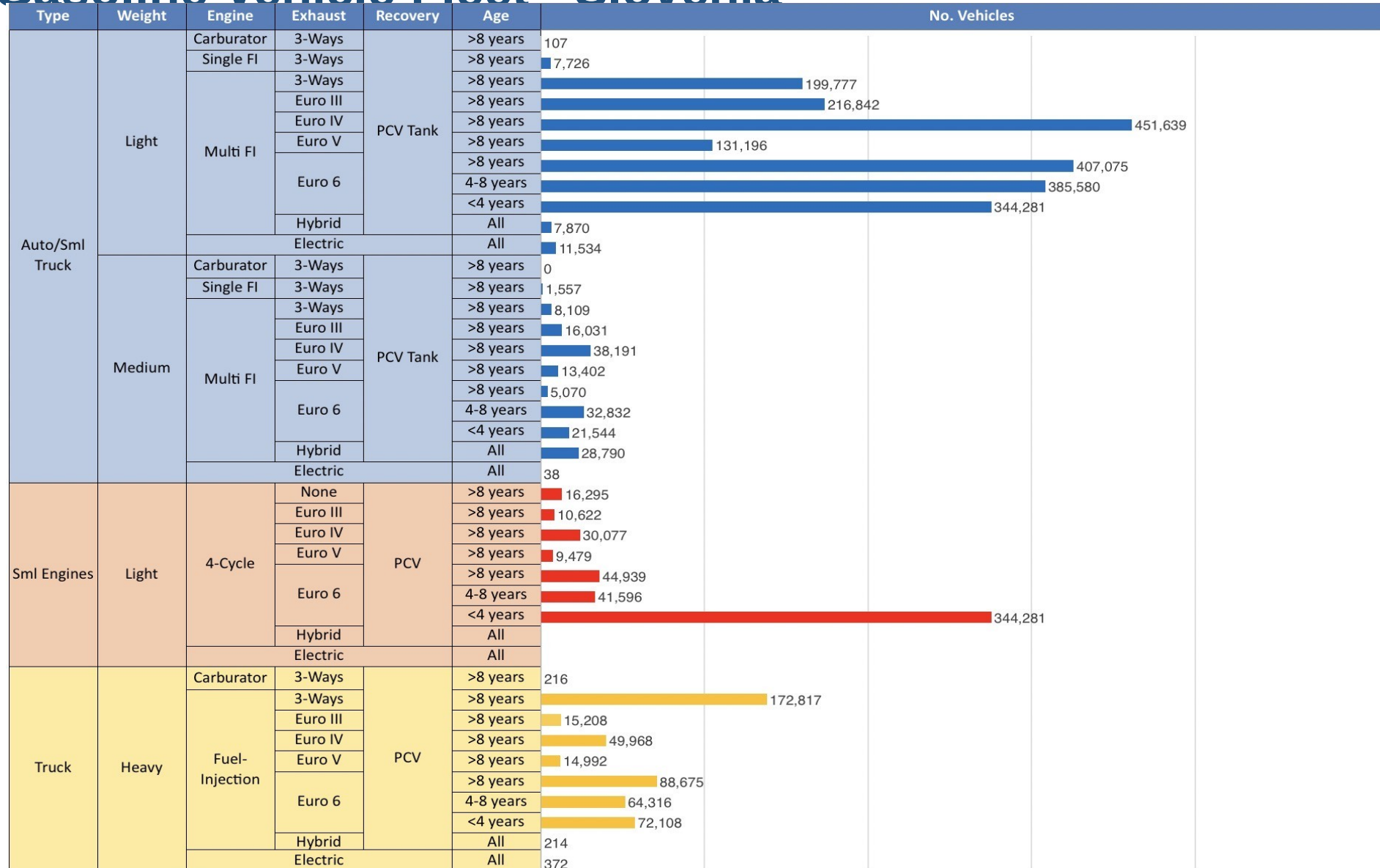


Average CIF 2025
Prices

Octane (RON)	98.0	99.2	101.3	102.6	104.0	104.7
Price (US\$/gal)	\$ 2.164	\$ 2.164	\$ 2.164	\$ 2.164	\$ 2.164	\$ 2.164

Source:
Faro90

Gasoline Vehicle Fleet - Slovenia

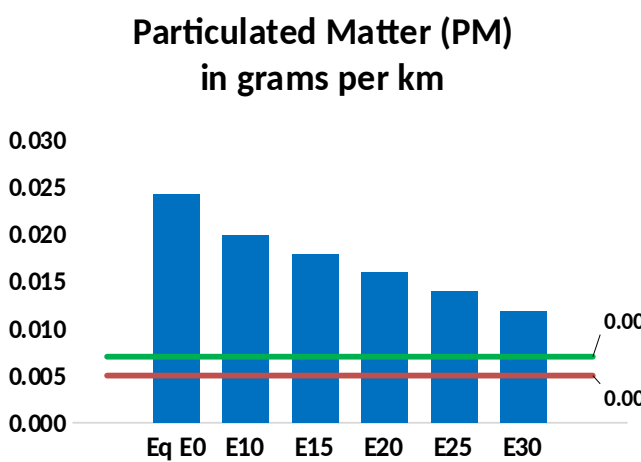
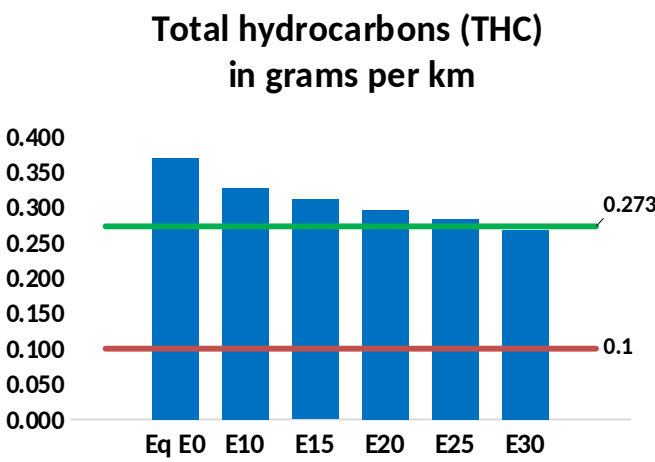
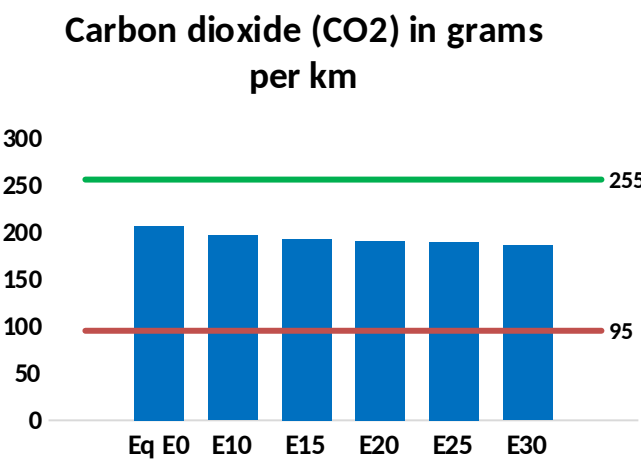
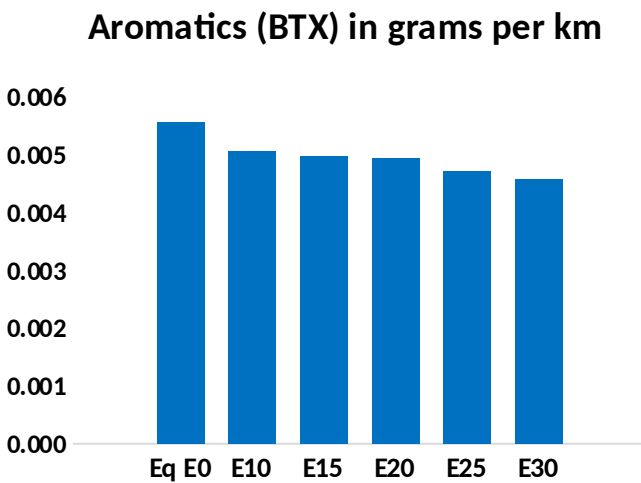
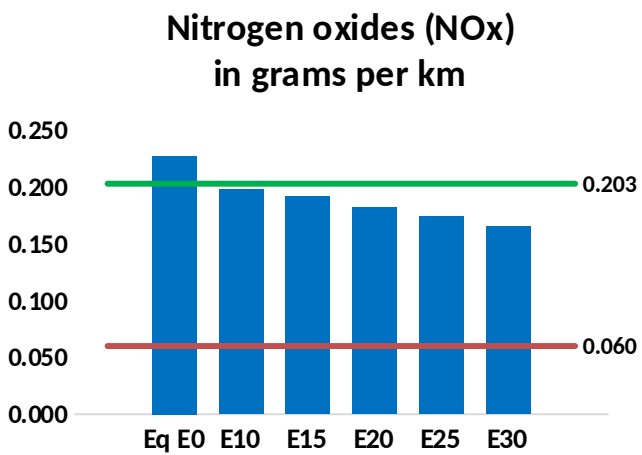
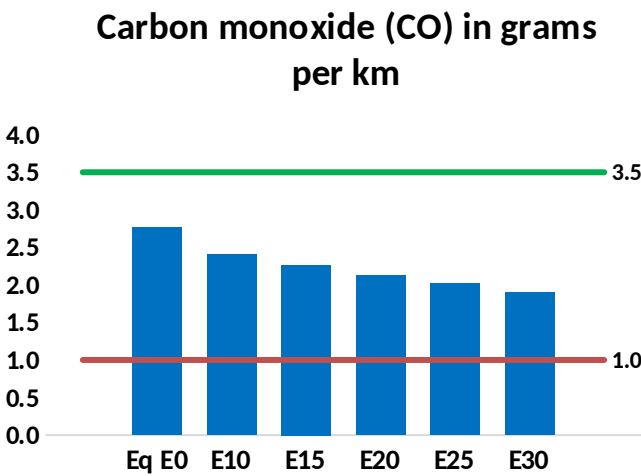
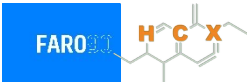


Vehicle Fleet: 3,305,370
Gasoline Vehicle Fleet: 1,146,777

Source: Eurostat, ACEA

Slovenia – Vehicular Emissions

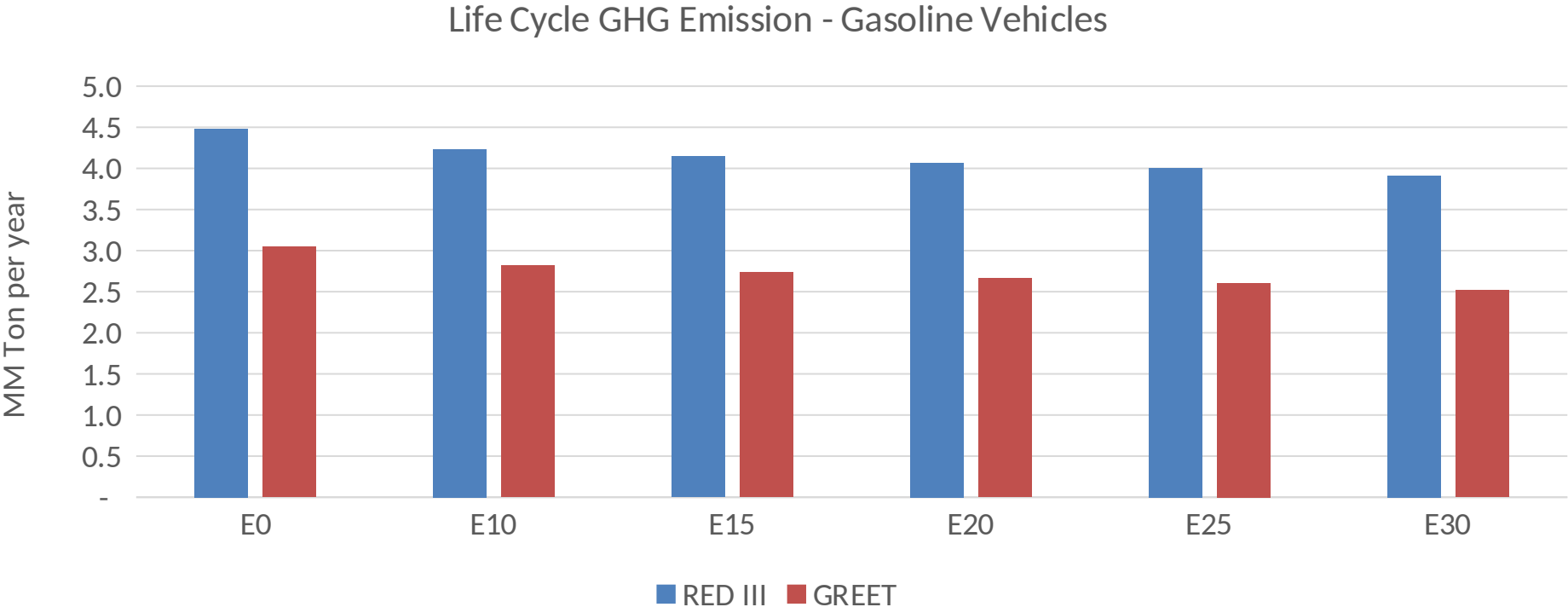
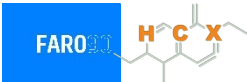
TIER III BIN 125 United States
Euro 6 Standard



Slovenia – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	75,651	70,723	65,794	61,929	58,118	55,212	51,895	-13%	-23%
CO2	5,643,424	5,498,307	5,361,253	5,253,464	5,200,125	5,171,959	5,076,625	-5%	-8%
VOC	10,120	9,537	8,954	8,514	8,089	7,767	7,319	-12%	-20%
NOx	6,220	5,743	5,415	5,248	4,979	4,768	4,520	-13%	-20%
SOx	64	59	54	50	46	42	38	-15%	-28%
NH3	2,008	1,988	1,968	1,977	1,999	2,015	2,028	-2%	0%
N2O	62	62	61	62	63	64	64	-1%	2%
CH4	2,206	2,206	2,206	2,239	2,287	2,329	2,369	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	78	76	74	73	72	72	71	-6%	-8%
Acetaldehyde	193	222	326	496	676	772	912	68%	249%
Formaldehyde	647	629	733	861	902	998	1,086	13%	39%
Benzene	152	146	139	136	135	129	125	-9%	-11%
PM2.5	341	310	279	252	225	197	167	-18%	-34%
PM10	322	293	264	238	213	186	157	-18%	-34%

Slovenia – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	16.5
Target @ 2035 (MMT/year)	9.3
Delta	7.2

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	7.5%	10.2%	12.6%	14.4%	17.3%
RED II/III	0.0%	5.5%	7.5%	9.2%	10.7%	12.7%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	3.2%	4.3%	5.3%	6.1%	7.3%
RED II/III	0.0%	3.4%	4.7%	5.8%	6.6%	7.9%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

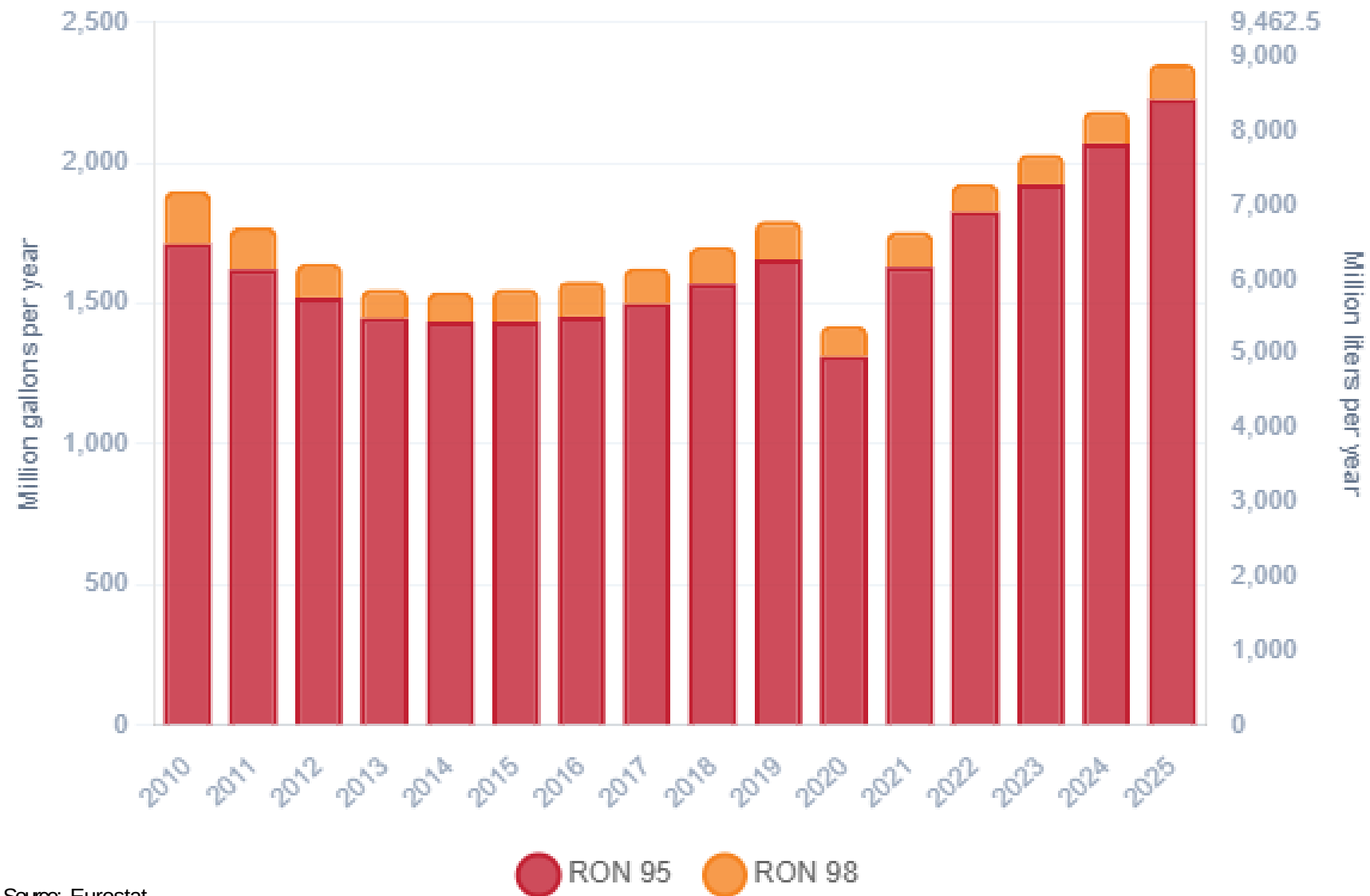
Ethanol Gasoline Blending in Spain



In 2025, gasoline consumption in Spain was 8,903 million liters (2,352 million gallons) and growth rate was 1.4%. Gasoline production and net imports were 10,808 and -1,906 million liters (2,855 and -503 million gallons) correspondingly. Spain complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

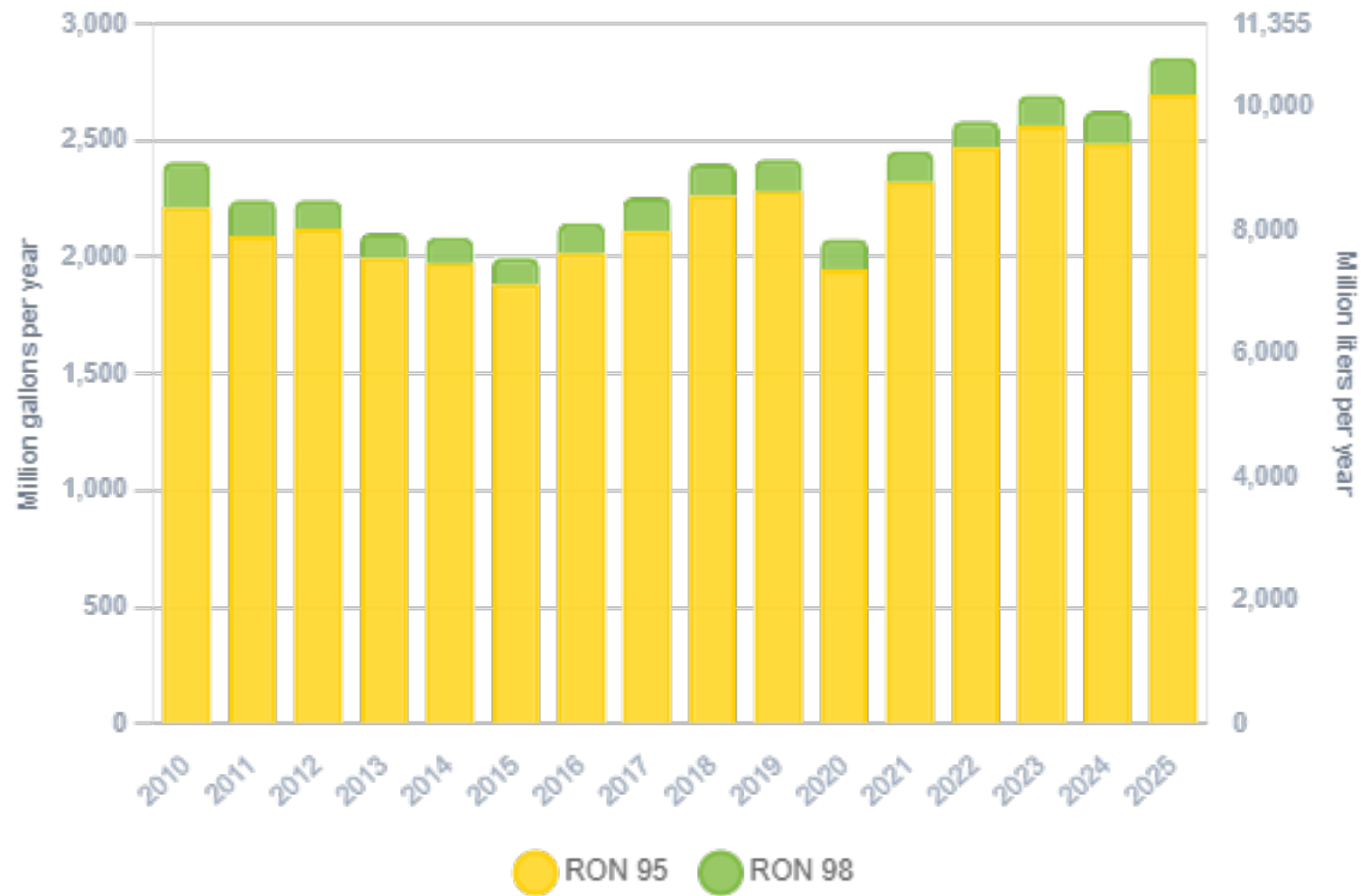
In 2025, ethanol consumption in Spain was 315 million liters (83 million gallons) representing 3.5% of gasoline demand. Ethanol production and Net Imports were 540 and -333 million liters (143 and -88 million gallons) correspondingly. Ethanol sources are Corn 60%, Cereals 25% and Sugar Beet/Wine 40%.

Gasoline Demand in Spain



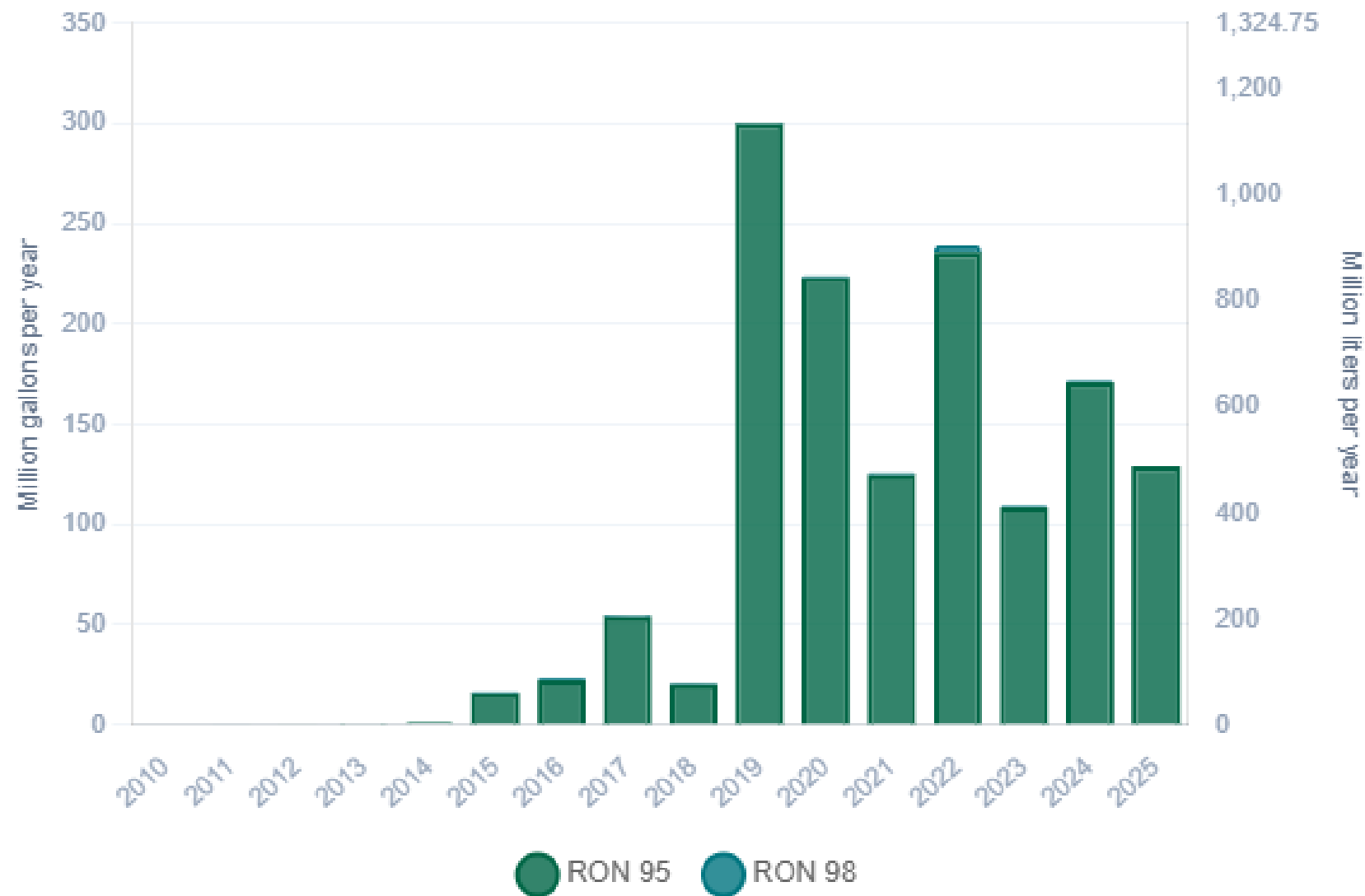
Source: Eurostat

Gasoline Production in Spain



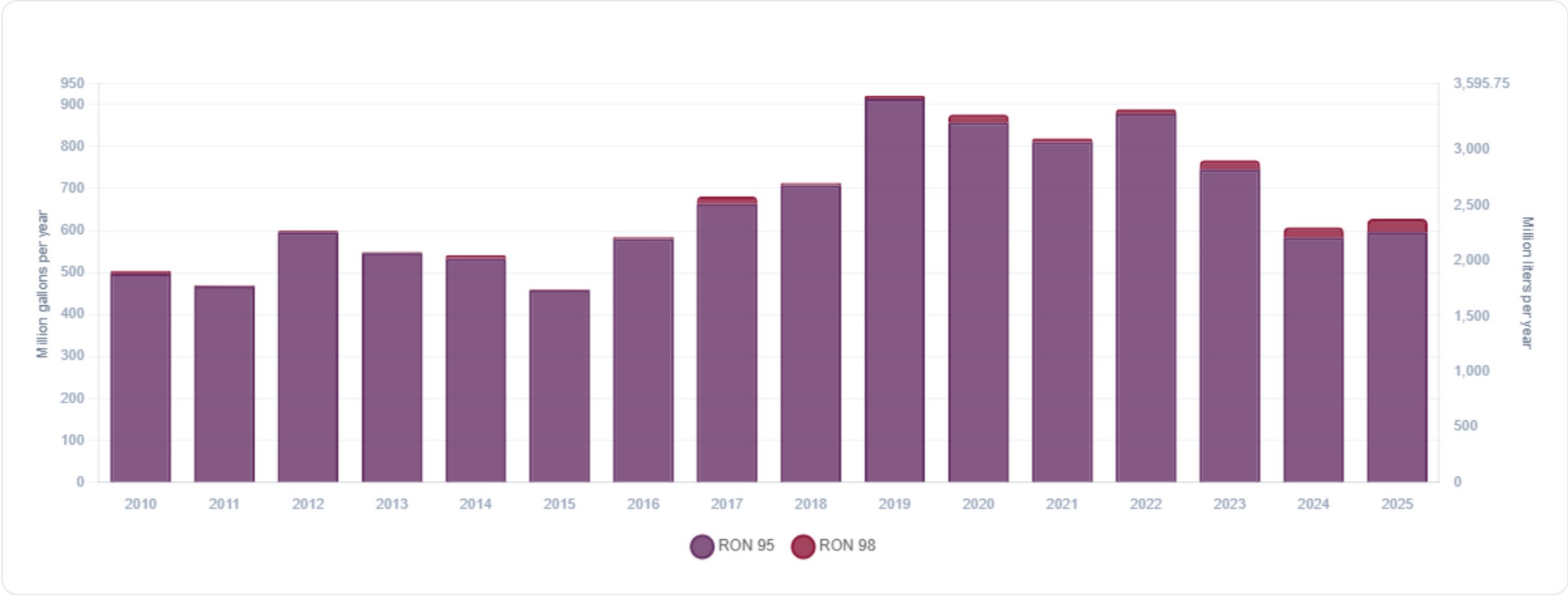
Source: Eurostat

Gasoline Imports in Spain



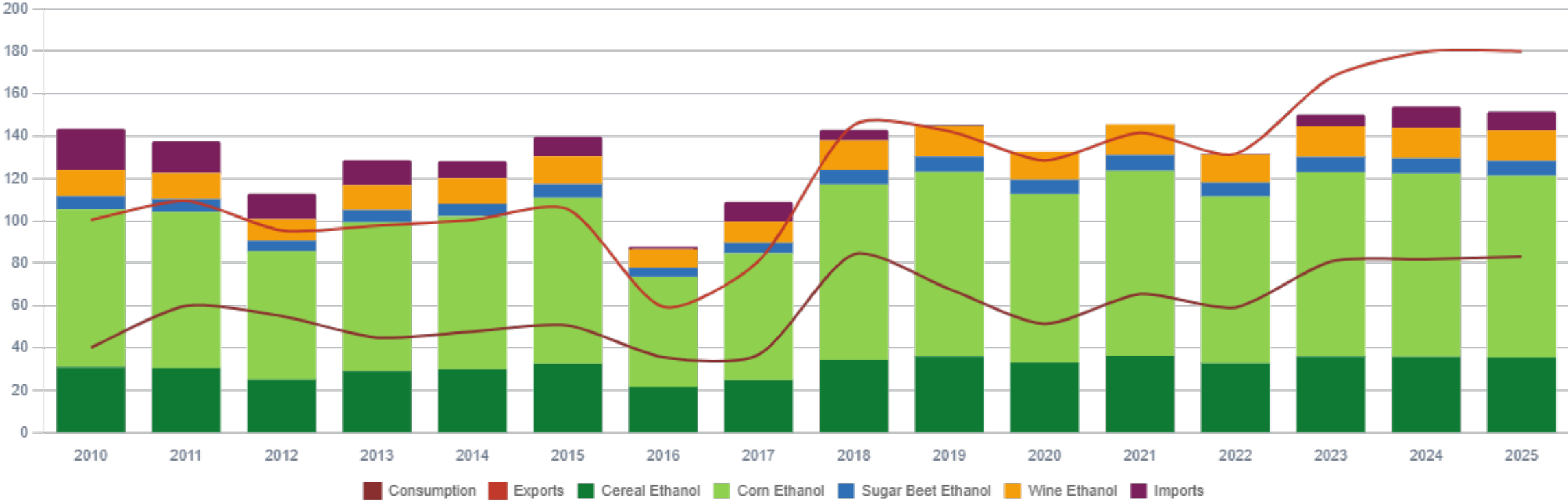
Source: Eurostat

Gasoline Exports in Spain



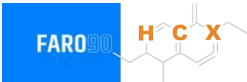
Source: Eurostat

Ethanol Balance in Spain



Source: Eurostat

Gasoline Quality in Spain

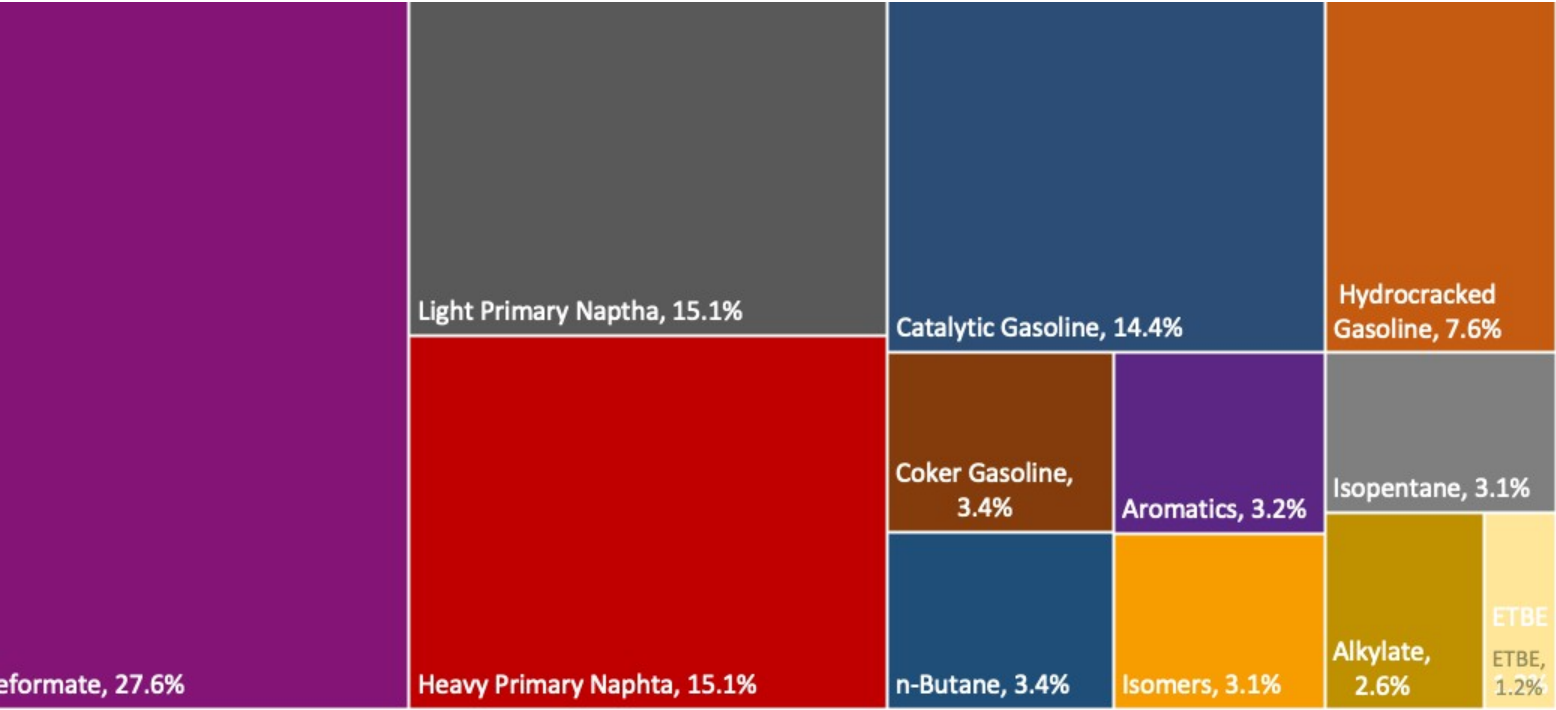


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	11.5							
Advanced Biofuels (%cal)	1.0							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

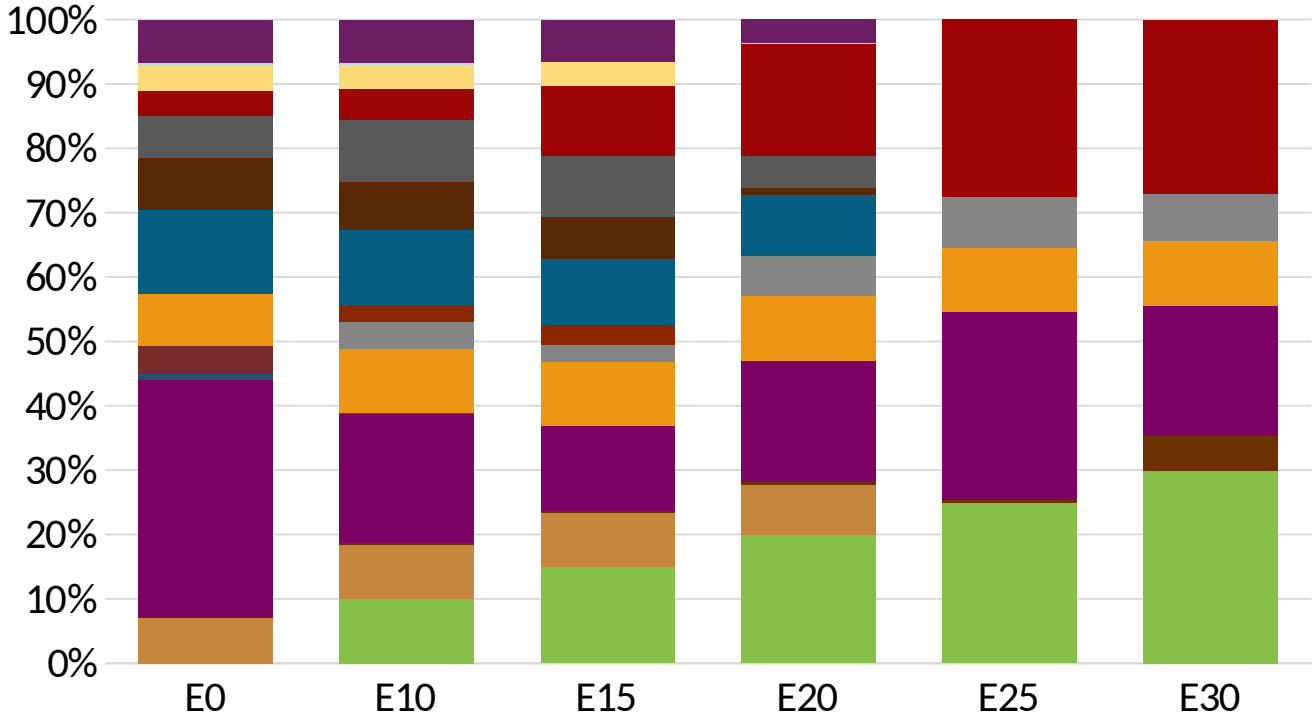
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Spain Available Blending Components



Spain – Gasoline 95 – Constant Octane

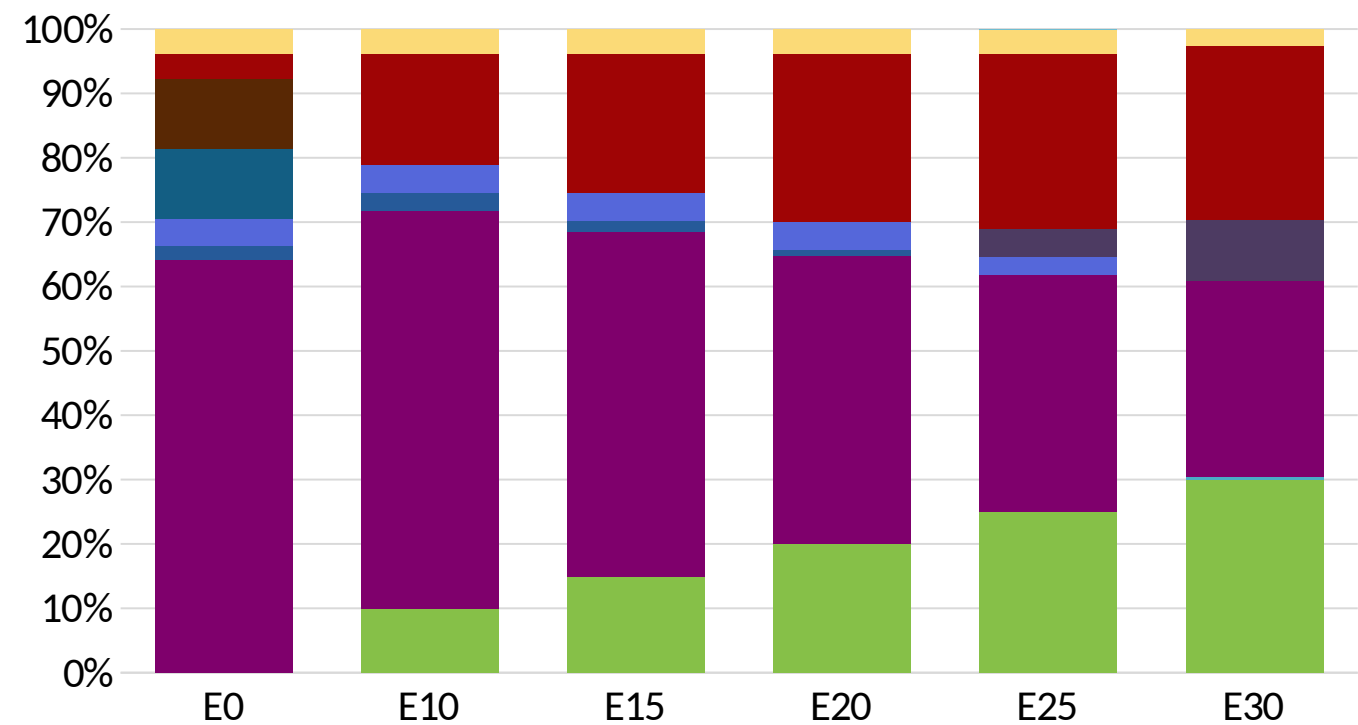


Average CIF 2025 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.3	95.8
Price (US\$/gal)	\$ 2.077	\$ 1.965	\$ 1.886	\$ 1.796	\$ 1.680	\$ 1.617

Source:
Faro90

Spain - Gasoline 98 - Constant Octane

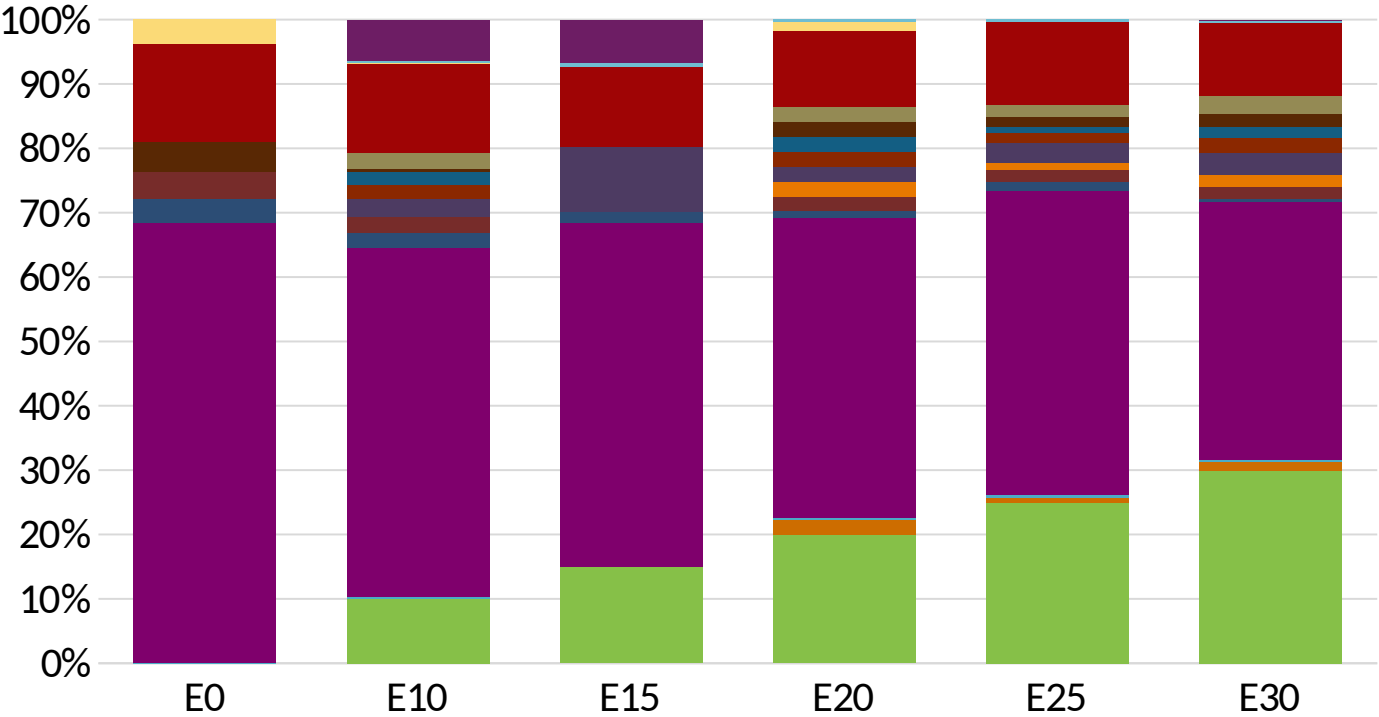


Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.146	\$ 2.022	\$ 1.953	\$ 1.878	\$ 1.800	\$ 1.721

Average CIF 2025 Prices

Source:
Faro90

Spain – Gasoline 95 – Increased Octane

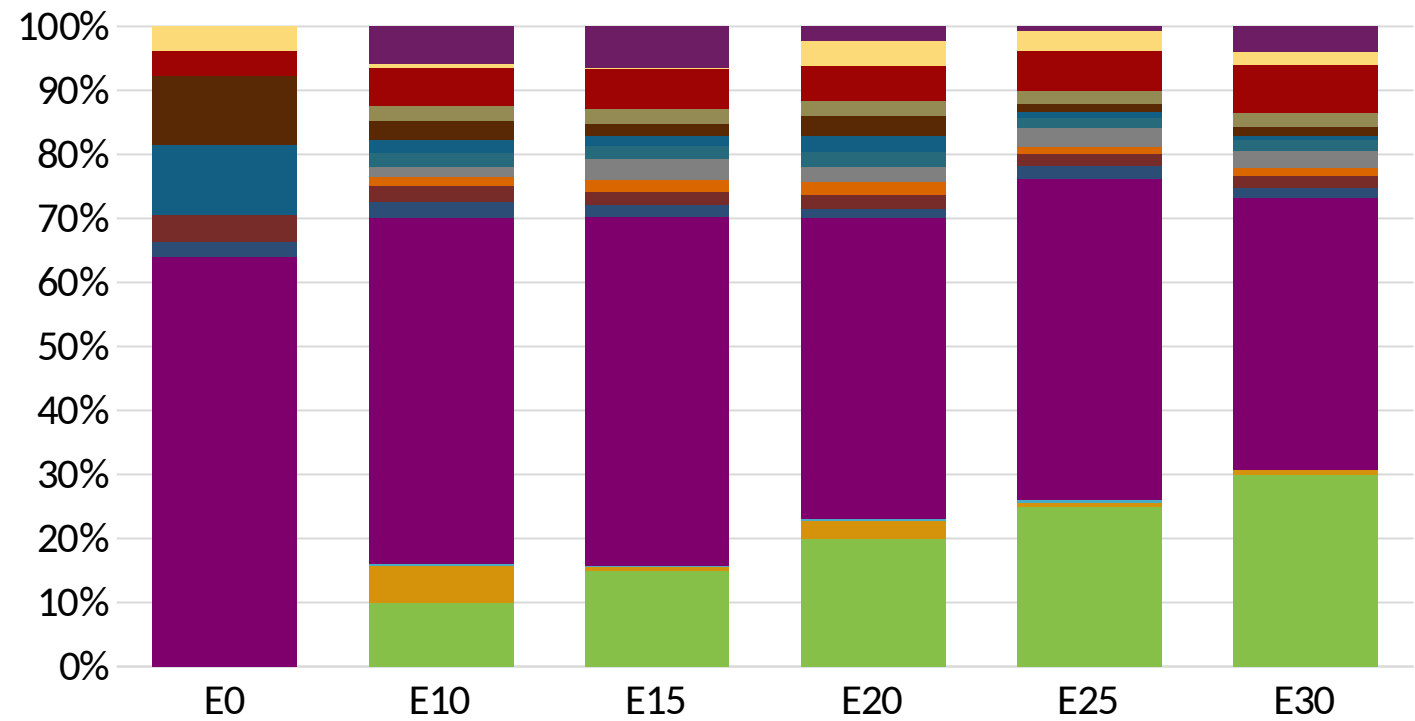


Average CIF 2025
Prices

Octane (RON)	95.0	96.8	98.2	99.9	101.4	102.6
Price (US\$/gal)	\$ 2.009	\$ 2.009	\$ 2.009	\$ 2.009	\$ 2.009	\$ 2.009

Source:
Faro90

Spain – Gasoline 98 – Increased Octane

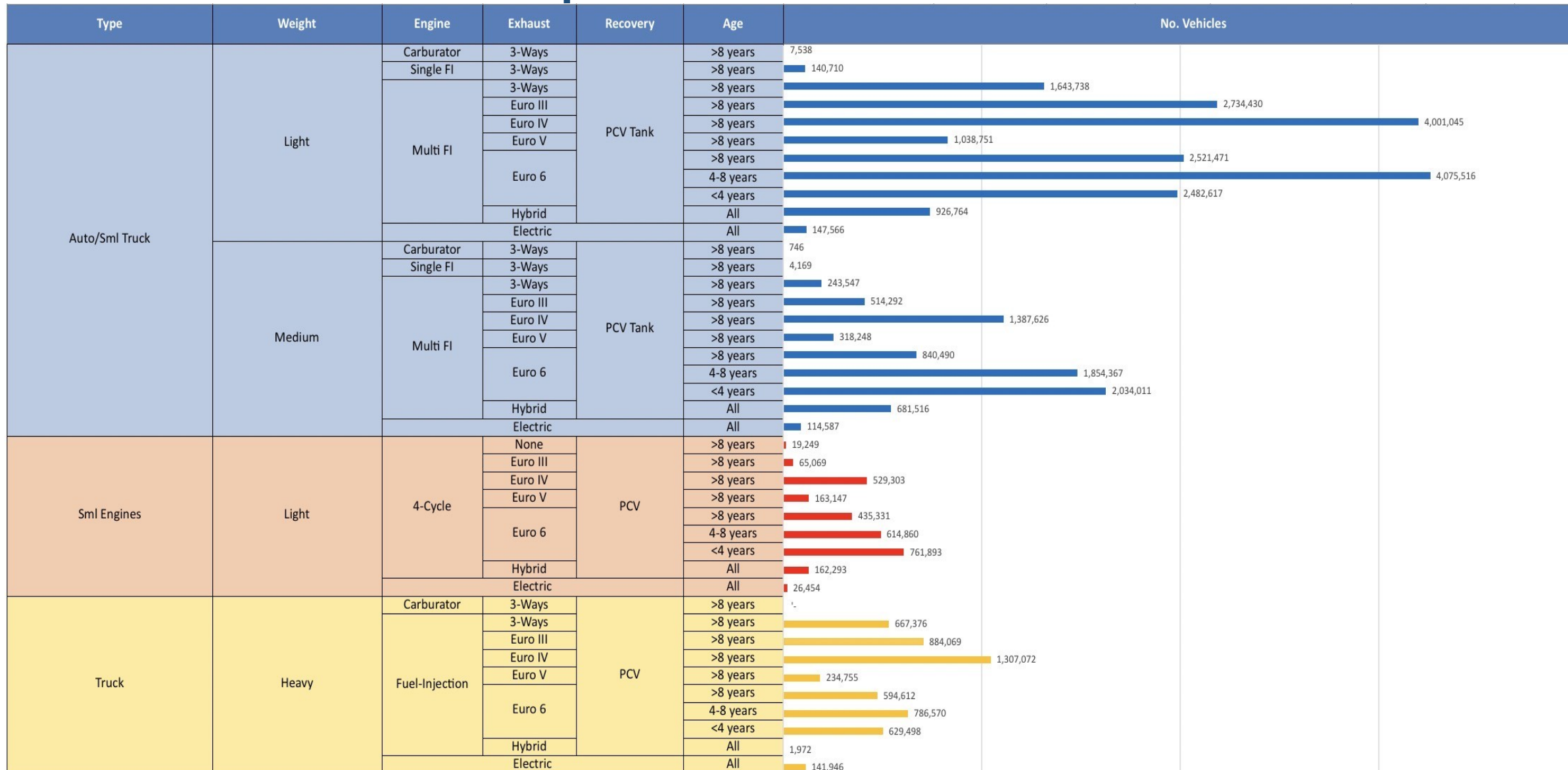


Average CIF 2025
Prices

Octane (RON)	98.0	99.3	100.9	102.5	104.0	105.0
Price (US\$/gal)	\$ 2.146	\$ 2.146	\$ 2.146	\$ 2.146	\$ 2.146	\$ 2.146

Source:
Faro90

Gasoline Vehicle Fleet - Spain

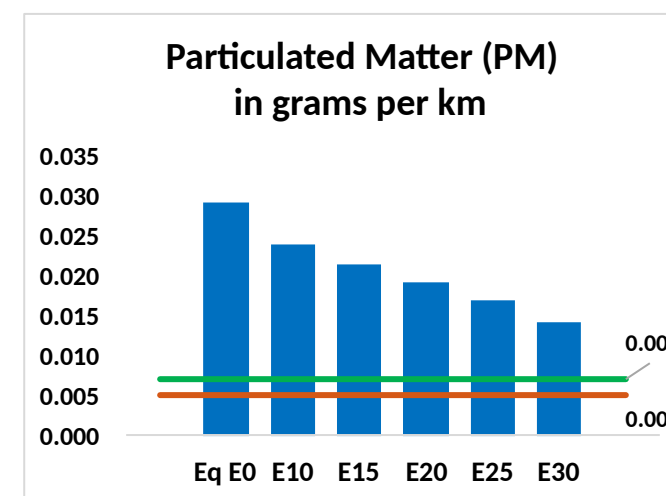
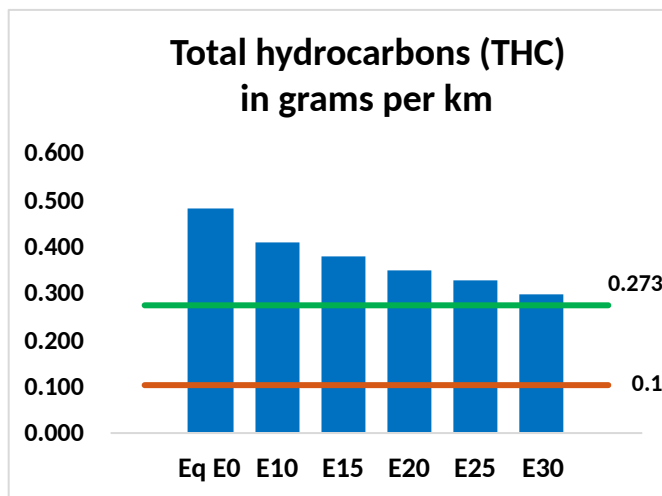
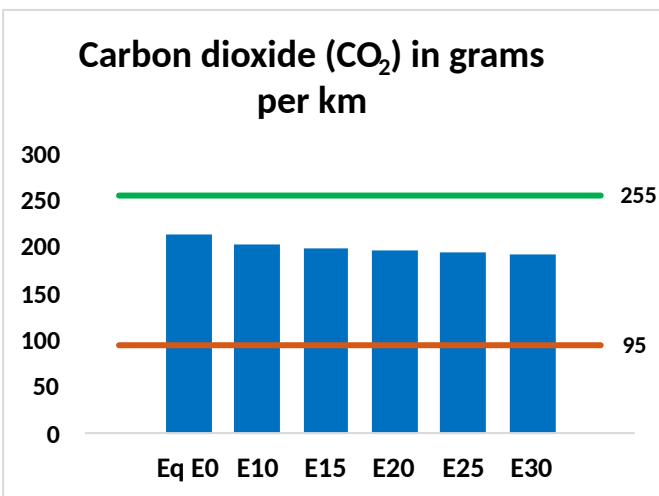
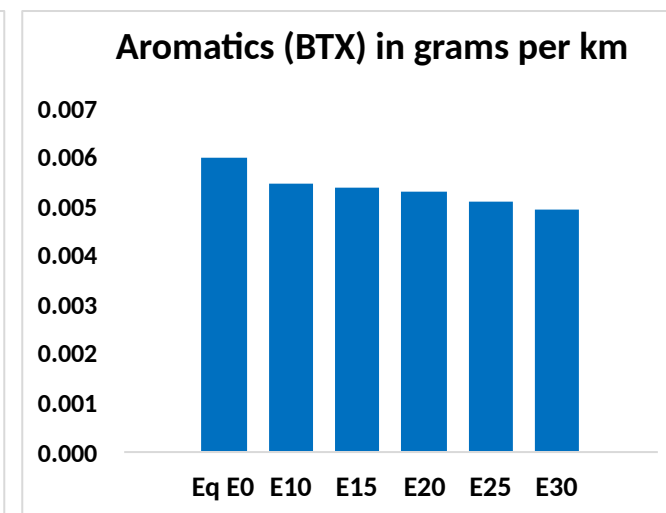
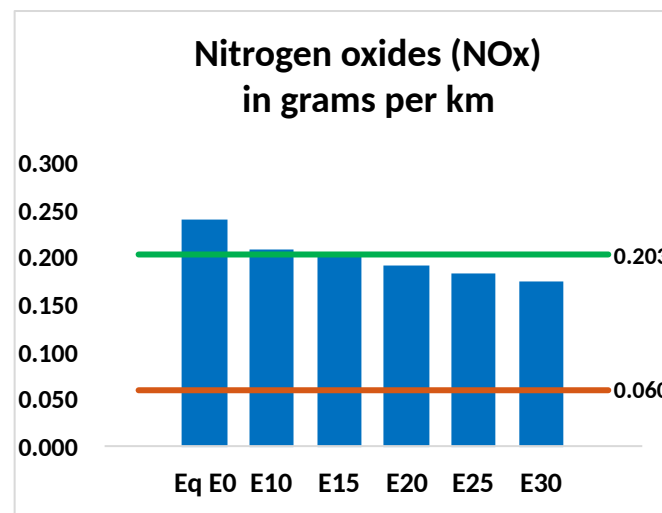
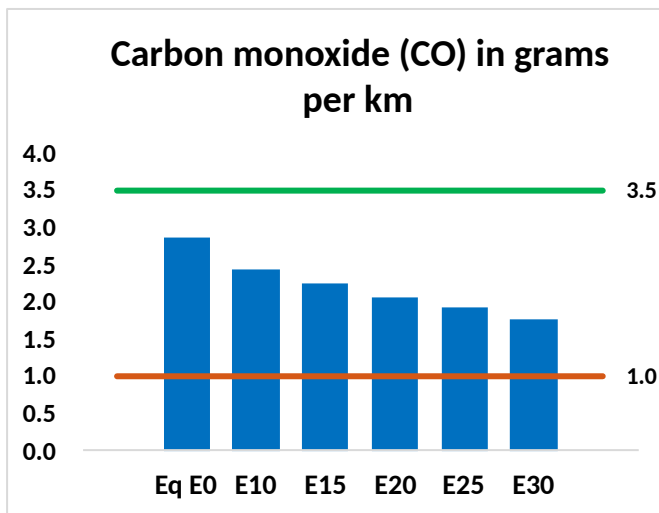
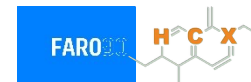


Vehicle Fleet: 35,739,214
Gasoline Vehicle Fleet: 11,668,578

Source: Eurostat, ACEA

Spain – Vehicular Emissions

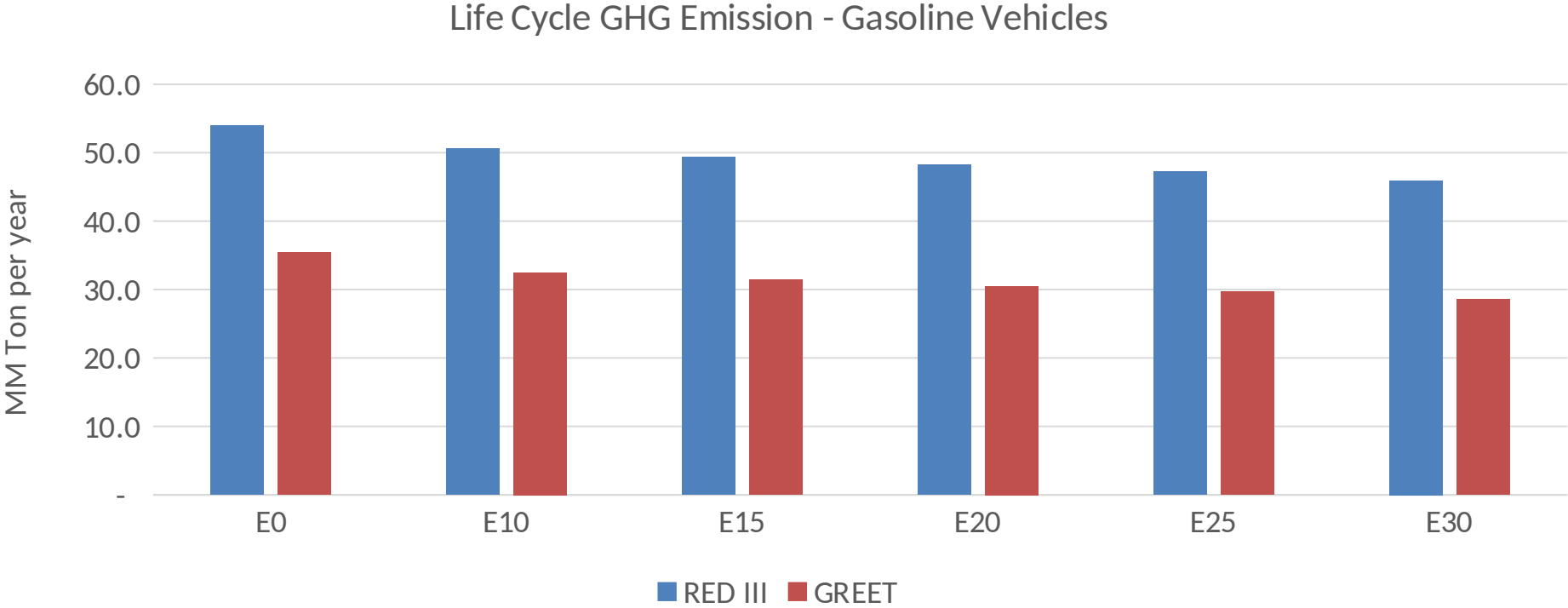
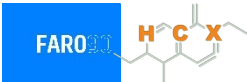
— TIER III BIN 125 United States
— Euro 6 Standard



Spain – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	844,467	778,638	712,809	658,688	604,142	562,175	516,809	-16%	-28%
CO2	62,637,573	61,026,893	59,505,695	58,309,317	57,717,302	57,380,467	56,322,780	-5%	-8%
VOC	140,959	130,441	119,923	111,366	102,784	96,193	87,367	-15%	-27%
NOx	70,807	65,378	61,657	59,762	56,709	54,316	51,498	-13%	-20%
SOx	712	659	605	559	515	468	418	-15%	-28%
NH3	22,136	21,917	21,698	21,801	22,046	22,219	22,362	-2%	0%
N2O	1,043	1,037	1,033	1,038	1,059	1,071	1,083	-1%	2%
CH4	29,799	29,799	29,799	30,245	30,901	31,467	32,004	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	816	786	755	737	721	710	692	-7%	-12%
Acetaldehyde	2,342	2,688	3,944	6,006	8,183	9,351	11,049	68%	249%
Formaldehyde	8,410	8,177	9,532	11,192	11,726	12,972	14,121	13%	39%
Benzene	1,765	1,696	1,607	1,582	1,568	1,500	1,452	-9%	-11%
PM2.5	3,664	3,331	2,998	2,705	2,418	2,111	1,789	-18%	-34%
PM10	4,939	4,490	4,041	3,646	3,260	2,845	2,411	-18%	-34%

Spain – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	269.9
Target @ 2035 (MMT/year)	151.9
Delta	118.0

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.2%	11.2%	13.9%	16.1%	19.3%
RED II/III	0.0%	6.3%	8.6%	10.7%	12.5%	14.9%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	2.5%	3.3%	4.2%	4.8%	5.8%
RED II/III	0.0%	2.9%	3.9%	4.9%	5.7%	6.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

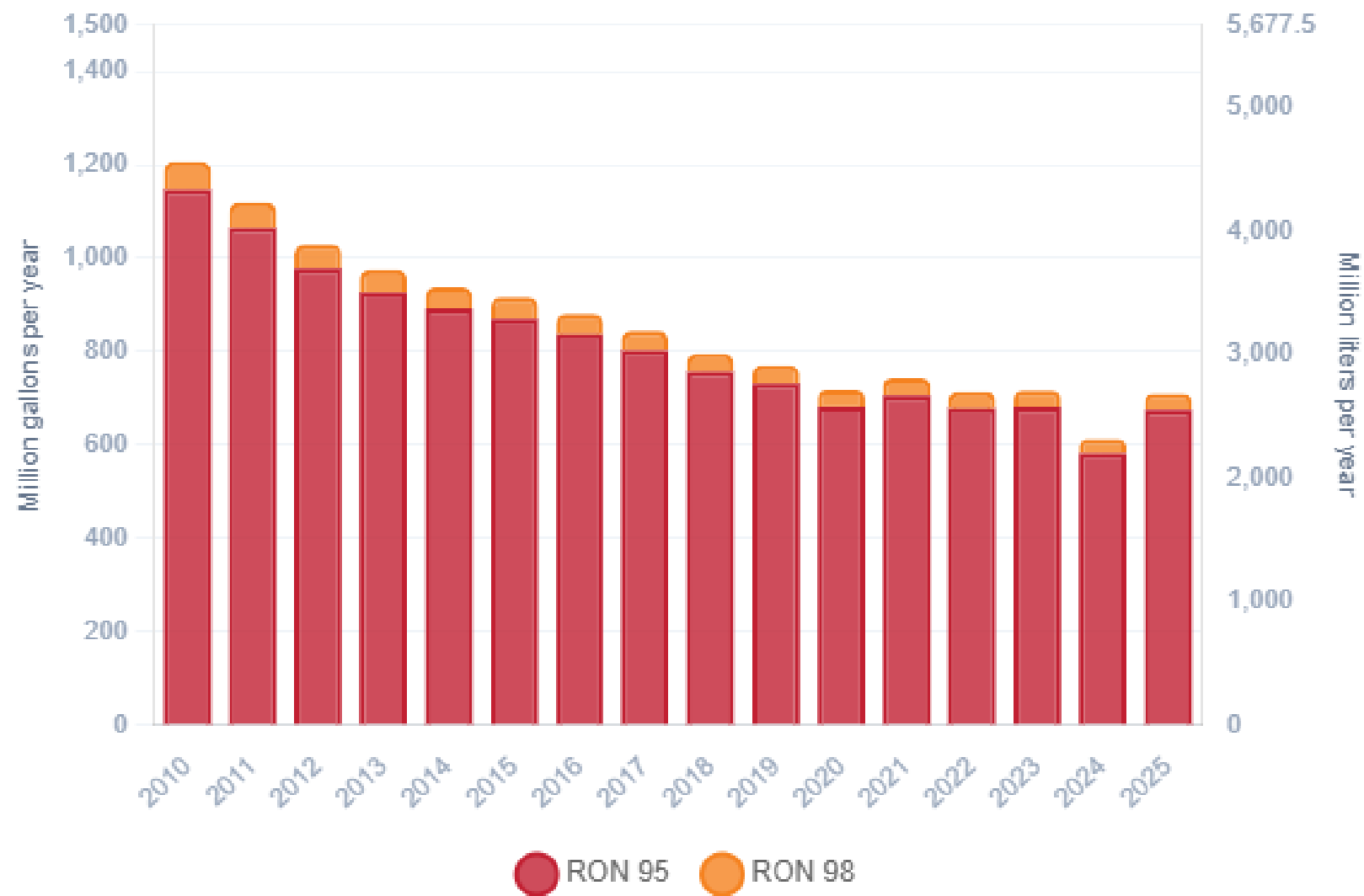
Ethanol Gasoline Blending in Sweden



In 2025, gasoline consumption in Sweden was 2,688 million liters (710 million gallons) and growth rate was -3.5%. Gasoline production and net imports were 6,801 and -4,113 million liters (1,797 and -1,087 million gallons) correspondingly. Sweden complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

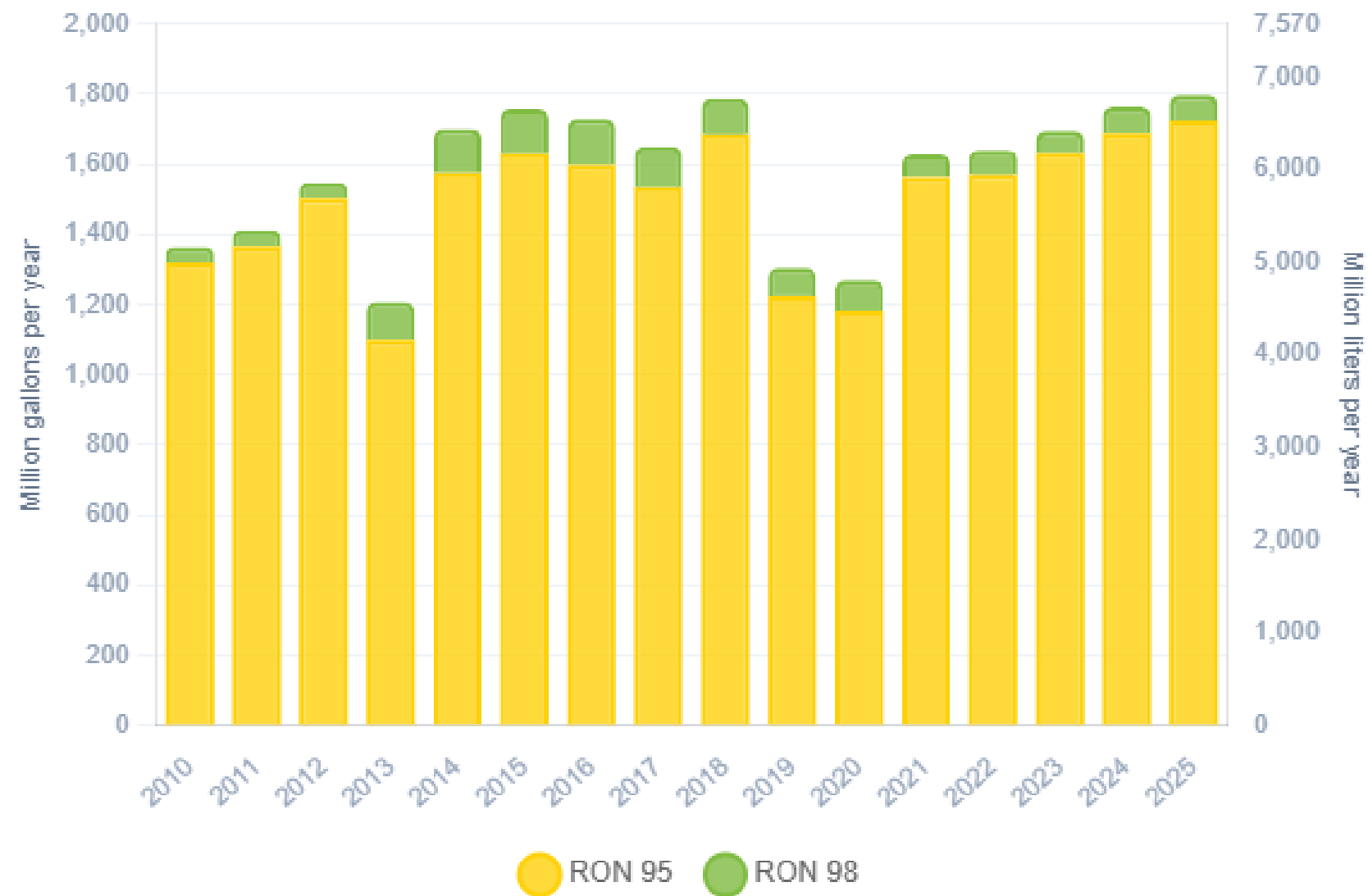
In 2025, ethanol consumption in Sweden was 335 million liters (89 million gallons) representing 12.5% of gasoline demand. Ethanol production and Net Imports were 62 and 273 million liters (16 and 72 million gallons) correspondingly. Ethanol sources are Wood 25% and Cereals 75%.

Gasoline Demand in Sweden



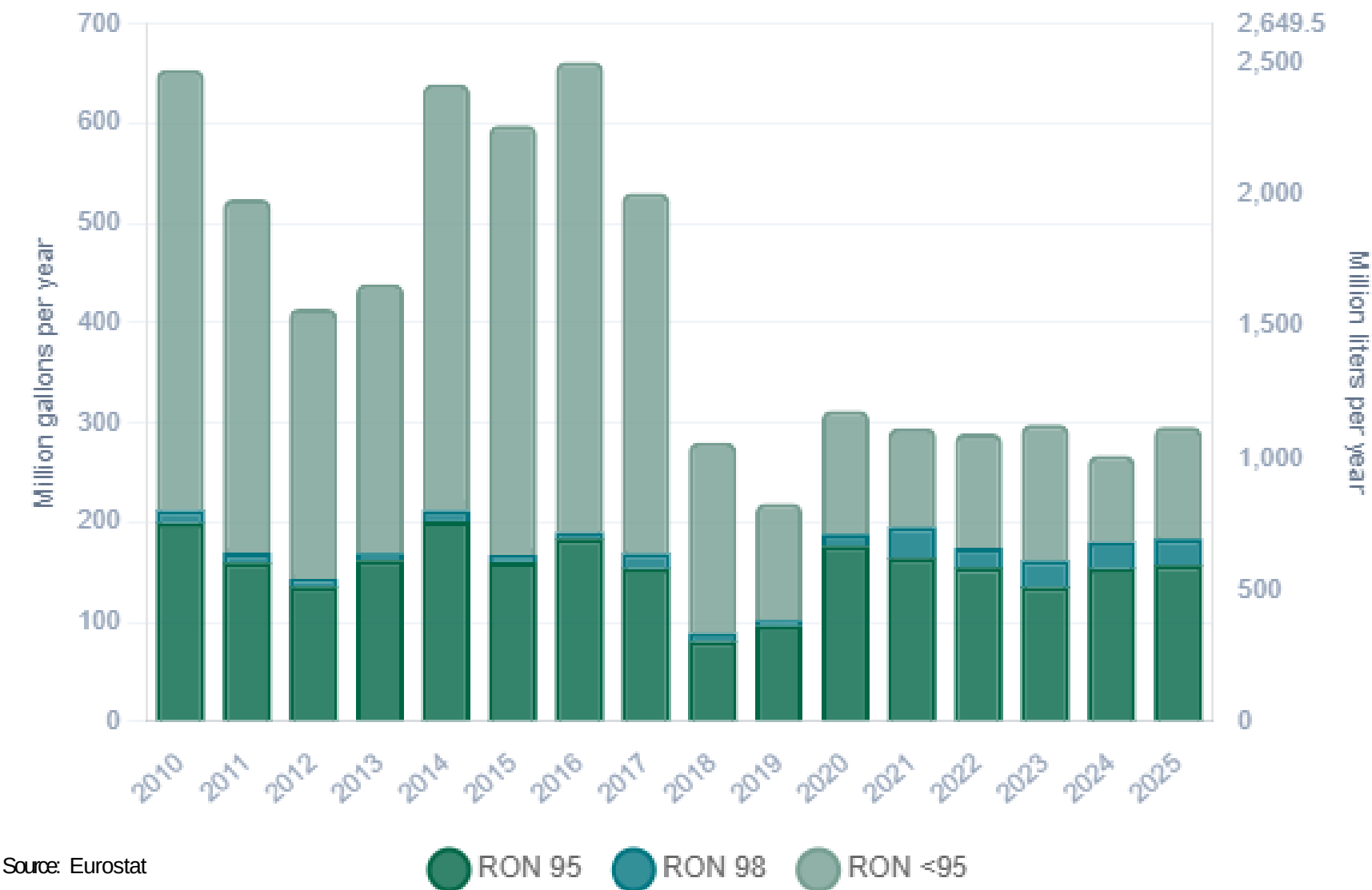
Source: Eurostat

Gasoline Production in Sweden

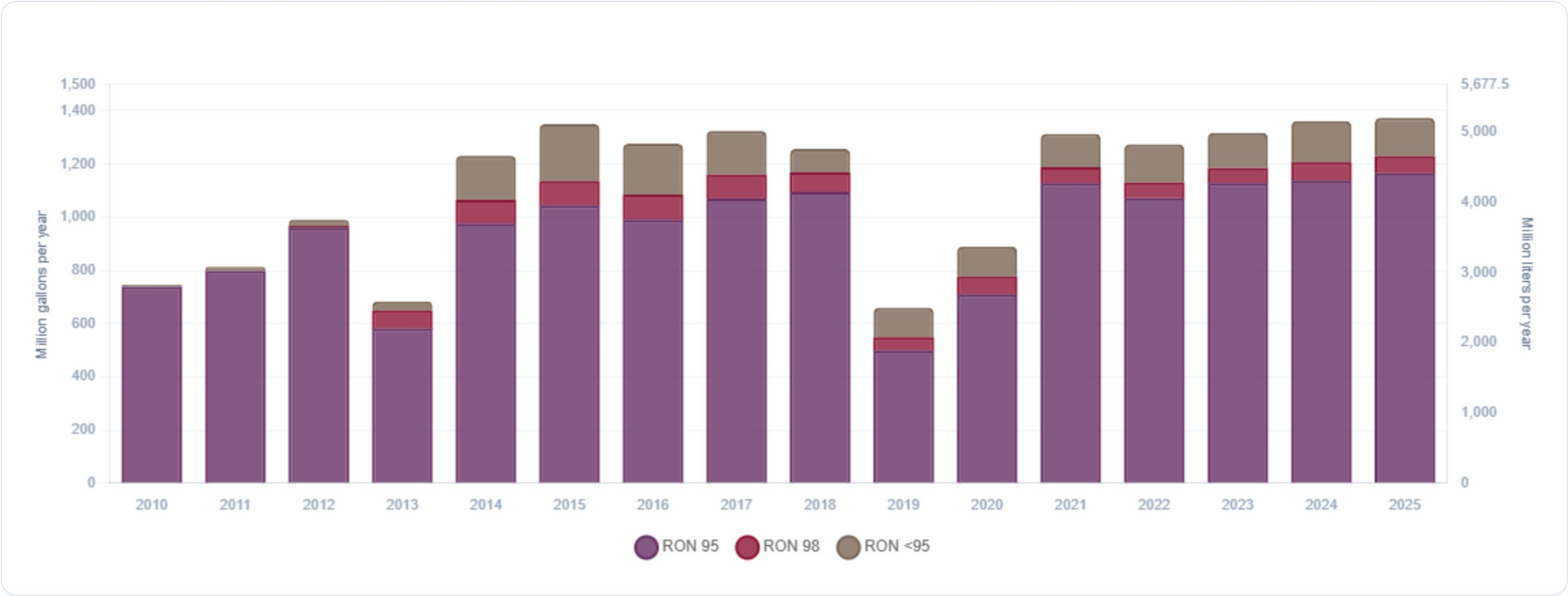


Source: Eurostat

Gasoline Imports in Sweden

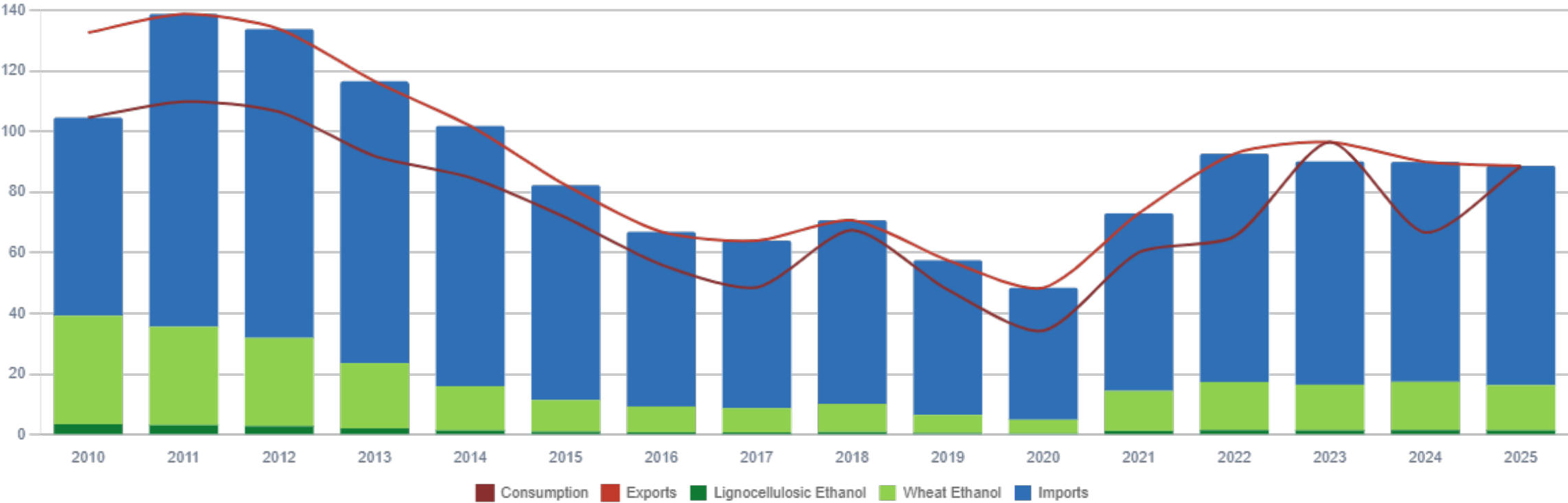


Gasoline Exports in Sweden



Source: Eurostat

Ethanol Balance in Sweden



Source: Eurostat

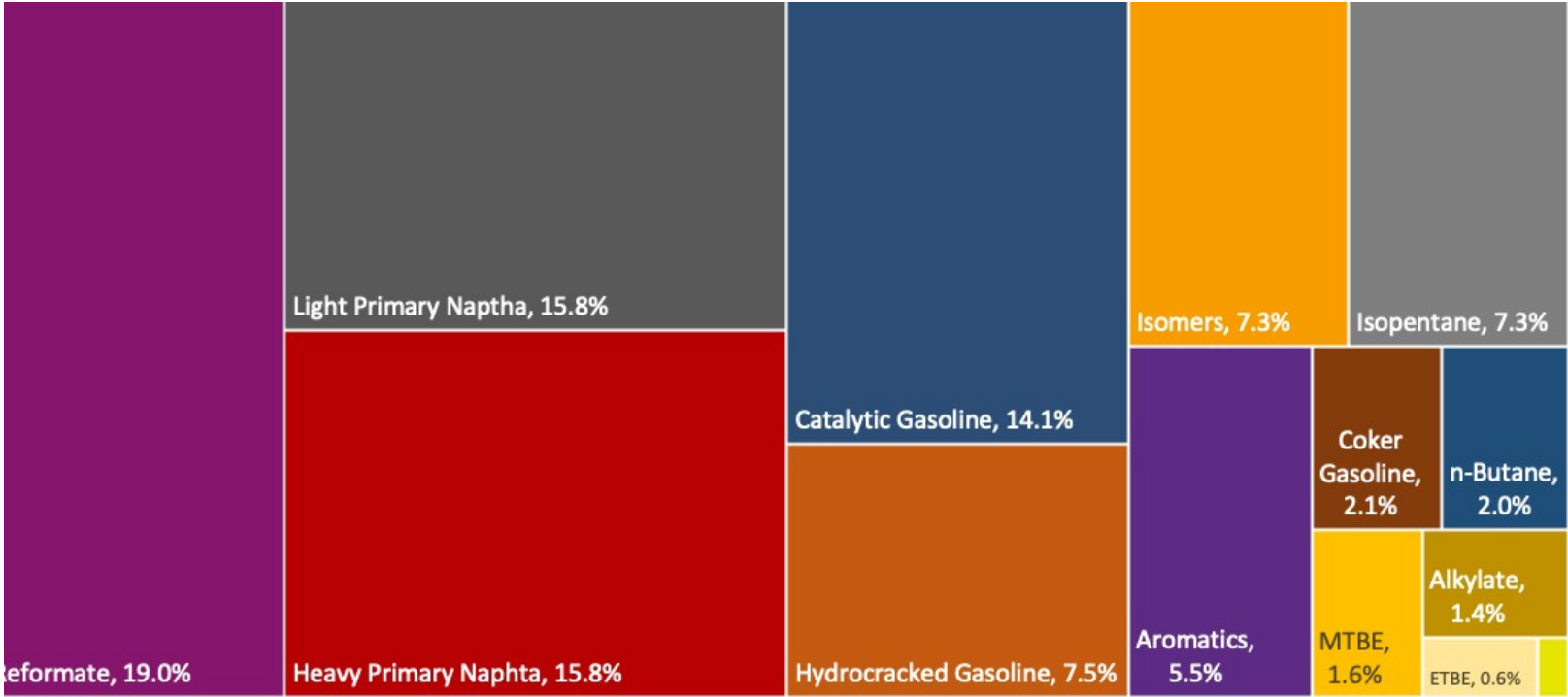
Gasoline Quality in Sweden

Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	6.0							

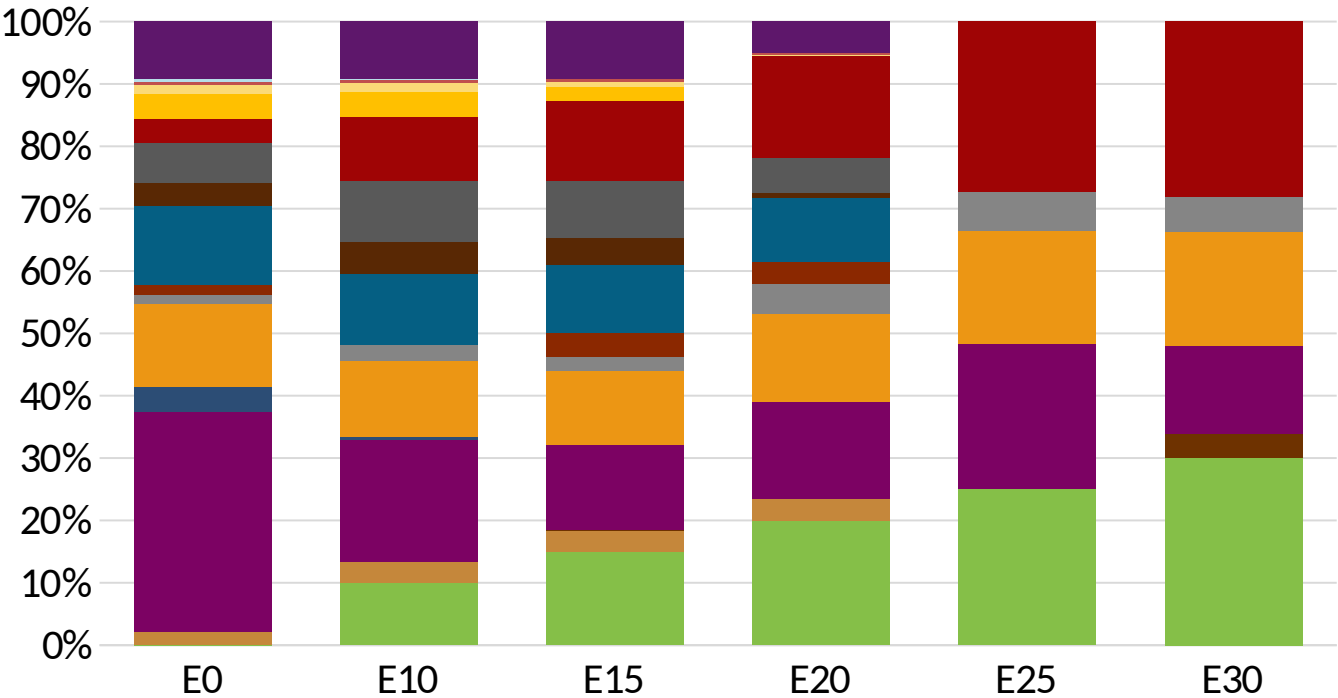
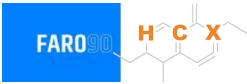
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

Sweden Available Blending Components



Sweden – Gasoline 95 – Constant Octane

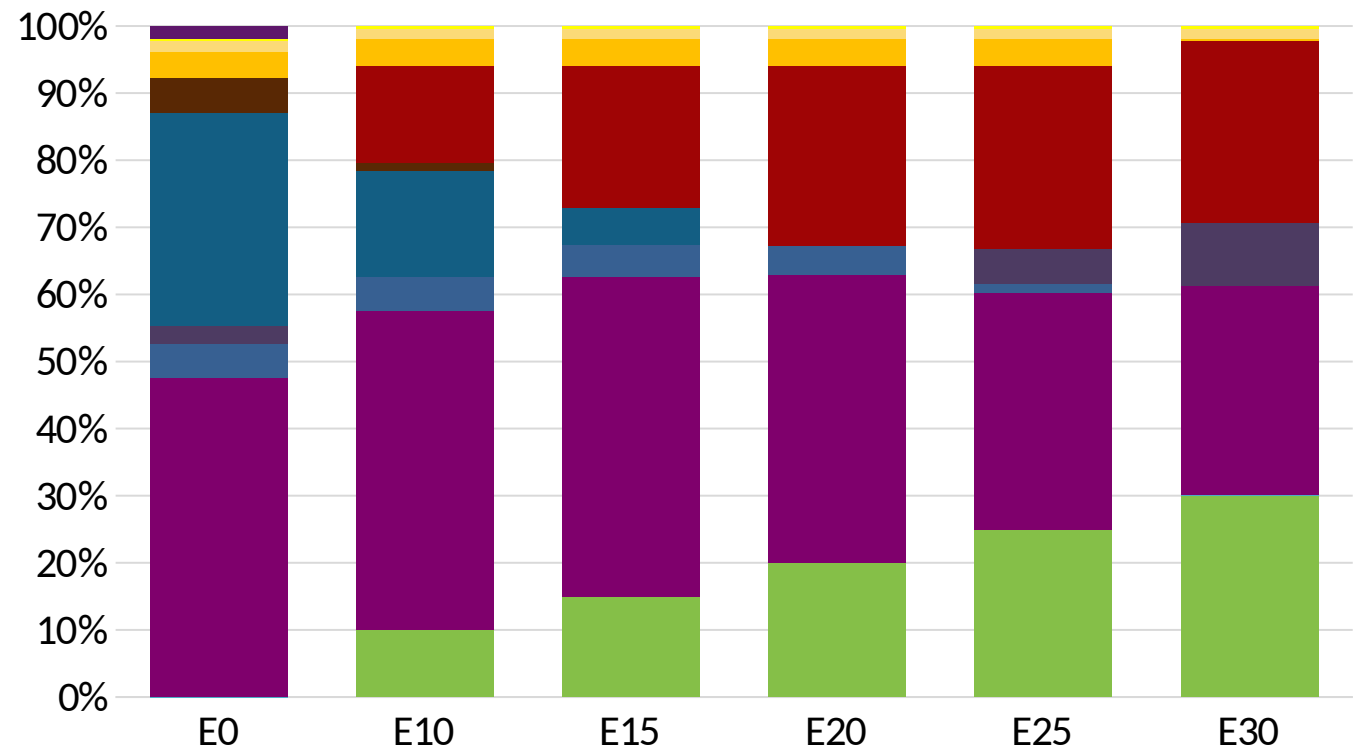
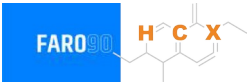


Average CIF 2025 Prices

Octane (RON)	95.0	95.0	95.0	95.0	95.1	95.6
Price (US\$/gal)	\$ 2.109	\$ 1.981	\$ 1.920	\$ 1.839	\$ 1.717	\$ 1.652

Source:
Faro90

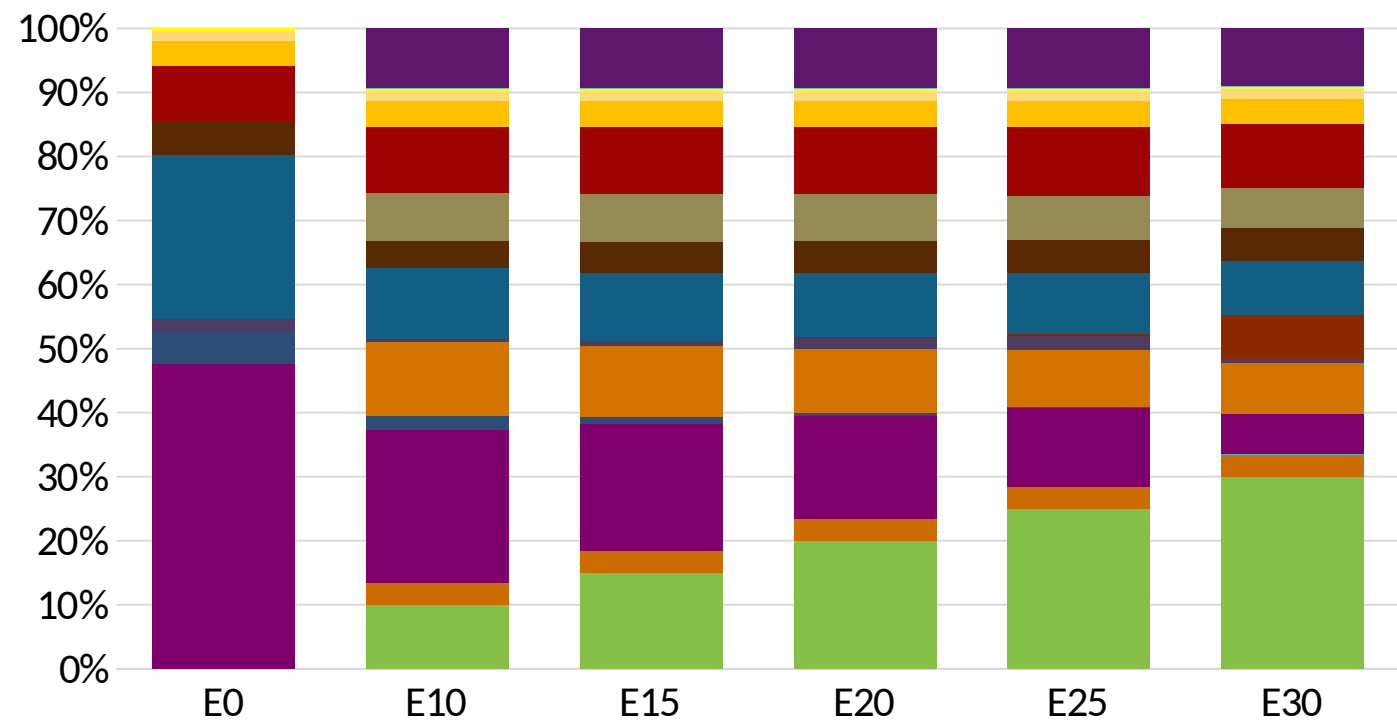
Sweden - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.184	\$ 2.055	\$ 1.983	\$ 1.906	\$ 1.829	\$ 1.762

Average CIF 2025
Prices

Sweden – Gasoline 95 – Increased Octane

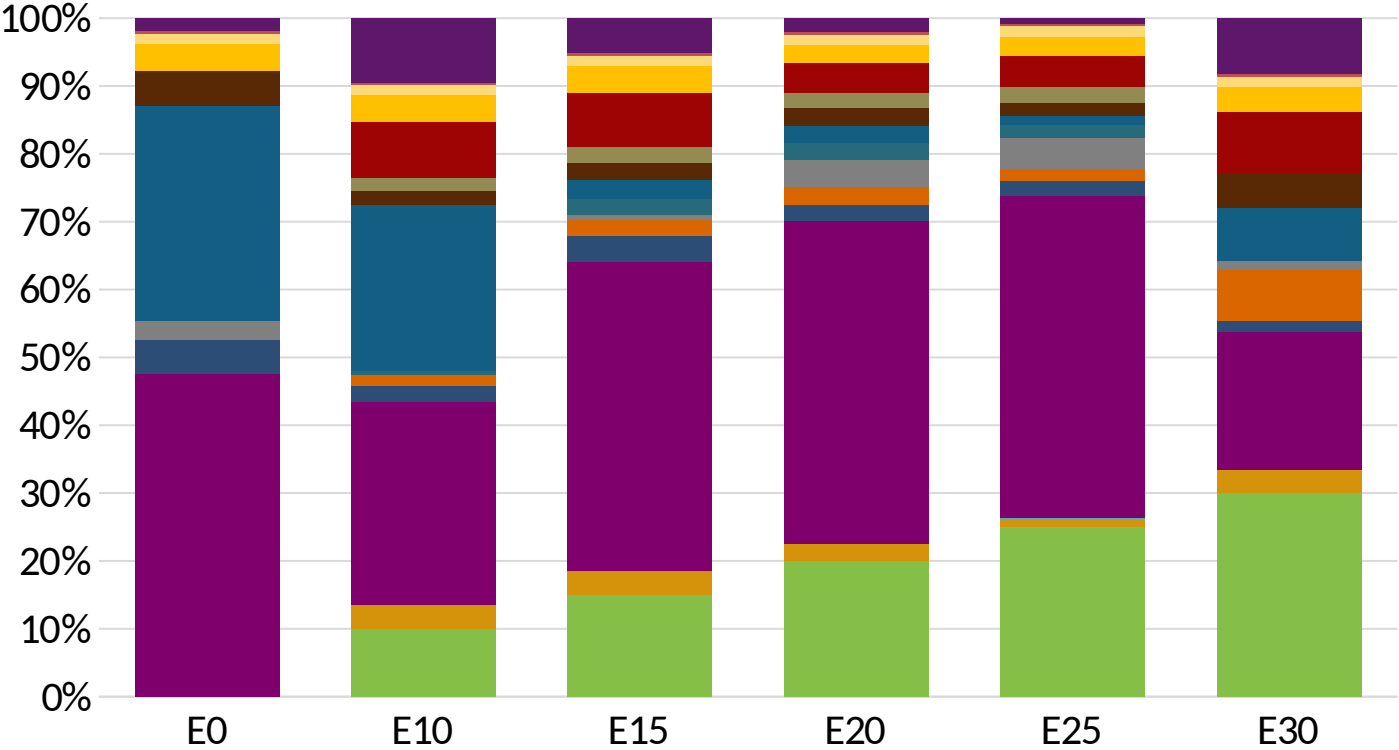


Average CIF 2025
Prices

Octane (RON)	95.0	96.5	97.9	99.4	100.8	102.0
Price (US\$/gal)	\$ 2.046	\$ 2.046	\$ 2.046	\$ 2.046	\$ 2.046	\$ 2.046

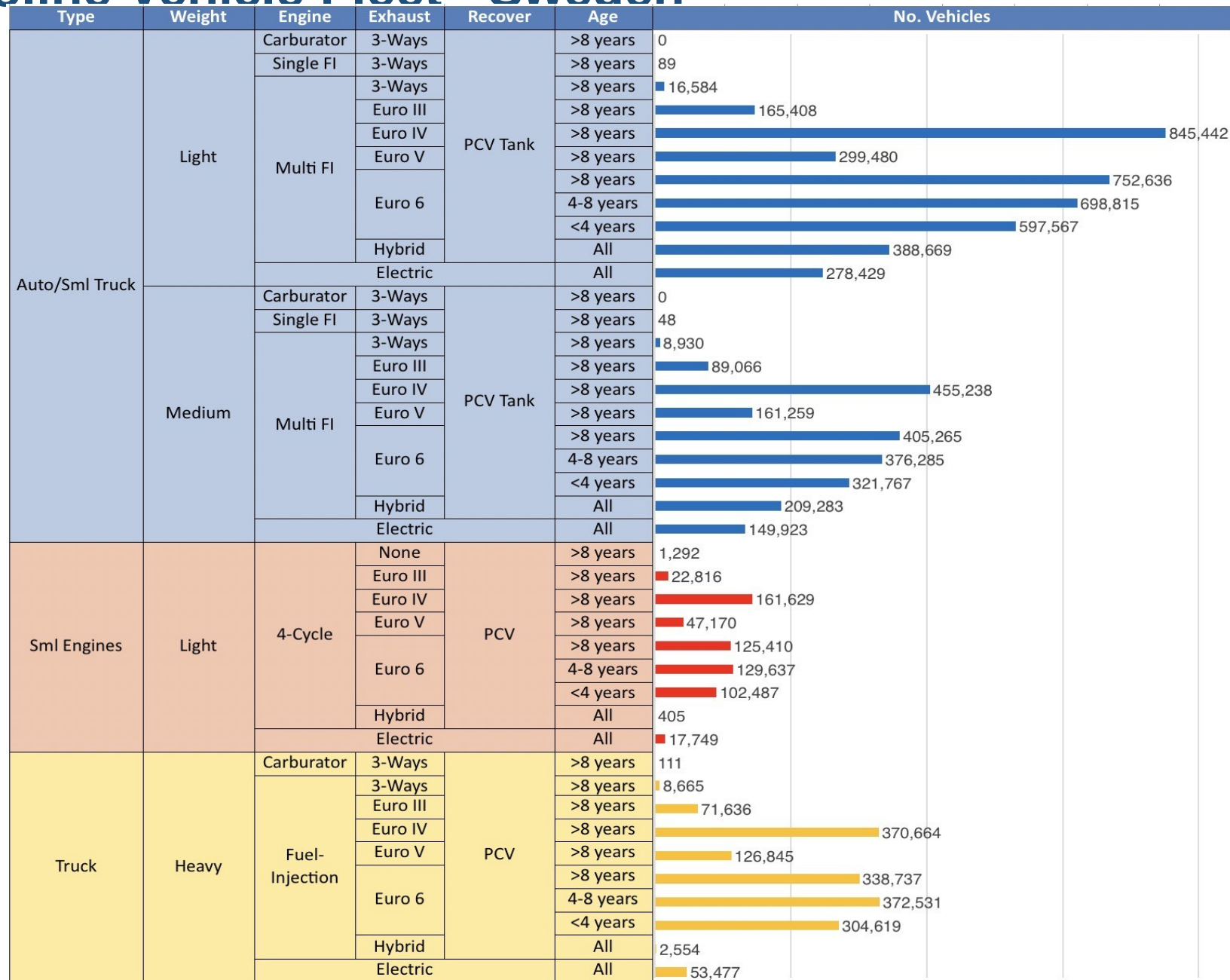
Source:
Faro90

Sweden – Gasoline 98 – Increased Octane



Average CIF 2025
Prices

Gasoline Vehicle Fleet - Sweden



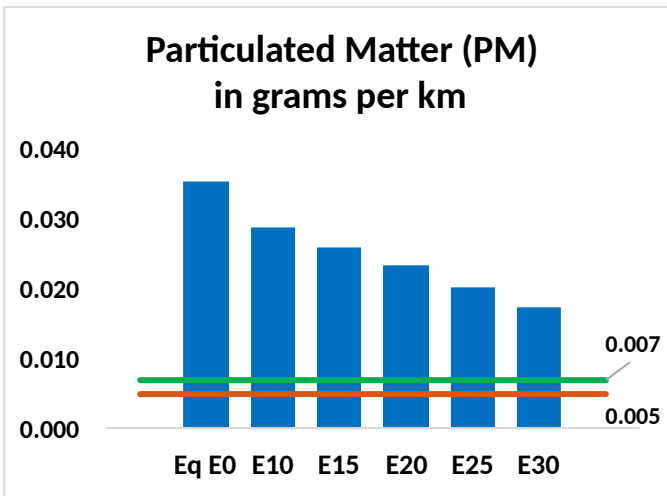
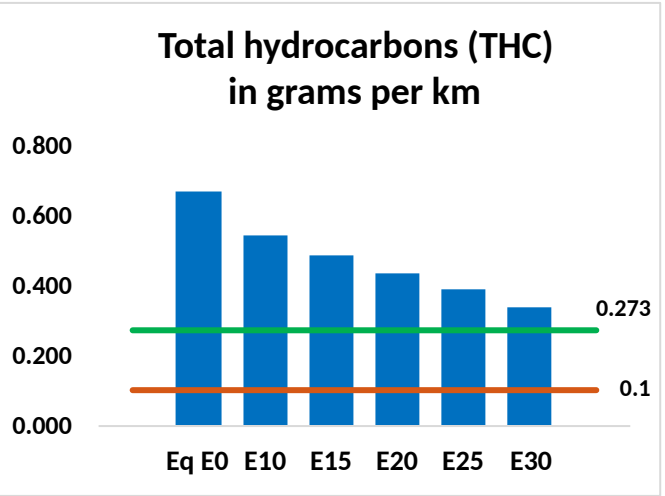
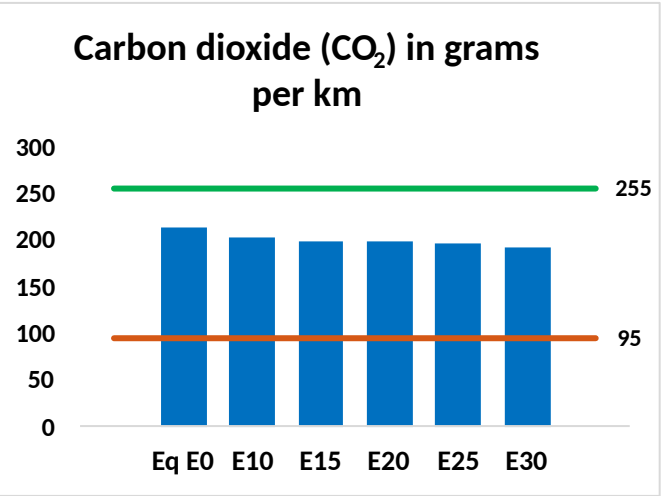
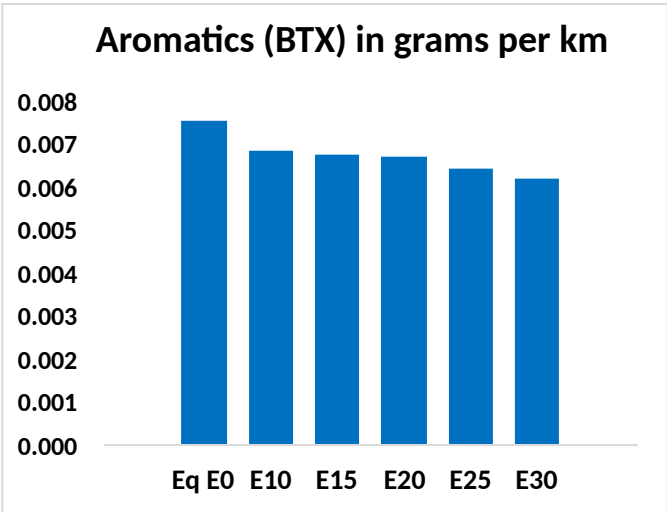
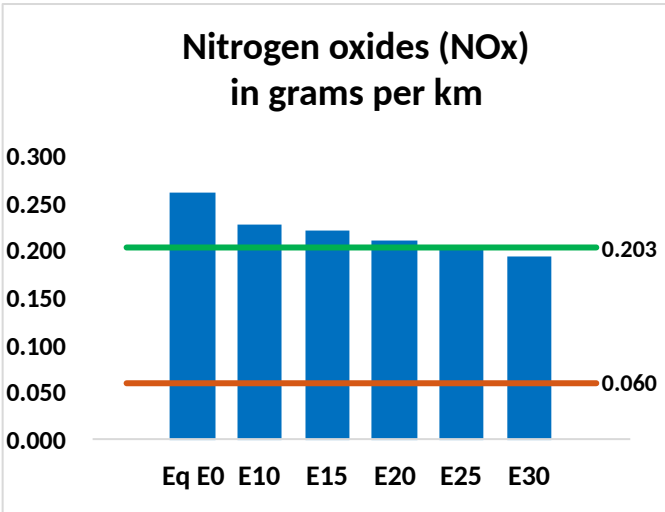
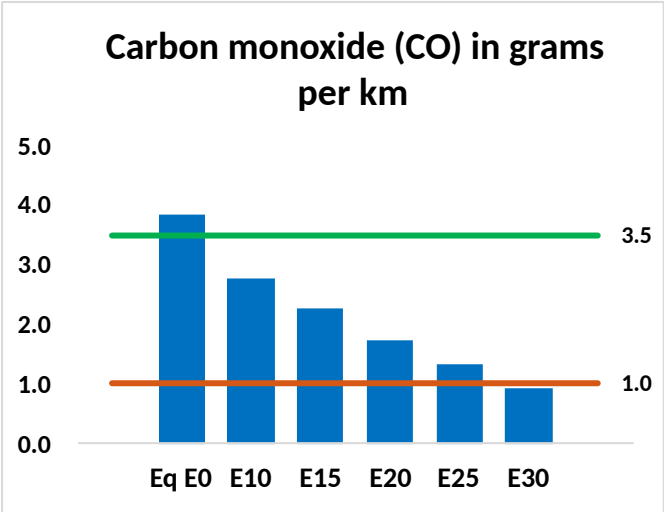
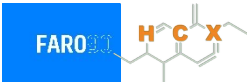
Vehicle Fleet: 8,478,616

Gasoline Vehicle Fleet: 3,347,360

Source: Eurostat, ACEA

Sweden – Vehicular Emissions

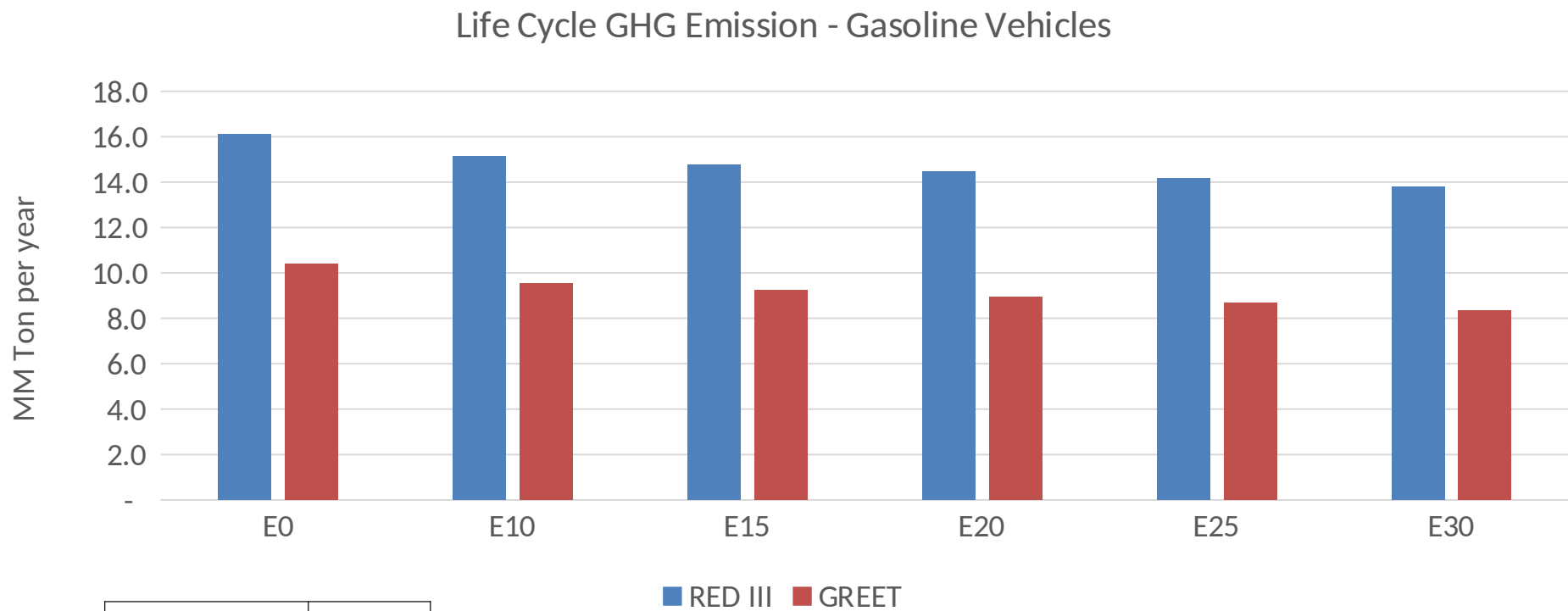
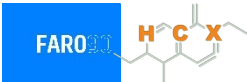
TIER III BIN 125 United States
Euro 6 Standard



Sweden – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	321,658	275,757	229,855	188,109	144,197	109,857	76,308	-29%	-55%
CO2	17,889,118	17,429,112	16,994,662	16,652,980	16,483,902	16,388,726	16,086,635	-5%	-8%
VOC	55,718	50,501	45,284	40,827	36,258	32,720	28,087	-19%	-35%
NOx	21,849	20,174	19,086	18,563	17,685	17,014	16,213	-13%	-19%
SOx	205	190	174	161	149	135	121	-15%	-28%
NH3	6,443	6,380	6,316	6,346	6,417	6,467	6,509	-2%	0%
N2O	319	318	316	318	325	328	332	-1%	2%
CH4	11,289	11,289	11,289	11,459	11,707	11,921	12,125	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	270	251	231	215	199	186	170	-15%	-26%
Acetaldehyde	831	954	1,399	2,131	2,903	3,318	3,920	68%	249%
Formaldehyde	3,153	3,065	3,573	4,196	4,396	4,863	5,294	13%	39%
Benzene	630	605	574	565	560	535	518	-9%	-11%
PM2.5	1,029	935	842	760	679	593	502	-18%	-34%
PM10	1,917	1,743	1,569	1,415	1,265	1,104	936	-18%	-34%

Sweden – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	35.9
Target @ 2035 (MMT/year)	20.2
Delta	15.7

	RED III GREET					
%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.4%	11.3%	14.1%	16.4%	19.7%
RED II/III	0.0%	6.0%	8.2%	10.2%	12.0%	14.2%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	5.5%	7.5%	9.3%	10.9%	13.0%
RED II/III	0.0%	6.2%	8.4%	10.5%	12.3%	14.6%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90

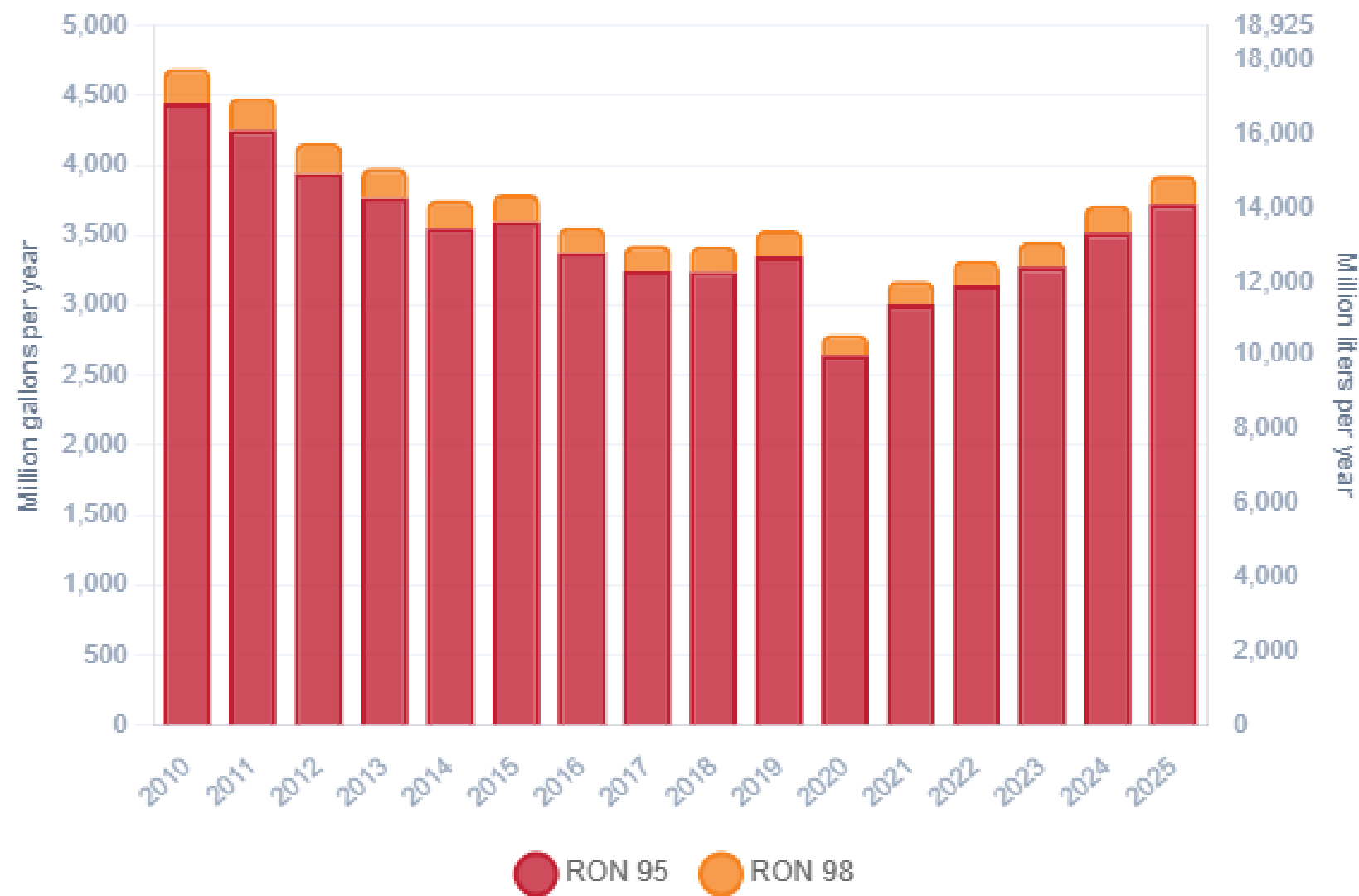
Ethanol Gasoline Blending in United Kingdom



In 2025, gasoline consumption in United Kingdom was 14,873 million liters (3,929 million gallons) and growth rate was -1.2%. Gasoline production and net imports were 19,506 and -4,633 million liters (5,152 and -1,224 million gallons) correspondingly. United Kingdom complies with the European gasoline standards and offers two main grades: RON 95 and RON 98 with an average octane of 95.2 RON.

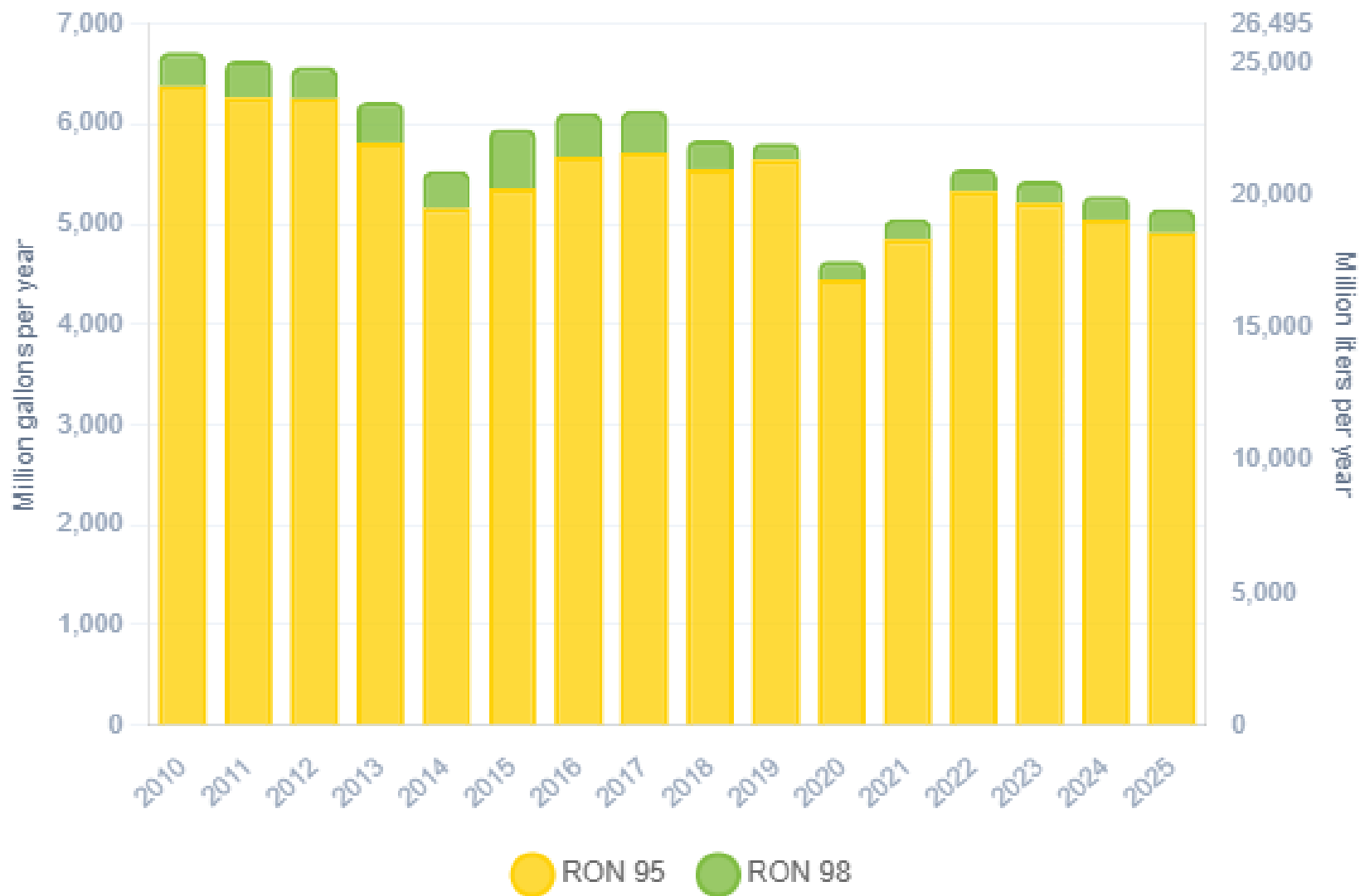
In 2025, ethanol consumption in United Kingdom was 1,551 million liters (410 million gallons) representing 10.4% of gasoline demand. Ethanol production and Net Imports were 480 and 1,071 million liters (127 and 283 million gallons) correspondingly. Ethanol sources are Corn 84%, Cereals 13% and Sugar Beet 3%.

Gasoline Demand in United Kingdom



Source: Eurostat

Gasoline Production in United Kingdom



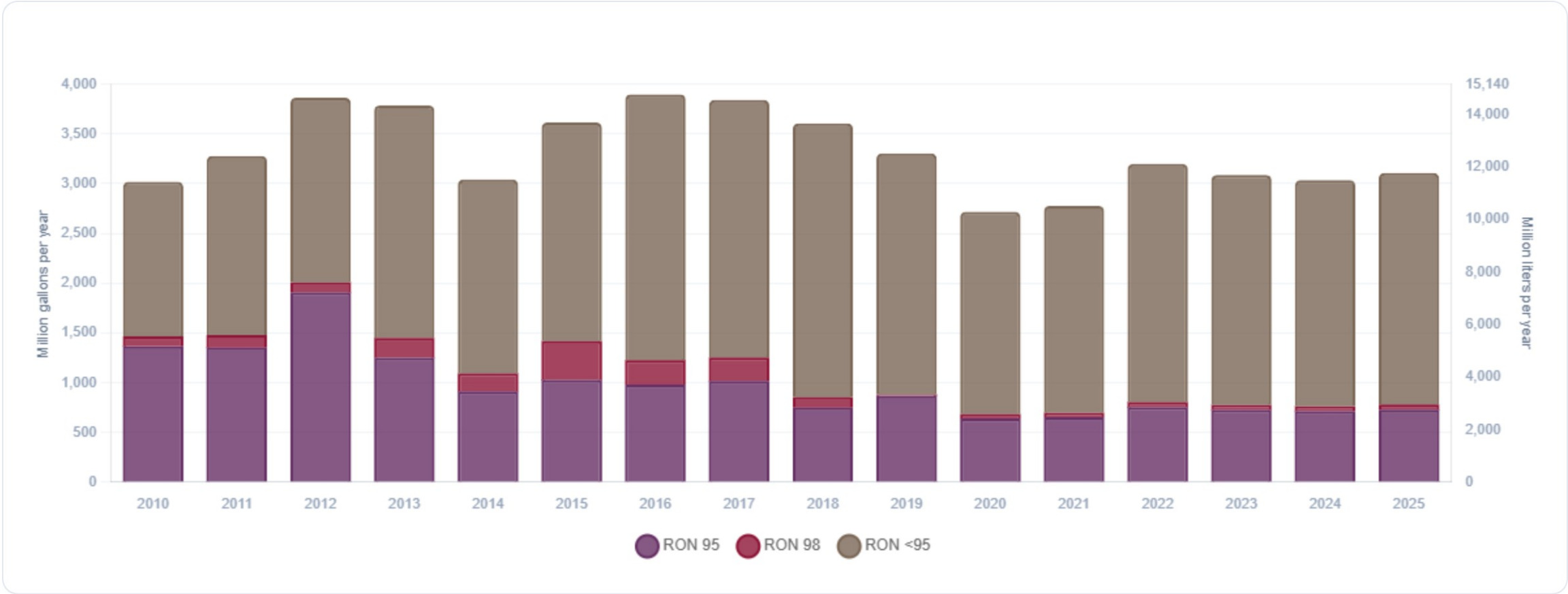
Source: Eurostat

Gasoline Imports in United Kingdom



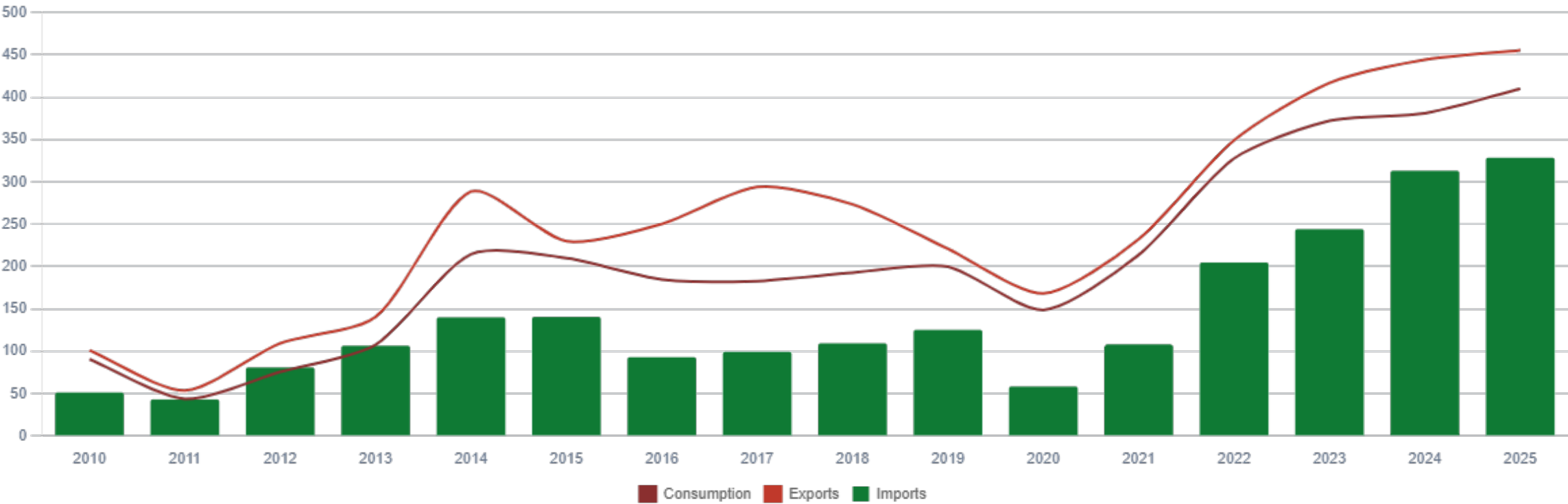
Source: Eurostat

Gasoline Exports in United Kingdom



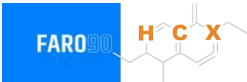
Source: Eurostat

Ethanol Balance in United Kingdom



Source: Eurostat

Gasoline Quality in United Kingdom

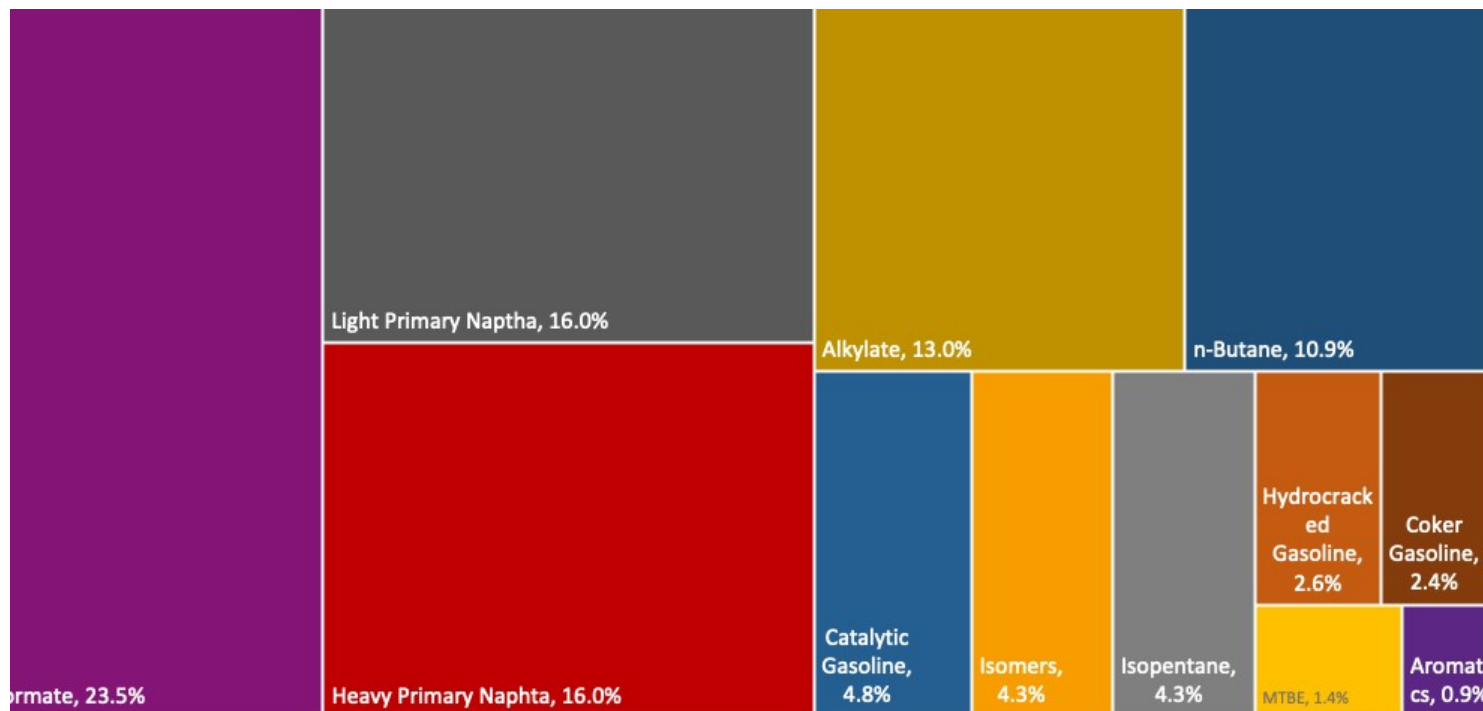


Name	EN 228:2025 (Euro 6); Country Level Fuel Ordinances							
Implementation Date	2025							
Applicability	All EU Contries + UK							
Grade	RON 95 E5		RON 95 E10		RON 98 E5		RON 98 E10	
	Min	Max	Min	Max	Min	Max	Min	Max
RON	95.0		95.0		98.0		98.0	
MON	> 85	> 88	> 85	> 88	> 85	> 88	> 85	> 88
AKI								
Benzene (%vol)	< 1							
Aromatics (% vol)	< 35							
Olefins (%vol)	< 18							
Lead (g/L)	< 0.005							
Manganese (mg/L)	< 2,0							
Sulfur (ppm)	10.0							
Oxygen (%w)*	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7	< 2,7	< 3,7
Ethanol (%vol)	< 5	< 10	< 5	< 10	< 5	< 10	< 5	< 10
PVR 37.8°C (psi)	<> 45 - 100 kPa in 6 Volatility Classes							
Overall Biofuels (%cal)	-							
Advanced Biofuels (%cal)	-							
Bioethanol in Gasoline Sales (%cal)	-							
GHG Emission Reduction (%)	-							

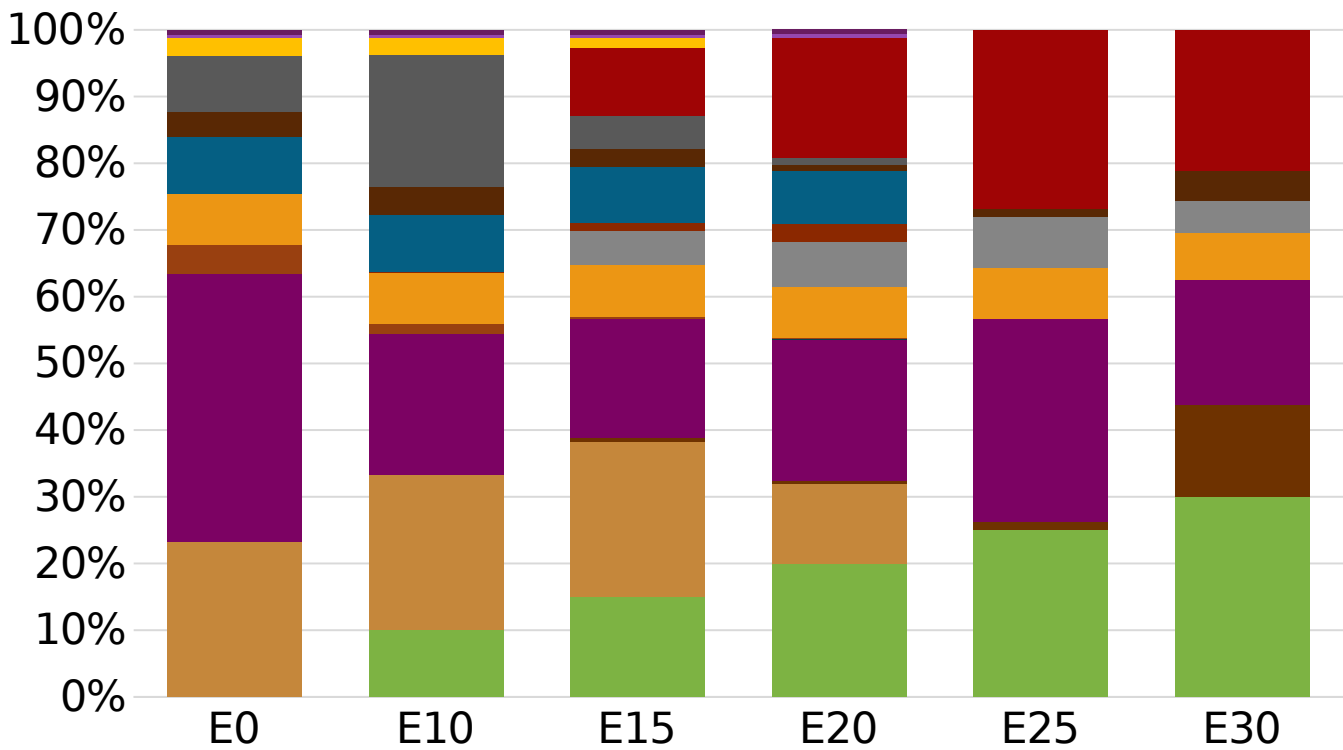
* Methanol 3%, Iso-propyl alcohol 12%, Ter-butyl alcohol 15%, Iso-butyl alcohol 15%, Ethers +5 Carbons 22%, Other Oxygenates 15%

Source: European Commision

United Kingdom Available Blending Components

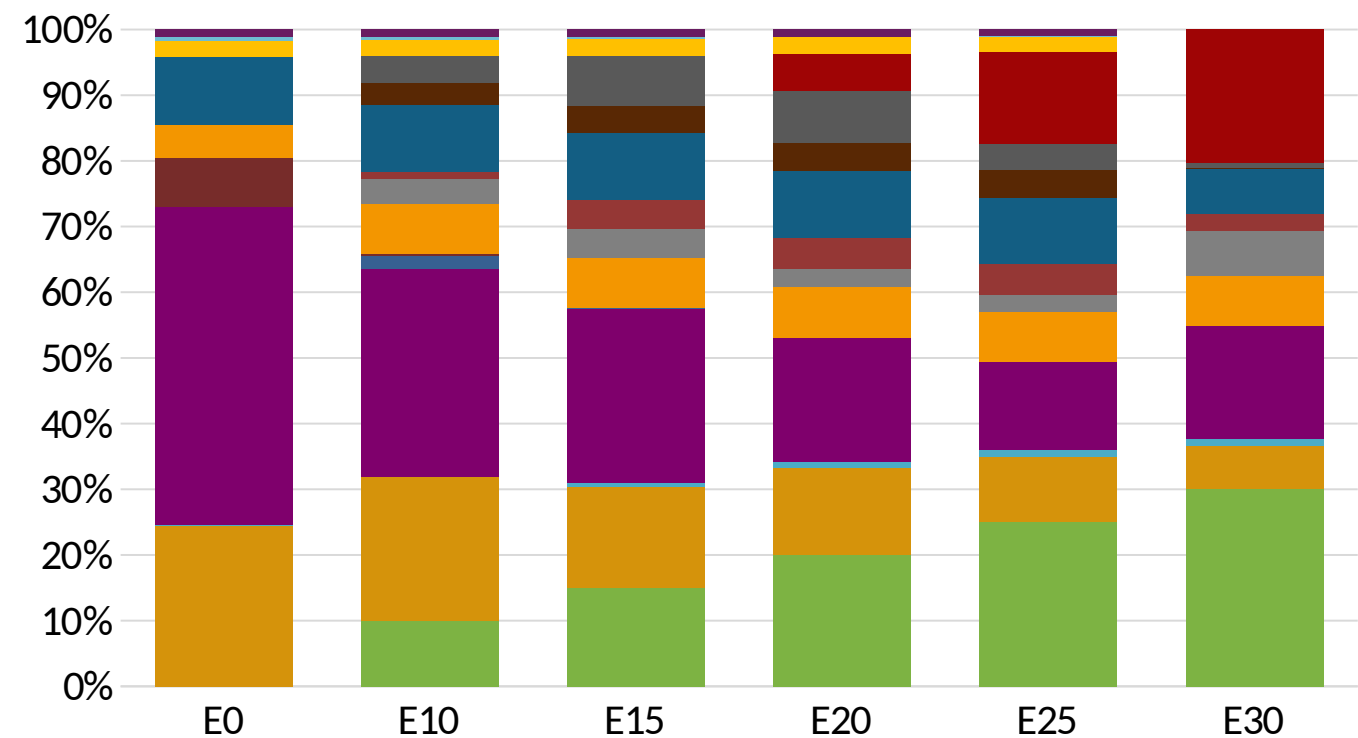


United Kingdom – Gasoline 95 – Constant Octane



Average CIF 2025
Prices

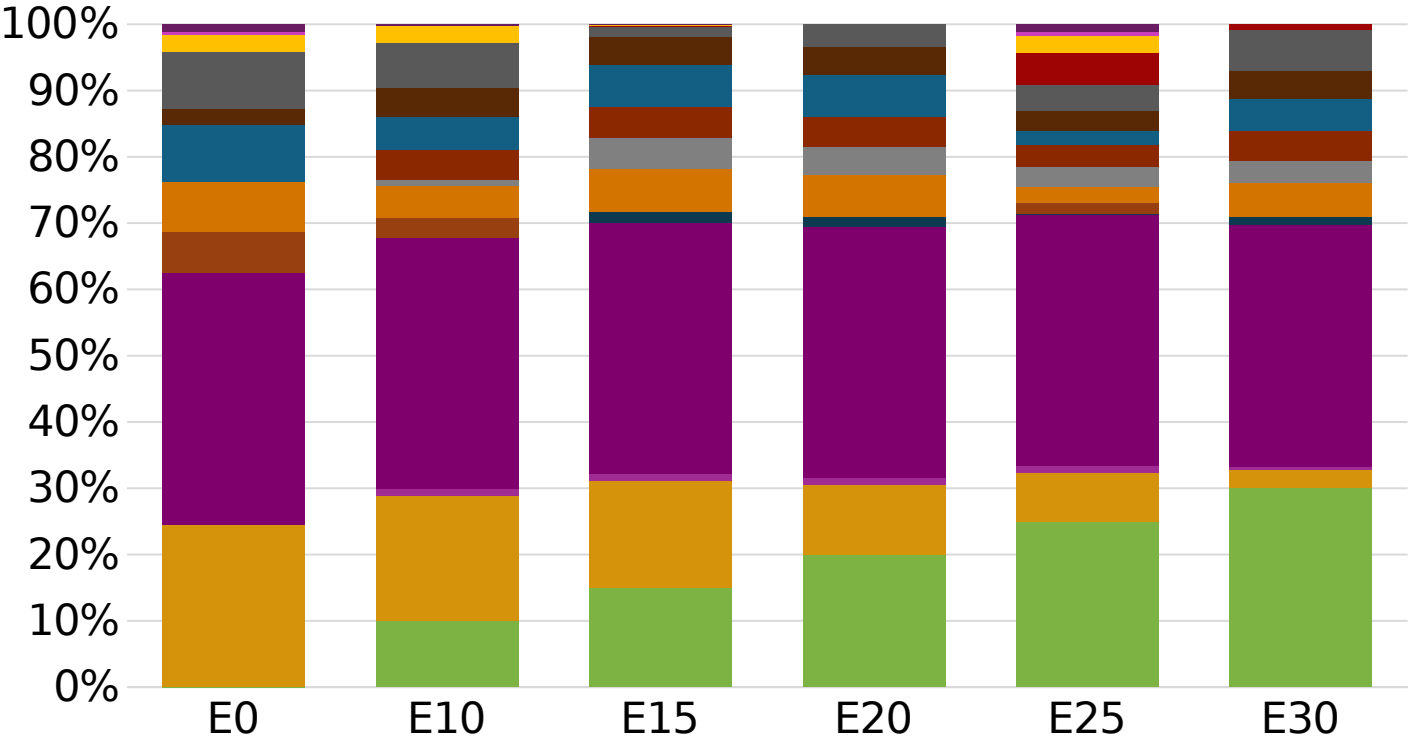
United Kingdom - Gasoline 98 - Constant Octane



Octane (RON)	98.0	98.0	98.0	98.0	98.0	98.0
Price (US\$/gal)	\$ 2.237	\$ 2.114	\$ 2.037	\$ 1.954	\$ 1.859	\$ 1.764

Average CIF 2025
Prices

United Kingdom – Gasoline 95 – Increased Octane

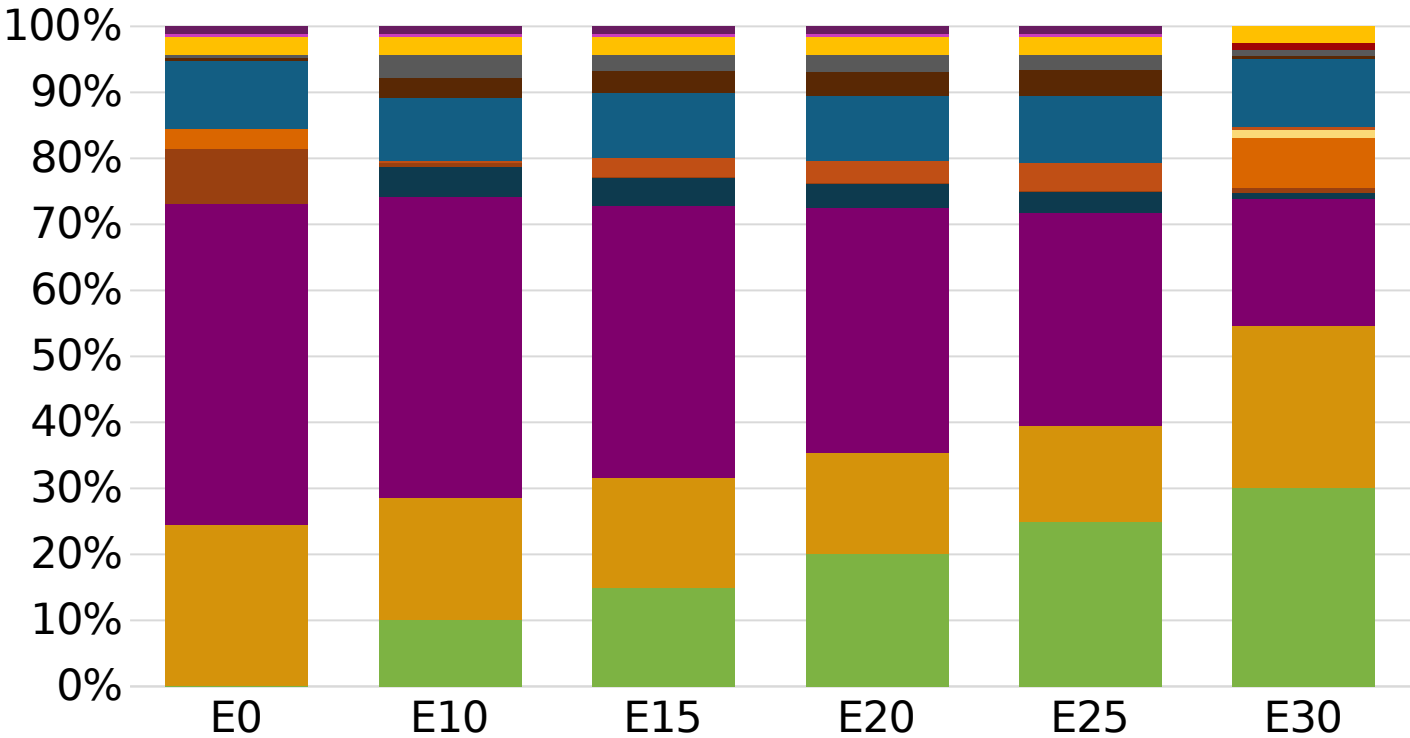


Average CIF 2025
Prices

Octane (RON)	95.0	97.9	99.3	100.9	102.6	103.9
Price (US\$/gal)	\$ 2.098	\$ 2.098	\$ 2.098	\$ 2.098	\$ 2.098	\$ 2.098

Source:
Faro90

United Kingdom – Gasoline 98 – Increased Octane

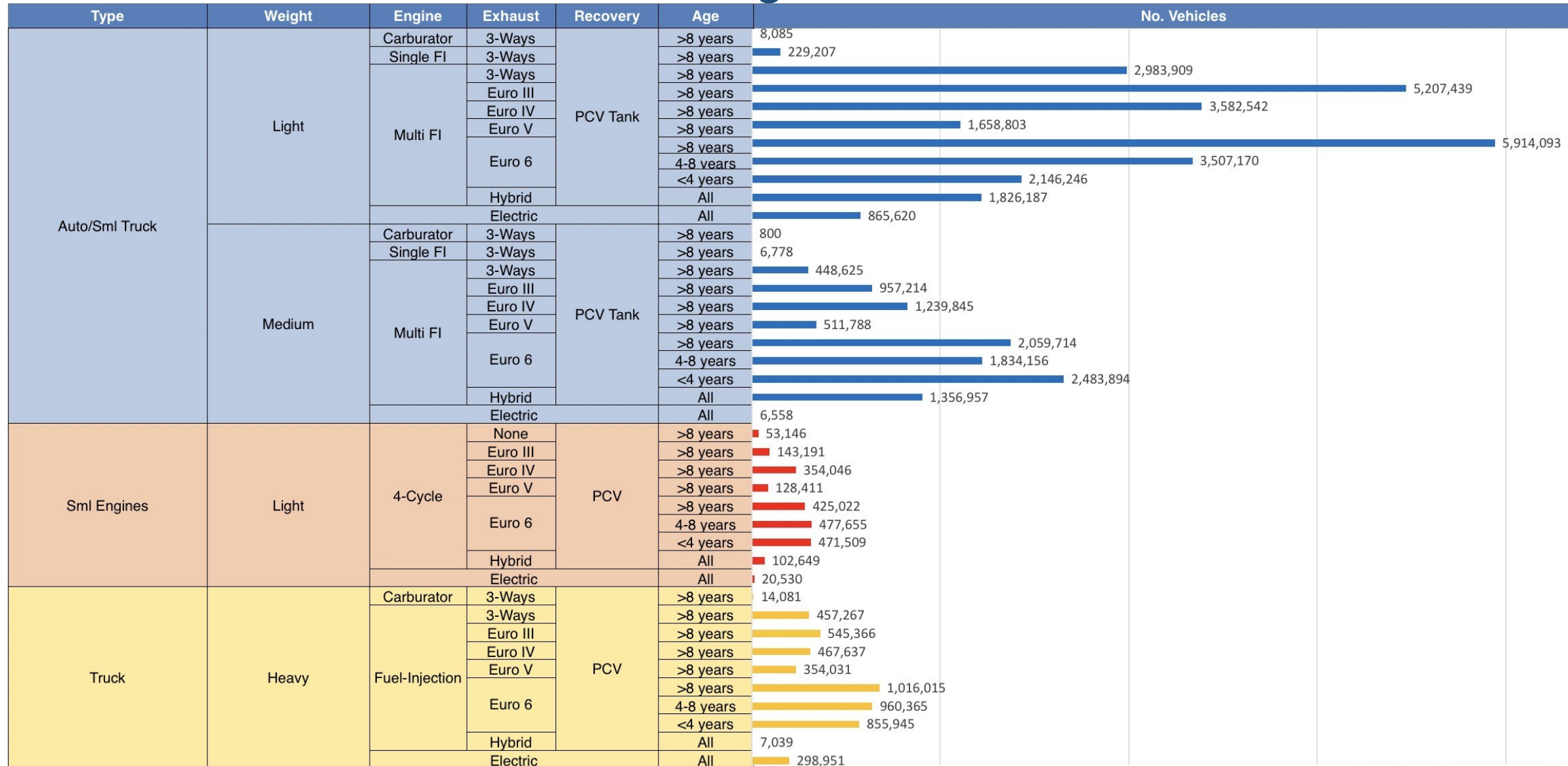


Average CIF 2025
Prices

Octane (RON)	98.0	100.7	102.0	103.3	104.5	105.0
Price (US\$/gal)	\$ 2.222	\$ 2.222	\$ 2.222	\$ 2.222	\$ 2.222	\$ 2.222

Source:
Faro90

Gasoline Vehicle Fleet - United Kingdom

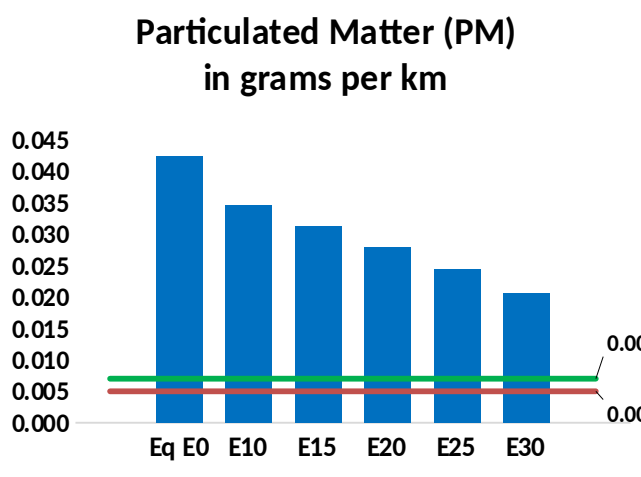
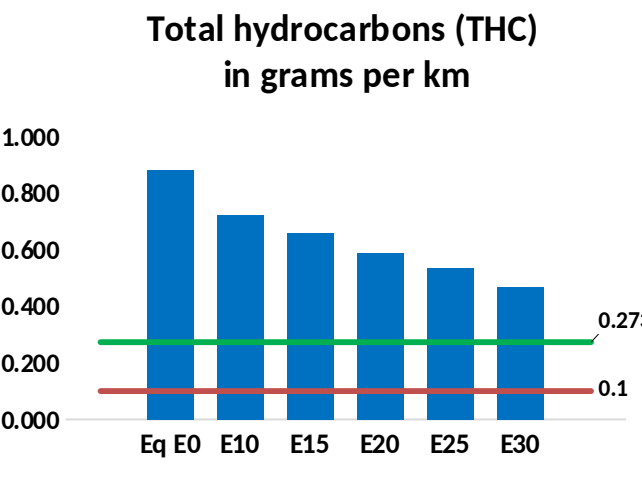
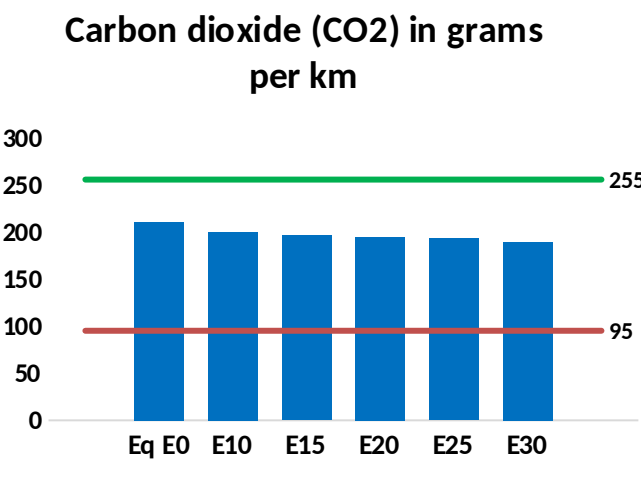
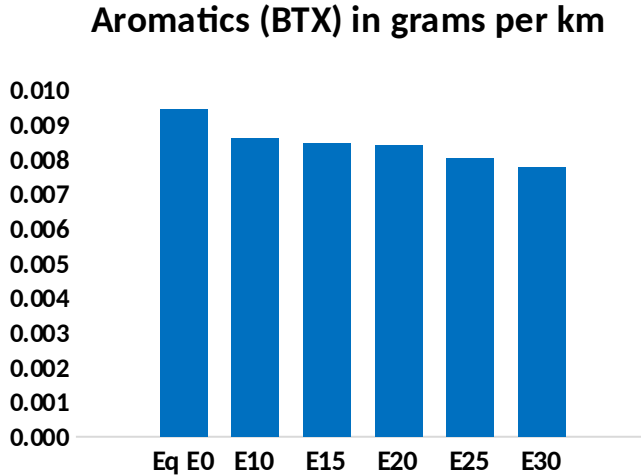
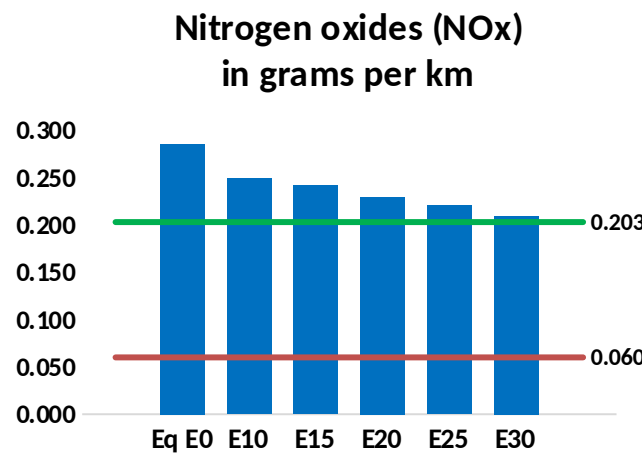
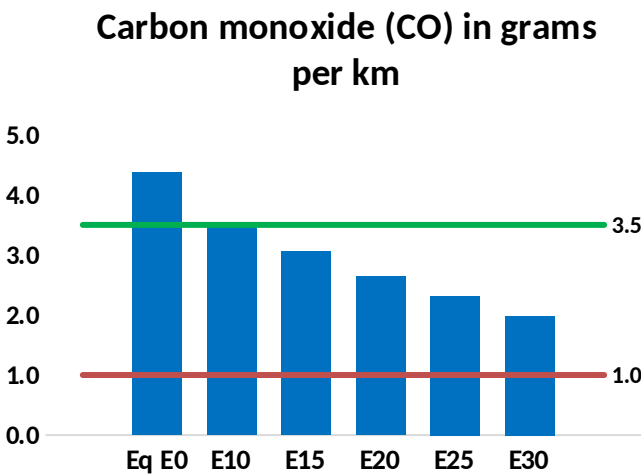
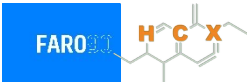


Vehicle Fleet: 45,988,484
Gasoline Vehicle Fleet: 23,441,918

Source: Eurostat, ACEA

United Kingdom – Vehicular Emissions

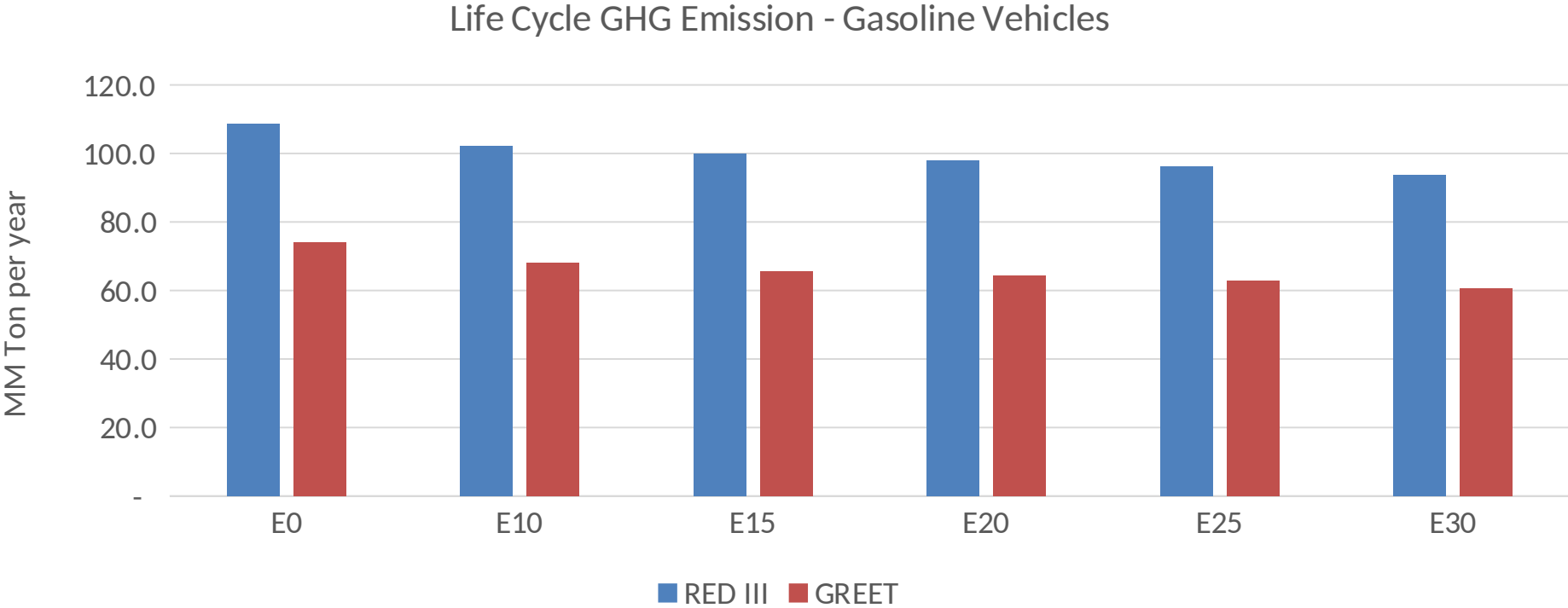
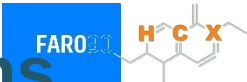
TIER III BIN 125 United States
Euro 6 Standard



United Kingdom – Vehicular Emissions

Vehicular Fleet Emissions (kg/day)	Eq E0	E5	E10	E15	E20	E25	E30	Reduction (%)	
								E10	E20
CO	2,708,873	2,423,550	2,138,226	1,889,376	1,632,028	1,432,054	1,227,312	-21%	-40%
CO2	130,366,312	127,014,036	123,847,997	121,358,000	120,125,852	119,403,382	117,202,434	-5%	-8%
VOC	545,921	497,111	448,301	406,968	364,760	332,118	289,214	-18%	-33%
NOx	176,765	163,213	154,116	149,587	142,173	136,416	129,604	-13%	-20%
SOx	1,494	1,382	1,269	1,173	1,081	982	877	-15%	-28%
NH3	48,691	48,209	47,727	47,954	48,493	48,873	49,188	-2%	0%
N2O	2,316	2,303	2,294	2,305	2,352	2,379	2,406	-1%	2%
CH4	108,612	108,612	108,612	110,242	112,631	114,695	116,650	0%	4%
Lead	-	-	-	-	-	-	-	0%	0%
Butadiene	2,536	2,401	2,266	2,167	2,072	2,001	1,901	-11%	-18%
Acetaldehyde	8,657	9,936	14,579	22,202	30,248	34,564	40,841	68%	249%
Formaldehyde	32,812	31,900	37,186	43,665	45,746	50,607	55,091	13%	39%
Benzene	5,853	5,625	5,330	5,245	5,202	4,975	4,814	-9%	-11%
PM2.5	7,602	6,911	6,220	5,612	5,017	4,379	3,711	-18%	-34%
PM10	18,606	16,915	15,223	13,735	12,278	10,717	9,083	-18%	-34%

United Kingdom – Gasoline Vehicle Fleet Life Cycle GHG Emissions



Country Emissions 2025 (MMT/año)	402.0
Target @ 2035 (MMT/year)	264.3
Delta	137.7

%Reduction vs E0	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	8.1%	11.7%	13.3%	15.4%	18.4%
RED II/III	0.0%	6.0%	8.0%	9.9%	11.5%	13.7%

% 2035 Target	E0	E10	E15	E20	E25	E30
GREET 2024	0.0%	4.3%	6.3%	7.2%	8.3%	9.9%
RED II/III	0.0%	4.7%	6.3%	7.8%	9.1%	10.8%

Source: Greet, RED II/III, IPCC, climateactiontracker.org, F90